

YOUR PERSONAL COACH



PHYSICS

for JEE Main and Advanced

Magnetism and Electromagnetic Induction



Mc
Graw
Hill

Shashi Bhushan Tiwari

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*All those who are behind technological innovations
to fight climate change*



Preface

This book forms a part of the series “Your Personal Coach”. Like other books in this series, this book has been written on two core principles:

- (i) a text book must have continuity and flow in what it discusses.
- (ii) nothing contributes more in understanding Physics than a good example.

I have tried to unfold the concepts gradually, one-by-one; illustrating each of them with examples. The main aim is to make the students learn the basic principles of Physics independently.

In this book we will explore the close linkage between electricity and magnetism which will help us understand the nature of light (electromagnetic wave). The story that will evolve at the end will truly fascinate you.

I shall be grateful to everyone who would provide feedback or help me with useful suggestions.

S.B. Tiwari

How to use this Book

To make full use of this book one must go through the topics sequentially while working through the examples and in-chapter problems given under heading “**Your Turn**”. By doing this you will have a fair amount of grasp over all the essentials in a chapter.

Miscellaneous examples given at the end of each chapter have problems which involve multiple concepts or have some mathematical complexity or are tricky. *If you are studying the subject for the first time or are hard pressed for time, you may skip the section on miscellaneous examples.*

Almost every solved example starts with explanation of physical situation and basic principles involved. This feature comes under heading “**Concepts**” at the beginning of each example.

I have highlighted the important points of learning under the heading “**In short**”. Here, I have also taken important learning points from the examples. While going through the chapter it is essential to go through these points.

Physics cannot be mastered without practice. Keeping this in view I have given three Worksheets (exercises) after every chapter. **Worksheet 1** has multiple choice objective type questions with single correct answer. **Worksheet 2** has multiple choice questions having *one or more than one correct answers*. **Worksheet 3** has subjective problems. A good number of problems has been given in the Worksheets to give you a good practice on concepts learnt.

After few chapters, at regular intervals, you will find separate assignments on miscellaneous type problems. These are problems based on latest trend of competitive examinations and contain **Match the Column type questions and problems based on a given paragraph**. Attempt these questions only after you gain enough confidence in the related chapters. I have kept these problems in separate chapters so that you have no bias or hint about the equation/s to use.

In the last chapter, you will find a collection of questions asked in competitive examinations since 2005. This is an ideal collection of problems for revision.

In the end of the book, solutions to all questions has been given. Solutions are quite descriptive and easy to understand.

Those who desire to practice at even higher level, I recommend my book – “*Problems in Physics for JEE Advanced*”.

I hope you will enjoy this book.

S.B. Tiwari

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I thank to all who helped me in preparation of this book. My special thanks to –

- My students, who have taught me a lot.
- The management at McGraw-Hill which has shown a lot of patience.

S.B. Tiwari

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Magnetic Field

“What magnetism is, no one knows. We can only think of it as a peculiar condition created in space by the motion of electricity.”

—Sydney Evershed (1925)

1. INTRODUCTION

More than 2000 years ago, certain stones (called lodestones) were found in the coastal district of Magnesia, in ancient Greece. These stones had the unusual property of attracting iron pieces. The term ‘magnet’ originated from the name ‘Magnesia’. Magnets were first crafted into compasses and used for navigation by the Chinese in the twelfth century.

The subjects of electricity and magnetism developed almost independently till 1820, when a Danish physicist Oersted discovered that an electric current exerts force on a magnetic compass. Thereafter, a series of experiments and discoveries established that electricity and magnetism are very closely related and it is not wise to study them separately. Ampere proposed that electric currents are the source of all magnetic phenomena. In 1867, Maxwell expressed that both the phenomena—electricity and magnetism—go hand in hand and one can produce another. The subject matter is now appropriately known as ‘electromagnetism’.

In this book we will explore the close linkage between the electricity and magnetism. In this chapter we will study about the magnetic force on a moving charge and a current carrying wire. But before we do so, we will introduce ourselves to magnetic field produced by a bar magnet.

2. BAR MAGNET

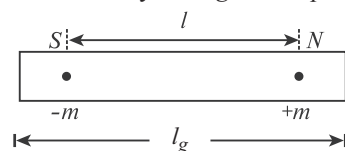
A magnet exerts force on another magnet, even without touching it. We can say that a magnet creates a magnetic field in its surrounding and this field exerts force on other magnets.

Simplest form of a magnet is a bar magnet. It has two ends of opposite nature. When it is suspended freely by tying a string at its centre, it acts like a compass. One end which points northward is called the *north pole* of the magnet. The other end, which points southward, is known as the *south*

pole of the magnet. When the north pole of one magnet is brought near the north pole of another magnet, they repel. The same is true for a south pole placed near another south pole. However, opposite poles attract one another.

Physicists have speculated for a long time about the existence of discrete *magnetic charges* (we can call them *magnetic monopoles*), just like positive and negative electric charges. But all our attempts to find tiny particles which will carry either a single north or a single south pole, has proved to be futile. If we go on cutting a magnet into many small pieces, each small piece has its own north pole and south pole. Even when one piece is one atom thick, there are two poles. This suggests that the atoms, in a magnet, themselves are magnets.

Though we have never found an isolated magnetic pole (i.e., a magnetic charge), we often think of a bar magnet as having two magnetic charges separated by a distance. The north pole is assumed to carry a positive magnetic charge ($+m$) and the south pole is assumed to have a negative magnetic charge ($-m$). The unit of magnetic charge (or, pole strength) is $A-m$. The distance between the two poles of a bar magnet is 0.84 times the geometrical length of the magnet. A bar magnet is essentially a *magnetic dipole*.



$$l = \text{magnetic length, } l_g = \text{geometrical length, } l = 0.84 l_g$$

The *dipole moment* of a bar magnet is defined to be a vector quantity directed from the south pole to the north pole of the magnet. Its magnitude is

$$M = ml \quad (1)$$

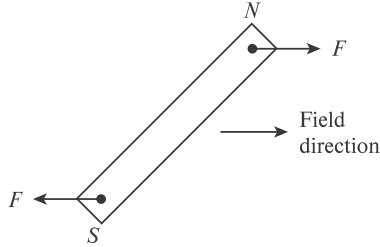
where m is pole strength and l is the magnetic length. A magnet with large magnetic dipole moment (M) produces

1.2 Magnetism and Electromagnetic Induction

large field in its surrounding. The dipole moment is measure of strength of a magnet.

2.1 Direction of magnetic field

The direction of magnetic field at a point is defined as the direction of magnetic force acting on a magnetic north pole (positive magnetic charge) placed at that point. A south pole, kept in a magnetic field, experiences a force opposite to the direction of the field.



A north pole experiences force in the direction of field.
A south pole experiences force opposite to the field.

2.2 Strength of magnetic field

We can define the strength of magnetic field at a point as magnetic force experienced by a unit magnetic charge kept at that point. If a magnetic pole of strength m experiences a force F when kept in a magnetic field, the strength of the field at the location is

$$B = \frac{F}{m}$$

Unit of magnetic field (B) is $\frac{N}{A-m}$, which is commonly

known as tesla (T) or weber/ m^2 (Wb/m^2). Another common unit is gauss.

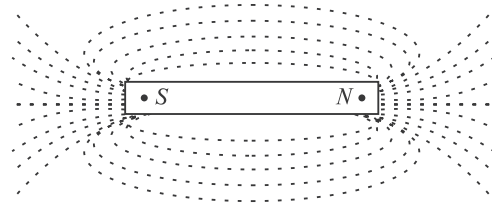
$$1T = 10^4 \text{ gauss}$$

Now, we can say that a magnetic pole of strength m placed in a magnetic field $\vec{B}$ experiences a force given by

$$\vec{F} = m \vec{B} \quad (2)$$

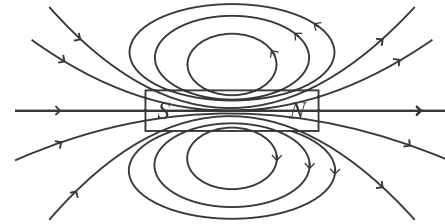
2.3 Magnetic field lines due to a bar magnet

Assume a bar magnet placed on a table. If we wish to know the direction of field at a point due to this bar magnet, we need a small magnetic compass. Place the compass at a point on the table. The direction in which the north pole of the compass points is the direction of field at that point. We can keep the compass at various points and know the direction of field at each point. Another method is to spread small iron filings around the magnet and tap the table so that the iron pieces can overcome friction. An iron piece, when placed near a magnet, develops magnetic poles in it. It becomes a small magnet. We say that it has been magnetised [we will learn about the reason in a later chapter]. Every iron piece becomes a magnetic compass and points in the direction of the field at that point. We can see the direction of field at every point.



Iron filings around a bar magnet.

To represent the variation of a magnetic field in space we can draw field lines just like the electric field lines. A tangent to a field line must represent the direction of field at a point and density of lines is proportional to the strength of the field. The next figure shows that field lines due to a bar magnet. Note that the magnetic field lines are always closed loops. Density of lines is very high inside the magnet.



Field lines due to a bar magnet. All lines are closed loops.

2.4 Magnetic field strength due to a bar magnet

A magnetic charge (m) produces a magnetic field (B) at a distance r which is given by a law similar to the Coulomb's law.

$$B = \left(\frac{\mu_0}{4\pi} \right) \frac{m}{r^2} \quad (3)$$

μ_0 is a constant known as *magnetic permeability of free space*. In SI system its value is

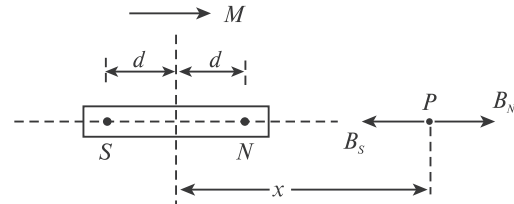
$$4\pi \times 10^{-7} \frac{T-m}{A}$$

The field is radially outward if the magnetic charge is positive and is radially inward if it is negative.

Magnetic field obeys principle of superposition.

Field on the axis of a bar magnet

The line joining the two poles of a magnet is known as its axis. Consider a bar magnet of magnetic length $2d$ having pole strength m . The dipole moment of the magnet is a vector from south pole to its north pole and its magnitude is $M = m(2d)$.



P is a point at a distance x from the centre of the magnet, on its axis. Distance of P from the positive magnetic charge (i.e., the north pole) is $(x - d)$ and its distance from the negative charge is $(x + d)$.

Field at P due to the north pole is

$$B_N = \frac{\mu_0}{4\pi} \frac{m}{(x-d)^2} \text{ directed towards right.}$$

Field at P due to the south pole is

$$B_S = \frac{\mu_0}{4\pi} \frac{m}{(x+d)^2} \text{ directed towards left.}$$

The resultant field is

$$B = B_N - B_S = \frac{\mu_0 m}{4\pi} \left[\frac{1}{(x-d)^2} - \frac{1}{(x+d)^2} \right]$$

$$\therefore B = \frac{\mu_0}{4\pi} \frac{m(4xd)}{(x^2 - d^2)^2}$$

$$\text{or, } B = \frac{\mu_0}{4\pi} \frac{2Mx}{(x^2 - d^2)^2} \quad [\because M = m(2d)] \quad (4)$$

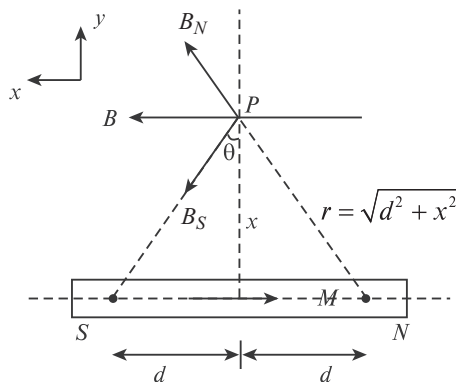
When $x \gg d$, the above result can be approximated as

$$B \simeq \frac{\mu_0}{4\pi} \frac{2M}{x^3} \quad (5)$$

Compare this result with the expression of electric field due to an electric dipole at a point on its axis.

Field at broadside – on position

A position on the perpendicular bisector of a bar magnet is known as broadside-on position. Let P be a point at a distance x from a magnet on its broadside-on position.



Field at P due to the north pole is

$$B_N = \frac{\mu_0}{4\pi} \frac{m}{r^2} \text{ (direction is along } \overrightarrow{NP})$$

Field at P due to the south pole is

$$B_S = \frac{\mu_0}{4\pi} \frac{m}{r^2} \text{ (direction is along } \overrightarrow{PS})$$

Field at P is vector sum of B_N and B_S . It is easy to see that components of the two fields along y direction will cancel out. Their components in x direction add.

Field at P is directed opposite to the dipole moment vector ($\vec{M}$) of the dipole and its magnitude is

$$\begin{aligned} B &= B_N \sin \theta + B_S \sin \theta = \frac{2\mu_0}{4\pi} \frac{m}{r^2} \sin \theta \\ &= \frac{\mu_0}{4\pi} \frac{2md}{r^3} \\ \text{or, } B &= \frac{\mu_0}{4\pi} \frac{2md}{(d^2 + x^2)^{3/2}} \\ &= \frac{\mu_0}{4\pi} \frac{M}{(d^2 + x^2)^{3/2}} \end{aligned} \quad (6)$$

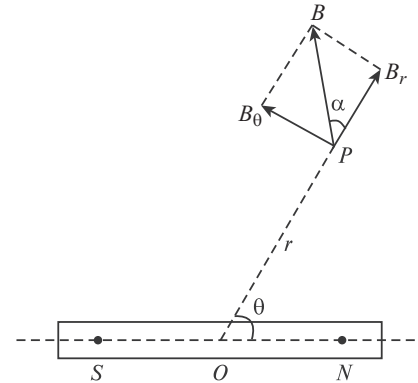
For $x \gg d$,

$$B \simeq \frac{\mu_0}{4\pi} \frac{M}{x^3} \quad (7)$$

Again, compare this result with that for an electric dipole.

Field at a general point

Consider a point P at a distance r from the centre (O) of a magnetic dipole (i.e., a bar magnet). The line OP makes an angle θ with the axis of the dipole. Let $r \gg d$, where $2d$ is distance between the poles.



Field at P can be worked out by using any of the methods learnt in the chapter of electrostatics. We are not presenting the derivation here, just stating the result.

The component of field along OP is given by

$$B_r = \frac{\mu_0}{4\pi} \frac{2M \cos \theta}{r^3} \quad (8)$$

$M = m(2d)$ is the dipole moment of the magnet. The component of field in a direction perpendicular to OP is given by

$$B_\theta = \frac{\mu_0}{4\pi} \frac{M \sin \theta}{r^3} \quad (9)$$

Resultant field at P is

$$B = \sqrt{B_r^2 + B_\theta^2}$$

1.4 Magnetism and Electromagnetic Induction

$$\text{or, } B = \frac{\mu_0}{4\pi} \frac{M}{r^3} \sqrt{1 + 3 \cos^2 \theta} \quad (10)$$

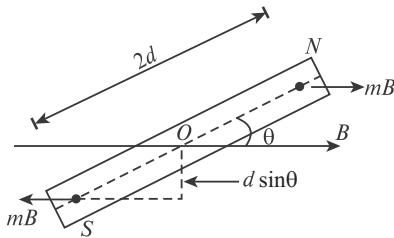
The angle made by B with the line OP is given by

$$\tan \alpha = \frac{B_\theta}{B_r} = \frac{1}{2} \tan \theta$$

$$\Rightarrow \alpha = \tan^{-1} \left(\frac{1}{2} \tan \theta \right) \quad (11)$$

2.5 Torque on a bar magnet in a uniform magnetic field

Consider a bar magnet of magnetic length $2d$ and pole strength m placed in a uniform magnetic field B . Angle between the direction of the field and the magnetic axis is θ .



Force acting on the north pole is mB in the direction of the external field and the force on the south pole is mB in a direction opposite to the field. These two forces form a couple and their torque is

$$\tau = mBd \sin \theta + mBd \sin \theta$$

$$\text{or, } \tau = 2mdB \sin \theta$$

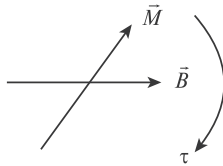
$$\text{or, } \tau = MB \sin \theta$$

$$M = m(2d) = \text{Magnetic dipole moment.}$$

The torque tries to rotate the dipole so as to align it parallel to the direction of the field. In vector notation, we can write the expression of the torque as

$$\vec{\tau} = \vec{M} \times \vec{B} \quad (12)$$

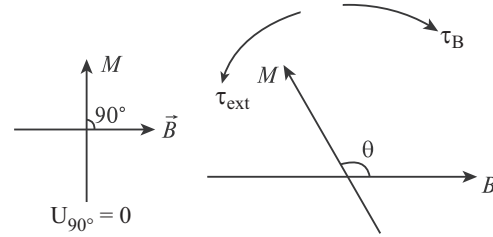
You must compare this to the expression of torque on an electric dipole placed in a uniform electric field ($\vec{\tau} = \vec{P} \times \vec{E}$).



A magnetic dipole experiences a torque in an external field which tries to align $\vec{M}$ parallel to $\vec{B}$.

2.6 Potential energy of a magnetic dipole in a magnetic field

Potential energy of a magnetic dipole is considered to be zero when it is held perpendicular to a magnetic field. If we wish to rotate it to some other angle θ , we must do work against the magnetic torque. Consider the dipole at an angle θ to the direction of the field.



For slowly rotating the dipole $\tau_{\text{ext}} = \tau_B$.

Magnetic torque on it is $\tau_B = MB \sin \theta$

We wish to rotate the dipole slowly, we must apply an equal torque in opposite direction.

$$\tau_{\text{ext}} = \tau_B = MB \sin \theta$$

Work done by the external agent in rotating the dipole further by $d\theta$ is

$$dW_{\text{ext}} = \tau_{\text{ext}} \cdot d\theta = MB \sin \theta d\theta$$

Total work done by the agent in rotating the dipole from initial angle $\theta = 90^\circ$ to final angle θ is

$$W_{\text{ext}} = \int_{90^\circ}^{\theta} dW_{\text{ext}} = MB \int_{90^\circ}^{\theta} \sin \theta d\theta$$

$$\text{or, } W_{\text{ext}} = -MB (\cos \theta - \cos 90^\circ) = -MB \cos \theta$$

This work done is change in potential energy of the dipole.

$$U_\theta - U_{90^\circ} = -MB \cos \theta$$

$$\therefore U_{90^\circ} = 0$$

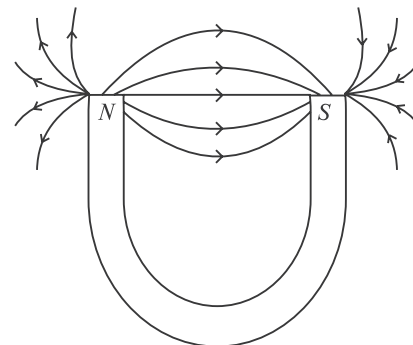
$$\therefore U_\theta = -MB \cos \theta$$

$$= -\vec{M} \cdot \vec{B} \quad (13)$$

Recall that the potential energy of an electric dipole in an electric field is $-\vec{P} \cdot \vec{E}$.

NOTES

- The above description of a magnet with two opposite magnetic charges is a useful model in understanding its magnetic behaviour. In practice, nobody has ever found an isolated magnetic charge.

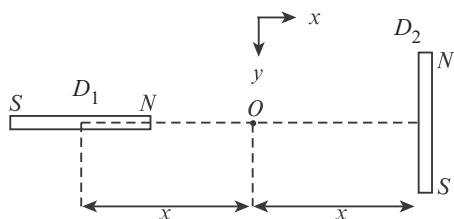


A horseshoe magnet.

2. A common horseshoe magnet is simply a bar magnet that has been bent into a U shape. The figure shows a rough sketch of field lines due to such magnet. All field lines are closed loops. We have not shown the field lines inside the material of the magnet.

Example 1 Two identical magnetic dipoles (D_1 and D_2) are placed at a separation $2x$ with their axes perpendicular.

The dipole moment of each dipole is M and their lengths is short compared to x . Find the strength of magnetic field at a point O that is midway between them.



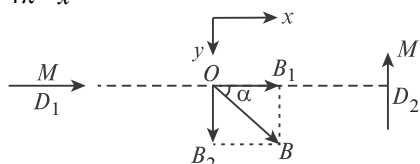
Solution

Concepts

- The magnetic field obeys the principle of superposition.
- Field due to the dipole D_1 is along x direction at point O .
- Field due to the dipole D_2 is along y direction (antiparallel to the dipole moment vector of D_2).

Field at O due to dipole D_1 is

$$B_1 = \frac{\mu_0}{4\pi} \frac{2M}{x^3} \text{ (along } x\text{)}$$



Field at O due to D_2 is

$$B_2 = \frac{\mu_0}{4\pi} \frac{M}{x^3} \text{ (along } y\text{)}$$

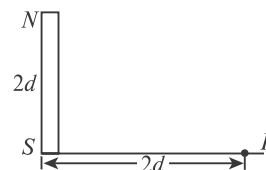
Note that O is at a broadsideon position for D_2 .

$$\begin{aligned} \therefore B &= \sqrt{B_1^2 + B_2^2} \\ &= \frac{\mu_0 M}{4\pi x^3} \sqrt{4 + 1} \\ &= \frac{\sqrt{5} \mu_0 M}{4\pi x^3} \end{aligned}$$

Direction of B makes an angle α with the x direction, where

$$\begin{aligned} \tan \alpha &= \frac{B_2}{B_1} = \frac{1}{2} \\ \Rightarrow \alpha &= \tan^{-1} \left(\frac{1}{2} \right) \end{aligned}$$

Example 2 A bar magnet of length $2d$ has magnetic dipole moment M . It is placed vertically on a horizontal table with its south pole on the table. Find the horizontal component of magnetic field at a point P on the table due to the magnet. The point P is at a distance $2d$ from the lower end of the magnet. Assume that the magnetic poles are at the geometrical ends of the magnet.



Solution

Concepts

- In practice, the magnetic poles are not exactly at the geometric ends of a magnet. But the question asks us to assume so.
- Field at P due to the south pole is horizontal along $\overrightarrow{PS}$. Field at P due to the north pole is along $\overrightarrow{NP}$. We will find its horizontal component and then add it vectorially to the horizontal field produced by the south pole.
- We will assume the pole strength to be m and write $m(2d) = M$

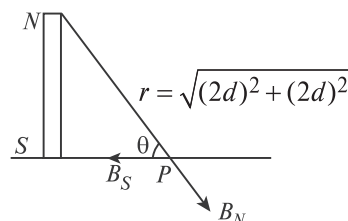
Field due to the north pole is

$$B_N = \frac{\mu_0}{4\pi} \frac{m}{r^2} = \frac{\mu_0}{4\pi} \frac{m}{(2\sqrt{2}d)^2}$$

From geometry, $\theta = 45^\circ$

$\therefore$ Horizontal component of B_N is

$$\begin{aligned} B_{NH} &= B_N \cos 45^\circ \\ &= \frac{\mu_0}{4\pi} \frac{m}{8\sqrt{2}d^2} \quad (\rightarrow) \end{aligned}$$



1.6 Magnetism and Electromagnetic Induction

Field at P due to the south pole is

$$B_S = \frac{\mu_0}{4\pi} \frac{m}{(2d)^2} (\leftarrow)$$

$\therefore$ Resultant horizontal field at P is

$$\begin{aligned} B_H &= B_S - B_{NH} (\leftarrow) \\ &= \frac{\mu_0}{4\pi} \frac{m}{4d^2} \left[1 - \frac{1}{2\sqrt{2}} \right] (\leftarrow) \\ &= \frac{\mu_0}{4\pi} \frac{m(2d)}{8d^3} \left(1 - \frac{1}{2\sqrt{2}} \right) (\leftarrow) \\ &= \frac{\mu_0 M}{32\pi d^3} \left(1 - \frac{1}{2\sqrt{2}} \right) (\leftarrow) \end{aligned}$$

Example 3 A bar magnet has a magnetic dipole moment 2.0 A-m^2 . It is released in a uniform magnetic field $B = 25 \mu\text{T}$ from a position where its dipole moment makes an angle of 120° with the direction of the field. Find the maximum kinetic energy of the dipole after its release. Assume no other force apart from the magnetic force.

Solution

Concepts

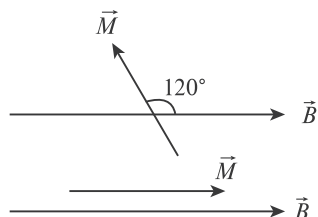
- The bar magnet has a potential energy in a magnetic field. It loses potential energy to gain Kinetic Energy.
- Kinetic Energy is maximum at position where the potential energy is minimum. Potential Energy is given by $U = MB \cos\theta$. It is minimum when $\theta = 0^\circ$.

$$U_i = -MB \cos 120^\circ = \frac{MB}{2}$$

$$U_f = -MB \cos 0^\circ = -MB$$

$$\therefore \text{Loss in potential energy} = \frac{3}{2} MB$$

$$\begin{aligned} \text{Gain in kinetic energy} &= \frac{3}{2} MB \\ &= \frac{3}{2} \times 2 \times 25 \times 10^{-6} \\ &= 75 \mu\text{J} \end{aligned}$$



Potential Energy is minimum in this position.

In Short

- A bar magnet has two poles (magnetic charges)—north pole (positive magnetic charge) and south pole (negative charge). A magnetic charge is denoted by 'm' and its unit is A-m .
- A pole of a magnet can never be separated. However small a magnet is sliced, it always has a north and a south pole.
- The poles of a magnet are slightly inside the geometrical edges. The pole to pole distance, known as magnetic length, is 0.84 times the geometrical length of the magnet.
- Magnetic dipole moment of a magnet is a vector directed from the south pole to the north pole. Its magnitude is $M = ml$, where l is the magnetic length. Unit of magnetic dipole moment is A-m^2 or J/T .
- The magnetic field due to a magnetic pole of strength m at a distance r from it is given by $B = \frac{\mu_0}{4\pi} \frac{m}{r^2}$

Field is radially outward if the source is a north pole. For a south pole it is directed towards the pole. Unit of B is tesla (T)

- The magnetic field obeys principle of superposition.
- The field at a distance x on the axis of a bar magnet is

$$B = \frac{\mu_0}{4\pi} \frac{2Mx}{(x^2 - d^2)^2}$$

The field at a distance x on the perpendicular bisector of the dipole is

$$B = \frac{\mu_0}{4\pi} \frac{M}{(d^2 + x^2)^{3/2}}$$

- When a bar magnet of dipole moment M is placed in a uniform field B , it experiences a torque given by $\vec{\tau} = \vec{M} \times \vec{B}$
- The potential energy of a dipole placed in a magnetic field is given by $U = -\vec{M} \cdot \vec{B}$

assuming the potential energy to be zero when $\vec{M}$ is perpendicular to $\vec{B}$.

Potential energy is minimum when $\vec{M}$ is parallel to $\vec{B}$ and the potential energy is maximum when $\vec{M}$ is anti parallel to $\vec{B}$.



YOUR TURN

Q.1 In what way are magnetic poles very different from electric charges?

Q.2 In what way are magnetic field lines different from electrostatic field lines?

Q.3 A bar magnet has a geometric length of 10 cm and its pole strength is 12A-m. Find the magnetic dipole moment of the magnet.

Q.4 Which property of a magnet has a unit of J/T?

Q.5 A bar magnet of length 10 cm has a magnetic dipole moment of 1.0 J/T. Find the magnetic field produced by the magnet at a point on its axis which is at a distance of 10 cm from its centre.

Q.6 In the last question, find the field on the perpendicular bisector of the magnet at a distance of 10 cm from its centre.
[Take $(1.25)^{3/2} = 1.4$]

Q.7 In which position the potential energy of a magnetic dipole is maximum when it is placed in a uniform magnetic field?

3. SOURCES OF MAGNETIC FIELD

A magnet is a practical source of magnetic field. However, it is not the most fundamental source. *Fundamental source of a magnetic field is moving charge (i.e., an electric current).* We will study about the magnetic effect of current in next chapter. A magnet is able to produce magnetic field due to moving charges (i.e., current) in its atoms. We will explore this further in a later chapter.

In the chapter of electromagnetic waves, we will learn that a time changing electric field can also create a magnetic field.

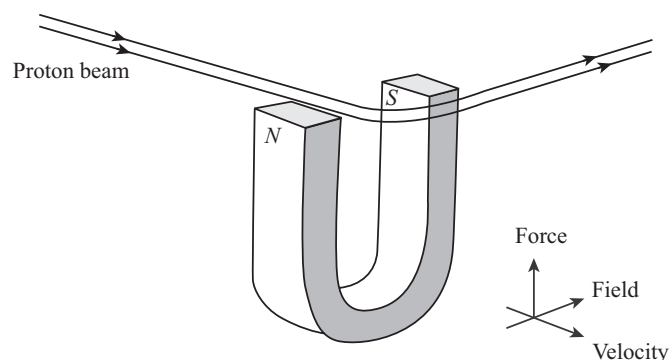
The Earth itself behaves like a magnet. The strength of magnetic field on the surface of the Earth ranges from 25 to 65 μT . In New Delhi the strength of the Earth's magnetic field is about 48 μT .

In practice, strong magnetic fields can be created by electromagnets. People have successfully created field of 100T also.

In the present chapter we will not worry much about the source of magnetic field. We will study the effect of a magnetic field on a moving charge and on a current carrying wire.

4. MAGNETIC FORCE ON MOVING CHARGED PARTICLES

A static charge does not experience any force due to a static magnetic field. But a moving charge, in general experiences a force in a magnetic field. If a proton beam is passed through a region between the poles of a magnet, it gets deflected as shown in the figure below. The magnetic field in the region between the poles exerts force on the moving protons. A similar force on moving charges can be seen when they pass near a current carrying wire. The magnetic field produced by the current is responsible for this force.



A proton beam gets deflected in a magnetic field.

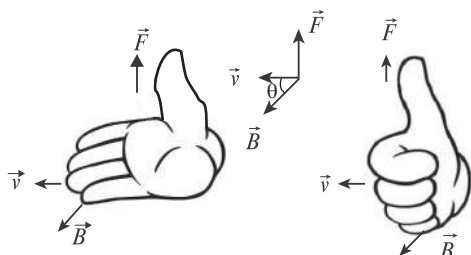
The magnetic force experienced by a charged particle is observed to be *greatest* when the particle moves perpendicular to the direction of the magnetic field. At other angles, the force is smaller and *becomes zero* when the velocity of the charge is parallel or antiparallel to the field. Also, the force is proportional to the speed of the particle and the strength of the field. In one aspect this force is very different from other interactions like the gravitational force or the electrostatic force. The magnetic force on a moving charge is not along the line joining the charge to the source of the field. *The force acts perpendicularly to both the magnetic field and velocity of the charge particle.*

In the diagram shown above, if we replace the proton beam with an electron beam the direction of the force on the beam reverses. This shows that the direction of magnetic force depends on the sign of the charge.

All the experimental observations can be explained if we define the magnetic force acting on a moving charge by the equation

$$\vec{F} = q \vec{v} \times \vec{B} \quad (14)$$

If θ is the angle between the vectors $\vec{v}$ and $\vec{B}$, the magnitude of the magnetic force is $F = qvB \sin \theta$.



Stretch your right hand palm along the direction of velocity so that the direction of field ($\vec{B}$) is outward to your palm. You must be able to curl your fingers from the direction of $\vec{v}$ to the direction of $\vec{B}$. The stretched thumb gives the direction of the force.

The direction of force on a positive charge is along $\vec{v} \times \vec{B}$. The diagram given here will help you recall how to get the direction of $\vec{v} \times \vec{B}$ using the right hand rule. Remember that *force on a negative charge will be directed opposite to the direction of $\vec{v} \times \vec{B}$*

Because the magnetic force on a moving charge is always perpendicular to the velocity of the charge, the force can only deflect the direction of motion of the charge and will never change its speed. Magnetic force on a moving charge never performs work.

Example 4 A positively charged particle is moving horizontally in north-east direction with a velocity of 2×10^5 m/s. It enters into a region where exists a uniform, vertically downward, magnetic field of strength 0.02T. Charge on the particle is $q = 3.2 \times 10^{-19}$ C. Find the direction and magnitude of the magnetic force that acts on the particle.

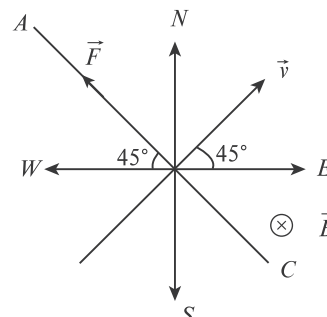
Solution

Concepts

- (i) Velocity of the particle is horizontal and the magnetic field is vertically downward. Angle between the two is 90° . Magnitude of the magnetic force on the particle will be $F = qvB \sin 90^\circ = qvB$.

- (ii) The direction of force must be perpendicular to $\vec{v}$ as well as $\vec{B}$. If you keep your note book horizontal, the magnetic field is directed perpendicularly into it. Force vector will be in the horizontal plane of your note book so that it is perpendicular to $\vec{B}$. The force vector is also perpendicular to the velocity vector.

The $\otimes$ in the figure indicates the direction of magnetic field vector. This symbol $\otimes$ represents a direction into the plane of the figure.



Force ($\vec{F}$) must be in the plane of the figure so that it is perpendicular to $\vec{B}$. The force must also be perpendicular to $\vec{v}$. This means it will be along the line AC. Now, use right hand rule. stretch your right hand palm along $\vec{v}$ so that you can curl your fingers downwards, towards B . The thumb points towards A . Therefore, the force is in *north-west* direction.

Magnitude of the force is

$$F = qvB \sin 90^\circ = 3.2 \times 10^{-19} \times 2 \times 10^5 \times 0.02 \\ = 1.28 \times 10^{-15} \text{ N}$$

NOTE

If the particle has a negative charge, it will experience force in south-west direction.



YOUR TURN

Q.8. A particle having charge q is projected towards north in a region where there is a magnetic field in the south direction. Field strength is B . Find the force on the particle.

Q.9. A charged particle having mass $M = 1$ mg and charge $q = 1 \mu\text{C}$ is projected in a magnetic field $B = 1 \text{ mT}$. Find the maximum possible acceleration of the charge if its speed is $1 \times 10^6 \text{ ms}^{-1}$

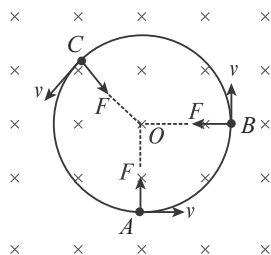
Q.10. A charged particle is projected vertically up in a region where a horizontal magnetic field exists in north direction. In which direction will the charge deflect (east or west) if it is (a) positive, (b) Negative?

Q.11. A negative charge $-q$ enters a region with velocity $\vec{v} = v_0 \hat{i} + v_0 \hat{j}$. A magnetic field $\vec{B} = B_0 \hat{k}$ exists in the region. Find the magnetic force on the particle.

Q.12. Write the dimensional formula of magnetic field (B).

4.1 Circular motion of a charged particle in a magnetic field

Imagine that there is a uniform magnetic field (B) directed into the plane of this page. A charged particle, having mass m and charge q , is projected from point A. The particle is given a velocity v in the plane of this page (see figure). In whichever direction you project the particle in the plane of this page, its velocity will be perpendicular to the direction of the magnetic field.



A charged particle projected perpendicularly into a magnetic field.

Immediately after receiving the velocity the particle starts experiencing a magnetic force. The force is perpendicular to the velocity in the plane of this figure (i.e., perpendicular to $\vec{B}$). The use of right hand rule tells us that the force on the particle (when it is at A) is in the direction shown in the figure. This force does not change the speed but deflects the particle in the plane of this paper. Let's say the particle reaches a point B after some time. Magnitude of the force on the particle has not changed. Its direction will be perpendicular to the velocity at B. This force will further deflect the particle and so on. The speed does not change and the force deflecting the particle also stays constant in magnitude. Obviously the particle will follow a path of constant curvature. It must move in a circle.

Magnitude of the magnetic force on the particle is

$$F = qvB \sin 90^\circ = qvB$$

This force provides the necessary centripetal force. If the particle moves in a circle of radius R , we can write

$$\frac{mv^2}{R} = qvB$$

$$\text{or, } R = \frac{mv}{qB} \quad (15)$$

The above equation can also be written as

$$R = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB} \quad (16)$$

where $p = mv$ = momentum of the particle.
and K = Kinetic energy of the particle.

The angular speed of the particle can be written as

$$\omega = \frac{v}{R} = \frac{qB}{m} \quad (17)$$

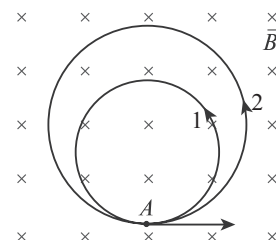
The time period of revolution can be written as

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB} \quad (18)$$

The frequency of circular motion is

$$f = \frac{1}{T} = \frac{qB}{2\pi m} \quad (19)$$

Note that the time period (or the frequency) of circular motion is independent of the speed of the particle. Whatever be the speed, the particle takes same amount of time to complete the circle. Actually, if you double the speed of the particle, the radius of circular path doubles (See equation 15). This means that doubling the speed doubles the circumference. Obviously, time period will not change.



Particle projected from A at smaller speed will follow path 1. Same particle projected with higher speed will follow path 2.

Time period will be same in both the cases.

Example 5 A proton and a deuteron

A proton and a deuteron have same kinetic energy. They enter perpendicularly into a uniform magnetic field.

Find the

- ratio of radii of their circular paths, and
- ratio of time period of their circular motion.

Solution

Concepts

- A deuteron is basically a deuterium nucleus. It has one proton and a neutron. Mass of a deuteron will be nearly twice that of a proton, charge on both the particles is same.
- Equation 16 and 18 shall be used to get the required ratios.

Let mass of the proton be m . Mass of the deuteron is $2m$. Charge on both of them is e . Both have same kinetic energy (Say K).

$$(a) \text{ Radius of the circular path is } R = \frac{\sqrt{2mK}}{qB}$$

K , q and B are same for both the particles.

$$\therefore R \propto \sqrt{m}$$

$$\therefore \frac{R_p}{R_d} = \sqrt{\frac{m}{2m}} = \frac{1}{\sqrt{2}}$$

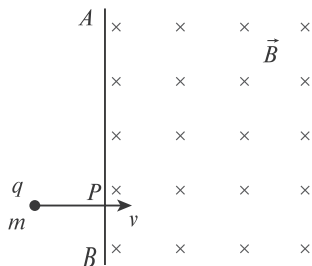
(b) Time period of circular motion is $T = \frac{2\pi m}{qB}$

$$\therefore T \propto m$$

$$\therefore \frac{T_p}{T_d} = \frac{m}{2m} = \frac{1}{2}$$

Example 6 Semicircular path in magnetic field

Region to the right of plane AB has a uniform magnetic field of strength B . There is no field in the region to the left of AB . A particle having mass m and charge q enters the region of magnetic field at point P while moving with a velocity v perpendicular to AB . Show the path of the particle and calculate the distance that it will travel inside the magnetic field.



Solution

Concepts

Path of the particle inside the field will be circular. Once the particle crosses AB while moving to the left, it will travel in a straight line in absence of any force.

As soon as the particle enters into the region of magnetic field, it begins to move in a circular path. Radius of the, circular path is

$$R = \frac{mv}{qB}$$

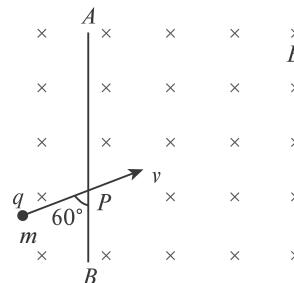
To fix the location of the centre of the circle, you should note that the velocity at P is along the tangent to the circle. Radius of the circle must be perpendicular to the velocity. Magnetic force on the particle (at P) is upward in the figure. This force is towards the centre of the circle. Therefore, the centre is a point (O) on the line AB such that $PO = R$.

After completing the semicircle, when the particle reaches point Q the force on it becomes zero (as there is no field to the left of the point Q). The particle moves in a straight line after this. The direction of motion of the particle has been changed by 180° .

$$\text{Distance travelled in the field region} = \pi R = \frac{\pi mv}{qB}$$

Example 7 One third of a circle in magnetic field

In the last question, the particle is made to enter the magnetic field at an angle of 60° with AB . Everything else is same as in the last question. Find the time spent by the particle in the field region.



Solution

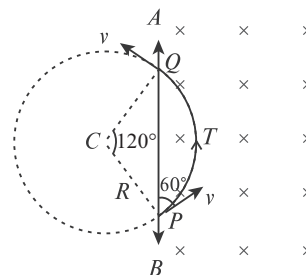
Concepts

- Circular motion begins as soon as the particle enters the field region. After leaving the field region the particle will travel in a straight line.
- Centre of the circle will not lie on line AB . A radius of the circle can be obtained by drawing a perpendicular on the direction of the velocity. The magnetic force is directed towards the centre.
- One easy way to figure out the path is to assume that the field exists everywhere (i.e., in the region to the left of AB also). Path will be a complete circle in this case. Now, think which point on this circle could be the point P given in the question. Draw line AB passing through the point P . Now you can see the arc of the circle along which the particle must have moved before leaving the field region.

Radius of the circular path will be

$$R = \frac{mv}{qB}$$

Had there been a field everywhere, the particle would have moved in a circle as shown in the figure. The point of projection P is such that velocity at the point makes an angle of 60° with the line AB .



The particle moves along the circular arc PTQ in the field region and exits at Q . Thereafter it goes straight.

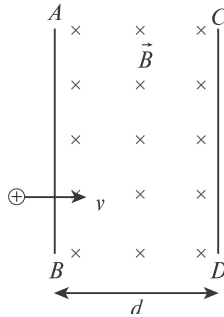
One can easily show that $\angle PCQ = 120^\circ$

Time needed to complete one third circle is obviously.

$$t = \frac{T}{3} = \frac{2\pi m}{3qB}$$

Example 8 Deflection in a narrow field.

The region between the parallel planes AB and CD contains a uniform magnetic field of strength B directed into the plane of the figure. Width of the region is d . A charged particle having mass m and charge q enters into the field region moving perpendicular to AB . Find the angular deviation in the path of the particle caused by the magnetic field if $d < \frac{mv}{qB}$.

**Solution****Concepts**

- Radius of the circular path is $R = \frac{mv}{qB}$. It is given that $d < R$. The particle will fail to complete a semicircular path as in example 6. It will definitely exit to the right of CD .
- The deviation angle is the angle between the initial and final velocities of the particles.

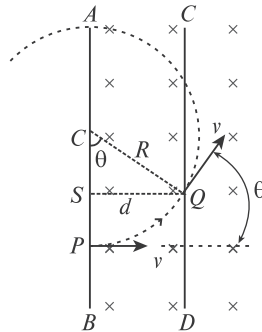
Particle moves along the arc PQ and exits the field region at Q .

θ is the angle between the initial and the final velocities. CP is perpendicular to the velocity at P and CQ is perpendicular to the velocity at Q . It is easy to see that $\angle PCQ = \theta$.

$$\text{In triangle } CSQ : \sin\theta = \frac{SQ}{CQ}$$

$$\therefore \sin\theta = \frac{d}{R} = \frac{d \cdot qB}{mv}$$

$$\theta = \sin^{-1} \left(\frac{dqB}{mv} \right)$$

**NOTE**

If angle between two lines is θ , then angle between their respective perpendiculars is also θ .

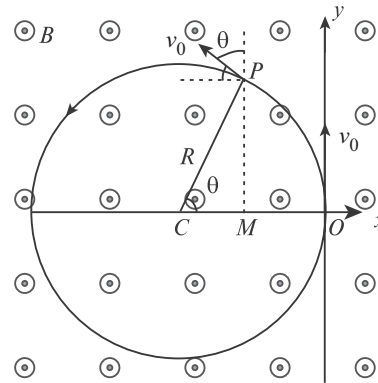
Example 9 Writing velocity and co-ordinates

A particle having mass m and charge $-q$ is projected with a velocity $\vec{v} = v_0 \hat{j}$ from the origin at $t = 0$. There exists a uniform magnetic field $\vec{B} = B_0 \hat{k}$ in the region. Write the velocity and co-ordinates of the particle as a function of time(t).

Solution**Concepts**

- Show the circular path in the xy plane. Remember that the magnetic field is always perpendicular to the plane of the circular motion of the particle.
- The particle rotates through an angle $\theta = \omega t$ in time t , where angular speed ω is given by equation 17.

The symbol $\odot$ indicates a direction out of the plane of the figure. Let this be our Z direction. Magnetic field exists in this direction.



$\vec{B}$ is directed out of the plane of the figure.

The charge is projected in y direction and the direction of initial force on it is towards negative x direction. [The force is directed along $-(\vec{v} \times \vec{B})$ as the charge is negative]

Therefore, the centre of the circle is a point C on the negative x axis.

$$\text{Angular speed of the particle is } \omega = \frac{qB}{m}.$$

$$\text{In time } t \text{ the particle rotates through an angle } \theta \text{ given by } \theta = \omega t = \frac{qB}{m} t.$$

Magnitude of the velocity (i.e., speed) does not change. Hence, velocity at time t can be written as

$$\vec{v} = -v_0 \sin\theta \hat{i} + v_0 \cos\theta \hat{j}$$

$$\text{or, } \vec{v} = -v_0 \sin \left(\frac{qBt}{m} \right) \hat{i} + v_0 \cos \left(\frac{qBt}{m} \right) \hat{j}$$

The co-ordinates of the particle are

$$\begin{aligned} x &= -OM = -(OC - MC) \\ &= -[R - R \cos\theta] = -R \left[1 - \cos \left(\frac{qBt}{m} \right) \right] \end{aligned}$$

$$y = MP = R \sin\theta = R \sin \left(\frac{qBt}{m} \right)$$



YOUR TURN

Q.13 Alpha particles are projected with a speed of 10 kms^{-1} in a direction perpendicular to a uniform magnetic field of magnitude 1.0 T . Find the radius of the circular path and frequency of revolution of the particle.

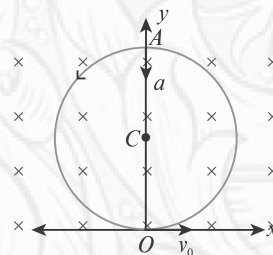
Q.14 A particle having charge q and mass m is accelerated through a potential difference of V and then made to enter a uniform magnetic field B . Find the radius of the circular trajectory of the particle if the particle moves perpendicular to the field.

Q.15 A proton enters perpendicularly into a uniform magnetic field with a velocity of $4 \times 10^6 \text{ ms}^{-1}$. The magnetic field exists in a region that is just less than 10 cm wide in the direction of initial velocity of the proton. Find the angular deviation in the path of the proton caused by the field. Take the charge on the proton to be $e = 1.6 \times 10^{-19} \text{ C}$ and its mass to be $m = 1.6 \times 10^{-27} \text{ kg}$. Magnitude of the magnetic field is $B = 0.4 \text{ T}$.

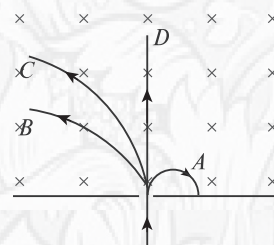
Q.16 Redo the question in example 7 considering that the charge on the particle is negative.

Q.17 A charged particle, when projected in uniform electric field, moves on a curved path. The same particle when projected perpendicularly into a uniform magnetic field, again moves in a curved path. What is the nature of the two paths?

Q.18 A particle having mass m and charge $+q$ is projected with a velocity $\vec{v} = v_0 \hat{i}$, from the origin of the co-ordinate system, into a region of uniform magnetic field $\vec{B} = -B_0 \hat{k}$. Find the time after which its acceleration will be in negative y direction for the first time.

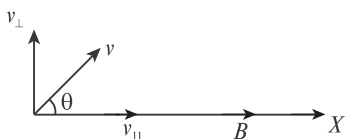


Q.19 A beam of charged particles enter perpendicularly into a magnetic field. The beam contains neutrons, electrons, protons and alpha particles. The particles follow the tracks A, B, C and D as shown in figure. Which track corresponds to which particle?



4.2 Helical path of a charged particle in a magnetic field

Consider a region having a uniform magnetic field of strength B directed along positive x direction. A particle having charge q and mass m is projected with a velocity v making an angle θ with the direction of the field. To understand the motion of the particle, let's divide its velocity into two components.



$v_{||}$ = component of velocity in the direction of field = $v \cos \theta$

$v_{\perp}$ = component of velocity in a direction perpendicular to the direction of the field = $v \sin \theta$

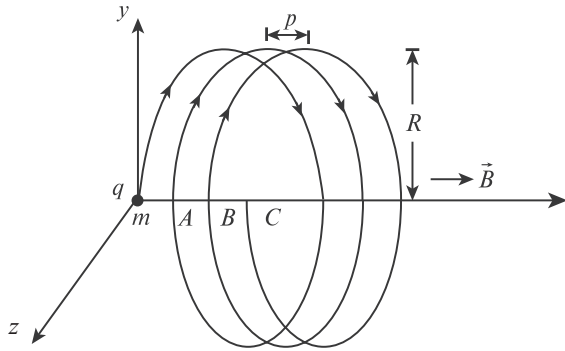
Due to the velocity component $v_{\perp}$ the particle will experience a force in the magnetic field and will describe a circle in yz plane (or a plane parallel to the yz plane). The radius of this circle and the time period of circular motion is given by

$$R = \frac{mv_{\perp}}{qB} = \frac{mv \sin \theta}{qB} \quad (20)$$

$$T = \frac{2\pi m}{qB} \quad (21)$$

The particle has another velocity component $v_{||}$ which remains constant. Due to this velocity component the particle continuously moves in x direction.

By superimposing the motions due to $v_{\perp}$ and $v_{||}$, one can easily see that the particle will describe a helical path. The radius of the helix is given by equation 20. The time needed for the particle to rotate by 360° in yz plane is the time period of the motion and it is given by equation 21. Carefully understand the figure on the next page with the comments written under it.



A charge moving in a helical path in a uniform B . The figure shows three completed rotation of the charge. The particle rotates in a plane perpendicular to $\vec{B}$. The z co-ordinate of the particle is never positive in this figure. The particle touches the axis at A , B and C only.

The *pitch* of the helical path is defined as the distance moved by the particle in the direction of $\vec{B}$ by the time it makes one full rotation in plane perpendicular to $\vec{B}$. The pitch is given by

$$P = v_{||} \cdot T = \left(\frac{2\pi m}{qB} \right) v \cos \theta \quad (22)$$

Example 10 A proton enters into a region having a uniform magnetic field $\vec{B} = (0.3\text{T}) \hat{k}$. The proton enters the field region at the origin of the co-ordinate system with a velocity $\vec{v} = (\sqrt{2}\hat{i} + \sqrt{2}\hat{j} + 2\hat{k}) \times 10^4 \text{ m/s}$.

- Find the time after which the proton will touch the z axis for the first time after entering into the field region.
- At what distance from the origin will the proton touch the z -axis for the first time?

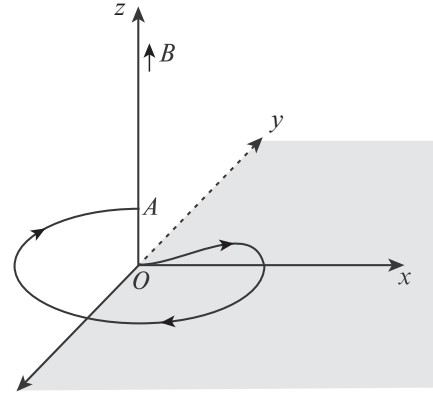
It is given that charge on a proton is $q = 1.6 \times 10^{-19} \text{ C}$ and its mass is $m = 1.6 \times 10^{-27} \text{ kg}$.

Solution

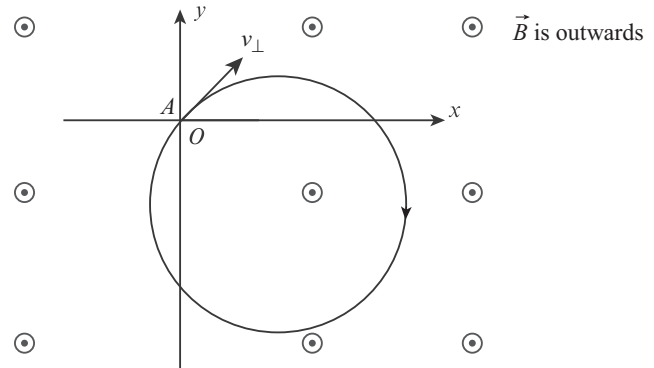
Concepts

- Field is in z direction. The velocity component in z direction is $v_{||} = 2 \times 10^4 \text{ m/s}$. The velocity component perpendicular to the field is $v_{\perp} = \sqrt{v_x^2 + v_y^2} = (\sqrt{2+2}) \times 10^4 = 2 \times 10^4 \text{ m/s}$.
- The particle will move in a helical path and will touch the z -axis after completing one full rotation

The figure below shows the path for one rotation. The second diagram shows how the path will look like, to an observer location on the z -axis. The first diagram shows that the particle rises from O to A along the z direction as it completes one rotation.



The particle touches the z -axis at A . OA is the pitch.



This figure shows the path in xy plane as seen by an eye located on the z -axis. The particle is coming out of the paper as it rotates.

- Time of motion from O to A = time period of the helical path

$$\text{or, } T = \frac{2\pi m}{qB} = \frac{2 \times 3.14 \times 1.6 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.3} = 2.1 \times 10^{-7} \text{ s}$$

- Distance OA = Pitch
 $= T \cdot v_{||} = 2.1 \times 10^{-7} \times 2 \times 10^4 = 4.2 \times 10^{-3} \text{ m} = 4.2 \text{ mm}$

Example 11 Position and velocity in helical path.

A region of space has a uniform magnetic field $B\hat{k}$. A particle having mass m and charge $+q$ enters into the region at the origin with a velocity $v\hat{i} + v\hat{k}$

- Write the velocity of the particle at time t after it enters the field.
- Write the co-ordinates of the particle as a function of time.

Solution**Concepts**

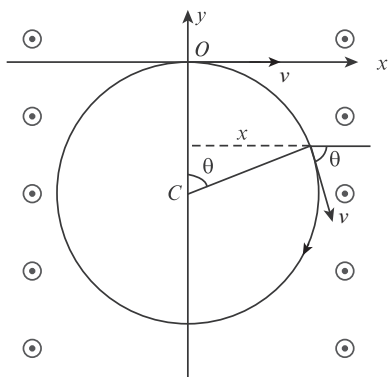
- Particle will describe a circular path in xy plane with speed $v_{\perp} = v$. Apart from this it will move uniformly in z direction with velocity component. $v_{||} = v_z = v$.
- We will first consider its motion in xy plane. Adopting an approach similar to that in example 9, we will be able to write the x and y component of velocity and x and y co-ordinates as a function of time. v_z remain constant at v and the displacement in z direction in time t is simply $z = v_z t = vt$

First consider the circular motion in xy plane. This motion takes place due to $v_{\perp}$ which is, initially, equal to $v\hat{i}$

$$\text{Angular speed, } \omega = \frac{qB}{m}$$

Angular displacement in time t is

$$\theta = \omega t = \frac{qB}{m} t$$



This is projection in xy plane for the path taken by the particle. The particle is moving out of the plane of the figure in a helical path

- x and y velocity components at time t are

$$v_x = v \cos \theta = v \cos \left(\frac{qBt}{m} \right)$$

$$v_y = -v \sin \theta = -v \sin \left(\frac{qBt}{m} \right)$$

z component of velocity remains unchanged.

$$v_z = v$$

$$\therefore \vec{v} = v \left[\cos \left(\frac{qBt}{m} \right) \hat{i} - \sin \left(\frac{qBt}{m} \right) \hat{j} \right] + v \hat{k}$$

- Co-ordinates at time t are: (see figure)

$$x = R \sin \theta = \frac{mv}{qB} \sin \left(\frac{qBt}{m} \right)$$

$$y = -(R - R \cos \theta) = -\frac{mv}{qB} \left[1 - \cos \left(\frac{qBt}{m} \right) \right]$$

$$z = vt$$

In Short

- Electric current is the basic source of magnetic field. A magnet exhibits magnetism due to current inside it at atomic scale. Later we will study that a time changing electric field also induces a magnetic field.
- A moving charge experiences force in a magnetic field. The force is given by $\vec{F} = q(\vec{v} \times \vec{B})$.
- Magnetic force on a moving charge is directed perpendicular to $\vec{B}$ as well as $\vec{v}$.
- Magnetic force does not change the speed of a charge. It can only change its direction of motion. Work done by a magnetic force on a moving charge is zero.
- When a charge is projected perpendicularly into a uniform magnetic field, it moves in a circular path. Plane of the circle is perpendicular to the magnetic field. Radius of circular path, time period of motion and angular speed are given by

$$R = \frac{mv}{qB} = \frac{p}{qB} = \sqrt{\frac{2mK}{qB}}$$

$$T = \frac{2\pi m}{qB} \quad \text{and} \quad \omega = \frac{2\pi}{T} = \frac{qB}{m}$$

- When a charge is projected into a uniform magnetic field with its initial velocity v making an angle θ with $\vec{B}$, the particle moves in a helical path ($\theta \neq 0^\circ, 90^\circ, 180^\circ$). The radius, time period and pitch for the helical path are

$$R = \frac{mv \sin \theta}{qB}, \quad T = \frac{2\pi m}{qB}, \quad p = \left(\frac{2\pi m}{qB} \right) v \cos \theta$$



YOUR TURN

Q.20 An electron with a kinetic energy of 90 eV moves into a region of uniform magnetic field of strength $B = 3.14 \times 10^{-4}$ T. The angle between the direction of $\vec{B}$

and initial velocity of the electron is 53° . Find the radius and pitch of the helical path of the electron.

[Mass of electron = 9.0×10^{-31} kg; $1\text{eV} = 1.6 \times 10^{-19}$ J]

5. MOTION OF A CHARGED PARTICLE IN ELECTRIC AND MAGNETIC FIELD

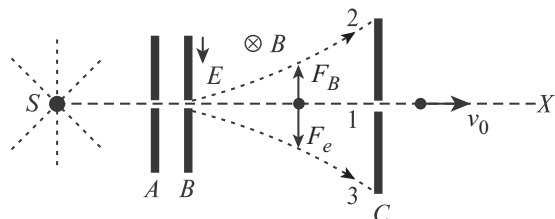
When a charge enters a region having both an electric field ($\vec{E}$) and a magnetic field ($\vec{B}$) it experiences an electric force as well as a magnetic force. The electromagnetic force on the particle is sometimes known as Lorentz force. It can be expressed as

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B}) \quad (23)$$

By suitable choice of the electric and magnetic field we can make a charged particle move on various kind of paths. We can also make interesting devices using the two fields. Below, we present the basic principle of a velocity selector, a mass spectrometer and a cyclotron

5.1 Velocity selector

This device can be used to obtain a parallel beam of charged particles with all the particles moving at same velocity. In the figure shown, S is a source throwing out charged particles having different speeds. The parallel slits A and B act as a collimator. All the particles which are able to cross through the two slits are moving along a straight line. However, they may have different speeds. Our intention is to have a beam in which all the particles are moving with same velocity.



Velocity Selector: Only particles having a particular speed are able to travel in a straight line and emerge out of slit C.

After crossing the slit B, the particles enter into a region where we have uniform magnetic and electric fields perpendicular to each other. In the diagram shown, the magnetic field is directed into the plane of the figure and

the electric field is directed downwards. A positive charge moving along the straight line SX experiences an upward magnetic force equal to

$$F_B = qvB$$

The electric force on the particle is independent of its speed. It is directed downward. Its magnitude is

$$F_e = qE$$

A particle will continue along the straight line path if net force on it is zero. Let this happen for a particle having speed v_0 .

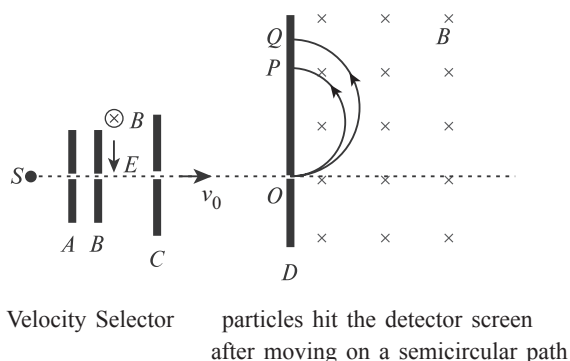
$$qv_0B = qE$$

$$\text{or, } v_0 = \frac{E}{B} \quad (24)$$

This means that all the particles moving with velocity $v_0 = \frac{E}{B}$ will experience no force and move in a straight line to come out of the slit C. All other particles having velocity greater than or less than v_0 will get deflected and hit the screen C. A particle having $v > v_0$ will experience a magnetic force higher than the electric force and will follow a path like 2 shown in the figure. A particle having $v < v_0$ will experience a magnetic force smaller than the electric force and will get deflected along a path like 3. Therefore, we have a beam of particles, all moving with velocity $v_0 = \frac{E}{B}$, to the right of slit C. We can choose the velocity that we want by adjusting the ratio $\frac{E}{B}$.

5.2 Mass spectrometer

This device can separate particles based on their masses and can be used to identify different isotopes of an element. Consider a source S which gives out singly ionised atoms of an element. These ions are passed through a velocity selector to obtain a beam of charged particles in which each particle has same speed v_0 . If needed we can accelerate the particles by applying a suitable potential difference. The beam of ions is now made to enter into another region having a uniform magnetic field (B).



The charged particles move on semicircular paths and hit the detector screen (D). Let's assume that the detector screen records that the particles strike it at two locations P and Q . This implies that there are two type of particles in the beam which moved on circular paths of diameters OP and OQ . Since all the particles were singly ionised and were travelling with the same speed, the only difference must be in their mass. The beam has particles with two different masses (say m_1 and m_2). The radii of circular paths for these particles will be

$$R_1 = \frac{m_1 v_0}{qB}$$

$$\text{and } R_2 = \frac{m_2 v_0}{qB}$$

$$\therefore \frac{R_1}{R_2} = \frac{m_1}{m_2} \quad (25)$$

By measuring the distance OP and OQ we can find $\frac{R_1}{R_2} = \frac{OP}{OQ}$. Therefore, we know the ratio of masses of the two types of particles. If they are ions of the same element, they are the two isotopes. Had there been three isotopes of the element the detector would have recorded particles at three locations.

5.3 Cyclotron

Many applications, particularly experiments in nuclear physics, require charged particles moving at high speeds. A cyclotron is a device which can be used to accelerate charged particles to impart them a kinetic energy of the order of few *MeV*. E.O. Lawrence built the first cyclotron in 1931 and it delivered protons with kinetic energy of 1.25 *MeV*.

A charged particle can be accelerated by applying a potential difference. But the method is not practical when we need to accelerate them to very high kinetic energies. If you wish to accelerate a proton to a kinetic energy of 10 *MeV*, you need a potential difference of 10^7 V.

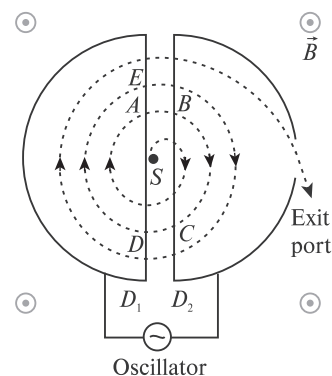
A cyclotron uses a moderate potential difference to accelerate a charged particle. The charge is made to follow a circular (rather spiral) path using a magnetic field and the same

potential difference is used again and again to impart energy to the particle. If a proton is made to cross a 10000 V potential difference 1000 times (everytime accelerating it), the kinetic energy will increase by $10^4 \times 10^3 = 10^7$ eV!

A cyclotron has two semicircular metallic hollow discs. Because of their shape the two pieces are often called as 'dees' of the cyclotron. They are kept with a small gap between them. A high frequency oscillator is connected across the two dees. For now, just understand that an oscillator produces a potential difference which keeps changing its polarity at regular intervals. An electromagnet is used to produce a uniform magnetic field perpendicular to the plane of the dees. Note that the dees are hollow and a charge can go inside it and come out. Assume that a charge comes out of the dee D_1 at point A (see figure). Also assume that at this moment the polarity of the oscillator is such that the dee D_1 is at higher potential than D_2 . There is an electric field in the gap between the two dees. Field is directed from D_1 to D_2 . This accelerates the charge as it crosses that gap to enter the second dee D_2 . Because the speed of the particle has increased, it travels on a path of higher radius inside the dee D_2 . By the time the charge reaches C, and again enters the gap, the polarity of the oscillator changes and now D_2 is positive. The electric field is directed from D_2 to D_1 and again accelerates the charge. Now, the charge moves on an even bigger circle inside D_1 . By the time it emerges into the gap at E, the polarity of the oscillator changes once again. The process continues and the charge moves on a spiral like path of growing radius. When the radius of the path becomes equal to the radius of the disc, it is made to exit out of the dees.



'Dees' of a cyclotron



Magnetic field is directed outward in this diagram.

The key thing to understand is that the time period of the circular motion of a charge in a magnetic field is independent of its speed or the radius of the path. It is given by

$$T = \frac{2\pi m}{qB}$$

The frequency of circular motion is

$$f = \frac{1}{T} = \frac{qB}{2\pi m}$$

The time period (or the frequency) of the oscillator is set equal to the time period (or frequency) of circular motion of the charge. The frequency given by the above equation is often called as *cyclotron frequency*.

What this means is that the polarity of the oscillator changes at regular intervals of $\frac{T}{2}$ and the charge always experience acceleration when it enters the gap between the dees.

The speed of the charge when radius of its circular path becomes equal to the radius (R_0) of the dee is given by

$$R_0 = \frac{mv_0}{qB}$$

$$\text{or } v_0 = \frac{qBR_0}{m}$$

This is the maximum speed that we can impart to the charged particle. The main limitation is to create a uniform magnetic field over a circular space of large radius (R_0).

NOTE

For very light particles like electrons, the speed changes a lot even for a small change in their kinetic energies. Theory of relativity says that with increase in speed, the mass of a particle increases. For a particle like an electron a small change in kinetic energy causes a big change in speed which results in a big change in its mass. This causes the time period of circular motion to increase ($T = \frac{2\pi m}{qB}$). The synchronisation between the circular motion of the particle and the polarity changes of dees gets disturbed. The cyclotron fails to accelerate such light particles to great kinetic energies.

In today's world, the Large Hadron Collider (LHC) in Europe, is the largest particle accelerator. It lies in a tunnel that is 27 km long and can accelerate particles upto energies of few TeV (tera electron volt). It is the largest machine in the world.

Example 12 Magnetic field in a cyclotron

Oscillator used in a cyclotron has a frequency 10 MHz. The device has dees of radius 50cm. This cyclotron is used to accelerate protons. Calculate

- The magnetic field (B) needed to operate this cyclotron
- The kinetic energy of the proton beam produced by the cyclotron.

Solution

Concepts

- The magnetic field should be chosen so that the frequency of circular motion of the proton is same as the oscillator frequency.
- The kinetic energy is limited by the radius of the dees. The final kinetic energy of the proton beam is attained when radius of its circular path becomes equal to the radius of the dee.

- (a) Frequency of circular motion of proton = Frequency of the oscillator

$$\Rightarrow \frac{Bq}{2\pi m} = 10 \times 10^6 \text{ Hz}$$

$$\begin{aligned} \text{or, } B &= \frac{2\pi m}{q} \times 10^7 \\ &= \frac{2 \times 3.14 \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19}} \times 10^7 \\ &= 0.67 \text{ T} \end{aligned}$$

- (b) Final speed of a proton is given by

$$R_0 = \frac{mv}{Bq}$$

$$\Rightarrow v = \frac{BqR_0}{m}$$

$$\therefore \text{Kinetic energy, } K = \frac{1}{2} mv^2$$

$$\begin{aligned} \text{or, } K &= \frac{1}{2} m \left(\frac{BqR_0}{m} \right)^2 = \frac{B^2 q^2 R_0^2}{2m_0} \\ &= \frac{(0.67)^2 \times (1.6 \times 10^{-19})^2 \times (0.5)^2}{2 \times 1.67 \times 10^{-27}} \end{aligned}$$

$$= 8.6 \times 10^{-13} \text{ J}$$

$$= \frac{8.6 \times 10^{-13}}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 5.37 \times 10^6 \text{ eV}$$

$$= 5.37 \text{ MeV.}$$



YOUR TURN

Q.21 A proton moves at a constant velocity of $(100 \text{ m/s})\hat{i}$ in a region having a uniform electric and magnetic fields. The magnetic field in the region is $\vec{B} = (4 \times 10^{-3} \text{ T})\hat{j}$. Find the electric field.

Q.22 A cyclotron accelerates deuterons. The radius of the dee is 0.8 m and the frequency of the cyclotron's oscillator is 1 MHz.

- (a) Find the number of times a deuteron enters into the gap between the dees in one second.

- (b) Find the speed of the deuteron as it leaves the cyclotron.

Q.23 In a mass spectrometer, the detector detects two spots at a distance of 1.2 cm and 1.4 cm measured from the slit through which the singly ionised carbon atoms enter into the magnetic field. What can you predict about the ratio of masses of two isotopes of carbon?

In Short

- (i) Force on a charge moving in an electric as well as a magnetic field is given by

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})$$

- (ii) Net force on a charge can be zero if

$$q\vec{E} + q(\vec{v} \times \vec{B}) = 0$$

$$\Rightarrow \vec{E} = -(\vec{v} \times \vec{B})$$

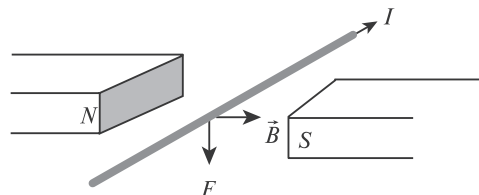
$$\text{or, } \vec{E} = \vec{B} \times \vec{v}$$

- (iii) In a cyclotron, a charge is made to move on a spiral kind of path. Its energy is increased after every half circle by pushing it using an electric force. The particle gains speed but time period of its circular motion does not change.

- (iv) In a cyclotron, the time period of the oscillator is same as the time period of circular motion of the charge and is given by $T = \frac{2\pi m}{qB}$.

- (v) If R_0 is the radius of the dees of a cyclotron, the maximum speed that it can impart to a charge is given by

$$v_{\max} = R_0 \omega = R_0 \frac{qB}{m}$$



A current carrying wire in a magnetic field experiences a force that is perpendicular to its length as well as $\vec{B}$.

A current carrying conductor has electrons moving inside it. These moving electrons experience force in a magnetic field. This force acts sideways on the electrons and get transmitted to the wire as the electrons collide with atoms.

Consider a straight wire carrying current I placed perpendicularly into a uniform magnetic field. Let us consider a length L of the wire and assume that the drift speed of the electrons is v_d . The wire is made of a material having n number of free electrons in its unit volume.

Magnetic force on an electron is

$$\vec{F} = -e(\vec{v}_d \times \vec{B})$$

In the diagram shown, this force is towards left and has magnitude

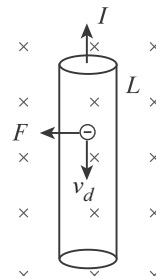
$$F = ev_d B$$

Total number of free electrons in L length of the wire is $= n(AL)$, where A is area of cross section.

Sum of forces acting on all free electrons is the resultant force on the wire. Therefore, the force on the wire is to the left and has magnitude.

$$F_B = (ev_d B)(nAL) = (nev_d A) BL$$

$$\text{or, } F_B = ILB$$



$$(26)$$

6. MAGNETIC FORCE ON A CURRENT CARRYING WIRE

When a current carrying wire is placed between the poles of a magnet, it experiences a lateral force. Experiments show that the force is in a direction given by the right hand rule with the thumb pointing in the direction of the force when the fingers are curled from the direction of the current to the direction of $\vec{B}$.

Here, $nev_d A = I$ is the current.

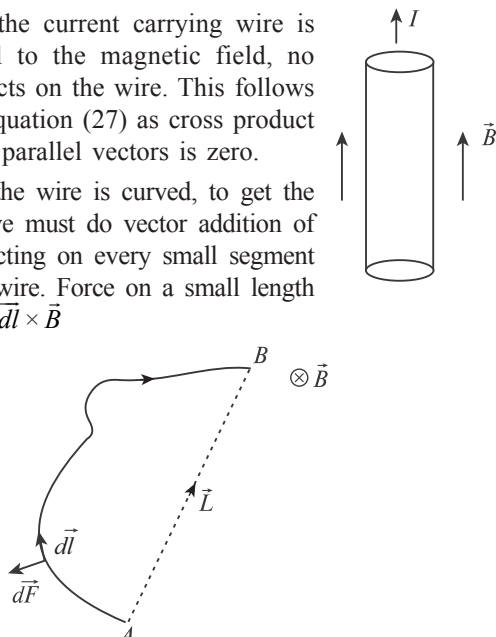
If the magnetic field is not perpendicular to the wire, the force is given by general form of the above equation

$$\vec{F}_B = I \vec{L} \times \vec{B} \quad (27)$$

$\vec{L}$ is a length vector that has magnitude L and is directed along the current.

Following situations are worth noting.

- (A) When the current carrying wire is parallel to the magnetic field, no force acts on the wire. This follows from equation (27) as cross product of two parallel vectors is zero.
- (B) When the wire is curved, to get the force we must do vector addition of force acting on every small segment of the wire. Force on a small length $d\vec{l} = I d\vec{l} \times \vec{B}$



$d\vec{F}$ is force on a small segment of length $d\vec{l}$. By adding forces on all the segments we get the resultant.

Force on the entire wire is

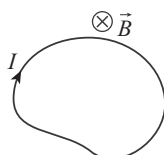
$$\begin{aligned} \vec{F} &= \int d\vec{F} \\ &= I \left[\int_A^B d\vec{l} \right] \times \vec{B} \end{aligned} \quad (i)$$

$\int_A^B d\vec{l}$ is vector sum of all small displacements along the curved wire AB. It must be equal to the vector $\vec{L}$ obtained by joining the end A to the end B of the wire.

$$\therefore \vec{F} = I \vec{L} \times \vec{B}$$

In short, wherever you encounter a curved current carrying wire kept in a uniform magnetic field $\vec{B}$, replace the wire with a straight wire by joining its ends. Magnetic force on this straight wire is same as on the curved wire.

- (C) Consider a closed current loop in a uniform magnetic field B . Force on the entire wire can again be written using above equation (i).



Here, $\int d\vec{l} = 0$ for a closed curve.

$$\therefore F = 0$$

A closed current loop experiences no force in a uniform magnetic field.

Example 13 Sliding a wire on a rough floor

A straight wire of length L and mass M carries a current I . It is placed on a horizontal rough surface and a vertical magnetic field B is switched on. The coefficient of friction between the wire and the surface is μ . Find the least value of B which can make the wire translate. Give your answer for $L = 1.0\text{m}$, $M = 50\text{ g}$, $I = 5\text{A}$ and, $\mu = 0.5$. How will your answer change if direction of B is reversed?

Solution

Concepts

- (i) Since field is vertical, the direction of the magnetic force on the wire must be horizontal ($\because \vec{F} = I \vec{L} \times \vec{B}$). Force is also perpendicular to the length of the wire.
- (ii) To draw a simple picture we can assume that the wire is going into the plane of this paper and show the cross section of the wire.

In the diagram shown, the current in the wire is directed into the plane of the figure. The circle shown is the cross section of the wire. The magnetic force $I \vec{L} \times \vec{B}$ is directed to the right. Its magnitude is

$$\begin{aligned} F &= ILB \sin 90^\circ \\ &= 5 \times 1 \times B \times 1 = 5B. \end{aligned}$$

The maximum friction that can act on the wire is

$$\begin{aligned} f_{\max} &= \mu N = \mu Mg \\ &= 0.5 \times \frac{50}{1000} \times 10 \\ &= 0.25\text{N} \end{aligned}$$

The wire will begin to move if

$$F \geq f_{\max} \Rightarrow 5B \geq 0.25\text{N}$$

$$\text{or, } B \geq 0.05\text{T}$$

If the direction of $\vec{B}$ is reversed (see figure), the magnetic force on it will also reverse in direction. To prevent the motion, friction will be directed rightwards. But all this will make no difference to the above calculations and our answer remains unchanged.

Example 14 A V-shaped wire

A V shaped wire has 90° angle between its arms and length of each of its arms is L . It is carrying a current I , as shown in

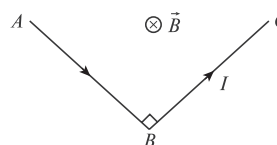
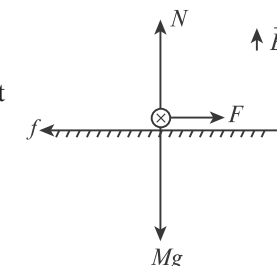


figure. A uniform magnetic field B is applied perpendicular to its plane. Find the magnetic force on the wire.

Solution

Concepts

For writing force in a uniform field, the wire can be replaced by a straight wire AC . There is no need of finding the forces separately on the arms AB and BC .

We can replace the V shaped wire by a straight wire AC . Length of AC is $AC = 2(L \sin 45^\circ) = \sqrt{2}L$

Force can be written as

$$\vec{F} = I (\vec{AC} \times \vec{B})$$

$$\therefore F = I (\sqrt{2}L)(B) \sin 90^\circ$$

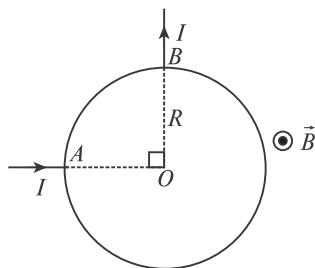
$$\sin 90^\circ = \sqrt{2}ILB$$

Direction of the force is perpendicular to AC in upward direction.

Example 15 A circular wire having two parallel paths for current.

A circular conducting wire of radius r is kept in a plane perpendicular to a uniform magnetic field B (see figure). A current I enters into the wire at A and leaves it at point B such that AB is quarter of the circle.

Find magnetic force on the circle.



Solution

Concepts

- (i) Assuming the wire to be uniform we can say that resistance of the longer path from A to B is $3r$ if the resistance of the quarter arc is r . The current I gets divided into two parallel paths at A . The ratio of currents in the two paths will be in inverse ratio of the resistances of the parallel paths.

- (ii) For writing the magnetic force on a curved wire we can replace it with a straight wire.

Current I divides as I_1 and I_2 into two parallel paths at A .

$$\frac{I_1}{I_2} = \frac{3}{1}$$

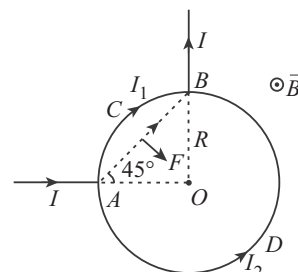
$$\therefore I_1 = \frac{3I}{4} \text{ and } I_2 = \frac{I}{4}$$

Force on arc ACB is same as force on straight wire AB , having current I_1 . Force on arc ADB is also same as force on straight wire AB , having current I_2 .

$$\therefore F = I_1(AB)B + I_2(AB)B = (I_1 + I_2)(AB)B$$

$$= I(\sqrt{2}R)B = \sqrt{2}IRB$$

The force is directed perpendicular to AB in the direction shown in the figure. Therefore, force on the circle is $F = \sqrt{2}IRB$ in a direction making 45° with AO , as shown in the figure.



NOTE

A common mistake in such questions is to write the force as zero when the diagram shows a closed loop. You must see if the current is forming a closed loop or not. In the above example, the current is not forming a closed loop.



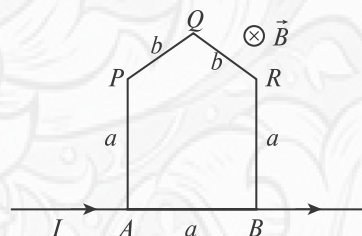
YOUR TURN

Q.24 A wire of length L carries a current I along the positive x direction. A uniform magnetic field $\vec{B} = B_0(2\hat{j} + \hat{k})$ exists in space. Find the force on the wire.

Q.25 A wire carries current in vertically upward direction. It is kept in a horizontal magnetic field that is directed in north west direction. Find the direction of magnetic force on the wire.

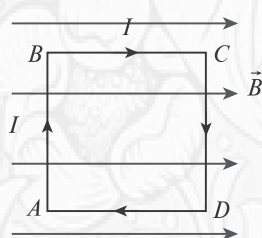
Q.26. A pentagon shaped wire frame ($APQRB$) is located in a uniform magnetic field B which is perpendicular to its plane. The wire is uniform and various arms have lengths

as shown in figure. A current I enters the pentagon at A and leaves it at B . Find the magnetic force on the pentagon.



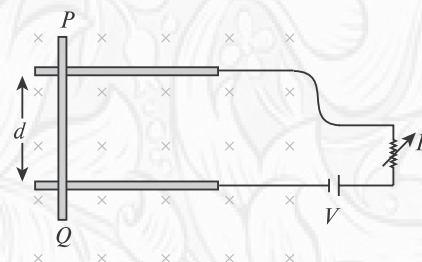
Q.27 A square frame of side length L carries a current I as shown in the figure. A uniform magnetic field B exists in the space parallel to the square frame. Find

- Force on arm AB
- Force on arm BC
- Force on the square loop



Q.28 A metal wire PQ of mass 20 g lies at rest, perpendicularly on two parallel horizontal conducting rails. The rails are separated by a distance $d = 10$ cm. A vertically

downward magnetic field $B = 0.5$ T exists in the space. The ends of the rails are connected to a 10V cell through a variable resistance R . When the variable resistance is made to decrease, the wire PQ begins to slide once R goes below 20Ω . Neglect resistance of all other parts of the circuit and find the coefficient of friction between the wire and the rails.



7. CURRENT LOOP AS A MAGNETIC DIPOLE

A current loop behaves like a magnetic dipole though there are no visible poles in it. In this chapter, as well as in the next one, we will come across many similarities in behaviour of a current loop and a magnet.

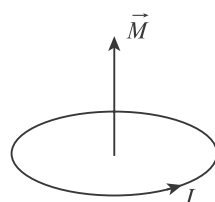
Magnetic dipole moment of a current loop is defined as a vector directed perpendicular to the plane of the loop. This direction is taken in the direction of outstretched right hand thumb when the fingers are curled in the sense of the current. Magnitude of the magnetic dipole moment of a current loop is

$$M = NIA \quad (28)$$

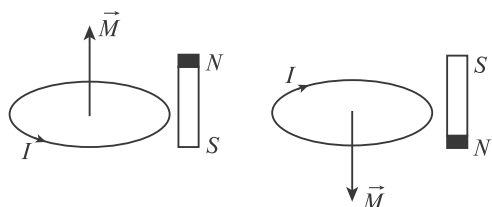
where N is number of turns in the loop, I is the current and A is the area of the loop

A current loop having dipole moment M is equivalent to a bar magnet having dipole moment M .

As said earlier, we will encounter several similarities to support this statement. In the next chapter we will learn that a current coil can produce magnetic field like a permanent magnet.



Magnetic dipole moment vector of a current loop.



The figure shows current loop and its equivalent bar magnet. Note the direction of north and south poles.

So, far we have said that a current carrying coil and a magnet are magnetic dipoles. You will learn later that subatomic particles, including the electron, the proton and the neutron, have magnetic dipole moments. Yes, they are tiny magnets! In magnetic resonance imaging (MRI) a patient is subjected to a strong external magnetic field which interacts with the protons in each hydrogen atom in a tissue. This interaction is possible only because a proton has a magnetic dipole moment.

Example 16 Gyromagnetic ratio

An electron (charge $-e$, mass m) is revolving in a hydrogen atom. Find the ratio of magnetic dipole moment (M) associated with this motion of an electron to its angular momentum (L) about the centre of the circular path. This ratio is known as gyromagnetic ratio.

Solution

Concepts

- If an electron moves at frequency f in a circular path, the current associated with its motion is $I = ef$. The current is directed opposite to the direction of motion of the electron as it has negative charge.

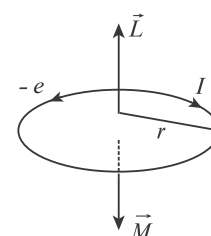
We can write the magnetic dipole moment corresponding to this current.

- Angular momentum is $L = I\omega$

Current associated with the motion of the electron is

$$I = ef = \frac{e\omega}{2\pi}$$

where f is frequency of revolution and ω is angular speed



1.22 Magnetism and Electromagnetic Induction

Magnetic dipole moment corresponding to this current is

$$M = I \cdot \pi r^2 \quad (r = \text{radius of the circular path})$$

$$\text{or, } M = \frac{e\omega}{2\pi} \pi r^2 = \frac{1}{2} e\omega r^2$$

In the figure shown, the direction of $\vec{M}$ vector is downward. Angular momentum of the electron about the centre is

$$L = mr^2\omega$$

Direction of $\vec{L}$ is upward, in the figure shown.

Ratio of magnitudes of M and L is

$$\frac{M}{L} = \frac{e}{2m} \quad (29)$$

In vector notation $\vec{M} = -\left(\frac{e}{2m}\right) \vec{L}$.

For a positive charge, both $\vec{M}$ and $\vec{L}$ will be in the same direction.

Example 17 Non conducting charged ring and disc

- (a) A non conducting ring of mass M and radius R has charge Q on it. It is spinning about its central axis (normal to its plane) with an angular speed ω . Find the magnetic dipole moment of the ring. Also find the ratio of its magnetic dipole moment to its angular momentum about the rotation axis.
- (b) Repeat the above exercise for a uniform non conducting disc having mass M and charge Q .

Solution

Concepts

- (i) A charged ring, that is spinning is just like a current loop.
- (ii) A disc can be divided into a large number of thin rings, magnetic dipole moment of each ring is in same direction and adds.

- (a) When the ring makes one rotation, a charge Q passes through any point on its circumference. If it makes f rotations in a second, the total charge that passes through a point in a second is Qf . This is the associated current.

$$I = Qf = Q \frac{\omega}{2\pi}$$

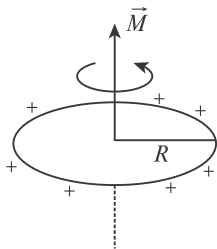
Magnetic dipole moment is

$$M = I \cdot \pi R^2 = \frac{1}{2} Q\omega R^2 \quad (i)$$

The angular momentum of the ring about its rotation axis is

$$L = mR^2\omega$$

$$\therefore \frac{M}{L} = \frac{Q}{2m}$$



This result is same as that obtained for a particle in the last example.

- (b) Figure shows a charged spinning disc. Charge per unit area on the disc is

$$\sigma = \frac{Q}{\pi R^2}$$

Consider a ring of radius x and thickness dx on the disc. Charge on the ring is

$$dQ = \sigma (2\pi x dx)$$

Magnetic dipole moment

for this ring is given by equation (i).

$$dM = \frac{1}{2} (dQ)\omega x^2 = \pi\sigma\omega x^3 dx$$

Dipole moment for all such rings are in same direction and add. Therefore, dipole moment of the disc is

$$M = \int dM = \pi\sigma\omega \int_0^R x^3 dx = \frac{\pi}{4} \sigma\omega R^4$$

$$= \frac{\pi}{4} \cdot \frac{Q}{\pi R^2} \omega R^4 = \frac{1}{4} QR^2\omega$$

The angular momentum for the disc is $L = \frac{1}{2} mR^2\omega$.

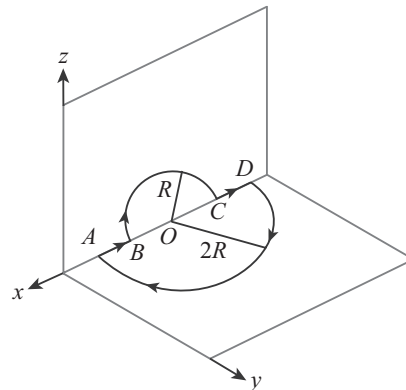
$$\therefore \frac{M}{L} = \frac{Q}{2m}$$

NOTE

You may try to prove that the ratio $\frac{M}{L}$ is $\frac{Q}{2m}$ for a uniformly charged spherical shell or a solid sphere.

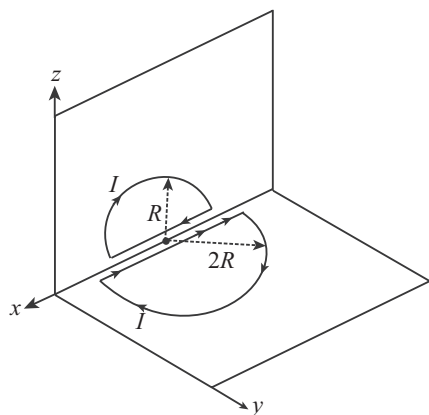
Example 18 Current loop that is not in a plane

A current loop has the shape shown in the figure. It has two semicircular parts of radii R and $2R$ which lie in two perpendicular planes. Two straight sections (AB , CD), each of length R , join these semicircular parts. Current in the loop is I . Find the magnetic dipole moment of the current loop.



Solution**Concepts**

- (i) We can consider the given loop as combination of one semicircular loop in xz plane and another semicircular loop in xy plane.
- (ii) We will write the dipole moments for the two loops separately and add them.



Consider two circular current loops as shown in the figure. The superposition of these two loops gives us the given loop. Current in the section BC is equal and opposite in the two loops and cancel out.

For the loop in xz plane, dipole moment is

$$\vec{M}_1 = \frac{\pi R^2}{2} I (-\hat{j})$$

For the loop in xy plane, dipole moment is

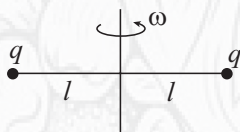
$$\vec{M}_2 = \frac{\pi(2R)^2}{2} I (-\hat{k}) = 2\pi R^2 I (-\hat{k})$$

Dipole moment of the give loop is

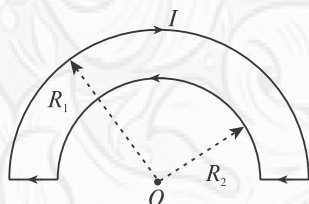
$$\vec{M} = \vec{M}_1 + \vec{M}_2 = -\pi R^2 I \left(\frac{1}{2} \hat{j} + 2 \hat{k} \right)$$

**YOUR TURN**

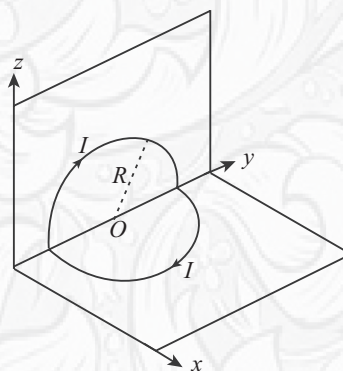
Q. 29 A non conducting rod of length $2l$ has two particles attached to its ends. Each particle has charge q and mass m . The rod has negligible mass. The system rotates with an angular speed ω about an axis passing through the centre of the rod and perpendicular to its length. Find the ratio of magnetic dipole moment to the angular momentum of the system.



Q. 30 Find the magnetic dipole moment of the current loop shown in the figure.

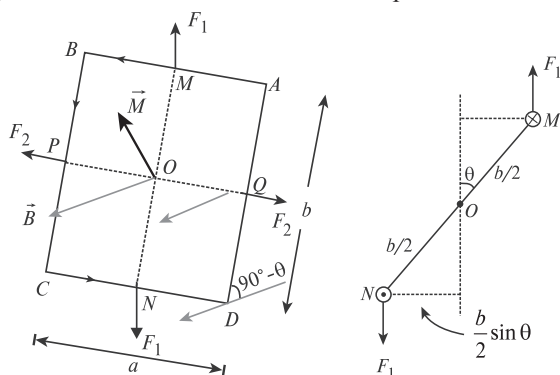


Q. 31 A circular current loop is folded along one of its diameter such that the two semicircular halves lie in xy and yz planes. The radius of the circle is R and it carries a current I . Write the magnetic dipole moment of the loop.



8. CURRENT LOOP IN A UNIFORM MAGNETIC FIELD

Consider a rectangular loop having sides of length a and b . It carries a current I . It is placed in a uniform magnetic field ($\vec{B}$) and the magnetic dipole moment ($\vec{M}$) of the loop makes an angle θ with the direction of the field. The dipole moment vector is normal to the surface of the loop with its direction being given by the stretched right hand thumb when fingers are curled in the sense of the current. Angle between $\vec{B}$ and the plane of the loop is $90^\circ - \theta$. For easy visualisation, assume that the magnetic field is horizontal and the loop has been tilted by



angle θ from its vertical position. When the loop is vertical, direction of $\vec{M}$ is along $\vec{B}$. When the loop is tilted by θ , direction of $\vec{M}$ also tilts by θ . Now, the angle between $\vec{B}$ and $\vec{M}$ becomes θ .

Magnetic force on arm AB is

$$F_1 = IaB$$

The force is directed perpendicular to AB and $\vec{B}$. In our discussion, AB is horizontal and therefore $\vec{F}_1$ is vertical. Force on the arm CD is also F_1 , directed downward.

Magnetic force on the arm BC is

$$F_2 = IbB \sin(90^\circ - \theta) = IbB \cos\theta$$

Force on DA is also of same magnitude. Both these forces have same line of action passing through the centre of the loop.

It is appropriate to show all the force acting at midpoints (M , N , P and Q) of the respective arms.

Obviously, the resultant force on the current loop is zero. We already know this result – *magnetic force on a closed current loop in a uniform field is zero.*

The pair of forces on the arms AB and CD form a couple. There is a resultant torque due to this pair. The second diagram shows the line MN and the two forces acting at its ends. Resultant torque is

$$\tau = F_1 \frac{b}{2} \sin\theta + F_1 \frac{b}{2} \sin\theta$$

$$\text{or, } \tau = F_1 b \sin\theta = (IaB) b \sin\theta$$

$$\text{or, } \tau = IAB \sin\theta \quad [A = ab = \text{area of the loop}]$$

$$\text{or, } \tau = MB \sin\theta$$

$$[M = IA = \text{magnetic dipole moment of the loop}]$$

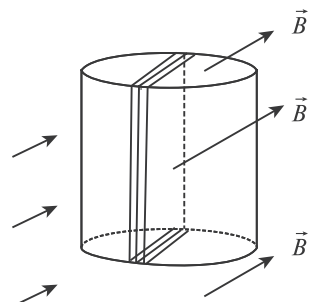
The torque is directed so as to bring the dipole moment vector ($\vec{M}$) parallel to the magnetic field ($\vec{B}$). In the shown diagram, the torque vector is along $\vec{PQ}$. This is the direction of the vector $\vec{M} \times \vec{B}$. Therefore, we can write

$$\vec{\tau} = \vec{M} \times \vec{B} \quad (30)$$

This equation is same as equation (12) that we wrote for a bar magnet in a uniform field. This is the first evidence that a current loop does behave like a bar magnet. It indeed is a magnetic dipole. The above relation is valid for a current loop of any shape, though we have derived it for a special case of a rectangular loop.

Example 19 Coil wrapped on a cylinder

A solid uniform cylinder has mass M , radius R and length L . A coil is wrapped on it in vertical plane as shown in figure. Number of turns in the coil is N . A current i is established in the coil. A magnetic field of strength B is switched on in horizontal direction parallel to the plane of the coil. Find the initial angular acceleration of the cylinder. Neglect the inertia of the wire used to make the coil.



Solution

Concepts

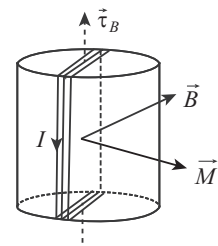
- The magnetic dipole moment of the coil is normal to the plane of the coil. Both $\vec{M}$ and $\vec{B}$ are horizontal with an angle $\theta = 90^\circ$ between them.
- The magnetic torque $\vec{\tau}_B = \vec{M} \times \vec{B}$ is directed vertical and causes the cylinder to rotate about its vertical axis.
- $\mu = \frac{\tau_B}{I_0}$, where I_0 is moment of inertia of the cylinder about its central vertical axis.

Magnetic dipole moment of the coil

$$M = N \cdot I \cdot A = NI (2RL)$$

Dipole moment vector is normal to the plane of the coil. $\vec{B}$ is parallel to the plane of the coil. Angle between them is 90°

$$\therefore \tau_B = MB \sin 90^\circ = 2NIR \cdot L \cdot B$$

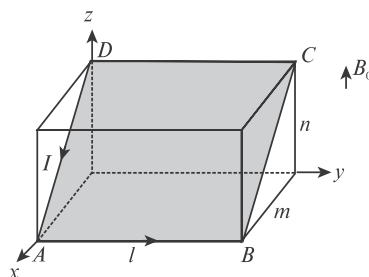


Moment of inertia of the cylinder $I_0 = \frac{1}{2}MR^2$

$$\therefore \alpha = \frac{\tau_B}{I_0} = \frac{4NILB}{MR}$$

Example 20 A method to find dipole moment

A conducting loop $ABCD$ carries a current I . An indicative box has been drawn to make you understand the dimensions of the loop and its orientation. Write torque on the loop if a magnetic field B_0 is switched on in vertical direction.



Solution

Concepts

- (i) If $\vec{a}$ and $\vec{b}$ represent two adjacent sides of a parallelogram, then $\vec{a} \times \vec{b}$ represents its area vector. This vector is directed perpendicular to the plane of the parallelogram and its proper direction is given by the right hand thumb when fingers are curled from $\vec{a}$ to $\vec{b}$.
- (ii) In the given diagram $\vec{AB} \times \vec{AD}$ gives the area vector of the surface of the coil, pointing in the direction of its magnetic dipole moment.

Magnetic dipole moment of the loop is

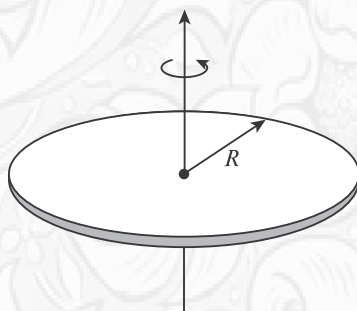
$$\begin{aligned}\vec{M} &= I (\vec{AB} \times \vec{AD}) \\ &= I [(l\hat{j}) \times (-m\hat{i} + n\hat{k})] = I [lm\hat{k} + ln\hat{i}] \\ &= Il [n\hat{i} + m\hat{k}]\end{aligned}$$

$$\therefore \vec{\tau} = \vec{M} \times (B_0\hat{k}) = -IlB_0\hat{j}$$

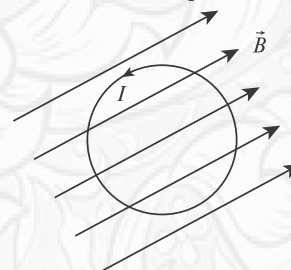


YOUR TURN

Q.32 A non-conducting disc of mass M and radius R has uniformly distributed charge of surface density σ . It is spinning about its vertical central axis. A vertical magnetic field B is switched on. Find the magnetic torque on the disc.



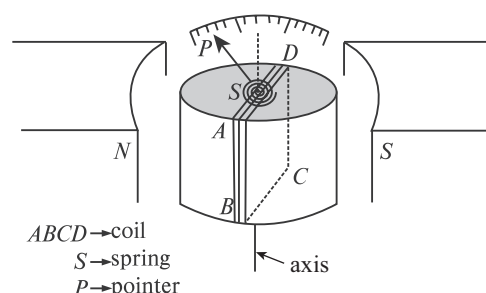
Q.33 A circular current loop of radius R has current I in it. A uniform magnetic field B is applied parallel to its plane. Write the magnitude of magnetic torque on the loop and indicate the direction of torque vector.

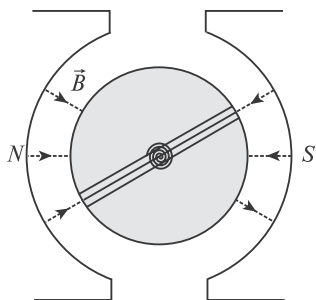


9. MOVING COIL GALVANOMETER

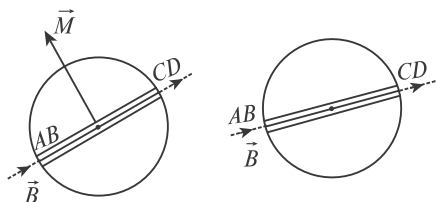
A rectangular coil of several turns is wound over a soft iron cylindrical core. The cylinder coil system is free to rotate about its axis on frictionless bearings. The wire of the coil is coated with an insulating material so that the turns are insulated from one another and also insulated from the iron cylinder. The arrangement is surrounded by two semicylindrical pole pieces of a magnet. A coil spring is attached to the cylinder and its axis. When the cylinder rotates by an angle θ , the spring exerts a torque $\tau_s = C\theta$ on it in opposite direction. C is known

as torsional constant of the spring. A pointer attached to the cylinder rotates with it on a graduated scale.





Top view. Field lines are radial. Field at the arms AB and CD are parallel to the plane of the coil.



Even when the coil rotates the field is parallel to the plane of the coil at side arms AB and CD .

The semi cylindrical pole pieces create a radial magnetic field. Inside the iron cylinder the field gets multiplied many times due to the magnetisation of iron (we will learn about this phenomenon in a later chapter). A torque on the coil $ABCD$ acts when a current is passed through it. The torque arises due to magnetic forces on the arms AB and CD . The forces on the other two arms are zero or always acting along the same line producing no torque. The key thing to realise is that whatever be the orientation of the coil, the magnetic field always remains normal to the sides AB and CD and parallel to the plane of the coil. Inside the iron cylinder, the field lines get distorted and are not perfectly radial (Field lines cannot intersect. Exactly radial lines will mean that they intersect). But that shall not bother us when we wish to find torque on the coil. The forces on the arms DA and BC do not contribute to the torque.

If B is strength of the radial field at the periphery of the cylindrical core (that is where arms AB and CD are located), then torque on the coil at any orientation is

$$\tau_B = MB \sin 90^\circ = NIAB$$

where N is number of turns in the coil, A is its area and I is current flowing in it.

When a current is passed, the magnetic torque causes the coil to rotate. The spring gets twisted and exerts a counter

torque. After making few oscillations, the coil comes to rest in its equilibrium position. Let θ be the angular rotation of the coil, from its original position, when it is in equilibrium.

$$\begin{aligned}\tau_B &= \tau_s \\ NIAB &= C\theta \\ \text{or, } I &= \left(\frac{C}{NAB} \right) \theta\end{aligned}\quad (31)$$

The pointer attached to the cylinder also deflects by an angle θ . The current I is proportional to the deflection angle θ . The constant $k = \frac{C}{NAB}$ is known as the *galvanometer constant*. This constant may be measured directly by passing a known current and measuring the deflection.

When a current is passed through the galvanometer coil, it begins to oscillate due to the variable spring torque. For fast damping of oscillations we have to make appropriate modifications. We will learn later that Eddy currents help in damping.

Sensitivity

The sensitivity of a galvanometer is the deflection produced when a unit current is passed through it. It can be written as

$$S = \frac{\theta}{I} = \frac{NAB}{C} \quad (32)$$

The inverse of sensitivity (S) is known as *figure of merit* for the galvanometer.

For high sensitivity, the magnetic field (B) must be high. Presence of iron core increases the field and hence sensitivity.

Voltage sensitivity of a galvanometer can be defined as the deflection produced when a unit potential difference is applied across it.

$$S_v = \frac{\theta}{V} = \frac{\theta}{IR_g} = \frac{NAB}{CR_g} \quad (33)$$

Here R_g is resistance of the coil.

Example 21 Will increasing the number of turns in a galvanometer coil, increase its voltage sensitivity?

Solution No. the voltage sensitivity is directly proportional to the number of turns N but inversely proportional to the coil resistance (R_g). Value of R_g is directly proportional to the length of the wire, i.e. N . Hence, S_v will not depend on N .



YOUR TURN

Q. 34 A moving coil galvanometer has 100 turns and area of each turn is 2.0 cm^2 . The magnetic field at the location of the iron cylinder is 0.01 T . The coil suffers a deflection

of 0.05 radian when 10 mA current is passed through it. Find the torsional constant of the spring used in the device.

In Short

- (i) A current carrying wire experiences force in a magnetic field. This force arises due to the magnetic forces acting on the moving electrons inside the wire.
- (ii) Force on a straight current carrying wire in a uniform magnetic field $\vec{B}$ is

$$\vec{F} = I \vec{L} \times \vec{B}$$
 Here, $\vec{L}$ is the length vector in the direction of the current.
- (iii) If a straight wire has its length along $\vec{B}$, no magnetic force will act on it.
- (iv) Magnetic force on a curved wire placed in a uniform magnetic field can be obtained by replacing the wire with a straight one joining its ends. *This is true, in general, for a uniform field only.*
- (v) Magnetic force on a closed current loop in a uniform magnetic field is zero.
- (vi) A current loop behaves like a magnetic dipole.
- (vii) Magnetic dipole moment of a current loop is a vector perpendicular to its plane having magnitude $M = NIA$. Here, N is number of turns in the loop, A is its area and I is the current in the loop. The proper direction of dipole moment vector ($\vec{M}$) is along the right hand thumb when fingers are curled in the sense of current.
- (viii) When placed in a uniform magnetic field, a current loop experiences torque.

$$\vec{\tau}_B = \vec{M} \times \vec{B}$$
 This torque is always about an axis in the plane of the loop and tries to align $\vec{M}$ parallel to $\vec{B}$.
- (ix) A particle having mass m and charge q , revolving in a circular orbit, has a constant ratio of magnetic dipole moment and angular momentum, irrespective of its rotation speed. The ratio is

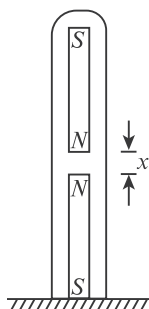
$$\frac{M}{L} = \frac{q}{2m}$$
 The ratio is known as gyromagnetic ratio.
- (x) In a moving coil galvanometer, current (i) through the coil is proportional to the deflection (θ) of the coil.

$$i \propto \theta \Rightarrow i = k\theta$$
 The constant k is known as galvanometer constant and its value is

$$k = \frac{C}{NAB}$$
 Where C is torsional constant of the spring, N is number of turns in the coil, A is area of the coil and B is the magnetic field at the position of the coil.
- (xi) $\frac{\theta}{i} = \frac{1}{k}$ is known as current sensitivity of the galvanometer.

Miscellaneous Examples

Example 22 Two identical magnets, with length l and mass M each are arranged vertically inside a glass tube as shown in the figure. The upper magnet remains suspended in air above the lower one so that the distance between the nearest poles of the magnets is x ($\ll l$). Find the pole strength (m) of the magnets. Assume that the magnetic length, for each magnet, is same as the geometric length.



Solution

Concepts

- (i) The force applied by the lower magnet on the upper magnet balances the weight of the upper magnet.
- (ii) Equation 4 can be used to write the field due to lower magnet at the position of the two poles of the upper magnet.

First, let's write the force due to lower magnet on the north pole of the upper magnet. Force due to south pole of the lower magnet can be neglected compared to the force applied by the north pole ($\because l \gg x$)

$$\therefore F_N = \frac{\mu_0}{4\pi} \frac{m \cdot m}{x^2} (\uparrow)$$

Force on the south pole of the upper magnet due to the lower magnet can be written as

$$\begin{aligned} F_S &= \frac{\mu_0}{4\pi} \frac{m \cdot m}{l^2} - \frac{\mu_0}{4\pi} \frac{m \cdot m}{(2l)^2} \\ &= \frac{3}{4} \frac{\mu_0}{4\pi} \frac{m \cdot m}{l^2} (\downarrow) \end{aligned}$$

For equilibrium

$$F_N = F_S + Mg$$

$$\text{or, } \frac{\mu_0}{4\pi} \frac{m^2}{x^2} = \frac{3}{4} \frac{\mu_0}{4\pi} \frac{m^2}{l^2} + Mg$$

$$\text{or, } \frac{\mu_0}{4\pi} m^2 \left[\frac{1}{x^2} - \frac{3}{4} \frac{1}{l^2} \right] = Mg$$

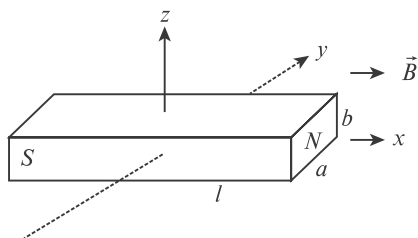
Again, $x \ll l$, permits us to neglect the second term inside the bracket. In effect, we are saying that $F_S \ll F_N$.

$$\Rightarrow \frac{\mu_0}{4\pi} \frac{m^2}{x^2} = Mg$$

$$\Rightarrow m = x \cdot \sqrt{\frac{4\pi Mg}{\mu_0}}$$

Example 23 Oscillating magnet

A bar magnet has the shape shown in the figure. Its mass is m . It is kept in a uniform magnetic field B that is parallel to its axis (along x direction). The magnet is given a small angular displacement and released find time period of its oscillations if it was displaced about (a) z axis (b) y axis. Its magnetic dipole moment is M . Neglect all other forces apart from the magnetic force.



Solution

Concepts

- The magnetic torque ($\vec{\tau} = \vec{M} \times \vec{B}$) is always directed so as to rotate the magnet back to its original equilibrium position. This restoring torque causes oscillations.
- The moment of inertia of the magnet is different about z axis and y axis.

Torque on the magnet in the position shown is

$$\tau = MB \sin\theta$$

For small angular displacement, this torque can be written as

$$\tau = MB\theta.$$

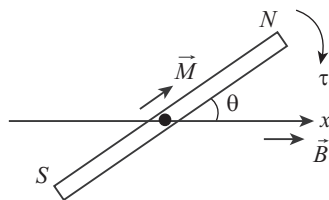
The torque remains same whether the magnet is rotated about the y axis or z axis.

- (a) Moment of inertia about the z axis is

$$I_z = \frac{m}{12} (l^2 + a^2)$$

$$\therefore I_z \frac{d^2\theta}{dt^2} = -MB\theta$$

$$\text{or, } \frac{d^2\theta}{dt^2} = -\left(\frac{MB}{I_z}\right)\theta$$



This is the standard equation for SHM. The magnet performs SHM.

$$\omega = \sqrt{\frac{MB}{I_z}}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{I_z}{MB}}$$

$$\text{or, } T = 2\pi \sqrt{\frac{m(l^2 + a^2)}{12MB}}$$

- (b) Moment of inertia about the y axis is

$$I_y = \frac{m}{12} (l^2 + b^2)$$

$$\therefore I_y \frac{d^2\theta}{dt^2} = -MB\theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\left(\frac{MB}{I_y}\right)\theta$$

$$\therefore \omega = \sqrt{\frac{MB}{I_y}}$$

$$\text{or, } T = 2\pi \sqrt{\frac{I_y}{MB}} = 2\pi \sqrt{\frac{m(l^2 + b^2)}{12MB}}$$

Example 24 Another question on writing velocity as a function of time

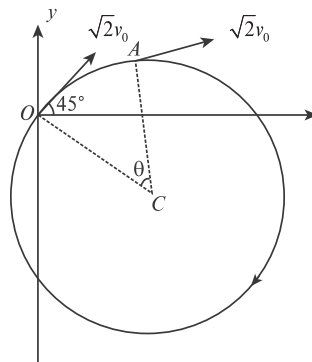
A particle having mass m and charge q is projected from the origin with a velocity $\vec{v} = v_0(\hat{i} + \hat{j})$. The region has a uniform magnetic field $\vec{B} = B_0\hat{k}$. Find the x and y component of the particle's velocity as a function of time assuming that it was projected at time $t = 0$.

Solution

Concepts

- It is important to draw the path correctly. The initial velocity makes 45° with the positive x direction and it must be tangential to the circle. Initial force on the particle is directed towards a point in the 4th quadrant. Thus the centre of the circle is located in the 4th quadrant.
- The angular speed of circular motion is $\omega = \frac{qB}{m}$. The particle rotates through an angle $\theta = \omega t$ in time t .

The circular path of the particle is as shown in figure. The speed of the particle is $\sqrt{2}v_0$ and its initial velocity is inclined at 45° to the x axis. In time t the particle rotates by an angle θ to reach point A and its velocity now makes an angle $45^\circ - \theta$ with the x direction. Recall that the speed does not change.



$$\therefore v_x = \sqrt{2} v_0 \cos\left(\frac{\pi}{4} - \omega t\right) = \sqrt{2} v_0 \cos\left(\frac{\pi}{4} - \frac{qBt}{m}\right)$$

$$v_y = \sqrt{2} v_0 \sin\left(\frac{\pi}{4} - \omega t\right) = \sqrt{2} v_0 \sin\left(\frac{\pi}{4} - \frac{qBt}{m}\right)$$

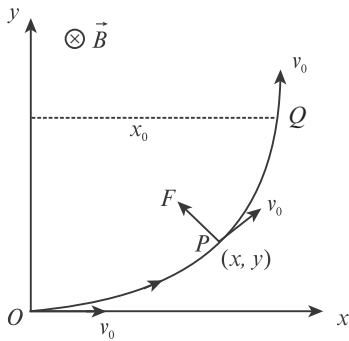
Example 25 Variable magnetic field

A particle having mass m and charge q is projected from the origin with a velocity $\vec{v} = v_0 \hat{i}$ in a non uniform magnetic field $\vec{B} = -(B_0 x) \hat{k}$, Here, v_0 and B_0 are positive constants of proper dimensions. Find the maximum positive x -coordinate of the particle during the motion.

Solution**Concepts**

- Magnetic field is directed along negative z . Initial velocity of the particle is along x . There is no z component in velocity. Therefore, the particle will move in xy plane only. Its speed will never change as magnetic force is always perpendicular to the instantaneous velocity. But the particle will not move in a circle as the magnetic field is changing.
- At the instant the x co-ordinate is maximum, its x component of velocity will be zero and the y component of velocity will have a magnitude v_0 .

The particle starts at O , moves along a curved path and its velocity at Q becomes parallel to y direction. The x -coordinate (x_0) for Q is our required answer.



P is an intermediate point having coordinates (x, y) . The magnetic force on the particle at P is

$$\vec{F} = q (v_x \hat{i} + v_y \hat{j}) \times (-B_0 x \hat{k})$$

$$= qB_0 v_x x \hat{j} - qB_0 v_y x \hat{i}$$

$$\therefore m \frac{dv_x}{dt} = -qB_0 v_y x$$

$$\text{and } m \frac{dv_y}{dt} = qB_0 v_x x$$

$$\text{or, } \frac{dv_y}{dx} \cdot \frac{dx}{dt} = \frac{qB_0 v_x}{m} \cdot x$$

$$\text{or, } \frac{dv_y}{dx} \cdot v_x = \frac{qB_0 v_x}{m} \cdot x$$

$$\text{or, } \frac{dv_y}{dx} = \frac{qB_0}{m} \cdot x \Rightarrow dv_y = \frac{qB_0}{m} x dx$$

$$\text{or, } \int_0^{v_0} dv_y = \frac{qB_0}{m} \int_0^{x_0} x dx$$

$$\text{or, } v_0 = \frac{qB_0}{m} \frac{x_0^2}{2}$$

$$\Rightarrow x_0 = \sqrt{\frac{2mv_0}{qB_0}}$$

Example 26 Helix with increasing pitch

A vertically downward magnetic field $\vec{B} = B \hat{j}$ exists in a region. A particle having charge q and mass m is projected with velocity $\vec{v} = v_0 \hat{i}$ from a point that is at a height h above the ground. Acceleration due to gravity is g .

- Write the x , y and z components of velocity of the particle after time t .
- Take the point of projection as the origin and write the coordinates of the particle at time t .
- In how much time will the particle hit the ground?

Solution**Concepts**

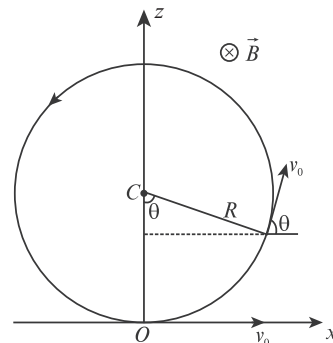
- Due to the magnetic field the particle will move in circular path in xz plane (horizontal plane). Due to gravity it will move in y direction with increasing speed. Therefore, it will trace a helical path with increasing pitch.
- The velocity component in xz plane will remain unchanged. It will only change direction in the plane. v_y goes on increasing due to acceleration g .

First consider the motion in horizontal xz plane.

$$\text{Angular speed, } \omega = \frac{qB}{m}$$

Angular displacement in time t

$$\theta = \omega t = \frac{qBt}{m}$$



$$(a) \quad v_x = v_0 \cos \theta = v_0 \cos \left(\frac{qBt}{m} \right)$$

$$v_z = v_0 \sin \theta = v_0 \sin \left(\frac{qBt}{m} \right)$$

In y direction (the vertically downward direction), the particle moves with a constant acceleration g .

$$\therefore v_y = gt$$

$$(b) \quad x = R \sin \theta = \frac{mv_0}{qB} \sin \left(\frac{qBt}{m} \right)$$

$$z = R - R \cos \theta = R \left[1 - \cos \left(\frac{qBt}{m} \right) \right]$$

$$y = \frac{1}{2} gt^2$$

$$(c) \quad h = \frac{1}{2} gt^2 \quad \Rightarrow \quad t = \sqrt{\frac{2h}{g}}$$

NOTE

If we replace the gravitational field with electric field, the motion of the particle will be identical.

Example 27 Cycloidal path in crossed $\vec{E}$ and $\vec{B}$

Consider a region where a uniform electric field E_0 is directed along x direction alongwith a uniform magnetic field B_0 directed in y direction. A particle having mass m and charge q is released at the origin.

- Write the velocity and position of the particle as function of time.
- Find the maximum x -coordinate of the particle.
- What is the velocity of the particle when it is at its maximum x -coordinate.

Solution

Concepts

- Initially, the electric force on the particle causes it to move in x direction. Now a magnetic force acts on it that deflects it. Since the magnetic force will be in xz plane, the particle will move in xz plane only.
- We will write the net force on the particle and then we will write its acceleration components to setup two differential equations.
- The solution of $\frac{d^2x}{dt^2} = -\omega^2x$ is $x = A \sin(\omega t + \delta)$. Values of constants A and δ can be calculated from the known initial conditions.

- The particle will move in xz plane. Let its velocity be $\vec{v} = (v_x \hat{i} + v_z \hat{k})$ at the instant its position is (x, z) . The net force on the particle is $\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$

$$= qE_0 \hat{i} + q(v_x \hat{i} + v_z \hat{k}) \times (B_0 \hat{j})$$

$$= q(E_0 - v_z B_0) \hat{i} + qv_x B_0 \hat{k}$$

$$\therefore F_x = q(E_0 - v_z B_0)$$

$$\therefore a_x = \frac{q}{m}(E_0 - B_0 v_z)$$

$$\text{or, } \frac{dv_x}{dt} = \frac{q}{m}(E_0 - B_0 v_z) \quad (i)$$

$$\text{And } F_z = qv_x B_0$$

$$\text{or, } a_z = \frac{qB_0}{m} v_x$$

$$\text{or, } \frac{dv_z}{dt} = \frac{qB_0}{m} v_x \quad (ii)$$

Differentiating equation (i) with respect to time gives

$$\frac{d^2 v_x}{dt^2} = -\left(\frac{qB_0}{m}\right) \frac{dv_z}{dt}$$

Substituting for $\frac{dv_z}{dt}$ using (ii) gives

$$\frac{d^2 v_x}{dt^2} = -\left(\frac{qB_0}{m}\right)^2 v_x \quad (iii)$$

Solution of this differential equation is

$$v_x = v_{x0} \sin(\omega t + \delta) \text{ where } \omega = \frac{qB_0}{m}$$

To find the constants v_{x0} and δ we observe two things:

- At $t = 0$, $v_x = 0$
 $\therefore 0 = v_{x0} \sin(0 + \delta) \Rightarrow \delta = 0$
 $\therefore v_x = v_{x0} \sin \omega t \quad (iv)$
- At $t = 0$, speed of the particle is zero and force on it is qE_0 in x direction. The x component of its acceleration is

$$a_x = \frac{qE_0}{m} \quad \text{at} \quad t = 0$$

Differentiating (iv) with respect to time gives

$$a_x = \frac{dv_x}{dt} = v_{x0} \omega \cos \omega t$$

$$\text{At } t = 0, a_x = \frac{qE_0}{m}$$

$$\therefore \frac{qE_0}{m} = v_{x0} \omega$$

$$\Rightarrow v_{x0} = \frac{qE_0}{m\omega} = \frac{E_0}{B_0} \quad [\because \omega = \frac{qB_0}{m}]$$

Therefore equation (iv) can now be written as

$$v_x = \frac{E_0}{B_0} \sin \left(\frac{qB_0}{m} t \right) \quad (v)$$

Substituting for v_x in equation (i) given

$$\frac{E_0}{B_0} \frac{qB_0}{m} \cos\left(\frac{qB_0}{m}t\right) = \frac{q}{m} (E_0 - B_0 v_z)$$

$$\text{or, } \frac{qB_0}{m} v_z = \frac{qE_0}{m} \left[1 - \cos\left(\frac{qB_0}{m}t\right)\right]$$

$$\text{or, } v_z = \frac{E_0}{B_0} \left[1 - \cos\left(\frac{qB_0}{m}t\right)\right] \quad (\text{vi})$$

The equations (v) and (vi) give the components of velocity.

To write the coordinates of the particle, consider equation (v).

$$\frac{dx}{dt} = \frac{E_0}{B_0} \sin\left(\frac{qB_0}{m}t\right)$$

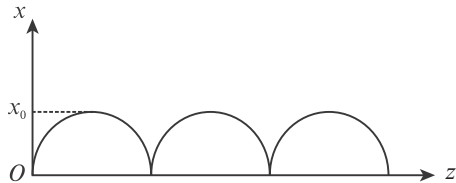
$$\text{or, } \int_0^x dx = \frac{E_0}{B_0} \int_0^t \sin\left(\frac{qB_0}{m}t\right) dt$$

$$\text{or, } x = \frac{mE_0}{qB_0^2} \left[1 - \cos\left(\frac{qB_0}{m}t\right)\right] \quad (\text{vii})$$

Similarly integrating equation (vi) gives

$$z = \frac{mE_0}{qB_0^2} \left[\frac{qB_0}{m}t - \sin\left(\frac{qB_0}{m}t\right)\right] \quad (\text{viii})$$

The above equations (vii) and (viii) represent a cycloid. The path appears as shown in the figure.



(b) From (vii) the maximum value of x occurs when

$$\cos\left(\frac{qB_0}{m}t\right) = -1$$

Maximum x -coordinate is

$$x_0 = \frac{2mE_0}{qB_0^2}$$

(c) When $\cos\left(\frac{qB_0}{m}t\right) = -1$

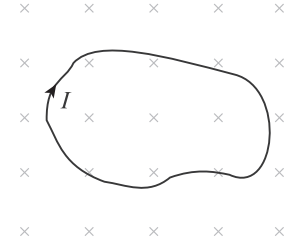
$$\sin\left(\frac{qB_0}{m}t\right) = 0$$

$\therefore$ From (v), $v_x = 0$ and

$$\text{From (vi), } v_z = \frac{2E_0}{B_0}$$

Example 28 Tension in a current loop

A loop is made of a flexible conducting wire of length L . It carries a current I . A uniform magnetic field (B) perpendicular to the plane of the loop is switched on. How will the shape of the loop change? Find tension in the loop.

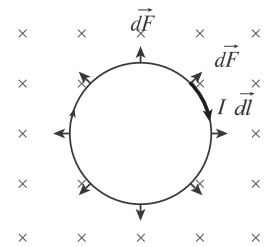


Solution

Concepts

- Magnetic force on a small length of the loop will be $d\vec{F} = I d\vec{l} \times \vec{B}$. This force is in the plane of the figure (i.e. perpendicular to $\vec{B}$) and perpendicular to $d\vec{l}$.
- The resultant of tension due to neighbouring elements will balance the magnetic force on an element and keep it in place.

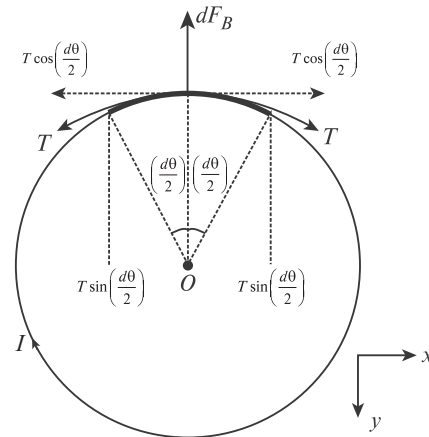
A small element of length $d\vec{l}$ will experience an outward magnetic force normal to it. Being pulled uniformly from all sides, the loop will take the shape of a circle.



Consider a small element in the loop that subtends an angle $d\theta$ at its centre. The outward magnetic force on it is

$$dF_B = I dl B = I (R d\theta) B$$

Here $R = \frac{L}{2\pi}$ is radius of the circular loop.



The tensions at the two ends of the elements are as shown in figure. The components of tension in x direction cancels out and sum of their components in y direction balances dF_B .

$$2T \sin\left(\frac{d\theta}{2}\right) = dF_B$$

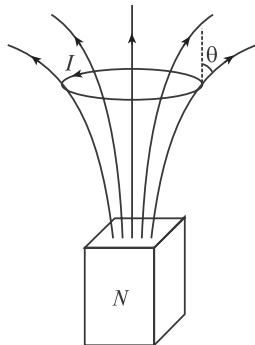
$$\text{or, } 2T \frac{d\theta}{2} = IR d\theta B$$

[For small angle $\sin\theta = \theta$]

$$\text{or, } T = IRB = \frac{ILB}{2\pi}$$

Example 29 Current loop in a uniform field

A ring of radius r carries a current I . It is placed horizontally above the north pole of a magnet. The magnetic field makes an angle θ with the vertical at all points on the circumference of the ring. Strength of the field at circumference of the ring is B .



- Find the magnetic force on the ring.
- Find the tension in the ring.

Solution**Concepts**

Resolve the magnetic field at the circumference into horizontal and vertical components and study their effect separately.

The current loop is in a non uniform field. Net force on it may not be zero.

The vertical component of the magnetic field is $B_v = B \cos\theta$ and its horizontal component is $B_H = B \sin\theta$.

First we will consider force due to the horizontal field. Force on a current element of length dl due to this component of the field is

$$dF = B_H Idl \sin 90^\circ$$

Direction of force on every small element is vertically downward. Resultant downward force is

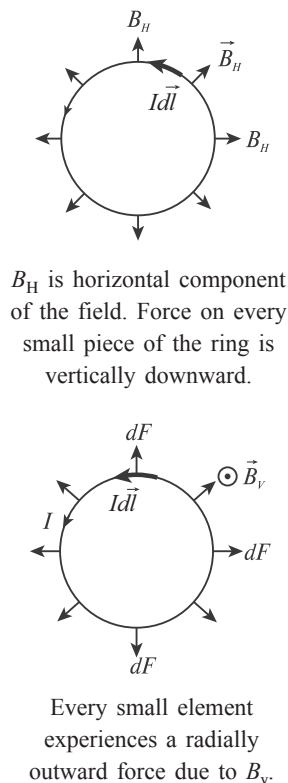
$$F = B_H I \int dl \\ = B_H I \cdot 2\pi r = 2\pi B I r \sin\theta$$

Now, consider the effect of vertical component of the field (B_v). Every small current element experiences a radially outward force. The vector sum of all these forces is zero. But a tension is developed in the ring due to this radial pull. From the last example, tension developed in the ring is

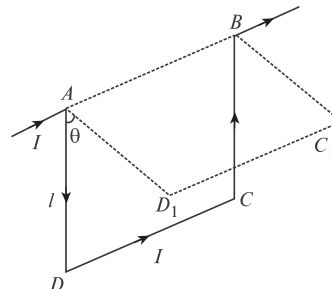
$$T = IrB_v = IrB \cos\theta$$

Obviously, the resultant force on the ring is

$$F = 2\pi B I r \sin\theta$$



Example 30 A wire is bent into U shape ($ADCB$) and is pivoted so that it can swing about the horizontal axis AB . Length of each arm of the wire frame is l . Mass per unit length of the wire is λ . There exists a uniform vertical magnetic field. When a current I is passed through the frame it deflects by an angle θ in its equilibrium position.



- Find the strength of magnetic field B .
- Find the force applied by the pivot on the frame in equilibrium position.

Solution**Concepts**

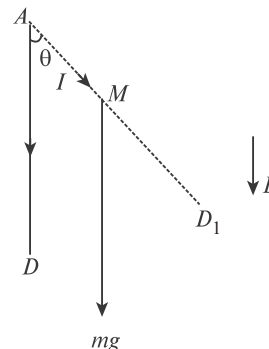
- In equilibrium the magnetic torque on the frame balances the torque due to gravity.
- The magnetic torque on the frame about AB is obtained by considering forces on the three arms separately.

(We can also write this torque in a slightly different way. Consider a closed loop $ADCB$. Torque on this loop can be written using $\vec{\mu} \times \vec{B}$. This is sum of torques due to forces on all arms. But the force on arm BA will not produce any torque about AB itself. Therefore, $\vec{\mu} \times \vec{B}$ will give us torque due to forces on the remaining three arms.

- In equilibrium the net force on the frame is zero. This will give us the force applied by the pivot on the frame.

- To keep the diagram two dimensional, let's draw one arm AD of the frame. The axis AB is perpendicular to the plane of the figure. Assume that you are looking from side BC .

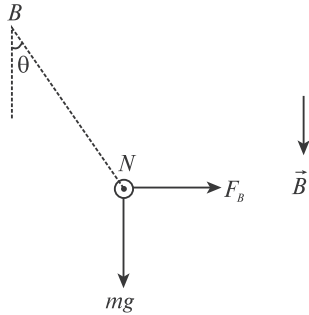
AD_1 is the equilibrium position. M is mid point of the arm AD . The magnetic force acting on the arm is directed into the plane of this figure. This force is parallel to the axis AB and has no torque about it.



Torque due to gravity is

$$\begin{aligned}\tau_{gAD} &= (\lambda l g) (AM \sin \theta) = (\lambda l g) \left(\frac{l}{2} \sin \theta \right) \\ &= \frac{1}{2} \lambda l^2 g \sin \theta\end{aligned}$$

Similarly, torque (about AB) due to magnetic force on BC_1 is zero. Gravitational torque is same as the above.



Observer is standing on BC side. He is looking towards BA .

Now, let's consider the horizontal arm DC . The diagram shows the cross section of the wire at mid point N . The current is towards the observer when he is looking from B to A .

Torque due to gravity

$$T_{gDC} = (\lambda l g) (l \sin \theta) = \lambda l^2 g \sin \theta$$

Torque due to the magnetic force is (about AB)

$$\begin{aligned}\tau_B &= F_B l \cos \theta \\ &= (IlB) (l \cos \theta) = I l^2 B \cos \theta.\end{aligned}$$

In equilibrium, the magnetic torque balances the gravitational torque

$$I l^2 B \cos \theta = \frac{1}{2} \lambda l^2 g \sin \theta + \frac{1}{2} \lambda l^2 g \sin \theta + \lambda l^2 g \sin \theta$$

$$\text{or, } IB \cos \theta = 2\lambda g \sin \theta$$

$$\text{or, } B = \frac{2\lambda g}{B} \tan \theta.$$

- (b) In equilibrium, the magnetic force on AD and BC are horizontal and oppositely directed. They cancel out. Magnetic force on DC is horizontal equal to

$$F_B = IlB$$

The pivot must apply equal and opposite horizontal force to balance this. Apart from this, the pivot must apply a vertical force to balance the weight of the frame. This force is

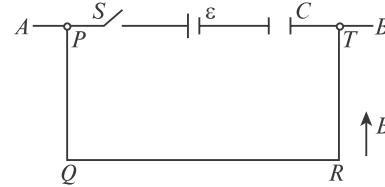
$$F_g = 3\lambda l g$$

$\therefore$ Resultant force by the pivot is

$$F = \sqrt{F_B^2 + F_g^2} = \sqrt{I^2 l^2 B + 9\lambda^2 l^2 g}$$

Example 31 Impulsive magnetic torque.

A conducting wire is bent in U shape. Mass of each of the segments PQ and TR is m and their length is L . Mass of segment QR is $2m$ and its length is $2L$.



The conducting frame can rotate freely about the horizontal axis AB . Between P and T there is a capacitor of capacitance C , a cell of emf ϵ and a switch S . Resistance of the loop $PQRT$ is nearly zero. A uniform vertical magnetic field B exists in the entire space. The switch S is closed. Find the angular speed acquired by the wire frame a short time after the switch is closed.

Solution

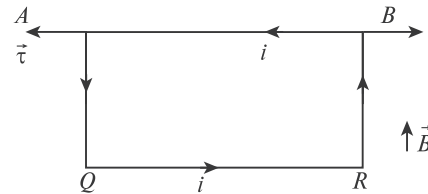
Concepts

- In absence of resistance, the capacitor will get charged in almost no time. Once the capacitor gets charged, there will be no current after that.
- When a current flows in the loop, a torque acts on it. This torque lasts for a very small time. The angular impulse of the torque ($\tau \Delta t$) is equal to change in angular momentum of the loop. In such a small time interval we can assume that the position of the loop has not changed much.

Charge flow in the loop is $Q = C\epsilon$

When current in the loop is i , its magnetic dipole moment is

$$M = iA = i(L \cdot 2L) = 2L^2 i$$



Direction of $\vec{M}$ is out of the plane of the figure. Direction of magnetic torque vector is along BA .

$$\tau = MB \sin 90^\circ = 2L^2 Bi$$

The current lasts for a very small interval Δt . Impulse of the torque is

1.34 Magnetism and Electromagnetic Induction

$$\int_0^{\Delta t} \tau dt = 2L^2 B \int_0^{\Delta t} \tau dt$$

$$= 2L^2 BQ = 2L^2 BC\epsilon$$

Now, Let's write the moment of inertia of the loop about axis AB.

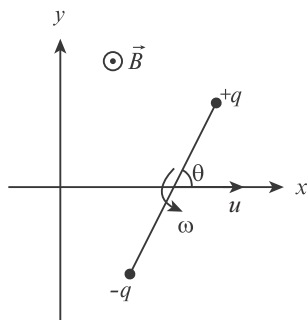
$$I = \frac{mL^2}{3} + \frac{mL^2}{3} + 2m(L)^2 = \frac{8}{3} mL^2$$

Angular impulse = Change in angular momentum of the loop.

$$\Rightarrow 2L^2 BC\epsilon = \frac{8}{3} mL^2 \cdot \omega$$

$$\text{or, } \omega = \frac{3}{4} \frac{BC\epsilon}{m}$$

Example 32 A non-conducting rigid rod of length L has particles attached to its ends. The particles have charge $+q$ and $-q$ on them and each has mass m . The rod is moving in xy plane in a space which has a uniform magnetic field $B\hat{k}$. At a particular instant the rod makes an angle θ with the positive x axis, its centre is moving with velocity $u\hat{i}$ and it is rotating with an angular velocity $\omega\hat{k}$. Find the net force and the net torque acting on the rod at this instant.



Solution

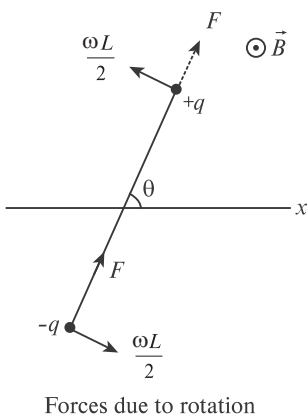
Concepts

- The particles at the ends of the rod have velocities which can be calculated by superposition of two velocities - (a) velocity perpendicular to the rod due to rotation and (b) the translational velocity $u\hat{i}$.
- The problem can be solved easily by writing forces on the particles, for two velocities, separately.

Consider, only rotation. Velocities of the particles are as shown in figure. Force on $+q$ charge is $F = q \frac{\omega L}{2} B$ directed along the rod as shown in figure. Same force acts on the other charge.

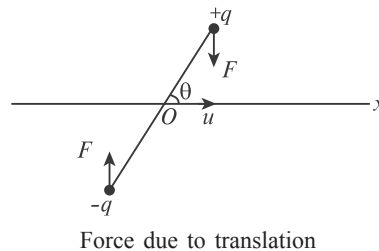
These forces do not produce a torque but have a resultant along the rod. The resultant force is

$$F_0 = q\omega LB$$



Forces due to rotation

Now, consider translational motion only. Forces on the two particles are $F = quB$ in the directions shown in figure. Resultant of these two forces is zero but they produce a torque.



Force due to translation

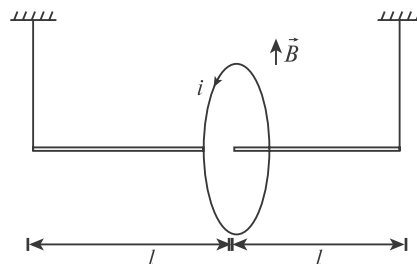
$$\tau = 2 F \frac{L}{2} \cos\theta = (quB)L \cos\theta = quBL \cos\theta.$$

When we superimpose the effects of rotation and translation, the net force and the net torque can be written as

$$F_{\text{net}} = q\omega LB$$

$$\tau_{\text{net}} = quBL \cos\theta$$

Example 33 A ring has mass m and radius R . A wire around the ring carries a current i . The ring is mounted on a massless horizontal axle. The ring-axle arrangement is suspended using two light threads tied to the ends (see figure). A uniform vertical magnetic field exists in the entire space. Find the tension in the two threads.



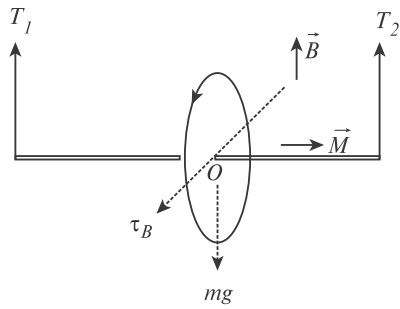
Solution

Concepts

- A magnetic torque acts on the current loop. There is no magnetic force on the loop.
- The key thing is to understand the direction of the torque. The torque tries to overturn the ring about a horizontal axis that is perpendicular to the axis.

Net force on the ring axle arrangement is zero.

$$\therefore T_1 + T_2 = mg \quad (i)$$



Magnetic torque is

$$\begin{aligned}\tau_B &= i \cdot \pi R^2 \cdot B \sin 90^\circ \\ &= i \pi R^2 B\end{aligned}$$

Direction of the torque is as shown in figure.

Torque of T_2 (about O) is also in same direction. Torque due to T_1 opposes the resultant of torques $T_2 l$ and τ_B .

$$\therefore T_1 l = T_2 l + i \pi R^2 B$$

$$\text{or, } T_1 - T_2 = \frac{i \cdot \pi R^2 B}{l} \quad (\text{ii})$$

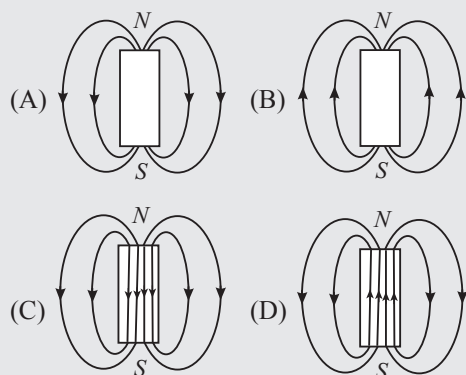
Solving (i) and (ii) gives

$$T_1 = \frac{mg}{2} + \frac{i \cdot \pi R^2 B}{2l}$$

$$\text{and } T_2 = \frac{mg}{2} - \frac{i \cdot \pi R^2 B}{2l}$$

Worksheet 1

1. The magnetic field lines due to a bar magnet are correctly shown in



2. A small bar magnet has a magnetic moment $1.2 \text{ A} \cdot \text{m}^2$. The magnetic field at a distance 0.1 m on its axis will be ($\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$)
 (a) $1.2 \times 10^{-4} \text{ T}$ (b) $2.4 \times 10^{-4} \text{ T}$
 (c) $2.4 \times 10^4 \text{ T}$ (d) $1.2 \times 10^2 \text{ T}$
3. Two identical thin bar magnets each of length l and pole strength m are placed at right angle to each other with north pole of one touching south pole of the other. Magnetic moment of the system is
 (a) ml (b) $2ml$
 (c) $\sqrt{2} ml$ (d) $\frac{1}{2} ml$
4. If a bar magnet of magnetic moment M is freely suspended in a uniform magnetic field of strength B , the work done in rotating the magnet through an angle θ is
 (a) $MB (1 - \sin \theta)$ (b) $MB \sin \theta$
 (c) $MB \cos \theta$ (d) $MB (1 - \cos \theta)$
5. A bar magnet of magnetic moment $3.0 \text{ A} \cdot \text{m}^2$ is placed in a uniform magnetic induction field of $2 \times 10^{-5} \text{ T}$. If each pole of the magnet experiences a force of $6 \times 10^{-4} \text{ N}$, the geometric length of the magnet is
 (a) 0.5 m (b) 0.3 m
 (c) 0.2 m (d) 0.12 m
6. A bar magnet with its poles 25 cm apart and of pole strength $24.0 \text{ A} \cdot \text{m}$ rests with its centre on a frictionless pivot. A force F is applied on the magnet, perpendicular to its length, at a distance of 12 cm from the pivot so that it is held in equilibrium at an angle of 30° with respect to a magnetic field of induction 0.25 T . The value of force F is
 (a) 5.62 N (b) 2.56 N
 (c) 6.52 N (d) 6.25 N

7. A particle having charge 1 C is moving in a magnetic field of 0.5 T with a velocity of 10 ms^{-1} making an angle of 30° with the magnetic field. Magnetic force experienced by the particle is
 (a) 5 N (b) 10 N
 (c) 2.5 N (d) 0 N
8. If a charged particle enters a uniform magnetic field, neither perpendicular, nor parallel, then its path will be
 (a) Straight line (b) Elliptical
 (c) Helical (d) Parabola
9. A proton and an alpha particle, after being accelerated through same potential difference, enter a uniform magnetic field the direction of which is perpendicular to their velocities. Find the ratio of radii of the circular paths of the two particles.
 (a) $\frac{1}{2}$ (b) $\frac{\sqrt{2}}{1}$
 (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{1}$
10. An electron is moving along positive x-axis. A uniform electric field exists towards negative y-axis. What should be the direction of magnetic field of suitable magnitude so that net force on the electron is zero?
 (a) positive z-axis (b) negative z-axis
 (c) positive y-axis (d) negative y-axis
11. A particle of charge q and velocity v passes undeflected through a space with non-zero electric field E and magnetic field B . The undeflecting conditions will hold if
 (a) signs of both q and E are reversed
 (b) signs of both q and B are reversed
 (c) both B and E are changed in magnitude, but keeping the product of and fixed
 (d) both B and E are doubled in magnitude.
12. An electron having kinetic energy T is moving in a circular orbit of radius R perpendicular to a uniform magnetic induction $\vec{B}$. If kinetic energy is doubled and magnetic induction is tripled, the radius will become
 (a) $\frac{3R}{2}$ (b) $\sqrt{\frac{3}{2}} R$
 (c) $\sqrt{\frac{2}{9}} R$ (d) $\sqrt{\frac{4}{3}} R$
13. A deuteron of kinetic energy 50 keV is describing a circular orbit of radius 0.5 m in a plane perpendicular

to magnetic field $\vec{B}$. The kinetic energy of a proton that describes a helical path of radius 0.5 m in the same magnetic field is certainly

- less than 25 keV
 - less than 50 keV
 - less than 200 keV
 - greater than 100 keV
14. Three ions H^+ , He^+ and O^{+2} having same kinetic energy pass through a region in which there is a uniform magnetic field perpendicular to their velocity, then
- H^+ will be least deflected
 - He^+ and O^{+2} will be deflected equally
 - O^{+2} will be deflected most
 - All will be deflected equally
15. Electrons moving with different speed enter a uniform magnetic field in direction perpendicular to the field. They will move along circular paths
- of same radius with same time period
 - with larger radius for the faster electrons but will have same time period.
 - with smaller radius and smaller time period for the faster electrons
 - with smaller radius and larger time period for the faster electrons
16. A particle of mass m and charge q moves with a constant velocity v along the positive x direction. It enters a region containing a uniform magnetic field B directed along the negative z direction, extending from $x = a$ to $x = b$. The minimum value of v required so that the particle can just enter the region $x > b$ is
- qbB/m
 - $q(b - a) B/m$
 - qaB/m
 - $q(b + a) B/2m$
17. A particle of charge q and mass m starts moving from the origin under the action of an electric field $\vec{E} = E_0\hat{i}$ and $\vec{B} = B_0\hat{i}$ with velocity $\vec{v} = v_0\hat{j}$. The speed of the particle will become $2v_0$ after a time
- $t = \frac{2mv_0}{qE_0}$
 - $t = \frac{2Bq}{mv_0}$
 - $t = \frac{\sqrt{3}Bq}{mv_0}$
 - $t = \frac{\sqrt{3}mv_0}{qE_0}$
18. An electron is projected with velocity v_0 in a uniform electric field E perpendicular to the field. Again it is projected with velocity v_0 perpendicular to a uniform magnetic field B . If r_1 is initial radius of curvature just after entering in the electric field and r_2 is initial radius of curvature just after entering in magnetic field then the ratio r_1/r_2 is equal to

- $\frac{Bv_0^2}{E}$
- $\frac{B}{E}$
- $\frac{Ev_0}{B}$
- $\frac{Bv_0}{E}$

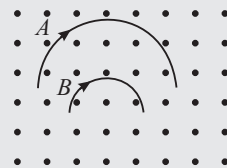
19. Two identical charged particles enter a uniform magnetic field with same speed but at angles 30° and 60° with field. Let a , b and c be the ratio of their time periods, radii and pitches of the helical paths. Which of the following relations is correct?

- $abc = 1$
- $abc > 1$
- $abc < 1$
- $ab = c$

20. An ionised gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the $+x$ direction and a magnetic field along the $+z$ direction, then

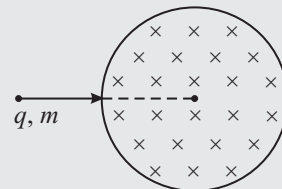
- positive ions deflect towards $+y$ direction and negative ions towards $-y$ direction
- all ions deflect towards $+y$ direction.
- all ions deflect towards $-y$ direction
- positive ions deflect towards $-y$ direction and negative ions towards $+y$ direction.

21. Two particles A and B of masses m_A and m_B respectively and having the same charge are moving in a plane. A uniform magnetic field exists perpendicular to this plane. The speeds of the particles are v_A and v_B respectively and the trajectories are as shown in the figure. Then



- $m_A v_A < m_B v_B$
- $m_A v_A > m_B v_B$
- $m_A < m_B$ and $v_A < v_B$
- $m_A = m_B$ and $v_A = v_B$

22. A particle having mass m , charge q enters a cylindrical region having uniform magnetic field B in the inward direction as shown. If the particle is deviated by 60° as it emerges out of the field then what is the time spent by it in the field?



- $\frac{2\pi m}{qB}$
- $\frac{2\pi m}{3qB}$
- $\frac{\pi m}{3qB}$

- It depends on the speed of particle.

23. A particle of specific charge (charge/mass) α starts moving from the origin under the action of an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{k}$. Its velocity at $(x_0, y_0, 0)$ is $(4\hat{i} - 3\hat{j})$. The value of x_0 is:

(a) $\frac{13}{2} \frac{\alpha E_0}{B_0}$ (b) $\frac{16\alpha E_0}{E_0}$
 (c) $\frac{25}{2\alpha E_0}$ (d) $\frac{5\alpha}{2B_0}$

24. A particle of specific charge (q/m) is projected from the origin of coordinates with initial velocity $u\hat{i} - v\hat{j}$. Uniform electric and magnetic fields exist in the region along the $+y$ direction, of magnitude E and B . The particle will definitely return to the origin once if
- (a) $[vB/2\pi E]$ is an integer
 (b) $(u^2 + v^2)/2 [B/\pi E]$ is an integer
 (c) $[vB/\pi E]$ is an integer
 (d) $[uB/\pi E]$ is an integer

25. A particle of mass M and charge Q moving with velocity $\vec{v}$ describes a circular path of radius R when subjected to a uniform transverse magnetic field of induction B . The work done by the field when the particle completes one full circle is

(a) $BQv2\pi R$ (b) $\left(\frac{Mv^2}{R}\right)2\pi R$
 (c) Zero (d) $BQ2\pi R$

26. An electron moving with a velocity $\vec{V}_1 = 2\hat{i}$ at a point in a magnetic field experiences a force $\vec{F}_1 = -2\hat{j}$. If the electron is moving with a velocity $\vec{V}_2 = -2\hat{j}$ at the same point, it experiences a force $\vec{F}_2 = +2\hat{j}$. The force the electron would experience if it were moving with a velocity $\vec{V}_3 = -2\hat{k}$ at the same point is (forces and velocities have appropriate units)

(a) zero (b) $2\hat{k}$
 (c) $-2\hat{k}$ (d) information is insufficient

27. Two particles of charges $+Q$ and $-Q$ are projected from the same point with a velocity v in a region of uniform magnetic field B such that the velocity vector makes an angle θ with the magnetic field. Their masses are M and $2M$, respectively. Then, they will meet again for the first time at a point whose distance from the point of projection is

(a) $2\pi Mv \cos\theta/QB$ (b) $8\pi Mv \cos\theta/QB$
 (c) $\pi Mv \cos\theta/QB$ (d) $4\pi Mv \cos\theta/QB$

28. A block of mass m and charge q is released on a long smooth inclined plane where magnetic field B is constant, uniform, horizontal and parallel to surface as shown in

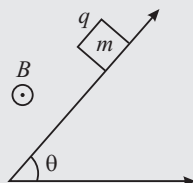


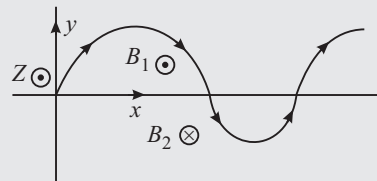
figure. Find the time from start when block loses contact with the surface.

(a) $\frac{m \cos\theta}{qB}$ (b) $\frac{m \operatorname{cosec}\theta}{qB}$
 (c) $\frac{m \cot\theta}{qB}$ (d) none

29. When a proton is released from rest in a gravity free room, it starts with an initial acceleration a_0 towards west. When it is projected towards north with a speed, it moves with an initial acceleration $3a_0$ towards west. The electric field and the maximum possible magnetic field in the room respectively are

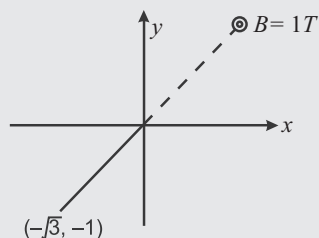
(a) $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ down
 (b) $\frac{ma_0}{e}$ east, $\frac{2ma_0}{ev_0}$ down
 (c) $\frac{ma_0}{e}$ east, $\frac{2ma_0}{ev_0}$ up
 (d) $\frac{2ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ up

30. At $t = 0$ a charge q is at the origin and moving in the y -direction with velocity $\vec{v} = v\hat{j}$. The charge moves in a magnetic field that is directed out of the page for $y > 0$ and given by $B_1 \hat{z}$. For $y < 0$ the field is directed into the page and is given by $-B_2 \hat{z}$. The charge's subsequent trajectory is shown in the sketch. From this information, we can deduce that



(a) $q > 0$ and $|B_1| < |B_2|$
 (b) $q < 0$ and $|B_1| < |B_2|$
 (c) $q > 0$ and $|B_1| > |B_2|$
 (d) $q < 0$ and $|B_1| > |B_2|$

31. A uniform magnetic field of magnitude 1 T exists in the region $y \geq 0$. Field is directed along $\hat{k}$ direction as shown. A particle of charge 1 C is projected from point $(-\sqrt{3}, -1)$ towards origin with speed 1 ms^{-1} . If the mass of the particle is 1 kg, then co-ordinates of centre of circle, in which particle moves, are



- (a) $(1, \sqrt{3})$ (b) $(1, -\sqrt{3})$
 (c) $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ (d) $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

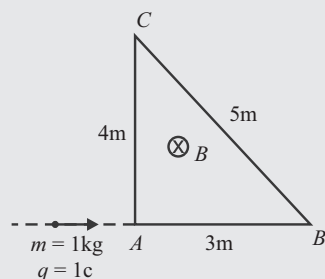
32. A charged particle of specific charge α is projected from origin at time $t = 0$ with velocity $\vec{V} = V_0\hat{i} + V_0\hat{j}$ in magnetic field $\vec{B} = B_0\hat{i}$. The coordinates of the particle at time $t = \frac{\pi}{B_0\alpha}$ are (specific charge $\alpha = q/m$)

- (a) $\left(\frac{V_0}{2B_0\alpha}, \frac{\sqrt{2}V_0}{\alpha B_0}, \frac{-V_0}{B_0\alpha}\right)$ (b) $\left(\frac{-V_0}{2B_0\alpha}, 0, 0\right)$
 (c) $\left(0, \frac{2V_0}{B_0\alpha}, \frac{V_0\pi}{2B_0\alpha}\right)$ (d) $\left(\frac{V_0\pi}{B_0\alpha}, 0, -\frac{2V_0}{B_0\alpha}\right)$

33. There is a constant homogeneous electric field of 100Vm^{-1} within the region $x = 0$ and $x = 0.167\text{m}$ pointing in the positive x-direction. There is a constant homogeneous magnetic field B within the region $x = 0.167\text{m}$ and $x = 0.334\text{m}$ pointing in the z-direction. A proton is released from rest at the origin ($x = 0$, $y = 0$). Find the minimum strength of the magnetic field B , so that the proton is deflected back at $x = 0$, $y = -0.167\text{m}$. (mass of the proton = $1.67 \times 10^{-27}\text{kg}$)

- (a) $5\sqrt{2}\text{mT}$ (b) $15\sqrt{2}\text{mT}$
 (c) $15\sqrt{3}\text{mT}$ (d) $10\sqrt{3}\text{mT}$

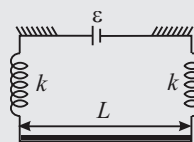
34. A small particle of mass $m = 1\text{kg}$ and charge of 1C enters perpendicularly in a triangular region of uniform magnetic field of strength 2T as shown in figure.



Calculate maximum velocity of the particle with which it should enter so that it completes a half-circle in magnetic region

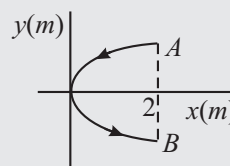
- (A) 2ms^{-1} (B) 2.5ms^{-1}
 (C) 3ms^{-1} (D) 4ms^{-1}

35. A straight wire of mass m and length L is suspended using two identical springs as shown in the figure. The springs are stretched by x_0 due to the weight of the wire. The circuit has total resistance R . When the magnetic field perpendicular to the plane of the paper is switched on, springs are observed to extend further by x_0 . The magnetic field is



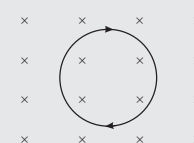
- (a) $\frac{mgR}{\epsilon L}$; directed outward from the plane of the paper
 (b) $\frac{mgR}{2\epsilon x_0}$; directed outward from the plane of the paper
 (c) $\frac{mgR}{\epsilon L}$; directed into the plane of the paper
 (d) $\frac{mgR}{\epsilon x_0}$; directed into the plane of the paper

36. A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current $I = 2\text{A}$ as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}\text{T}$. The magnetic force on the wire is (in newton)



- (a) $-16\hat{i}$ (b) $32\hat{i}$
 (c) $-32\hat{i}$ (d) $16\hat{i}$

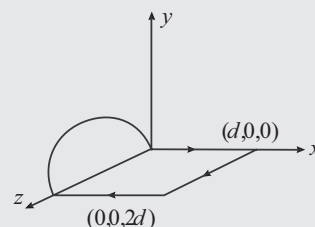
37. A circular coil carrying current I is placed in a region of uniform magnetic field acting perpendicular to a coil as shown in the figure. Mark correct option



- (a) coil expands
 (b) coil contracts
 (c) coil moves left
 (d) coil moves right

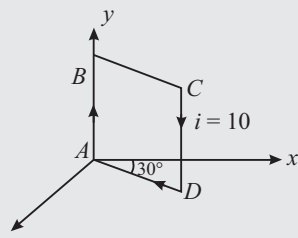
38. The current loop shown in figure has a semicircular segment in yz plane and a rectangular part in xz plane. The magnetic dipole moment of the current loop is

- (a) $id^2\left(\frac{x}{2} + 2\right)\hat{k}$
 (b) $id^2\left(-\frac{x}{2}\hat{i} - 2\hat{j}\right)$
 (c) $id^2\left(-\frac{x}{2}\hat{i} + 2\hat{j}\right)$

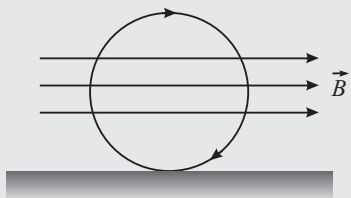


- (d) none of the above

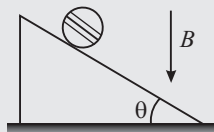
39. Figure shows a square current carrying loop $ABCD$ of side 10cm and current $i = 10A$. The magnetic moment $\vec{M}$ of the loop is



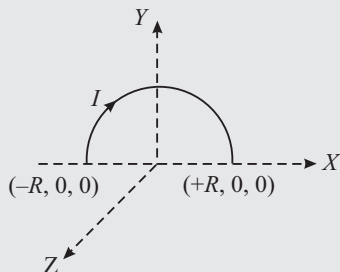
- (a) $(0.05)(\hat{i} - \sqrt{3}\hat{k}) A - m^2$
 (b) $(0.05)(\hat{j} - \hat{k}) A - m^2$
 (c) $(0.05)(\sqrt{3}\hat{i} + \hat{k}) A - m^2$
 (d) $(\hat{i} - \hat{k}) A - m^2$
40. A conducting ring of mass 2kg and radius 0.5m is standing on a smooth horizontal plane. The ring carries a current $i = 4A$ horizontal magnetic field $B = 10T$ is switched on at time $t = 0$ as shown in figure. The initial angular acceleration of the ring will be



- (a) $40\pi \text{ rads}^{-2}$ (b) $20\pi \text{ rads}^{-2}$
 (c) $5\pi \text{ rads}^{-2}$ (d) $15\pi \text{ rads}^{-1}$
41. In the figure shown a coil of three turns is wound on a sphere of radius R and mass m . The plane of the coil is parallel to the incline plane and lies in the equatorial plane of the sphere. Current in the coil is i . The value of B if the sphere is in equilibrium is

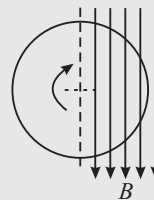


- (a) $\frac{mg \cos \theta}{3\pi i R}$ (b) $\frac{mg}{3\pi i R}$
 (c) $\frac{mg \tan \theta}{\pi i R}$ (d) $\frac{mg \sin \theta}{3\pi i R}$
42. A semi circular current carrying wire having radius R is placed in x - y plane with its centre at the origin. There is a non-uniform magnetic field $\vec{B} = \frac{B_0 x}{2R} \hat{k}$ (here B_0 is positive constant) existing in the region.

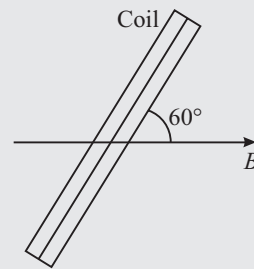


The magnetic force acting on semicircular wire will be along

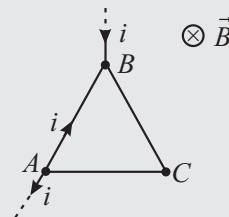
- (a) $-x$ -axis (b) $+y$ -axis
 (c) $-y$ -axis (d) $+x$ -axis
43. A thin non conducting disc of radius R is rotating clockwise (see figure) with an angular velocity ω about its central axis, which is perpendicular to its plane. Both its surfaces carry positive charges of uniform surface density. Half the disc is in a region of a uniform magnetic field B parallel to the plane of the disc, as shown. Then,
- (a) The net torque on the disc is zero.
 (b) The net torque vector on the disc is directed leftwards.
 (c) The net torque vector on the disc is directed rightwards.
 (d) The net torque vector on the disc is parallel to B .



44. The figure shows the side view of a rectangular coil having $2n$ turns, an area A and a current I . The plane of the coil is at 60° to a horizontal uniform magnetic field of intensity B . The torque on the coil due to magnetic force is



- (a) $BnAI$
 (b) $2BnAI$
 (c) $2nAI B \sin 60^\circ$
 (d) none
45. An equilateral triangle ABC , of side l , is part of an electric circuit. Figure shows the currents. It is placed in a uniform magnetic field B perpendicular to its plane. The magnitude of magnetic force on the triangle is

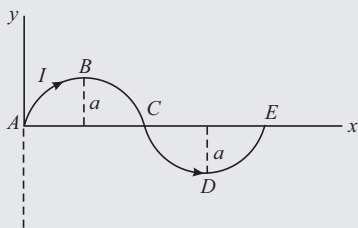


- (a) ilB (b) $2ilB$
 (c) $3ilB$ (d) zero

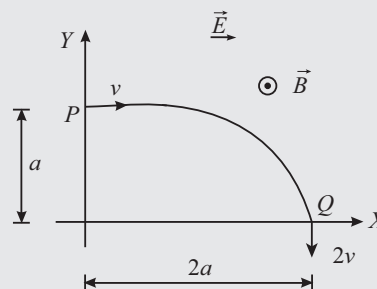
Worksheet 2

- A charged particle is projected in a plane perpendicular to a uniform magnetic field. The areal velocity (area swept per unit time by the position vector of the particle relative to the centre) of the particle is :
 - directly proportional to kinetic energy of particle
 - directly proportional to momentum of the particle
 - inversely proportional to magnetic field strength
 - inversely proportional to charge on particle
- Which of the following statements are correct (given : $\tan 81^\circ = 6.28$)
 - A charged particle enters a region of uniform magnetic field at an angle 85° to magnetic lines of force. The path of the particle is a helix with pitch larger than radius.
 - A charged particle enters a region of uniform magnetic field at an angle 85° to magnetic lines of force. The path of the particle is a helix with pitch smaller than radius.
 - A charged particle enters a region of uniform magnetic field at an angle 45° to magnetic lines of force. The path of the particle is a helix with pitch larger than radius.
 - A charged particle enters a region of uniform magnetic field at an angle 45° to magnetic lines of force. The path of the particle is a helix with pitch equal to radius.
- A proton enters in a region having a uniform electric and magnetic field $\vec{E}$ and $\vec{B}$ respectively. Velocity of proton is $\vec{V}$. All the three vectors are mutually perpendicular. It is observed that the proton is deflected along positive x -axis when either of the fields or both are switched on simultaneously. Then
 - $\vec{V}$ may be along positive y -axis
 - $\vec{E}$ is along positive axis
 - $\vec{B}$ may be along positive z -axis
 - $\vec{B}$ may be along negative y -axis
- A conductor $ABCDE$ in xy plane, has a shape which can be described as

$$y = a \sin\left(\frac{2\pi}{\lambda}x\right)$$
 It carries current I .



- A uniform magnetic field of magnitude B exists in the region. The force acting on it will be
 - zero, if B is in the x -direction
 - λBI in the z -direction, if B is in the y -direction
 - λBI in the negative y -direction, If B is in the negative z -direction
 - $\lambda a BI$ if B is in the x -direction.
- A charged particle is released from rest under gravity. A uniform horizontal magnetic field exists in the space. Which of the statements are true ?
 - Starting with zero velocity, the particle will eventually acquire a circular path if gravity is also there apart from the magnetic field.
 - The speed of the charge increases continuously.
 - Magnetic force alone is not responsible for deflection in the path of the particle subsequently.
 - magnetic force alone is not responsible for deflection in the path of the particle subsequently.
- A charged particle with velocity $\vec{v} = x\hat{i} + y\hat{j}$ moves in a magnetic field $\vec{B} = y\hat{i} + x\hat{j}$. The magnitude of magnetic force on the particle is F .
 - $F = 0$ if $x = y$
 - $F \propto (x^2 - y^2)$ if $x > y$
 - F is along z -axis if $x > y$
 - F is along y -axis if $y > x$
- A particle of charge $+q$ and mass m moving under the influence of a uniform electric field $E\hat{i}$ and a uniform magnetic field $B\hat{k}$ follows a trajectory from P to Q as shown in the figure. The velocities at P and Q are $v\hat{i}$ and $-2v\hat{j}$. Which of the following statement(s) is/are correct?



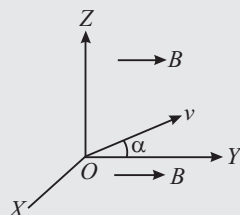
- $E = \frac{3}{4} \left(\frac{mv^2}{qa} \right)$
- Rate of work done by the electric field at P is $\frac{3}{4} \left(\frac{mv^3}{a} \right)$

- (c) Rate of work done by the electric field at P is zero
 (d) Rate of work done by the fields at Q is zero

8. Consider three quantities $x = E/B$, $y = \sqrt{l/\mu_0\epsilon_0}$ and $z = \frac{l}{CR}$. Here, l is the length of a wire, C is a capacitance and R is a resistance. All other symbols have standard meanings.

- (a) x, y have the same dimensions
 (b) y, z have the same dimensions
 (c) z, x have the same dimensions
 (d) none of the three pairs have the same dimensions.

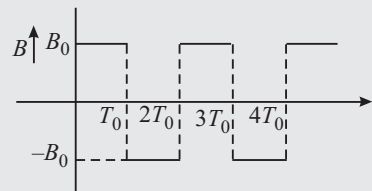
9. In a region of space, a uniform magnetic field B exists in the y direction. A proton is fired from the origin, with its initial velocity v making a small angle α with the y direction in the yz plane. In the subsequent motion of the proton,



- (a) its x coordinate can never be positive
 (b) its x and z coordinates cannot both be zero at the same time
 (c) its z coordinate can never be negative
 (d) its y coordinate will be proportional to the square of its time of flight

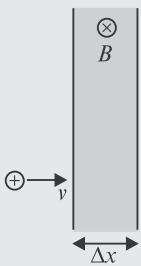
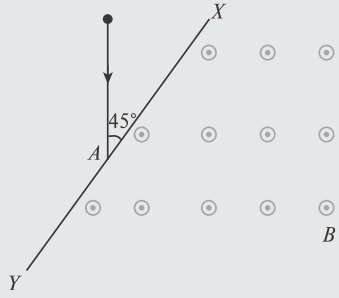
10. In a region magnetic field along x axis changes with time according to the given graph. A charged particle is projected from origin at an angle θ_0 with the positive x -axis in x - y plane.

If time period, pitch and radius of the helical path are T_0 , P_0 and R respectively then which of the followings are incorrect



- (a) At $t = \frac{T_0}{2}$, co-ordinates of charge are $\left(\frac{P_0}{2}, 0, -2R_0\right)$.
 (b) At $t = \frac{3T_0}{2}$, co-ordinates of charge are $\left(\frac{3P_0}{2}, 0, 2R_0\right)$.
 (c) The maximum distance of the particle from the x -axis is $2R_0$
 (d) The maximum distance of the particle from the x -axis is $4R_0$

Worksheet 3

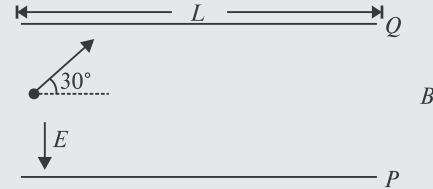
1. A charged particle is projected in a magnetic field $\vec{B} = (0.03\hat{i} + 0.02\hat{j} - 0.1\hat{k})$ T. The acceleration of the particle is found to be $\vec{a} = (x\hat{i} + 4\hat{j} + 2\hat{k})$ ms⁻². Find x .
2. A particle having charge $q = 10\mu\text{C}$ moves in a uniform magnetic field with velocity $v_1 = 10^6\text{ms}^{-1}$ at an angle 45° with x axis in xy plane and experiences a force $F_1 = 5\sqrt{2}$ mN along the negative z axis. When the same particle is projected with velocity $v_2 = 10^6\text{ms}^{-1}$ along the z axis it experiences a force F_2 in y direction. Find the magnitude and direction of the magnetic field. Also find the magnitude of the force F_2 .
3. A singly charged ion of mass m is accelerated from rest by a potential difference ΔV . It enters into a uniform magnetic field perpendicularly and moves in a circle of radius R . Now, a doubly charged ion of mass M is accelerated through the same potential difference and made to enter the same field perpendicularly. The particle moves in a circle of radius $2R$. What is the ratio of the masses of the ions?
4. In a TV picture tube electron are first accelerated through a potential difference of V . Then these electrons travel through a region of length D having uniform transverse magnetic field (B). Thereafter, the electrons impinge on a screen.
 - (a) Find the distance (d) by which the point of incidence of the beam on the screen shifts due to presence of the magnetic field. Assume that $D \ll \frac{mv}{eB}$, where m is mass of an electron, v is its speed and e is charge on it. Express your answer in terms of B , D , e , m and V .
 - (b) Find d when $V = 750$ V, $D = 50$ cm and $B = 0.5\mu\text{T}$
5. A particle of mass m has charge q . It is moving with a velocity v . It enters into a region having a uniform magnetic field perpendicular to its direction of motion. The width of the region is Δx . Find the impulse of the magnetic force that acted on the particle as it crossed the field region. Take $\Delta x \ll \frac{mv}{qB}$

6. A uniform magnetic field B exists, out of the plane of the figure, in the space to the right of the XY plane. A particle having mass m and positive charge q enters the field region at point A , travelling as shown in figure.
 

Calculate the distance travelled by the charge particle in the magnetic field. Speed of the particle is v .

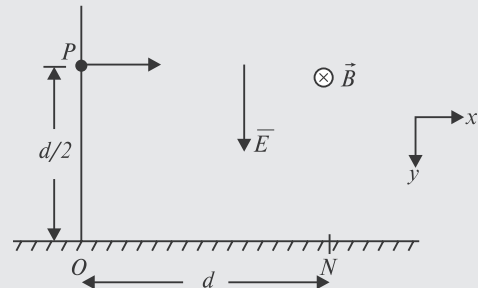
7. In the figure shown, G is an electron gun that shoots out electrons of energy 2keV in positive z -direction. The electrons are required to hit the spot S where $GS = 0.1\text{m}$ and the line GS makes an angle of 60° with the z -direction. A uniform magnetic field B parallel to GS exists in the region outside the gun. Find the minimum value of B needed to make the electrons hit the spot S . What is the next higher value of B for which the electrons will again start hitting the spot S ?
8. In a region of space an electric field exists and is given by $\vec{E} = -E_0\hat{i}$. There is also a magnetic field given by $\vec{B} = B_0\hat{k}$. A particle having mass m and charge $-q$ is projected from the origin with a velocity $\vec{v} = v_0\hat{j}$. Find the radius of curvature of the path of the particle immediately after projection if $v_0 = \frac{E_0}{2B_0}$. Express your answer in terms of v_0 , m , q and E_0 .
9. The region between $x = 0$ and $x = b$ is filled with a uniform steady magnetic field $-B_0\hat{k}$. A particle having mass m and charge q , travelling along positive x direction with speed v_0 , enters the region. Neglect gravity and find the
 - (a) value of b if the particle emerges from the region of the field with its velocity making an angle 30° to its initial velocity.
 - (b) final velocity of the particle and the time spent by it in the magnetic field, if the magnetic field is now expanded upto $2.6b$.

10. A particle with specific charge σ leaves the origin in the direction of x axis with an initial velocity v_0 . Uniform electric and magnetic field exist in the entire space. Their strengths are E_0 and B_0 respectively and both are directed along positive y -direction. Find the
- y co-ordinate of the particle when it crosses the y axis for the n^{th} time, and
 - angle between the particle's velocity and the axis at that moment.
11. In a certain region uniform electric field $\vec{E} = -E_0 \hat{k}$ and magnetic field $\vec{B} = B_0 \hat{k}$ are present. A particle having mass m and charge $+q$ is imparted a velocity $u(\hat{j} + \hat{k})$ at time $t = 0$. Find the time when the speed of the particle becomes smallest and also find the minimum speed.
12. A charge particle is released from rest at the origin. The region has an electric field $\vec{E} = E_0 \hat{i}$ and a magnetic field $\vec{B} = B_0 \hat{j}$. Mass of the particle and charge on it are m and q respectively. The maximum value of x -coordinate of the particle is observed to be d . Find the maximum speed of the particle during its course of motion.
13. A conducting wire of length l carrying current I , is placed on a rough horizontal surface, where a uniform horizontal magnetic field B perpendicular to the length of the wire exists. The smallest horizontal force needed to move the wire in a direction perpendicular to its length is F_1 . When the direction of current in the wire is reversed, the least force needed to move the wire perpendicular to its length is $F_2 (< F_1)$. Find the weight of the rod.
14. Two parallel conducting rails form an incline of inclination angle $\theta = 37^\circ$. Separation between them is $L = 0.5\text{m}$. At the lower end they are connected to a constant current source which supplies a current $I = 2.5\text{A}$. A smooth conducting bar is placed across the rails and a vertically upward magnetic field B is switched on. Mass of the bar is 50g and it slides with constant velocity on the rails. Find B .
15. In the last question assume that the coefficient of friction between the bar and the inclined rails is $\mu = 0.5$. Find the maximum and minimum values of B for which the bar can remain in equilibrium.
16. Two parallel plates have length $L = 0.17\text{m}$ and separation between them is $d = 0.1\text{m}$. A uniform electric field $E = 10^{-3} \text{ NC}^{-1}$ is present between the plates in a direction perpendicular to the plates (see figure). A

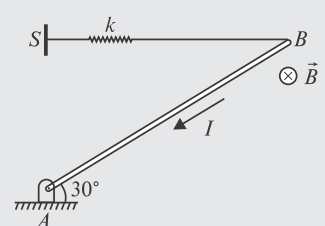
uniform magnetic field is present outside the plates directed perpendicular to the plane of the figure. A particle having mass $m = 10^{-10} \text{ kg}$ and charge $1\mu\text{C}$ is fired from the midpoint of the left edges of the plates making an angle of 30° with the plane of the plates. The particle leaves the region of electric field grazing the plate at Q and re-enters the plate region grazing the plate at P . Find the velocity of projection of the particle and the strength of magnetic field outside the plates. Neglect gravity.



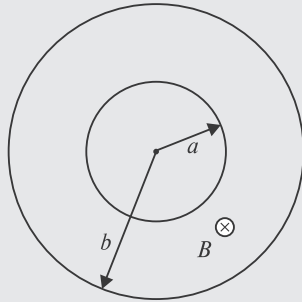
17. A particle of mass m and charge q enters a region having uniform electric field E as shown in the figure. The particle's initial velocity is v and the electric field is in y -direction. The particle moves from P to N and hits a flat smooth surface. The collision is elastic. At the instant the particle hits the surface, the electric field is switched off and a uniform magnetic field is switched on. Magnetic field is directed into the plane of the figure. The particle again hits the surface at point O . Express B in terms of E , m , q and d . Neglect gravity.



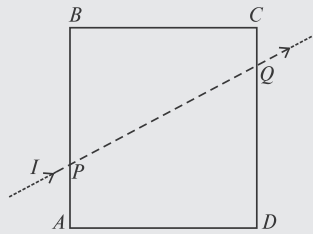
18. A thin, uniform conducting rod with negligible mass has length l . It is hinged at its lower end at A . A horizontal spring with a force constant K connects the other end B of the rod to a vertical support wall S . There is a uniform horizontal magnetic field directed into the plane of the figure. A current I flows in the rod in the direction shown and it stays in equilibrium making 30° with the horizontal.
- Is the spring stretched or compressed?
 - Find the energy stored in the spring.



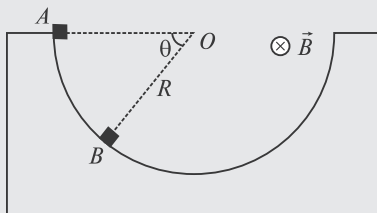
19. Two coaxial cylinders have radius a and b ($b > a$). A uniform magnetic field exists in the space between the two cylinders parallel to their axis. An electron (charge $-e$, mass m) shoots out of the outer surface of the inner cylinder in radial direction with a velocity u . Find values of magnetic field for which the electron will not hit the outer cylinder.



20. A uniform rectangular wire frame has side lengths $AB = 2L$ and $BC = L$. A long straight wire carrying current I joins the frame at P such that $AP = \frac{L}{2}$. The current exits into another long wire at Q such that $CQ = \frac{L}{2}$. A uniform magnetic field (B) exists in the entire space parallel to the plane of the frame. The direction of $\vec{B}$ is parallel to the two long wires. If the frame is released from this position, what will be its instantaneous acceleration? Mass of the frame is m . Assume no other force apart from magnetic force.



21. A fixed non conducting block has a hemispherical cut of radius R in it. A small block of mass m having charge Q is released from point A . A uniform horizontal magnetic field B is present directed into the plane of the figure. There is no friction. Find the normal force applied by the block on the hemispherical surface when it reaches point B after rotating through an angle θ .

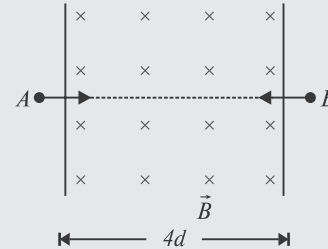


22. Two parallel conducting rails are parabolic in shape. Their shapes can be described as
 Rail 1 : $y = x^2$ (and $z = 0$)
 Rail 2 : $y = x^2$ (and $z = a$)

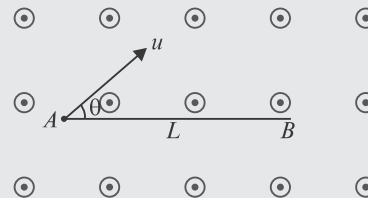
The x -direction is horizontal and y is vertical. A uniform magnetic field $B_0 \hat{j}$ exists in the entire space. A horizontal conducting bar rests on the two rails. It

carries a current i . What should be y co-ordinate of the bar for it to remain in equilibrium? Mass of the bar is m .

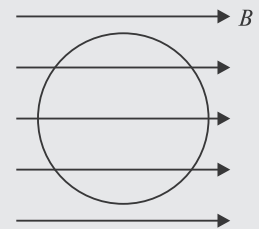
23. Two identical charged particles, A and B , enter perpendicularly into a region of uniform magnetic field B . The particles enter the field region while moving along the same line in opposite directions (see figure). The width of the field region is $4d$ where $d = \frac{mv}{qB}$. Here, m , q and v are mass, charge and speed of each particle, respectively. Find the angular speed of particle B relative to the particle A at the instant the two particles come out of the field region.



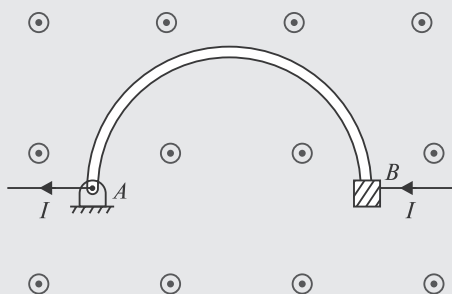
24. A particle having mass m and charge q is projected with velocity u , perpendicularly into a uniform magnetic field. The particle starts from point A and hits point B . Distance $AB = L$. Find the strength (B_0) of the magnetic field.



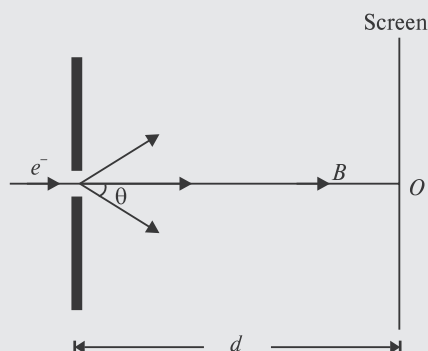
25. A ring is made of a wire of resistance R . Its radius is a . It is placed in a uniform magnetic field B directed parallel to the plane of the ring. You are provided with a source that provides constant current I . How will you connect this battery across two points on the ring so as to obtain maximum magnetic force on the ring? How much is this force?



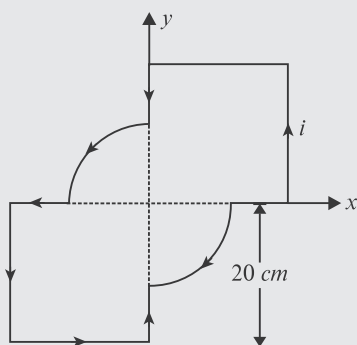
26. A conducting ring of mass m and radius R is hinged at its end A so that it can rotate in a vertical plane about the horizontal axis passing through A . The other end rests on a metallic contact B . The ring is connected in an electric circuit through the hinge A and the metallic contact B . A uniform horizontal magnetic field B_0 exists in the region. Find the value of current I at which the circuit gets opened. This device is a magnetic current limiter.



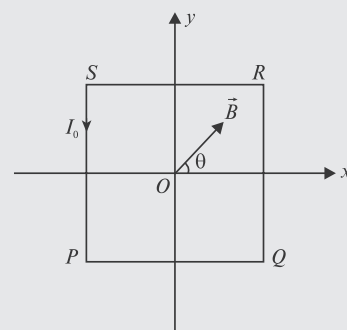
27. A beam of electrons has electrons accelerated from rest, through a potential difference ΔV . The beam passes through a narrow hole, as shown in the figure. The electron velocity vectors lie within a narrow cone of half angle θ oriented along the beam axis. A uniform magnetic field B exists along the beam axis. A screen is held normal to the beam axis at a distance d from the hole. Find the smallest distance d between the hole and the screen so that the electrons are focused at a point O on the screen (see figure). Charge on an electron is e and its mass is m .



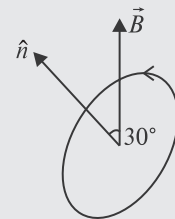
28. A current $i = 5\text{A}$ is established in a loop shown in the figure. The side length of the square is 20cm and radius of the circular arcs is 10cm . An external magnetic field $\vec{B} = (14\hat{i} + 14\hat{j})\text{ T}$ exists in the space. Find the torque acting on the loop.



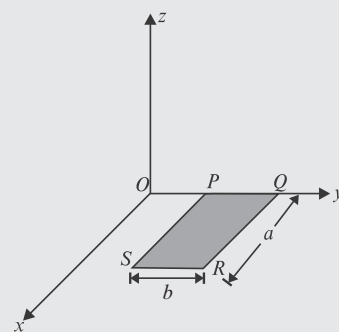
29. A uniform magnetic field B is directed at an angle of $\theta = 45^\circ$ to the x axis in the xy plane. $PQRS$ is a rigid wire frame of square shape and carries a current I_0 . It has its centre at the origin. Each side has mass M and length L . At time $t = 0$ the frame is at rest with its sides parallel to the x and y axis



- (a) Find torque on the frame (about point O) due to magnetic field.
(b) Find the angle by which the frame rotates under the action of this torque in a short interval of time Δt . Assume the angle to be so small that the variation in the value of torque during this interval can be neglected.
30. An electron in the ground state of a hydrogen atom is revolving in anticlockwise direction in a circular orbit of radius R (see figure).
(a) Write an expression for the orbital magnetic moment of the electron.
(b) The atom is in a uniform magnetic field B such that the plane normal to the electron orbit makes an angle of 30° with the direction of field. Find the torque experienced by the orbiting electron,
[m = mass of electron, e = charge on electron]



31. A rectangular loop $PQRS$ made from a uniform wire has length a , width b and mass m . It is free to rotate about the arm PQ , which remains hinged along a horizontal line taken as y -axis (see figure).



- Take the vertically upward direction as the z -axis. A uniform magnetic field $\vec{B} = (3\hat{i} + 4\hat{k}) B_0$ exists in the region. The loop is held in the xy plane and a current I is passed through it. The loop is now released and is found to stay in the horizontal position in equilibrium.
(a) Find the direction of current I in the arm PQ .
(b) Find the magnetic force on arm RS
(c) Find the value of current I .

Answer Sheet

Your Turn

- There are no magnetic mono poles.
- Magnetic field lines are always closed loops. Electric field lines, due to static charges, are always open ended.
- $1.008 \text{ A}\cdot\text{m}^2$
- Magnetic dipole moment
- $3.6 \times 10^{-5} \text{ T}$
- $7.1 \times 10^{-5} \text{ T}$ directed opposite to the magnetic dipole moment
- When the dipole moment is opposite to the external field.
- Zero
- 1000 m/s^2
- (a) West (b) East
- $qv_0 B_0 (\hat{j} - \hat{i})$
- $M^1 T^{-2} A^{-2}$
- $R = 0.21 \text{ mm}; f = 7.6 \times 10^6 \text{ Hz.}$
- $R = \frac{1}{B} \sqrt{\frac{2mV}{q}}$
- 90°
- $\frac{4\pi m}{3qB}$
- In electric field the path is parabolic. In magnetic field the path is circular
- $\frac{\pi m}{qB}$
- A $\rightarrow$ electron, B $\rightarrow$ proton, C $\rightarrow$ alpha, D $\rightarrow$ neutron
- 38 cm
- $(0.4 \text{ Vm}^{-1}) \hat{k}$
- (a) 2×10^6 (b) $5 \times 10^6 \text{ ms}^{-1}$
- Ratio is $\frac{14}{12} = \frac{7}{6}$
- $ILB_0 (2\hat{k} - \hat{j})$
- South west
- IaB
- (a) $IaB \otimes$ (b) Zero (c) Zero]
- $\frac{1}{8}$
- $M = q\omega l^2, \frac{M}{L} = \frac{q}{2m}$
- $\frac{\pi I}{2} (R_1^2 - R_2^2)$ directed inward
- $-\frac{\pi R^2 I}{2} (\hat{i} + \hat{k})$
- zero
- I. $\pi R^2 B$
- $4.0 \times 10^{-5} \text{ Nm rad}^{-1}$

Worksheet 1

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (b) | 3. (c) | 4. (d) | 5. (d) | 6. (c) | 7. (c) | 8. (c) | 9. (c) |
| 10. (b) | 11. (d) | 12. (c) | 13. (d) | 14. (b) | 15. (b) | 16. (b) | 17. (d) | 18. (d) |
| 19. (a) | 20. (c) | 21. (b) | 22. (c) | 23. (c) | 24. (d) | 25. (c) | 26. (a) | 27. (d) |
| 28. (c) | 29. (a) | 30. (a) | 31. (d) | 32. (d) | 33. (a) | 34. (c) | 35. (a) | 36. (b) |
| 37. (a) | 38. (b) | 39. (a) | 40. (a) | 41. (b) | 42. (a) | 43. (b) | 44. (a) | 45. (a) |

Worksheet 2

- | | | | | |
|--------------|--------------|-----------------|-----------|------------|
| 1. (a, c, d) | 2. (a, b, c) | 3. (a, b, c, d) | 4. (a, b) | 5. (c) |
| 6. (a, b, c) | 7. (a, b, d) | 8. (a, b, c) | 9. (a) | 10. (b, d) |

Worksheet 3

1. 4
2. $\vec{B} = (10^{-3} \text{ T}) \hat{i}$, $F_2 = 0.01 \text{ N}$
3. $\frac{M}{m} = 8$
4. (a) $d = \frac{D^2 B}{2} \sqrt{\frac{e}{2mV}}$, (b) $d = 6.7 \text{ cm}$
5. $\Delta x qB$
6. $\frac{\pi m v}{2qB}$
7. $4.71 \times 10^{-3} \text{ T}$, $9.42 \times 10^{-3} \text{ T}$
8. $\frac{2mv_0^2}{qE_0}$
9. (a) $\frac{mv_0}{2qB_0}$ (b) $\vec{v} = -v_0 \hat{i}$, $t = \frac{\pi m}{qB_0}$
10. (a) $y = \frac{2\pi^2 n^2 E_0}{\sigma B_0^2}$ (b) $\tan^{-1} \left(\frac{v_0 B_0}{2\pi n E} \right)$
11. $t = \frac{mu}{qE_0}$, u
12. $\sqrt{\frac{2qE_0 d}{m}}$
13. $B l \left(\frac{F_1 + F_2}{F_1 - F_2} \right)$
14. 0.3 T
15. $B_{\min} = \frac{mg(\sin \theta - \mu \cos \theta)}{IL(\cos \theta + \mu \sin \theta)}$; $B_{\max} = \frac{mg(\sin \theta + \mu \cos \theta)}{IL(\cos \theta - \mu \sin \theta)}$
16. $u = 1.98 \text{ ms}^{-1}$, $B = 3.4 \text{ mT}$
17. $B = 2 \sqrt{\frac{mE}{qd}}$
18. (a) stretched (b) $\frac{I^2 l^2 B^2}{2K}$
19. $B > \frac{2mu}{e\sqrt{b^2 - a^2}}$
20. 0
21. $N = 3 mg \sin \theta - Q.B. \sqrt{2gR \sin \theta}$
22. $\left(\frac{iaB_0}{2mg} \right)^2$
23. $\frac{v}{4d}$
24. $\frac{2mu}{qL} \sin \theta$
25. $2laB$
26. $I > \frac{mg}{2RB_0}$
27. $\frac{2\pi}{B} \sqrt{\frac{2m\Delta V}{e}}$
28. $\vec{\tau} = 6.7 (\hat{j} - \hat{i}) \text{ Nm}$
29. (a) $\vec{\tau} = \frac{I_0 L^2 B}{\sqrt{2}} (\hat{j} - \hat{i})$ (b) $\Delta \theta = \frac{3}{4} \frac{I_0 B}{M} (\Delta t)^2$
30. (a) $M = \frac{eh}{4\pi m}$ (b) $\frac{ehB}{8\pi m}$
31. (a) P to Q (b) $IbB_0 (3\hat{k} - 4\hat{i})$ (c) $\frac{mg}{6bB_0}$

Magnetic Effect of Current

“The whole, theory and experiment, seems as if it had leaped, full grown and full armed, from the brain of the ‘Newton of electricity’.”

— Maxwell (Praising Ampere’s Work)

1. INTRODUCTION

In the early part of 19th century, Oersted discovered that a current carrying wire produces a magnetic field in its surrounding that exerts force on a magnet. On receiving this news, Andre-Marie Ampere started his work to establish a quantitative law for this effect. At the same time, the team of Jean-Baptiste Biot and Felix Savart (both French) also worked on the same problem. They performed experiments and established a law which became the standard starting point for calculating the magnetic field due to steady currents. Ampere was successful in obtaining a formula for force between two current carrying wires. The results obtained by Ampere could be easily explained on the basis of the law given by Biot and Savart.

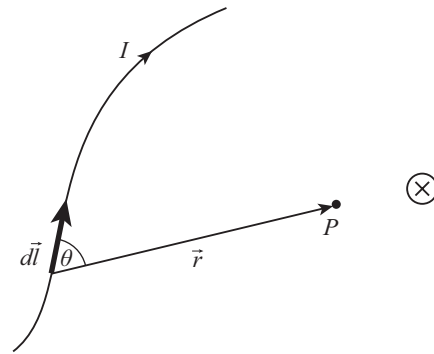
In this chapter we will basically study the Biot-Savart law and its applications. We will also study Ampere’s circuital law which is useful in certain situations for calculating magnetic field produced by a current.

2. BIOT-SAVART LAW

Biot and Savart performed very intricate experiments of measuring time period of oscillations of a small magnetic needle kept at a distance from a straight and long wire carrying a steady current. From the measurement of time period they were able to calculate the force experienced by each pole of the magnetic dipole. The result obtained was soon converted in the modern form involving the expression of magnetic field.

Consider a small segment of length dl in a current carrying wire having current I . P is a point at a distance r from this current element. The line joining the current element to the point P makes an angle θ with $d\vec{l}$ ($d\vec{l}$ is a vector in the direction of current). The Biot-Savart law tells us how to write the magnetic field ($d\vec{B}$) at point P due to

the current element ($I d\vec{l}$). The field at P due to the current element is given as



Biot-Savart law tells us how to write field at a Point P due to a current element $I d\vec{l}$.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3} \quad (1)$$

The magnitude of this field is

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2} \quad (2)$$

Direction of the field is along $d\vec{l} \times \vec{r}$, i.e. it is perpendicular to the plane containing the current element and the point P .

The constant μ_0 is known as *permeability of vacuum* and its value is

$$\mu_0 = 4\pi \times 10^{-7} \text{ T-m/A}$$

Note: The constants ϵ_0 and μ_0 have been chosen so articulately that the speed of light (c) in vacuum is given by

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \quad (3)$$

2.2 Magnetism and Electromagnetic Induction

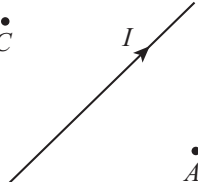
In practice, to find the magnetic field at a point due to a current carrying wire, we need to add the contributions made by each current element in the wire.

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \vec{r}}{r^3} \quad (4)$$

Understanding how to evaluate the direction of field and performing the above integral will be the key in learning to use the Biot-Savart law.

Example 1 Direction of field

A straight wire carries current I as shown. What is the direction of magnetic field produced by the wire at points A and C ?



Solution

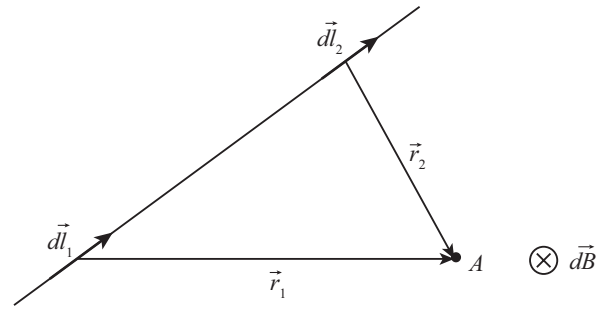
Concepts

The field at a point due to a current element $I d\vec{l}$ has the direction given by $I d\vec{l} \times \vec{r}$. Here, $\vec{r}$ is position vector of the point with respect to the current element.

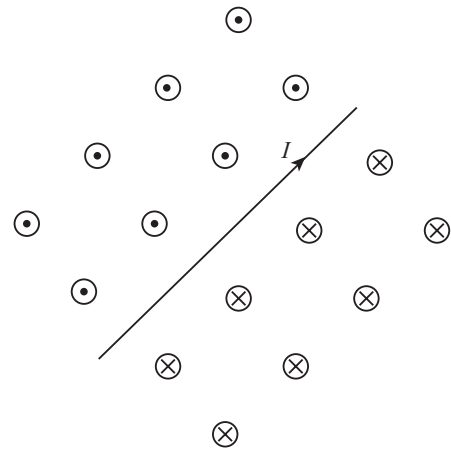
Consider a segment dl_1 in the wire. Field due to this segment at point A is along $d\vec{l}_1 \times \vec{r}_1$. In the figure shown, this direction is into the plane of the figure.

Now, consider another segment of length dl_2 . Due to this element the field at A is in the direction of $d\vec{l}_2 \times \vec{r}_2$. This is again directed into the plane of the figure. One can see that every element contributes field in the same direction. Therefore, the resultant field at A is directed into the plane of the figure.

Similarly, one can show that the field at the point C is directed out of the plane of the figure.



In fact, the magnetic field at all points to the right of the current is directed into ($\otimes$) the plane of the figure and at points to the left of the current the field is directed out ($\odot$) of the plane of the figure.

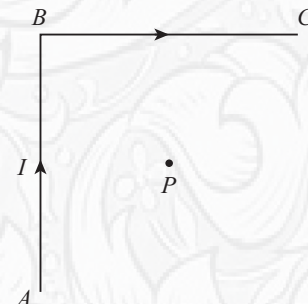


Direction of magnetic field due to current carrying wire.



YOUR TURN

Q.1 A current carrying wire is bent in L shape as shown. Find the direction of magnetic field at point P due to the current in the wire.

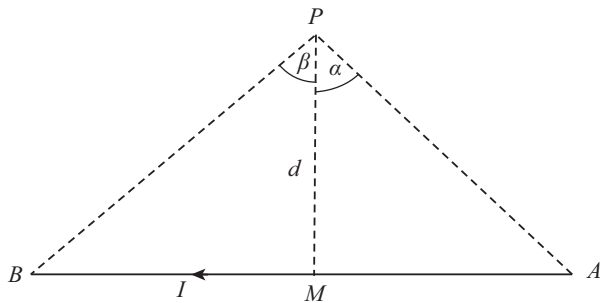


3. CALCULATION OF MAGNETIC FIELD USING BIOT-SAVART LAW

Now, we will take up few calculations to illustrate the use of the Biot-Savart law.

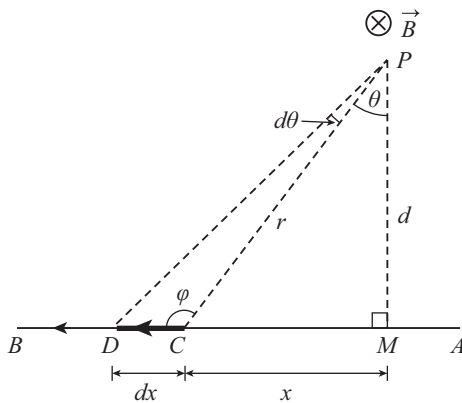
3.1 Field due to a straight current carrying wire

A straight wire AB carries current I . We wish to find the magnetic field at point P due to this wire. The point P is at a perpendicular distance d from the wire and the angles $\angle MPA$ and $\angle MPB$ are α and β respectively. d , α and β are three convenient numbers which fix the position of the point P and also tells us the length of the wire AB .



d , α and β fix the position of point P . They also tell us the length of the wire AB .

Consider a small segment (CD) of the wire having length dx , subtending an angle $d\theta$ at P (see figure). Let's write magnetic field at P due to this segment.



$$dB = \frac{\mu_0}{4\pi} \frac{(Idx)\sin\phi}{r^2} \otimes \quad (i)$$

Here, $CP = r$ is distance of the point P from the current element (CD). Angle between the current element and the vector $\vec{CP}$ (i.e., $\vec{r}$) is ϕ .

We will express the above expression in terms of θ . It means we will express dx , $\sin\phi$ and r in terms of constant d and angle θ .

$$\frac{x}{d} = \tan\theta \Rightarrow x = d \tan\theta$$

$$\text{or, } \frac{dx}{d\theta} = d \sec^2\theta \Rightarrow dx = d \sec^2\theta \cdot d\theta \quad (ii)$$

$$\text{From figure } r = d \sec\theta \quad (iii)$$

$$\text{Also, } \phi = 90^\circ + \theta$$

$$\therefore \sin\phi = \sin(90^\circ + \theta) = \cos\theta \quad (iv)$$

Substituting for dx , r and $\sin\phi$ into equation (i) we get

$$dB = \frac{\mu_0 I}{4\pi} \frac{(d \sec^2\theta d\theta)(\cos\theta)}{d^2 \sec^2\theta}$$

$$\text{or, } dB = \frac{\mu_0 I}{4\pi d} \cos\theta d\theta \quad (v)$$

The direction of field $\vec{dB}$ at P due to the element $\vec{CD}$ is in the direction of $\vec{CD} \times \vec{CP}$ (i.e. $I\vec{dx} \times \vec{r}$). This direction is into the plane of the figure. Every segment on the straight wire contributes field in the same direction. We need to add contributions made by all the segments to get the resultant field. This addition can be done by integrating the equation (v) with the angle θ changing from $-\alpha$ to β . [Note that we are measuring angle θ with PM as reference line and clockwise as positive.]

$$\therefore B = \int dB = \frac{\mu_0 I}{4\pi d} \int_{-\alpha}^{\beta} \cos\theta d\theta$$

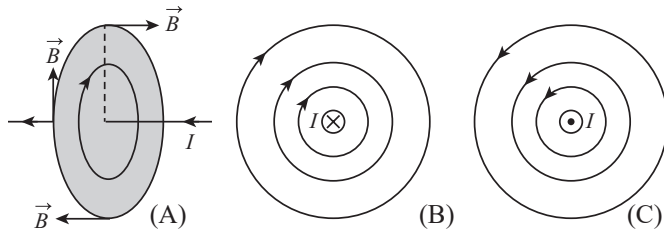
$$\text{or, } B = \frac{\mu_0 I}{4\pi d} [\sin\theta]_{-\alpha}^{\beta} = \frac{\mu_0 I}{4\pi d} [\sin\beta - \sin(-\alpha)]$$

$$\text{or, } B = \frac{\mu_0 I}{4\pi d} (\sin\alpha + \sin\beta) \quad (5)$$

Direction of the field

In the above figure the direction of the field at the point P is perpendicular to the plane containing the wire and the point. Consider the point P to be above the plane of the paper at same distance d from the wire (just think that line PM is rotated by 90° about the wire). Field at the point is still given by equation (5) but its direction is horizontal parallel to this page. If the point P is below the line AB in the figure shown above, the direction of the field will be out of the plane of the figure. With a little bit of imagination it is easy to see that the direction of field at any point near the wire is along the tangent drawn on a circle surrounding the wire. The magnetic field lines surrounding a straight current carrying wire are circles. The sense of the lines is given by the curling right hand fingers when the wire is held in the palm with the thumb pointing in the direction of the current.

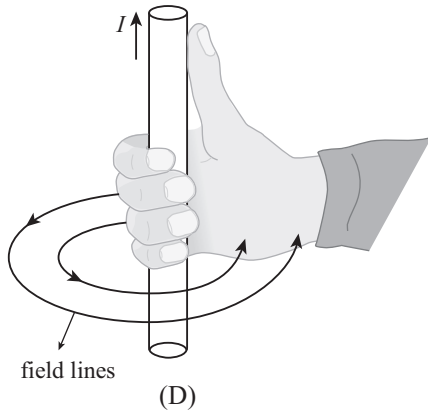
2.4 Magnetism and Electromagnetic Induction



Field line around a straight current carrying wire is circular. Tangent to the circle gives the direction of $\vec{B}$.

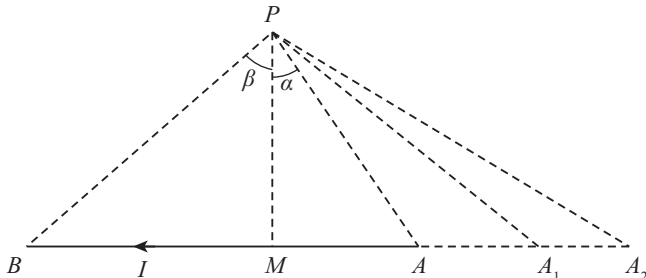
These two figures show the cross section of a current carrying wire having current perpendicular to the plane of the figure. The circles are the magnetic field lines.

You must convince yourself about the sense of circular field lines shown in the figures (B) and (C) above. Figure (D) will help you in doing so.



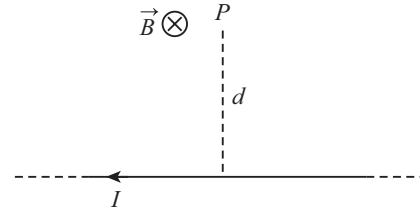
Special cases

- (i) AB is a straight wire carrying current I . We know that the magnetic field at point P is given by equation (5). If the wire is made longer to the right, the end A moves to $A_1, A_2, \dots$. One can see that angle α goes on increasing. When the wire is made too long, the end A is at a large distance from M and $\alpha \rightarrow \frac{\pi}{2}$. Similarly, when the wire is made longer to the left, the angle β also tends to $\frac{\pi}{2}$.



Angle α increases as point A moves to right.

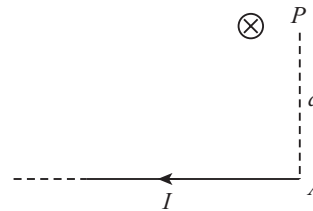
For a long current carrying wire, field at P can be obtained by putting $\alpha = \beta = \frac{\pi}{2}$ in equation (5).



$$B = \frac{\mu_0 I}{4\pi d} \left(\sin \frac{\pi}{2} + \sin \frac{\pi}{2} \right)$$

$$\text{or, } B = \frac{\mu_0 I}{2\pi d} \quad (6)$$

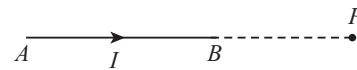
- (ii) Consider the case of a semi-infinite straight wire carrying current I . One end of the wire is at A and the other end is at a large distance from A . P is a point at a perpendicular distance d from A . Magnetic field can be obtained by putting $\alpha = 0$ and $\beta = \frac{\pi}{2}$ in equation (5).



$$B = \frac{\mu_0 I}{4\pi d} \quad (7)$$

The value of field is exactly half of the field given in equation 6. This is what anybody will guess.

- (iii) AB is a straight current carrying wire and P is a point on the line AB (see figure).



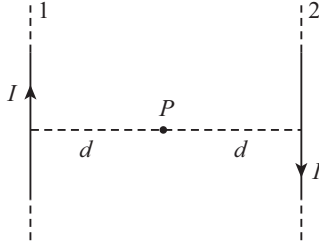
Magnetic field at P due to the wire is zero. This follows straight from the Biot-Savart law. If you take a small element on the wire, field at P due to it will be

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin 0^\circ}{r^2} = 0$$

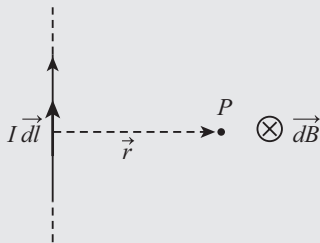
In the end, please take a note that the magnetic field lines around a straight current carrying wire of any length—infinite, semi infinite or finite—are circles with the wire along the x axis.

Example 2 Parallel wires in the plane of the paper

Two parallel wires, 1 and 2, carry equal current (I) in opposite direction. They are at separation $2d$. Find the magnetic field due to the wires at a point P (see figure). The wires and the point P lie in the plane of the paper.

**Solution****Concepts**

- (i) The only way one can commit a mistake is by assuming wrong directions for the fields due to the two wires.
- (ii) You can decide the direction of the field at point P due to wire 1 (or the wire 2) in two ways.
 - (a) Field lines (due to current in wire 1 only) are circles around the wire. Hold the wire in your right palm keeping thumb in the direction of current. The curling fingers are pointing into the plane of the figure at P . This is the direction of field at P due to wire 1.
 - (b) In a situation where the straight wire and the concerned point (P) are in the plane of the figure, you may find this second method more convenient. Just consider any small segment in the wire and find the direction of field at P due to this segment using the Biot-Savart law. This is also the direction of field at P due to the entire wire.



Field at P due to wire 1 is

$$B_1 = \frac{\mu_0 I}{2\pi d} \otimes$$

Field at P due to wire 2 is

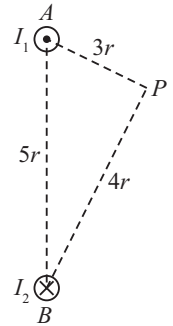
$$B_2 = \frac{\mu_0 I}{2\pi d} \otimes$$

Resultant field at the point P is

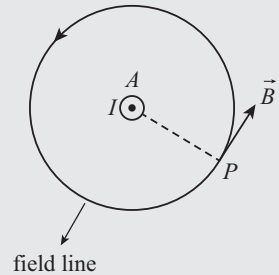
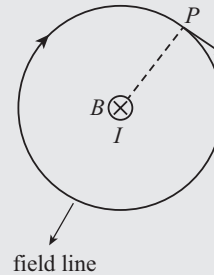
$$B = B_1 + B_2 = \frac{\mu_0 I}{\pi d} \otimes$$

Example 3 Parallel wires perpendicular to the plane of the figure

The small circles at A and B are cross sections of two long straight wires perpendicular to the plane of the figure. The wire at A carries current $I_1 = I$ directed out of the plane of the figure (indicated by $\odot$) and the wire at B carries current $I_2 = 2I$ directed into the plane of the figure. Find the strength of magnetic field at the point P .

**Solution****Concepts**

- (i) Again, getting correct direction of the field is the core concept. The following figures are helpful when the current carrying wire is perpendicular to the plane of the figure.

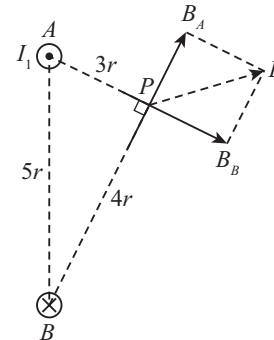


Field at a point P is directed along the tangent (i.e., it is along a direction that is perpendicular to the radius BP or AP).

- (ii) A careful look at the diagram reveals that $\angle APB = 90^\circ$

Field at P , due to the current carrying wire at A , is directed perpendicular to AP . Its is directed along $\overrightarrow{BP}$. The strength of the field is

$$B_A = \frac{\mu_0 I_2}{2\pi(4r)} = \frac{\mu_0 I}{4\pi r}$$



Similarly, field due to the wire at B is perpendicular to BP . It is along $\overrightarrow{AP}$.

2.6 Magnetism and Electromagnetic Induction

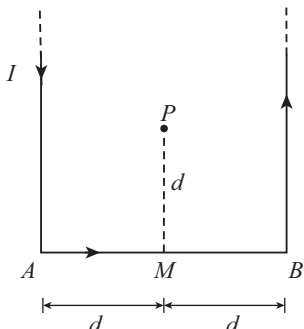
$$B_B = \frac{\mu_0 I_2}{2\pi(4r)} = \frac{\mu_0(2I)}{8\pi r} = \frac{\mu_0 I}{4\pi r}$$

Resultant field at P is vector sum of B_A and B_B .

$$B = \sqrt{B_A^2 + B_B^2} = \frac{\mu_0 I}{2\pi r} \sqrt{\frac{1}{9} + \frac{1}{4}} = \sqrt{\frac{13}{36}} \frac{\mu_0 I}{2\pi r}$$

Example 4 A U shaped wire

A long wire, having current I , is bent into the shape shown in the figure. Find the magnetic field at point P . The lengths $AM = MB = PM = d$.

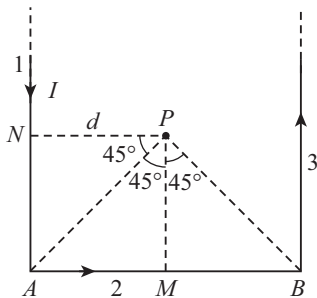


Solution

Concepts

- We will find the field due to each of the three segments separately and add them.
- As learnt in example 2, you can decide the direction of field at P due to each segment.

Field due to segment 1 can be obtained by putting $\alpha = 45^\circ$ and $\beta = 90^\circ$ in equation (5).



$$B_1 = \frac{\mu_0 I}{4\pi d} (\sin 45^\circ + \sin 90^\circ) \odot$$

$$= \frac{\mu_0 I}{4\pi d} \left(\frac{1}{\sqrt{2}} + 1 \right) \odot$$

[One end of the wire is at A , hence $\alpha = \angle APN = 45^\circ$. The other end goes to infinity. Therefore, $\beta = 90^\circ$]

Field due to segment 3 is same as B_1 . The direction of the field is also same.

$$B_3 = \frac{\mu_0 I}{4\pi d} \left(\frac{1}{\sqrt{2}} + 1 \right) \odot$$

Field due to segment 2 (i.e., AB) is

$$B_2 = \frac{\mu_0 I}{4\pi d} (\sin 45^\circ + \sin 45^\circ) = \frac{\mu_0 I}{4\pi d} \frac{2}{\sqrt{2}} \odot$$

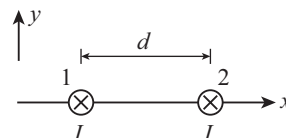
The direction of the field due to each segment is out of the plane of this figure. The three fields will add.

$$\therefore B = B_1 + B_2 + B_3$$

$$= \frac{\mu_0 I}{4\pi d} \left(\frac{4}{\sqrt{2}} + 2 \right) \odot$$

$$= \frac{\mu_0 I}{2\pi d} (\sqrt{2} + 1) \odot$$

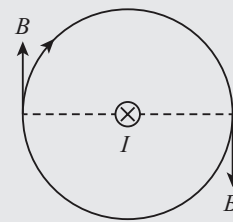
Example 5 Two long thin parallel wires carry same current I . The wires are perpendicular to the plane of this paper. The figure shows the cross sections of the wire. Plot the variation of magnetic field on the line marked as x axis.



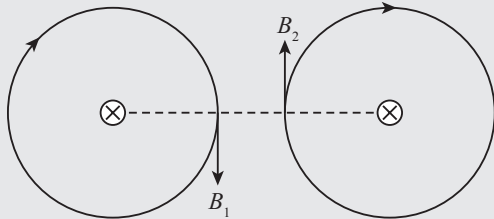
Solution

Concepts

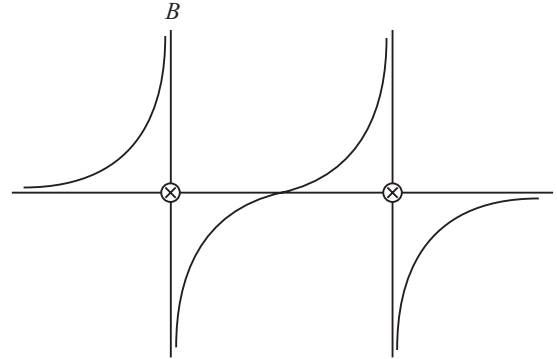
- Field at a distance ' d ' from a long wire is $B = \frac{\mu_0 I}{2\pi d}$. When $d \rightarrow 0$, the field $B \rightarrow \infty$. It means the field at points very close to the wire is very high.
- At points lying to the left of wire 1, the fields due to both the wires is directed in positive y direction. Both the fields add. At a point at a distance much larger than d , the field will resemble the field produced due to a single wire having current $2I$.
- To the right of wire 2, field due to both the wires are in negative y direction. Again the field decreases with increasing distance.



- (iv) In the region between the two wires the fields are in opposite directions. At the midpoint the two fields cancel out. To the left of the midpoint, field due to 1 is higher. Resultant is in negative y direction. To the right of the midpoint the field due to 2 is higher. Resultant is in positive y direction.

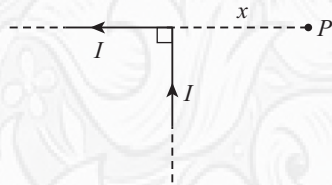


We will take the field to be positive when it is in positive y direction. On the x axis, the field is either in positive y direction or in the negative y direction.



YOUR TURN

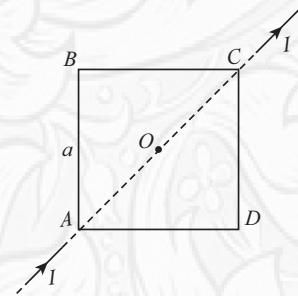
Q.2 A long wire, carrying current I , is bent in L shape. Find the magnetic field at point P .



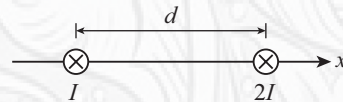
Q.3 A square current loop of side length a has current I . Find the magnetic field at the centre of the square.



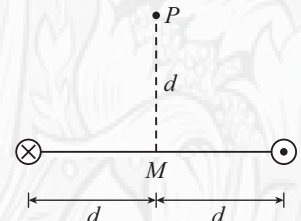
Q.4 A square loop of side length a is made of a uniform wire. A current I enters the loop at point A and leaves at C . The wire carrying the current to A and that taking the current out of C are along the diagonal. Find the magnetic field at the centre (O) of the square.



Q.5 The figure shows two long straight current carrying wires having current I and $2I$. The wires are normal to the plane of the figure and are at separation d . Find a point on the x -axis where the magnetic field is zero.



Q.6 The figure shows two parallel wires having current I in opposite direction. The wires are perpendicular to the plane of the figure and separation between them is $2d$. Find the magnetic field produced at point P due to the two wires. Distance $MP = d$.



3.2 Magnetic field at the centre of a current carrying circular arc

A wire ACB is bent in the shape of a circular arc of radius R . It subtends an angle θ at the centre of the circle. The wire carries a current I . We wish to write the magnetic field produced at the centre (O) due to the circular arc.

Consider a small segment of length dl at point P on the circular arc. The field at O due to this element is

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin 90^\circ}{R^2} = \frac{\mu_0 I}{4\pi R^2} dl$$

The direction of the field is along $\vec{dl} \times \vec{PO}$. In the figure shown, this direction is into the plane of the figure.

Every segment of the arc contributes the field in same direction. Therefore, the resultant field at O is

$$B = \int dB = \frac{\mu_0 I}{4\pi R^2} \int_A^B dl = \frac{\mu_0 I}{4\pi R^2} L \otimes$$

Here, L is length of the arc.

$$L = R\theta$$

$$\therefore B = \frac{\mu_0 I}{4\pi R} \theta \quad (8)$$

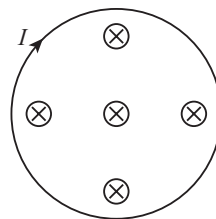
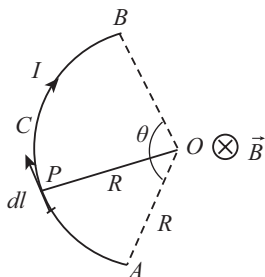
If we have a complete circle carrying a current I , field at its centre can be obtained by putting $\theta = 2\pi$ in equation (8)

$$B = \frac{\mu_0 I}{2R} \quad (9)$$

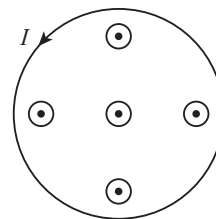
For a semicircle, the fields at the centre of curvature is obtained by substituting $\theta = \pi$ in equation (8)

$$B = \frac{\mu_0 I}{4R} \quad (10)$$

One can easily get the direction of magnetic field due to a circular loop, at any inside point, by using the right hand rule. Curl your right hand fingers in the sense of the current and the thumb gives the direction of magnetic field at all inside points. In other words, *the field at an inside point is in the direction of magnetic dipole moment of the loop.*



At all inside points the field is directed into the plane of the figure



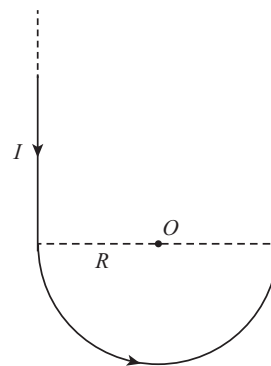
At all inside points the field is directed out of the plane of the figure

If you have a circular coil of N turns carrying a current I , Then the magnetic field at the centre is given by N times the field expressed by the equation (9).

$$B = \frac{\mu_0 NI}{2R} \quad (11)$$

Example 6 Again a U shaped wire

A long wire, carrying a current I , is bent in the shape shown in the figure. It has two straight segments at a separation $2R$ and a semicircular part joining them. Find the magnetic field at the centre (O) of the circular part.



Solution

Concepts

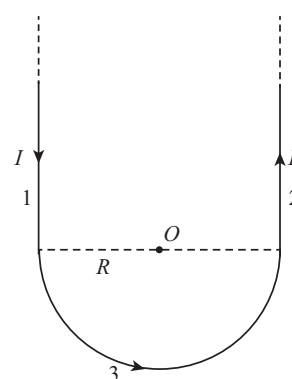
Divide the wire into three segments- two straight and one semicircular and write field due to each of them. Correctly mark the directions of the three fields and add them.

Straight segment 1 is a semi infinite wire. Field due to this wire at O is

$$B_1 = \frac{\mu_0 I}{4\pi R} \odot$$

Field due to segment 2 is also same and has the same direction.

$$B_2 = \frac{\mu_0 I}{4\pi R} \odot$$



Field due to semicircle is

$$B_3 = \frac{\mu_0 I}{4R} \odot$$

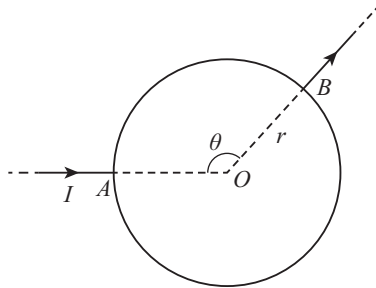
The direction of B_3 is obtained by curling the right hand fingers in the sense of the current. The thumb gives the direction of the field.

Resultant field at O is

$$\begin{aligned} B &= B_1 + B_2 + B_3 = \frac{\mu_0 I}{4R} \left(\frac{1}{\pi} + \frac{1}{\pi} + 1 \right) \\ &= \frac{\mu_0 I}{4R} \left(\frac{2}{\pi} + 1 \right) \odot \end{aligned}$$

Example 7 Current in parallel arcs

current I enters a uniform circular conducting loop at point A . The current divides into two parallel paths and leaves the circle at B . The straight wires carrying the current at A and B are along the radius of the circle. Find the magnetic field at the centre of the circle. Radius of the circle is r .

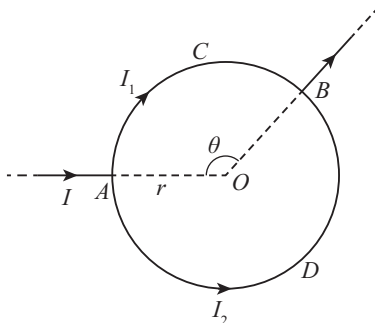


Solution

Concepts

- (i) The resistance of two circular arcs will be proportional to their respective lengths.
- (ii) Current will divide into two paths in inverse ratio of resistances.
- (iii) Field due to the two circular arcs will be oppositely directed at the centre (O).
- (iv) Field, at O , is zero due to the straight wires.

Let the length of arc ACB be l_1 and the length of arc ADB be l_2 . The resistances of the two area are in ratio



$$\frac{R_{ACB}}{R_{ADB}} = \frac{l_1}{l_2} = \frac{r\theta}{r(2\pi - \theta)} = \frac{\theta}{2\pi - \theta}$$

$\therefore$ Currents I_1 and I_2 are in the ratio

$$\frac{I_1}{I_2} = \frac{R_{ADB}}{R_{ACB}} = \frac{2\pi - \theta}{\theta} \quad (i)$$

Magnetic field at O due to current I_1 in the arc AB is

$$B_1 = \frac{\mu_0 I_1}{4\pi r} \theta \otimes$$

Field at O due to current I_2 in the arc ADB is

$$B_2 = \frac{\mu_0 I_2}{4\pi r} (2\pi - \theta) \odot$$

From (i) $I_1 \theta = I_2 (2\pi - \theta)$

$\therefore B_1 = B_2$

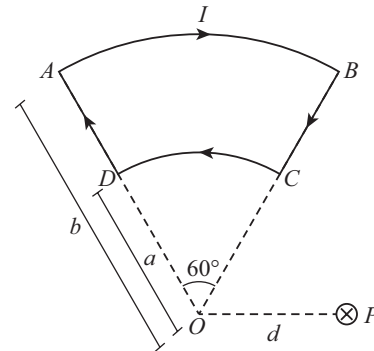
Therefore, resultant field at O is zero. This result is independent of θ !

Note: The result could have been guessed. The currents I_1 and I_2 are in ratio $\frac{l_2}{l_1}$. The field due to each arc is proportional to the current and arc length. It means $B \propto Il$. This implies

$$\frac{B_1}{B_2} = \frac{l_1 I_1}{l_2 I_2} = \frac{l_1 l_2}{l_2 l_1} = 1.$$

Example 8 Circular arcs and radial wires

Figure shows a current loop having two concentric circular arcs of radii a and b joined by two radial arms BC and DA . The loop carries a current I . A long wire carrying current I is perpendicular to the plane of the loop and is located at a distance $OP = d$ from the centre of curvature of the circular arcs. Find the strength of magnetic field at O .



Solution

Concepts

- (i) The radial wires BC and DA do not produce any field at O .
- (ii) The circular arcs AB and CD produce field in opposite directions normal to the plane of the figure.
- (iii) Field due to the straight wire at P is in the plane of the figure perpendicular to PO .

2.10 Magnetism and Electromagnetic Induction

Field at O due to arc AB is $\frac{1}{6}$ of the field due to a circular current loop of radius b . This is a much easier way to interpret equation (8).

$$B_1 = \frac{1}{6} \frac{\mu_0 I}{2b} = \frac{\mu_0 I}{12b} \otimes$$

Similarly, field due to the arc CD is

$$B_2 = \frac{\mu_0 I}{12a} \odot$$

Resultant field due to the loop is

$$B_L = \frac{\mu_0 I}{12} \left(\frac{1}{a} - \frac{1}{b} \right) \odot$$

Field due to the straight wire at P will be in the plane of the figure, perpendicular to the line PO .

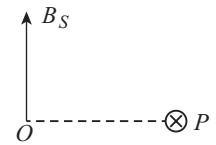
Its magnitude is

$$B_S = \frac{\mu_0 I}{2\pi d}$$

Resultant field is vector sum of two perpendicular fields B_L and B_S .

$$\therefore B = \sqrt{B_L^2 + B_S^2}$$

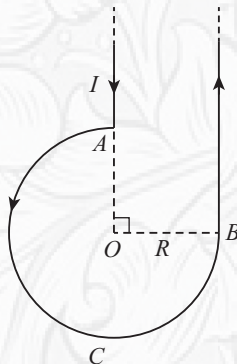
$$= \frac{\mu_0 I}{2} \sqrt{\frac{1}{36} \left(\frac{1}{a} - \frac{1}{b} \right)^2 + \frac{1}{\pi^2 d^2}}$$



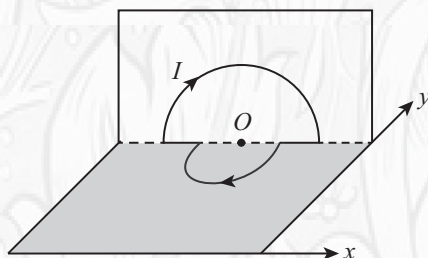
YOUR TURN

Q.7 An insulated straight wire of length L carries current I . It is bent to form a coil of n turns. Find the magnetic field at the centre of the coil.

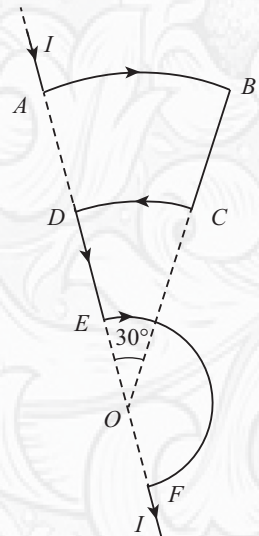
Q.8 A long wire carrying current I is bent into the shape shown in the figure. ACB is $\frac{3}{4}$ of a circle of radius R and the remaining two arms of the wire are parallel. Find the magnetic field at the centre (O) of the circle.



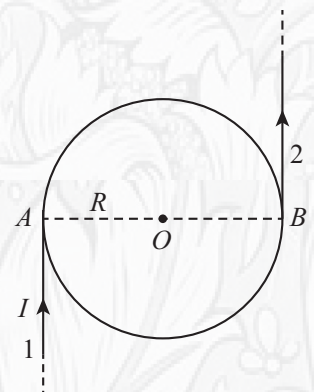
Q.9 A current loop consists of two semicircular loops of radii 0.05m and 0.1m . The two semicircular parts lie in xy plane and yz plane respectively. The semicircular parts are joined by two radial wire pieces. A current $I = 5\text{A}$ is established in the loop in the direction shown. Find the magnetic field at the common centre of curvature of the two loops (at O).



Q.10 A long wire carrying current I has been bent into the shape shown in the figure. AB and CD are circular arcs centred at O , having radii $3r$ and $2r$ respectively. EF is a semicircle of radius r centred at O . All other segments are straight and radial pointing towards O . Find the magnetic field at the point O .



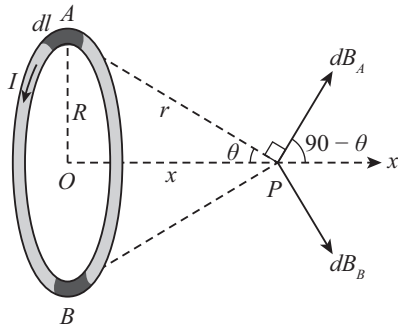
Q.11 A long straight wire 1 brings a current I to A where it enters into a uniform circular wire frame of radius R . The current is taken out at B by the straight long conductor 2. Conductors 1 and 2 are tangential to the circle and parallel to one another. Find the magnetic field intensity at the centre O .



3.3 Magnetic field on the axis of a current carrying circular loop

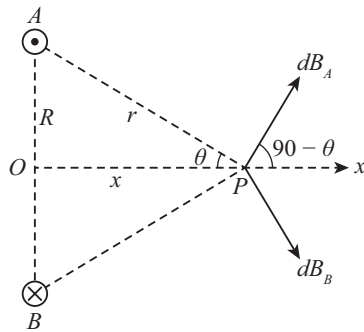
Consider a circular loop of radius R carrying a current I . O is the centre of the loop and P is a point on its central axis. Distance OP is x . We wish to calculate the magnetic field due to the circular current loop at point P .

Consider a small segment (A) of the loop having length dl . Using the Biot-Savart law we can write the magnetic field at the point P due to this segment.



$$dB_A = \frac{\mu_0 I}{4\pi} \frac{dl \sin 90^\circ}{r^2} \quad \text{where } r = AP = \sqrt{R^2 + x^2}$$

You must visualise clearly that dl is perpendicular to the plane OAP and angle between dl and AP is 90° . Direction of $\vec{dB}_A$ is along $\vec{dl} \times \vec{AP}$. This direction is perpendicular to $\vec{dl}$. Hence it is in the plane OAP . This direction is also perpendicular to $\vec{AP}$. Therefore, $\vec{dB}_A$ is directed as shown. Curling your right finger from $\vec{dl}$ to $\vec{AP}$ and looking at the direction of the thumb will help you in correctly fixing the direction of $\vec{dB}_A$. Still confused? well try it this way. AOB is the diameter of the ring and P is the point on the axis. The plane of this figure is the plane formed by line AOB and point P . Half the ring is below this paper and half of it is above this paper. The small current element at A is normal to the paper and the current in it is directed towards you. An identical element at B has current going into the plane of the paper. Field due to element A is perpendicular to the element, thereby is in the plane of the figure. It is also normal to AP . Identical element at B creates field dB_B in a direction perpendicular to BP . These two fields are equal in magnitude and their resultant will be along the axis. We can always choose such identical current elements at diametrically opposite ends of the ring which will together



create field along the axis. Therefore, resultant field at the point P is along the axis, directed along OP .

Resultant field can be obtained by adding, for all elements, the components of dB along the axis

$$\begin{aligned} \therefore B &= \int \frac{\mu_0 I}{4\pi} \frac{dl}{r^2} \sin \theta \\ &= \frac{\mu_0 I}{4\pi r^2} \sin \theta \int dl \quad [r \text{ and } \sin \theta \text{ are same for all elements}] \\ &= \frac{\mu_0 I}{4\pi r^2} \frac{R}{r} 2\pi R \quad [\sin \theta = \frac{R}{r} \text{ and } \int dl = 2\pi R] \\ &= \frac{\mu_0}{4\pi} \frac{2\pi R^2 I}{(R^2 + x^2)^{3/2}} \quad [r = \sqrt{R^2 + x^2}] \end{aligned}$$

If there are N turns in the circular current loop, the field is equal to N times the above value.

$$B = \frac{\mu_0}{4\pi} \frac{2 \cdot N \pi R^2 I}{(R^2 + x^2)^{3/2}} = \frac{\mu_0}{2} \frac{NIR^2}{(R^2 + x^2)^{3/2}} \quad (12)$$

Here $N \cdot \pi R^2 \cdot I = M$ (Magnetic dipole moment of the coil)

$$\therefore B = \frac{\mu_0}{4\pi} \frac{2M}{(R^2 + x^2)^{3/2}} \quad (13)$$

In vector form, we can write

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{2\vec{M}}{(R^2 + x^2)^{3/2}} \quad (14)$$

The direction of the field at any point on the axis is in the direction of $\vec{M}$.

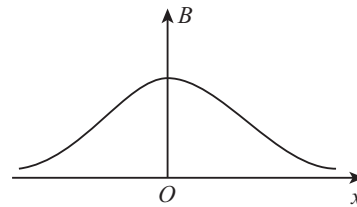
Following points are worth noting in view of the above result.

- (i) The magnetic field at the centre of the ring can be obtained by putting $x = 0$ in the above equation.

$$B = \frac{\mu_0}{4\pi} \frac{2N\pi R^2 I}{R^3} = \frac{\mu_0 NI}{2R}$$

This is same as equation (11).

- (ii) Field is maximum at the centre and decreases as we move away along the axis. Direction of the field is same everywhere on the axis.



2.12 Magnetism and Electromagnetic Induction

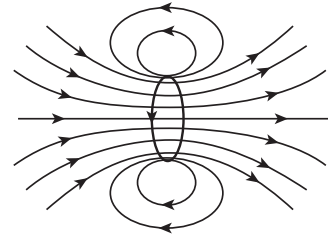
(iii) For $x \gg R$, the equation (13) can be approximated as

$$B \simeq \frac{\mu_0}{4\pi} \frac{2M}{x^3}$$

This is same as the expression of the magnetic field due to bar magnet at a distant point on its axis. *Once again we have an evidence to believe that a current loop is indeed a magnetic dipole.*

(iv) The magnetic field lines representing the field produced by a current loop are somewhat like those

shown in the figure. Note the similarity between these lines and the lines due to a bar magnet.



Field lines due to a current loops



YOUR TURN

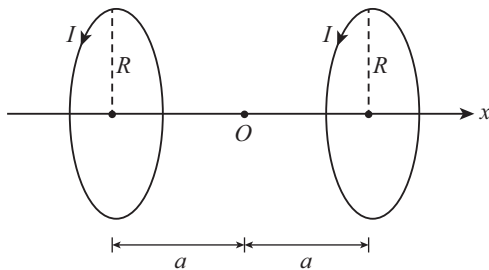
Q.12 The magnetic field due to a current carrying circular loop of radius 3 cm at its centre is 5×10^{-5} T. Find the

magnetic field due to this loop at a point on its axis located at a distance of 4cm from the centre.

3.4 Helmholtz Coils

Consider two identical circular coils of radius R having N turns. The coils are held parallel with their common axis lying along x axis. The centres of the coils are located at $x = -a$ and $x = +a$. Both the coils are given same current I in the same direction.

Consider a point P having x -coordinate equal to x on the axis. The field at this point can be obtained by adding the fields due to both the coils [using equation (12)]. Note that the distance of point P from the centre of one coil is $(x - a)$ and its distance from the other coil is $(x + a)$.



$$B_x = \frac{\mu_0 N I R^2}{2} \left[\frac{1}{[(x-a)^2 + R^2]^{3/2}} + \frac{1}{[(x+a)^2 + R^2]^{3/2}} \right] \quad (i)$$

At $x = 0$ (i.e., at a point exactly midway between the coils), the field is

$$B_0 = \frac{\mu_0 N I R^2}{(a^2 + R^2)^{3/2}}$$

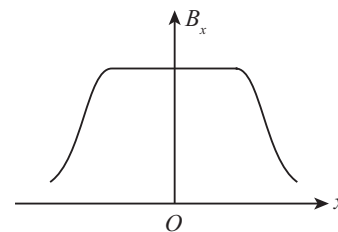
When the separation between the coils is equal to their radius (i.e., $2a = R$), the field at the midpoint becomes

$$B_0 = \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 N I}{R} \quad (ii)$$

If we differentiate equation (i) to get $\frac{dB_x}{dx}$, it is observed that $\frac{dB_x}{dx} = 0$ at $x = 0$.

Also, it can be shown that for $2a = R$, $\left(\frac{d^2 B_x}{dx^2}\right)_{x=0} = 0$.

This analysis shows that, for separation between the two coils equal to their radius, the strength of magnetic field is fairly constant around $x = 0$. A rough plot of B_x is as shown. Such pair of coils are known as Helmholtz coils.

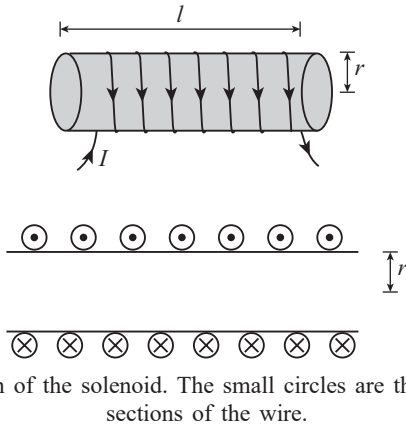


Field around $x = 0$ is constant.

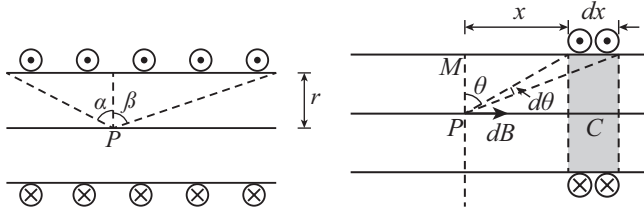
3.5 Solenoid

An insulated wire wound in a closely spaced spiral over a cylindrical core is known as a solenoid. The figure shows a solenoid and its section. The small circles with $\odot$ and $\otimes$ are cross sections of the wire. The $\odot$ represents current

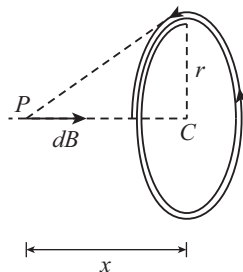
coming out of the paper towards you and the $\otimes$ represents current going into the paper.



We will write the magnetic field at a point P on the axis of the solenoid when a current I is given to the solenoid. For calculating this we will consider the solenoid as combination of number of thin circular coils. The location of point P (on the axis) and the length of the solenoid gets fixed if we are given the two angles α and β . Let the number of turns on unit length of the solenoid be uniform and equal to n .



Consider a ring of width dx on the solenoid. This ring has a total of ndx turns and field due to this ring at point P can be written using equation (12).



A circular coil having width dx .

$$dB = \frac{\mu_0}{2} \frac{(ndx)Ir^2}{(r^2 + x^2)^{3/2}} \quad (i)$$

Direction of the field, in the diagram shown, is along the axis towards right.

$$\text{From the figure, } x = r \tan \theta \quad (ii)$$

$$\text{or, } dx = r \sec^2 \theta d\theta \quad (iii)$$

Substituting these values in (i)

$$dB = \frac{\mu_0}{2} \frac{(nr \sec^2 \theta d\theta)I \cdot r^2}{(r^2 + r^2 \tan^2 \theta)^{3/2}} = \frac{\mu_0 n I}{2} \cos \theta d\theta$$

Field due to all circular loops are in same direction. Hence the resultant field at P is

$$B = \frac{\mu_0 n I}{2} \int_{-\alpha}^{\beta} \cos \theta d\theta$$

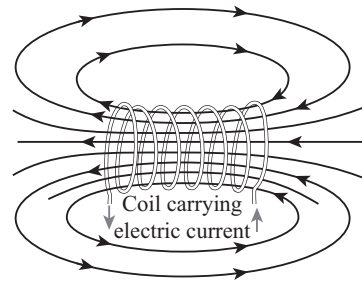
$$\text{or, } B = \frac{\mu_0 n I}{2} (\sin \beta + \sin \alpha) \quad (15)$$

When the length of the solenoid is large compared to its radius ($l \gg r$) and the point P is well within the solenoid, both α and β are close to 90° .

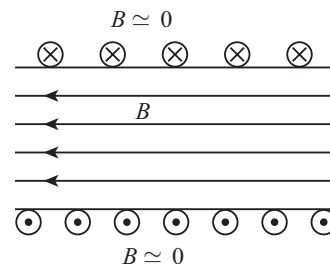
$$\therefore B = \frac{\mu_0 n I}{2} (\sin 90^\circ + \sin 90^\circ)$$

$$\text{or, } B = \mu_0 n I \quad (16)$$

When the radius of the solenoid is small (compared to l) and number of turns per unit length (n) is large, the magnetic field inside the solenoid tends to get uniform. Its magnitude is given by equation (16). Such a solenoid is known as an *ideal solenoid*. The magnetic field outside an ideal solenoid is very small. We will treat it to be zero.



A real solenoid with large number of turns and small radius. Field lines resemble that of a bar magnet



Field outside an ideal solenoid is close to zero. Field is uniform inside the solenoid.

2.14 Magnetism and Electromagnetic Induction

Example 9 A solenoid has length $l = 2.0\text{ m}$ and its radius is $r = 0.05\text{ m}$. There are $N = 2000$ turns of coil uniformly wound over it. Find the magnetic field at

- a point well within the solenoid
- a point at the edge of the solenoid.

Current in the coil is $I = 5\text{ A}$.

Solution

Concepts

- The solenoid can be treated as ideal as $l \gg r$.
- Field at an inside point is $B = \mu_0 n I$
- At the edge, it will not be appropriate to write the field as $\mu_0 n I$. It will be more appropriate to use equation (15) with $\alpha = 0$ and $\beta \approx 90^\circ$.

$$n = \frac{N}{l} = \frac{2000}{2} = 1000 \text{ turns/m.}$$

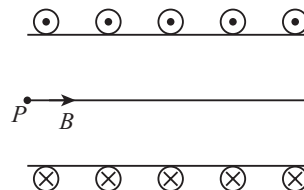
$$(a) B = \mu_0 n I = 4\pi \times 10^{-7} \times 1000 \times 5 = 2\pi \text{ mT} = 6.28 \text{ mT}$$

- At point P on the edge, the field is

$$B = \frac{\mu_0 n I}{2} (\sin 0^\circ + \sin 90^\circ)$$

$$= \frac{\mu_0 n I}{2}$$

$$= 3.14 \text{ mT.}$$

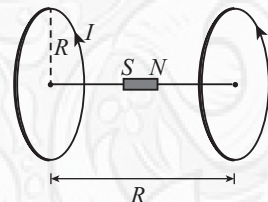


YOUR TURN

Q.13 An ideal solenoid is constructed using a wire of cross sectional diameter $d = 1\text{ mm}$. The turns are close and touch one another. Find the field inside the solenoid when current is $I = 1\text{ A}$.

Q.14 A pair of Helmholtz coils have radius R and are placed at a separation R . Both the coils carry current I . A small magnetic needle is kept on the axis midway between

the two coils. Find the force on the needle if its dipole moment is M .



In Short

- Biot-Savart law gives magnetic field due to a small current element ($I d\vec{l}$) at a point having position vector $\vec{r}$ with respect to the current element.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{r^3}$$

$$|d\vec{B}| = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

θ is angle between $d\vec{l}$ and $\vec{r}$.

- Magnetic field at a distance d from a straight long wire carrying current I is $B = \frac{\mu_0 I}{2\pi d}$.
- Magnetic field at a point at a distance d from one end of a semi infinite wire is $B = \frac{\mu_0 I}{4\pi d}$.

- Magnetic field lines are circles around a straight current carrying wire. The sense of the circle is in the direction of curling right hand fingers, around the wire, when the thumb points in the direction of the current.

- Magnetic field at a point on the axis of a circular current loop is $\vec{B} = \frac{\mu_0}{4\pi} \frac{2\vec{M}}{(R^2 + x^2)^{3/2}}$.

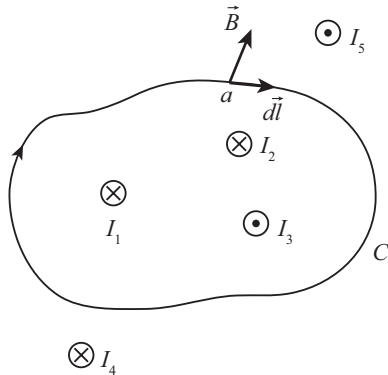
Here, $\vec{M}$ is magnetic dipole moment of the loop, R is its radius and x is distance of the concerned point from the centre of the loop.

- Field at the centre of a circular current loop of N turns is $B = \frac{\mu_0 N I}{2\pi}$. Direction of $\vec{B}$ is along the magnetic dipole moment of the loop.

- (vii) A circular arc of radius R subtending an angle θ at its centre of curvature (O) and having a current I , creates a magnetic field $B = \left(\frac{\mu_0 I}{2R}\right)\left(\frac{\theta}{2\pi}\right)$ at O .
- (viii) A pair of identical parallel coils mounted on the same axis at separation equal to their radius is known as Helmholtz coil. The magnetic field midway between them, is fairly constant.
- (ix) An ideal solenoid has large number of equally spaced turns and small radius. The field created by such a solenoid is uniform and parallel to axis inside the solenoid. Strength of the field is $B = \mu_0 nI$. Here n is number of turns per unit length. Field outside an ideal solenoid is nearly zero.

4. AMPERE'S CIRCUITAL LAW

Ampere's law gives us another method to calculate magnetic field due to a given current distribution. However, its usefulness is limited to certain symmetrical situations only. This law can be established using the Biot-Savart law and, in principle, we can establish the Biot-Savart law using the Ampere's law. Here, we will just try to understand the statement of the Ampere's law and its applications for calculating magnetic field.



The figure shows cross sections of many current carrying wires piercing through the plane of the paper. They have currents I_1, I_2, I_3, I_4 and I_5 . A closed curve (C) encloses the currents I_1, I_2 and I_3 and the remaining two currents are outside the area bound by the curve.

Assign a sense to the closed curve C , say clockwise. When we curl our right hand fingers in the sense of the curve C , the thumb points downward into the plane of the curve. Current like I_1 , which is pointing downward is taken as positive and a current like I_3 , which is pointing upward is taken to be negative.

Let the magnetic field (resultant of the fields contributed by all the currents) at a point 'a' on the curve be $\vec{B}$. Let $d\vec{l}$ be a small length element on the curve at 'a'. We take the scalar product $\vec{B} \cdot d\vec{l}$. We move around the curve C and for every small element $d\vec{l}$ we find the dot products $\vec{B} \cdot d\vec{l}$. Add all the dot products. The resultant is known as *line integral* or *circulation of $\vec{B}$ along the curve*. This is appropriately represented as

$$\oint \vec{B} \cdot d\vec{l}$$

The Ampere's law states that *the line integral of magnetic field $\vec{B}$ along a closed curve is equal to μ_0 times the total current piercing through the area bound by the curve*.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i \quad (17)$$

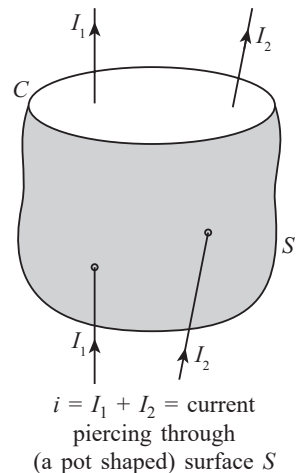
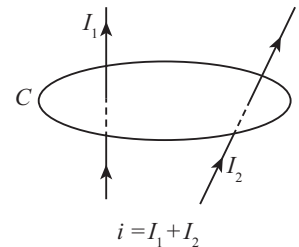
Here, i is the total current piercing through the area bound by the curve C . In the figure shown,

$$i = I_1 + I_2 - I_3$$

Any current outside the curve C is not included in i , but the magnetic field $\vec{B}$ is the resultant field due to all the currents existing anywhere.

Ampere's law, in above stated form, is true only when there is no time changing electric field at the area bound by the curve C . We will study about this aspect in a later chapter. For the time being, Just ignore this point.

Before we proceed to the application of this law, here is one more clarification. The figure shows two current carrying wires piercing the plane of a closed curve (C). One of the wires is normal to the plane of the loop and the other is at some angle. The line integral of magnetic field around the closed loop is given by equation (17). Value of i is $I_1 + I_2$. It makes no difference whether the current carrying wire is normal to the plane of the loop or not. To remove this confusion we may say that the current i in equation (17) is the net current piercing through any surface S having the curve C as its boundary. Imagine a three dimensional pot shaped surface with the curve C as its mouth. Current i is the net current that pierces through the surface S . In the applications to follow, we will take the plane of the curve C to be our surface and any current piercing through it – normal or inclined – will be considered for writing i in equation (17).



5. APPLICATIONS OF AMPERE'S LAW

As far as calculation of magnetic field is concerned, the usefulness of the Ampere's law is limited. It can be used in certain symmetrical situations only. Below we present some prominent examples to highlight the uses of this law. Existence of symmetry and a proper choice of closed curve (we will call it Amperian loop) are essential for successful use of this law.

5.1 Field due to a long straight current carrying wire

Consider a long straight wire carrying current I . We wish to find the magnetic field at a point P at a distance d from the wire. We have already derived the result in equation (6) using the Biot-Savart law. Now, we want to establish the same result using the Ampere's law.

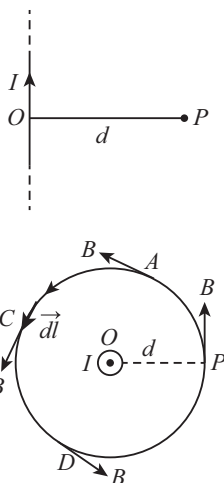
The second figure on the right shows the cross section of the wire in the plane of the figure and the current is coming out of the plane. Draw a circle of radius d , passing through the point P and centred at O . This circle is our chosen closed curve (the Amperian loop) for application of the Ampere's law.

Due to symmetry, we can assume that the strength of the magnetic field at all points on the circle is same, say B . There is no reason for the magnitude of the field to be different at point P , A , C , or D . The direction of magnetic field at all the points on the circle must be alike. Well, you can say that field at all points can be radial. Such a choice of direction supports the symmetry. But the magnetic field lines cannot be radial – they are always closed loops. Therefore, we are left with one choice only – the field must be tangential to the circle everywhere. The circle shown is itself a field line. Arguing on the basis of symmetry we conclude that the field at every point on the Amperian loop is tangential and has same magnitude (B).

We will move anticlockwise on the circle for calculating the line integral of $\vec{B}$. A small segment $d\vec{l}$ of the circular path is always along $\vec{B}$. Angle between $\vec{B}$ and $d\vec{l}$ is 0 everywhere.

$$\therefore \oint \vec{B} \cdot d\vec{l} = \oint B dl \cos 0^\circ = B \oint dl$$

$$\text{or, } \oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi d$$



In this figure the wire is normal to the plane of the figure. The circle shown is the Amperian loop.

The current piercing through the area bound by the circle is I . Thus, from Ampere's law

$$B \cdot 2\pi d = \mu_0 I$$

$$\text{or, } B = \frac{\mu_0 I}{2\pi d} \quad [\text{Same as equation (6)}]$$

Example 10 Field inside and outside a long cylindrical current carrying wire

A long, cylindrical wire of radius b carries a current I . The current is distributed uniformly over its entire cross section. Find the magnetic field produced by the current at a point that is at a distance x from the axis of the wire, when

- (a) $x < b$ (b) $x > b$.

Also plot a graph showing variation of the field (B) with the distance x . Ignore the effect of material of the wire on the value of field inside it.

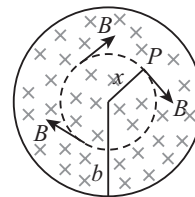
Solution

Concepts

- A circle as Amperian loop will be suitable for finding field at $x < b$ as well as $x > b$.
- We need to carefully write the value of current piercing through the area of the chosen Amperian loop.
- Current density is uniform. This implies that the current through unit area of the cross section of the wire is $j = \frac{I}{\pi b^2}$ everywhere.

- (a) The figure shows the cross section of the wire. Current is directed into the plane of the figure. We wish to find field at a point P at a distance x from the axis ($x < b$).

The dotted circle through P is the Amperian loop. Every point on this circle has same strength (B) of the field and the field is tangential.



$$\oint \vec{B} \cdot d\vec{l} = \oint B dl \cos 0^\circ = B \oint dl = B 2\pi x.$$

The current piercing through the circle of radius x is

$$i = j \cdot \pi x^2 \quad \left[j = \frac{I}{\pi b^2} \right]$$

Using Ampere's law, we can write

$$B \cdot 2\pi x = \mu_0 i$$

$$\text{or, } B \cdot 2\pi x = \mu_0 j \cdot \pi x^2$$

$$\text{or, } B = \left(\frac{\mu_0 j}{2} \right) x \quad (i)$$

$$\text{or, } B = \left(\frac{\mu_0 I}{2\pi b^2} \right) x \quad \left[\because j = \frac{I}{\pi b^2} \right]$$

- (b) P is a point outside the wire at a distance x from the axis of the wire. Again the symmetry of the situation tells us that field is tangential and of same strength at every point on a circular Amperian loop.

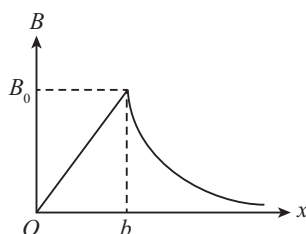
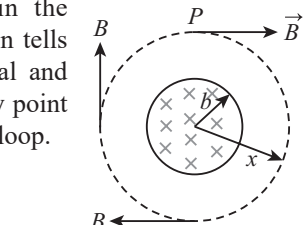
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$\text{or, } B \cdot 2\pi x = \mu_0 I$$

[Complete current I passes through the loop]

$$\text{or, } B = \frac{\mu_0 I}{2\pi x} \quad (\text{ii})$$

For the graphical plot, we must observe that the field varies linearly with x inside the wire [equation (i)] and it varies inversely with x for $x > b$ [equation (ii)]



Field on the surface of the wire is $B_0 = \frac{\mu_0 I}{2\pi b}$

Note: In the above example, if we do not choose a circular Amperian loop and rather choose a rectangular one (or any other shape), the Ampere's law will still remain true but we will not be able to find the field B . The symmetry will be lost for any other shape of the loop. We will not be able to claim that the field has same strength at all the points on the loop.

5.2 Field inside an ideal solenoid

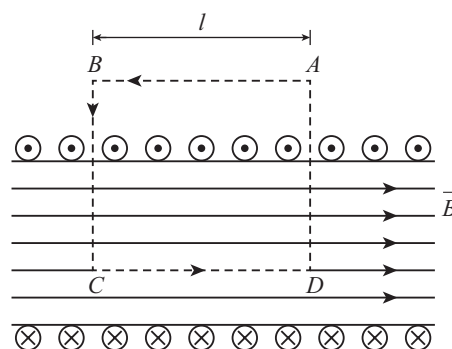
We will derive equation (16) using the Ampere's law. An ideal solenoid has large length compared to its radius and the magnetic field inside it is uniform. The field is parallel to the axis. Let the strength of the field be B . Outside the solenoid the field is zero.

This assumption is nearly correct when number of turns are large.

Consider a rectangle $ABCD$ as our Amperian loop. The side CD is parallel to the solenoid axis and hence parallel to the field inside the solenoid.

$$\begin{aligned} \oint \vec{B} \cdot d\vec{l} &= \int_C^D \vec{B} \cdot d\vec{l} + \int_D^A \vec{B} \cdot d\vec{l} + \int_A^B \vec{B} \cdot d\vec{l} + \int_B^C \vec{B} \cdot d\vec{l} \\ &= Bl + 0 + 0 + 0 = Bl \end{aligned}$$

The value of $\vec{B} \cdot d\vec{l}$ is zero everywhere on the sides DA , AB and BC , because either $\vec{B}$ is zero (outside the solenoid) or is perpendicular to $d\vec{l}$ (inside the solenoid).



The dotted rectangular loop is the Amperian loop.

If n is number of turns per unit length of the solenoid, then a total of nl turns pierce through the rectangle $ABCD$. Thus, the total current crossing through the area $ABCD$ is nIl , where I is the current given to the solenoid coil.

Using Ampere's law,

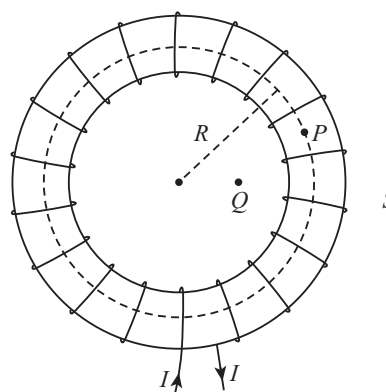
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 nIl$$

$$\text{or, } Bl = \mu_0 nIl$$

$$\text{or, } B = \mu_0 nI \quad [\text{Same as equation (16)}]$$

5.3 Toroid

Consider a long, ideal solenoid. Bend it in a circle and join the ends. Now, you get a coil as shown in the figure. A coil in this shape is known as a toroid.



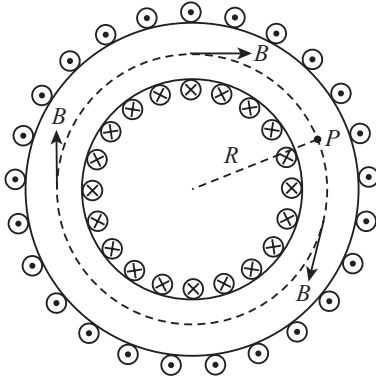
We will assume that the radius (r) of the cross section of the toroid (i.e., radius of the solenoid which was bent) is small compared to the radius (R) of the circular ring. Number of turns on the toroid is N . The number of turns per unit length along the ring is

$$n = \frac{N}{2\pi R}$$

We wish to write magnetic field at points P , Q and S when a current I is given to the coil.

2.18 Magnetism and Electromagnetic Induction

The figure shows that section of a toroid. Consider a circular Amperian loop of radius R passing through point P inside the toroid. Due to symmetry the field at all the points on this circle is tangential and must have a common strength B .



Total number of $\odot$ is same as number of $\otimes$. Both are equal to number of turns N .

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

or, $B \cdot 2\pi R = \mu_0 NI$ [Currents passing through the circle of radius R are those indicated by $\otimes$]

$$\text{or, } B = \mu_0 \left(\frac{N}{2\pi R} \right) I$$

$$\text{or, } B = \mu_0 n I \quad (18)$$

This is same as field inside the original solenoid which was bent to form this toroid. As you bend the solenoid, the magnetic field lines also get bent along the axis and become circular in shape.

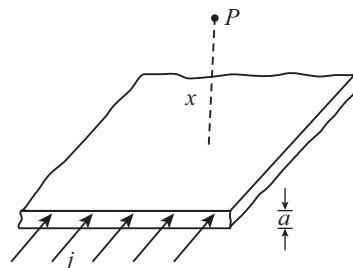
If a circle is drawn passing through point Q and we take this as our Amperian loop, we can write

$$B \cdot 2\pi x = 0 \quad [\because \text{current passing through the circle is } 0] \\ \Rightarrow B = 0$$

A similar Amperian circle drawn through the outside point S will give the field at S to be zero. For this Amperian loop the total $\odot$ current is same as total $\otimes$ current. Therefore, net current through the loop is zero.

5.4 Field due to a large current sheet

The figure shows a large conducting sheet having a uniform thickness a . It carries a current with the current density (j) being uniform over its cross section. We wish to find magnetic field at a point P at a distance x from the sheet.



Consider two identical strips M and N located symmetrically with respect to the point P . Both the strips are like long wires carrying current dI . The figure (b) shown above shows the two dimensional view of the situation. Field at P due to the strip M is dB_M , directed perpendicular to MP . Similarly, the field at P due to the strip N is dB_N , directed perpendicular to NP . Obviously, $dB_M = dB_N$.

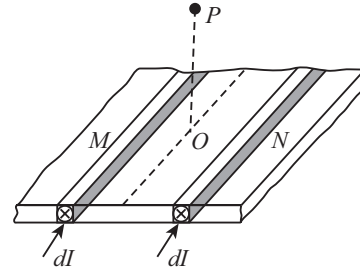


Fig (a)

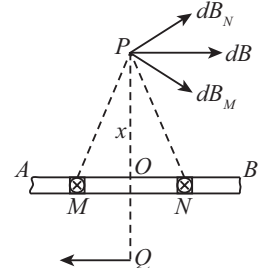
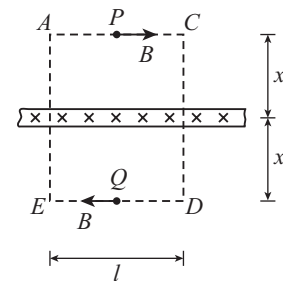


Fig (b)

The resultant of the two fields is (shown as dB) parallel to the conducting sheet (parallel to AB). We can divide the entire sheet into pair of such strips which are symmetrically located on two sides of OP . Therefore, the field at a point like P is parallel to the sheet. A similar argument can show that the field at a point like Q is also parallel to the sheet but is directed opposite to the field at P .

Let the field at a distance x from the sheet be having a magnitude B . To find this magnitude, consider a rectangular Amperian loop as shown

$$\oint \vec{B} \cdot d\vec{l} = \int_A^C \vec{B} \cdot d\vec{l} + \int_C^D \vec{B} \cdot d\vec{l} + \int_D^E \vec{B} \cdot d\vec{l} + \int_E^A \vec{B} \cdot d\vec{l} \\ = B \cdot l + 0 + B \cdot l + 0 = 2Bl$$



The line integrals along CD and EA are zero as $\vec{B}$ is perpendicular to $d\vec{l}$.

Using the Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

or, $2Bl = \mu_0 (j al)$ [cross sectional area of the strip intercepted by the rectangle is al]

$$\text{or, } B = \frac{\mu_0 j a}{2} \quad (19)$$

Note that the result is independent of x . The field is uniform on either side of the sheet.



YOUR TURN

Q.15 Four long, straight wires are parallel to one another. They carry equal current I perpendicular to the plane of the figure. The cross sections of the wire in the plane of the figure, lie on the vertices of a square. A student considers a circular Amperian loop as shown. He assumes the field at every point on the circle to be B and writes

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

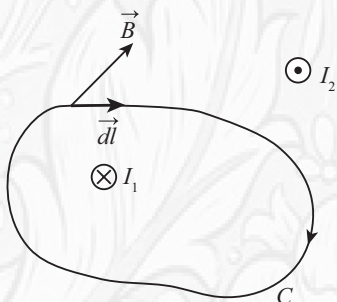
or, $B \cdot 2\pi r = 0$ [There is no current piercing through the circle]

or, $B = 0$

Has the student committed a mistake?

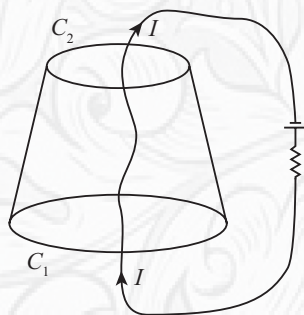
Q.16 There are two current carrying wires having currents I_1 and I_2 as shown. C is a closed loop. While writing Ampere's law $\oint_C \vec{B} \cdot d\vec{l} = \mu_0 i$, a student claims that $i = I_1$.

While calculating the line integral, the student assumes $\vec{B}$ as magnetic field due to current I_1 only. Is he right?

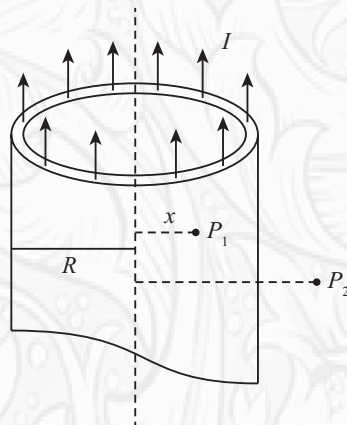


Q.17 A loose wire carrying current I passes through a frustum. The wire passes through the centre of the base circle C_1 and through the centre of the other circular face C_2 .

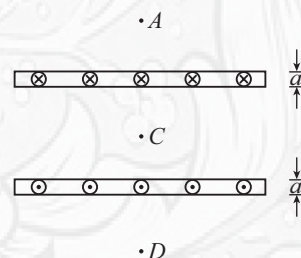
The line integral of magnetic field $\oint_{C_1} \vec{B} \cdot d\vec{l}$ calculated for the circle C_1 has value K_1 and the line integral $\oint_{C_2} \vec{B} \cdot d\vec{l}$ for circle C_2 has value K_2 . Which is larger – K_1 or K_2 – and why?



Q.18 A thin walled, long, hollow, cylindrical tube carries a current I along its length. The current is uniformly distributed along the tube wall. Radius of the tube is R . Find the magnetic field at a point P_1 inside the tube, which is at a distance x from the axis. Also write the field at an outside point P_2 which is at a distance x ($>R$) from the axis.



Q.19 Two large metal sheets are held parallel to each other at a separation. They carry currents in opposite directions. Current density in each sheet is j through their cross sections of thickness a . Find the magnitude of magnetic field at the points A , C and D .



Q.20 A solenoid of length $l = 2.0$ m has cross sectional radius of 5 cm. It has a total of 3000 turns. It is bent to form a toroid. A current of 5A is given to the coil.

- Imagine a closed loop running inside the toroid along its circular axis. Find $\oint \vec{B} \cdot d\vec{l}$ along this loop.
- Find the strength of magnetic field inside the toroid.

6. FORCE BETWEEN PARALLEL CURRENTS

Consider two long wires 1 and 2 kept parallel to one another. They carry currents I_1 and I_2 in the same direction. Separation between the wires is d .

The magnetic field (B_1) due to the wire 1 at the location of the wire 2 is

$$B_1 = \frac{\mu_0 I_1}{2\pi d} \otimes$$

Magnetic force on a segment (AB) of length L of wire 2 due to the field B_1 is

$$\vec{F} = I_2 \vec{L} \times \vec{B}_1$$

$$\text{or, } F = I_2 L \frac{\mu_0 I_1}{2\pi d} \quad (\text{directed towards left perpendicular to the wire})$$

The force on unit length of wire 2 is

$$\frac{F}{L} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d} \quad (20)$$

If we calculate force on unit length of wire 1 due to the field created by 2, it is again given by the above equation.

If the wires carry currents in opposite directions, they repel one another. The force is again given by the above equation (20).

Defining the unit of current – the ‘ampere’

Consider the wires, in the above discussion, to have a current $I_1 = I_2 = 1\text{A}$ and let the separation between the wires be $d = 1\text{m}$. The mutual force on unit length of the wires is

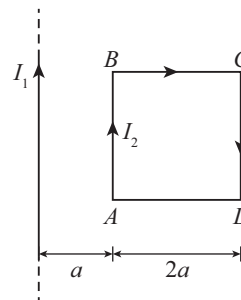
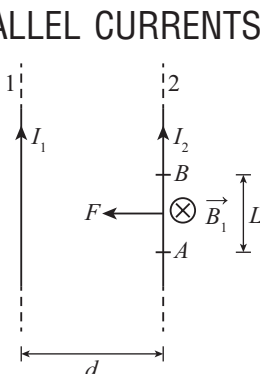
$$\frac{F}{L} = \frac{\mu_0}{2\pi} \times \frac{1 \times 1}{1} = 2 \times 10^{-7} \text{ N/m.}$$

This result is used to formally define the *SI* unit of current.

One ampere is the current in each of the two parallel long wires which exert a force of $2 \times 10^{-7} \text{ N}$ on 1 m length of each other, when placed at a separation of 1m.

Example 11 A square loop and a straight wire

A square loop $ABCD$ has side length $2a$ and carries a current I_2 . In the plane of the loop there is a long straight wire carrying current I_1 , as shown. Find the magnetic force on the square loop due to the current I_1 .



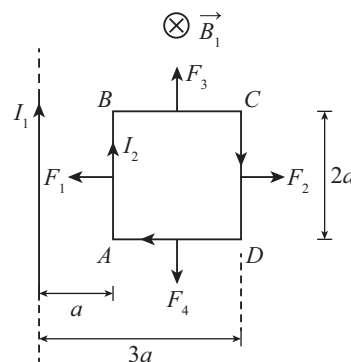
Solution

Concepts

- The magnetic field, due to I_1 , is directed into the plane of the figure at the location of the square loop. This field exerts a leftward force on arm AB and a rightward force on arm CD of the loop. But the two forces are not equal in magnitude because the field is different at the locations of AB and CD .
- Direction of force on arm BC is upward and that on DA is downward. These two forces must have equal magnitude as BC and DA are at identical positions relative to the current I_1 .

The forces on each arm are shown in the figure. The directions are decided using $\vec{F} = I\vec{L} \times \vec{B}$.

Forces on BC and DA balance out ($F_3 = F_4$).



Force on unit length of wire AB , due to I_1 , is

$$\frac{\mu_0}{2\pi} \frac{I_1 I_2}{a}$$

$\therefore$ Force on the wire AB due to I_1 is

$$F_1 = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{a} (2a) = \frac{\mu_0 I_1 I_2}{\pi} \quad (\leftarrow)$$

Force on wire CD due to current I_1 is

$$F_2 = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{3a} (2a) = \frac{\mu_0 I_1 I_2}{3\pi} \quad (\rightarrow)$$

$\therefore$ Resultant force on the loop is

$$F = F_1 - F_2 = \frac{2\mu_0 I_1 I_2}{3\pi} \quad (\leftarrow)$$

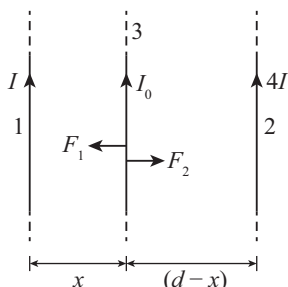
Example 12 Three parallel wires

Two parallel wires separated by a distance d carry currents I and $4I$ along the same direction. Where shall we place a third current carrying wire so that it experiences no force?

Solution**Concepts**

- The third wire shall be placed in the plane of the two wires, otherwise the force applied on it by the two wires cannot be along a line (in opposite directions).
- The third wire shall be placed parallel to the two wires. This will ensure that the forces on the third wire are perpendicular to it (i.e., along the same line)
- The third wire must be placed somewhere between the two wires so that it feels the two forces in opposite directions.

Let the third wire carry a current I_0 in the same direction as the other two wires. F_1 and F_2 are forces on unit length of wire 3 due to the currents I and $4I$. The two forces are opposite. Even if the direction of I_0 is reversed, the two forces will remain oppositely directed. Now, F_1 will be towards right and F_2 will be towards left.



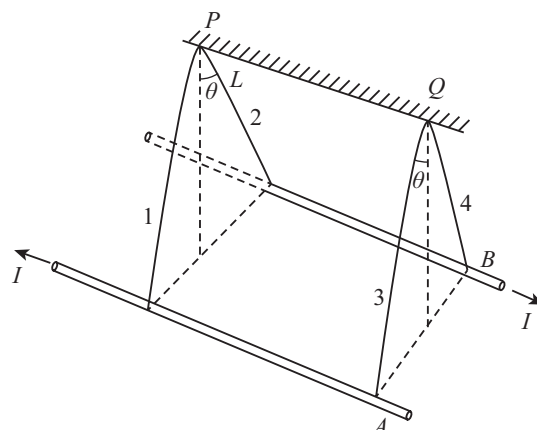
$$F_1 = F_2$$

$$\text{or, } \frac{\mu_0 I I_0}{2\pi x} = \frac{\mu_0 I_0 (4I)}{2\pi(d-x)}$$

$$\text{or, } d - x = 4x \quad \Rightarrow x = \frac{d}{5}$$

Example 13 Long suspended wires

A and B are two long, identical, parallel wires carrying equal current I in opposite directions. These wires are suspended using four identical strings (1, 2, 3 and 4) from two points P and Q on the ceiling. Due to repulsion the two wires stay apart at a separation in their equilibrium positions. In this position all the strings make angle θ with vertical. If length of each string is L , find the linear mass density (λ) of the two wires.

**Solution****Concepts**

The magnetic force between the wires is repulsive. The resultant of the magnetic force, string tensions and the weight on a wire is zero.

The figure shows the cross sections of the two wires.

Let T be tension in each string. Magnetic force on a length L_0 of the wire is

$$F = \frac{\mu_0}{2\pi} \frac{I^2}{d} L_0 = \frac{\mu_0}{2\pi} \frac{I^2}{(2L \sin \theta)} L_0$$

$$\text{or, } F = \frac{\mu_0 I^2 L_0}{4\pi L \sin \theta}$$

Weight of length L_0 of the wire is

$$mg = \lambda L_0 g$$

For equilibrium,

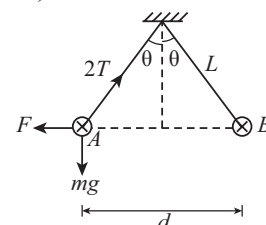
$$2T \sin \theta = F \quad (i)$$

$$\text{And } 2T \cos \theta = mg \quad (ii)$$

Dividing (i) by (ii) gives

$$\tan \theta = \frac{F}{mg} = \frac{\mu_0 I^2 L_0}{(4\pi L \sin \theta)(\lambda L_0 g)}$$

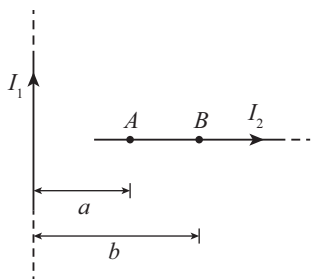
$$\text{or, } \lambda = \frac{\mu_0 I^2}{4\pi L g \sin \theta \tan \theta}$$



Note: The length L_0 of the two wires make no difference to our answer, as long as they are long enough so that equation (20) holds.

Example 14 Force between perpendicular wires

A long, straight wire carries current I_1 . Another wire, carrying current I_2 is kept perpendicular to it. Find the force on segment AB of the wire due to the magnetic field created by I_1 . Both the wires lie in the plane of the figure.



Solution

Concepts

- Field due to I_1 is directed into the plane of the figure at the location of the second wire. The magnetic field is perpendicular to the second wire. But the strength of the field decreases as one moves from A to B .
- We will write force on a small segment of length dx of the second wire and then integrate to get the resultant force on the segment AB .

Field due to I_1 at a distance x from the wire is

$$B_1 = \frac{\mu_0 I_1}{2\pi x} \otimes$$

Consider an elemental length dx on the second wire. Force on this segment is

$$d\vec{F} = I_2 d\vec{x} \times \vec{B}_1$$

or, $dF = \frac{\mu_0 I_1 I_2}{2\pi} \frac{dx}{x}$ Directed perpendicular to AB as shown.

$\therefore$ Force on the segment AB is

$$F = \frac{\mu_0 I_1 I_2}{2\pi} \int_a^b \frac{dx}{x}$$

$$\text{or, } F = \frac{\mu_0 I_1 I_2}{2\pi} \ln \left(\frac{b}{a} \right)$$

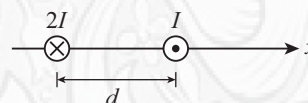


YOUR TURN

Q.21 A straight long wire carries a current $I_1 = 10\text{A}$. It repels another parallel wire with a force of 10^{-5} N/m , when the distance between the two wires is 10cm . A third parallel wire carrying a current $I_3 = 2\text{A}$ is placed parallel to the two wires, midway between them. All three wires are in a plane. Find the net magnetic force on 1m length of the third wire.

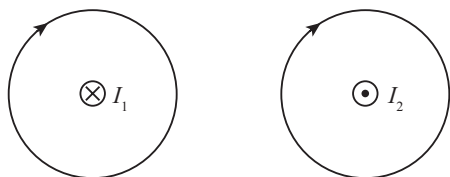
Q.22 Two long wires carry currents $2I$ and I in opposite directions. The two wires lie in xy plane at a separation

d . The figure shows their cross sections. A third current carrying wire is to be placed in xy plane parallel to the two wires, so that it experiences zero net force. Where shall we place the third wire?



In Short

- The line integration of the magnetic field on a closed loop is equal to μ_0 times the total current piercing through the loop. The sense in which we move on the loop to perform the integration is used to decide the positive and negative directions of the current. This law is known as Ampere's law.



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

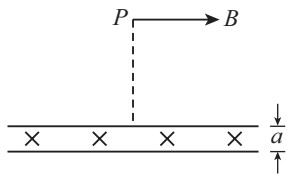
In the two figure, if we walk clockwise to evaluate $\oint \vec{B} \cdot d\vec{l}$, then the current I_1 is positive (it points in the direction of the right hand thumb when fingers are curled clockwise) and the current I_2 is negative.

- Ampere's law can be used to find the magnetic field due to a current distribution, only under certain symmetrical situations.

Often, a circular or a rectangular Amperian loop is chosen to calculate B .

- Field due to a large current sheet at a point P near it is given by

$$B = \frac{\mu_0 j a}{2}$$



Here, j is the current density and 'a' is thickness of the sheet. The direction of the field is parallel to the sheet and perpendicular to the direction of the current.

- (iv) Field inside an ideal solenoid as well as inside an ideal toroid is $B = \mu_0 ni$, where n is number of turns per unit length and i is the current in the coil. The direction of the field is along the axis.
- (v) Two long parallel currents (I_1 and I_2) attract each other when they are in same direction and repel when they are oppositely directed. The force experienced by a segment of length L of each of the wires is

$$F = \frac{\mu_0 I_1 I_2}{2\pi d} L$$

- (vi) The magnetic field inside a long hollow cylindrical conductor is zero when the current density is uniform on its cross section. For outside points the conductor behaves like a thin wire along its axis and the field is given by

$$B = \frac{\mu_0 I}{2\pi r}$$

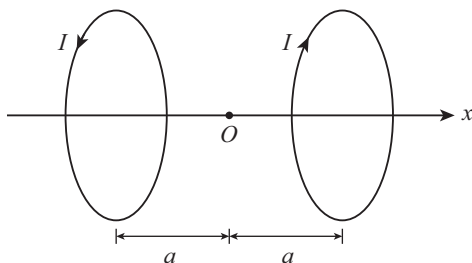
- (vii) Consider a cylindrical conductor of radius a . Current is uniformly distributed on its circular cross section of area πa^2 . Current density is $j = \frac{I}{\pi a^2}$. The magnetic field at a point at a distance r from the axis is

$$B = \left(\frac{\mu_0 j}{2} \right) r \text{ for } r \leq a$$

$$= \frac{\mu_0 I}{2\pi r} \text{ for } r \geq a$$

Miscellaneous Examples

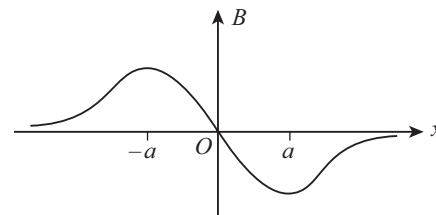
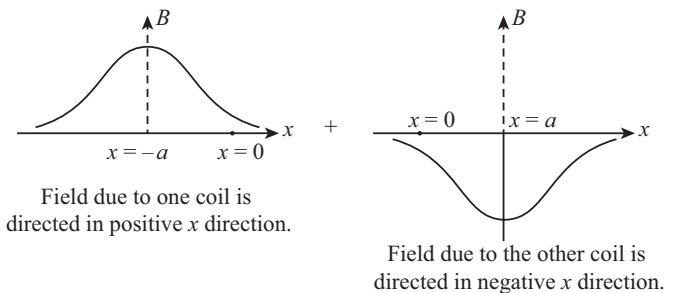
Example 15 Two identical circular coils are held parallel to each other with their common axis being the x axis. The centres of the two coils lie at $x = -a$ and $x = a$. Draw a rough graph representing the variation of magnetic field along the x axis.



Solution

Concepts

- Fields due to the two coils are oppositely directed at all points on the axis.
- We can draw the variation of field due to individual coils. The resultant field is obtained as superposition of the two fields.



Variation of the resultant field.
The field is zero at the origin.

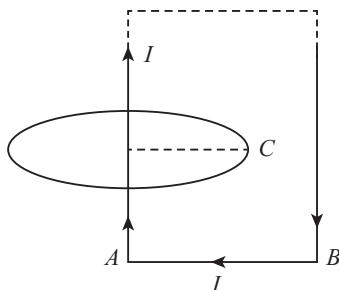
Example 16 A semi infinite wire has current I and one of its end is at A . To find the magnetic field at a point (P) at a distance x from the wire, a student assumes a circular Amperian loop of radius x , around the wire. The student argues that every point on the circular loop is symmetrically placed with respect to the wire and must have same strength of the field. Using Ampere's law he writes

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I \quad \Rightarrow \quad B = \frac{\mu_0 I}{2\pi x}$$

where has the student gone wrong?

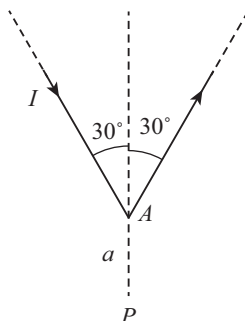
Solution

We cannot just have a wire in which a current enters at A and leaves at infinity. There must be some other wire bringing the current to A . Presence of other nearby wire distorts the symmetry of magnetic field on the loop C . For example the segment BA also contributes to magnetic field at C . Its contribution is not same everywhere.



Example 17

A long wire, carrying a current I , has been bent into a V shape with the vertex angle 60° . A point P is located on the angle divider at a distance ' a ' from the vertex. Find the magnetic field at P .



Solution

Concepts

- In the formula of field due to a straight wire, d stands for the perpendicular distance of the concerned point from the wire.
- Both the segments of the wire are smaller than a semi-infinite wire with regard to point P .

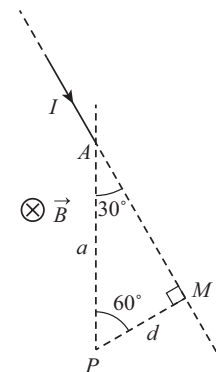
The figure shows the position of the point P with respect to the one arm of the V -shaped wire.

Had the wire extended upto the point M , it would have been a semi infinite wire. Field at P due to this wire can be written as

$B =$ (Field due to a semi infinite wire extending upto M) –

(Field produced due to the segment AM of the wire)

$$\begin{aligned} &= \frac{\mu_0 I}{2\pi d} - \frac{\mu_0 I}{4\pi d} (\sin 0^\circ + \sin 60^\circ) \\ &= \frac{\mu_0 I}{\pi a} - \frac{\mu_0 I}{2\pi a} \frac{\sqrt{3}}{2} \quad \left[\because d = a(\sin 30^\circ = \frac{a}{2}) \right] \\ &= \frac{\mu_0 I}{\pi a} \left(1 - \frac{\sqrt{3}}{4} \right) \otimes \end{aligned}$$

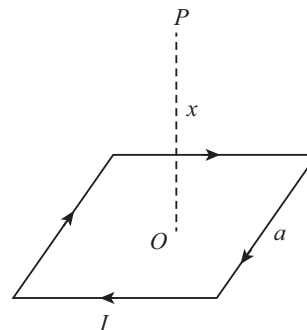


The other arm of the V also contributes same amount of field at P and in the same direction. Therefore, the resultant field at P is double the value of B obtained above.

$$B_P = \frac{2\mu_0 I}{\pi a} \left(1 - \frac{\sqrt{3}}{4} \right) \otimes$$

Example 18

A square loop of side length a carries a current I . consider a point P on a line perpendicular to the plane of the loop and passing through its centre (O). Find the magnetic field at P when $OP = x$.



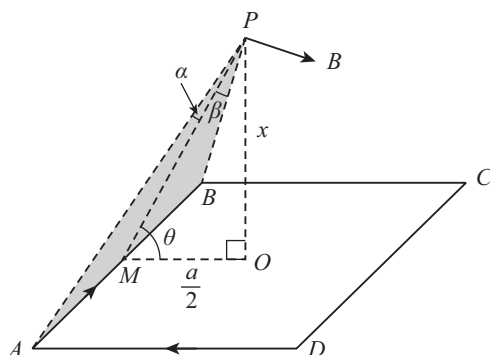
Solution

Concepts

- A side of the square loop is a wire of finite length. To write the field at P due to such a wire we must correctly figure out α , β and d in the equation.

$$B = \frac{\mu_0 I}{4\pi d} (\sin \alpha + \sin \beta)$$

- Imagination of a plane containing the point P and a side of the square will help. The magnetic field at P due to the current in the side will be normal to this plane.



We will write the field at P due to the current in the side AB of the square loop. M is the midpoint of the side AB .

First consider the right angled triangle PMO .

$$PM = \sqrt{x^2 + \frac{a^2}{4}} \quad \text{and}$$

$$\tan \theta = \frac{2x}{a} \quad \Rightarrow \quad \cos \theta = \frac{a}{\sqrt{4x^2 + a^2}}$$

Now, consider the triangle PAB . PM is the perpendicular distance of the point P from the wire AB . The two angles α and β are equal, given by

$$\tan \alpha = \tan \beta = \frac{MB}{PM} = \frac{\frac{a}{2}}{\sqrt{x^2 + \frac{a^2}{4}}} = \frac{a}{\sqrt{4x^2 + a^2}}$$

$$\Rightarrow \sin \alpha = \sin \beta = \frac{a}{\sqrt{4x^2 + 2a^2}}$$

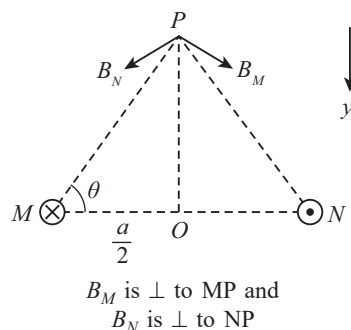
The magnetic field at P due to the wire AB is

$$B = \frac{\mu_0 I}{4\pi d} (\sin \alpha + \sin \beta) \quad \text{where } d = PM$$

$$\begin{aligned} \text{or, } B &= \frac{\mu_0 I}{2\pi\sqrt{4x^2 + a^2}} \cdot \frac{2a}{\sqrt{4x^2 + 2a^2}} \\ &= \frac{\mu_0 I a}{\pi\sqrt{4x^2 + a^2}\sqrt{4x^2 + 2a^2}} \end{aligned}$$

Direction of B is perpendicular to MP as shown.

When field due to each side is added, the horizontal components will cancel out. Symmetry of the situation tells us that the resultant must be along PO . To make you visualise this I have drawn a diagram showing the cross sections of the wires AB and CD at their mid points M and N .



Component of B along PO is

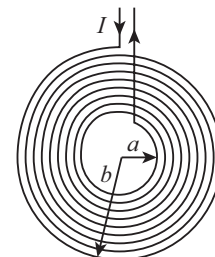
$$\begin{aligned} B_y &= B \cos \theta = B \frac{a}{\sqrt{4x^2 + a^2}} \\ &= \frac{\mu_0 I a^2}{\pi(4x^2 + a^2)\sqrt{4x^2 + 2a^2}} \end{aligned}$$

The y components of the field due to all the wires add.

$$\therefore \text{Resultant field} = \frac{4\mu_0 I a^2}{\pi(4x^2 + a^2)\sqrt{4x^2 + 2a^2}}$$

Example 19 A Spiral

A thin insulated wire forms a plane spiral of N turns carrying a current I . The inner and the outer radii of the spiral are a and b respectively. Find the magnetic field at the centre of the spiral.



Solution

Concepts

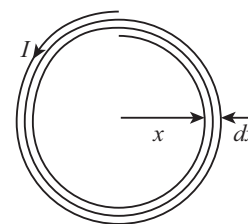
- (i) We will consider a ring of radius x and width dx . We will write the field at the centre due to such a ring using $B = \frac{\mu_0 NI}{2R}$
- (ii) The entire spiral is made of large number of such rings. Each ring contributes field in the same direction. The resultant field is obtained by adding contributions from all such rings.
- (ii) N turns of the wire are spread over a width $(b - a)$.

Number of turns in unit length along a radial line is

$$\frac{N}{b - a}$$

Number of turns in a ring having radial width dx is

$$\frac{N dx}{(b - a)}$$



Field due to the current in the ring, at its centre is

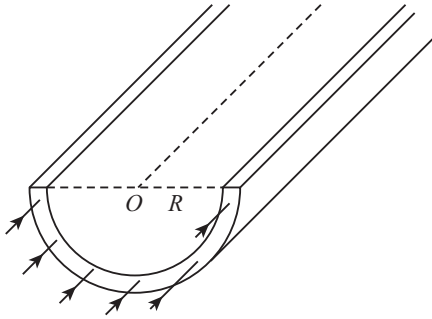
$$dB = \frac{\mu_0 I}{2x} \frac{N dx}{(b - a)} \odot$$

Adding contributions from all rings gives the resultant

$$\begin{aligned} B &= \frac{\mu_0 I}{2(b - a)} \int_a^b \frac{dx}{x} \\ &= \frac{\mu_0 I}{2(b - a)} \ln \left(\frac{b}{a} \right) \odot \end{aligned}$$

Example 20 A thin current sheet bent to form a half cylinder

A long, thin current sheet carries a current I . It is bent to form a conductor having semicircular cross section of radius R . The current is uniformly distributed on the entire cross section. Find the magnetic field at a point O (the centre of curvature of a semicircular section).



Solution

Concepts

- The entire sheet can be divided into number of thin straight wires. Adding the contributions by all such wires will give the field.
- Considering two thin straight elements, located symmetrically on the two sides of O will help us visualise the direction of the field at O .
- Current through a unit length along the circumference of the semicircular section is $\frac{I}{\pi R}$.

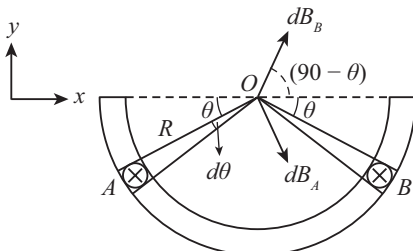
Consider two long strips of current at symmetrical locations relative to O . Both the strips subtend an angle $d\theta$ at O . Current in each of them is

$$dI = \frac{I}{\pi R} R d\theta = \frac{Id\theta}{\pi}$$

Field due to the strip A is

$$dB_A = \frac{\mu_0(dI)}{2\pi R} = \frac{\mu_0 I}{2\pi^2 R} d\theta$$

Direction of $\vec{dB}_A$ is perpendicular to AO (see figure).



Current is going into the plane of the figure.

The strip B also creates field of same strength in a direction perpendicular to BO as indicated in the figure. Obviously, the y components of the two fields cancel out and the resultant is in x direction.

Resultant field at O is obtained by adding x component of the field due to each element.

$$\therefore B = \frac{\mu_0 I}{2\pi^2 R} \int_0^\pi \sin \theta d\theta$$

$$\text{or, } B = \frac{\mu_0 I}{2\pi^2 R} [-\cos \theta]_0^\pi$$

$$\text{or, } B = \frac{\mu_0 I}{\pi^2 R}$$

Example 21 Polygon to a circle

An n sided conducting regular polygon has side length L . It carries a current I .

- Find the magnetic field at the centre of the polygon.
- Use the above result to find the magnetic field at the centre of a circular current loop having radius r .

Solution

Concepts

- A side of the polygon subtends an angle $\frac{2\pi}{n}$ at the centre.
- Each side contributes the magnetic field in the same direction.
- A circle is a limiting case of a polygon with $n \rightarrow \infty$.

- AB is a side of the polygon.

$$\angle ACB = \frac{\pi}{n}$$

$$\therefore \theta = \frac{\pi}{2n}$$

Field at C due to the side AB is

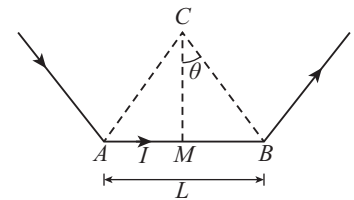
$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta + \sin \theta)$$

$$\text{Here, } d = CM = \frac{L}{2 \tan \theta} = \frac{L}{2 \tan \left(\frac{\pi}{n} \right)}$$

$$\therefore B = \frac{\mu_0 I}{\pi L} \tan \left(\frac{\pi}{n} \right) \sin \left(\frac{\pi}{n} \right)$$

The total field at the centre is nB .

$$B_0 = nB = n \frac{\mu_0 I}{\pi L} \tan \left(\frac{\pi}{n} \right) \sin \left(\frac{\pi}{n} \right) \quad (i)$$



- (b) When $n \rightarrow \infty$, the polygon approaches a circular shape. Circumference of the circle = nL

$$\text{or, } 2\pi r = nL$$

$$\text{or, } L = \frac{2\pi r}{n}$$

Putting in (i) gives

$$B_0 = \frac{n^2 \mu_0 I}{2\pi^2 r} \tan\left(\frac{\pi}{n}\right) \sin\left(\frac{\pi}{n}\right)$$

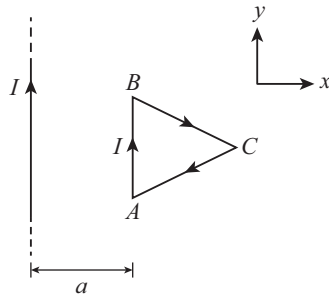
when $n \rightarrow \infty$

$$B_0 = \frac{\mu_0 I}{2r} \lim_{n \rightarrow \infty} \left(\frac{\tan \pi/n}{\pi/n} \right) \left(\frac{\sin \pi/n}{\pi/n} \right)$$

$$= \frac{\mu_0 I}{2r} \left[\because \lim_{n \rightarrow \infty} \frac{\tan \pi/n}{\pi/n} = 1 \text{ and } \lim_{n \rightarrow \infty} \frac{\sin \pi/n}{\pi/n} = 1 \right]$$

Example 22 A triangular frame in the field of a straight wire

An equilateral triangular wire frame has side length a and carries a current I . A long straight wire, carrying a current I , is held parallel to one of the sides of the triangle. The wire is at a distance a from the parallel side of the triangle. Find the force on the triangle.



Solution

Concepts

- You should not write the force as zero. The triangular loop is not in a uniform field.
- We need to find forces on each side separately and add them.
- Sides BC and CA see variable field. We need to choose a small element and write force on it. Then we will be integrating to obtain the force on BC (or CA).

$$\text{Force on side } AB \text{ is } F_{AB} = \frac{\mu_0 I^2}{2\pi a} \cdot a = \frac{\mu_0 I^2}{2\pi}$$

This force is directed perpendicular to AB in negative x direction.

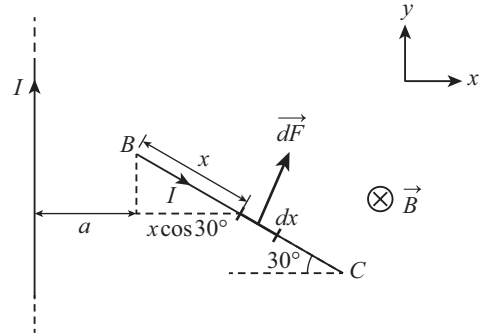
$$\therefore \vec{F}_{AB} = \frac{\mu_0 I^2}{2\pi} (-\hat{i})$$

Now, consider side BC of the triangular loop. On this side assume a small length dx . Force on this small length due to the field (B) created by the current in the straight wire is

$$dF = (Idx) B$$

$$\text{where } B = \frac{\mu_0 I}{2\pi(a + x \cos 30^\circ)}$$

$$\therefore dF = \frac{\mu_0 I^2 dx}{2\pi \left(a + \frac{\sqrt{3}}{2} x \right)}$$

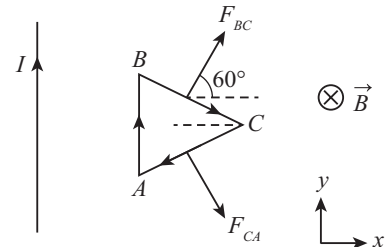


This force is normal to BC as shown. We can write force on arm BC as

$$F_{BC} = \frac{\mu_0 I^2}{2\pi} \int_0^a \frac{dx}{\left(a + \frac{\sqrt{3}}{2} x \right)} = \frac{\mu_0 I^2}{2\pi \frac{\sqrt{3}}{2}} \left[\ln \left(a + \frac{\sqrt{3}}{2} x \right) \right]_0^a$$

$$= \frac{\mu_0 I^2}{\sqrt{3}\pi} \ln \left(1 + \frac{\sqrt{3}}{2} \right)$$

Force on the arm CA will also be of same magnitude perpendicular to CA as shown. Obviously, the y components of F_{BC} and F_{CA} will cancel out. Resultant force is in x direction only.



$$(F_{BC})_x = F_{BC} \cos 60^\circ = \frac{\mu_0 I^2}{2\sqrt{3}\pi} \ln \left(1 + \frac{\sqrt{3}}{2} \right)$$

Sum of F_{BC} and F_{CA} is twice the above value.

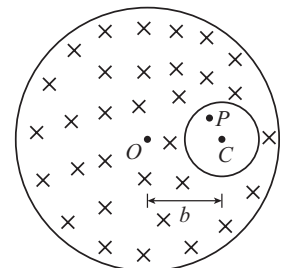
$\therefore$ Resultant force on the loop is

$$\vec{F} = \left[\frac{\mu_0 I^2}{\sqrt{3}\pi} \ln \left(1 + \frac{\sqrt{3}}{2} \right) - \frac{\mu_0 I^2}{2\pi} \right] (\hat{i})$$

$$= \frac{\mu_0 I^2}{\pi} \left[\frac{1}{\sqrt{3}} \ln \left(\frac{2 + \sqrt{3}}{2} \right) - \frac{1}{2} \right] (\hat{i})$$

Example 23 Cylindrical cavity

A long cylindrical conductor has a cylindrical cavity running all along its length. Apart from the cavity, the entire cross section has a uniform current density j . Find the direction and magnitude of the magnetic field at a point P inside the cavity. It is given that



Cross section of the conductor

the distance between the axis of the conductor and the axis of the cavity is b .

Solution

Concepts

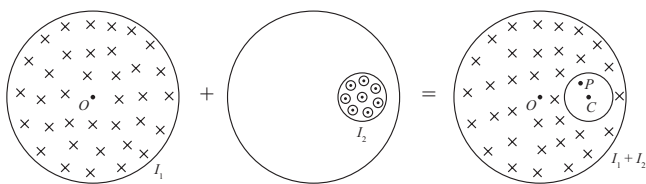
- (i) The magnetic field at a distance x from the axis of a cylindrical current carrying wire (for $x <$ radius of the wire) is given by equation (i) in example 10.

$$B = \left(\frac{\mu_0 J}{2} \right) x$$

- (ii) We will assume a cylinder without cavity having a uniform current density directed into the plane of the figure. We will write field $\vec{B}_1$ due to this current at the point P .

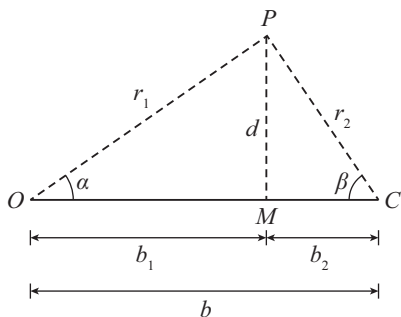
Then we will assume an additional current density j inside the cavity region directed out of the plane of the figure. We will write field ($\vec{B}_2$) due to this current at P .

Resultant of $\vec{B}_1$ and $\vec{B}_2$ will be the desired field.



Let distance of the point P from O and C be r_1 and r_2 respectively. The distance between O and C can be expressed as

$$b = r_1 \cos \alpha + r_2 \cos \beta \quad (i)$$



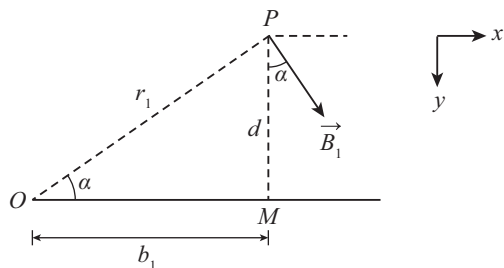
Field at P due to the current I_1 in the entire cross section is

$$B_1 = \left(\frac{\mu_0 J}{2} \right) r_1$$

Direction of this field is perpendicular to OP as shown. Its x and y components are

$$B_{1x} = B_1 \sin \alpha = \frac{\mu_0 J}{2} r_1 \sin \alpha$$

$$\text{or, } B_{1x} = \frac{\mu_0 J}{2} d$$



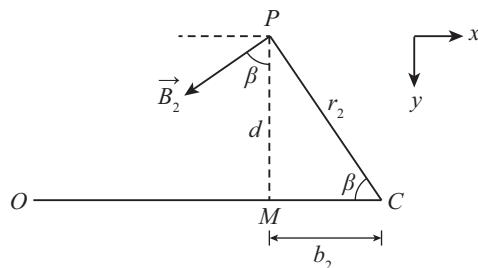
$$\text{And } B_{1y} = B_1 \cos \alpha = \frac{\mu_0 J}{2} r_1 \cos \alpha$$

$$\text{or, } B_{1y} = \frac{\mu_0 J}{2} b_1$$

Field at P due to the current I_2 in the cavity region is

$$B_2 = \frac{\mu_0 J}{2} r_2$$

Direction of this field is perpendicular to CP as shown. Its x and y components are



$$B_{2x} = - \frac{\mu_0 J}{2} r_2 \sin \beta$$

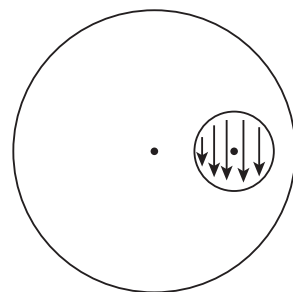
$$\text{or, } B_{2x} = - \frac{\mu_0 J}{2} d$$

$$\text{And } B_{2y} = \frac{\mu_0 J}{2} r_2 \cos \beta = \frac{\mu_0 J}{2} b_2$$

Obviously, B_{1x} and B_{2x} are equal and opposite and they cancel out. B_{1y} and B_{2y} add to produce a resultant in y direction.

$$B = \frac{\mu_0 J}{2} (b_1 + b_2) = \left(\frac{\mu_0 J}{2} \right) b$$

This field is independent of the position of the point P inside the cavity. The field is uniform perpendicular to OC .

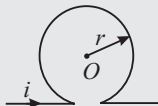


Field is uniform inside the cavity.

Worksheet 1

- A wire carrying a constant current, produces:
 - electric field due to the charge on free electrons inside it
 - magnetic field due to moving electrons
 - both electric and magnetic field
 - zero magnetic field if protons in the atoms are also taken into account

- Which of the followings is correct?
 - A magnet may get attracted by a current carrying loop.
 - A magnet will never get repelled by a current carrying loop.
 - A current carrying wire can attract or repel another current carrying wire but will exert no force on a magnet.
 - A current loop can never repel another current loop.

- An infinitely long straight conductor, carrying a current i is bent into a shape as shown in the figure. The circular part is nearly complete and has a radius r . Then the magnetic induction field at its centre will be
 

- $\frac{\mu_0}{4\pi} \frac{2i}{r} (\pi+1)$
- $\frac{\mu_0}{4\pi} \frac{2i}{r} (\pi-1)$
- Zero
- Infinite

- An alpha particle is rotating in a circle of radius 0.8 metre. It makes one rotation in two microseconds. The value of the magnetic field B at the centre of the circle will be

- $\frac{10^{-19}}{\mu_0}$
- $10^{-13} \mu_0$
- $2 \times 10^{-10} \mu_0$
- $\frac{2 \times 10^{-10}}{\mu_0}$

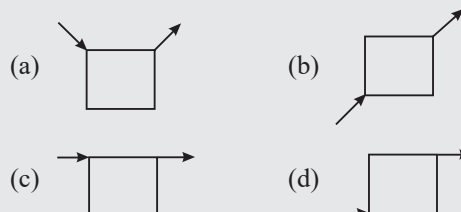
- A solenoid of 1.5 metre length and 4.0 cm diameter has 10 turn per cm. A current of 5 ampere is flowing through it. The magnetic field inside the solenoid is

- $2\pi \times 10^{-3}$ Tesla
- $2\pi \times 10^{-5}$ Tesla
- $4\pi \times 10^{-2}$ Gauss
- $2\pi \times 10^{-5}$ Gauss

- A copper wire of diameter 1.6 mm carries a current i . The maximum magnetic field due to this wire is 5×10^{-3} T. The value of i is

- 40 A
- 5 A
- 20 A
- 2 A

- Current flows through uniform, square frames as shown. In which case is the magnetic field at the centre of the frame not zero?



- A circular current coil has a radius R . The distance from the centre of the coil on the axis where the magnetic induction will be 1/8th to its value at the centre of the coil, is

- $\frac{R}{\sqrt{3}}$
- $R\sqrt{3}$
- $R\sqrt{3}R$
- $\frac{2}{\sqrt{3}}R$

- Two concentric coils X and Y of radii 16 cm and 10 cm lie in the same vertical plane containing north-south direction. X has 20 turns and carries 16 A. Y has 25 turns & carries 18A. X has current in anticlockwise direction and Y has current in clockwise direction for an observer, looking at the coils while himself facing the west. The magnitude of net magnetic field at their common centre is

- $5\pi \times 10^{-4} T$ towards West
- $13\pi \times 10^{-4} T$ towards East
- $13\pi \times 10^{-4} T$ towards West
- $5\pi \times 10^{-4} T$ towards East

- Two very long, straight, parallel wires carry steady current I in opposite directions. The distance between the wires is d . At a certain instant of time, a point charge q is at a point equidistant from the two wires, in the plane of the wires. Its instantaneous velocity v is perpendicular to this plane. The magnitude of the force due to the magnetic field acting on the charge at this instant is:

- $\frac{\mu_0 I q v}{2\pi d}$
- $\frac{\mu_0 I q v}{\pi d}$
- $\frac{2\mu_0 I q v}{\pi d}$
- 0

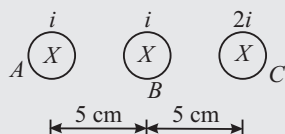
11. Two long mutually perpendicular conductors carrying currents I_1 and I_2 lie in one plane. Locus of the points at which the magnetic induction is zero, is a

- (a) circle with centre as the point of intersection of the conductor.
 (b) parabola with vertex as the point of intersection of the conductors.
 (c) straight line passing through the point of intersection of the conductors.
 (d) rectangular hyperbola

12. A long wire carries a steady current. It is bent into a circle of one turn and the magnetic field at the centre of the coil is B_0 . It is then bent into a circular loop of n turns. The magnetic field at the centre of the coil will now be

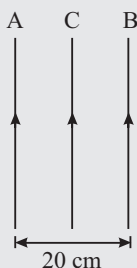
- (a) nB_0 (b) n^2B_0
 (c) $2nB$ (d) $2n^2B_0$

13. Three infinitely long conductors A , B and C carrying current are as shown in figure. The position of the point lying on the straight line AC where magnetic field is zero, is given by-



- (a) Between A and B at a distance of 3.2 cm from B
 (b) Between B and C at a distance of 3.2 cm from C
 (c) Between A and B at a distance of 1.3 cm from B
 (d) Between B and C at a distance of 1.3 cm from B

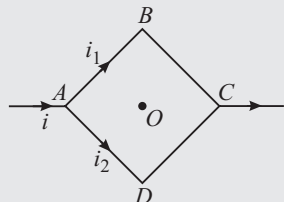
14. In the adjoining figure, two very long, parallel wires A and B carry currents of 10 ampere and 20 ampere respectively and are at a distance 20 cm apart. If a third wire C (length 15 cm) having a current of 10 ampere is placed midway between them then how much force will act on C ?



The direction of current in all the three wires is same

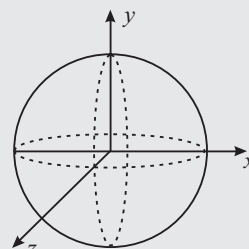
- (a) 3×10^{-5} N (left) (b) 3×10^{-5} N (right)
 (c) 6×10^{-5} N (left) (d) 6×10^{-5} N (right)

15. Figure shows a square loop $ABCD$ with edge length a . The resistance of the wire ABC is r and that of ADC is $2r$. The value of magnetic field at the centre of the loop is



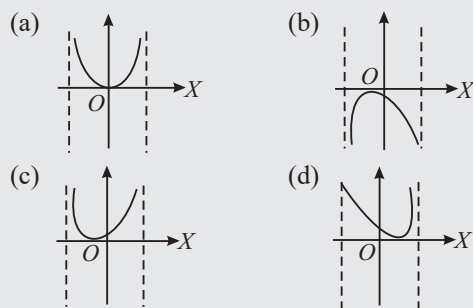
- (a) $\frac{\sqrt{2}\mu_0 i}{3\pi a} \odot$ (b) $\frac{\sqrt{2}\mu_0 i}{3\pi a} \otimes$
 (c) $\frac{\sqrt{2}\mu_0 i}{\pi a} \odot$ (d) $\frac{\sqrt{2}\mu_0 i}{\pi a} \otimes$

16. Three rings, each having equal radius R , are placed mutually perpendicular to each other and each having its centre at the origin of co-ordinate system. If current I is flowing through each ring then the magnitude of the magnetic field at the common centre is

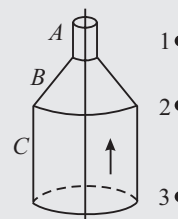


- (a) $\sqrt{3} \frac{\mu_0 I}{2R}$ (b) Zero
 (c) $(\sqrt{2}-1) \frac{\mu_0 I}{2R}$ (d) $(\sqrt{3}-\sqrt{2}) \frac{\mu_0 I}{2R}$

17. Two very long straight parallel wires, parallel to y -axis, carry currents $4I$ and I along $-y$ direction and $+y$ directions. The wires intersect the x -axis at the point $(d, 0, 0)$ and $(-d, 0, 0)$ respectively. The graphical variation of the z -component of the magnetic field as one moves along the x -axis from $x = -d$ to $x = +d$, is best given by :

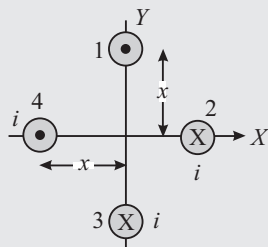


18. A long, straight, hollow conductor (tube) carrying a current has two sections A and C of unequal cross-sections joined by a conical section B . 1, 2 and 3 are points on a line parallel to the axis of the conductor. The magnetic fields at 1, 2 and 3 have magnitudes B_1 , B_2 and B_3 . Then



- (a) $B_1 = B_2 = B_3$
 (b) $B_1 = B_2 \neq B_3$
 (c) $B_1 < B_2 < B_3$
 (d) B_2 cannot be found unless the dimensions of the section B are known.

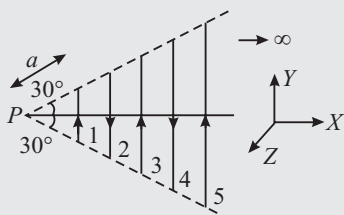
19. Figure shows the cross section of four infinite length parallel wires. Each wire produces magnetic field ' B ' at origin. What will be the resultant magnetic field at origin due to four wires?



- (a) $4B$
 (b) $\sqrt{2}B$
 (c) $2\sqrt{2}B$
 (d) Zero

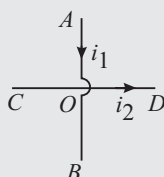
20. Infinite number of straight wires each carrying current I are placed at regular separations as shown in the figure. Adjacent wires have current in opposite direction. Net magnetic field at point P is -

(Given : $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$)



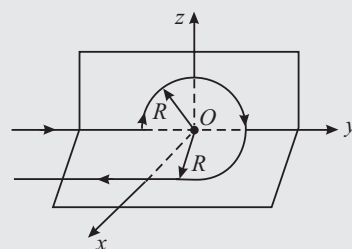
- (a) $\frac{\mu_0 I}{4\pi} \frac{\ln 2}{\sqrt{3}a} \hat{k}$ (b) $\frac{\mu_0 I}{4\pi} \frac{\ln 4}{\sqrt{3}a} \hat{k}$
 (c) $\frac{\mu_0 I}{4\pi} \frac{\ln 4}{\sqrt{3}a} (-\hat{k})$ (d) zero

21. Two straight and long conductors AOB and COD are perpendicular to each other and carry current of I_1 and I_2 . The magnitude of the magnetic field at a point P at a distance a from point O in a direction perpendicular to the plane $ABCD$ is :



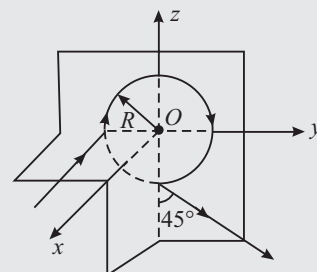
- (a) $\frac{\mu_0}{2\pi a} (I_1 + I_2)$ (b) $\frac{\mu_0}{2\pi a} (I_1 - I_2)$
 (c) $\frac{\mu_0}{2\pi a} (I_1^2 + I_2^2)^{1/2}$ (d) $\frac{\mu_0}{2\pi a} \frac{I_1 I_2}{(I_1 + I_2)}$

22. A long wire carrying current I has been given the shape shown in the figure. Apart from two straight portions, it has a semicircular part in yz plane and a quarter of a circular arc in xy plane. O is the centre of curvature of both the curved part. The magnetic field at point O is-



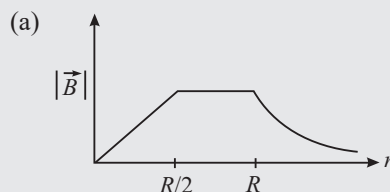
- (a) $B = \frac{-\mu_0 I}{4R} \hat{i} - \frac{\mu_0 I}{8\pi R} (\pi + 2) \hat{k}$
 (b) $B = +\frac{\mu_0 I}{4R} \hat{i} - \frac{\mu_0 I}{8\pi R} (\pi + 2) \hat{k}$
 (c) $B = \frac{\mu_0 I}{4R} \hat{i} - \frac{\mu_0 I}{4\pi R} (\pi + 2) \hat{k}$
 (d) none

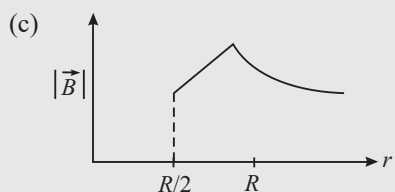
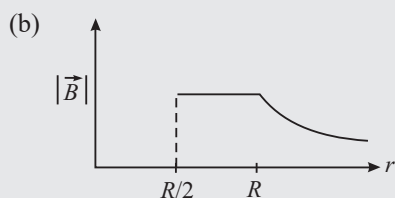
23. A long wire carrying current I has been given the shape shown in the figure. Apart from two straight portions, it has a part in yz plane that is three fourth of a circle. O is the centre of curvature of the curved part. The straight parts are very long. The magnetic field at point O is



- (a) $B = \frac{\mu_0 I}{4R} \hat{i} - \frac{\mu_0 I}{4\pi R} (\pi + 2) \hat{k}$
 (b) $B = +\frac{\mu_0 I}{4R\pi} (\sqrt{2} - 1) \hat{i} - \frac{\mu_0 I}{4\pi R} \hat{k} - \frac{3\mu_0 I}{8R} \hat{i}$
 (c) $B = \frac{\mu_0 I}{R} \hat{i} - \frac{\mu_0 I}{4\pi R} (\pi + 2) \hat{k}$
 (d) none

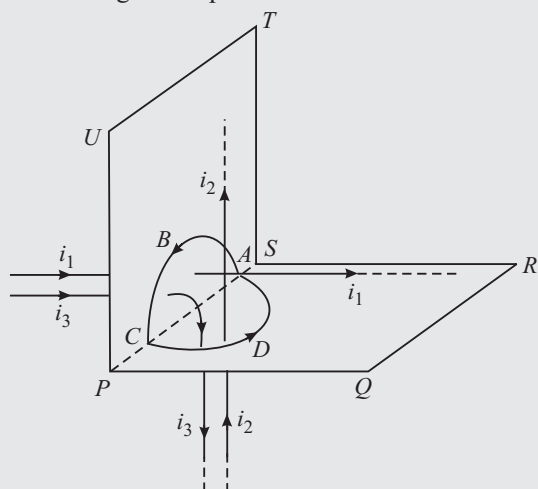
24. An infinitely long hollow conducting cylinder with inner radius $R/2$ and outer radius R carries a uniform current density on its cross section. The magnitude of the magnetic field, $|\vec{B}|$ as a function of the radial distance r from the axis is best represented by:





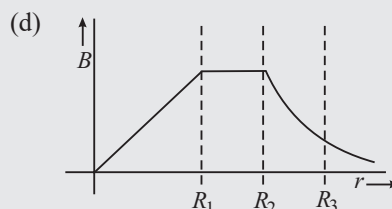
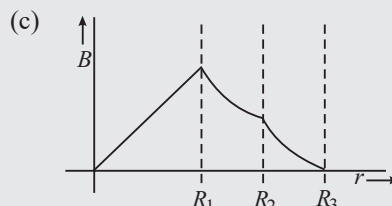
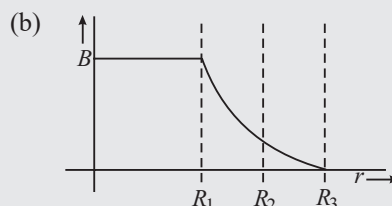
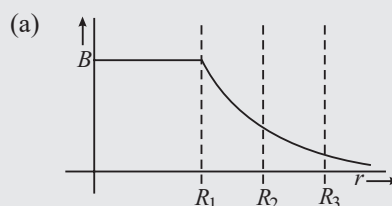
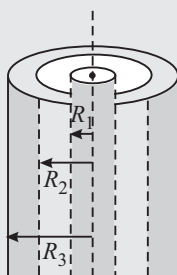
(d) none of the above

25. Figure shows an amperian path $ABCD$. Part ABC is in vertical plane $PSTU$ while part CDA is in horizontal plane $PQRS$. Direction of circulation along the path is shown by an arrow near point B and at point D . The value of $\oint \vec{B} \cdot d\vec{l}$ for this path according to Ampere's law will be:



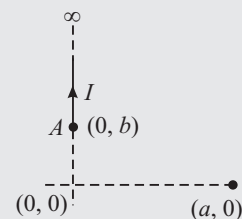
- (a) $(i_1 - i_2 + i_3) \mu_0$
 (b) $(-i_1 + i_2) \mu_0$
 (c) $i_3 \mu_0$
 (d) $(i_1 + i_2) \mu_0$

26. A coaxial cable is made up of two conductors. The inner conductor is solid and is of radius R_1 & the outer conductor is hollow of inner radius R_2 and outer radius R_3 . The space between the conductors is filled with air. The inner and outer conductors are carrying currents of equal magnitudes and in opposite directions. Then the variation of magnetic field with distance from the axis is best plotted as:



27. An infinitely long wire carrying current I is along Y axis and its one end is at point $A(0, b)$ while the wire extends upto $+\infty$. The magnitude of magnetic field strength at point $(a, 0)$ is

- (a) $\frac{\mu_0 I}{4\pi a} \left(1 + \frac{b}{\sqrt{a^2 + b^2}} \right)$
 (b) $\frac{\mu_0 I}{4\pi a} \left(\frac{b}{\sqrt{a^2 + b^2}} \right)$
 (c) $\frac{\mu_0 I}{4\pi a} \left(1 - \frac{b}{\sqrt{a^2 + b^2}} \right)$



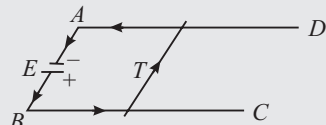
(d) None of these

28. A straight solenoid of length 40 cm and radius 0.1 cm is wound with two layers of wires. The inner layer has 350 turns and the outer layer has 250 turns. The current is 4A in the same direction in both the layers. The magnetic field at the centre of the solenoid will be:

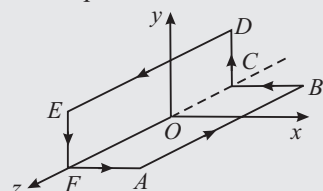
- (a) $7.54 \times 10^{-2} T$ (b) $1.54 \times 10^{-3} T$
 (c) $4.54 \times 10^{-3} T$ (d) $7.54 \times 10^{-3} T$

Worksheet 2

1. AD and BC are smooth, parallel and horizontal rails on which a conductor T can slide. A cell E , drives current I through the rails and T . then:

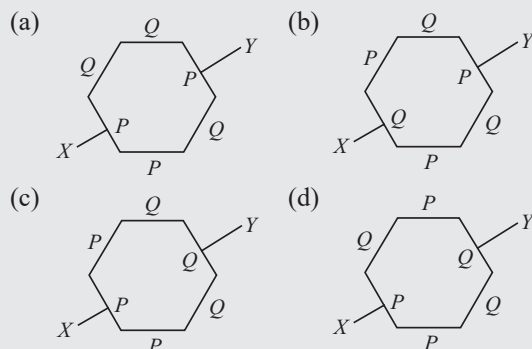


- the current in the rails will set-up a magnetic field over T
 - T will experience a force to the right side
 - T will experience a force to the left side
 - T will not experience any force.
2. Two parallel conductors carrying current in the same direction attract each other, while two parallel beams of electrons moving in the same direction repel each other. Which of the following statements are necessary to provide reasons for this?
- The conductors are electrically neutral
 - The conductors produce magnetic fields on each other
 - The electron beams do not produce magnetic fields on each other
 - The magnetic forces caused by the electron beams on each other are weaker than the electrostatic forces between them
3. In the figure shown, $ABCDEF$ represents a square loop of side length l folded in two equal parts so that half of it lies in xz plane and the other half lies in the yz plane. The origin ' O ' is the centre of the frame. The loop carries current ' i '.

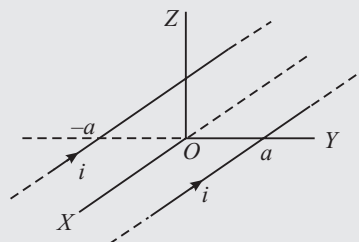


- The magnetic field at ' O ' due to the segment BC is $\frac{\mu_0 i}{2\sqrt{2}\pi l} \hat{j}$
- The magnetic field at ' O ' due to the segment DE is $\frac{\mu_0 i}{2\pi l} (-\hat{i} + \hat{j})$
- The magnetic field at ' O ' is $\frac{\sqrt{2}\mu_0 i}{\pi l} (\hat{i} + \hat{j})$
- The magnetic field at ' O ' is $\frac{\mu_0 i}{\sqrt{2}\pi l} (\hat{i} + \hat{j})$

4. The following figure show hexagons made of wires of two different materials P and Q . A current enters the hexagon at point X and leaves at Y . X and Y are midpoints of the respective sides. In which case the magnetic field at the centre is not zero?



5. Consider the magnetic field produced by a finitely long current carrying wire.
- The lines of field will be concentric circles with centres on the wire.
 - There can be two points in the same plane, perpendicular to the wire, where magnetic fields are same.
 - There can be number of points where the magnetic field is same.
 - The magnetic field at a point is inversely proportional to the distance of the point from the wire.
6. Two long and thin, parallel wires carrying equal currents in the same direction are fixed parallel to the x -axis, one passing through $y = a$ and the other through $y = -a$. The resultant magnetic field due to the two conductors at any point is B . Which of the followings are correct?

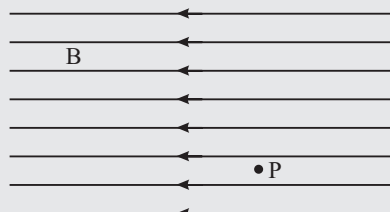


- $B = 0$ for all points on the x -axis
- At all points on the y -axis, excluding the origin, B has only a z -component.
- At all points on the z -axis, excluding the origin, B has only a y -component.
- B cannot have an x -component.

7. A long straight wire carries a current i along the x -axis. Consider the points $A(0, 1, 0)$, $B(0, 1, 1)$, $C(1, 0, 1)$ and $D(1, 1, 1)$.

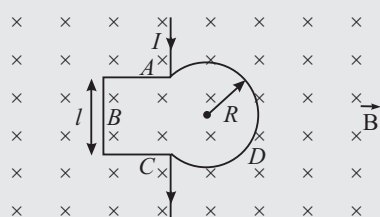
- The magnetic field at D has magnitude $B = \frac{\mu_0 i}{2\sqrt{2}\pi}$
- Points A and C have magnetic fields of same magnitude
- Points B and C have magnetic fields of same magnitude
- The magnetic field at D makes an angle of 45° with the xy plane

8. Consider the uniform magnetic field shown:



Starting from point P and without leaving the region of magnetic field, is it possible to choose a closed path (that is, a path that returns to P) for which the line integral of the magnetic field is non-zero?

- Yes, but the line integral can only be positive
 - Yes, but the line integral can only be negative
 - Yes
 - No
9. The figure shows a conducting loop $ABCD$ placed in a uniform magnetic field perpendicular to its plane. The part ABC is the $(3/4)$ th portion of the square of side length l . The part ADC is a circular arc of radius R . The points A and C are connected to a battery which supply a current I to the circuit. The magnetic force on the loop due to the field B is



- The magnetic force on the loop due to the field B is BIl .
- The magnetic force on the loop due to the field B is $2BIl$.
- The magnetic force on the loop due to the field B is $\frac{BIIR}{l+R}$.
- The magnetic field due to the loop is zero everywhere outside it.

10. A current-carrying ring is placed inside an ideal solenoid. The plane of the ring is perpendicular to the axis of the solenoid. When a current is passed through the solenoid

- There is no net force on the ring.
- There is no torque on the ring.
- If the ring is rotated so as to make its plane normal to the axis of the solenoid, it may tend to contract.
- If the ring is rotated so as to make its plane normal to the axis of the solenoid, it may tend to expand.

11. A region of space has no currents. The static magnetic field in the region

- cannot be both unidirectional and non-uniform
- can be unidirectional
- can be non-uniform
- can be both unidirectional and non-uniform

12. A thin hollow cylinder has infinite length and carries a uniform current that is circulating along its surface as shown. The current per unit length of the cylinder is λ .



- Magnetic field inside the cylinder is $\frac{\mu_0 \lambda}{2}$
- Magnetic field inside the cylinder is $\mu_0 \lambda$
- There is no magnetic field inside the cylinder
- The magnetic field just outside the cylinder is larger than the field just inside it.

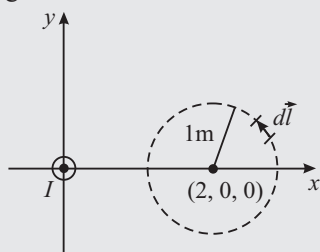
13. A thin sheet of width L and thickness t has a long length. It is bent into the shape of a long thin walled pipe having a semicircular cross section. It carries a current I along its length. The current density is uniform over the entire cross section of area Lt . O is the centre of curvature of the middle section (perpendicular to length) of the conductor. Field at O is B_1 . Now, the same sheet is bent so as to have a cross section in the shape of a quarter of a circle. It is given a current $I/2$ which distributes uniformly on its cross section. O is the centre of curvature of the middle cross section of the conductor. Field at O has magnitude B_2 .

- $B_2 = B_1$
- $B_2 = \frac{B_1}{2\sqrt{2}}$
- Direction of B_1 is parallel to the extreme radius of the semicircular cross section of the conductor.
- Direction of B_2 is parallel to the extreme radius of the cross section of the conductor.

14. A metal ring of radius $r = 0.5$ m with its plane normal to a uniform magnetic field B of induction 0.2 T carries a current $I = 100$ A. Assume that the magnetic field produced by the current does not distort the uniform magnetic field and there is no magnetic interaction between the various segments of the ring.

- The force applied by one half of the ring on the other equal half is 20 N
- The force applied by one third of the ring on the other two third part is 10 N
- No tension is developed in the ring
- The tension (or the compression force) developed in the ring is 10 N

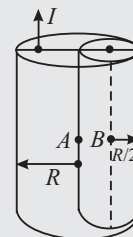
15. An infinitely long wire lying along z -axis carries a current I , flowing in positive z -direction. There is no other current. Consider a circle in x - y plane with centre at $(2, 0, 0)$ m and radius 1 m. Divide the circle in small segments and let dl denote the length of a small segment in anticlockwise direction, as shown.



- The line integral $\oint \vec{B} \cdot d\vec{l}$ of the magnetic field $\vec{B}$ along the perimeter of the given circle moving from the point $(1,0,0)$ to $(3,0,0)$ is 0 .
- The line integral $\oint \vec{B} \cdot d\vec{l}$ of the magnetic field $\vec{B}$ along the perimeter of the given circle moving from the point $(1,0,0)$ to $(3,0,0)$ is $\frac{\mu_0 I}{8}$
- Consider two points $T(3, 0, 0)$ and $P(2, 1, 0)$ on the given circle. The magnitude of the path integral $\int_T^P \vec{B} \cdot d\vec{l}$ of the magnetic field $\vec{B}$ along the perimeter of the given circle from T to P is $\frac{\mu_0 I}{8}$
- The value of $\int_A^B \vec{B} \cdot d\vec{l}$ between any two points A along the perimeter of the given circle is 0

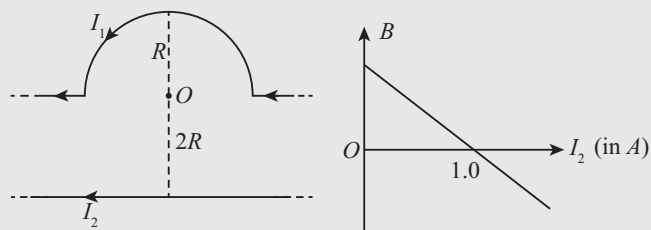
16. From a long cylindrical conductor of radius R , a cylinder of radius $R/2$ is removed, as shown. Current flowing in the remaining cylinder is I . Current is uniformly distributed on the cross section. Magnetic field strength is:

- zero at point A
- zero at point B
- $\frac{\mu_0 I}{3\pi R}$ at point A
- $\frac{\mu_0 I}{3\pi R}$ at point B

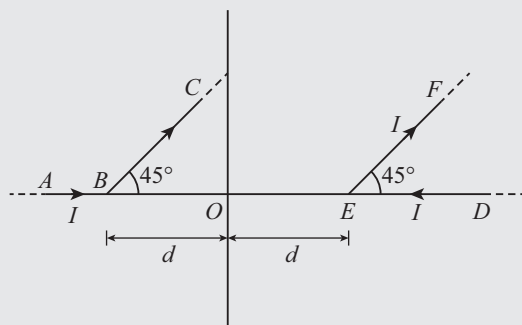


Worksheet 3

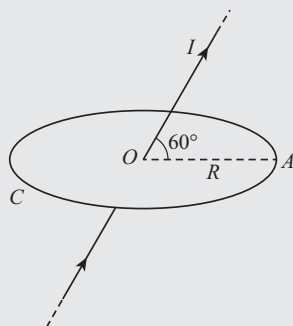
1. A long straight wire is laid along the x axis. It carries current I in positive x direction. Write the magnetic field vector at a point having co-ordinates (x, y, z) .
2. A long wire has a semicircular part of radius R (see figure) and carries a current I_1 . Another long wire carries a current I_2 and runs parallel to the first wire. This wire is at a distance $2R$ from the centre of curvature (O) of the semicircle. The two wires lie in the same plane. When current I_2 is increased from zero, the magnetic field at O changes as represented in the graph. Find the value of current I_1 .



3. Two long wires ABC and DEF carry equal current I as shown. The wires are bent at B and E respectively. The distance $OB = OE = d$. Find the direction and magnitude of magnetic field at O .



4. A circle C has radius R . A long and straight current carrying wire pierces through its centre making an angle of 60° with the plane of the circle. The wire carries a current I .



- (a) Does the magnetic field at all points on the circumference of C has the same magnitude?

- (b) Find the line integral $\oint \vec{B} \cdot d\vec{l}$ around the circle C .

- (c) Find the magnitude of the magnetic field at the point A indicated in the figure.

5. A short solenoid has its cross sectional radius r large compared to its length l ($r \gg l$). The number of turns on unit length of the solenoid is n . Find the approximate value of the field at the centre of the solenoid when the current is I .

6. A long, straight wire lies in the xy plane along the line $y + x - 3 = 0$. It carries a current $I = 10A$. Find the magnetic field produced by the wire at points.

- (a) $(0, 0, 0)$

- (b) $(6, 0, 0)$

- (c) $(1, 2, 2)$.

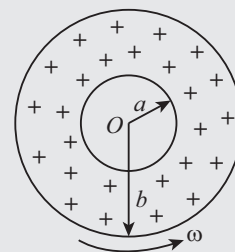
All co-ordinates are in meter. The direction of current is along $\overrightarrow{AB}$, where A and B are the points where the wire intersects the x and y axes.

7. Two straight, long and thin parallel wires are spaced $d = 0.1m$ apart and each, carry a current of $10A$. A point P is located at a distance d from both the wires. Find the magnetic field at P when

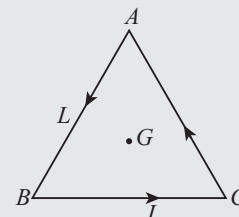
- (a) both the wires have current in the same directions.

- (b) both the wires have current in the opposite directions.

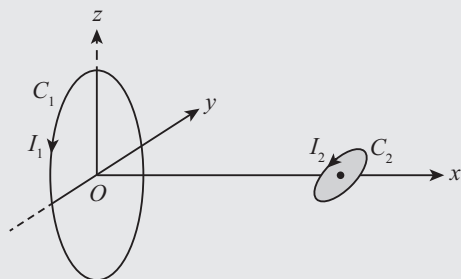
8. A non conducting disc carries a uniform surface charge density σ in the region defined by $a < x < b$, where x is the radial distance of a point on the surface of the disc from its centre. The disc is made to rotate with an angular speed ω about its central axis. Find the magnetic field at the centre (O) of the disc.



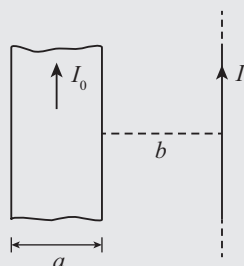
9. A current loop, having current I , is in the shape of an equilateral triangle of side length L . Find the magnetic field at the centroid (G) of the triangle.



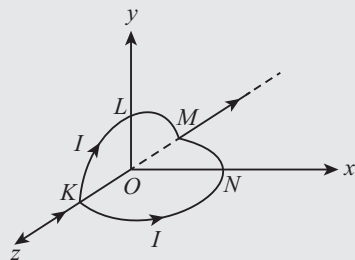
10. A circular coil (C_1) has radius R and it carries a current I_1 . It lies in the yz plane. A small circular loop (C_2) of radius r ($\ll R$) is placed at a distance x from the centre of the loop C_1 . The loop C_2 has its centre on the x axis and lies in xy plane. Estimate the magnetic torque on C_1 when a current I_2 is given to it.



11. A long, thin strip has a width a . It carries a current I_0 along the length which is distributed uniformly along the width. In the plane of the strip, there is another wire carrying a parallel current I . The distance of the wire from the edge of the strip is b . Find the magnetic force on a segment of length L of the wire.

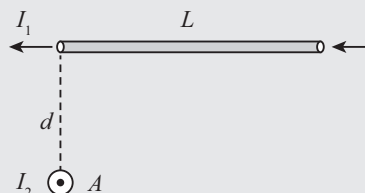


12. A circular loop of radius R is bent along a diameter and given a shape as shown in the figure. One of the semicircles (KNM) lies in the xz plane and the other semicircular part lies in the yz plane with their centre of curvature at the origin. Current $2I$ enters the loop at K , divides equally into two parts along the semicircles and leaves the loop at M .



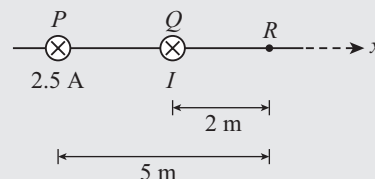
- (a) A particle having charge q is projected from the origin with a velocity $\vec{v} = -v_0 \hat{i}$. Find the instantaneous force $\vec{F}$ on the particle. Neglect gravity
- (b) An external magnetic field $B_0 \hat{j}$ is applied. Write the magnetic force on the loop.
13. A thin copper wire of length L carries a current I_1 . It is held at a height d above another long wire (A) carrying current I_2 . The wire A is exactly below one of the ends of the copper wire and is directed perpendicular to the

orientation of the copper wire. Find the magnetic force experienced by the copper wire.

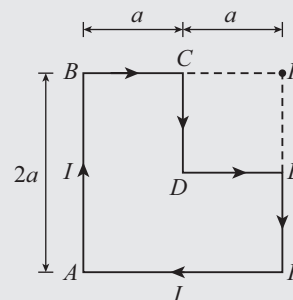


$$\text{Given: } \int \frac{2xdx}{a^2 + x^2} = \ln(a^2 + x^2) + C$$

14. Two long parallel wires carrying currents 2.5A and I in the same direction (directed into the plane of the paper) are held at point P and Q respectively such that they are parallel to each other. The points P and Q are located at a distance of 5m and 2m respectively from a collinear point R .

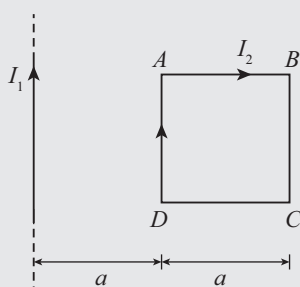


- (a) An electron moving with a velocity of $4 \times 10^5 \text{ m/s}$ along the positive x direction experiences a force of $3.2 \times 10^{-20} \text{ N}$ at the point R . Find the current I .
- (b) Find all the positions at which a third long parallel wire carrying a current of magnitude 2.5A may be placed, so that the magnetic field at the point R is zero.
15. Find the magnetic field at the point P due to the current loop shown in the figure.



16. A long cylindrical conductor of radius R carries a current. The current density (j) is a function of radius r and varies as $j = ar$, where a is a positive constant. Find the expression for magnetic field B
- (a) at a distance r from the axis such that $r < R$, and
- (b) at a distance r from the axis such that $r \geq R$.
- Ignore the effect of the material of the conductor on the value of field.

17. A square loop of side length a carries a current I_2 . In the plane of the loop there is a long straight wire having current I_1 . This wire is parallel to the side DA of the wire. Find the magnetic force on the side AB of the square loop. What is magnetic torque on the loop?

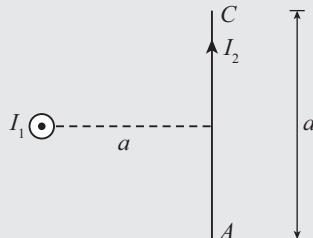


18. A long horizontal wire AB , which is free to move in a vertical plane and carries a steady current of 20A, is in equilibrium at a height of 0.01m above another parallel long wire CD which is fixed in a horizontal position and carries a current of 30A, as shown. Show that when AB is slightly depressed and released, it executes simple harmonic motion. Find the time period of oscillations.

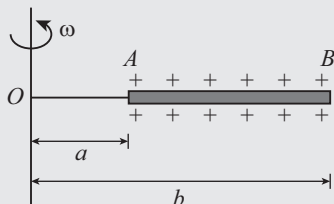


19. Three long, thin wires each carrying current i in the same direction, are in xy plane. The central wire passes through the origin and is along the y axis. The other two wires are along the lines $x = \pm d$. Find the locus of all the points in xy plane where the magnetic field is zero, apart from the y axis.
20. In the last question, the central wire is displaced slightly in z direction and released. Show that it performs SHM. If the linear mass density of the wire is λ find the frequency of oscillations. Neglect gravity.

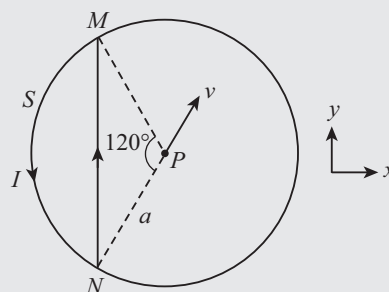
21. A long straight wire is perpendicular to the plane of the figure. It carries current I_1 . Another straight wire AC is in the plane of the figure. It has a length a and it is placed symmetrically with respect to the other wire at a distance a . Find the force experienced by wire AC in the magnetic field of I_1 .



22. A thin nonconducting rod AB carries a linear charge density λ . It is rotated with an angular velocity ω about point O that is at a distance a from its end (see figure). Find the magnetic field at O .

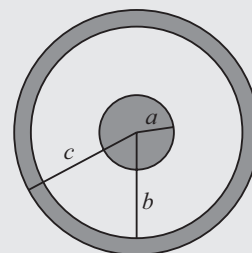


23. A wire loop has a straight arm NM and a circular arc MSN . It carries current I as shown.



- (a) If a particle having mass m and charge $+Q$ is placed at the centre P of the circular arc and given a velocity v along NP , find its instantaneous acceleration.
- (b) If an external magnetic field $\vec{B} = B\hat{i}$ is applied, find the force and the torque acting on the loop due to this field.

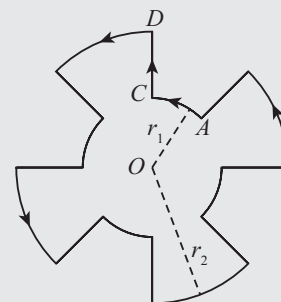
24. A long coaxial cable consist of an inner wire of radius a and an outer shell of inner and outer radii b and c respectively. The inner wire of radius a carries a current I directed into the plane of the figure and the current returns through the outer shell (coming out of the plane of the figure). The current density is uniform for the inner wire as well as the outer shell. Find the magnetic field at a distance x from the axis for



Cross section of the cable

- (a) $x < a$ (b) $a < x < b$
(c) $b < x < c$ (d) $x > c$

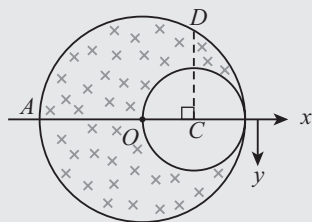
25. A closed current loop has eight alternating arcs of radii $r_1 = 0.08$ m and $r_2 = 0.12$ m. Each arc has a common centre of curvature at O . Each arc subtends the same angle at O . The current in the loop is $I = 10$ A.



- (a) Find the magnetic field produced by the loop at O .
- (b) An infinitely long straight wire carrying a current $I_0 = 10$ A is kept perpendicular to the plane of the figure, passing through O . The current is directed out of the plane of the paper. What is the force acting on this straight wire due to the current in the loop? What is the force acting on

the arc AC and the straight segment CD due to the current I_0 ? [$\ln(1.5) = 0.41$]

26. Figure shows the cross section of a long cylindrical current carrying wire of radius a . The wire has a cylindrical cavity of radius $\frac{a}{2}$ as shown.



The axis of the wire passes through O and the axis of the cavity passes through C . Current density is j .

- Find the magnetic field at a point A on the surface of the wire.
 - Find the y component of the magnetic field at a point D on the surface of the wire. CD is perpendicular to OC .
27. A current I along the square edge of a cube produces a magnetic field of magnitude B_0 at the centre of the cube (Fig 1). Consider another identical cube. A current I flows along the various edges forming a loop $ABCDEF$. Find the magnetic field at the centre of the cube. Express your answer in unit vector $\hat{i}$, $\hat{j}$ and $\hat{k}$.

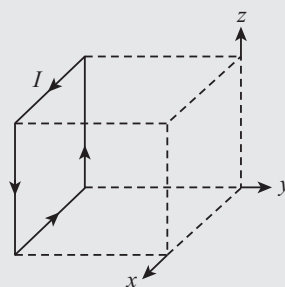


Fig (1)

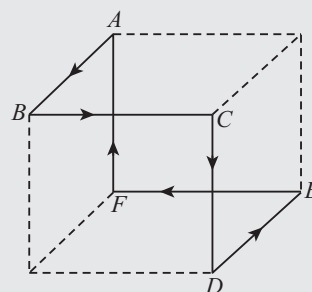
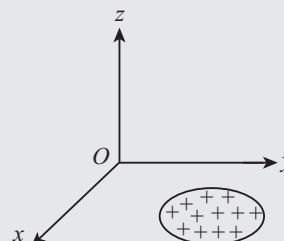


Fig (2)

28. An elliptical plate is made of plastic and carries some charge q on it. The plate is placed in xy plane. The electrostatic potential at the origin due to charge on the plate is V . Now the plate is made to rotate about z -axis with angular speed ω . Find the magnetic field at the origin due to charge on the plate.



Answer Sheet

Your Turn

1. Directed into the plane of the paper
2. $\frac{\mu_0 I}{4\pi x}$
3. $\frac{2\sqrt{2}\mu_0 I}{\pi a} \otimes$
4. 0
5. At a distance $\frac{d}{3}$ from the wire having current I
6. $\frac{\mu_0 I}{4\pi d}$ along $\vec{PM}$
7. $\frac{\mu_0 \pi n^2 I}{L}$
8. $\frac{\mu_0 I}{4R} \left(\frac{3}{2} + \frac{1}{\pi} \right)$
9. $5\pi (\hat{j} - 2\hat{k}) \mu T$
10. $\frac{35}{144} \frac{\mu_0 I}{r} \otimes$
11. Zero
12. $1.08 \times 10^{-5} T$
13. $4\pi \times 10^{-4} T$
14. Zero
15. Yes
16. No
17. $K_1 = K_2$
18. Field at P_1 is zero. Field at P_2 is $\frac{\mu_0 I}{2\pi x}$
19. $B_A = B_D = 0$; $B_C = \mu_0 j a$
20. (a) $6\pi \times 10^{-3} T\text{-m}$ (b) $3\pi \times 10^{-3} T$
21. $9.6 \times 10^{-5} N/m$
22. At a distance d to the right of wire having current I

Worksheet 1

1. (b)
2. (a)
3. (b)
4. (b)
5. (a)
6. (c)
7. (c)
8. (b)
9. (a)
10. (d)
11. (c)
12. (b)
13. (a)
14. (b)
15. (b)
16. (a)
17. (b)
18. (a)
19. (c)
20. (b)
21. (c)
22. (a)
23. (b)
24. (d)
25. (d)
26. (c)
27. (c)
28. (d)

Worksheet 2

1. (a, b)
2. (a, b, d)
3. (a, c)
4. (a)
5. (a, c)
6. (a, b, c, d)
7. (a, b, c, d)
8. (d)
9. (a)
10. (a, c, d)
11. (a, b, c)
12. (b)
13. (b, c)
14. (a, d)
15. (a, c)
16. (c, d)

Worksheet 3

1. $\frac{\mu_0 I}{4\pi(y^2 + z^2)} (-z\hat{j} + y\hat{k})$
2. 0.16A
3. $\frac{\mu_0 I}{2\pi d} \otimes$
4. (a) No (b) $\mu_0 I$ (c) $\frac{\mu_0 I}{\sqrt{3}\pi R}$
5. $\frac{\mu_0 n I l}{2r}$
6. (a) $(0.94\mu T)\hat{k}$ (b) $(-0.94\mu T)\hat{k}$ (c) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})\mu T$
7. (a) $2\sqrt{3} \times 10^{-5} T$ (b) $2 \times 10^{-5} T$
8. $\frac{\mu_0 \sigma \omega}{2}(b - a)$
9. $\frac{9\mu_0 I}{2\pi L} \odot$
10. $\frac{\mu_0 \pi I_1 I_2 R^2 r^2}{2(R^2 + x^2)^{3/2}}$
11. $\frac{\mu_0 I_0 I L}{2\pi a} \ln \left(1 + \frac{a}{b} \right)$
12. (a) $\vec{F} = -\left(\frac{\mu_0 q v_0 I}{4R} \right) \hat{k}$ (b) $(4BIR) \hat{i}$
13. $F = \frac{\mu_0 I_1 I_2}{4\pi} \ln \left(\frac{d^2 + L^2}{d^2} \right)$ in the direction of current I_2
14. (a) 4A (b) At 1 m from R on either side of it

$$15. \frac{\mu_0 I}{4\sqrt{2}\pi a} \odot$$

$$16. \text{(a)} \frac{\mu_0 a r^2}{3} \quad \text{(b)} \frac{\mu_0 a R^3}{3r}$$

$$17. \frac{\mu_0 I_1 I_2}{2\pi} \ln(2) \quad \text{Torque is zero.}$$

$$18. T = 0.2\text{s}$$

$$19. \text{The field is zero along the lines } x = \pm \frac{d}{\sqrt{3}}$$

$$20. f = \frac{i}{2\pi d} \sqrt{\frac{\mu_0}{\pi\lambda}}$$

$$21. \text{zero}$$

$$22. \frac{\mu_0 \lambda \omega}{4\pi} \ln\left(\frac{b}{a}\right)$$

$$23. \text{(a)} \frac{0.11\mu_0 I Q v}{2am} (\hat{j} - \sqrt{3} \hat{i}) \quad \text{(b)} 0.6Ia^2 B \hat{j}$$

$$24. \text{(a)} \frac{\mu_0 I x}{2\pi a^2} \quad \text{(b)} \frac{\mu_0 I}{2\pi x} \quad \text{(c)} \frac{\mu_0 I}{2\pi x} \frac{(c^2 - x^2)}{(c^2 - b^2)} \quad \text{(d)} 0$$

$$25. \text{(a)} 6.54 \times 10^{-5} \text{ T } \odot \quad \text{(b)} \text{Zero, } F_{AC} = 0, F_{CD} = 8.2 \times 10^{-6} \text{ N}$$

$$26. \text{(a)} \frac{5\mu_0 j a}{12} \text{ in y direction} \quad \text{(b)} \frac{\mu_0 j a}{4}$$

$$27. \vec{B} = B_0 (-\hat{i} + \hat{j} + \hat{k})$$

$$28. \mu_0 \epsilon_0 \omega V$$

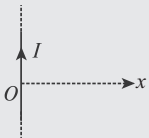
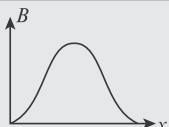
Miscellaneous Problems on Chapters 1 and 2

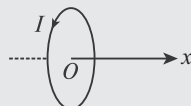
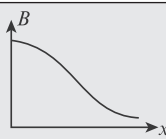
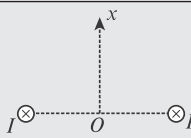
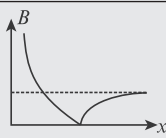
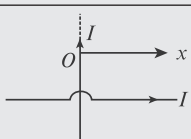
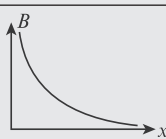
MATCH THE COLUMNS

1. Column-I gives various combinations of gravitational field, electrostatic field and magnetic field that can exist in a region. Column-II gives the possible paths of a charge particle in these fields. Match them.

Column-I		Column-II	
(A)	A uniform magnetic field and a uniform gravitational field	(P)	Straight line with acceleration
(B)	A uniform electric field and a uniform gravitational field	(Q)	Straight line without acceleration
(C)	Electric field of a point charge	(R)	Circle
(D)	A uniform electric field parallel to a magnetic field	(S)	Helix with variable pitch

2. The entries in Column-I show the source current producing magnetic field. The graphs in column II show the variation of the magnitude of magnetic field along the x axis. Match the entries in Column-I with the corresponding graph in Column-II.

Column-I		Column-II	
(A)	Long straight current carrying wire 	(P)	

(B)	Circular current carrying wire 	(Q)	
(C)	 Two long current carrying wires having current into the plane of the figure	(R)	
(D)	 Two long perpendicular current carrying wires	(S)	

3. Column-I shows a long wire carrying current I . The curved parts are area of a circle of radius r . O is the centre of curvature of the circular arc in each figure. Magnetic field at O is represented by B . Match the figures in Column-I with the possible values of B in Column-II.

[Take $\frac{1}{\pi} \simeq 0.3$]

3.2 Magnetism and Electromagnetic Induction

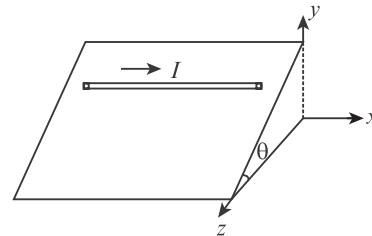
Column-I		Column-II	
(A)		(P)	$B < 0.65 \frac{\mu_0 I}{r}$
(B)		(Q)	$B < 0.4 \frac{\mu_0 I}{r}$
(C)		(R)	$B > 0.15 \frac{\mu_0 I}{r}$
(D)		(S)	$0.15 \frac{\mu_0 I}{r} < B < \frac{0.45 \mu_0 I}{r}$

4. Match the current system in Column-I to the observed results in Column-II. The straight wires are very long.

Column-I		Column-II	
(A)		(P)	Torque on the loop is non-zero
(B)		(Q)	1 and 2 repel each other
(C)		(R)	1 and 2 do not exert force on one another

(D)		(S)	Magnetic fields at point P due to wires 1 and 2 are perpendicular to each other
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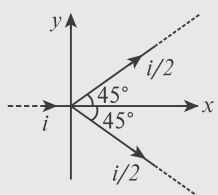
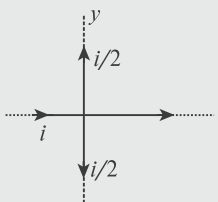
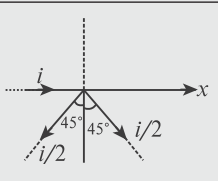
5. A straight horizontal conductor is placed on a rough incline. It carries current I in x direction as indicated. Inclination angle θ is higher than the angle of repose. A uniform magnetic field B is applied. Column-I gives direction of the field and Column-II gives their effects on the conductor. Match them. (Cross-section of the rod is rectangular and it does not roll)



Column-I		Column-II	
(A)	$\vec{B} = B_0 \hat{j}$	(P)	The conductor may remain in equilibrium
(B)	$\vec{B} = -B_0 \hat{j}$	(Q)	The conductor cannot remain in equilibrium
(C)	$\vec{B} = B_0 \hat{k}$	(R)	Frictional force on the rod can be up along the incline
(D)	$\vec{B} = -B_0 \hat{k}$	(S)	Frictional force on the rod can be down along the incline

6. Column-I gives four situations in which three or four current carrying, straight semi-infinite wires are arranged as shown. Column-II talks about magnetic field at a point P having co-ordinates $(0, 0, d)$. Match the entries in the two columns.

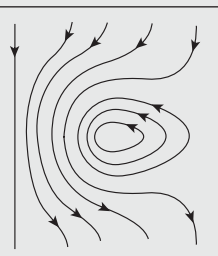
Column-I		Column-II	
(A)		(P)	The x component of the magnetic field at P is zero

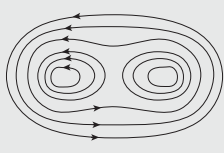
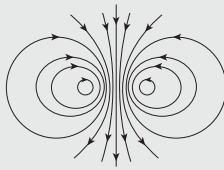
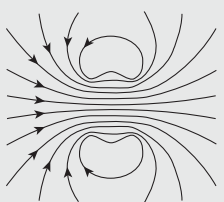
(B)		(Q)	The z component of the magnetic field at P is zero
(C)		(R)	The magnitude of the magnetic field at point P is $\frac{\mu_0 i}{4\pi d}$
(D)		(S)	The magnitude of the magnetic field at the point P is less than $\frac{\mu_0 i}{2\pi d}$

7. Column-I lists the changes made in design of a moving coil galvanometer. Column-II contains the corresponding effects. Match the changes (in Column-I) to their effects in Column-II.

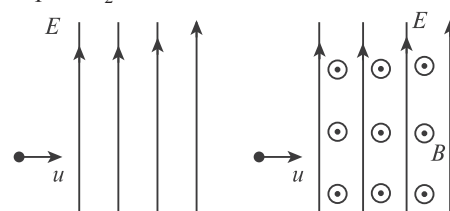
Column-I		Column-II	
(A)	Using aluminium wire instead of copper wire of same cross-section and length	(P)	Current sensitivity increases
(B)	Doubling the number of turns in the coil	(Q)	Voltage sensitivity decreases
(C)	Doubling the length of the rectangular coil	(R)	The range to measure current decreases
(D)	Replacing the spring with a much stiffer spring	(S)	The range to measure voltage increases

8. Column-I shows magnetic field lines in the plane of the figure. Column-II lists the possible sources. Match them.

Column-I		Column-II	
(A)		(P)	A straight current carrying wire going out of the plane of the paper placed in a uniform external field

(B)		(Q)	Two long straight current carrying wires perpendicular to the plane of the figure carrying current in same direction.
(C)		(R)	Two long straight current carrying wires perpendicular to the plane of the figure. Wires carrying current in opposite direction
(D)		(S)	Two circular co-axial coils placed at a separation

9. A particle having charge $+q$ enters into a region having a uniform electric field E , while moving perpendicular to the field at a speed u . It exits the field region with speed v_1 . Now, the experiment is repeated with a uniform magnetic field B superimposed on the electric field keeping everything else unchanged. The magnetic field is perpendicular to the electric field as shown. This time the particle exited the field region on the opposite side with speed v_2 .



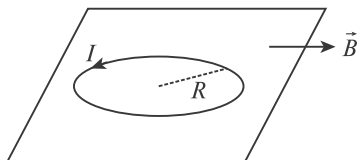
Match the conditions given in Column-I with the results in Column-II.

Column-I		Column-II	
(A)	$u = \frac{E}{B}$	(P)	$v_2 = v_1$
(B)	$u < \frac{E}{B}$	(Q)	$v_2 < v_1$
(C)	$u > \frac{E}{B}$	(R)	$v_2 > v_1$
(D)	The direction of magnetic field is reversed		

Passage Based Problems

Passage 1

A uniform circular current loop of radius R lies on a horizontal surface.



A uniform magnetic field, of strength B , exists in the entire space parallel to the horizontal surface. The value of B is increased from zero to a value B_0 at which the current loop just begins to show some movement. Mass of the loop is M and acceleration due to gravity is g .

- As B is increased from zero to B_0 , the normal force applied by the horizontal surface on the loop
 - increases
 - decreases
 - remains constant in magnitude but its point of application shifts
 - finally becomes zero
- The loop is held on the horizontal surface by an external agent and the value of B is made $2B_0$. Now, the loop is released. Immediately after the loop is released, its angular acceleration is

(a) $\frac{2g}{3R}$	(b) $\frac{g}{2R}$
(c) $\frac{3g}{2R}$	(d) $\frac{g}{R}$
- In the last question, the normal force applied by the surface on the loop immediately after it is released is

(a) $\frac{Mg}{3}$	(b) $\frac{2Mg}{3}$
(c) $\frac{5Mg}{3}$	(d) Mg

Passage 2

A moving coil galvanometer has coil of area A and number of turns N . The moment of inertia of the coil and cylinder assembly about the rotation axis is I . The radial magnetic field in the space has strength B . It is observed that the coil rotates by an angle of $\frac{\pi}{2}$ in equilibrium position when a current i_0 is passed through it. The spring attached to the cylinder has an unknown torsional constant (C). When the spring is twisted by an angle θ , the torque that is developed is $C\theta$.

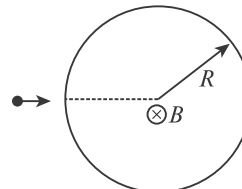
- The torsional constant (C) of the spring can be expressed as

(a) $\frac{NBi_0A}{\pi}$	(b) $\frac{2NBi_0A}{\pi}$
(c) $\frac{NBi_0A}{2\pi}$	(d) $\frac{2NBi_0A}{3\pi}$
- If a charge Q is passed suddenly through the galvanometer coil, the maximum deflection that the coil will suffer before its oscillation is damped is

(a) $Q\sqrt{\frac{\pi NBA}{2Ii_0}}$	(b) $Q\sqrt{\frac{NBAi_0}{\pi I}}$
(c) $Q\sqrt{\frac{2NBA}{\pi Ii_0}}$	(d) $Q\sqrt{\frac{3NBA}{\pi I}}$

Passage 3

A cylindrical region has a uniform magnetic field parallel to its axis. Field strength is B . Several particles, each having mass m and charge q , enter the field radially and perpendicular to the axis of the cylindrical region. The cylindrical region has radius R .



- Which of the following are true?
 - Slower the particle, lesser is the time spent in the magnetic field
 - Faster the particle, lesser is the time spent in the magnetic field
 - Slower the particle, greater is the radius of the circular path that it follows
 - Faster the particle, higher is the distance travelled in the field region and higher is the time spent
- A particle spent $t = \frac{2\pi m}{3qB}$ time inside the magnetic field. The speed of the particle must be

- | | |
|-----------------------------|------------------------------|
| (a) $\frac{qBR}{m}$ | (b) $\frac{\sqrt{3}qBR}{m}$ |
| (c) $\frac{qBR}{\sqrt{3}m}$ | (d) $\frac{2qBR}{\sqrt{3}m}$ |

Passage 4

In a three dimensional space, the region above the plane $z = 0$ has a uniform magnetic field B in positive y direction and below the plane $z = 0$ there is a uniform magnetic field B in positive x direction. A positively charged particle having charge q and mass m is given a velocity $v\hat{k}$ at the origin.

8. The co-ordinates (x, y) of the point where the particle crosses the $z = 0$ plane for the first time is

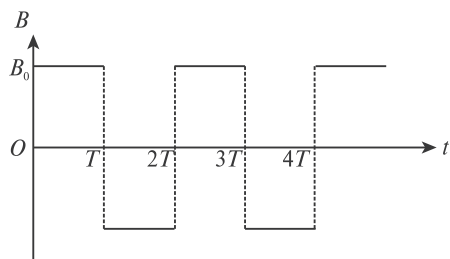
- (a) $\left(0, -\frac{2mv}{qB}\right)$ (b) $\left(0, \frac{mv}{qB}\right)$
 (c) $\left(-\frac{2mv}{qB}, 0\right)$ (d) $\left(\frac{2mv}{qB}, \frac{-2mv}{qB}\right)$

9. The co-ordinates of the point where the particle crosses the $z = 0$ plane for the second time is

- (a) $\left(\frac{mv}{qB}, \frac{mv}{qB}\right)$ (b) $\left(\frac{2mv}{qB}, \frac{2mv}{qB}\right)$
 (c) $\left(-\frac{mv}{qB}, \frac{-mv}{qB}\right)$ (d) $\left(\frac{-2mv}{qB}, \frac{-2mv}{qB}\right)$

Passage 5

In a region of space, a uniform magnetic field changes direction at regular intervals of T . The magnitude of the field remains constant, only its direction reverses. At $t = 0$, a particle of charge q and mass m enters the region with velocity v perpendicular to the field. No other force acts on the particle apart from the magnetic force.



10. If $T = \frac{2\pi m}{qB_0}$, then the path of the particle will be like

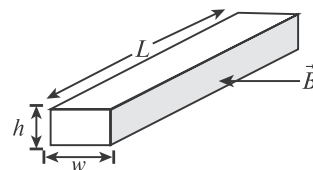
- (a) (b)
 (c) (d)

11. If $T = \frac{\pi m}{qB_0}$, then the path of the particle will be like

- (a) (b)
 (c) (d)

Passage 6

Liquid sodium is an excellent conductor. It is used in some nuclear reactors to cool the reactor core. It is moved through pipes in a reactor by use of electromagnetic pump. Such a pump works on the principle that a moving charge experiences force in a magnetic field. Consider a liquid metal to be in an electrically insulated pipe that has a rectangular cross-section of width w and height h .

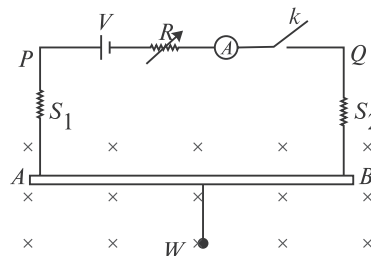


A uniform magnetic field is applied perpendicular to the shaded face of the pipe, as shown. A potential difference is applied across the top and bottom faces (see figure) to pass a current through the liquid. The current density is j (directed downward in the figure).

12. The liquid experiences a force
 (a) along the pipe length (L)
 (b) perpendicular to the pipe length
 (c) upward direction in the figure
 (d) equal to zero
13. The pressure created on a liquid segment of length ΔL , along the axis of the pipe is
 (a) $\frac{j h \Delta L B}{w}$ (b) $\frac{j h \Delta L B}{h}$
 (c) $j \Delta L B$ (d) zero

Passage 7

The figure shows the construction of a hypothetical magnetic balance. AB is a massless conducting bar that is suspended in horizontal position with the help of two springs S_1 and S_2 . Length of the bar AB is L .



3.6 Magnetism and Electromagnetic Induction

The weight (W) to be measured is suspended at the midpoint of the bar. The springs are also conducting. A circuit is completed by joining the conducting supports (P and Q) of the springs to a strong battery, a rheostat, an ammeter (A) and a switch (K). A uniform magnetic field (B) is switched on which is horizontal and perpendicular to the bar. The switch is closed and the resistance of the rheostat is decreased to a point at which the tension in the springs becomes zero. The ammeter is properly calibrated to give the reading of weight (W) directly. The ammeter has 50 divisions and its current sensitivity is 5 division per ampere. Values of other quantities are

$B = 0.5\text{ T}$, length of the bar $L = 0.5\text{ m}$

14. The reading is taken when the spring has zero tension to ensure
 - (a) that weight is balanced by the magnetic force only.
 - (b) that there is no torque on the bar
 - (c) that the springs offer least resistance to the current flow
 - (d) none of the above
15. The maximum weight that can be measured using this device is

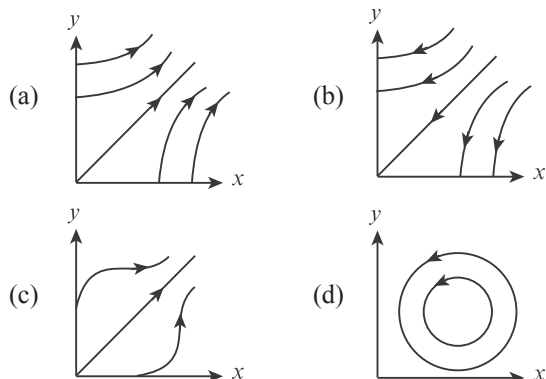
(a) 25N	(b) 0.25N
(c) 2.5 N	(d) 1.0N

Passage 8

A region of space has magnetic field given by $\vec{B} = \left(\frac{B_0 y}{L}\right) \hat{i} + \left(\frac{B_0 x}{L}\right) \hat{j}$, where B_0 is a positive constant of

appropriate dimensions. A square loop lies in xy plane with its corners at $(0, 0)$, $(0, L)$, $(L, 0)$ and (L, L) . A current I is there in the loop such that its magnetic dipole moment is along negative z direction.

16. The magnetic field lines can be approximately drawn as



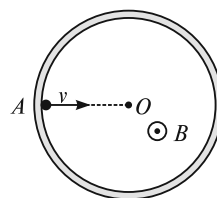
17. The torque on the square loop about x axis has a magnitude

- | | |
|-----------------|----------------------------|
| (a) $B_0 IL^2$ | (b) $\frac{1}{4} B_0 IL^2$ |
| (c) $2B_0 IL^2$ | (d) $\frac{1}{2} B_0 IL^2$ |
18. The torque on the loop about y axis has a magnitude

(a) $B_0 IL^2$	(b) $\frac{1}{4} B_0 IL^2$
(c) $2B_0 IL^2$	(d) $\frac{1}{2} B_0 IL^2$

Passage 9

There is no gravity inside a rigid hollow insulating cylinder of radius R . A uniform magnetic field B exists inside the cylinder, directed parallel to its axis. A small ball having mass m and charge q is projected radially from point A on the inner surface of the cylinder. The ball collides elastically with the inner cylinder wall. The ball is projected with a finite speed v and returns back to the projection point A .



19. The ball hits the cylindrical surface normally at two positions before returning to A . The speed of projection v_0 is

(a) $\frac{RBq}{m}$	(b) $\frac{\sqrt{2}RBq}{m}$
(c) $\frac{2RBq}{m}$	(d) $\frac{\sqrt{3}RBq}{m}$
20. The ball hits the cylindrical surface normally at three positions before returning to A . The time after which the ball returns to A is

(a) $\frac{\pi m}{qB}$	(b) $\frac{2\pi m}{qB}$
(c) $\frac{\sqrt{3}\pi m}{2qB}$	(d) $\frac{\pi m}{3qB}$

Passage 10

A particle, having mass m and charge $+q$, is accelerated by a potential difference V . This particle is moving along x axis. It enters a region having electric field at the origin. The field is uniform (Strength = E) directed in positive y direction and exists in the space between $x = 0$ and $x = b$. The region $x > b$ contains a uniform magnetic field of strength B directed in positive y direction.

21. The distance from the x axis where the particle enters the magnetic field is

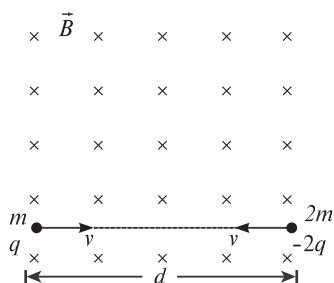
(a) $\frac{Eb^2}{V}$ (b) $\frac{Eb^2}{2V}$
(c) $\frac{Eb^2}{3V}$ (d) $\frac{Eb^2}{4V}$

22. Pitch of the helical path in the region of magnetic field is

(a) $\frac{\pi Eb}{B} \sqrt{\frac{2m}{eV}}$ (b) $\frac{2\pi Eb}{B} \sqrt{\frac{2m}{eV}}$
(c) $\frac{\pi Eb}{B} \sqrt{\frac{m}{2eV}}$ (d) zero

Passage 11

A charged particle having mass m and charge $+q$ is placed at a distance d from another particle having mass $2m$ and charge $-2q$. A uniform magnetic field B exists in the region as shown. The particles are projected towards one another with speed v . The particles do not exert any electromagnetic force on one another.



23. Find maximum value of projection speed v_0 for which the particles will not collide.

(a) $\frac{qBd}{2m}$ (b) $\frac{qBd}{m}$
(c) $\frac{3qBd}{m}$ (d) $\frac{qBd}{3m}$

24. If the particles are projected with speed $2v_0$, the time after which they collide is

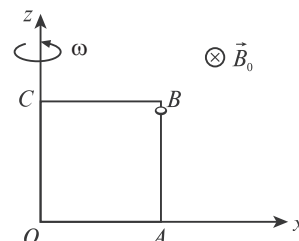
(a) $\frac{\pi m}{3qB}$ (b) $\frac{\pi m}{12qB}$
(c) $\frac{\pi m}{6qB}$ (d) $\frac{\pi m}{2qB}$

25. The particles are projected with speed $2v_0$ and they stick when they collide. The radius of the circular path followed by the combined particle is

(a) $\sqrt{2}d$ (b) $\sqrt{3}d$
(c) d (d) $\frac{d}{\sqrt{3}}$

Passage 12

A frame made of a smooth insulating wire has side length a . It can rotate freely about its side marked as z axis. A magnetic field $\vec{B} = B_0 \hat{j}$ exists in the entire region. A small bead having mass m and charge q can move along side BA . The bead is at B when the frame is in xz plane. The frame is rotating with angular speed ω . Neglect gravity. Take $t = 0$ when the bead starts from B .



26. The z component of acceleration of the bead as function of time is

(a) $-\frac{qa\omega B_0}{m} \sin(\omega t)$ (b) $-\frac{qa\omega B_0}{m} \cos(\omega t)$
(c) $\frac{qa\omega B_0}{m} (1 - \cos \omega t)$ (d) $\frac{qa\omega B_0}{m} (1 - \sin \omega t)$

27. If the bead reaches corner A at time T , the value of B_0 in terms of T is

(a) $\frac{m\omega}{q(\omega T + \sin \omega T)}$ (b) $\frac{m\omega}{q(\omega T - \sin \omega T)}$
(c) $\frac{m\omega}{q(\omega T + 2 \sin \omega T)}$ (d) $\frac{m\omega}{q(\omega T + \cos \omega T)}$

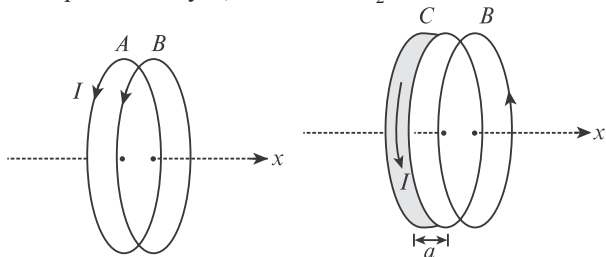
Passage 13

Two infinitely long, parallel wires A and B lie in yz plane. The two wires are parallel to z axis and intersect the y axis at $y = -a$ and $y = +a$, respectively. Wire A carries a current I_0 in positive z direction. A third current carrying wire is laid along the x axis and carries a current I in positive x direction. F is force experienced by a segment OP of this wire due to the field created by the wires A and B . O is the origin and the point P has coordinate $(L, 0, 0)$.

28. The value of current in the wire B and its direction, for which F is zero, are
- $2I_0$ in negative z direction
 - I_0 in positive z direction
 - I_0 in negative z direction
 - $2I_0$ in positive z direction
29. The value of F if the current in the wire B is I_0 in the positive z direction is
- $\frac{\mu_0 I_0 I}{2\pi} \ln \left(\frac{L^2 + a^2}{a^2} \right)$
 - $\frac{\mu_0 I_0 I}{\pi} \ln \left(\frac{L^2 + a^2}{L^2} \right)$
 - $\frac{3\mu_0 I_0 I}{2\pi} \ln \left(\frac{a^2}{L^2 + a^2} \right)$
 - $\frac{3\mu_0 I_0 I}{\pi} \ln \left(\frac{L^2 + a^2}{2a^2} \right)$

Passage 14

Two identical thin rings A and B are held parallel to one another with the x axis being their common axis. Both the rings have radius R and carry current I in the same direction. The centres of the rings A and B are at $x = 0$ and $x = a$, respectively. It is given that $R \gg a$. Under this condition every small segment of the ring B is essentially subjected to the magnetic field created by a long and nearly straight segment of the ring A . With this approximation, we can easily calculate the magnetic force experienced by B due to the magnetic field set by A . We will call this force as F_1 . Now, the ring A is replaced by a somewhat wider conducting ring C . Ring C has radius R and it also carries a current I . The face of the ring C which is nearer to B has its centre located at $x = 0$. The width of the ring C is a . Now, the force experienced by B , due to C is F_2 .



30. Value of F_1 is nearly

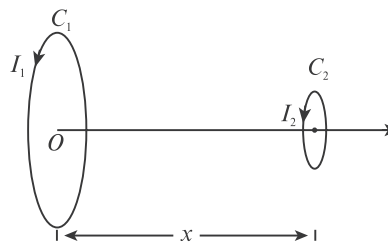
- $\frac{\mu_0 I^2 R}{a}$
- $\frac{\mu_0 I^2 R}{4a}$
- $\frac{\mu_0 I^2 R}{2R}$
- $\frac{\mu_0 I^2 R}{2a}$

31. Value of F_2 is nearly

- $\frac{\mu_0 I^2 R (\ln 2)}{2a}$
- $\frac{\mu_0 I^2 R}{a}$
- $\frac{\mu_0 I^2 R}{2R}$
- $\frac{\mu_0 I^2 R (\ln 2)}{2a}$

Passage 15

A current loop (C_1) is a magnetic dipole and has an energy associated with its orientation in a magnetic field. Consider a fixed circular current loop of radius a in yz plane with its centre at the origin of the co-ordinate system. The loop carries a current I_1 . Another circular current loop (C_2), carrying current I_2 , is placed parallel to the first loop at a distance x from the first loop. Both the loops have x axis as their common central axis. Radius of the second loop is b ($\ll a$). The magnetic field due to current I_1 is not uniform on the circumference of the loop C_2 . However, for writing the energy associated with the orientation of C_2 in the magnetic field of C_1 , we will consider that the field is nearly uniform and parallel to the x axis over the entire area of loop C_2 . This is justified because $a \gg b$. We will assume the orientation potential energy to be zero when the dipole moment is normal to the direction of the field.



32. Potential energy of the dipole C_2 , owing to its orientation in the field created by C_1 is

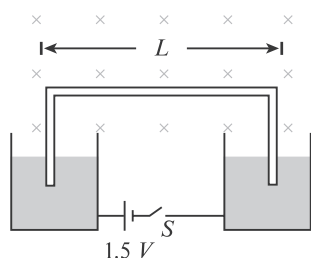
- $-\frac{\pi\mu_0}{2} \frac{I_1 I_2 a^2 b^2}{(a^2 + x^2)^{3/2}}$
- $-\frac{\pi\mu_0}{2} \frac{I_1 I_2 ab}{(a^2 + b^2)}$
- $-\frac{\pi\mu_0}{2} \frac{I_1 I_2 a^2 b}{(a^2 + b^2)^2}$
- zero

33. The force experienced by C_2 due to the current in C_1 is

- $\frac{\pi\mu_0 I_1^2 a^2 b^2 x}{2(a^2 + x^2)^{3/2}}$
- $\frac{3\pi\mu_0 I_1 I_2 a^2 b^2 x}{2(a^2 + x^2)^{5/2}}$
- $\frac{5\pi\mu_0 I_1 I_2 a^2 b^2 x^2}{2(a^2 + x^2)^{5/2}}$
- $\frac{5\pi\mu_0 I_1 I_2 a^2 b^2 x}{7(a^2 + x^2)^{5/2}}$

Passage 16

An insulated wire is bent into the shape of an inverted U . The horizontal part has length $L = 15\text{cm}$ and mass of the entire wire is $m = 5.4 \times 10^{-5}\text{ kg}$. The vertical arms of the wire are partially immersed in two tubs of mercury with 2.5cm length of each arm dipped in the mercury. The wire is subjected to a horizontal magnetic field perpendicular to the horizontal arm of the wire. The strength of the field is $B = 6.5\text{mT}$. On closing the switch S , the circuit gets completed and the wire jumps $h = 35\text{ cm}$ into the air. Neglect buoyancy.

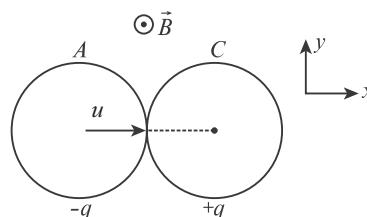


34. Speed of the wire as it leaves mercury is
 (a) 2.52 ms^{-1} (b) 1.26 ms^{-1}
 (c) 5.04 ms^{-1} (d) 1 ms^{-1}
35. Assuming that the current (I) in the circuit remains constant from the instant the switch was closed to the moment the wire left contact with mercury, value of I is
 (a) 2.6 A (b) 3.6 A
 (c) 5.6 A (d) 7.6 A

Passage 17

A uniform magnetic field exists in the entire space perpendicular to the plane of the paper. An insulating disc (C) of radius r lies

at rest in the plane. It carries a charge $+q$ at its centre. Another identical disc (A) also carries a charge $-q$ at its centre. The disc A hits C while travelling in positive x direction with velocity u . The collision is head on and elastic. Mass of each disc is m and the strength of the magnetic field is $B = \frac{mu}{2qr}$. Assume no electrostatic force between the two charges and disregard other forces like gravity and friction.



36. Time after the first collision when the disc C will hit A for the first time is
 (a) $\frac{2r}{4}$ (b) $\frac{3r}{4}$
 (c) $\frac{r}{4}$ (d) $\frac{\sqrt{2}r}{34}$
37. Time after the first collision when A will again hit C is
 (a) $\frac{2r}{4}$ (b) $\frac{3r}{4}$
 (c) $\frac{\sqrt{2}r}{34}$ (d) $\frac{6r}{4}$

Answer Sheet

Match the Columns

1. (A) P, Q, S (B) P, Q (C) P, R (D) P, S
2. (A) S (B) Q (C) P (D) R
3. (A) P, R, S (B) P, Q, R, S (C) P, Q, R, S (D) P, R, S
4. (A) R, S (B) Q (C) Q (D) P, S
5. (A) Q, R (B) P, R, S (C) Q, R (D) P, R, S
6. (A) P, Q, S (B) P, Q, S (C) P, Q, R, S (D) Q, S
7. (A) Q, S (B) P (C) R (D) Q, S
8. (A) P (B) Q (C) R (D) S
9. (A) Q (B) Q (C) Q (D) R

Passage Based Problems

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (a) | 3. (c) | 4. (b) | 5. (a) | 6. (b) | 7. (c) | 8. (c) | 9. (d) |
| 10. (b) | 11. (a) | 12. (a) | 13. (c) | 14. (a) | 15. (c) | 16. (a) | 17. (d) | 18. (d) |
| 19. (d) | 20. (b) | 21. (d) | 22. (a) | 23. (a) | 24. (c) | 25. (b) | 26. (a) | 27. (b) |
| 28. (c) | 29. (a) | 30. (d) | 31. (a) | 32. (a) | 33. (b) | 34. (a) | 35. (d) | 36. (b) |
| 37. (d) | | | | | | | | |

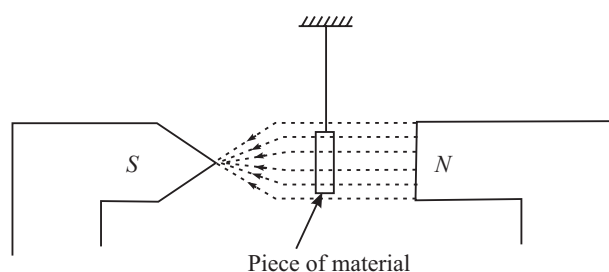
Magnetic Properties of Matter and Earth's Magnetism

“What magnetism is, no one knows. We can only think of it as a peculiar condition created in space by motion of electricity.”

– Sydney Evershed (1925)

1. INTRODUCTION

Figure shows an electromagnet with a flat pole and a pointed pole. This makes the field lines between the poles to be non-uniform. If a piece of iron, nickel or cobalt is suspended between the poles, it gets strongly attracted towards the pointed pole. The material has a strong tendency to move from a region of weak magnetic field to a region of strong magnetic field. Such a material is known as *ferromagnetic material*. If a small aluminium piece is suspended between the poles, it also experiences force towards the pointed pole, but a weak force. Such a material is called *paramagnetic*. When a small piece of bismuth is placed between the poles, it is repelled from the pointed pole. However, the force is very weak. Such a material which experiences a weak force towards a region of weak field is known as *diamagnetic*.



A small piece of material experiences force in a non-uniform field

In this chapter, we shall attempt to understand the properties of paramagnetism, ferromagnetism and diamagnetism. Truth is that these things can be explained properly only by an advanced branch of physics known as quantum mechanics. Here, we will try to understand things in classical ways; picking up facts from the quantum mechanical description wherever needed. No justification of such facts can be presented here.

At the end of this chapter, we will present a model to explain the essential characteristics of the Earth's magnetic field.

2. ATOMS AS SMALL MAGNETS

An electron is often pictured as moving in a circular orbit around a nucleus in an atom. There is an angular momentum ($\vec{L}$) associated with this orbital motion of an electron. The orbiting electron constitutes a current and there is also a magnetic dipole moment ($\vec{\mu}$) associated with it. In previous chapter, we have shown that

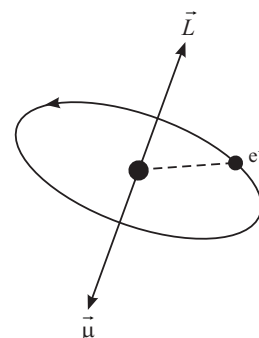
$$\vec{\mu} = - \left(\frac{e}{2m} \right) \vec{L} \quad (1)$$

The negative sign indicates that the magnetic dipole moment ($\vec{\mu}$) is directed opposite to $\vec{L}$. The above result was obtained classically but is also true quantum-mechanically.

This is not the only magnetic dipole moment that an electron has! An electron has an intrinsic angular momentum (called its spin angular momentum or just spin, $\vec{S}$) and an intrinsic magnetic dipole moment (called as spin magnetic dipole moment $\vec{\mu}_s$). Yes, we are saying that $\vec{S}$ or $\vec{\mu}_s$ are basic properties of an electron like its mass and charge. For an easy visualisation, you can think of an electron rotating about its axis just like the Earth. The value of $\vec{S}$ or $\vec{\mu}_s$ cannot be measured. However, their component along any axis (say z-axis) can be measured. The z components of $\vec{S}$ and $\vec{\mu}_s$ are related as

$$\mu_{sz} = - \left(\frac{e}{m} \right) S_z \quad (2)$$

Any electron has this permanent magnetic moment and its value is $9.285 \times 10^{-24} \text{ JT}^{-1}$. This quantity is known as *Bohr magneton*.



An orbiting electron has an angular momentum ($\vec{L}$) and an oppositely directed $\vec{\mu}$

4.2 Magnetism and Electromagnetic Induction

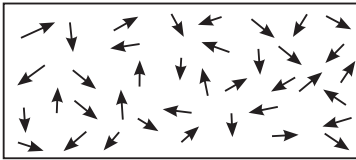
In any atom there are, generally, several electrons. The combination of their angular momentum and magnetic moment builds up a total angular momentum and a total magnetic moment. It can be shown that the total dipole moment and the angular momentum of an atom are oppositely directed. The nucleons (protons and neutrons) also have magnetic dipole moments. But their contribution to the overall dipole moment of an atom is negligible.

There are materials whose atoms have a non-zero magnetic dipole moment. These atoms are tiny magnets! There are other materials whose atoms have a zero net magnetic dipole moment.

An atom with a magnetic dipole moment can be visualised as a tiny magnet or a tiny current loop.

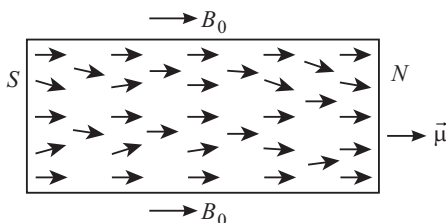
3. PARAMAGNETISM

Each atom of a paramagnetic material has a permanent resultant magnetic dipole moment. Due to thermal agitation, the magnetic dipole moments of the atoms in a piece of material are randomly oriented. Even a small piece of material has large number of atoms and the vector sum of their magnetic dipole moments is zero.



Each arrow represents a tiny atomic magnetic dipole. Sum of all these dipole moments is zero.

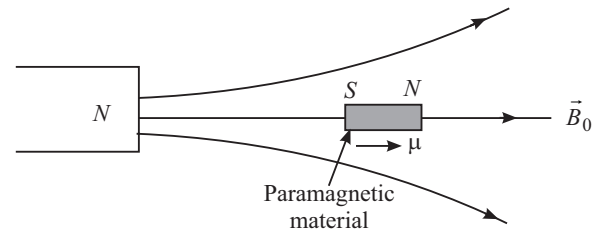
If such a material is placed in an external magnetic field ($\vec{B}_0$), then the individual atomic dipole experiences torque and tends to align parallel to $\vec{B}_0$. The piece of material now has a magnetic moment parallel to the external field. The piece is a weak magnet. We say that the piece is *weakly magnetised*. The tiny atomic dipoles produce magnetic field which makes the field strength inside the material slightly stronger than the outside field. The piece of material shown in the figure is like a weak bar magnet with its right face as the north pole and left face as the south pole.



In external field ($\vec{B}_0$), the atomic dipoles tend to align parallel to $\vec{B}_0$. The piece of material has a net magnetic dipole moment.

When a paramagnetic material is placed near a magnet, it gets magnetised in the field created by the magnet. This magnetised piece is *weakly attracted by the magnet*. You can

also say that a paramagnetic material moves from a region of low magnetic field to a region of high magnetic field.



The magnetic field ($\vec{B}_0$) of the magnet magnetises the piece of material and the magnetised piece gets attracted to the magnet.

When we increase the external field, the alignment of atomic dipoles gets better and the magnetic dipole moment of the piece of material increases. We say that the material gets more magnetised. With the removal of the external field the dipoles again get randomly oriented and the piece of material loses its dipole moment. Another important factor which decides the extent of alignment is temperature. Higher the temperature, more is the thermal agitation and lesser is the alignment.

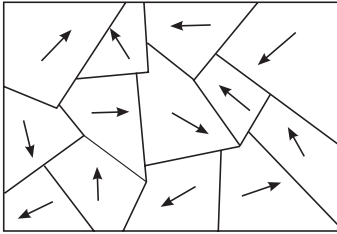
Paramagnetism normally occurs in atoms or molecules with an odd number of electrons. The dipole moments of the paired electrons tend to cancel out but that of the unpaired electron remains unbalanced. Aluminium, lithium, magnesium, etc., are paramagnetic.

4. FERROMAGNETISM

When a paramagnetic (or diamagnetic, as we shall study shortly) material is placed in an external magnetic field, it acquires a dipole moment. This dipole creates its own field but this field is so weak that it produces insignificant change to the value of overall field. For ferromagnetic materials, however, the magnetic dipole moment induced is very large when they are placed in an external field. The magnetic field created by the dipole moment of the material can change significantly the value of overall field.

Like paramagnetic material, an atom of a ferromagnetic material also has a net dipole moment. However, one thing makes ferromagnetism very different from paramagnetism. That is interaction between nearby atomic dipoles. Each atomic dipole is linked with the neighbours such that all of them point in the same direction. (The reason for this can be explained only by quantum mechanics). All tiny atomic dipoles are aligned perfectly parallel to one another. If that is true then any iron piece will be a strong magnet! The fact is alignment of atomic dipole occurs in relatively small patches, called *domains*. A domain can have a linear dimension as large as a millimeter. Each domain contains large number of dipoles, all lined up. These domains can be seen under a microscope using suitable techniques. A domain has a large magnetic dipole moment. An object of usual size, made of iron, will have enormous number of

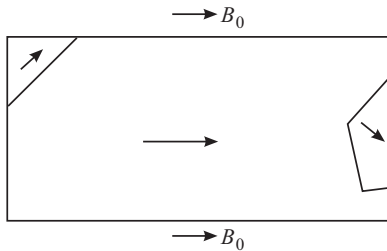
domains. These domains have random orientations. Magnetic moments of all individual domains sum up to zero in an ordinary piece of iron.



Ferromagnetic domains

All atomic dipoles inside a domain are in perfect alignment. Every domain has a large dipole moment. But the sum of dipole moments of all the domains is zero. There is no large scale magnetisation.

When a ferromagnetic material is placed in an external magnetic field, the domain boundaries move. Domains having dipole moment parallel to the external field grow in size and the others shrink. If the external field is strong enough, one domain takes over entirely, and the piece of material is said to be saturated. The material now has a strong dipole moment and is said to be intensely magnetised. The piece of material is a strong magnet in this state. When the external field is removed, all the original domains are not reformed. Overall the material remains magnetised (with dipole moment less than that in saturated state). The piece of material is now a permanent magnet.



In an external field, the domain which is parallel to $(\vec{B}_0)$ grows and all others shrink. Many domains merge as one. If $\vec{B}_0$ is increased further the entire piece will get occupied by one domain.

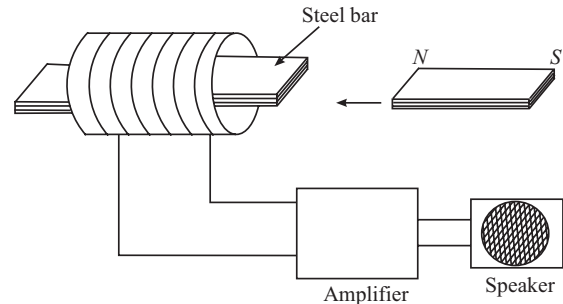
All ferromagnetic materials are strongly attracted by a magnet. When placed in an external field, due to intense magnetisation, the material itself produces a large field. The strength of magnetic field inside the material is far higher than the external field.

Demonstration of breakage of domain boundaries

(Read after going through the next chapter)

A coil having large number of turns is connected to a speaker via an amplifier. Silicon steel bar (ferromagnetic material used in a transformer) is inserted inside the coil. A strong bar magnet is pushed towards the coil. As the magnets gets

closer, the field strength at the steel bar increases and it gets magnetised. Its magnetisation does not increase very smoothly. It changes abruptly when a domain boundary gets violated. These sudden changes in magnetisation will produce sudden change in flux linked with the coil and sudden impulses of emf are produced in the coil. Everytime a domain boundary breaks, there is an impulsive current in the speaker and a distinct sound is produced by it. As the magnet gets closer to the steel we hear a series of click sound from the speaker. As the magnet gets very close the noise subsides. Why? Nearly all domain walls have moved as far as possible.

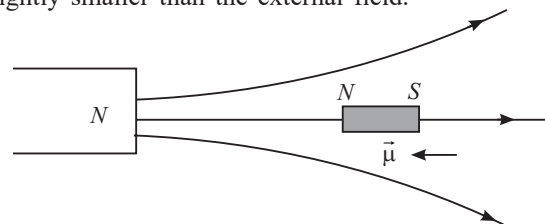


5. DIAMAGNETISM

Think of a material whose individual atoms do not have magnetic dipole moment. [Atoms having even number of electrons mainly show this behaviour]. When a piece of material is placed in an external magnetic field, a small dipole moment gets induced in every atom. This induced dipole moment points opposite to the external field. Any classical model used to explain this phenomenon is flawed and it can be explained only by quantum mechanics.

A piece of diamagnetic material acquires a very small dipole moment when placed in an external field. The induced dipole moment is opposite to the direction of the external field. We say that the piece is very weakly magnetised opposite to the external field. This causes a diamagnetic material to be weakly repelled by a magnet. The material has a tendency to move from a region of high field to a region of low field.

Due to the magnetic field created by its own atomic dipoles, the strength of the magnetic field inside the material is slightly smaller than the external field.



A diamagnetic piece is kept near the north pole of a magnet. The piece develops a small $\vec{\mu}$ and gets repelled by the magnet.

4.4 Magnetism and Electromagnetic Induction

Magnetic moment gets induced in atoms of all materials when they are placed in an external field. *Diamagnetism is a property of all materials.* However, the properties of paramagnetism and ferromagnetism are much stronger and these materials apparently do not exhibit diamagnetism. For example, when a paramagnetic material is placed in an external field, a dipole moment is induced in every atom opposite to the field. But the atom was already having a much higher dipole moment and its overall dipole moment does not change much due to diamagnetism. The material will show paramagnetic behaviour only.

Example 1 A uniform magnetic field exists in a region. A small rectangular piece of material is placed in the region with its longer sides parallel to the field. Draw the magnetic field lines if the material is

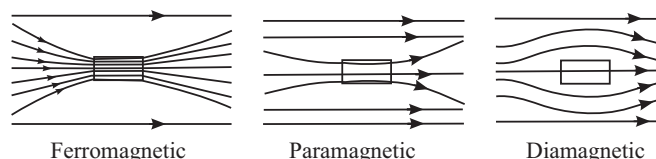
- (a) Ferromagnetic,
- (b) Paramagnetic,
- (c) Diamagnetic.

Solution

Concepts

- (i) The magnetic field strength inside the ferromagnetic material is much higher than the outside field. The atomic dipoles of the material create intense field inside it.
- (ii) The field strength is slightly greater than the external field inside a paramagnetic material.
- (iii) Field inside a diamagnetic material is slightly less than the external field.

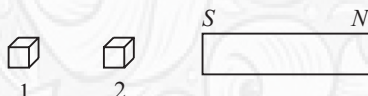
The density of field lines is very high inside ferromagnetic material. Density of lines is slightly higher (than outside density) inside the paramagnetic material. The density of field lines inside a diamagnetic material is slightly less than the external density.



YOUR TURN

Q.1 Why cannot we make permanent magnets from paramagnetic material?

Q.2 Two cubes made of same paramagnetic material are kept near the south pole of a bar magnet.



- (a) Which cube has higher magnetic dipole moment?
- (b) What is the direction of force experienced by the cubes due to the magnet?

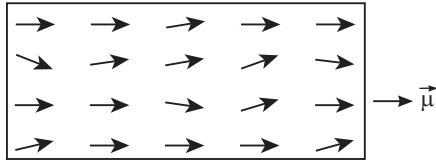
Q.3 Is it true to say that a diamagnetic material exhibits magnetism only when placed in an external magnetic field?

In Short

- (i) Electrons have an intrinsic magnetic dipole moment and another dipole moment associated with their orbital motion in an atom.
- (ii) The magnetic moments of the electrons in an atom have a tendency to cancel in pairs. A material whose atoms have zero net dipole moment are known as diamagnetic. If the atoms have a non-zero dipole moment, the material is known as paramagnetic.
- (iii) In a ferromagnetic material, the dipole moment of neighbouring atoms are locked parallel to one another. A region in which all the atoms have their dipole moments in the same direction is known as a domain. A piece of material has many such randomly oriented domains.
- (iv) When placed in an external field, a paramagnetic material acquires a small dipole moment parallel to the field. A ferromagnetic material acquires a very large dipole moment parallel to the field. A diamagnetic material, kept in an external field acquires a small dipole moment opposite to the external field.
- (v) A ferromagnetic material is strongly attracted by a magnet. A paramagnetic material is weakly attracted by a magnet and a diamagnetic material is weakly repelled by a magnet.
- (vi) Magnetic field lines are dense inside a ferromagnetic material. Field strength inside the material is far higher than the outside field. Density of lines inside the paramagnetic material is slightly higher than outside. Density of field lines inside a diamagnetic material is slightly less than outside.

6. MAGNETISATION ($\vec{I}$)

When the atomic dipole in a piece of material are aligned, fully or partially, there is a net magnetic dipole moment in the direction of alignment. How good is the alignment – this is expressed by *magnetisation vector* $\vec{I}$. It is defined as magnetic dipole moment per unit volume. It is also known as *intensity of magnetisation* or simply *magnetisation*.



Higher $\vec{I}$ implies better alignment of atomic dipoles

If the magnetic dipole moment of a piece of volume V is $\vec{\mu}$, then its magnetisation is

$$\vec{I} = \frac{\vec{\mu}}{V} \quad (3)$$

The unit of I is $\frac{\text{A-m}^2}{\text{m}^3} = \text{Am}^{-1}$

Example 2 Magnetisation of a bar magnet

A bar magnet has length l and cross-sectional area A . The pole strength of the magnet is m . Find its magnetisation.

Solution

Dipole moment of the magnet is, $\mu = ml$

Volume of the magnet is, $V = Al$

$$\therefore \text{Magnetisation, } I = \frac{\mu}{V} = \frac{m}{A}$$

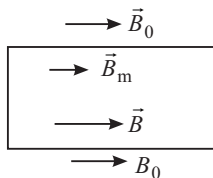
7. MAGNETISING FIELD INTENSITY ($\vec{H}$)

Consider a region having an external magnetic field $\vec{B}_0$. A material is placed in this region. The material gets magnetised and produces its own field. The actual field inside the material is

$$\vec{B} = \vec{B}_0 + \vec{B}_m \quad (4)$$

where $\vec{B}_m$ is the field contributed by the material itself. It turns out that $\vec{B}_m$ is proportional to the intensity of magnetisation ($\vec{I}$) of the material and is given as

$$\vec{B}_m = \mu_0 \vec{I} \quad (5)$$



$\vec{B}_0$ is external field. $\vec{B}_m$ is field produced by the material itself and $\vec{B}$ is the resultant field

The external field, responsible for magnetisation of the material, is often expressed in terms of a new vector $\vec{H}$. It is defined as

$$\vec{H} = \frac{\vec{B}_0}{\mu_0} \quad (6)$$

Engineers often use $\vec{H}$ in place of $\vec{B}_0$. $\vec{H}$ is known as *magnetising field intensity* or simply *magnetic intensity*.

Using equations (5) and (6) we can write equation (4) as

$$\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{I}$$

$$\text{or, } \vec{B} = \mu_0 (\vec{H} + \vec{I}) \quad (7)$$

$$\text{or, } \vec{H} = \frac{\vec{B}_0}{\mu_0} - \vec{I} \quad (8)$$

Example 3 $\vec{H}$ inside a solenoid

An ideal solenoid has n turns per metre length and it carries a current i . What is magnetic intensity inside it?

Solution

Magnetic field inside the solenoid is $B_0 = \mu_0 ni$

$$\therefore H = \frac{B_0}{\mu_0} = ni$$

Note: It is easy to see from the above expression that unit of H is Am^{-1} .

8. MAGNETIC SUSCEPTIBILITY (χ)

Magnetic susceptibility is the measure of how a material responds to an external field. When a material is placed in a magnetising field intensity $\vec{H}$, the magnetisation ($\vec{I}$) acquired by it can be written as

$$\vec{I} = \chi \vec{H} \quad (9)$$

χ is known as susceptibility of the material. It is dimensionless. Aluminium has $\chi = +2.1 \times 10^{-5}$. This means that when an aluminium piece is kept in an external field, it gets weakly magnetised in the direction of the external field. Copper has $\chi = -0.96 \times 10^{-5}$. When placed in an external field, it gets very weakly magnetised opposite to the direction of the field.

Diamagnetic materials have a small negative value of χ and paramagnetic materials have a small positive value of χ . For ferromagnetic materials, χ is a large positive number of the order of thousands.

9. MAGNETIC PERMEABILITY

When a material is kept in an external field, the resultant field inside it is combination of the external field and the contribution made by the material. It is given by equation (7).

$$\vec{B} = \mu_0 (\vec{H} + \vec{I})$$

4.6 Magnetism and Electromagnetic Induction

Using equation (9)

$$\vec{B} = \mu_0 (\vec{H} + \chi \vec{H})$$

$$\text{or, } \vec{B} = \mu_0 (1 + \chi) \vec{H} \quad (10)$$

$$\text{or, } \vec{B} = \mu \vec{H} \quad (11)$$

$$\text{where } \mu = \mu_0 (1 + \chi) \quad (12)$$

$\mu = \mu_0 (1 + \chi)$ is a constant called *permeability of the material*.

Permeability of vacuum is μ_0 .

$$\text{The constant } \mu_r = \frac{\mu}{\mu_0} = 1 + \chi \quad (13)$$

is known as *relative magnetic permeability* of the material. Equation (10) can also be written as

$$\vec{B} = \mu_0 \mu_r \vec{H}$$

$$\text{or, } \vec{B} = \mu_r \vec{B}_0 \quad (14)$$

The meaning of all this is not difficult to understand. Consider an ideal solenoid having number of turns per metre

length equal to n and carrying current i . The field inside the solenoid is

$$B_0 = \mu_0 n i \text{ (along the axis)}$$

$$\text{Magnetic intensity is } H = \frac{B_0}{\mu_0} = n i \text{ (along the axis).}$$

Now, assume that a material is inserted into the solenoid. Due to magnetisation of the material, the field inside the material becomes

$$B = \mu H = \mu_0 \mu_r H = \mu_r B_0.$$

Thus, the field changes by a factor μ_r when a material is brought into the field.

Using equation (13), it is easy to see that value of μ_r is a small positive number (less than 1) for a diamagnetic material. This is because χ is a small negative number greater than -1 .

Value of μ_r is slightly more than 1 for paramagnetic substance and it is of the order of thousands for a ferromagnetic material.



YOUR TURN

Q.4 A piece of material has a magnetic dipole moment of 1.0 A-m^2 . Its volume is $1.0 \times 10^{-6} \text{ m}^3$. Find the magnetisation of the material.

Q.5 The magnetic susceptibility of a material is -2.0×10^{-5} .

- What kind of material is this?
- Find the relative permeability of the material.
- Find the magnetic field inside the material if it is placed in an external field of 1 mT .

Q.6 An alloy has permeability equal to 0.252 Tm A^{-1} . Find the relative permeability and the susceptibility.

Q.7 A toroid has 1000 turns per metre length and carries a current. The magnetic intensity inside the toroid is observed to be 1000 Am^{-1} . A rod is inserted inside the toroid and the magnetisation of the rod is found to be 0.1 Am^{-1} .

- Find current in the toroid.
- Find susceptibility of the material of the rod. What kind of material is it?

9. TEMPERATURE AND MAGNETISATION

The external field (B_0) tends to align the atomic dipoles whereas increasing temperature enhances the randomisation of individual atomic magnetic dipoles. Towards the end of nineteenth century, Pierre Curie discovered experimentally that the magnetisation (I) of a paramagnetic sample is directly proportional to the magnitude of the external field (B_0) and inversely proportional to the temperature (in kelvin).

$$I = C \frac{B_0}{T} \quad (15)$$

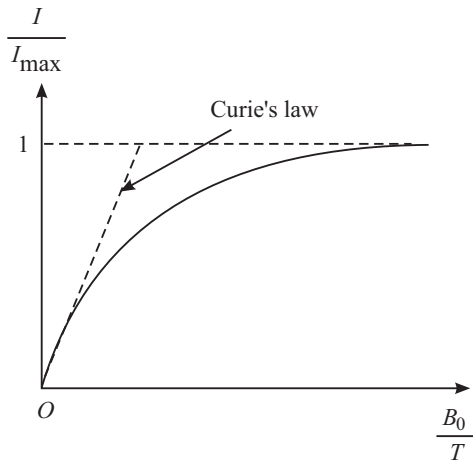
Putting $I = \chi H$ and $B_0 = \mu_0 H$, we get

$$\chi = C \frac{\mu_0}{T} \quad (16)$$

The above law is known as Curie's law and is known to be true when the material is far from saturation. The constant C is known as *Curie Constant*.

The figure shows the plot of $\frac{I}{I_{\max}}$ versus $\frac{B_0}{T}$ for a paramagnetic salt. $I_{\max}$ is the maximum magnetisation when the material is saturated. One can see that Curie's law ($I \propto \frac{B_0}{T}$) is obeyed when the material is far from saturation.

The law is valid only when $\frac{B_0}{T}$ is not too large.



Effect of $\frac{B_0}{T}$ on magnetisation of a paramagnetic material

When a ferromagnetic material is heated, and the temperature is high enough, the domain structure gets destroyed. Now the material becomes paramagnetic. The temperature at which a ferromagnetic material converts into a paramagnetic material is known as *Curie point* or *Curie temperature*. The Curie temperature for iron is $T_C = 1043$ K (or 770°C). The susceptibility above the Curie temperature, i.e., in paramagnetic phase changes with temperature as

$$\chi = \frac{C'}{T - T_C} \quad (T > T_C) \quad (17)$$

Abrupt changes in properties of a substance, occurring at sharply defined temperatures, is known as phase transition (in a branch of Physics which is known as Statistical mechanics).

Example 4 Temperature of an iron specimen is changed from 20°C to 220°C . How will its susceptibility change?

Solution

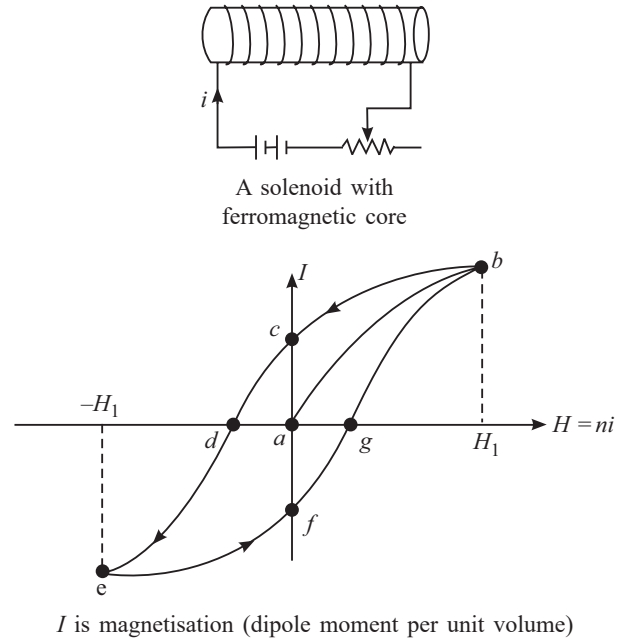
The susceptibility will not change. The domain structure does not change as long as the temperature remains below the Curie temperature ($= 770^\circ\text{C}$). Only above the Curie temperature, a rise in temperature will decrease χ .

11. HYSTERESIS

The word *hysteresis* derives from a Greek verb meaning 'to lag behind'. The magnetisation of a ferromagnetic material depends on the history of magnetisation of the specimen, apart from the external field.

Consider an ideal solenoid having n turns per unit length. When a current i is passed through it, the magnetic intensity inside it is

$$H = \frac{B_0}{\mu_0} = ni \quad (i)$$



Now, imagine that initially there is no current in the solenoid and a core of a ferromagnetic material is inserted into the solenoid. There is no magnetisation (i.e., there is no dipole moment) in the core. Current (i) in the solenoid is increased gradually from zero. This produces a magnetic intensity (H) given by equation (i). Note that $B_0 = \mu_0 H$ is magnetic field produced by the current. It is an external field for the core. The core gets magnetised. As the current i (i.e., the external magnetising field) is increased, the domain boundaries move and the magnetisation grows (i.e., the core becomes stronger and stronger magnet). Ultimately, a saturation point is reached, with all the dipoles aligned and a further increase in current has no effect on the value of I (point b in the figure). Now assume that the current is gradually reduced. The magnetisation (I) decreases but the path ab is not retraced. When $H = 0$ (i.e., current in solenoid $i = 0$), the magnetisation does not become zero. It means that there is only a partial return to randomly oriented domains in the material. The remaining value of I at C is known as *retentivity* of the material. The material has become a permanent magnet with magnetisation equal to value of I at C . Now the question that comes to our mind is — can this magnetisation be killed? The answer is yes. To do so, we change the direction of the current in the solenoid. This produces a field opposite to the initial direction. In the graph, we have shown this magnetic intensity (H) as negative. When the reverse current is increased, the domains get forcibly disaligned and the material loses its entire magnetisation at d . The value of H at d is known as *coercive force*. If we keep on increasing the current in reverse direction, the material starts getting magnetised in opposite direction. This has been represented by negative I

4.8 Magnetism and Electromagnetic Induction

in the graph. As the current is increased, the material again reaches saturation (point *e*) – all dipoles are aligned in the other direction. The core is a strong magnet, like at *b*, with polarity reversed. If we reduce the current, the magnetisation reduces following the path *ef*. At *f* there is no external field and the core is a permanent magnet. Now if we again reverse the current to its original direction and increase it, the magnetisation follows the path *fgb* to again attain saturation. If we keep on changing the value of H from H_1 to $-H_1$ and back to H_1 , the magnetisation of the material keeps following the course *b c d e f g b*.

The fact that the material does not regain its original magnetisation when H is changed and brought back to its original value, is known as hysteresis. The closed loop *bcd efgb* is known as the *hysteresis loop*. If a material is repeatedly moved on its hysteresis loop, it gets heated. The heat developed per unit volume of the material is proportional to the area of the hysteresis loop.

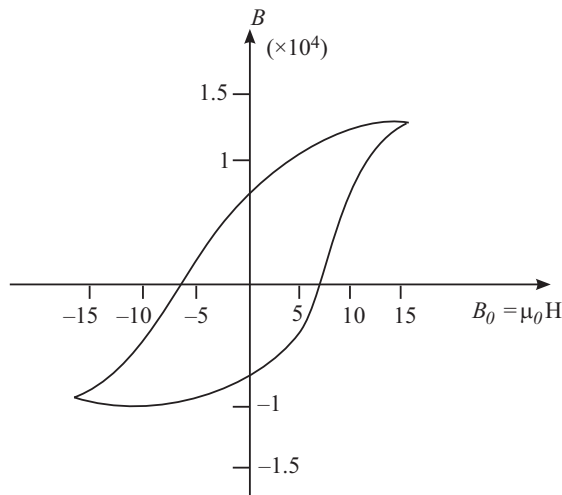
The field inside the material of the core is

$$B = \mu_0 (H + I)$$

In practice, I is very large compared to H for a ferromagnetic material and

$$B \simeq \mu_0 I.$$

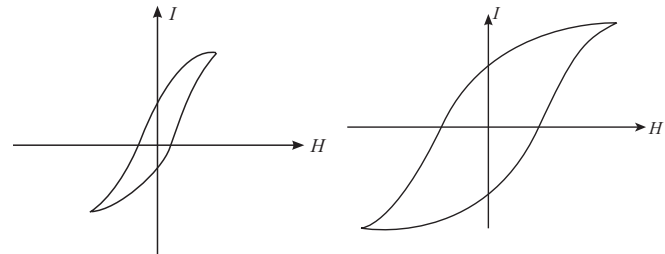
In many cases, we draw the hysteresis curve as plot of B versus H , rather than I against H . To make the unit on both axes as tesla, the given graph is a plot of B versus $\mu_0 H$ for a ferromagnetic material. Look at the magnitudes. The values of B are 10^3 order higher than values of $\mu_0 H$. This means that field inside the material is thousands times higher than the external field.



Hysteresis curve for steel and soft iron

The area of hysteresis loop is larger for steel than that for soft iron. The retentivity and coercive force is also higher for steel. Soft iron gets saturated (fully magnetised) by application of small external field and only a small magnetisation is retained when the external field is removed.

Due to small loop area, the heat dissipation in the material when it is taken through periodic variations in magnetising field, is small. Materials like soft iron are suitable for use in electromagnets.



Soft iron: Low retentivity and low coercive force

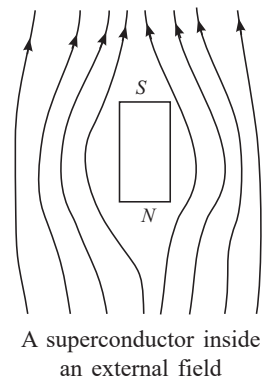
Steel: High retentivity and high coercive force

The so called *transformer irons* (alloy of iron with a small amount of silicon) have very narrow hysteresis loops and are used in transformer cores, electric motors, etc.

Steel needs large external field for complete magnetisation. But once magnetised, it can retain it to larger extent even after the magnetising field is removed. Due to large coercive force, its magnetisation is not easily destroyed even if it is placed in some reverse field of ordinary magnitude. Such a material is suitable for making a permanent magnet.

12. PERFECT DIAMAGNETISM

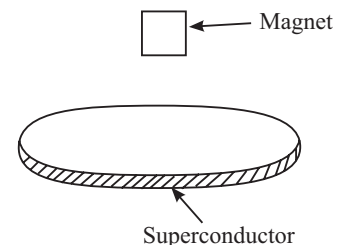
Metals when cooled below a certain temperature, known as critical temperature, lose their resistivity and become superconductors. The superconductors have an astonishing magnetic property. When placed in an external field they expel out all the field lines. It means that the induced dipole moment is large and the field produced by this dipole moment is able to perfectly cancel out the external field inside the material. In an external field, a superconductor is intensely magnetised opposite to the external field. For a superconductor $\chi = -1$ and $\mu_r = 0$. This is perfect diamagnetism.



A superconductor inside an external field

A superconductor repels a magnet. A magnet can remain levitated when placed above a superconductor. The principle is used in running magnetically levitated (MAGLEV) superfast trains.

The phenomenon of perfect diamagnetism in superconductors is called *Meissner effect*.



A magnet remains levitated above a metal which has been cooled below its critical temperature.

13. COOLING BY ADIABATIC DEMAGNETISATION

Suppose a paramagnetic salt is cooled to very low temperature (using liquid helium) in presence of a strong magnetic field. All atomic dipoles are aligned and magnetisation is saturated. Now, the salt is thermally isolated by removing the liquid helium and creating a good vacuum. The magnetic field is gradually turned off. The temperature of the salt falls further taking it to extremely low temperature.

As the liquid helium is removed, the atomic dipole begins to flip over due to thermal motion. Ultimately with no external field, they all will go back to their random orientations. But as they try to rotate from their initial position, there is still some external magnetic field (field is decreased a bit slowly) and they must do work against the magnetic torque. This takes energy from the thermal motion and lowers the temperature. So if the strong external field is not removed too suddenly, the temperature of the salt falls to an extremely low value. This is cooling by demagnetisation.

In Short

(i) Magnetisation (I) of a sample is defined as magnetic dipole moment per unit volume. Its unit is Am^{-1} .

(ii) Magnetic intensity or magnetising field intensity ($\vec{H}$) is defined as

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{I}$$

when there is no material present, $\vec{H} = \frac{\vec{B}}{\mu_0}$

(iii) $\vec{I} = \chi \vec{H}$

χ is known as magnetic susceptibility of the material.

(iv) $\mu_r = 1 + \chi \rightarrow$ relative magnetic permeability

(v) When a material is kept in an external field (B_0), the field inside the material is

$$B = \mu_r B_0$$

(vi) Magnetic permeability of a material is $\mu = \mu_0 \mu_r$

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Susceptibility (χ)	Lies between -1 and 0	A small positive number less than 1	Far greater than 1
Relative permeability ($\mu_r = 1 + \chi$)	Lies between 0 and 1	A positive number slightly greater than 1	Far greater than 1

(vii) A superconductor is a perfectly diamagnetic material with $\chi = -1$.

(viii) The susceptibility of a paramagnetic material is inversely proportional to the temperature.

(ix) A ferromagnetic material converts into a paramagnetic material above a certain temperature known as Curie temperature.

(x) A ferromagnetic material, which has been magnetised by an external field, does not lose its entire magnetisation on removal of the external field. The amount of magnetisation that is retained is known as retentivity of the material. The phenomenon is known as hysteresis.

(xi) If a ferromagnetic material is subjected to an external field which keeps reversing direction, then the magnetisation of the material follows a cycle. The graphical plot of I versus H is known as hysteresis loop.

(xii) Soft iron is preferred (over steel) for making electromagnets due to its low retentivity and steel is used to make permanent magnets because of its high retentivity.

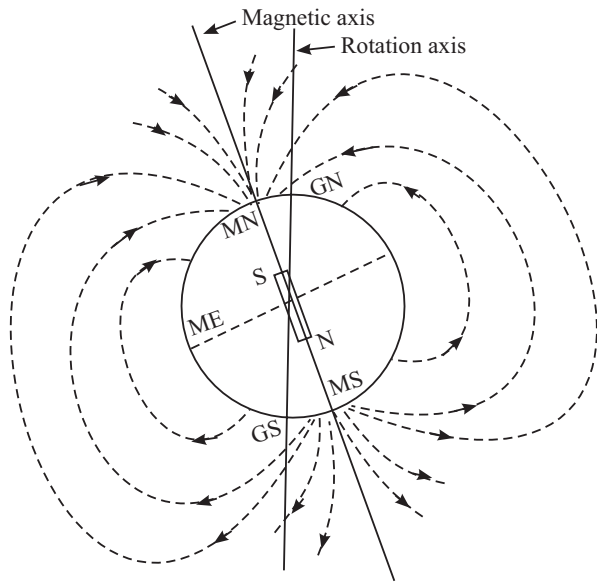
14. EARTH'S MAGNETISM

We have magnetic field present everywhere near the surface of the Earth. Its magnitude ranges between $25\mu T$ to $65\mu T$, at various places on the surface. The exact source of the Earth's magnetism is not very well understood. It is believed that the magnetic field is generated by electric currents in the molten material inside the core of the Earth. The current arises due to convective motion of metallic fluid consisting mostly of ionised iron and nickel.

The characteristics of the Earth's magnetic field can be understood by assuming a *hypothetical* magnetic dipole located at the centre of the Earth. Magnetic dipole moment of this dipole is nearly $8 \times 10^{22} \text{ A-m}^2$. The axis of the dipole is known as magnetic axis of the Earth. The magnetic axis makes about 11° angle with the rotation axis of the Earth. The magnetic axis intersects the surface of the Earth at a point in Ellesmere Island in Canada and at a point off the coast of Antarctica. The former point is known as the *geomagnetic north pole* and the later point is known as the *geomagnetic south pole*.

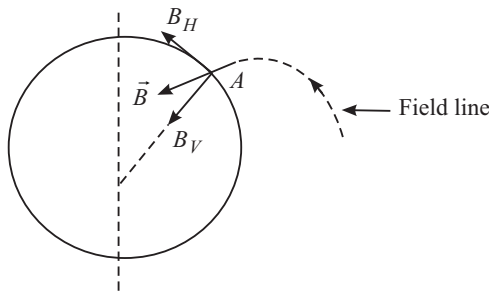
If the dipole inside the Earth is assumed to be a bar magnet, the south pole is towards the geomagnetic north pole and the north pole is towards the geomagnetic south pole. The figure below shows the magnetic field lines. The field lines go into the Earth at the geomagnetic north pole and come out of the Earth's surface at the geomagnetic south pole. A freely suspended bar magnet near the surface of the Earth will align along the field with its north pole pointing exactly in the direction of the field. A freely suspended bar magnet at the geomagnetic north pole will become vertical with its north pole pointing downwards. At geomagnetic south pole a freely suspended magnet will stay in vertical

position with the north pole pointing up. At a place like Delhi a freely suspended bar magnet will point in the direction of field and remain inclined to the horizontal.

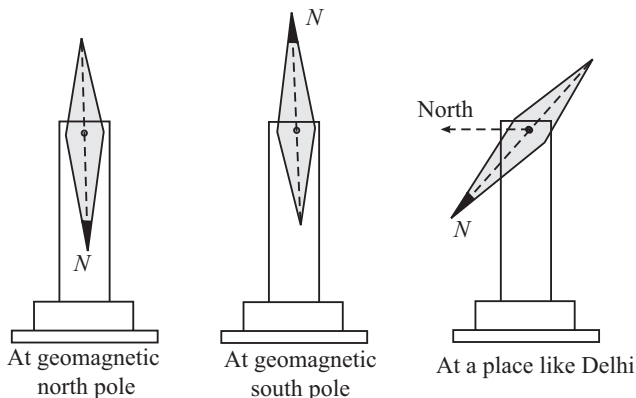


Magnetic field of the Earth

GN - Geographic north
GS - Geographic south
MN - Magnetic north
MS - Magnetic south
ME - Magnetic equator



Direction of magnetic field at a place like A is indicated by a tangent on the field line



A magnetic needle that is free to rotate about an axis passing through its centre of mass

The Earth's magnetic field is known to change with time. There are proofs that the magnetic field may even reverse its direction. Many such reversals have happened in the past. It is believed that the magnetic north pole of the Earth is moving towards Siberia from Canada. Recent studies show that it is presently shifting at an alarming rate of over 50 km per year. World's magnetic model is released every 5 years. Last time it was released in 2015 and was expected to last till 2020. But the magnetic pole has begun to shift very quickly and it has been realised that a new model is needed as soon as possible. You must note that an error in the model causes appreciable errors in navigation.

15. ELEMENTS OF THE EARTH'S MAGNETIC FIELD

What is the magnitude and direction of the Earth's magnetic field at Delhi? If you ask this question, a scientist will give you three numbers. These three numbers tell you the direction and magnitude of the field at a place and are known as the elements of the Earth's magnetic field. These are (a) declination angle, (b) dip angle and (c) horizontal component of the field.

Declination (θ)

Consider a point P on the surface of the Earth. The longitude circle passing through the point determines the geographic north-south direction. If you imagine yourself walking along this circle, you will reach the geographic north pole, keep walking and you will be at the south pole and then back to your original location. This circle (which has the rotation axis as a diameter) is known as the *geographic meridian plane* at the point P . In short, it is a vertical plane at a place with its horizontal direction along true north (the geographic north).

Similarly, imagine a circle drawn on the surface of the Earth passing through the point P and the two geomagnetic poles. The plane of this circle is known as the *magnetic meridian plane* at P . In other words, the magnetic meridian is a vertical plane through point P that contains the geomagnetic poles. *The magnetic field at a place lies in the magnetic meridian plane* because this plane contains the magnetic dipole of the Earth.

The angle between the magnetic meridian plane and the geographic meridian plane at a point is known as declination (θ) at that point.

Suppose you are at a place on the surface of the Earth. These are infinite horizontal directions in which you can walk. Mark the horizontal direction which is headed towards the geographic north pole. Then mark the horizontal direction that will take you to the magnetic north pole. The angle between these two directions is the declination.

In India, the declination angle is less than 1° . In Delhi, it is about 0.7° . Suppose you are in Delhi and you know the direction of true north. Stand facing that direction. Now

turn 0.7° and stretch your arm horizontally in front of you. Imagine a vertical plane through your stretched arm. This plane is magnetic meridian. The magnetic field lies in this plane.

Dip (δ)

The angle made by the Earth's magnetic field with the horizontal direction in the magnetic meridian plane is known as dip angle (δ) at the point.

In the magnetic northern hemisphere, the field has a vertically downward component. A magnetic needle which can rotate in vertical plane will point in the direction of the field if it is kept in the magnetic meridian plane.

Knowledge of θ and δ specifies the direction of the Earth's magnetic field at a point.

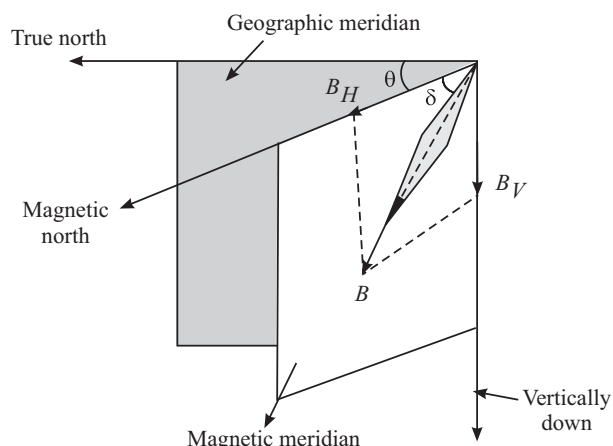


Figure shows declination (θ) and dip (δ). A magnetic needle points in the direction of magnetic field in the magnetic meridian plane.

Horizontal component of field (B_H)

This is the component of the Earth's magnetic field at a place in horizontal direction in the magnetic meridian plane.

We can easily find the magnitude of the magnetic field (B) of the Earth if we know B_H and δ . From the figure shown above

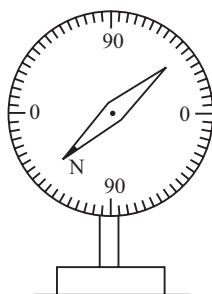
$$B_H = B \cos \delta \quad (18)$$

The relation between the horizontal component and the vertical component of the field is

$$\frac{B_V}{B_H} = \tan \delta \quad (19)$$

Example 5 Apparent dip

A magnetic needle is free to rotate on a vertical circle about a horizontal axis passing through its centre. When the plane of the circle is in the magnetic meridian, the needle points in the direction of the magnetic field at the

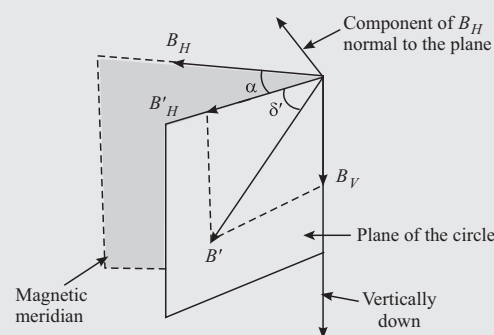


place. Inclination of the needle with the horizontal can be read directly from the scale marked on the circle. This angle is true dip (δ) at the place. If the circle is kept in a vertical plane that makes an angle α with the magnetic meridian plane, the needle is inclined at an angle δ' to the horizontal. This angle is known as apparent dip at the place. Relate α , δ and δ' .

Solution

Concepts

- (i) The horizontal component of the Earth's magnetic field can be resolved into two components— one in the plane of the circle (B'_H) and the other normal to the plane.



- (ii) The magnetic needle is free to rotate in the plane of the circle only. The component of B_H normal to the plane will try to turn the needle out of the circular plane. But that rotation is not allowed. This means that B'_H is the effective horizontal field that will try to rotate the needle.
- (iii) Vertical component of the field (B_V) remains same in the magnetic meridian and the circular plane of the needle.

Component of horizontal field in the plane of the circular scale is

$$B'_H = B_H \cos \alpha$$

The needle will point along B' , the effective field in the plane.

$$\therefore \tan \delta' = \frac{B_V}{B'_H} = \frac{B_V}{B_H \cos \alpha}$$

But $\frac{B_V}{B_H} = \tan \delta$ where δ is true dip

$$\therefore \tan \delta' = \frac{\tan \delta}{\cos \alpha}$$

$$\text{or, } \cos \alpha \tan \delta' = \tan \delta$$

This is the desired relation.

$$(20)$$



YOUR TURN

Q.8 At a place, the horizontal component of the Earth's magnetic field is $20\mu\text{T}$ and the dip angle is 37° . Find the magnitude of the magnetic field at the place.

Q.9 A magnet is suspended by a vertical string attached to its midpoint. At what angle to the horizontal, the magnet can stay in equilibrium? It is given that the horizontal and vertical component of the magnetic field at the place is $25\mu\text{T}$ and $18.75\mu\text{T}$.

16. AURORA

Aurora is a mysterious display of light in the night sky, specially in the regions near the Earth's magnetic poles. The aurora occurring near the north pole is known as *aurora borealis* or the *northern lights* and those occurring near south pole are known as *aurora australis* or *southern lights*. Usually the auroras are of milky greenish colour though they can be red, blue, violet and pink also.

A typical aurora is caused by collision of fast moving electrons from space with oxygen and nitrogen in the upper atmosphere of the Earth. The electrons which come from *magnetosphere* (a region of space where the Earth's magnetic field manipulates and guides the charged particles) collide with oxygen and nitrogen atoms and excite them. As the atoms return to their ground state, they emit light. When large number of charge particles come from the magnetosphere and excite a large number of atmospheric atoms, enough light is emitted for us to detect. This makes the sky colourful. When many oxygen atoms are excited, the light is predominantly green. Excitation of nitrogen gives blue light.

The Sun keeps throwing high energy charged particles. Many of these particles are headed towards the Earth. The magnetic field of the Earth, in a way, acts as a shield and prevents these high energy particles from entering our atmosphere. Many of these particles are trapped in the magnetic field and are guided towards the polar region. When these particles interact with the atoms in the outer atmosphere, auroras are produced.

In Short

- (i) The magnetic field due to the Earth can be modelled by assuming that a giant bar magnet is located at its centre. The axis of this magnet is known as magnetic axis of the Earth.
- (ii) The intersection points of the magnetic axis with the surface of the Earth are known as the geomagnetic poles.
- (iii) Magnetic axis is inclined at 11° to the rotation axis of the Earth.
- (iv) A freely suspended magnet will remain vertical at the geomagnetic poles.
- (v) Three quantities needed to specify the direction and magnitude of the Earth's magnetic field at a place are known as the element's of the Earth's field. They are—
 - (a) declination
 - (b) dip and
 - (c) horizontal component of the field.
- (vi) Declination at a place is the angle between the direction pointing towards the magnetic north (i.e., the direction indicated by a compass needle) and the direction of true geographic north.
- (vii) The vertical plane which contains the magnetic field (B) at a place is known as the magnetic meridian plane.
- (viii) The angle that the magnetic field makes with horizontal in the magnetic meridian plane is known as dip angle (δ).
- (ix) $\tan \delta = \frac{B_V}{B_H}$

B_V and B_H are vertical and horizontal components of the Earth's field at a point.

Miscellaneous Examples

Example 6 A sample of paramagnetic material contains 2×10^{24} atoms. Each atom has a dipole moment $1.5 \times 10^{-23} \text{ JT}^{-1}$. The sample is kept in a uniform magnetic field of $B_1 = 0.64 \text{ T}$ and cooled to a temperature of 4.2 K . The degree of magnetic saturation achieved is 15%. What is the dipole moment of the sample when it is placed in a field of strength $B_2 = 0.98 \text{ T}$ and is at a temperature 2.8 K ? Assume Curie's law to be applicable.

Solution

Concepts

- (i) 15% saturation means that the magnetisation attained is 15% of the maximum possible value.
- (ii) Curie's law (equation 15) is $I = C \frac{B_0}{T}$

With 15% saturation, the dipole moment of the sample is

$$\mu_1 = 1.5 \times 10^{-23} \times 2 \times 10^{24} \times \frac{15}{100} = 4.5 \text{ J T}^{-1}$$

From Curie's law

$$\begin{aligned} \frac{I_1}{I_2} &= \frac{B_1/T_1}{B_2/T_2} \\ \Rightarrow \frac{\mu_1}{\mu_2} &= \frac{B_1}{B_2} \times \frac{T_2}{T_1} \quad [\because \mu = I \times \text{Volume}] \\ \text{or, } \frac{4.5}{\mu_2} &= \frac{0.64}{0.98} \times \frac{2.8}{4.2} \\ \therefore \mu_2 &= 7.9 \text{ J T}^{-1} \end{aligned}$$

Example 7 An electromagnet is constructed by inserting a long cylindrical iron core of cross section 10 cm^2 into a long solenoid having 1000 turns/m and carrying a current 2.0 A . The magnetic field inside the core is found to be $\frac{\pi}{2} \text{ T}$. Neglect end effects and calculate the pole strength of the magnet so formed.

Solution

Concepts

- (i) $B = \mu_0 (H + I)$
- (ii) From example (2), $m = IA$

Magnetising field intensity inside the solenoid is

$$H = ni = 1000 \times 2 = 2000 \text{ Am}^{-1}$$

Magnetic field inside the solenoid is

$$B = \mu_0 (H + I)$$

$$\text{or } I = \frac{B}{\mu_0} - H$$

$$\begin{aligned} &= \frac{\pi/2}{4\pi \times 10^{-7}} - 2000 = 1.25 \times 10^6 - 2000 \\ &\approx 1.25 \times 10^6 \text{ Am}^{-1} \end{aligned}$$

Note that $I \gg H$ for a ferromagnetic core. Pole strength of the magnet formed is

$$m = IA = 1.25 \times 10^6 \times 10 \times 10^{-4} = 1250 \text{ A-m}$$

Example 8 Neutral point

A circular current carrying coil has a dipole moment equal to 10^{-1} Am^2 . It is placed with its plane in a vertical plane that is perpendicular to the magnetic meridian. At what distance on its axis there is a neutral point? Radius of the coil is $R = 5 \text{ cm}$ and B_H at the place is $20 \mu\text{T}$.

Solution

Concepts

A neutral point is a point where the horizontal field due to the coil cancels the horizontal component of the Earth's magnetic field. A compass needle placed at such a point can stay in any direction.

The field at P due to the coil balances B_H .

$$\therefore B = B_H$$

$$\text{or, } \frac{\mu_0}{4\pi} \frac{2\mu}{(R^2 + x^2)^{3/2}} = 20 \times 10^{-6} \quad \begin{array}{c} \text{Diagram: A circular coil of radius } R \text{ is shown with a point } P \text{ on its axis at a distance } x \text{ from the center. The magnetic field } B_H \text{ is indicated as a horizontal arrow pointing right. The distance } x \text{ is marked with a double-headed arrow below the axis.} \end{array}$$

$$\text{or, } \frac{10^{-7} \times 2 \times 10^{-1}}{20 \times 10^{-6}} = (R^2 + x^2)^{3/2}$$

$$\Rightarrow 10^{-3} = [(0.05)^2 + x^2]^{3/2}$$

$$\Rightarrow 0.01 = 0.0025 + x^2$$

$$\text{or, } x^2 = 0.0075$$

$$\Rightarrow x = 0.087 \text{ m} = 8.7 \text{ cm}$$

Example 9 Ampere's law and magnetic intensity (H)

The magnetic intensity ($\vec{H}$) depends only on the actual current responsible for the field and is independent of the medium. Ampere's law written in terms of $\vec{H}$ can be stated as

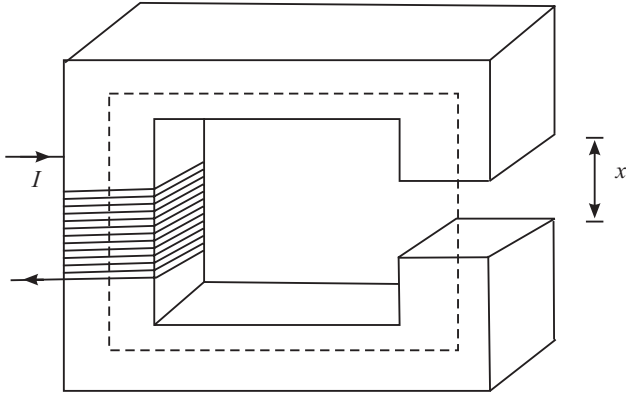
$$\oint \vec{H} \cdot d\vec{l} = i$$

where i is the total current crossing the closed loop on which the line integral has been calculated. Based on this, solve the following problem.

An iron core has cross-sectional area A and its mean perimeter (the perimeter of the dotted rectangle) is L . The air gap has width x . Relative permeability of iron is μ_r ($\mu_r \gg \mu_0$). A coil having N turns is wound on the core and is given a current I . Ignore the fringing of the field at the

4.14 Magnetism and Electromagnetic Induction

edges of the gap and the leakage of the magnetic flux. Find the magnetic field strength B in the gap.

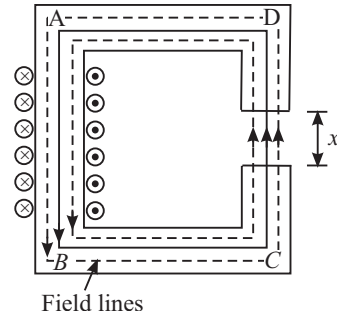


Solution

Concepts

- (i) The field lines that are present inside the core are also present in the air gap. Therefore, value of B is same inside the core as well as the gap.
- (ii) H inside the core and the air gap is not same.
- (iii) We will take an amperian loop in the shape of the dotted rectangle shown in the figure. The total current piercing this loop is NI .

Magnetic field lines are closed loops. All lines present inside the iron core are also present inside the air gap. Field strength (B) is same inside the core as well as the gap.



Value of H inside the core is

$$H_1 = \frac{B}{\mu_r \mu_0}$$

Value of H inside the air gap is $H_2 = \frac{B}{\mu_0}$

Ampere's law says $\oint \vec{H} \cdot d\vec{l} = i$

Applying this law along the path $ABCD$,

$$H_1 (L - x) + H_2 x = NI$$

$$\text{or, } \frac{B}{\mu_r \mu_0} (L - x) + \frac{B}{\mu_0} x = NI$$

$$\text{or, } B = \frac{\mu_0 \mu_r NI}{L + (\mu_r - 1)x}$$

Worksheet 1

- When a diamagnetic substance is brought near the north or the south pole of a bar magnet, it is
 - attracted by both the poles
 - repelled by both the poles
 - repelled by the north pole and attracted by the south pole
 - attracted by the north pole and repelled by the south pole
- The atoms of a material do not have magnetic dipole moment.
 - The material is paramagnetic
 - The material is diamagnetic
 - The material is ferromagnetic
 - No such material exists
- A superconductor shows
 - diamagnetism
 - ferromagnetism
 - paramagnetism
 - none
- Which of the following property is present in all materials?
 - diamagnetism
 - ferromagnetism
 - paramagnetism
 - none
- The susceptibility of a paramagnetic substance
 - first increases and then decreases with rise of temperature
 - increases with rise of temperature
 - decreases with rise of temperature
 - does not vary with temperature
- An electromagnet uses
 - soft iron core
 - steel core
 - nickel core
 - copper core
- The angle between geographical and magnetic meridian is called
 - angle of dip
 - angle of declination
 - latitude
 - longitude
- The ratio of the intensity of magnetisation to the magnetising field is called
 - magnetic intensity
 - magnetic susceptibility
 - magnetic permeability
 - magnetic induction
- Curie temperature is the temperature above which
 - a ferromagnetic material becomes paramagnetic
 - a paramagnetic material becomes diamagnetic
 - a ferromagnetic material becomes diamagnetic
 - a paramagnetic material becomes ferromagnetic
- Electromagnets are made of soft iron because soft iron has
 - low retentivity and low coercive force
 - high retentivity and low coercive force
 - low retentivity and high coercive force
 - high retentivity and high coercive force
- An iron rod of volume 10^{-4} m^3 and relative permeability 1000 is placed inside a long solenoid wound with 5 turns/cm. If a current of 0.5 A is passed through the solenoid, then the magnetic moment of the rod is
 - 10 Am^2
 - 15 Am^2
 - 20 Am^2
 - 25 Am^2
- A magnetising field of $2 \times 10^3 \text{ Am}^{-1}$ produces a magnetic flux density of 8π Tesla in an iron rod. The relative permeability of the rod will be
 - 10^2
 - 10^0
 - 10^4
 - 10^1
- If the Earth's field induction at a place is 0.36 G and the angle of dip is 60° . The horizontal and vertical components of the field are B_H and B_V , respectively. [1 gauss = 10^{-4} T]
 - $B_H = 0.18 \text{ G}, B_V = \frac{0.18}{\sqrt{3}} \text{ G}$
 - $B_H = 0.18\sqrt{3} \text{ G}, B_V = 0.18 \text{ G}$
 - $B_H = 0.18 \text{ G}, B_V = 0.36 \text{ G}$
 - $B_H = 0.18 \text{ G}, B_V = 0.18\sqrt{3} \text{ G}$
- A compass needle has magnetic moment 60 Am^2 . It is pointing towards geographical north at a certain place where the horizontal component of Earth's magnetic field is $40 \mu\text{T}$. It experiences a torque of $1.2 \times 10^{-3} \text{ Nm}$. The declination of the place is
 - 60°
 - 45°
 - 30°
 - 15°

Worksheet 2

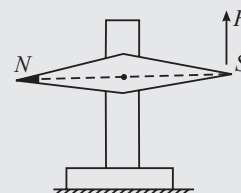
1. A ferromagnetic material is placed in an external magnetic field. A magnetic domain
 - (a) must grow in size (b) must decrease in size
 - (c) may grow in size (d) may decrease in size
2. A piece of iron is being carried over its hysteresis loop. The magnetic susceptibility
 - (a) at an instant may be zero
 - (b) at an instant may be negative
 - (c) at an instant may be infinite
 - (d) has always a fixed value
3. The desired properties of a material for making a permanent magnet from it is
 - (a) high retentivity (b) low retentivity
 - (c) high coercive force (d) low coercive force
4. A paramagnetic specimen is kept in a uniform magnetic field. The temperature as well as the external field begins to change with time.
 - (a) The magnetisation of the specimen will not change with time if the plot of the external magnetic field versus the temperature is a straight line passing through the origin.
 - (b) The magnetisation of the specimen will not change with time if the plot of the external magnetic field versus the temperature is a hyperbola.
 - (c) The magnetisation of the specimen will not change with time if the plot of the external magnetic field versus the temperature is a parabola.
 - (d) The magnetisation of the specimen will decrease if the temperature doubles and the external field becomes 1.5 times.
5. A magnetic needle is free to rotate in a vertical plane about a horizontal axis passing through its centre of mass and perpendicular to its length. The place of experiment is Delhi.
 - (a) The needle remains horizontal when the axis of rotation is in magnetic meridian plane.
 - (b) The needle remains horizontal when the axis of rotation is perpendicular to the magnetic meridian plane.
 - (c) The needle remains vertical when the axis of rotation is in magnetic meridian plane.
 - (d) The needle remains vertical when the axis of rotation is perpendicular to the magnetic meridian plane.
6. The horizontal component of the Earth's magnetic field measured in a vertical plane is B directed at an angle β with the horizontal. The vertical plane makes an angle α with the magnetic meridian.
 - (a) The horizontal component of the Earth's magnetic field at the place is $B_H = \frac{B \cos \beta}{\cos \alpha}$
 - (b) The horizontal component of the Earth's magnetic field at the place is $B_H = \frac{B \cos \alpha}{\cos \beta}$
 - (c) The dip angle at the place is $\delta = \tan^{-1} \left(\frac{\sin \beta \cos \alpha}{\cos \beta} \right)$
 - (d) The dip angle at the place is $\delta = \tan^{-1} \left(\frac{\sin \beta \sin \alpha}{\cos \beta} \right)$

Worksheet 3

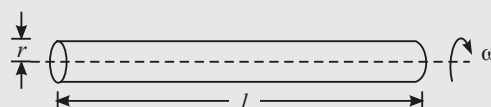
1. Would the theoretical limit of magnetisation of a paramagnetic sample be of the same order of magnitude as that of a ferromagnet? Explain.
2. Space within a current carrying solenoid is filled with tungsten having susceptibility 6.8×10^{-5} . Find the percentage change in the magnetic field inside the solenoid.
3. A bar magnet has magnetisation I . It is cut into two parts along a plane perpendicular to the length of the magnet. If the lengths of the two parts are in ratio 1 : 2, what will be ratio of magnetisation of the two parts?
4. A domain in an iron specimen is cubical in shape with side length $1 \mu\text{m}$. Find the maximum possible dipole moment of the domain using the following data:
Molar mass of iron = 55 g mol^{-1} , density of iron = 7.9 g/cc ,
Dipole moment of each iron atom = $9.27 \times 10^{-24} \text{ Am}^2$.
5. The susceptibility of a specimen at 300 K is 1×10^{-5} . At what temperature will the susceptibility become 1.5×10^{-5} ?
6. The coercive force for a magnet is $2 \times 10^4 \text{ Am}^{-1}$. The magnet is placed inside an ideal solenoid and a magnetic field is setup, by passing current in the solenoid, so as to demagnetise the specimen. Find the number of turns required per metre length of the solenoid if the current needed is 5A.

7. The Earth's magnetic field at a place on the magnetic equator is nearly $4 \times 10^{-5} \text{ T}$. Estimate the dipole moment of the Earth. Radius of the Earth is $R = 6.4 \times 10^6 \text{ m}$.
8. The magnetic field at geomagnetic poles has magnitude close to $60 \mu\text{T}$. Find the strength of the field at a point on the magnetic equator.

9. The magnetic needle shown in the figure can rotate about the horizontal axis passing through its centre. When it lies in the magnetic meridian plane it makes an angle of 45° with the horizontal. The



- horizontal component of the Earth's field at the location is $B_H = 18 \mu\text{T}$. The needle is kept horizontal by applying a vertical force F at one of its ends. Find F . It is given that the magnetic dipole moment of the needle is 0.16 A-m^2 and its length is 0.1 m .
10. A long cylindrical rod has radius r and length l ($l \gg r$). It is made of a dielectric material and carries a uniform surface charge density σ . It is rotated about its long axis with constant angular speed ω . Find the magnetic field intensity at the centre of the rod. The susceptibility of the material is χ .



Answer Sheet

Your Turn

2. (a) cube 2 (b) towards the magnet 3. yes 4. 10^6 Am^{-1}
5. (a) Diamagnetic (b) 0.99998 (c) 0.99998 mT 7. (a) 1 A (b) 10^{-4} , paramagnetic
6. 2×10^5 ; 2×10^5 9. 37° with the horizontal
8. $25 \mu\text{T}$

Worksheet 1

1. (b) 2. (b) 3. (a) 4. (a) 5. (c) 6. (a) 7. (b) 8. (b) 9. (a)
10. (a) 11. (d) 12. (c) 13. (d) 14. (c)

Worksheet 2

1. (c, b) 2. (a, b, c) 3. (a, c) 4. (a, d) 5. (c) 6. (a)

Worksheet 3

1. Yes 2. $6.8 \times 10^{-3}\%$ 3. 1 4. $8.0 \times 10^{-13} \text{ Am}^2$
5. 200K 6. 4000/m 7. $1.05 \times 10^{23} \text{ Am}^2$ 8. $30 \mu\text{T}$
9. $5.8 \times 10^{-5} \text{ N}$ 10. $\mu_0(1 + k) \sigma r l \omega$

Electromagnetic Induction

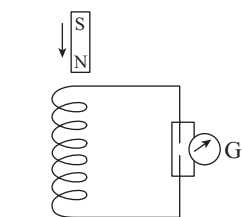
“Nothing is too wonderful to be true, if it is consistent with the laws of nature.”

– Michael Faraday

1. INTRODUCTION

Before Michael Faraday’s discovery of the law of electromagnetic induction, the only current producing devices were voltaic cells which produced small currents. In 1830s, Faraday in England and Joseph Henry in United States discovered the basic principle which is used today in mass production of electricity.

Consider a magnet placed near a coil. When the magnet is pushed towards the coil, a galvanometer connected to the coil detects a current in the coil. If the magnet is pulled away from the coil, then also there is a current in the coil, but this time in opposite direction. The galvanometer pointer swings in opposite direction.



When the magnet is pushed towards or away from the coil, the galvanometer (G) shows deflection

The phenomenon is known as electromagnetic induction. If the magnet is held at rest and the coil is moved, towards or away from the magnet, then also there is a current induced in the coil. There is no current in the coil when the relative positions of the coil and the magnet is not changing. The ‘magic’ of induction of a current in the coil happens only when something is changing or there is some movement. It never happens while everything is static. Faraday was the first to find out which physical quantity must change for the ‘magic’ to happen. This physical quantity is magnetic flux. Let’s introduce ourself to this new quantity before we give the formal statement of the Faraday’s law.

2. MAGNETIC FLUX

Consider a surface in a region of magnetic field (B). A small patch on the surface has area dS . Represented as a vector, the area vector $\vec{dS}$ is directed normal to the surface at the location of the patch. Let the magnetic field at the location of the patch be $\vec{B}$. The flux of the magnetic field through the patch of area ds is defined as

$$d\phi = \vec{B} \cdot \vec{dS}$$

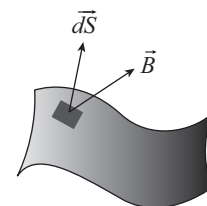
Flux through the entire surface is obtained by adding the flux through all such small patches on the surface. Mathematically, we can write it as

$$\phi = \int \vec{B} \cdot \vec{dS} \quad (1)$$

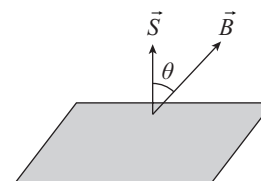
Unit of magnetic flux is called weber (which is same as $T - m^2$). When the surface is flat and $\vec{B}$ is uniform, the flux through the surface is simply

$$\phi = \vec{B} \cdot \vec{S} = BS \cos\theta$$

Here, θ is angle between the area vector $\vec{S}$ and the magnetic field $\vec{B}$.



Flux through the small path is $\vec{B} \cdot \vec{dS}$. To get the flux through the entire surface we add flux through all such patches.



$\phi = BS \cos\theta$ when B is uniform and the surface is flat

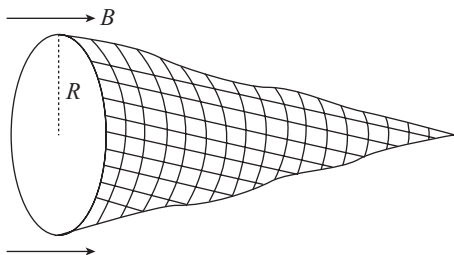
5.2 Magnetism and Electromagnetic Induction

Flux is zero when the direction of $\vec{B}$ is parallel to the surface and it is maximum when $\vec{B}$ is normal to the surface.

Basically, *magnetic flux through any surface held in a magnetic field is measure of number of field lines piercing through the surface*. When no field line pierces through the surface, flux is zero.

Example 1 *A fly net*

A fly net has a circular mouth of radius R . It is placed in a uniform magnetic field B with the direction of the field perpendicular to the circular surface. Find the flux of magnetic field through the curved surface of the net.



Solution

Concepts

The number of field lines passing through the circular base = number of lines passing through the curved net surface.

Any field line that enters through the circular mouth will come out through the curved surface.

Flux through the circular base is

$$\phi = B \cdot \pi R^2 \cos 0^\circ = B \cdot \pi R^2$$

This is also the flux through the curved surface.

3. FARADAY'S LAW OF ELECTROMAGNETIC INDUCTION

The law can be stated as:

When the flux of magnetic field through the area of a closed conducting loop is made to change, an emf is induced in the loop. The emf is equal to negative of the rate of change of the flux.

$$\varepsilon = - \frac{d\phi}{dt} \quad (2)$$

If the coil has N loops, the induced emf in the coil is N times higher.

$$\varepsilon = - N \frac{d\phi}{dt} \quad (3)$$

In our discussions, we will call the quantity $N\phi$ as the 'total flux' through a coil where ϕ is the flux through single turn.

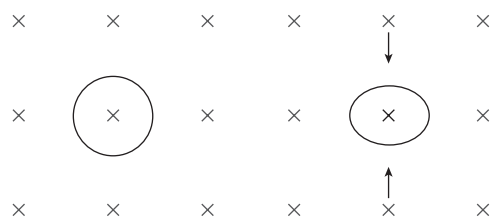
The emf induced will drive a current in the loop. This current is known as *induced current*. If R is the resistance of the loop, the induced current is

$$i = \frac{\varepsilon}{R} = - \frac{1}{R} \frac{d\phi}{dt} \quad (4)$$

Ways to change flux ($\phi = BScos\theta$)

An emf is induced in a closed conducting loop only when the magnetic flux linked to it is changed. As long as the flux keeps changing, the emf persists. We can change the flux linked to a loop in following ways.

- By varying the magnetic field $\vec{B}$ with time. When you push a magnet towards a coil, you are actually changing $\vec{B}$ at the location of the coil.
- By changing the area of the loop.



A loop in a magnetic field

If the loop is deformed (by applying force) its area changes Flux linked to the area also changes.

- By varying the angle between the magnetic field $\vec{B}$ and area vector ($\vec{S}$) of the loop. This can be done easily by rotating a loop in a magnetic field.

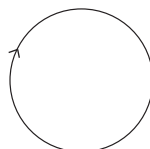
Of course one can change more than one quantity (among B , S and θ) simultaneously.

Direction of induced emf/current

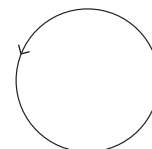
Assume a positive direction of current in the loop. The choice is arbitrary. Curl your right hand fingers in the sense of the positive direction. The stretched thumb gives the direction of area vector ($\vec{S}$) of the loop. Now, calculate the flux $\phi = \int \vec{B} \cdot d\vec{S}$ through the area of the loop. If ϕ is increasing

$\frac{d\phi}{dt}$ is positive and, according to equations (2) and (4), the

induced emf and induced current is negative. A negative current means a current in a direction opposite to the chosen positive direction. When $\frac{d\phi}{dt}$ is negative, the induced emf/current is positive. The induced current is in the positive direction marked on the loop.



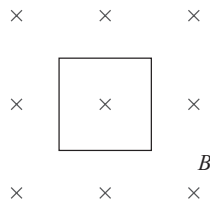
If clockwise is positive, the direction of area vector $\vec{S}$ is inwards.



If anticlockwise is positive, the direction of area vector $\vec{S}$ is outwards.

Example 2 Changing B

Consider a conducting loop of area $S = 0.2\text{m}^2$ kept in a uniform magnetic field. The plane of the loop is perpendicular to the direction of the field. The magnetic field changes with time according to $B = (0.2t)T$, where t is time in second.



- Find the magnitude of emf induced in the loop.
- Find the direction of induced current in the loop.
- If the loop is made of two turns of wire, rather than one, how will the induced current change?

Solution**Concepts**

- $|\varepsilon| = \left| \frac{d\phi}{dt} \right|$
- The negative sign in the expression $\varepsilon = - \frac{d\phi}{dt}$ is essential to fix the direction of induced emf/current.
- If number of turns is doubled, the induced emf doubles (equation (3)) but the resistance of the loop also doubles.

Assume that clockwise direction is positive. It means we are taking the direction of area vector of the loop to be directed into the plane of the figure. [Curl your right hand fingers in the clockwise sense, the thumb points inwards]

(a) $\phi = B \cdot S \cdot \cos 0^\circ = (0.2t)(0.2) = 0.04t$

An emf is induced in the loop as the flux is changing.

Induced emf is:

$$\varepsilon = - \frac{d\phi}{dt} = -0.04 \text{ V} \quad (i)$$

Magnitude of the emf is 0.04 V.

- (b) We will represent the induced emf with the symbol of a cell (though, physically there is no cell) in the loop. The emf cannot be as shown in figure (a). Such an emf will drive a current in positive direction. Equation (i) tells us that the emf is negative. Therefore, the emf represented in figure (b) is correct. The current is anticlockwise.

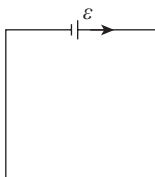


Fig (a)
Incorrect

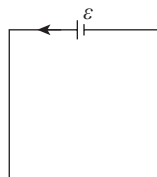


Fig (b)
Correct

- (c) For N turns in the loop, the induced emf is

$$\varepsilon = -N \frac{d\phi}{dt}$$

For $N = 2$, $\varepsilon = -0.08 \text{ V}$.

Doubling the turns, doubles the length of wire used. This doubles the resistance of the loop.

$\therefore$ Induced current does not change.

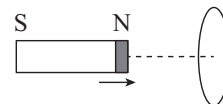
4. LENZ'S LAW

This law was formulated by Emil Lenz in 1830s. The law gives us a simpler method of deciding the direction of induced current in the phenomenon of electromagnetic induction. The law can be stated as:

The direction of the induced current is so as to oppose the change that has induced it.

If a current is induced in a loop due to an increasing flux, the magnetic field produced by the induced current will be directed so as to weaken the flux. If a current is induced by a decreasing flux, it will produce a magnetic field that will strengthen the original flux. Following example will illustrate the application of this law.

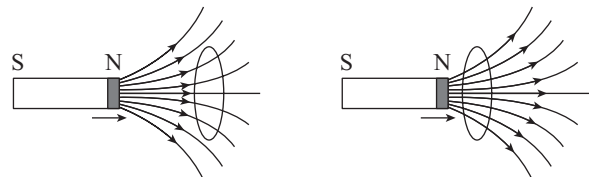
Example 3 A magnetic approaching a conducting ring
A bar magnet is placed along the axis of a fixed conducting ring. The magnet is moved towards the ring (see figure).



- Find the direction of the induced current in the ring.
- Will the magnet experience a force due to the induced current in the ring? Is the force attractive or repulsive?

Solution

- (a) As the magnet approaches the ring, the magnetic flux through the ring increases. The figure given below demonstrates this in terms of the magnetic field lines. You may simply think that the magnetic field at the location of the ring strengthens as the magnet approaches it.

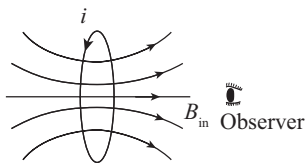


Number of field lines passing through the ring increases as the magnet gets closer.

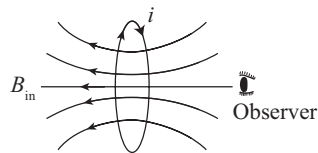
5.4 Magnetism and Electromagnetic Induction

The changing flux induces an electric current in the ring (Faraday's law). Let us assume, for a moment, that the induced current is anti-clockwise as seen from the right side of the ring. Such a current will produce its own magnetic field as shown in the first figure given below. This will further increase the already increasing flux. Therefore, the rate of change of flux will become higher and induced current will increase further. The increased current will increase the rate of flux change even more, and so on. This will violate the *law of conservation of energy*. Lenz's law says that this will never happen.

Lenz's law says that the current induced in the ring will be in clockwise direction (seen from right). Such a current will produce a magnetic field that will weaken the increasing flux due to the magnet.



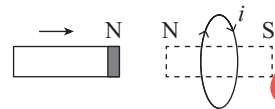
If induced current is anticlockwise, it produces a field (B_{in}) which will help the already increasing flux. This is not possible.



If induced current is clockwise, it produces a B_{in} which opposes the increasing of B due to the magnet getting closer.

- (b) The cause of induction can be stated as – “the magnet getting closer to the ring”. The induced current must oppose this cause. The induced current will definitely repel the magnet.

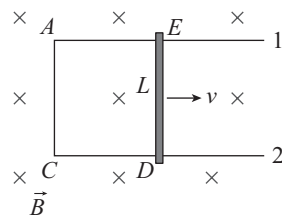
To look at it in slightly different way we can always think of a current loop as a magnetic dipole. With the induced current in it, the ring is like a dipole with its dipole moment directed towards left. This dipole repels the magnet.



Ring is a dipole that repels the approaching magnet

Example 4 Sliding wire

1 and 2 are long parallel conducting rails at separation L . The ends A and C are joined using a straight wire AC . DE is a conducting rod that can slide on the rails. It is placed perpendicular to both the rails and moved with constant velocity v perpendicular to its length. A uniform magnetic field B exists in the entire space directed perpendicular to the plane of the rails.



- (a) Find the emf induced in the closed loop $ACDE$.
(b) Find the direction of current in the loop.

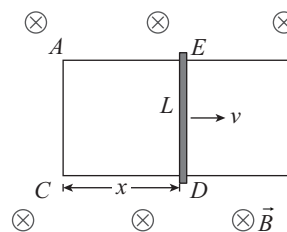
Solution

Concepts

- (i) We will use Faraday's law to write the magnitude of the induced emf and Lenz's law to predict the direction of the induced current.
(ii) The flux attached to the loop $ACDE$ is changing because the area of the loop is increasing.

- (a) Let $CD = x$

Area of the loop is $S = Lx$. Consider the area vector to be directed into the plane of the figure. Flux through the loop is



$$\phi = BS \cos 0^\circ = BLx$$

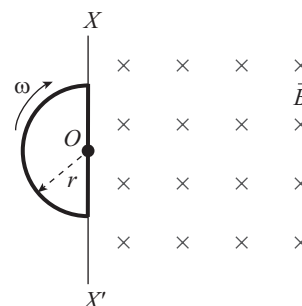
Magnitude of the induced emf is

$$\varepsilon = \left| \frac{d\phi}{dt} \right| = BL \left(\frac{dx}{dt} \right) = BLv$$

Note that to write the magnitude of ε , we need not fuss about the direction of area vector (whether inward or outward) or about the negative sign in the Faraday's law.

- (b) As the sliding conductor moves to right, the flux linked with the loop $ACDE$ increases. The induced current in the loop must be anti-clockwise. This current produces a magnetic field that is directed outwards and thereby weakening the flux linked with the loop.

Example 5 There is a uniform magnetic field B in the region to the right of XX' . The field is directed perpendicular to the plane of the figure. The region to the left of XX' is free of field. A semicircular conducting loop of radius r with its centre at O is located in the plane of the paper. The loop is rotated with a constant angular velocity ω about an axis through O and perpendicular to the plane of the paper.



The resistance of the loop is R .

- (a) Obtain an expression for the magnitude of the induced current in the loop.
(b) Show the direction of current when the loop is entering into the region of the field.
(c) Plot a graph between the induced emf and the time of rotation for two periods of rotation.

Solution**Concepts**

- (i) As the loop enters into the field region, the flux through it increases due to more and more area getting exposed to the field.
- (ii) The magnitude of emf is $\varepsilon = \left| \frac{d\phi}{dt} \right|$. The direction of induced current can be easily found by applying Lenz's law. As the loop enters into the field region, the induced current must produce a field that opposes the increasing flux.
- (iii) When the loop is completely inside the field region, flux is maximum. After this the flux decreases as the loop moves out of the field region.

At time t , $\theta = \omega t$

Area of the loop in the field region is

$$S = \frac{\pi r^2}{2} \left(\frac{\theta}{\pi} \right) = \frac{r^2}{2} \theta$$

$$= \frac{r^2}{2} \omega t$$

Flux through the loop at time ' t ' is

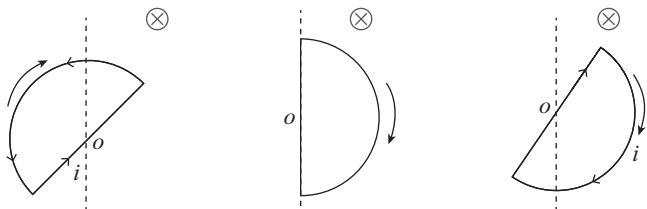
$$\phi = B \cdot S = \frac{Br^2}{2} \omega t$$

$\therefore$ Induced emf in the loop is $\varepsilon = \left| \frac{d\phi}{dt} \right| = \frac{1}{2} Br^2 \omega$. This is a constant.

- (a) Induced current is $i = \frac{\varepsilon}{R} = \frac{1}{2} \frac{Br^2 \omega}{R}$
- (b) Induced current in the loop is anti-clockwise. This current produces a field that is directed outward. This tries to weaken the field and thereby decrease the flux.
- (c) When the loop begins to come out, the direction of induced current/emf becomes clockwise. This current creates an inward field trying to increase the flux. We will show this emf as negative in our graph.

Time period of rotation is $T = \frac{2\pi}{\omega}$.

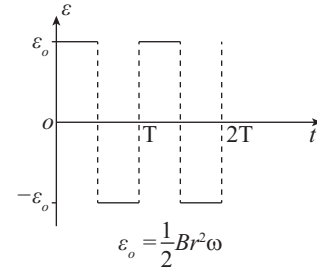
The graph has been plotted for two periods.



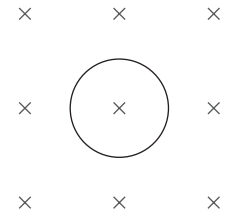
Flux increasing.
Induced current (i)
is anti-clockwise

Flux is maximum

Flux is decreasing
Induced current (i)
is clockwise.

**Example 6** Charge flow

A closed conducting loop of area S is placed perpendicular to a uniform magnetic field. The strength of the field is B_0 . How much charge will flow in the loop if the field is switched off? What is the direction of the charge flow? Resistance of the loop is R .

**Solution****Concepts**

- (i) The field decreases from B_0 to zero in a small interval Δt . Flux changes rapidly and an emf is induced. This causes a current to flow. This current lasts for interval Δt .
- (ii) The flux is decreasing. The induced current must be clockwise so as to produce a field directed inwards.

Flux at any instant is $\phi = B \cdot S$

$$\text{emf induced is } \varepsilon = \left| \frac{d\phi}{dt} \right| = S \frac{dB}{dt}$$

$$\text{Instantaneous current is } i = \frac{\varepsilon}{R} = \frac{S}{R} \frac{dB}{dt}$$

Charge flow in interval dt is

$$dq = idt = \frac{S}{R} dB$$

$\therefore$ Charge flow when the field changes by B_0 is

$$\Delta q = \int dB = \frac{S}{R} \int dB = \frac{S}{R} B_0$$

The direction of current is clockwise so as to produce a field in the direction of the original field.

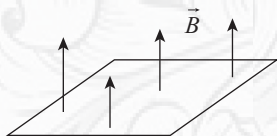
Note: If the magnetic flux through a coil changes by $\Delta\phi$, the quantity of charge that flows in the coil is given by

$\Delta q = \frac{\Delta\phi}{R}$. This does not depend on the time needed for the flux change.

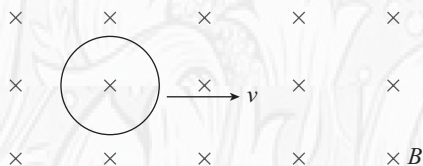


YOUR TURN

Q.1 Magnetic field (B) is uniform in a region. A square surface of area S held in the field region with the field directed normal to the surface. Now the surface is tilted by 60° about its diagonal. Find the magnetic flux through the surface.

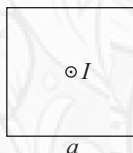


Q.2 A uniform magnetic field B exists in a region of space. A conducting loop of area S is placed in the field region with its plane normal to the field. The loop is moved in the field region in its own plane with velocity v . Find the emf induced in the loop. Find the emf induced if the loop is moved in the direction of the field (i.e., perpendicular to the plane of the figure).



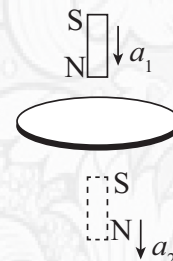
Q.3 A coil has 100 turns and has a resistance of $R = 10\Omega$. The area of the coil is $S = 0.1\text{m}^2$. A magnetic field perpendicular to the plane of the coil is turned on. The field changes at a uniform rate from 0 to 1.0T in 0.5s . Find the magnitude of induced emf in the coil and the current in it when the field is changing

Q.4 A conducting square loop has side length a . A long straight current carrying wire passes through the centre of the loop, perpendicular to its plane. The current (I) in the straight wire is decreased. Find the direction of induced current in the loop.



Q.5 A copper ring of radius 0.1m is placed perpendicular to a uniform magnetic field B . The magnetic field is increased at a rate of 0.2 T/s . Emf induced in the loop is ϵ_1 and the induced current is i_1 . Another aluminium ring of same size, made of wire of same thickness is also placed perpendicular to a uniform magnetic field. The field is decreasing at a rate of 0.2T/s . Emf induced in the ring is ϵ_2 and the induced current is i_2 . Find ϵ_1 and ϵ_2 . Which one is larger i_1 or i_2 ?

Q.6 A copper ring is held in horizontal position. A bar magnet is allowed to fall vertically along the axis of the ring. As the magnet approaches the ring its acceleration is a_1 . After it crosses the ring and moves away from it, its acceleration is a_2 . Which of the following two statements are true.



(a) $a_1 > g$

(b) $a_2 > g$

Q.7 A conducting loop is kept in a strong magnetic field that is nonuniform. Will a current get induced in the loop?

Q.8 A coil of area S is placed perpendicularly into a uniform magnetic field B . It is rotated slowly by 90° so that its surface gets parallel to the field. Find the quantity of charge that flows in the coil. Resistance of the coil is R .

In Short

- (i) Flux of magnetic field ($\vec{B}$) through a surface having area $\vec{S}$ is defined as

$$\phi = \int \vec{B} \cdot d\vec{S}$$

For uniform field and plane surface, $\phi = BS \cos\theta$. Here, θ is angle between $\vec{B}$ and $\vec{S}$.

- (ii) Faraday's law of electromagnetic induction says that an emf is induced in a closed conducting loop whenever the magnetic flux linked to it is made to change. The induced emf is given by

$$\epsilon = - \frac{d\phi}{dt}$$

The minus sign in the above expression is essential to decide the correct direction of induced emf/current. But we mostly decide the direction of emf by using another law known as Lenz's law.

- (iii) Lenz's law says that the induced current is directed so that it can produce a magnetic field which opposes the change in flux. If the cause of induction is *decreasing flux*, the induced current will produce a magnetic field that will increase the flux. If the cause of induction is *increasing flux*, the induced current produces a field that weakens the flux.

(iv) Cause of induction can often be stated using phrases like

- Magnet getting closer to the coil.
- Coil moving away from the magnet.
- Conductor moving to the right, etc.

In the above cases the direction of induced current shall be such that

- the magnet gets repelled by the coil
- the coil gets attracted by the magnet

- the conductor experiences a force to the left.

(v) When the magnetic flux changes through a closed loop by an amount $\Delta\phi$, then the quantity of charge that flows in the loop is given by

$$\Delta q = \frac{\Delta\phi}{R}$$

Here, R is resistance of the loop. The charge flow (Δq) is independent of the time needed for the change in the flux.

5. THE SOURCE OF INDUCED EMF

When you place a conductor in an electrostatic field, the charges move for a moment. They quickly rearrange themselves so as to nullify the external field inside the conductor. If we want to keep the charges moving in a conductor, we need some external mechanism to maintain the electric field inside it. A battery is such a device. *The amount of work done by the external mechanism (such as a battery) in moving a unit charge once in an electrical loop is known as emf in the loop.*

When a battery is connected across a conductor, it establishes an electric field inside it. This field keeps pushing the free electrons and maintains an electric current. What happens in electromagnetic induction? Which force keeps pushing the electrons to maintain the induced current? We will investigate this problem under two headings – *Motional emf and induced electric field*. In motional emf we will understand what happens when a conductor is moved in a static magnetic field. In the topic of induced electric field we will investigate the case when a conducting loop is kept in a time changing magnetic field.

5.1 Motional emf

Consider a straight conducting bar of length L placed perpendicular to a uniform magnetic field B . The free electrons are moving randomly inside the conductor. Individually, they experience magnetic force but collectively there is no net force. It is better to forget about this random motion of electrons.

Now, suppose that the bar is moved perpendicular to its length with velocity v . The electrons also move with the bar. Due to this velocity they experience a magnetic force.

$$\vec{F}_B = -e(\vec{v} \times \vec{B})$$

In the figure shown, this force is directed towards end B and has magnitude

$$F_B = evB$$

This magnetic force pushes the free electrons towards end B and the end develops a negative charge. The end A

of the rod acquires a positive charge. Due to this separation of charge an electric field is established inside the bar directed from A to B . This electric field exerts a force on a free electron towards the end A . Further accumulation of charges at the ends A and B stops when the magnetic and electric force on an electron balances. If E is the electric field inside the conductor in this situation, then

$$eE = evB$$

or, $E = vB$

The potential difference between the two ends is $EL = vBL$. This is the induced emf in the bar.

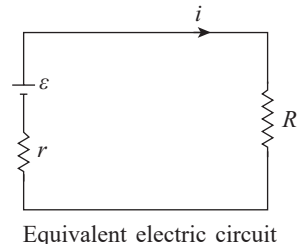
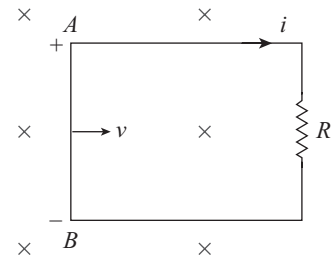
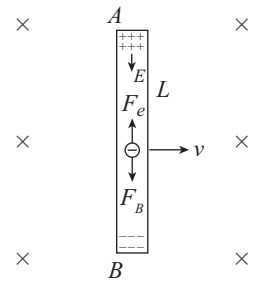
$$\varepsilon = vBL \quad (5)$$

If we connect a resistance (R) across the ends A and

B , the induced emf drives a current through it. As long as the bar keeps moving in the magnetic field, the emf persists. Due to the current, the charges at ends A and B tend to neutralise. Any reduction in the accumulated charge causes the electric field E to decrease. This makes the magnetic force (F_B), on a free electron, stronger. This imbalance in force will push more electrons to the end B and replenish the charge. This keeps the emf constant as long as the bar keeps moving uniformly.

To find the current, we can draw an equivalent circuit as shown. The resistance of the moving bar is r and you can treat the situation as if the bar is a cell with emf ε and internal resistance r .

$$i = \frac{\varepsilon}{R+r} = \frac{BLv}{R+r}$$



5.8 Magnetism and Electromagnetic Induction

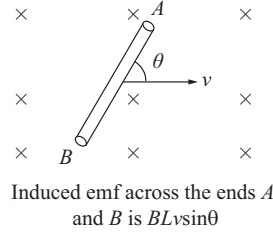
Following points are important with regard to the motional emf.

- Faraday's law is actually not needed to understand the origin of emf when a conductor is moved in a static magnetic field.
- In the above discussion, the three quantities B , v and L were mutually perpendicular. No emf is induced across the ends of the rod if any of these two are parallel. For example, no emf will be induced between the ends A and B of the bar if the bar is moved in a direction parallel to its length or if it is moved in the direction of $\vec{B}$.
- When angle between any two of B , v and L is θ and the third quantity is perpendicular to the two, then the induced emf is

$$\varepsilon = BLv \sin\theta \quad (6)$$

An example to this is shown in the figure.

One may think that the component of velocity parallel to the length ($= v \cos\theta$) contributes nothing to the emf and the component perpendicular to the length ($= v \sin\theta$) is responsible for the emf.



- The positive end of the bar is towards $\vec{v} \times \vec{B}$.

5.1.1 Motional emf in a curved wire

Consider a curved conducting wire AC in the plane of the paper. The wire is moved with a velocity v in the plane of the paper. There exists a uniform magnetic field (B) perpendicular to the plane of the figure.

Assume a small segment of length dl in the wire. The angle between this segment and the velocity (i.e., x -direction) is α . The emf induced in this segment is

$$\begin{aligned} d\varepsilon &= B dl v \sin\alpha \\ &= Bv (dl \sin\alpha) \\ &= Bv dy \end{aligned}$$

dy is projection of length dl along y direction.

In the diagram shown, the positive and negative ends have been marked on the segment.

The total emf induced in the wire can be obtained by adding the emfs in all such segments. You can think as if

large number of small cells having small emfs have been connected in series between the ends A and C . Therefore, emf in the wire is

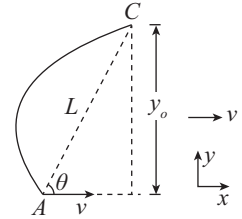
$$\varepsilon = \int d\varepsilon = Bv \int dy$$

or, $\varepsilon = Bvy_0$ (i)
Here, y_0 = length of projection of the curved wire on the y axis.

If we replace the entire curved wire AC with a straight wire AC (shown in dashed line), we can write the emf induced in it as

$$\varepsilon_{\text{straight (AC)}} = BvL \sin\theta = Bvy_0.$$

Therefore we conclude that for calculation of emf induced between the ends of a curved wire moving in a uniform field (B), the wire can be replaced with a straight wire joining its two ends.



5.1.2 Rotating conductor in a uniform field.

AC is a conducting rod of length L . It rotates about its end A with an angular velocity ω , as shown in the figure. There exists a uniform magnetic field B perpendicular to the plane of the paper. We wish to find emf induced between the two ends of the rod.

Consider a segment (MN) of length dx at a distance x from the end A . Velocity of this segment is $v = \omega x$ and is directed perpendicular to the element. An emf is induced in the segment given by

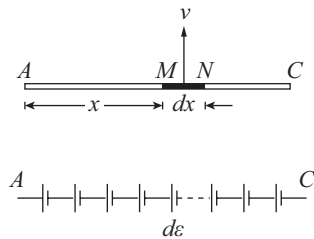
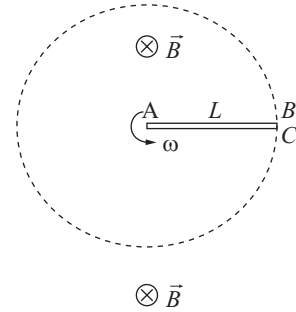
$$d\varepsilon = B(\omega x) dx \quad (i)$$

The positive end is directed along $\vec{v} \times \vec{B}$, i.e., towards end M . [You can think of an electron inside MN moving with velocity v . This electron will experience magnetic field towards end N .]

Every segment in the rod has an induced emf and all are in series – with their positive ends towards A . The total emf between A and C is

$$\varepsilon = \int_A^C d\varepsilon = B\omega \int_0^L x dx$$

$$\text{or, } \varepsilon = \frac{1}{2} B\omega L^2 \quad (7)$$



A small emf is induced in every small segment of the rod. These emf are in series.

This expression is approximately correct. Equation (5) [and hence (i)] is correct when the magnetic force is balanced by the electric force on an electron inside the conductor. In the case of a rotating rod, described above, an electron inside the rod must have a resultant centripetal force. If we assume that the electric field at a distance x from the end A is E , then the force on a free electron located at distance x from the end A is

$$F_e - F_B = eE - evB$$

This must be equal to the centripetal force needed for the circular motion.

$$\therefore eE - evB = m\omega^2 x \quad [m = \text{mass of an electron}]$$

$$\therefore E = vB + \frac{m\omega^2 x}{e}$$

$$\text{or, } E = B\omega x + \frac{m\omega^2}{e} x \quad (\text{ii}) \quad [v = \omega x]$$

The second term on the right side is small due to small value of $\frac{m}{e}$ for an electron. This can be neglected, in general.

$$\therefore E \simeq B\omega x$$

The electric field varies along the length of the rod. There is continuous distribution of the charge along the length of the rod with more negative charge towards C . The potential difference across the ends A and C can be calculated as

$$\int_A^C dV = - \int_{x=0}^{x=L} E dx$$

$$\text{or, } V_C - V_A = - B\omega \int_0^L x dx$$

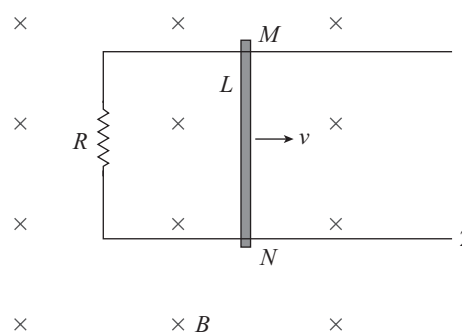
$$\text{or, } V_A - V_C = \frac{1}{2} B\omega L^2$$

This is the induced emf which is same as equation (7). If we do not neglect the centripetal force on an electron and take the electric field as given in equation (ii), then the induced emf will be

$$\varepsilon = \frac{1}{2} B\omega L^2 + \frac{1}{2} \frac{m\omega^2 L^2}{e}$$

Example 7 Revisiting the sliding wire system in example 4.

Two smooth, conducting parallel rails (1 and 2) are kept at a separation L . One ends of the rails are connected using a resistance R . The entire system is located in a uniform magnetic field B directed perpendicular to the plane of the figure. A sliding bar MN is placed perpendicularly on the rails and is moved with constant velocity v as shown. Except for R there is no resistance in the system.



- Calculate the current in the resistance R .
- Calculate the rate of heat dissipation in R .
- Calculate the power of the external agent to keep the bar moving uniformly.
- Give a suitable name to the device.

Solution

Concepts

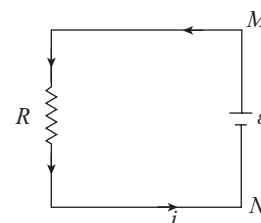
- By the concept of motional emf, the sliding bar will act like a source of emf. Its emf is given by equation (5) and is equal to $\varepsilon = BvL$.
- The positive terminal is directed towards $\vec{v} \times \vec{B}$.
- A current in the bar will cause it to experience a magnetic force. The external agent must apply an equal and opposite force to keep the bar moving uniformly.

- Induced emf in the bar is

$$\varepsilon = BvL$$

The equivalent circuit is as shown. The current is

$$i = \frac{\varepsilon}{R} = \frac{BvL}{R}$$



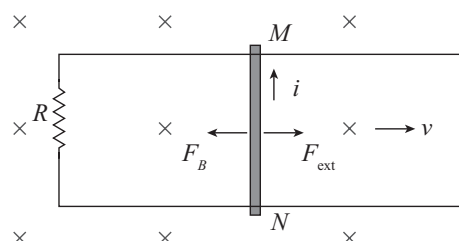
- Power dissipated in R will be

$$P_R = i^2 R = \frac{B^2 v^2 L^2}{R} \quad (\text{i})$$

- The current in the sliding bar is directed from N to M . This current carrying bar experiences a magnetic force to the left given by

$$F_B = iLB \quad [\because \vec{F} = i\vec{L} \times \vec{B}]$$

$$= \frac{B^2 L^2 v}{R}$$



5.10 Magnetism and Electromagnetic Induction

The external agent must apply equal force to the right.

$$F_{\text{ext}} = \frac{B^2 L^2 v}{R}$$

(Note that the magnetic force on the bar is to the left trying to slow it. This is in accordance to the Lenz's law.)

Power spent by the external agent is

$$P_{\text{ext}} = F_{\text{ext}} \cdot v = \frac{B^2 L^2 v^2}{R} \quad (\text{ii})$$

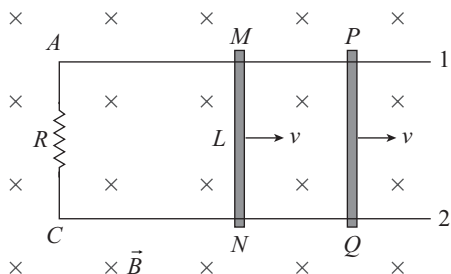
- (d) From (i) and (ii) $P_{\text{ext}} = P_R$.

The rate of work done by the external agent is same as rate of power dissipation in the resistance R . This device is basically converting mechanical work done by the agent into electrical energy. The device is a simple generator. Let's call it a *sliding wire generator*.

Note: The device described above is a 100% efficient generator. It converts the entire mechanical work into electrical energy. If we assume some friction between the sliding bar and the rails, the efficiency will not be 100%.

Example 8 A pair of sliding wires

In the system described in the previous example, let's add one more sliding rod PQ . Both the wires (MN and PQ) are moved uniformly with velocity v . Assume that each of the sliding rods has a resistance r .



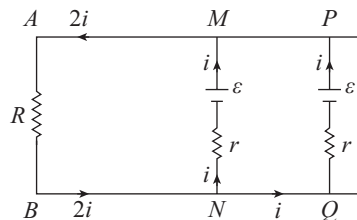
- Find the induced emf in the loop $MNQP$. Does the concept of motional emf contradict the Faraday's law?
- Find the potential difference between the ends M and N of the sliding bar.

Solution

Concepts

- As per the concept of motional emf, the emf induced in each of the sliding bars is $\varepsilon = BLv$ with ends M and P positive.
- Each bar can be treated like a cell of emf ε and resistance r . Once we know the current through a bar, we can find the potential difference between the two ends.

- According to Faraday's law the emf in the loop $MNQP$ is zero. The area of the loop is not changing and the flux linked to it is also not changing.



There is emf in each of the sliding bars. When you move along $MNQP$ the net emf is zero. Please note that the concept of motional emf and the Faraday's law are not contradicting one another.

- Due to symmetry, the current in both the sliding wires is same (say, i). The equivalent electric circuit is as shown. Using Kirchhoff's voltage law in the loop $ABNM$, we get

$$ri + R(2i) = \varepsilon \quad \Rightarrow \quad i = \frac{\varepsilon}{2R + r} \quad [\varepsilon = BLv]$$

Now,

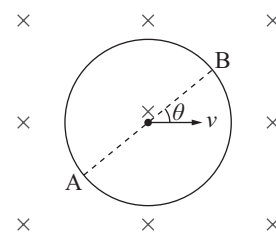
$$V_N - ir + \varepsilon = V_M$$

$$\Rightarrow V_M - V_N = \varepsilon - ir = \varepsilon - \frac{\varepsilon r}{2R + r}$$

$$= \frac{2\varepsilon R}{2R + r} = \frac{2BLvR}{2R + r}$$

Example 9 Emf between two points of a moving circular loop

A circular conducting loop has radius r . It is kept perpendicular to a uniform magnetic field B . The loop is moved with a velocity v as shown.



- What is induced emf in the loop?
- AB is a diameter of the loop making an angle θ with the direction of its velocity. If we connect a voltmeter across A and B , will the voltmeter show any reading? How much?

Solution

Concepts

- Use of Faraday's law tells us that there is no emf in the loop. The flux through the loop is not changing. But the use of the concept of motional emf tells us that there will be an electric field inside the conductor and we do expect an emf between two points on the loop.
- The emf induced in a curved conductor is same as that induced in a straight conductor obtained by joining the ends of the curved conductor.

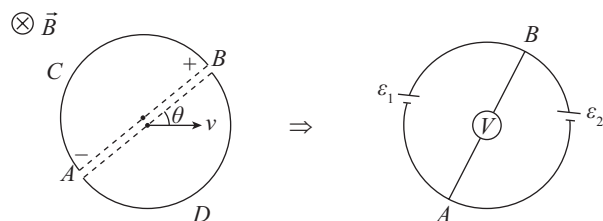
(a) Net emf in the loop is zero as there is no change in flux linked with the loop.

(b) Consider the semicircular part ACB of the loop. The emf induced in this part is same as that induced in a straight wire AB . It is equal to

$$\varepsilon_1 = Bv (2r) \sin \theta$$

Similarly the emf induced in semicircular part ADB is same as that in a straight wire AB . It is also equal to

$$\varepsilon_2 = Bv (2r) \sin \theta = \varepsilon_1$$

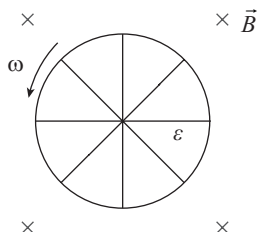


There is no current in the loop as the net emf is zero. When a voltmeter is connected between the points A and B it will read a voltage equal to

$$\varepsilon = \varepsilon_1 = \varepsilon_2 = Bv (2r) \sin \theta$$

Example 10 Rotating wheel

A wheel of radius R has 8 spokes. The wheel is made of conducting material. It is rotated with an angular speed ω in a uniform magnetic field B perpendicular to its plane. Find the emf developed between the fixed centre and the circumference of the wheel.



Solution

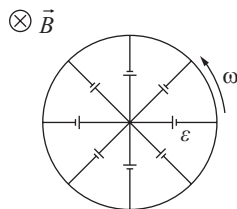
Concepts

- The emf developed in a spoke is given by the equation (7).
- The spokes are in parallel between the centre and the circumference.

The emf induced in each of the spokes is

$$\varepsilon = \frac{1}{2} B \omega R^2$$

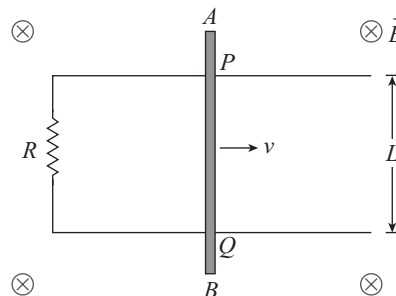
For the direction of rotation shown in the figure, the positive end of the emf developed is the centre. [Imagine a small segment in any of the spokes. The positive end is in the direction indicated by $\vec{v} \times \vec{B}$]



We can think of a cell of emf ε = $\frac{1}{2} B \omega R^2$ in each of the spokes. The resultant emf of all these emfs in parallel is also $\varepsilon = \frac{1}{2} B \omega R^2$.

Example 11 Longer sliding wire

In a sliding wire generator (see figure) the sliding wire AB has a length $2L$ and is placed perpendicularly on two parallel rails at separation L . The wire is moved with velocity V . Magnetic field (B) is uniform perpendicular to the plane of the figure.



- Find the emf induced in the sliding wire AB .
- Find the current in resistance R .

Solution

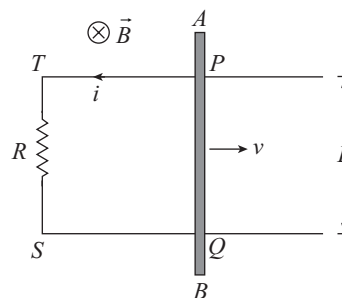
Concepts

- The emf induced in the wire AB is independent of presence of rails and the circuit.
- The entire emf induced in the rod is not used to drive the current.

- As the wire AB moves the end A becomes positive and B becomes negative. From the concept of motional emf, the induced emf is

$$\varepsilon_{AB} = B(2L)v = 2BLv$$

- There is an electric field inside the rod directed from A to B as it moves in the magnetic field.



The emf that is part of the loop $PTSQ$ is equal to

$$\varepsilon_{PQ} = E \cdot L$$

Where E is the electric field inside the wire AB . It is easy to see that

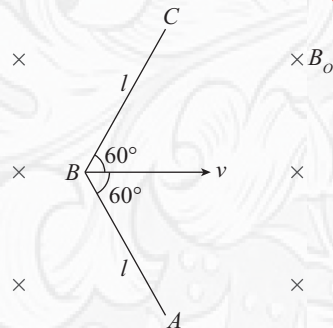
$$\varepsilon_{PQ} = BLv$$

$$\therefore i = \frac{BLv}{R}$$

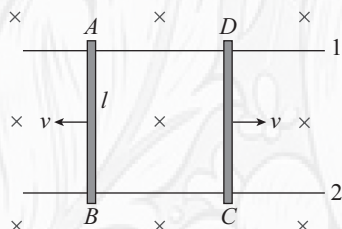


YOUR TURN

Q.9 A conducting wire ABC has been bent with its two arms AB and BC making 120° . Length of each arm is l . The bent wire is moving with velocity v along the angle bisector of the angle ABC . There exists a uniform magnetic field (B) in the region directed perpendicular to the plane of the wire. Find the emf between two ends A and C of the wire.

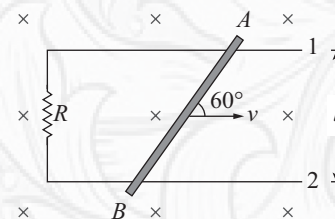


Q.10 Two perfectly conducting rails 1 and 2 lie perpendicular to a uniform magnetic field B . Separation between the rails is l . Two bars AB and CD are placed perpendicular to the rails and moved away from one another with velocity v . The resistance of each bar is R . Find the

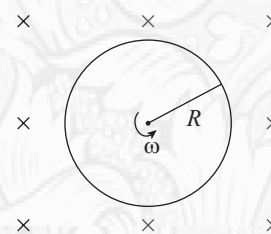


- Current in the bars
- Force needed to keep the bars moving uniformly.

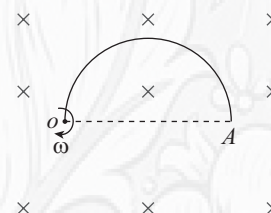
Q.11 Two parallel, conducting rails 1 and 2 are kept perpendicular to a uniform magnetic field (B). The rails are at separation l and are joined at their ends by a resistance R . A conducting bar AB is placed on the rails making 60° with them. The bar is pulled with a velocity v parallel to the rails. Find the current in the resistance R .



Q.12 A metal disc of radius R rotates about its central axis with angular speed ω . The disc is placed in a uniform magnetic field (B) directed perpendicular to its plane. Find the potential difference between the centre of the disc and its circumference.



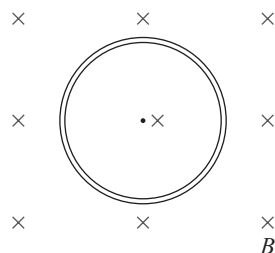
Q.13 A conducting wire is bent in the shape of a semicircle of radius r . It rotates about an axis perpendicular to its plane passing through its end O . The angular speed is ω . There is a uniform magnetic field (B) present in the entire space. Find the induced emf between the ends O and A .



5.2 INDUCED ELECTRIC FIELD

In the last section of 'motional emf' we learnt that when a conductor is moved through a magnetic field, the free electrons inside it experience force that leads to polarization. This produces an emf. In this section we will try to understand what happens when a conducting loop is placed in a time changing magnetic field.

Consider a conducting loop in a magnetic field B . The random motion of electrons inside the conductor do not produce a resultant magnetic force. It is better to forget the random thermal motion of the electrons. Suppose the magnetic field is increased. Flux linked with the loop changes and a current is induced in the loop. The question that arises is – which force causes the electrons to



move? Certainly it is not the magnetic force. A magnetic force never acts on a charge that is at rest. We explain the force on electrons inside the conductor by saying that a *time changing magnetic field induces an electric field*. This induced electric field drives the current in the conductor. Following point are worth noting about the induced electric field.

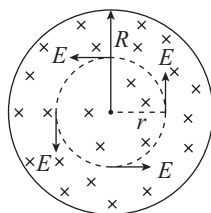
- This electric field is produced by time changing magnetic field and not by charges particles. It is different from electrostatic fields produced by static charges.
- Induced electric field is non-conservative in nature.
- The lines of the induced electric field are closed curves. There are no starting or terminating points of these lines.
- We cannot define a potential corresponding to this field.

- (e) In one aspect the induced electric field is similar to the electrostatic field — The force it applies on a charge q is $q\vec{E}$.

The presence of a conducting loop is not necessary for existence of the induced electric field. Wherever there is a time changing magnetic field it produces an electric field (E). If a conductor is placed in this region of E , the free electrons experience force and we see a current. In short, we are saying that a time changing magnetic field (B) is a source of electric field (E). This is a bit difficult to accept but we do not have any better explanation for existence of induced current in a loop kept in a changing B .

In certain symmetrical situations we can calculate the value of the induced electric field. Here is an example.

Consider a circular region of radius R in the plane of the paper. A uniform magnetic field B exists inside the circle directed perpendicular to the plane of the paper. The magnetic field is changed at a rate $\frac{dB}{dt}$. We wish to find the induced electric field at a radial distance r inside the circular region ($r < R$).



Time changing B induces E

Consider a circle of radius r . Due to symmetry of the situation, the induced electric field must be of same magnitude (E) at all points on this circle and it must be tangential in direction.

Read the first paragraph of article 5 in this chapter. The work done by the induced electric field on a unit charge in moving it by $\vec{dl}$ is $\vec{E} \cdot \vec{dl}$. The emf developed in the circle of radius r is therefore

$$\varepsilon = \oint \vec{E} \cdot \vec{dl}$$

$$\text{or, } \varepsilon = \oint E \cdot dl \cos 0^\circ = E \oint dl = E 2\pi r \quad (\text{i})$$

Using Faraday's law, we can write the emf in the circular loop as (imagine a conducting circular loop of radius r)

$$\varepsilon = \left| \frac{d\phi}{dt} \right| = \pi r^2 \frac{dB}{dt} \quad (\text{ii})$$

From (i) and (ii)

$$E \cdot 2\pi r = \pi r^2 \frac{dB}{dt}$$

$$\text{or, } E = \frac{r}{2} \frac{dB}{dt} \quad (8)$$

Direction of induced emf (and hence direction of $\vec{E}$) can easily be established using Lenz's law.

Does the region outside the magnetic field ($r > R$) also have induced electric field. The answer is, yes. Consider

a circle of radius r ($> R$) concentric to the field region. Again due to symmetry the field is tangential and has same magnitude at all the points on the circle. The induced emf in the circle is

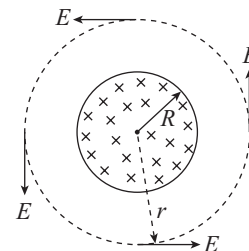
$$\varepsilon = \oint \vec{E} \cdot \vec{dl} = E \cdot 2\pi r.$$

From Faraday's law

$$\varepsilon = \left| \frac{d\phi}{dt} \right| = \frac{d}{dt} (B \cdot \pi R^2) \\ = \pi R^2 \frac{dB}{dt}$$

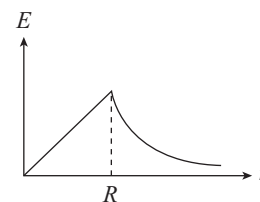
$$\therefore E \cdot 2\pi r = \pi R^2 \frac{dB}{dt}$$

$$\text{or, } E = \left(\frac{R^2}{2} \frac{dB}{dt} \right) \frac{1}{r} \quad (9)$$



The induced electric field is also present outside the region of B

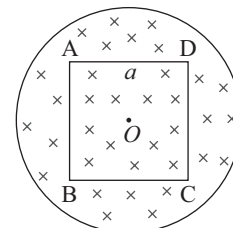
Equation (8) tells us that inside the circular region of R , the induced electric field is proportional to radial distance r . Equation (9) tells us that outside the circular region of radius R , the induced electric field is inversely proportional to r . At very large distance the electric field tends to zero. The variation of the induced field with r has been plotted in the figure.



Variation of induced electric field with r

Example 12 Emf in a square

A circular region centred at O has a uniform magnetic field (B) perpendicular to the plane of the figure. The magnetic field is increased at a rate of 2 T/s . A square loop $ABCD$ of side length a is placed in the region as shown. The centre of the square is at O . Side BC of the square has resistance $2R$ and the resistance of every other side is R . Given $a = 0.5 \text{ m}$, $R = 1 \Omega$.



- Find $\int_B^C \vec{E} \cdot \vec{dl}$ where $\vec{E}$ is the induced electric field and the integration is carried along the straight path BC . Will the value of integration remain same if the integration from B to C is performed along a different path?
- Find the current in the square loop.
- If a voltmeter is connected across B and C , what will be its reading?

5.14 Magnetism and Electromagnetic Induction

Solution

Concepts

(i) $\int_B^C \vec{E} \cdot d\vec{l}$ is nothing but the induced emf in the side BC .

(ii) The emf induced has nothing to do with the resistance of the sides. Therefore, the induced emf in each side of the square is same. (All sides are at identical locations in the magnetic field).

(iii) Performing the integration $\int_B^C \vec{E} \cdot d\vec{l}$ will involve a bit of mathematics but it can be evaluated very easily by the fact that it is $\frac{1}{4}$ of the total emf in the square loop.

(a) According to Faraday's law, emf induced in the square loop is

$$\varepsilon = \left| \frac{d\phi}{dt} \right| = a^2 \cdot \frac{dB}{dt} = (0.5)^2 \times 2 = 0.5 \text{ V.}$$

Emf induced in each of the 4 sides is $\frac{1}{4}$ of ε .

$$\therefore \varepsilon_{BC} = \frac{0.5}{4} = 0.125 \text{ V.}$$

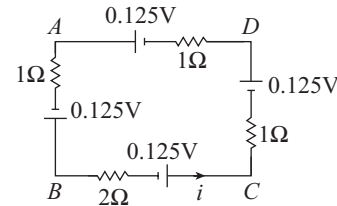
According to the Lenz's law the emf will drive an anticlockwise current in the square loop. In side BC , the induced electric field will drive a current from B to C .

$$\int_B^C \vec{E} \cdot d\vec{l} = 0.125 \text{ V.}$$

(b) Total resistance of the loop is $2\Omega + 3\Omega = 5\Omega$.

$$\therefore i = \frac{0.5V}{5\Omega} = 0.1A.$$

(c) The equivalent electric circuit is as shown.



$$V_B - 2i + 0.125 = V_C$$

$$\text{or, } V_B - V_C = 2 \times 0.1 - 0.125 = 0.075 \text{ V.}$$



YOUR TURN

Q.14 A long solenoid has 1000 turns per meter and its radius is $R = 2.5$ cm. Current in the solenoid is increased at a rate of $\frac{200}{\pi}$ A/s. Find the induced electric field

(a) at a point inside the solenoid at a distance 1 cm from its axis,

(b) at a point outside the solenoid at a distance 4 cm from its axis.

6. EDDY CURRENT

Consider a conductor placed in a time changing magnetic field $B(t)$. The induced electric field drives current along various possible paths inside the conductor. There is no definite loop to guide the induced current. The paths of induced current is difficult to predict. All we can say is that there will be current along many paths inside the conductor.

Figure (b) Shows a pendulum made of a metal plate. When such a pendulum enters into a region having a magnetic field, the flux linked through various loops inside the plate may change with time. Current gets induced along various paths.

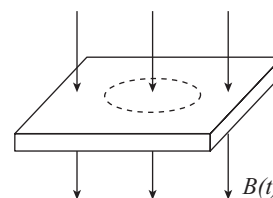


Fig (a)

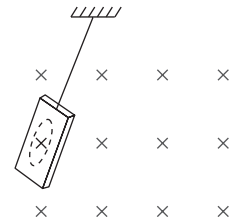


Fig (b)

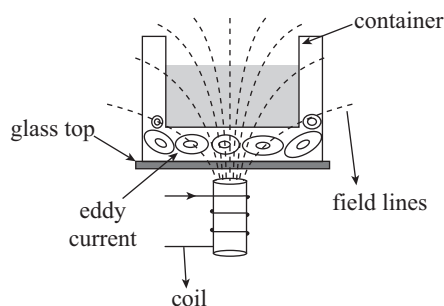
Flux changes through a closed path shown in dotted line. There are various such paths

The current in the entire volume of a conductor, along various unpredictable paths, when the conductor moves through a magnetic field or is placed in a time changing

magnetic field is known as eddy current. Such eddy current causes heating of the conductor and is responsible for energy loss in many electrical devices. However, it is useful in certain instances.

The eddy current induced in the pendulum in figure (b) above, will cause it to retard. According to Lenz's law the induced eddy current will always be so as to oppose the cause of induction. Here the cause of induction is the motion of the pendulum. The force/torque arising due to current in magnetic field will oppose the motion. The heat dissipated in the conductor comes at the cost of kinetic energy. Such damping of motion is often known as electromagnetic damping. Eddy current helps to quickly damp the oscillations of a galvanometer coil. This can be utilized in braking of electric trains. The braking force is turned on or off by activating and deactivating the electromagnets used to create the magnetic field around the wheels.

One more common use that has picked up pace in recent times is induction cooking. The basic principle has been illustrated in the figure.



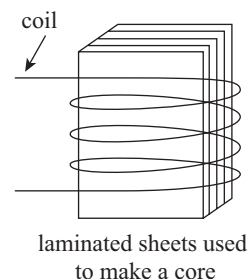
Induction cooking

There is a large copper coil beneath a glass top. A high frequency ac current is supplied to this coil. This current sets up a time changing magnetic field around the coil.

A pot having thick ferromagnetic base is placed on the glass top. The time changing magnetic field induces large eddy currents in the pot. The electrical resistance of the pot produces heat. This heat is conducted to the food being cooked. There is no flame. There is no heating of the glass top. It is the pot which directly gets heated. This method of cooking is quite energy efficient.

In many devices, to reduce energy loss due to eddy currents, we often use laminated sheets to make a core. Transformer is one such device.

Large number of thin sheets are pasted together with a layer of insulation between them. This decreases the number of possible paths of eddy currents. This reduces heating.

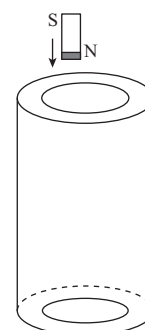


Example 13 A magnet falling through a metal tube

A magnet falls slower through a vertical metallic tube than it does through a nonmetallic tube. Explain.

Solution

When a magnet is dropped down a vertical metal tube, the magnetic field created by the magnet is changing because it is moving. This changing B induces eddy currents in the wall of the tube. These eddy currents have their own magnetic field that opposes the fall of the magnet. The magnet falls much slower than it does in ordinary fall inside a nonmetallic tube. This is a very easy experiment and you must try this. A tube with thick wall and small opening will ensure that you get amazed to see how slow the magnet falls.

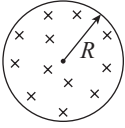


In Short

- (i) Mechanism for induction of emf in a conductor moving in a magnetic field is easily understood in terms of the force experienced by a free electron inside it. The magnetic force causes the electrons to accumulate on one side and the other electron deficient side becomes positive. This separation of charge produces an electric field inside the conductor. A cell also has charges of opposite polarities on its two electrodes. A conductor moving in a magnetic field is like a cell.
- (ii) When a straight conductor of length L moves with a velocity v in a uniform field B , the emf induced in it is given by $\varepsilon = BLv$. This is valid when L , v and B are mutually perpendicular.
- (iii) When two among the three – L , v and B – are parallel, there is no emf between the ends of the conductor.
- (iv) When angle between any two (among L , v and B) is θ then, $\varepsilon = BLv \sin \theta$
- (v) A curved conductor moving in a uniform magnetic field can be replaced with a straight conductor, obtained by joining its ends, for finding the induced emf.
- (vi) A thin conducting rod of length L is rotating with angular speed ω in a plane perpendicular to a uniform magnetic field (B). Emf induced between the ends of the rod is $\varepsilon = \frac{1}{2} B \omega L^2$

5.16 Magnetism and Electromagnetic Induction

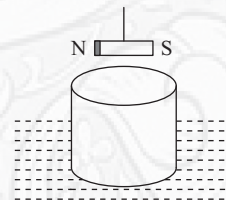
- (vii) When an emf is induced in a moving conductor we can draw an equivalent electric circuit by replacing the conductor with a cell having the same emf and internal resistance equal to the resistance of the conductor.
- (viii) A time changing magnetic field induces an electric field. This thought is used to explain the induced current in a closed conducting loop when it is placed in a time changing B .
- (ix) The induced electric field is non conservative. It is represented by closed field lines. But it exerts force on a charge just like an electrostatic field.
- (x) The emf in a closed loop can be defined as work done by the electric field in moving a unit positive test charge once along the loop.

$$\varepsilon = \int \vec{E} \cdot d\vec{l}$$
- (xi) Imagine a uniform B in a circular region of radius R . There is no field outside the circular region. Field is directed perpendicular to the plane of the figure and is being changed at a rate $\frac{dB}{dt}$. The induced electric field is given by
- $$E = \frac{r}{2} \left(\frac{dB}{dt} \right) \text{ for } r \leq R$$
- $$= \left(\frac{R^2}{2} \frac{dB}{dt} \right) \frac{1}{r} \text{ for } r \geq R.$$
- 
- (xi) When a conductor is moved (or rotated) in a magnetic field or when it is placed in a time changing magnetic field, current is induced along various paths in the entire volume of the conductor. Such current is known as eddy current.



YOUR TURN

Q.15 A metal can (like the one you get with coke or mountain dew) is kept floating in water. The upper lid has been removed. A magnet is tied to a string and lowered inside the can. Now the magnet is rotated by twisting the string. The can also rotates. Explain.

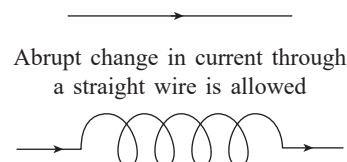


7. SELF INDUCTION

Consider a closed conducting loop (or a coil). A current is given to the loop. This current produced a magnetic field. This magnetic field has its flux through the loop. It means wherever a current flows in a loop or a coil there is a self created magnetic flux associated with it. If we change the current with time, the self flux also changes and hence an emf is induced in the loop/coil. *This process of induction of emf in a coil due to current changing through it is known as self induction.* According to the Lenz's law this induced emf is so as to oppose the change in current through the loop/coil.

A straight wire has property of resistance. This property does not oppose our attempt to change the current in the wire. If we double the potential difference the current doubles immediately. If we half the potential difference the current halves immediately. When the same wire is coiled, it acquires an additional property of self inductance apart from resistance. Due to this property it opposes any change in current through it. This property does not allow us to change the value of current suddenly. One cannot increase a current from 1A to 5A immediately by changing the potential

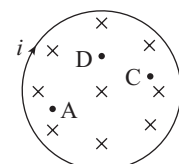
difference across a coil. The change will happen slowly. *The property of inductance is like electrical inertia.*



When the wire is coiled, it acquires electrical inertia (self inductance). It does not allow sudden change in current through it.

7.1 Self inductance and inductors

Consider a coil of single turn. A current i is given to the coil. The current produces a magnetic field at every point inside the coil. The magnetic field at any point is proportional to the current (i). Therefore, the magnetic flux (ϕ) through



Flux (B) due to current i at any point line A , C or D is proportional to i .

the coil is also proportional to the current. If i is doubled, the field (B) at every point doubles and ϕ also doubles.

$$\phi \propto i$$

$$\text{or, } \phi = Li \quad (10)$$

L is a constant for a given coil. It is known as self inductance of the coil and depends on its geometry. Unit of L is weber/ampere also known as henry (H).

The relationship given in equation (10) is valid for a coil having more than one turn also. In this case ϕ represents the total flux through the coil (= number of turns $\times$ flux through one turn).

When current i in a coil is changed, the self flux given by the equation (10) changes. According to Faraday's law an emf is induced in the coil. The induced emf is

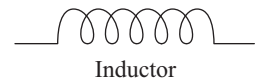
$$\varepsilon = -\frac{d\phi}{dt} = -L \frac{di}{dt} \quad (11)$$

The negative sign has significance in deciding the direction of the induced emf. The induced emf is directed so as to oppose the change in current.

A coil with large value of L develops a large opposing emf when current through it is changed. Higher value of L implies higher inertia. It is difficult to quickly increase or

decrease the current through a coil which has a large self inductance (L).

A coil designed to have a desired value of inductance is known as an inductor. An inductor is an important element in many electrical circuits. The symbol used to represent an inductor is shown in the figure. In practice, a coil will always have some resistance. In a circuit, we can show the properties of inductance and resistance in series.



Example 14 A coil carries a current. When the current is decreased at a rate of 2 A/s, the emf induced in the coil is measured to be 0.4 V. Find the self inductance of the coil.

Solution

Concepts

$$|\varepsilon| = L \left| \frac{di}{dt} \right|$$

$$|\varepsilon| = L \left| \frac{di}{dt} \right|$$

$$\Rightarrow 0.4 = L(2) \quad \Rightarrow L = 0.2 \text{ H.}$$



YOUR TURN

Q.16 Find the dimensions of self inductance (L)

Q.17 When current in a coil is 2A, the magnetic self-flux linked with it is 0.04 Wb. Find the self inductance of the coil.

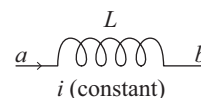
7.2 Inductor in a circuit

Consider a pure inductor in an electric circuit. The current in the branch containing the inductor is i . When i is steady, the property of inductance is not playing any role in the branch. There is no potential drop across it. It is not absorbing or dissipating any energy.

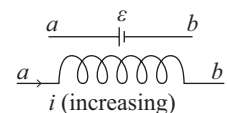
Now, assume that the current i is increasing. An emf is induced in the inductor coil that opposes the growth in current. The inductor absorbs energy in such a situation (we will discuss this later). There is a potential drop when we cross the inductor in the direction of the current. The drop is equal to

$$|\varepsilon| = L \left| \frac{di}{dt} \right|$$

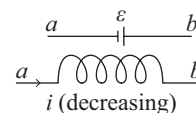
When the current i is decreasing, again an emf is induced in the inductor. The emf opposes the decay of current. There is a gain in potential when we cross the inductor in the direction of the current. We will learn later that the inductor loses energy in this situation.



The inductor can be replaced with a plain conducting wire between a and b



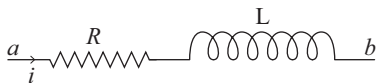
The inductor can be replaced with a source of emf $\varepsilon = L \left| \frac{di}{dt} \right|$ with polarity as shown



The inductor can be replaced with a source of emf

$$\varepsilon = L \left| \frac{di}{dt} \right| \text{ with polarity as shown}$$

Example 15 The figure shows a branch of an electric circuit. Current i in the branch is equal to 5A at an instant and is increasing at a rate of 2 A/s. Find the potential difference between the points a and b . It is given that $R = 1\Omega$, $L = 0.5H$.

**Solution****Concepts**

The emf induced in the inductor opposes the increasing current. We can replace the inductor by an emf source with positive terminal on the left side.



Emf induced in the inductor is

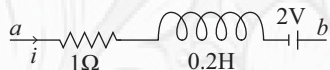
$$\varepsilon = \left| L \frac{di}{dt} \right| = 0.5 \times 2 = 1 \text{ V}$$

$$V_a - iR - \varepsilon = V_b$$

$$\text{or, } V_a - V_b = iR + \varepsilon = 5 \times 1 + 1 = 6 \text{ V.}$$

**YOUR TURN**

Q.18 The figure shows a branch of an electric circuit. The value of current i at an instant is 2A. Find $V_a - V_b$ if



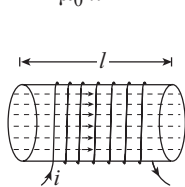
(a) i is constant

(b) i is decreasing at a rate of 1 A/s.

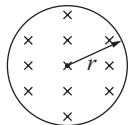
7.3 Self inductance of an ideal solenoid

Consider an ideal long solenoid of length l and cross sectional radius r . There are n turns per unit length on the solenoid. The magnetic field (B) inside the solenoid when a current i passes through it is

$$B = \mu_0 n i$$



Field inside an ideal solenoid is uniform and parallel to the axis



Cross-sectional view. Magnetic field (B) is uniform and perpendicular to the circular cross section

Flux through single turn of the coil is $= \pi r^2 \cdot B$ Total flux through nl turns in the coil is

$$\phi = (\pi r^2 \cdot B) (nl) = (\pi r^2 \cdot \mu_0 n i) (nl)$$

$$\text{or, } \phi = (\pi \mu_0 n^2 r^2 l) i$$

$$\text{But } \phi = Li$$

$$\therefore L = \pi \mu_0 n^2 r^2 l \quad (12)$$

The self inductance depends on geometrical factors of a coil. It may be mathematically tedious to calculate self inductance for various geometries of a coil. However, we can measure the inductance of a coil in laboratory.

Example 16 A tightly wound solenoid

A solenoid is made by tightly winding an insulated wire of diameter $d = 2$ mm. The turns are close enough to be touching one another. The radius of the solenoid is 5cm. The length of the wire used is 78.5m. Find its self inductance.

Solution**Concepts**

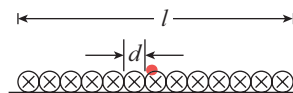
(i) Total number of turns, $N = \frac{\text{length of wire}}{2\pi r}$

Here r is cross sectional radius of the solenoid.

(ii) Length of the solenoid, $l = d \cdot N$

(iii) $L = \pi \mu_0 n^2 r^2 l$. In this equation n = number of turns per meter length

Length of wire used in one turn $= 2\pi r$



$\therefore$ Total number of turns is

$$N = \frac{78.5\text{m}}{2 \times 3.14 \times 0.05} = 250$$

Length of solenoid, $l = d \cdot N = (2 \times 10^{-3}) (250) = 0.5\text{m}$

$$\therefore L = \pi \mu_0 n^2 r^2 l = \pi \mu_0 \left(\frac{N}{l} \right)^2 r^2 l$$

$$= \pi \mu_0 \frac{N^2 r^2}{l} = \frac{3.14 \times 4 \times 3.14 \times 10^{-7} \times (250)^2 \times (0.05)^2}{0.5}$$

$$= 1.23 \times 10^{-3} \text{ H}$$

$$= 1.23 \text{ mH}$$



YOUR TURN

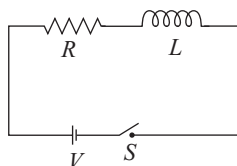
Q.19 An ideal solenoid has 500 turns per meter length and the volume of the solenoid is 0.02m^3 . Find the self inductance of the solenoid.

8. TRANSIENT CURRENT IN LR CIRCUIT

If a circuit has only resistors, the moment we put on the switch a steady current is set up. The moment we put off the switch, the current goes to zero. However, in a circuit having an inductor current cannot change suddenly. The current is variable for a short time before it becomes steady. We will explore the circuit during the short time during which the current is not steady. The word *transient* means *lasting for a short time*.

8.1 Growth of current

Consider an inductance L and a resistor R connected in series to a battery of emf V through a switch S . Initially, the switch is open. The switch is closed at $t = 0$. If there were no inductance, the current will immediately become $\frac{V}{R}$. But in presence of the inductance this cannot happen. Inductance is electrical inertia. It does not allow sudden change in value of current through it. The current will ultimately become $i_0 = \frac{V}{R}$, but it is going to take time.



Immediately after the switch is closed ($t = 0^+$), the current is zero but increasing.

The potential drop across R is zero.

$$V_L + V_R = V$$

$$\text{or, } V_L = V$$

The emf induced in the inductor is equal to V and is opposing the growth of the current.

$$L \frac{di}{dt} = V$$

$$\text{or, } \frac{di}{dt} = \frac{V}{L} \quad (\text{at } t = 0^+)$$

After some finite time t there will be a current (i) in the circuit. Now, $V_R = iR$. This means that the emf induced in L has decreased. As the current rises, V_R increases and V_L decreases.

$$V_L + V_R = V$$

$$\text{or, } L \frac{di}{dt} + iR = V \quad (i)$$

[Note that the net emf in the loop is $V - L \frac{di}{dt}$]

The above differential equation can be solved as given below.

$$L \frac{di}{dt} = V - Ri$$

$$\text{or, } \int_0^i \frac{di}{V - Ri} = \frac{1}{L} \int_0^t dt$$

$$\text{or, } -\frac{1}{R} [\ln(V - Ri)]_0^i = \frac{1}{L} t$$

$$\text{or, } \ln(V - Ri) - \ln V = -\frac{Rt}{L}$$

$$\text{or, } \ln \left(\frac{V - Ri}{V} \right) = -\frac{Rt}{L}$$

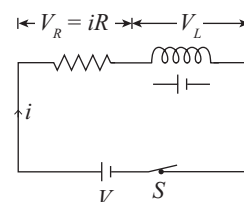
$$\text{or, } 1 - \frac{R}{V} i = e^{-\frac{Rt}{L}}$$

$$\text{or, } i = \frac{V}{R} \left(1 - e^{-\frac{Rt}{L}} \right)$$

$$\text{or, } i = i_0 (1 - e^{-t/\tau}) \quad (13)$$

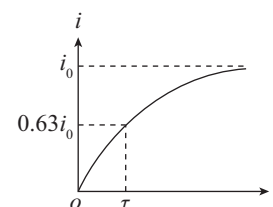
$$\text{Here, } i_0 = \frac{V}{R} \quad (14)$$

$$\text{and } \tau = \frac{L}{R} \quad (15)$$



Circuit at time t . As $V_R = iR$ increases, V_L decreases

The graphical plot of i versus t is as shown. The current rises from zero (at $t = 0$) to a maximum value i_0 after a long time ($t \rightarrow \infty$). After a long time, the circuit gets steady. The current becomes constant at $i_0 = \frac{V}{R}$. The inductor has no role to play in this state.



Growth of current in LR circuit. Smaller the time constant (τ), faster is the rise of the current.

Time constant (τ)

The quantity $\tau = \frac{L}{R}$ has dimension of time and is known as the time constant of the circuit. At time $t = \tau$ the value of current in the circuit is

$$i = i_0 (1 - e^{-1}) \simeq 0.63 i_0$$

At time $t = 2\tau$ the value of current in the circuit is

$$i = i_0 (1 - e^{-2}) \simeq 0.86 i_0$$

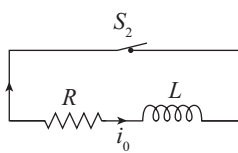
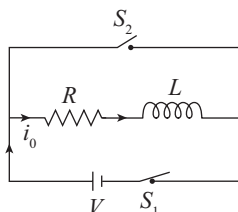
Thus, in one time constant the current reaches 63% of the final maximum value and in two time constants it reaches 86% of the maximum value. The time constant tells us how fast will the current grow. *If the time constant is small, the current rises fast.*

Theoretically, time needed for the current to attain its final maximum value is infinite. In practice, the current is almost at its peak value (more than 99% of i_0) after five time constants. If time constant of a circuit is 1 ms, we can assume it to be in steady state after 5 or 6 ms.

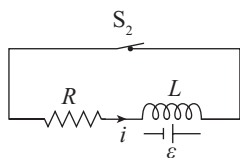
8.2 Decay of current

In the arrangement shown in the figure, switch S_1 is closed and S_2 is open. A steady current $i_0 = \frac{V}{R}$ is maintained through the series $R-L$ combination.

Suddenly at $t = 0$, the switch S_2 is closed and S_1 is opened [Practically, it may be difficult to operate the two switches precisely at the same instant. In such cases we need to ensure that S_2 is closed first and almost at the same instant S_1 is opened]. This operation of the switches completes the circuit through S_2 and disconnects the battery from the circuit. The current through an inductor cannot change suddenly. Immediately after the operation of the switches the current in the circuit is $i_0 = \frac{V}{R}$. The current begins to decrease. The emf induced in the inductor maintains the current. The polarity of the emf is as shown.



Current is i_0 at $t = 0^+$



Current is decreasing. Emf induced in the inductor maintains the current

$$-L \frac{di}{dt} = Ri$$

[We have already decided the polarity of the emf. We just need to write $|\varepsilon| = Ri$. Since the current is decreasing, $\frac{di}{dt}$ is a negative quantity. $-\frac{di}{dt}$ is a positive quantity. It is same as $|\varepsilon|$.]

$$\frac{di}{i} = -\frac{R}{L} dt$$

$$\text{or, } \int_{i_0}^i \frac{di}{i} = -\frac{R}{L} \int_0^t dt$$

$$\text{or, } \ln \left(\frac{i}{i_0} \right) = -\frac{Rt}{L}$$

$$\text{or, } i = i_0 e^{-\frac{Rt}{L}}$$

$$\text{or, } i = i_0 e^{-t/\tau}$$

(16)

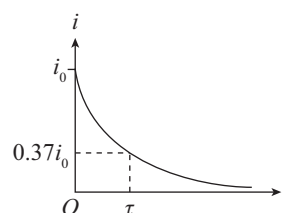
Here $\tau = \frac{L}{R}$ is time constant of the circuit.

At $t = \tau$,

$$i = i_0 e^{-1} = 0.37 i_0$$

The current reduces to 37% of its initial value in one time constant, i.e., 63% of the decay is completed. Smaller the value of τ , faster is the decay.

Theoretically, it takes infinite time for the current to become zero. Practically, the current can be assumed to be zero after few time constants.



Example 17 A coil of resistance 100Ω and inductance 0.1 H is connected to a 20V battery. The circuit is completed at $t = 0$. Find

- the current immediately after the circuit is completed.
- the current at $t = 1 \text{ ms}$
- the current at $t = 1.5 \text{ ms}$
- the current at $t = 5 \text{ s}$

[Given: $e^{-1.5} \simeq 0.22$]

Solution**Concepts**

All that we need to know are equations (13), (14) and (15).

- At $t = 0^+$; $i = 0$

$$(b) \quad i_0 = \frac{V}{R} = \frac{20}{100} = 0.2 \text{ A}$$

$$\text{Time constant, } \tau = \frac{L}{R} = \frac{0.1}{100} = 10^{-3} \text{ s} = 1 \text{ ms.}$$

$\therefore$ Current at $t = 1 \text{ ms}$ is

$$i = i_0 (1 - e^{-1}) = 0.63 i_0 = 0.63 \times 0.2 = 0.126 \text{ A}$$

$$(c) \quad i = i_0 (1 - e^{-t/\tau}) = 0.2 (1 - e^{-3/2}) \\ = 0.2 \times (1 - 0.22) \\ = 1.67 \text{ A}$$

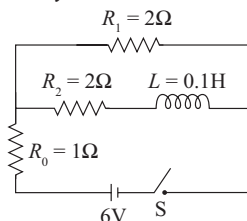
- $t = 5 \text{ s}$ is a long time after the circuit is completed (remember that the time constant is only 1 ms).

$$\therefore i \rightarrow i_0$$

$$\Rightarrow i = 0.2 \text{ A}$$

Example 18 Initial Current and Steady State Current

The switch in the circuit is closed at $t = 0$. Find the current through the cell immediately after the switch is closed and a long time after the switch is closed.

**Solution****Concepts**

- Current, in the branch having the inductor cannot change suddenly. At $t = 0^+$ (i.e. immediately after closing the switch) current through the branch having inductor is zero.
- After a long time the circuit attains the steady state and the inductor acts like a plain conducting wire.

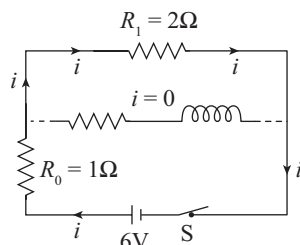
At $t = 0^+$, the circuit is effectively as shown.

$$i = \frac{6V}{(1+2)\Omega} = 2A.$$

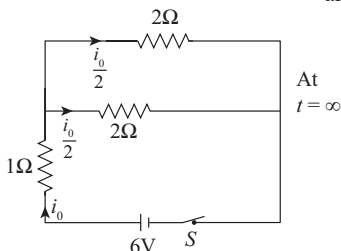
At $t = \infty$, the circuit attains steady state and we can replace the inductor with a plain wire. The equivalent resistance is

$$R_{eq} = \frac{2 \times 2}{2+2} + 1 = 2\Omega$$

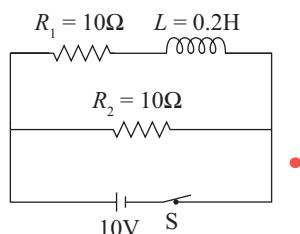
$$\therefore i_0 = \frac{6V}{2\Omega} = 3A.$$



At $t = 0^+$ there is no current in the branch containing L and R



Example 19 In the circuit shown in the figure, the switch (S) is closed since long. At $t = 0$, the switch is opened.

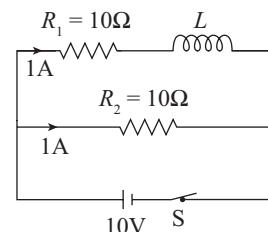


- Find the current through R_2 just before and just after the switch is opened.
- Find the time at which the current in R_1 falls to half its initial value.

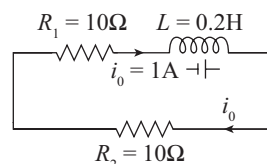
Solution**Concepts**

- Current through the branch having R_2 can change suddenly. But the current in the branch having L and R_1 cannot change suddenly.
- After S is opened the current in maintained in the loop containing R_1 , L and R_2 due to the emf induced in L .

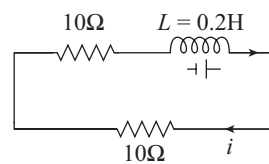
Just before opening the switch, the circuit is in steady state. The inductor is like a plain wire. Current in both resistors is 1A. Just after the switch is opened, the current in L remains 1A. Now the current circulates through R_1 , L and R_2 .



Just before opening 'S'



At $t = 0^+$ (immediately after S is opened)



At time t , the current is decreasing

The current in R_2 has suddenly changed direction. That is perfectly OK. It can happen. Current cannot change suddenly through the inductor.

Now, the current begins to decrease.

$$\tau = \frac{L}{R} = \frac{0.2H}{20\Omega} = 0.01 \text{ s.}$$

Using equation (16)

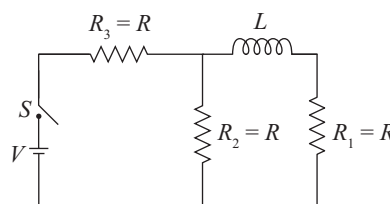
$$i = i_0 e^{-t/\tau}$$

$$\text{or, } \frac{i_0}{2} = i_0 e^{-t/\tau} \Rightarrow \frac{1}{2} = e^{-t/\tau}$$

$$\text{or, } \ln(1) - \ln(2) = -t/\tau$$

$$\text{or, } t = \tau \ln(2) = 0.01 \times 0.693 \text{ s} = 6.93 \text{ ms.}$$

Example 20 Consider the electric circuit shown in the figure. The switch (S) is closed at $t = 0$. Find the current through the battery as a function of time.



Solution

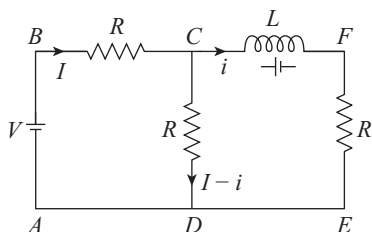
Concepts

We will use kirchhoff's loop law for two loops in the circuit and set up a differential equation for the current through L .

The current distribution at any time t is as shown. For loop $ABCD$ we can write

$$IR + (I - i) R = V$$

$$\text{or, } I = \frac{V}{2R} + \frac{i}{2} \quad (\text{i})$$



i is increasing

For the loop $ABFE$ we can write

$$IR + L \frac{di}{dt} + Ri = V$$

$$\text{or, } \frac{V}{2} + \frac{Ri}{2} + L \frac{di}{dt} + Ri = V$$

$$\text{or, } L \frac{di}{dt} = \frac{V}{2} - \frac{3}{2} Ri$$

$$\text{or, } \int_0^i \frac{di}{\frac{V}{2} - \frac{3}{2} Ri} = \frac{1}{L} \int_0^t dt$$

$$\text{or, } \left[\ln \left(\frac{V}{2} - \frac{3}{2} Ri \right) \right]_0^i = - \frac{3R}{2L} t$$

$$\text{or, } \ln \left(\frac{\frac{V}{2} - \frac{3}{2} Ri}{\frac{V}{2}} \right) = -\frac{3R}{2L} t$$

$$\text{or, } i = \frac{V}{3R} \left(1 - e^{-\frac{3Rt}{2L}} \right) \quad (\text{ii})$$

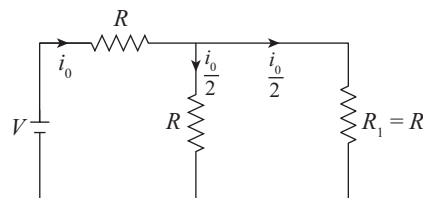
Substituting this in (i) gives

$$I = \frac{V}{2R} + \frac{V}{6R} \left(1 - e^{-\frac{3Rt}{2L}}\right)$$

$$= \frac{V}{6R} \left(4 - e^{-\frac{3Rt}{2L}}\right)$$

Note: A careful look at equation (ii) tells us that the current through the inductor can be written in following steps:

(i) Find the steady state current through L . In the present example:



Inductor replaced with a plain wire in steady state

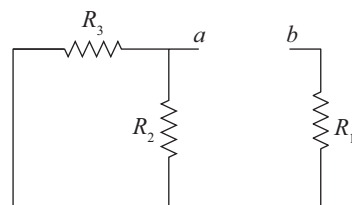
$$R_{eq} = \frac{3R}{2}$$

$$\therefore i_0 = \frac{V}{R_{eq}} = \frac{2V}{3R}$$

$\therefore$ Current through R_1 is

$$\frac{i_0}{2} = \frac{V}{3R}$$

(ii) To get the time constant (τ), short the cell and find the equivalent resistance across the inductor.



R_2 and R_3 are in parallel. Their equivalent resistance is in series with R_1

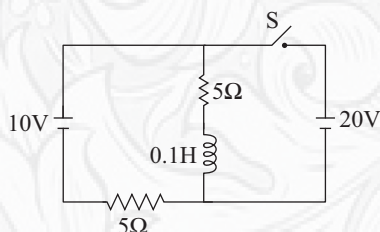
$$R_0 = \frac{3R}{2}$$

$$\therefore \tau = \frac{L}{R_0} = \frac{2L}{3R}$$



YOUR TURN

Q.20 The circuit shown in the figure is in steady state. The switch is closed at time $t = 0$. Find the current through the inductor



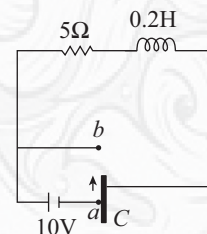
- (a) immediately after the switch is closed,
(b) A long time after the switch is closed.

Q.21 A series combination of an inductance $L = 0.01H$ and resistance 100Ω is connected to a cell of emf 10V. The circuit is completed at time $t = 0$. Find

- (a) The time after which the potential difference across the resistor is 6.3V.

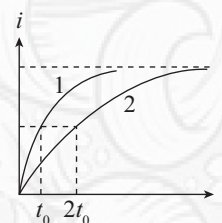
(b) The time after which the emf induced in the inductor is 5V.

Q.22 In the figure shown, C is a sliding switch. Initially, it is in contact with a . The circuit is in steady state. The switch C is pushed up so that it comes in contact with b and loses contact with a . Find the current in the circuit 0.08 s after the switch comes in contact with b .



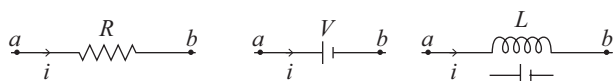
Q.23 A solenoid is connected across a cell of emf V and the growth of current is represented by the graph 1. Another solenoid is connected across a cell of emf 2V and the growth of current is as represented by graph 2. Which solenoid has

- (a) higher resistance?
(b) higher inductance?



9. ENERGY STORED IN AN INDUCTOR

When there is a current in a resistance, the charge loses electrical energy in crossing it. This lost energy is converted into heat. When a cell is getting recharged, the charge loses energy in crossing it and the energy gets converted into chemical energy. When current through an inductor is growing there is a potential difference across it. The charge loses energy when it crosses the inductor. Where does this energy go? [In case of resistor it got converted into heat. In case of a cell it got converted into chemical energy.]



In the three cases shown the electric charge loses energy as it moves from a to b . In a resistor this energy becomes heat. In a cell it gets converted into chemical energy. In an inductor this gets stored in the magnetic field inside the inductor coil.

It is believed that *energy gets stored in the magnetic field inside the inductor*.

Consider an instant when the current through an inductor is i and is increasing at a rate $\frac{di}{dt}$.

The emf induced is

$$|\varepsilon| = L \frac{di}{dt}$$

Amount of charge that flows in time dt is $dq = i dt$. Energy added to the inductor in time dt = electrical energy lost by the charge dq .

$$\text{or, } dU = (dq) E$$

$$\text{or, } dU = (i dt) \left(L \frac{di}{dt} \right) = L i di$$

Total energy that gets stored in the inductor as the current through it increases from 0 to I is

$$U = \int dU = L \int_0^I i di$$

$$\text{or, } U = \frac{1}{2} LI^2 \quad (17)$$

10. ENERGY DENSITY IN MAGNETIC FIELD

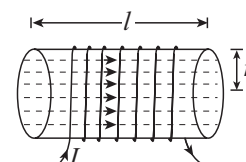
Consider an ideal solenoid carrying a current I . The inductance of the coil is

$$L = \mu_0 \pi r^2 n^2 l$$

Here, n = number of turns/m

r = radius

l = length



Field inside the solenoid is $B = \mu_0 n I$. Energy is assumed to be stored in this field.

5.24 Magnetism and Electromagnetic Induction

Energy stored in the coil is

$$U = \frac{1}{2}LI^2 = \frac{1}{2}\mu_0\pi r^2 n^2 l I^2$$

This energy is assumed to be stored in the magnetic field that exists inside the solenoid. Volume of the space where the field is present is $V = \pi r^2 l$.

Energy contained in unit volume of the space where the magnetic field exists is

$$u_B = \frac{U}{V} = \frac{\frac{1}{2}\mu_0\pi r^2 n^2 l I^2}{\pi r^2 l}$$

$$\text{or, } u_B = \frac{1}{2}\mu_0 n^2 I^2$$

$$\text{or, } u_B = \frac{1}{2\mu_0}(\mu_0 n I)^2$$

$$\text{or, } u_B = \frac{B^2}{2\mu_0} \quad (18)$$

The above expression has been derived for a special case of the magnetic field inside a solenoid, but it is true in general. Whenever there is a magnetic field, there is an energy associated with it. The amount of energy contained in unit volume of a space having a magnetic field B is given by equation (18). This is known as *energy density in magnetic field*.

Example 21 Energy stored in the magnetic field keeps the circuit alive

In example 19, how much heat is dissipated in the resistance R_2 after the switch is opened at $t = 0$.

Solution

Concepts

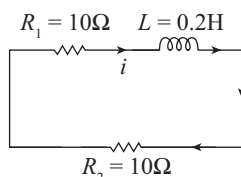
- (i) After the switch is opened, the energy stored in the inductor is used to maintain current in the circuit during the decay phase.
- (ii) Total heat dissipated in R_1 and R_2 is equal to the initial energy stored in the inductor.

Just after the switch is opened, the current in the circuit is

$$i_0 = 1\text{ A}$$

Energy stored in the inductor is

$$U = \frac{1}{2}L i_0^2 = \frac{1}{2} \times 0.2 \times 1^2 = 0.1\text{ J}$$



This entire energy gets dissipated as heat in R_1 and R_2 as the current falls to zero. As both the resistances are equal and current in them is also same (always), therefore equal energy is dissipated in them.

$$\text{Energy dissipated in } R_2 = \frac{0.1}{2} = 0.05\text{ J}$$

Example 22 Energy in space around a straight current carrying wire

A thin, long straight wire carries a current I . Find the energy stored in the magnetic field around the wire in the space between radial distance a and b ($b > a$). Consider the length of the region to be l .

Solution

Concepts

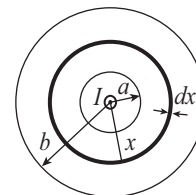
- (i) The magnetic field at a radial distance x from the wire is $B = \frac{\mu_0 I}{2\pi x}$.
- (ii) Energy stored in a volume dV at a location having field B is $dU = u_B \cdot dV = \frac{B^2}{2\mu_0} \cdot dV$

The figure shows the cross section of the long wire. The magnetic field at radial distance x is

$$B = \frac{\mu_0 I}{2\pi x}$$

Energy density of magnetic field at radial distance x is

$$u_B = \frac{B^2}{2\mu_0} = \frac{\mu_0 I^2}{8\pi^2 x^2}$$



Consider a cylindrical shell of thickness dx and length l . Volume of the shell is

$$dV = (2\pi x dx) l$$

Magnetic energy stored in the volume of the shell is

$$dU = u_B \cdot dV = \frac{\mu_0 I^2 l}{4\pi} \frac{dx}{x}$$

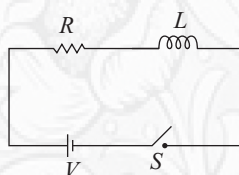
Energy stored in the space lying between $x = a$ to $x = b$ is obtained by adding the energy in all such shells.

$$U = \frac{\mu_0}{4\pi} I^2 l \int_a^b \frac{dx}{x} = \frac{\mu_0}{4\pi} I^2 l \ln\left(\frac{b}{a}\right)$$



YOUR TURN

Q.24 In the circuit shown in the figure, the switch (S) is closed at $t = 0$. Find



(a) The energy stored in the inductor at $t = \frac{L}{R}$.

(b) The rate of heat dissipation in R at $t = \frac{L}{R}$.

(c) The power absorbed by the inductor at $t = \frac{L}{R}$.

Q.25 An inductor coil has inductance L and is connected to a resistance R . Current is decaying. The value of current at time $t = 0$ is I . Find the time in which the magnetic field energy density in the coil becomes $\frac{1}{4}$ of its initial value.

11. INDUCTORS IN SERIES AND PARALLEL

We will discuss how to find the equivalent inductance for two or more pure inductors connected in series and parallel.

Series connection

Consider two inductance L_1 and L_2 in series. Current through them is i and is increasing. The emf induced in the two inductances are

$$\varepsilon_1 = L_1 \frac{di}{dt} \text{ and } \varepsilon_2 = L_2 \frac{di}{dt}$$

Total emf developed between a and b is

$$\varepsilon = \varepsilon_1 + \varepsilon_2 = (L_1 + L_2) \frac{di}{dt} \quad (i)$$

L is said to be the equivalent inductance, if the emf induced in it is ε when the current through it changes at a rate $\frac{di}{dt}$.

$$\varepsilon = L \frac{di}{dt} \quad (ii)$$

Comparing equations (i) and (ii)

$$L = L_1 + L_2 \quad (19)$$

The equivalent of two or more inductances in series is obtained by adding them.

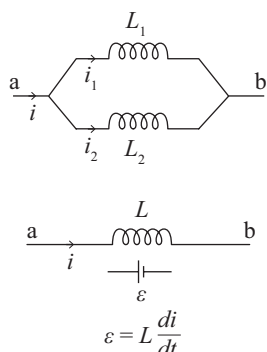
Parallel connection

L_1 and L_2 are two inductance in parallel. Current through them are i_1 and i_2 , respectively. The emf induced in them is always same.

$$\therefore \varepsilon = L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt} \quad (i)$$

$$i = i_1 + i_2$$

$$\Rightarrow \frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt}$$



$$\text{or, } \frac{\varepsilon}{L} = \frac{\varepsilon}{L_1} + \frac{\varepsilon}{L_2} \quad [\text{from (i)}]$$

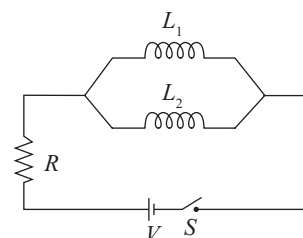
Here, L is the equivalent inductance. Its value must be such that

$$L \frac{di}{dt} = \varepsilon \quad \Rightarrow \quad \frac{di}{dt} = \frac{\varepsilon}{L}$$

$$\therefore \frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} \quad (20)$$

The above relation can be extended for any number of inductance.

Example 23 In the circuit shown in the figure, the switch S is closed at time $t = 0$. Find the current in L_1 and L_2 when steady state is attained.



Solution

Concepts

The induced emf in both L_1 and L_2 is always same as they are in parallel.

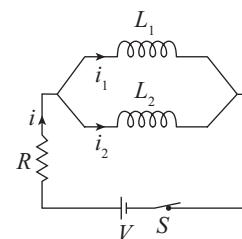
In steady state current through the cell is

$$i_0 = \frac{V}{R}$$

At any instant

$$L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt}$$

$$\text{or, } \frac{di_1}{dt} \bigg/ \frac{di_2}{dt} = \frac{L_2}{L_1}$$



5.26 Magnetism and Electromagnetic Induction

Since the two currents have their rate of growth in the ratio $\frac{L_2}{L_1}$, their final value will also be in the same ratio.

$$\frac{i_{10}}{i_{20}} = \frac{L_2}{L_1} \quad (i)$$

$$\text{And } i_{10} + i_{20} = i_0 = \frac{V}{R} \quad (ii)$$

Solving (i) and (ii) we get

$$i_{10} = \left(\frac{L_2}{L_1 + L_2} \right) \frac{V}{R} \text{ and } i_{20} = \left(\frac{L_1}{L_1 + L_2} \right) \frac{V}{R}$$

In Short

- (i) A coil has a property of opposing any change in current through it. This property of 'electrical inertia' is known as self inductance.
- (ii) A current (i) given to a coil produces a magnetic field and therefore, sets up a magnetic flux (ϕ) linked to the coil. The flux is always proportional to the source current (i).

$$\phi \propto i \Rightarrow \phi = Li$$

The constant L is known as self inductance of the coil and its unit is henry (H).

- (iii) L depends on geometry of the coil.
- (iv) When current through a coil is changed, an emf is induced in it. The induced emf is

$$\varepsilon = -\frac{d\phi}{dt} = -L \frac{di}{dt}$$

The induced emf is directed so as to oppose the change in current.

- (v) Self inductance of an ideal solenoid is

$$L = \pi \mu_0 n^2 r^2 l$$
- (vi) If a series L - R circuit is connected to a battery of emf V at time $t = 0$, the current changes with time as

$$i = i_0 (1 - e^{-t/\tau})$$

$$i_0 = \frac{V}{R} = \text{Final steady state current after a long time}$$

$$\tau = \frac{L}{R} = \text{Time constant of the circuit}$$
- (vii) Time constant (τ) is the time in which the current grows to 63% of its final value.
- (viii) When current decays in a LR circuit having no source of emf, it varies with time as

$$i = i_0 e^{-t/\tau}$$

$$i_0 = \text{initial current}$$
- (ix) Current through an inductor coil cannot change suddenly.

- (x) Energy stored in an inductor coil having inductance (L) carrying current I is

$$U = \frac{1}{2} LI^2$$

This energy is stored in the magnetic field inside the coil.

- (xi) Energy stored in unit volume of a magnetic field is

$$u = \frac{B^2}{2\mu_0}$$

This is known as magnetic field energy density.

- (xii) The equivalent of two or more inductance in series is given by

$$L = L_1 + L_2 + L_3 + \dots$$

- (xiii) The equivalent of two or more inductance in parallel is given by

$$\frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots$$

12. LC OSCILLATIONS

A capacitor having capacitance C is given a charge q_0 . It is connected to an inductance L by closing the switch (S) at time $t = 0$. We will assume that there is no resistance in the circuit.

Consider the capacitor to be discharging and charge on it to be q at time t . The current in the circuit is

$$i = -\frac{dq}{dt}$$

Using kirchhoff's voltage law, we can write

$$L \frac{di}{dt} = \frac{q}{C}$$

$$\text{or, } -L \frac{d^2q}{dt^2} = \frac{q}{C}$$

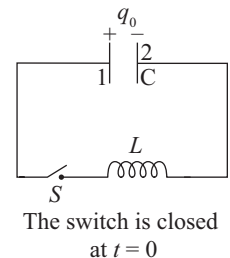
$$\text{or, } \frac{d^2q}{dt^2} = -\left(\frac{1}{LC}\right)q$$

$$\text{Let } \omega = \sqrt{\frac{1}{LC}} \quad (21)$$

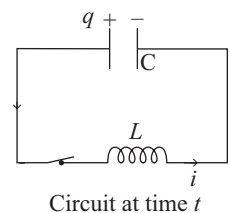
$$\frac{d^2q}{dt^2} = -\omega^2 q \quad (22)$$

This equation is similar to the standard equation of SHM and its solution is

$$q = q_0 \sin(\omega t + \delta)$$



(i)



q_0 is maximum value of the charge on the capacitor.

δ is a constant and its value can be found by knowing the initial condition.

At $t = 0$, $q = q_0$

$$\therefore \delta = \frac{\pi}{2}$$

$$\therefore q = q_0 \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$\text{or, } q = q_0 \cos \omega t \quad (23)$$

Using (i)

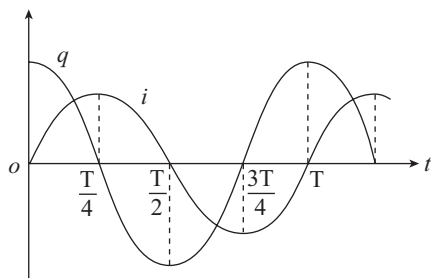
$$i = - \frac{dq}{dt} = q_0 \omega \sin \omega t$$

$$\text{or, } i = i_0 \sin \omega t \quad (24)$$

$$\text{where } i_0 = q_0 \omega = \frac{q_0}{\sqrt{LC}} \quad (25)$$

The above analysis shows that both the current in the circuit and the charge on the capacitor varies sinusoidally with time period given by

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{LC} \quad (26)$$



Variation of charge on capacitor (q) and current (i) versus time

Initially, at $t = 0$ charge on the capacitor is q_0 with plate 1 positive. At this instant the current is zero. Energy stored in the electric field between the capacitor plates is

$$U_E = \frac{q_0^2}{2C}$$

In the time interval $t = 0$ to $t = \frac{T}{4}$ the charge decreases and the current in the circuit increases. At $t = \frac{T}{4}$ the charge is $q = 0$ and current is maximum (i_0). The entire energy has been transferred from the capacitor to the inductor. At $t = \frac{T}{4}$ the energy stored in the magnetic field in the inductor is

$$U_B = \frac{1}{2} L i_0^2$$

Using equation (25) one can easily show that

$$\frac{1}{2} L i_0^2 = \frac{q_0^2}{2C} \quad (27)$$

In the interval $t = \frac{T}{4}$ to $t = \frac{T}{2}$, the charge changes from zero to $-q_0$. The negative value of charge simply means that the polarity of capacitor plate has changed.

Now plate 2 is positive. In the same interval the current changed from i_0 to zero. The magnetic field energy again got transferred to the capacitor.

In the interval $t = \frac{T}{2}$ to $t = \frac{3T}{4}$ the current is negative.

This means that the current has reversed in the direction and is clockwise. At $t = \frac{3T}{4}$ the current is maximum and $q = 0$.

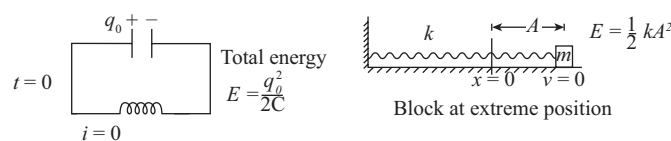
The entire energy in the circuit is again in form of magnetic field energy.

In the interval $t = \frac{3T}{4}$ to $t = T$ the capacitor again acquires full charge and the current becomes zero. The circuit is back to its original state at $t = T$. The capacitor has charge q_0 and plate 1 is positive. The entire energy is again with the capacitor.

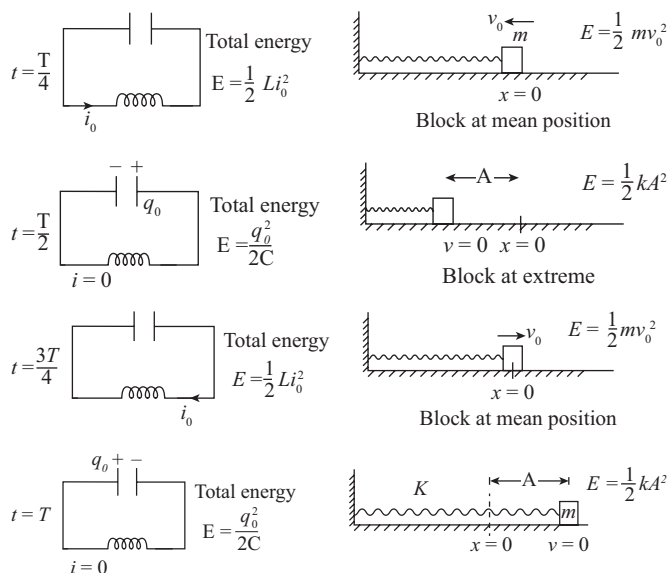
The oscillations in the LC circuit can be compared to the spring block oscillations.

Quantities in LC oscillation	Comparable quantities in spring block system
Charge on the capacitor (q)	Position of the block (x)
Current in the circuit (i)	Velocity of the block (v)
Inductance (L)	Mass (m)
Capacitance (C)	$\frac{1}{k}$ (inverse of force constant)
Energy in magnetic field $U_B = \frac{1}{2} L i^2$	Kinetic energy of the block $K = \frac{1}{2} m v^2$
Energy in electric field $U_E = \frac{q^2}{2C}$	Potential energy in the spring $U = \frac{1}{2} k x^2$
Angular frequency $\omega = \frac{1}{\sqrt{LC}}$	Angular frequency $\omega = \sqrt{\frac{k}{m}}$

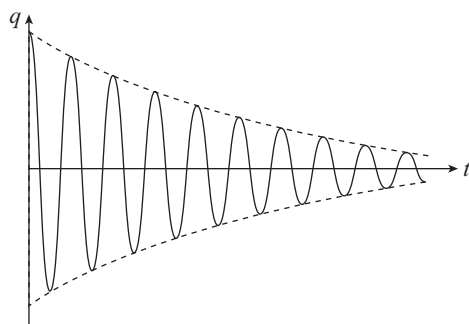
The diagrams given below show few of the comparable states of the two oscillations.



5.28 Magnetism and Electromagnetic Induction



An LC oscillation is an idealized system since it assumes no dissipation of energy. In practice, there will be some resistance in the circuit and energy will get dissipated. In a spring block system the energy gets dissipated due to friction. Such oscillations are known as damped oscillations. The amplitude of charge and current (i.e., q_0 and i_0) in LC circuit goes on decreasing if there is some resistance. The amplitude of oscillation of a spring block system also goes on decreasing due to friction.



Variation of q with time if there is some resistance. The current (i) also decreases similarly. This also represents the position-time graph of the block in a spring block system having some friction.

LC circuits find many applications in modern electronics. They are used to generate signals at a particular frequency, or picking out a signal of desired frequency from a mixture of frequencies.

Example 24 Be mindful of initial conditions

An LC circuit is oscillating. At $t = 0$, the current in the circuit is maximum (i_0) and is in negative direction.

- Write the variation of current (i) as a function of time.
- Find the maximum charge on the capacitor.
- Find the current when half the total energy is with the capacitor. When does this happen for the first time?

Solution

Concepts

- Both current and charge on the capacitor varies sinusoidally. The phase constant shall be decided by the initial condition.
- Energy is conserved in LC oscillation sum of the magnetic field energy and the electric field energy is constant.

- Let the current be $i = i_0 \sin(\omega t + \delta)$

At $t = 0$, $i = -i_0$

$$\therefore \delta = \pi$$

$$\therefore i = i_0 \sin(\omega t + \pi)$$

$$\text{or, } i = i_0 \sin\left(\frac{t}{\sqrt{LC}} + \pi\right) \quad (i)$$

- Energy in the circuit is $E = \frac{1}{2}Li_0^2$

Charge on the capacitor is maximum when all the energy is possessed by it.

$$\frac{q_0^2}{2C} = \frac{1}{2}Li_0^2 \Rightarrow q_0 = \sqrt{LC} i_0$$

- Half the total energy is with the capacitor when the remaining half is in the inductor. If i is the current at this instant, then

$$\frac{1}{2}Li^2 = \frac{1}{2}\left(\frac{1}{2}Li_0^2\right)$$

$$\text{or, } i = \pm \frac{i_0}{\sqrt{2}}$$

Using (i) we can see that $i = -\frac{i_0}{\sqrt{2}}$ when

$$\frac{t}{\sqrt{LC}} = \frac{\pi}{4} \Rightarrow t = \frac{\pi}{4} \sqrt{LC}$$

Example 25 In an ideal LC circuit, the voltage across the capacitor varies with time t as

$$v = v_0 \sin\left(\omega t + \frac{\pi}{3}\right)$$

Here, $\omega = \frac{1}{\sqrt{LC}}$ and v_0 is a constant.

- Find the maximum value of $\left|\frac{di}{dt}\right|$ where i is current in the circuit.
- Find charge on the capacitor when current is half its maximum value.
- Write the variation of current as a function of time. Take initial direction of the current to be positive.

Solution**Concepts**

- (i) $q = Cv$
 (ii) Emf induced in the inductor = $\left| L \frac{di}{dt} \right|$
 (iii) Current in the circuit is rate of change of charge on the capacitor.

Charge on the capacitor is

$$q = Cv = Cv_0 \sin \left(\omega t + \frac{\pi}{3} \right)$$

- (a) Maximum potential difference across the capacitor = Cv_0

This is also the maximum potential difference across the inductor.

$$\therefore \left| L \frac{di}{dt} \right|_{\max} = Cv_0$$

$$\therefore \left| \frac{di}{dt} \right|_{\max} = \frac{Cv_0}{L}$$

- (b) When current is $i = \frac{i_0}{2}$, energy stored in the inductor is $\frac{1}{4}$ of the total energy $\left(U_B = \frac{1}{2} Li^2 \right)$. Remaining $\frac{3}{4}$ of the total energy is in the capacitor.

$$\therefore \frac{q^2}{2C} = \frac{3}{4} \left(\frac{q_0^2}{2C} \right)$$

$$\text{or, } q = \frac{\sqrt{3}}{2} q_0 = \frac{\sqrt{3}}{2} Cv_0$$

- (c) $|i| = \left| \frac{di}{dt} \right| = Cv_0 \omega \cos \left(\omega t + \frac{\pi}{3} \right)$

It is given that the current is positive at $t = 0$. The equation for current will be $i = Cv_0 \omega \cos(\omega t + \pi/3)$

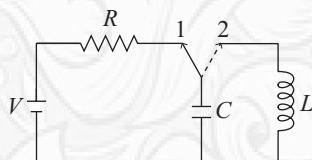
Note: If initial current were in negative direction, then we would have taken

$$i = -\frac{dq}{dt} = -Cv_0 \omega \cos \left(\omega t + \frac{\pi}{3} \right)$$

**YOUR TURN**

Q.26 An LC circuit has capacitance $C = 1 \mu F$. The circuit is oscillating at a frequency of 10 kHz. Find the energy of the circuit if the peak current is $i_0 = 1 A$. [Take $\pi^2 = 10$]

Q.27 In the circuit shown in the figure, the emf of the cell is $V = 10V$. The values of capacitance, inductance and the resistance are $10 \mu F$, $1mH$ and 10Ω respectively. The switch is held in position 1 for a long time and then it is shifted to position 2 at time $t = 0$

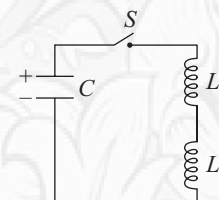


- (a) Find the time period of resulting LC oscillations.
 (b) Find maximum current in the LC circuit.
 (c) Write charge on the capacitor at time t . Take charge on the capacitor to be positive when its upper plate is positive.

Q.28 In an LC circuit the peak voltage across the inductor is V_0 . Find the maximum charge on the capacitor. The capacitance and inductance are C and L , respectively.

Q.29 In the circuit shown in the figure, capacitor is charged to a potential difference V_0 . The switch S is closed at $t = 0$.

- (a) Find the frequency of LC oscillations
 (b) Find the potential difference across L , when the charge on the capacitor is half the maximum.

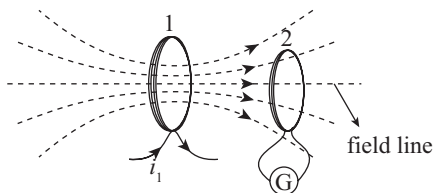


Q.30 In an LC oscillation the capacitor is fully charged at $t = 0$. Plot the variation of the electric field energy and the magnetic field energy with time.

Q.31 Explain why the LC oscillation continues even after the capacitor has been completely discharged.

13. MUTUAL INDUCTION

Figure shows two coils 1 and 2 placed close to one another. When the current (i_1) in the coil 1 is changed, an emf is induced in the coil 2. This event is known as mutual induction.



When i_1 is changed, the galvanometer (G) shows deflection

Consider a current i_1 in the coil 1. The current carrying coil sets up a magnetic field. The coil 2 intercepts the field lines (i.e., the field lines pass through the coil 2) and there is a flux linked to it. The flux linked with the coil 2 is proportional to the current in coil 1.

$$\phi_{21} \propto i_1$$

$$\text{or, } \phi_{21} = M_{21} i_1 \quad (\text{i})$$

M_{21} is a constant known as *mutual inductance of coil 2 with respect to the coil 1*. It is a geometrical constant. Unit of mutual inductance is henry (H).

If current in coil 1 is changed, the flux linked with the coil 2 changes and an emf gets induced in it.

$$\varepsilon_{21} = - \frac{d\phi_{21}}{dt} = - M_{21} \frac{di_1}{dt} \quad (\text{ii})$$

A galvanometer connected to the coil 2 shows deflection when the current (i_1) in coil 1 is changed.

Now, Assume that a current (i_2) is given to the coil 2 by attaching some source to it. This current sets up a magnetic field. The coil 1 placed in this field has a flux linked to it.

$$\phi_{12} = M_{12} i_2 \quad (\text{iii})$$

The constant M_{12} is known as *mutual inductance of the coil 1 with respect to coil 2*.

For a given pair of coils having fixed geometry

$$M_{12} = M_{21} = M \text{ (say)}$$

The constant M is simply known as the *mutual inductance* for the pair of coils. The coil which is given current from outside is known as the *primary coil* and the other coil in which we get induced emf is known as the *secondary coil*. The above equations can be written as

$$\phi_s = M i_p \quad (28)$$

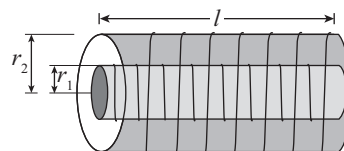
$$\text{and } \varepsilon_s = - M \frac{di_p}{dt} \quad (29)$$

Here, the subscripts p and s refer to primary and secondary coils, respectively.

In subsections 13.1 and 13.2 we present examples to illustrate the procedure for calculation of mutual inductance.

13.1 Mutual inductance of a pair of coaxial solenoids

Two coils are wrapped on two coaxial cylinders of radii r_1 and r_2 . Length of both the cylinders is l and the smaller one is completely inside the outer cylinder. Number of turns per unit length on the two solenoids are n_1 and n_2 for inner and outer one, respectively. We will consider the two solenoids to be ideal.



This is a geometry of two coils in which calculation of M is simple

Let the inner coil be primary. A source is attached to it and current in it is i_1 . This current produces a uniform magnetic field (B_1) inside the inner cylinder. There is no field in the space between the two cylinders.

$$B_1 = \mu_0 n_1 i_1$$

Flux linked with single turn of the outer coil (Secondary coil) is $B_1 \pi r_1^2$. The total flux linked with the outer coil is

$$\phi_2 = (B_1 \pi r_1^2) \cdot (\text{number of turns in outer coil})$$

$$\text{or, } \phi_2 = (\mu_0 n_1 i_1 \pi r_1^2) (n_2 l)$$

$$\text{or, } \phi_2 = (\pi \mu_0 n_1 n_2 r_1^2 l) i_1$$

Comparing the above equation to equation (28) gives

$$M = \pi \mu_0 n_1 n_2 r_1^2 l \quad (30)$$

The value of mutual inductance remains same even if the outer coil is made primary. Consider a current i_2 supplied to the outer coil. The magnetic field that it creates is uniform inside the outer cylinder and has magnitude.

$$B_2 = \mu_0 n_2 i_2$$

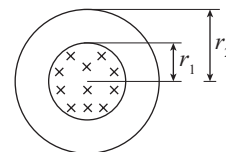
Flux linked with the single turn of the inner coil is $B_2 \cdot \pi r_1^2$. The total flux linked with the inner coil is

$$\phi_1 = (B_2 \cdot \pi r_1^2) (n_1 l)$$

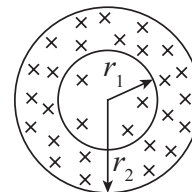
$$\text{or, } \phi_1 = (\mu_0 n_2 i_2 \pi r_1^2) (n_1 l) = (\pi \mu_0 n_1 n_2 r_1^2 l) i_2$$

$$\therefore M = \pi \mu_0 n_1 n_2 r_1^2 l$$

Therefore, the value of mutual inductance depends only on geometry of the coils and is independent of the choice of the primary coil.



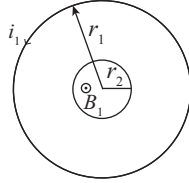
$B_1 = \mu_0 n_1 i_1$ exists inside the inner cylinder. This diagram is the cross sectional view.



When a current is given to the outer coil, the magnetic field exists in the entire cross section of radius r_2 .

13.2 Mutual inductance of two concentric coplanar loops

Consider two single turn, co-planar, concentric coils of radii, r_1 and r_2 , with $r_1 \gg r_2$ as shown. A current i_1 is given to the outer coil. The magnetic field at the centre due to this current is



Because the inner coil is very small, the magnetic field over its entire area can be treated as uniform

$$B_1 = \frac{\mu_0 i_1}{2r_1}$$

Since $r_1 \gg r_2$, the field can be approximated as B_1 for the entire area of the smaller coil. Flux linked to the inner coil is

$$\phi_2 = B_1 \cdot \pi r_2^2 = \frac{\mu_0 i_1}{2r_1} \cdot \pi r_2^2$$

$$\text{or, } \phi_2 = \left(\frac{\pi \mu_0 r_2^2}{2r_1} \right) i_1$$

Using equation (28)

$$M = \frac{\pi \mu_0 r_2^2}{2r_1} \quad (31)$$

Note that if we make the smaller coil as primary coil, the calculation of the mutual inductance will be difficult.

Example 26 Two coils on the same cylinder

A long cylinder with length l and a cross sectional area A has a wire wrapped over it in N_1 turns. Another insulated wire is wrapped over this solenoid in N_2 turns. The turns in both the coils are uniformly spaced over the entire length of the cylinder.

(a) Calculate the mutual inductance of the pair of the coils.

(b) Show that $M = \sqrt{L_1 L_2}$, where L_1 and L_2 are self inductances of the two coils.

Solution

Concepts

- The system is like a pair of co axial solenoids with $r_1 = r_2$.
- Self inductance of an ideal solenoid is $L = \pi \mu_0 n^2 r^2 l$.

- (a) The mutual inductance is given by equation (28) with $r_1 = r_2 = r$
- $$M = \pi \mu_0 n_1 n_2 r^2 l$$

$$= \mu_0 \frac{(n_1 l)(n_2 l)(\pi r^2)}{l} = \frac{\mu_0 N_1 N_2 A}{l}$$

$$(b) L_1 = \pi \mu_0 n^2 r^2 l = \mu_0 \frac{(nl)^2 (\pi r^2)}{l}$$

$$\text{or, } L_1 = \frac{\mu_0 N_1^2 A}{l}$$

$$\text{Similarly, } L_2 = \frac{\mu_0 N_2^2 A}{l}$$

$$L_1 L_2 = \left(\frac{\mu_0 N_1 N_2 A}{l} \right)^2 = M^2$$

$$\therefore M = \sqrt{L_1 L_2}$$

Note: In general, the mutual inductance of a pair of coils is given by

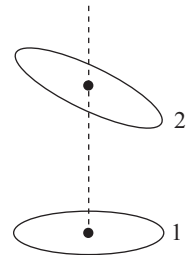
$$M = k \sqrt{L_1 L_2}$$

Here, L_1 and L_2 are self inductances of the two coils and k is a constant known as the *coupling coefficient*. In the present example $k = 1$. Value of k ranges from 0 to 1 for various geometries.

Example 27 Mutual inductance depends on geometry

Two coils 1 and 2 are placed as shown in the figure. How will the mutual inductance of the pair of coils change if

- the coil 2 is rotated to bring its plane parallel to the plane of coil 1?
- the coil 2 is moved down to bring it closer to the coil 1?
- the number of turns in the coil 2 is increased?



Solution

Concepts

When a current is given to one of the coils (say coil 1), it produces a magnetic field. The other coil placed in this field has a flux linked with it. Higher the flux linked with coil 2, for a given current in coil 1, higher is the value of M .

$$\phi_2 = M i_1$$

For a given current i_1 in coil 1, all the actions given in (a), (b) and (c) will cause the flux linked with coil 2 to increase. Therefore, the value of M will increase in all the three cases.



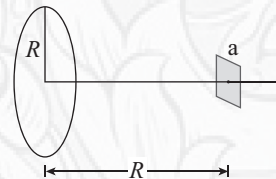
YOUR TURN

Q.32 Two coils are kept close to one another. When the current in one of them is changed at a rate of 2 A/s , emf induced in the other is 0.2 V . Find the mutual inductance for the pair of the coils.

Q.33 When current in coil A is changed at rate of 4 A/s , it induces an emf of 0.1 V in coil B . Now, the coil B is made primary and the current in it is changed at a rate of 2 A/s . Find the emf induced in A .

Q.34 Mutual inductance for a pair of coil is M . Number of turns in one of them is doubled. What is the new value of mutual inductance?

Q.35 A circular loop of single turn has radius R . Another square loop of one turn has side length a ($a \ll R$). The two loops are held parallel to one another with their axes coinciding. Distance between their centres is R . Find the mutual inductance of the pair of loops.



In Short

(i) A circuit consisting only of an inductor and a capacitor will keep oscillating forever with its energy remaining conserved.

(ii) The angular frequency of oscillation is $\omega = \frac{1}{\sqrt{LC}}$

(iii) Current in the circuit and the charge on the capacitor changes sinusoidally with angular frequency ω . When the charge on the capacitor is maximum the current is zero and vice versa.

The current in the circuit is $i = i_0 \sin(\omega t + \delta)$.

Value of δ is decided by the initial conditions. For example, $\delta = 0$ if $i = 0$ at $t = 0$ and the current is in positive direction immediately after $t = 0$.

(iv) The energy keeps oscillating between the magnetic energy and the electric energy.

$\frac{1}{2}Li_0^2 = \frac{q_0^2}{2C}$, where i_0 and q_0 are maximum values of the current and the charge.

(v) If the magnetic flux linked with coil 2 is ϕ_2 when a current i_1 is given to the coil 1, then $\phi_2 = Mi_1$.

M is a geometrical constant known as mutual inductance for the pair of coils.

(vi) If current in primary coil 1 is changed, an emf is induced in the secondary coil 2. The induced emf is

$$\varepsilon_2 = -M \frac{di_1}{dt}$$

The negative sign is in accordance with the Lenz's law.

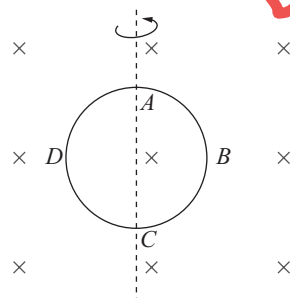
(vii) Value of M does not depend on the choice of primary coil.

(viii) The value of M for two coaxial long solenoids is $M = \mu_0 \pi n_1 n_2 r_1^2 l$, where r_1 = radius of inner solenoid.

Miscellaneous Examples

Example 28 Rotating coil

A circular conducting loop of area S can rotate about its vertical diameter. There is a uniform and horizontal magnetic field in the entire space. Initially the loop is in a position where its surface is normal to the field. The loop is made to rotate with a constant angular velocity ω in the sense shown in the figure.



When a current flows along $ABCD$ it is taken as positive.

- Write the emf induced in the loop as a function of time.
- Write the current in the loop at time $t = \frac{\pi}{6\omega}$, if its resistance is R . Is the current positive or negative?

Solution

Concepts

- $ABCD$ is positive sense. This means that the area vector of the loop is initially directed into the plane of the figure.
- As the loop rotates, the area vector rotates with angular speed ω . After time t the angle between $\vec{B}$ and $\vec{S}$ is $\theta = \omega t$.
- Use of Faraday's law, with its negative sign, will give us the emf with its direction.

- Flux through the loop at time t is

$$\phi = BS \cos(\omega t)$$

$$\varepsilon = - \frac{d\phi}{dt} = B\omega S \sin(\omega t)$$

Note: Such an emf is known as alternating emf. We will study more about it in the next chapter.

- At $t = \frac{\pi}{6\omega}$

$$\varepsilon = B\omega S \sin\left(\omega \cdot \frac{\pi}{6\omega}\right) = B\omega S \sin\left(\frac{\pi}{6}\right)$$

$$\text{or, } \varepsilon = \frac{B\omega S}{2}$$

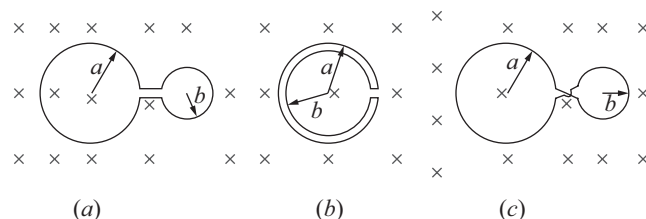
$$\therefore i = \frac{\varepsilon}{R} = \frac{B\omega S}{2R}$$

The current is positive. This means it is directed along $ABCD$.

Note: When $\frac{\pi}{\omega} < t < \frac{2\pi}{\omega}$, the emf/current is negative.

The current is along $DCBA$.

Example 29 The figures given below show two circular loops of radii a and b ($a > b$) joined together with short wires of negligible resistance. All the three loops are placed perpendicular to a uniform magnetic field that changes with time as $B = kt$. The resistance of the unit length of the wire making the loops is λ . Determine the direction and magnitude of the induced current in each loop. k is a positive constant.

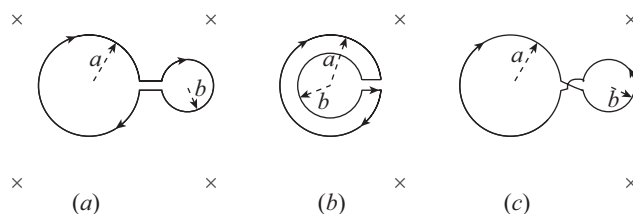


Solution

Concepts

- Mark a direction on the loop. Consider it positive direction.
- In case (b) and (c), when you curl your right hand fingers in the positive sense, the thumb points in opposite directions for the two loops. The area vectors of the two loops must be taken in opposite directions.

In case (a) The thumb points in same direction while the fingers are curled in the positive sense for both the loops.



The arrows show that assumed positive direction of current flow in the three situations.

In figure (a), the right hand thumb points inwards when the fingers are curled clockwise (in positive direction) along the loop. The area vector is

$$\vec{S} = (\pi a^2 + \pi b^2) \otimes$$

$$\therefore \phi = \vec{B} \cdot \vec{S} = \pi (a^2 + b^2) kt$$

$$\therefore \varepsilon = - \frac{d\phi}{dt} = - \pi (a^2 + b^2) k$$

$$\text{Resistance of the loop, } R = 2\pi (a + b) \lambda$$

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$$\therefore \text{Current } i = \frac{\varepsilon}{R} = -\frac{(a^2 + b^2)k}{2(a+b)\lambda}$$

The negative sign shows that the current is opposite to the assumed positive direction.

In the figure (b), the thumb points inwards when fingers are curled in the sense of the arrow in the outward loop but the thumb points outward when the fingers are curled in the sense of arrow marked in the inner loop.

$$\therefore \text{Area } \pi a^2 \text{ points inwards and } \pi b^2 \text{ points outwards.}$$

$$\therefore \phi = \pi a^2 B \cos 0^\circ + \pi b^2 B \cos 180^\circ$$

$$= \pi (a^2 - b^2)kt$$

$$\therefore \varepsilon = -\frac{d\phi}{dt} = -\pi (a^2 - b^2)k$$

$$\therefore i = \frac{\varepsilon}{R} = \frac{-\pi(a^2 - b^2)k}{2\pi(a+b)\lambda}$$

$$= -\frac{(a-b)k}{2\lambda}$$

The negative sign indicates that the current is directed opposite to the marked direction.

Similarly, in figure (c),

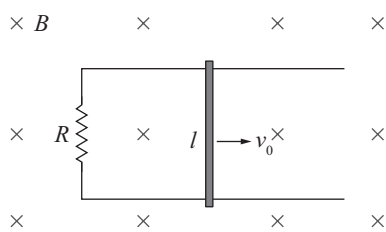
$$\phi = \pi(a^2 - b^2)kt$$

$$\therefore i = -\frac{(a-b)k}{2\lambda} \quad [\text{as done for figure (b)}]$$

The direction of the current is opposite to the marked direction.

Example 30 Decreasing emf in sliding wire generator

A conducting rod, having mass m and length l , moves on two smooth parallel, conducting rails. A uniform magnetic field (B) directed into the plane of the figure exists in the space. The rod is given an initial velocity v_0 to the right and released at $t = 0$. One ends of the rails are joined by a resistance R .



(a) Find the velocity of the rod and current in it as a function of time.

(b) Plot a graph of velocity versus time and magnitude of acceleration versus time for the rod.

Neglect the inductance of the circuit.

Solution

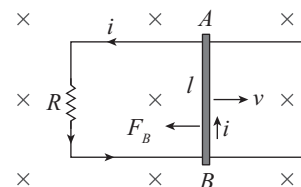
Concepts

- An emf is induced in the rod as it moves in the magnetic field. This causes a current to flow.
- The current carrying rod must experience a magnetic force that is opposite to its motion. This retards the rod.

(iii) As the rod retards, induced current decreases and the force on the rod falls further. It means that the rod moves with a decreasing retardation.

The mathematical expression that we are going to derive must support the above conclusions.

(a) Let the velocity of the rod be v at time t . Emf induced in the rod is



$$\varepsilon = Blv \quad [\text{end A is positive}]$$

Current in the loop is

$$i = \frac{\varepsilon}{R} = \frac{Blv}{R} \quad (i)$$

Magnetic force on the current carrying rod is

$$F_B = ilB = \frac{B^2 l^2 v}{R} \quad [\text{towards left}]$$

Using Newton's second law for the motion of the rod:

$$m \frac{dv}{dt} = -\frac{B^2 l^2}{R} v$$

$$\text{or, } \frac{dv}{dt} = -\frac{B^2 l^2}{mR} v$$

$$\text{or, } \frac{dv}{v} = -k dt \quad \left[k = \frac{B^2 l^2}{mR} \right]$$

$$\text{or, } \int_{v_0}^v \frac{dv}{v} = -k \int_0^t dt$$

$$\text{or, } \ln \left(\frac{v}{v_0} \right) = -kt$$

$$\text{or, } v = v_0 e^{-kt} \quad (ii)$$

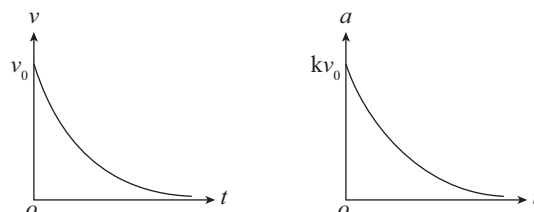
Substituting this value of v in (i) we get

$$i = \frac{Blv_0}{R} e^{-kt} \quad (iii)$$

$$(b) v = v_0 e^{-kt}$$

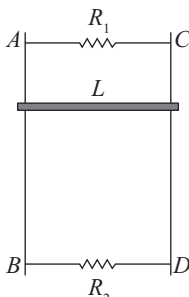
$$\therefore a = \frac{dv}{dt} = -v_0 k e^{-kt} \Rightarrow |a| = v_0 k e^{-kt}$$

Plot of v versus t and $|a|$ versus t are as shown below.



Example 31 Conducting rod falling in magnetic field

Two parallel, vertical, conducting rails AB and CD are separated by $L = 1\text{ m}$. They are connected at the ends by two resistances R_1 and R_2 , as shown. A horizontal metallic bar of mass $m = 0.2\text{ kg}$ slides, without friction, on the rails under gravity. There is a uniform horizontal magnetic field $B = 0.6\text{ T}$ perpendicular to the planes of the rails. It is observed that when the terminal velocity is attained, the power dissipated in R_1 and R_2 are 0.76 W and 1.2 W , respectively. Find the terminal velocity of the bar and the values of R_1 and R_2 . Neglect the inductance of the circuit.

**Solution****Concepts**

- Due to gravitational force the bar accelerates. As it gains velocity, an emf is induced in it. This drives a current.
- The direction of magnetic field is not stated clearly. Actually it makes no difference whether you take it directed out of the plane of the figure or directed into it. The induced current will always be in a direction so that the magnetic force opposes the motion of the bar.
- With increasing speed, the upward magnetic force on the bar goes on increasing. At a point the magnetic force becomes equal to the gravitational force. After this the velocity of the bar will become constant. This constant velocity is the terminal velocity.
- After the bar attains terminal velocity, the rate of work done by the gravitational force is same as the the power dissipated in the two resistances.

Let the magnetic field be directed inward. Let the velocity of the rod at any instant of time be v .

Induced emf $\varepsilon = Bvl$.

The current in the bar will be

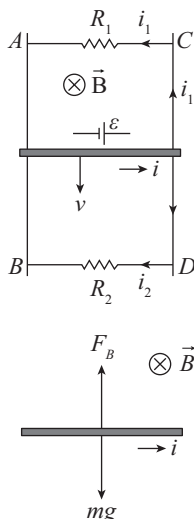
$$i = \frac{\varepsilon}{R_0} = \frac{Bvl}{R_0}$$

Here, R_0 is equivalent resistance of R_1 and R_2 .

$$\frac{1}{R_0} = \frac{1}{R_1} + \frac{1}{R_2}$$

Magnetic force on the bar is

$$F_B = ilB = \frac{B^2 l^2 v}{R_0}$$



When the rod attains terminal velocity ($= v_0$), net force on it is zero.

$$\therefore F_B = mg$$

$$\Rightarrow \frac{B^2 l^2 v_0}{R_0} = mg \Rightarrow \frac{1}{R_0} = \frac{mg}{B^2 l^2 v_0}$$

$$\text{or, } \frac{1}{R_0} = \frac{0.2 \times 9.8}{(0.6)^2 \times (1)^2 \times (1)} = \frac{49}{9} \Omega^{-1} \quad (\text{i})$$

Power dissipated in the two resistances are

$$P_1 = \varepsilon^2 / R_1 \text{ and } P_2 = \varepsilon^2 / R_2$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{R_2}{R_1} \text{ or, } \frac{R_2}{R_1} = \frac{0.76}{1.2} = \frac{19}{30} \quad (\text{ii})$$

$$\text{From (i)} \quad \frac{1}{R_1} + \frac{1}{R_2} = \frac{49}{9}$$

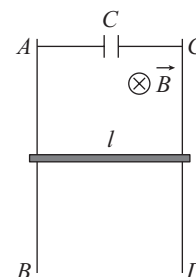
$$\text{or, } \frac{1}{R_1} + \frac{30}{19R_1} = \frac{49}{9} \quad [\text{Using (ii)}]$$

$$\Rightarrow R_1 = \frac{9}{19} \Omega$$

$$\text{And } R_2 = \frac{19R_1}{30} = \frac{3}{10} \Omega$$

Example 32 Sliding wire generator with a capacitor

AB and CD are two vertical, parallel, conducting rails at separation l . A horizontal conducting bar of mass m slides smoothly on them. The upper ends of the rails are connected through a capacitor of capacitance C . A uniform horizontal magnetic field (B) is switched on and the bar is released from rest. Write the acceleration of the bar. Assume that it remains horizontal and neglect the resistance and inductance of the loop.

**Solution****Concepts**

- The charge on the capacitor, at an instant when the induced emf is ε , is $q = C\varepsilon$.
- The current in the loop is $i = \frac{dq}{dt}$
- The magnetic force on the current carrying bar must be upward, opposing its motion.

Let the velocity of the bar be v at time t . The motional emf in the bar is

$$\varepsilon = Bvl$$

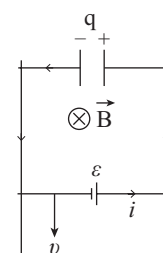
Charge on the capacitor is

$$q = C\varepsilon = C(Bvl)$$

Current in the loop is

$$i = \frac{dq}{dt} = CBl \frac{dv}{dt}$$

$$= CBl a$$



(i)

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For motion of the bar, we can write

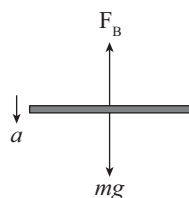
$$ma = mg - F_B$$

$$\text{or, } ma = mg - ilB$$

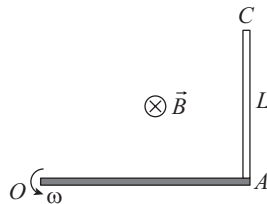
$$\text{or, } ma = mg - (CB^2l^2) a \text{ [using (i)]}$$

$$\text{or, } a = \frac{mg}{m + CB^2l^2}$$

The bar falls down with constant acceleration.



Example 33 Rod rotating about an axis away from it AC is a conducting rod of length L . OA is a non conducting rod joining AC at right angles. The L shaped body is rotated with a constant angular speed ω about an axis passing through O , perpendicular to the plane of the body. A uniform magnetic field B exists parallel to the axis of rotation. Find the emf induced in the rod AC .



Solution

Concepts

- Emf induced in a rod of length L moving with velocity v in a perpendicular field B is $\epsilon = BLv \sin \theta$, where θ is angle between the length (L) and velocity v .
- The velocity of a point in a rotating rigid body is $v = \omega r$ where r is distance of the point from the rotation axis. The direction of the velocity is perpendicular to r .

Consider a segment of length dx (at P) in the conducting rod AC . Velocity of this segment is $v = \omega r$, where $r = OP$.

Direction of the velocity is perpendicular to OP .

Emf induced in the segment is

$$d\epsilon = Bvdx \cdot \sin \theta \quad [\text{The end towards A is positive}]$$

$$\text{or, } d\epsilon = B(\omega r)(dx)(\sin \theta) = B\omega(r \sin \theta)dx$$

$$\text{or, } d\epsilon = B\omega x dx$$

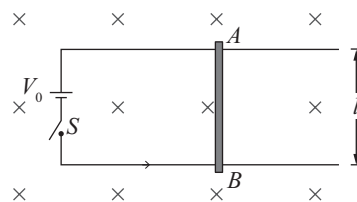
The total emf in the rod can be obtained by adding the emfs in every small segment.

$$\therefore \epsilon = \int d\epsilon = B\omega \int_0^L x dx$$

$$\text{or, } \epsilon = \frac{1}{2} B\omega L^2$$

Example 34 Conducting rod AB can slide on two parallel, smooth rails. The rod has resistance R and mass m . Separation between the rails is l and their ends are connected to a cell of emf V_0 through a switch S . A uniform magnetic field B exists in the entire space perpendicular to the plane

of the figure. The switch is closed at time $t = 0$. Find the velocity of the rod as a function of time. Draw a graph showing the variation of velocity of the rod with time.



Solution

Concepts

- A current flows through the rod after the switch is closed. The rod experiences a magnetic force and accelerates.
- An emf is induced in the moving rod which is opposite to the emf of the cell. The current reduces. As the current reduces the magnetic force on the rod decreases. The acceleration of the rod decreases.
- When the emf induced in the rod becomes equal to V , the net emf in the loop becomes zero. The current becomes zero. Force on the rod becomes zero and it moves with constant velocity (v_0)

$$Blv_0 = V_0 \Rightarrow v_0 = \frac{V_0}{Bl}$$

Let speed of the rod at time t be v .

Induced emf $\epsilon = Blv$

$$\text{Induced current is } i = \frac{V_0 - Blv}{R}$$

$$\text{Magnetic force, } F = i l B = \left(\frac{V_0 - Blv}{R} \right) l B$$

$$\therefore m \frac{dv}{dt} = \left(\frac{V_0 - Blv}{R} \right) l B$$

$$\text{or, } \int_0^v \frac{dv}{V_0 - Blv} = \frac{Bl}{mR} \int_0^t dt$$

$$\text{or, } [\ln(V_0 - Blv)]_0^v = -\frac{B^2 l^2}{mR} t$$

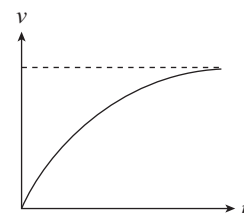
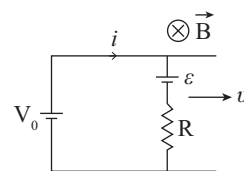
$$\text{or, } \ln \left(\frac{V_0 - Blv}{V_0} \right) = -\frac{B^2 l^2 t}{mR}$$

$$\text{or, } 1 - \frac{Blv}{V_0} = e^{-\frac{B^2 l^2 t}{mR}}$$

$$\text{or, } v = \frac{V_0}{Bl} \left(1 - e^{-\frac{B^2 l^2 t}{mR}} \right)$$

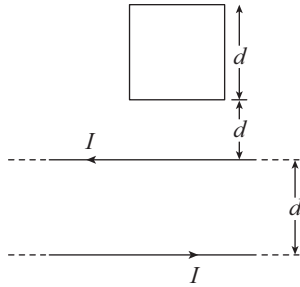
As $t \rightarrow \infty$

$$v \rightarrow \frac{V_0}{Bl}$$



Example 35 Two parallel wires and a square loop

Two fixed long parallel wires carry same current I in opposite directions, as shown. A square loop of side length d is fixed in the plane of the wires with its side parallel to the two wires. The separation between the square loop and the nearest wire is d . The current in the two wires is increased at a rate of $\frac{dI}{dt} = \alpha$



- (a) Calculate the induced emf in the loop.
 (b) What is the direction of force on the loop due to the parallel wires?

Solution**Concepts**

- The magnetic fields created by the two wires at the locations of the square loop are in opposite directions.
- We will first calculate the flux through the square loop and then find the induced emf using Faraday's law.
- The direction of current in the loop can be found using Lenz's law.
- Two parallel wires having currents in same direction attract each other and two wires having currents in opposite direction repel each other.

- (a) Consider the square loop at a distance a from a long wire, as shown. The field due to the straight wire at a distance x from it is

$$B = \frac{\mu_0 I}{2\pi x} \otimes$$

Flux through a strip of width dx is

$$d\phi = d(dx) \frac{\mu_0 I}{2\pi x}$$

Flux through the square loop is

$$\phi = \frac{\mu_0 I d}{2\pi} \int_a^{a+d} \frac{dx}{x} = \frac{\mu_0 I d}{2\pi} \ln\left(\frac{a+d}{a}\right)$$

$$\therefore \phi = \frac{\mu_0 I d}{2\pi} \ln\left(1 + \frac{d}{a}\right)$$

For the two wires given in the problem the distance a is equal to d and $2d$. Also the field produced by the two wires are oppositely directed. Therefore the net flux through the square loop is

$$\phi = \frac{\mu_0 I d}{2\pi} \ln\left(1 + \frac{d}{d}\right) - \frac{\mu_0 I d}{2\pi} \ln\left(1 + \frac{d}{2d}\right)$$

$$= \frac{\mu_0 I d}{2\pi} \left[\ln 2 - \ln \frac{3}{2} \right] = \frac{\mu_0 I d}{2\pi} \ln\left(\frac{4}{3}\right)$$

$$\therefore \varepsilon = \left| \frac{d\phi}{dt} \right| = \frac{\mu_0 d}{2\pi} \ln\left(\frac{4}{3}\right) \frac{dI}{dt}$$

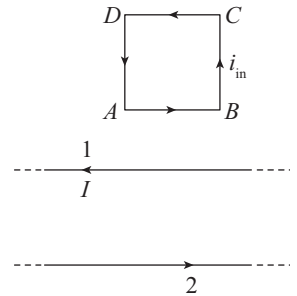
$$\text{or, } \varepsilon = \frac{\mu_0 \alpha d}{2\pi} \ln\left(\frac{4}{3}\right)$$

- (b) Net magnetic field at the location of the square loop is directed into the plane of the figure. With increasing I , the field increases. The induced current in the loop must oppose this. The induced current is anticlockwise which produces a magnetic field directed outward.

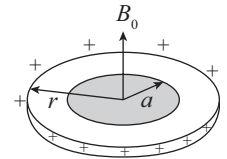
Force due to wire 1 on AB is repulsive and that on CD is attractive. The net force on the loop due to wire 1 is repulsive. Similarly, the net force on the loop due to wire 2 is attractive.

Since wire 1 is closer its effect is much stronger.

Therefore, the net force on the square loop due to the two wires is repulsive.

**Example 36** Disc rotates when magnetic field is switched off

A uniform disc has mass M and radius r . It is free to rotate about its central vertical axis. A light insulating belt carrying a charge λ per unit length, is wrapped along the circumference of the disc. A uniform magnetic field B_0 exists in vertically upward direction in the central region of radius a . The magnetic field is suddenly switched off. The disc begins to spin. Find its angular speed.



The magnetic field is suddenly switched off. The disc begins to spin. Find its angular speed.

Solution**Concepts**

- As the magnetic field changes from B_0 to zero it induces an electric field. The electric field lines are closed circles.
- A small charge element on the disc experiences a tangential force due to the induced electric field. This produces a torque. All charge elements on the circumference of the disc experience torque in the same direction.
- Change in angular momentum = $\tau \Delta t$

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The induced electric field on the circumference of the disc is given by equation (9).

$$E = \left(\frac{a^2}{2} \frac{dB}{dt} \right) \frac{1}{r}$$

A charge Δq experiences a tangential force

$$\Delta F = \Delta q E = \left(\frac{a^2}{2r} \frac{dB}{dt} \right) \Delta q$$

Torque (about the rotation axis) of this force is

$$\Delta \tau = \Delta F \cdot r = \left(\frac{a^2}{2} \frac{dB}{dt} \right) \Delta q$$

Sum of torque on all such charge elements is

$$\tau = \left(\frac{a^2}{2} \frac{dB}{dt} \right) \sum \Delta q = \frac{a^2}{2} \frac{dB}{dt} \lambda \cdot 2\pi r$$

Angular impulse due to the torque is

$$\int \tau dt = \pi a^2 \lambda r \int dB = \pi a^2 \lambda r B_0$$

[The magnetic field changes from B_0 to zero in small interval Δt . The torque acts on the disc for this short interval. Impulse of the torque is independent of Δt]

$$\text{Now, } \Delta L = \int \tau dt$$

$$\text{or, } I\omega = \pi a^2 \lambda r B_0$$

$$\left[I = \frac{1}{2} Mr^2 = \text{moment of inertia of the disc} \right]$$

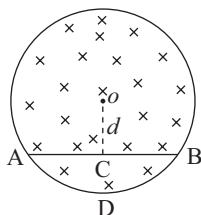
$$\text{or, } \frac{1}{2} Mr^2 \cdot \omega = \pi a^2 \lambda r B_0$$

$$\text{or, } \omega = \frac{2\pi a^2 \lambda B_0}{Mr}$$

Example 37 Induced emf in a chord of circular region having time changing B .

A uniform magnetic field (B) exists in a region of circular cross section of radius R . AB is a straight conductor along the chord of the circle at a distance $OC = d$ from the centre. Find the emf induced in the conductor when the magnetic field changes at a rate $\frac{dB}{dt}$. Is the value of

$\int \vec{E} \cdot d\vec{l}$ calculated along the arc ADB same as the emf in the conductor?



Solution

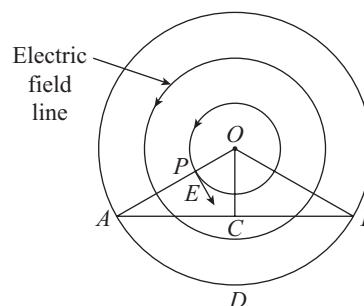
Concepts

- (i) The emf induced is $\int_A^B \vec{E} \cdot d\vec{l}$, where $\vec{E}$ is the induced electric field. Using equation (8) to write $\vec{E}$ and performing the integration will give us the

result. However, there is a much simpler way. We can easily write emf induced in the ΔOAB by using Faraday's law.

- (ii) Emf in ΔOAB is sum of emfs in OA , AB and BO . But emf in radial lines OA and OB will be zero as explained below.
- (iii) Induced electric field is not conservative. The line integral of $\vec{E}$ is path dependent and is not zero for a closed path.

Induced electric field is represented by circular lines. $\vec{E}$ is perpendicular to OA and OB .



$$\therefore \int_O^A \vec{E} \cdot d\vec{l} = \int_B^O \vec{E} \cdot d\vec{l} = 0$$

It means that no emf is induced along the radial lines OA and OB . Emf induced in ΔOAB is present entirely along the chord AB .

$$\text{Flux through } \Delta OAB \text{ is } \phi = B \cdot \frac{1}{2} (AB) (OC)$$

$$\text{or, } \phi = \frac{1}{2} B (2\sqrt{R^2 - d^2}) (d) = Bd \sqrt{R^2 - d^2}$$

$\therefore$ emf induced in ΔOAB is

$$\varepsilon = \left| \frac{d\phi}{dt} \right| = d\sqrt{R^2 - d^2} \cdot \frac{dB}{dt}$$

$$\therefore \varepsilon_{AB} = d\sqrt{R^2 - d^2} \cdot \frac{dB}{dt}$$

Emf induced along arc ADB is different.

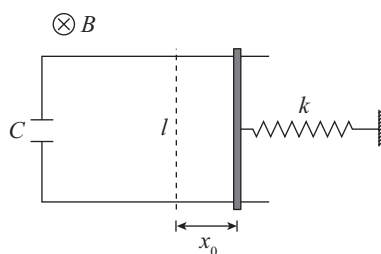
[We know the induced field at all points on the arc ADB is

$$E = \frac{R}{2} \frac{dB}{dt}$$

$$\begin{aligned} \therefore \int_{ADB} \vec{E} \cdot d\vec{l} &= \int_{ADB} E dl \cos 0^\circ \\ &= E \int dl = E (\text{arc length } ADB) \end{aligned}$$

Example 38 Sliding rod connected to a spring

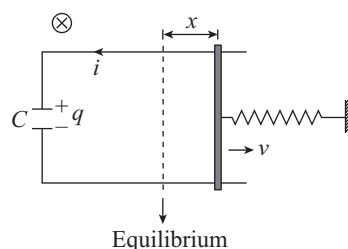
In the arrangement shown in the figure, a capacitor of capacitance C connects the ends of two long, smooth parallel conducting rails. The rails are separated by a distance l . A sliding connector of mass m is connected to a spring of force constant k , as shown. The connector is pushed so that the spring gets compressed by x_0 and is released. Find the time taken by the connector to reach its original position where the spring is unstretched. The entire space has a uniform magnetic field B perpendicular to the figure, there is no resistance in the circuit and the spring is non conducting.

**Solution****Concepts**

- The spring pushes the connector and an emf is developed in it. This emf charges the capacitor.
- The system is conservative. When the connector stops, there is no emf and therefore there is no charge in the capacitor. The entire energy is back with the spring.
- As shown below, the motion is SHM.

Equilibrium position is where the spring is relaxed consider the connector at a distance x from the equilibrium position.

Let the velocity of the connector be v , charge on the capacitor be q and the current in the circuit be i .



$$\frac{q}{C} = Blv \quad (i)$$

$$\text{or, } \frac{dq}{dt} = BCl \frac{dv}{dt}$$

$$\text{or, } i = BCl \cdot a \quad [a = \text{acceleration}]$$

$$\text{Using } ma = F$$

$$ma = -kx - ilB$$

$$\text{or, } ma = -kx - B^2 l^2 C a$$

$$\text{or, } a = - \left(\frac{k}{m + B^2 l^2 C} \right) x \quad (iii)$$

Therefore, the motion is SHM.

$$\omega = \sqrt{\frac{k}{m + B^2 l^2 C}}$$

$$\text{Time period of oscillation is } T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m + B^2 l^2 C}{k}}$$

Time needed to move from the extreme position to the mean position is

$$\frac{T}{4} = \frac{\pi}{2} \sqrt{\frac{m + B^2 l^2 C}{k}}$$

Alternative method

In place of force method, we can use energy method to prove that the motion is SHM.

The total energy of the system when the connector is at distance x from the mean position is

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \frac{q^2}{2C} = E \text{ (a constant)}$$

$$\text{or, } \frac{1}{2}m \left(2v \frac{dv}{dt} \right) + \frac{1}{2}k \left(2x \frac{dx}{dt} \right) + \frac{1}{2C} \left(2q \frac{dq}{dt} \right) = \frac{dE}{dt} = 0$$

$$\text{or, } mv \frac{dv}{dt} + kxv + \frac{1}{C} (BClv)(BCl a) = 0$$

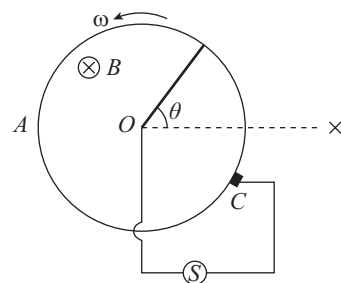
$$[\text{From (i) } q = BClv \text{ and from (ii) } i = BCl a]$$

$$\text{or, } a(m + B^2 l^2 C) = -kx$$

$$\text{or, } a = - \left(\frac{k}{m + B^2 l^2 C} \right) x$$

This equation is same as equation (iii).

Example 39 A metal rod of length L and mass m rotates in vertical plane about a horizontal axis through its end O . The other end of the rod remains in contact with a vertical conducting ring. A voltage source S is connected between the end O of the rod and a contact C which touches the ring. The rod has resistance R and everything else is having negligible resistance. A uniform horizontal magnetic field B exists in the entire space. The rod is initially horizontal along OX . It rotates with constant angular speed ω such that it traverses an angle $\theta = \omega t$ in time t . Find the time dependence of the voltage of the source so that the rod keeps rotating with constant angular speed without any external help.

**Solution****Concepts**

- When a current flows in the rod, it experiences a magnetic force. This force produces a torque. The weight of the rod produces another torque. The two torques should always balance to keep the rod rotating with constant angular speed.

- Also keep in mind that there is an emf in the rod.

$$\varepsilon = \frac{1}{2} B \omega L^2$$

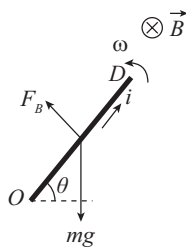
5.40 Magnetism and Electromagnetic Induction

Induced emf in the rod is

$$\varepsilon = \frac{1}{2} B \omega L^2$$

The end O of the rod is positive.

For the position shown, the direction of magnetic force (F_B) should be as indicated (so as to balance the torque due to mg). For this the direction of current should be from O to D .



If V is voltage of the source

$$V - \varepsilon = iR$$

$$\text{or, } i = \frac{V - \frac{1}{2} B \omega L^2}{R}$$

$$\tau_B = \tau_g$$

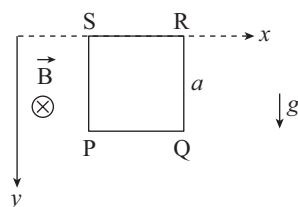
$$\Rightarrow (iLB) \left(\frac{L}{2} \right) = mg \frac{L}{2} \cos \theta$$

$$\Rightarrow \left(\frac{V - \frac{1}{2} B \omega L^2}{R} \right) LB = mg \cos \omega t \quad (\because \theta = \omega t)$$

$$\Rightarrow V = \frac{mgR}{LB} \cos \omega t + \frac{1}{2} B \omega L^2$$

Example 40 A horizontal magnetic field $\vec{B} = \left(\frac{B_0 y}{a} \right) \hat{k}$ is established in a large hall.

The y axis points vertically downward and B_0 and a are constants. A square loop of side length a , mass m and resistance R is released from rest with its upper side along the x axis at $t = 0$.



- Find the current in the loop as a function of its speed.
- Find the acceleration of the loop when it acquires a speed v .
- Find the speed of the loop as a function of time.

Solution

Concepts

- We can write the induced emfs in the sides PQ and SR by the concept of motional emf. The net emf in the loop is algebraic sum of these two emfs. One can also write the emf in the loop by calculating the rate of change of flux linked to the loop.
- Induced emf $\varepsilon = BLv$ is directly proportional to B which is directly proportional to y . It means the emfs induced in PQ and SR are directly proportional to their y co-ordinates.

(iii) The field is different at the location of PQ and SR . Therefore, magnetic forces on PQ and SR are not equal.

(iv) $ma = mg + F_B$

- (a) Let the velocity of the loop at time t be v and the y coordinate of side SR be y . The y coordinate of PQ will be $(y + a)$

Emf induced in SR is

$$\varepsilon_1 = B_1 a v = \left(\frac{B_0 y}{a} \right) a v$$

Emf induced in PQ is

$$\varepsilon_2 = B_2 a v = \frac{B_0}{a} (y + a) a v$$

Net emf in the loop is

$$\varepsilon = \varepsilon_2 - \varepsilon_1 = B_0 a v$$

$$\text{Induced current } i = \frac{\varepsilon}{R} = \frac{B_0 a v}{R}$$

- (b) Magnetic force on the loop is

$$\begin{aligned} F_B &= F_2 - F_1 \\ &= \frac{B_0}{a} (y + a) i a - \frac{B_0}{a} y i a \\ &= B_0 a i = \frac{B_0^2 a^2 v}{R} \end{aligned}$$

The resultant magnetic force is upwards.

$$\therefore ma = mg - F_B$$

$$\text{or, } a = g - \frac{B_0^2 a^2 v}{mR}$$

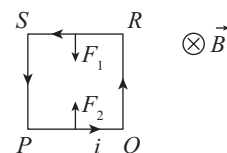
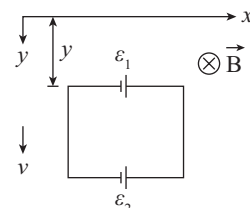
$$(c) \frac{dv}{dt} = g - \frac{B_0^2 a^2 v}{mR} \Rightarrow \int_0^v \frac{dv}{g - \frac{B_0^2 a^2 v}{mR}} = \int_0^t dt$$

$$\text{or, } \left\{ \ln \left[g - \frac{B_0^2 a^2 v}{mR} \right] \right\}_0^v = - \frac{B_0^2 a^2}{mR} t$$

$$\text{or, } \ln \left[1 - \frac{B_0^2 a^2 v}{mgR} \right] = - \frac{B_0^2 a^2 t}{mR}$$

$$\text{or, } 1 - \frac{v}{kg} = e^{-t/k} \quad \text{where } k = \frac{mR}{B_0^2 a^2}$$

$$\text{or, } v = kg (1 - e^{-t/k})$$

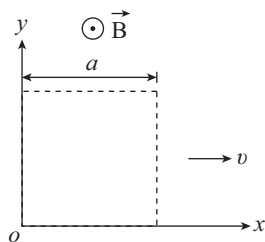


Magnetic forces on the arms SP and QR are equal and opposite

Example 41 Magnetic field changing with position and time

A magnetic field is directed perpendicular to the xy plane. The strength of the field at $x = 0$ is B_0 which is changing with time at a rate $\frac{dB_0}{dt} = \alpha$. The magnetic field increases

as one moves along the x axis at a rate $\frac{dB}{dx} = \beta$. The field does not change with change in y co-ordinate. A square loop of side length a is initially placed as shown. It is moved with a constant velocity v in positive x direction starting at $t = 0$. Find the emf induced in the loop at time t .

**Solution****Concepts**

- The magnetic field is changing with position as well as time. We will use Faraday's law for finding the emf.
- The field at the origin at time t is $B_0 = \alpha t + c$ where c is value of B_0 at $t = 0$.

Field at coordinate x at time t will be $B = B_0 + \beta x$

- We will divide the square loop into thin rectangular loops having length a parallel to y axis, for calculating the flux.

- We will find flux at time t and then use $\varepsilon = \left| \frac{d\phi}{dt} \right|$.

The figure shows the situation at time t . Field at co-ordinate x is

$$B = B_0 + \beta x \\ = (\alpha t + c) + \beta x$$

Flux through a strip of width dx is

$$d\phi = B \cdot (adx) \\ = (\alpha t + c + \beta x) adx$$

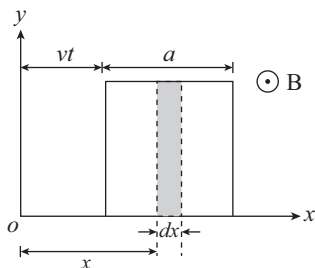
Flux through the loop at time t is

$$\phi = \int_{x=vt}^{(vt+a)} d\phi = \left[\alpha t a \int dx + ca \int dx + \beta a \int x dx \right]_{vt}^{vt+a} \\ = \left[\alpha t a x + ca x + \frac{\beta a x^2}{2} \right]_{vt}^{vt+a} \\ = (\alpha t + ca)(vt + a - vt) + \frac{\beta a}{2} [(vt + a)^2 - (vt)^2]$$

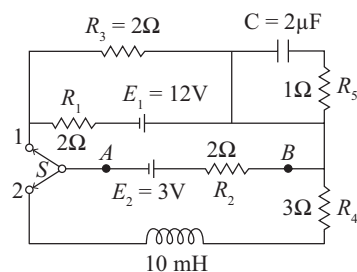
$$= (\alpha t + c)a^2 + \frac{\beta a}{2} (2vat + a^2)$$

$$\text{Induced emf } \varepsilon = \left| \frac{d\phi}{dt} \right|$$

$$= \alpha a^2 + \beta a^2 v = a^2 (\alpha + \beta v)$$



Example 42 The switch S shown in the circuit can be placed in position 1 or 2. Initially the switch is in position 1 and the circuit is in steady state.



- Find the potential difference $V_A - V_B$ and power dissipation in R_1 .
- If now the switch S is put in position 2 at $t = 0$, find
 - the steady state current in R_4
 - the time when the current in R_4 is half the steady value
 - the energy stored in the inductor when current in R_4 is half the steady value.

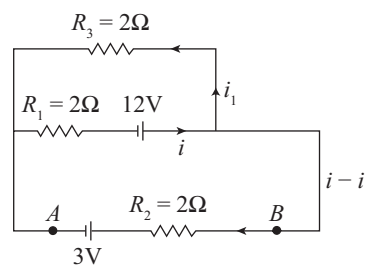
Solution**Concepts**

- With the switch in position 1 and the circuit in steady state, there will be no current through the capacitor. Kirchhoff's laws can be used to find current in different branches.
- With the switch in position 2, the current grows through the inductor with a time constant

$$\tau = \frac{L}{R_2 + R_4}$$

- The effective circuit, for calculation of current, is as shown writing the loop equation for the upper loop gives

$$2i + 2i_1 = 12$$



$$\text{or, } i + i_1 = 6 \quad (i)$$

For the lower loop we can write

$$2(i - i_1) + 3 + 2i = 12$$

$$\text{or, } 4i - 2i_1 = 9 \quad (ii)$$

Solving equations (i) and (ii) gives $i = 3.5\text{A}$, $i_1 = 2.5\text{A}$

$$V_A + 3 + (i - i_1)^2 = V_B$$

$$\text{or, } V_A - V_B = -5\text{V}$$

$$\text{Power dissipation in } R_1 \text{ is } i^2 R_1 = (3.5)^2 \times 2 = 24.5\text{W}$$

- (b) After the switch is pert into position 2, the effective circuit is as shown.

(i) The steady state current is

$$i_0 = \frac{3V}{5\Omega} = 0.6A$$

(ii) Time constant

$$\tau = \frac{10mH}{(2+3)\Omega} = 2 \text{ ms}$$

$$i = i_0 (1 - e^{-t/\tau})$$

When $i = \frac{i_0}{2}$, we have

$$\frac{1}{2} = 1 - e^{-t/\tau} \Rightarrow e^{-t/\tau} = \frac{1}{2}$$

$$\text{or, } \frac{t}{\tau} = \ln 2 \Rightarrow t = (\ln 2)\tau = 0.693 \times 2 \text{ ms}$$

$$= 1.386 \text{ ms}$$

$$(iii) U = \frac{1}{2} Li^2 = \frac{1}{2} \times (10 \times 10^{-3}) \times (0.3)^2 = 4.5 \times 10^{-4} \text{ J}$$

Example 43 Maximum rate at which the energy is transferred to the magnetic field

A solenoid of inductance L and resistance R is connected to a cell of emf V at time $t = 0$

- Find the time when the rate of growth of the magnetic field energy in the solenoid is maximum.
- Find the maximum rate of growth of the magnetic field energy.
- Find the time when the rate of heat dissipation in the circuit is same as the maximum rate of growth of the magnetic field energy.

Solution

Concepts

$$(i) U_B = \frac{1}{2} Li^2$$

$$(ii) \text{ The rate of growth of } U_B \text{ is } P_B = \frac{dU_B}{dt}$$

We have to find the time when P_B is maximum.

$$(a) i = i_0 (1 - e^{-t/\tau})$$

$$\frac{di}{dt} = \frac{i_0}{\tau} e^{-t/\tau} = \frac{Ri_0}{L} e^{-t/\tau}$$

$$U_B = \frac{1}{2} Li^2$$

$$P_B = \frac{dU_B}{dt} = Li \frac{di}{dt} = Li_0 (1 - e^{-t/\tau}) \frac{R}{L} i_0 e^{-t/\tau}$$

$$= i_0^2 R (e^{-t/\tau} - e^{-2t/\tau}) \quad (i)$$

$$P_B \text{ is maximum when } \frac{dP_B}{dt} = 0$$

$$\text{or, } -\frac{1}{\tau} e^{-t/\tau} + \frac{2}{\tau} e^{-2t/\tau} = 0$$

$$\text{or, } e^{-t/\tau} = \frac{1}{2} \Rightarrow \frac{t}{\tau} = \ln 2$$

$$\text{or, } t = \tau \ln 2 = \frac{L}{R} (\ln 2)$$

$$(b) P_B \text{ is maximum when } e^{-t/\tau} = \frac{1}{2}$$

Putting this in (i) gives the maximum value of P_B .

$$\therefore (P_B)_{\max} = i_0^2 R \left(\frac{1}{2} - \frac{1}{4} \right) = \frac{1}{4} i_0^2 R$$

$$= \frac{V^2}{4R} \quad \left[\because i_0 = \frac{V}{R} \right]$$

(c) When rate of heat dissipation is equal to $(P_B)_{\max}$.

$$i^2 R = \frac{V^2}{4R} \Rightarrow i = \frac{V}{2R} = \frac{i_0}{2}$$

$$\text{or, } i_0 (1 - e^{-t/\tau}) = \frac{i_0}{2}$$

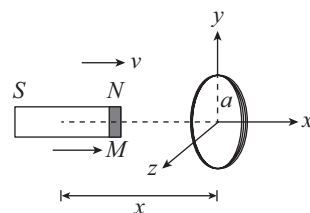
$$\text{or, } e^{-t/\tau} = \frac{1}{2} \Rightarrow \frac{t}{\tau} = \ln 2$$

$$\text{or, } t = (\ln 2) \frac{L}{R}$$

Example 44 Force on a magnet being pushed towards a coil

A small bar magnet of dipole moment M is moving with velocity v in the x -direction as shown in the figure. A circular conducting loop of radius a and resistance R lies in yz plane with its centre at $x = 0$ and its axis coinciding with x -axis.

Distance of the magnet from the loop is x ($x \gg a$). Find the magnetic force that opposes the motion of the magnet.



Solution

Concepts

- Magnetic field created by the bar magnet at the centre of the loop is nearly $B = \frac{\mu_0}{4\pi} \frac{2M}{x^3}$.

The field over the entire area of the loop can be assumed to be constant because $a \ll x$.

- As the magnet moves, the flux linked with the loop changes. An emf is induced in the loop which drives a current in it.
- The current carrying loop is a magnetic dipole.
- Force between two co axial magnetic dipoles kept at a large separation x is

$$F = \frac{\mu_0}{4\pi} \frac{6M_1 M_2}{x^4}$$

Magnetic field due to the magnet on its axis at a distance x is

$$B = \frac{\mu_0}{4\pi} \frac{2M}{x^3}.$$

This field is nearly uniform over the entire area of the loop. Hence, the flux linked with the loop is

$$\phi = B \cdot \pi a^2 = \frac{\mu_0}{4\pi} \frac{2M}{x^3} \cdot \pi a^2 = \frac{\mu_0}{2} \frac{Ma^2}{x^3}$$

Emf induced in the loop is

$$\varepsilon = \left| \frac{d\phi}{dt} \right| = \frac{3\mu_0}{2} \frac{Ma^2}{x^4} \frac{dx}{dt} = \frac{3\mu_0}{2} \frac{Ma^2}{x^4} v$$

Current induced in the loop is

$$i = \frac{\varepsilon}{R} = \frac{3\mu_0 Ma^2}{2x^4 R} v$$

The dipole moment induced in the loop is

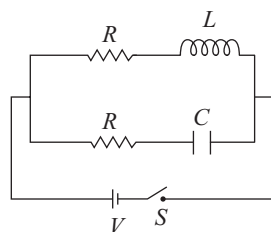
$$M' = i\pi a^2 = \frac{3\mu_0 \pi Ma^4}{2x^4 R} v$$

Force between the two dipole is

$$F = \frac{\mu_0}{4\pi} \frac{6MM'}{x^4} = \frac{\mu_0}{4\pi} \frac{6M}{x^4} \left(\frac{3\mu_0 \pi Ma^4 v}{2x^4 R} \right) \\ = \frac{9}{4} \frac{\mu_0^2 M^2 a^4}{R x^8} v$$

Example 45 A decreasing and an increasing current makes a constant current

In the circuit shown in the figure, the current through the battery remains constant forever after the switch (S) is closed. How are L , C and R related?



Solution

Concepts

- (i) Immediately after the switch is closed the current in the branch having the inductor is zero and that in the branch having the capacitor is $\frac{V}{R}$. After a long time the current in capacitor branch becomes zero and that in the branch having inductor becomes $\frac{V}{R}$.
- (ii) The current through the battery is sum of the currents in the two branches. It can remain constant at $\frac{V}{R}$ if the growth in the current in the inductor branch in a time t is same as the fall in the current in the capacitor branch in time t .

Current in the inductor branch at time t is

$$i_1 = \frac{V}{R} \left(1 - e^{-\frac{Rt}{L}} \right)$$

Current in the capacitor branch at time t is

$$i_2 = \frac{V}{R} e^{-t/RC}$$

Currents through the battery is

$$i = i_1 + i_2 = \frac{V}{R} \left[1 - e^{-\frac{Rt}{L}} + e^{-\frac{t}{RC}} \right]$$

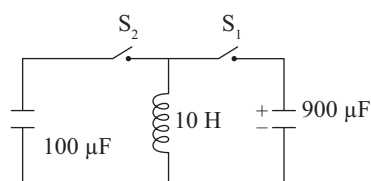
i is constant at $\frac{V}{R}$ if

$$-e^{-\frac{Rt}{L}} + e^{-\frac{t}{RC}} = 0$$

$$\text{or, } \frac{L}{R} = RC \Rightarrow R = \sqrt{\frac{L}{C}}$$

Example 46 Transferring energy from one capacitor to another using an inductor.

In the circuit shown, the $900\mu\text{F}$ capacitor is charged to 100V . Switch S_1 is closed for a time interval t_1 and then it is opened. At the same instant the other switch



S_2 is closed. It is kept closed for an interval t_2 and then it is opened. It is now found that the $100\mu\text{F}$ capacitor is charged to 300V . Find the smallest possible values of t_1 and t_2 . Take $\pi = \sqrt{10}$.

Solution

Concepts

- (i) As S_1 is closed, the $900\mu\text{F}$ capacitor discharges and the energy gets transferred to the inductor.
- (ii) When S_1 is opened and S_2 is closed, the energy begins to transfer from the inductor to the $100\mu\text{F}$ capacitor.
- (iii) Time period of oscillation in an LC circuit is $T = 2\pi\sqrt{LC}$ and entire energy from the capacitor gets transferred to the inductor in an interval $\frac{T}{4}$.

Initial energy stored in $900\mu\text{F}$ capacitor is

$$U_1 = \frac{1}{2} \times 900 \times 10^{-6} \times (100)^2 = 4.5\text{ J}$$

Final energy stored in the $100\mu\text{F}$ capacitor is

$$U_2 = \frac{1}{2} \times 100 \times 10^{-6} \times (300)^2 = 4.5\text{ J}$$

The entire energy got transferred from the $900\mu\text{F}$ capacitor to the $100\mu\text{F}$ capacitor.

First the energy from the $900\mu\text{F}$ capacitor gets transferred to the inductor. This takes a minimum time.

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$$t_1 = \frac{T_1}{4} = \frac{1}{4} \cdot 2\pi\sqrt{LC_1}$$

$$= \frac{\pi}{2} \sqrt{10 \times 900 \times 10^{-6}} = 0.15\text{s}$$

Now, at the moment the current through L is maximum (i.e., energy is entirely with the inductor) the switch S_1 is opened and S_2 is closed. The $100\mu\text{F}$ capacitor starts getting charged and will acquire the entire energy in a minimum time of

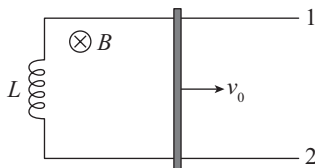
$$t_2 = \frac{T_2}{4} = \frac{1}{4} \cdot 2\pi\sqrt{LC_2}$$

$$= \frac{\pi}{2} \sqrt{10 \times 100 \times 10^{-6}} = 0.05\text{s}$$

Therefore, the smallest possible values of t_1 and t_2 are 0.15s and 0.05s respectively.

Example 47 An inductor in a sliding wire generator

1 and 2 are two perfectly conducting parallel rails at separation l . A pure inductance L is used to connect their ends. A conducting rod of mass m is placed perpendicular to the rails and can slide frictionlessly. A uniform magnetic field B exists in the entire region perpendicular to the plane of the figure. At time $t = 0$ the rod is imparted a velocity v_0 to the right. Express the displacement of the rod as a function of time neglect any resistance.



Solution

Concepts

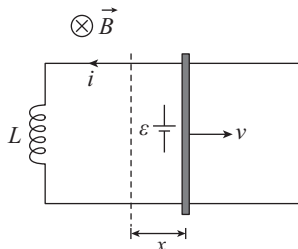
- The system is conservative. The kinetic energy of the rod will get transferred to the magnetic field energy inside the inductor and it will again go back to the rod as kinetic energy. One can easily guess that the motion of the rod will be oscillatory.
- We will consider the system at time t when the rod is displaced by x from its original position. We will use $F = ma$ for the motion of the rod and prove that its motion is SHM.

Let the velocity and displacement of the rod be v and x at time t . Induced emf in the rod is $\varepsilon = Blv$. If the current in the loop is i , then

$$Blv - L \frac{di}{dt} = 0$$

$$\text{or, } Bl \frac{dx}{dt} = L \frac{di}{dt}$$

$$\text{or, } L di = Bl dx$$



$$\text{or, } L \int_0^i di = Bl \int_0^x dx$$

$$\Rightarrow Li = Blx \quad (i)$$

[Immediately after the rod is projected, current will be zero as there is an inductor in the circuit]

$$\text{Magnetic force on the rod is } F_B = ilB = \frac{B^2 l^2 x}{L} \quad (ii)$$

[Using (i)]

Using $F = ma$ for the motion of the rod

$$m \frac{d^2 x}{dt^2} = - \frac{B^2 l^2}{L} x$$

$$\text{or, } \frac{d^2 x}{dt^2} = - \frac{B^2 l^2}{mL} x$$

$\therefore$ Motion of the rod is SHM.

$$\omega = \frac{Bl}{\sqrt{mL}}$$

We can express the displacement as

$$x = A \sin(\omega t + \delta)$$

At $t = 0$, $x = 0$ and the rod was moving in positive direction. Hence, $\delta = 0$.

$$\therefore x = A \sin(\omega t)$$

To find A we proceed as follows.

$$v = \frac{dx}{dt} = A\omega \cos(\omega t)$$

$$\text{At } t = 0, v = v_0$$

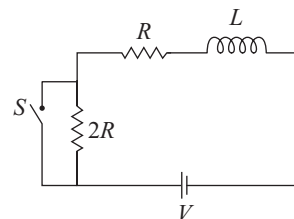
$$\therefore A\omega = v_0 \Rightarrow A = \frac{v_0}{\omega}$$

$$\therefore x = \frac{v_0}{\omega} \sin(\omega t)$$

$$\text{where } \omega = \frac{Bl}{\sqrt{mL}}$$

Example 48 Sudden change in resistance of the circuit

The circuit shown is in steady state. The switch (S) is closed at $t = 0$. Find the current in the circuit at time t .



Solution

Concepts

- Closing the switch changes the resistance of the circuit from $3R$ to R . The current begins to rise further.
- We will set up a differential equation in i and t using the Kirchhoff's loop law and integrate it to find i in terms of t .

With the switch (S) open, the current in the circuit is

$$i_0 = \frac{V}{3R}$$

At $t = 0$ the switch is closed and the resistance of the circuit becomes R . Now, the final steady state current will be $I = \frac{V}{R}$. We have to find an expression showing the rise of current from $\frac{V}{3R}$ to $\frac{V}{R}$.

Consider the circuit at time t .

$$L \frac{di}{dt} + iR = V$$

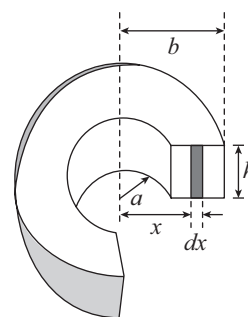
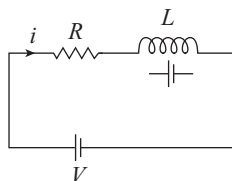
$$\text{or, } \int_{\frac{V}{3R}}^i \frac{di}{V - Ri} = \frac{1}{L} \int_0^t dt$$

$$\text{or, } [\ln(V - Ri)]_{\frac{V}{3R}}^i = -\frac{Rt}{L}$$

$$\text{or, } \ln\left(\frac{V - Ri}{2V/3}\right) = -\frac{Rt}{L}$$

$$\text{or, } \frac{3}{2} - \frac{3Ri}{2V} = e^{-\frac{Rt}{L}} \Rightarrow \frac{3Ri}{2V} = \frac{2}{3} - e^{-\frac{Rt}{L}}$$

$$\text{or, } i = \frac{2V}{3R} \left(\frac{3}{2} - e^{-\frac{Rt}{L}} \right)$$



When a current is given to the toroid, it produces a field that is directed perpendicular to the rectangular cross section.

Ampere's law gives as the strength of the field at a radial distance x .

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$\text{or, } B \cdot 2\pi x = \mu_0 Ni$$

$$\text{or, } B = \frac{\mu_0 Ni}{2\pi x}$$

$$\text{Flux through a strip of width } dx \text{ is } d\phi = \frac{\mu_0 Ni}{2\pi x} h dx$$

$$\begin{aligned} \text{Flux through the rectangular section is} &= \frac{\mu_0 N h i}{2\pi} \int_a^b \frac{dx}{x} \\ &= \frac{\mu_0 N h i}{2\pi} \ln\left(\frac{b}{a}\right) \end{aligned}$$

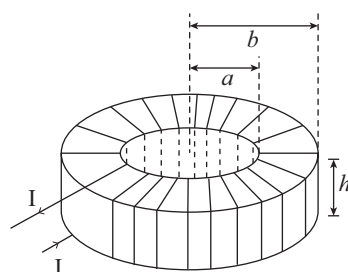
Total flux linked to N turns of the coil is

$$\phi = \left[\frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{b}{a}\right) \right] i$$

$$\therefore L = \frac{\mu_0 N^2 h}{2\pi} \ln\left(\frac{b}{a}\right)$$

Example 49 Self inductance of a toroid

A toroid has a rectangular cross section with height h and inner and outer radii a and b , respectively. It has a large number of turns N spaced uniformly. Find the self inductance of the solenoid.



Solution

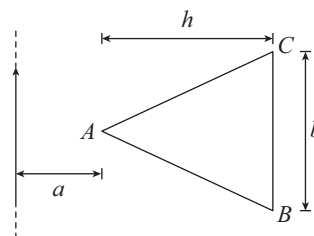
Concepts

- We will find the magnetic field at a radial distance r ($a < r < b$) using the Ampere's law.
- We need to perform an integration to find the magnetic flux through a rectangular section of the toroid. Multiplying this with N gives the total flux.
- Comparing the obtained expression with $\phi = Li$ gives the self inductance.

The diagram shows a rectangular cross section of the toroid. When a current i is given to it, it produces a magnetic field represented by circular field lines inside the toroid.

Example 50 Mutual Inductance of a long wire and a triangular coil

A triangular conducting loop has two sides AB and AC of equal lengths. In the plane of the loop, at a distance a from its vertex A , there is a long straight current carrying wire. Find the mutual inductance of the triangular loop and the straight wire. Base length and height of the triangle are b and h .



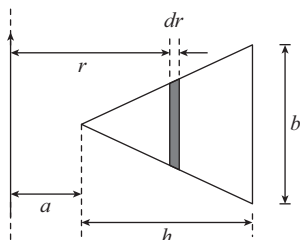
Solution

Concepts

- We can think of the long straight wire as an arm of a close loop of very large size. So there are two closed loops about which we are talking.

- (ii) Assume a current I in the large loop. Only the current along the straight wire shown in the figure will produce a meaningful magnetic field at the location of the triangle. Other arms of the loop are far away and do not produce any field at the location of the triangle.

Consider current I in the long wire. Field at a distance r from the wire is $B = \frac{\mu_0 I}{2\pi r}$



The length of the strip shown is $\frac{(r-a)b}{h}$.

Area of the strip is $dA = \frac{b}{h}(r-a)dr$

Flux through the strip is $d\phi = B \cdot dA = \frac{\mu_0 Ib}{2\pi h} \left(1 - \frac{a}{r}\right) dr$

Flux through the triangular loop is $\phi = \frac{\mu_0 Ib}{2\pi h} \int_a^{a+h} \left(1 - \frac{a}{r}\right) dr$

$$\text{or, } \phi = \frac{\mu_0 Ib}{2\pi h} \left[h - a \ln \left(1 + \frac{h}{a}\right) \right]$$

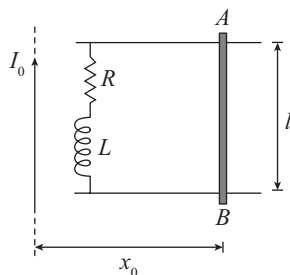
Comparing with $\phi = MI$ gives

$$\therefore M = \frac{\mu_0 b}{2\pi h} \left[h - a \ln \left(1 + \frac{h}{a}\right) \right]$$

Example 51 A metal bar AB can slide on two perfectly conducting parallel rails separated by a distance l . A resistance R and an inductance L are connected to the rails as shown. A long straight wire, carrying a constant current I_0 is placed in the plane of the rails and is perpendicular to them.

The bar AB is held at a distance x_0 from the long wire. At $t = 0$, it is made to slide on the rails away from the wire.

- (a) It is observed that at time $t = T$, the bar AB is at a distance $2x_0$ from the long wire and the resistance R carries a current I_1 . Obtain an expression for the net charge that has flown through R from $t = 0$ to $t = T$.



- (b) The bar is suddenly stopped at time T . The current through resistance R is found to be $\frac{I_1}{4}$ at time $2T$. Find the value of $\frac{L}{R}$ in terms of the other quantities.

Solution

Concepts

- (i) We will assume a current I in the circuit at time t and flux ϕ linked to the loop. Using Kirchhoff's law we can frame an equation involving $\frac{d\phi}{dt}$, RI and $L\frac{dI}{dt}$.
- (ii) Integration of $\int Idt$ will give us the charge flow.
- (iii) After the bar is stopped the circuit is a simple LR circuit with decaying current.

- (a) ϕ is the flux linked with the loop and it is increasing as the bar moves to right. Emf induced in the loop is $\varepsilon = \frac{d\phi}{dt}$ and it drives anticlockwise current. An emf is induced in the inductor which opposes this.

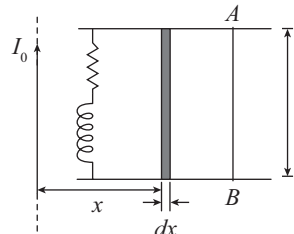
$$\therefore \frac{d\phi}{dt} = RI + L\frac{dI}{dt}$$

$$\therefore d\phi = RIdt + LdI$$

$$\text{or, } \int d\phi = R \int Idt + L \int dI$$

$$\text{or, } \Delta\phi = R\Delta q + LI_1$$

$$\text{or, } \Delta q = \frac{\Delta\phi}{R} - \frac{LI_1}{R}$$



(i)

$\Delta\phi$ can be calculated by summing the fluxes over thin stripes like the one shown in figure

$$\Delta\phi = \int_{x_0}^{2x_0} \frac{\mu_0 I_0}{2\pi x} l dx = \frac{\mu_0 I_0 l}{2\pi} \ln 2$$

$$\therefore \text{From (i) } \Delta q = \frac{\mu_0 I_0 l (\ln 2)}{2\pi R} - \frac{LI_1}{R}$$

- (b) The current decays with a time constant $\tau = \frac{L}{R}$
- $$I = I_0 e^{-\frac{\Delta t}{\tau}} \quad [\Delta t = 2T - T = T]$$

$$\therefore \frac{I_1}{4} = I_0 e^{-\frac{T}{\tau}}$$

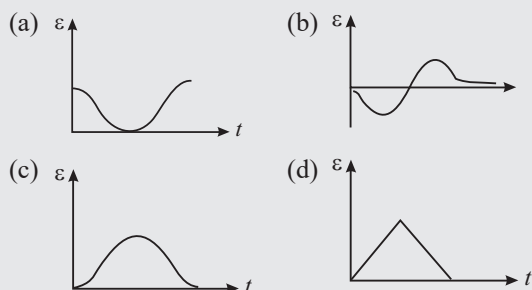
$$\text{or, } \frac{T}{\tau} = \ln 4 \quad \Rightarrow \quad \tau = \frac{T}{\ln(4)}$$

$$\therefore \frac{L}{R} = \frac{T}{\ln(4)}$$

Worksheet 1

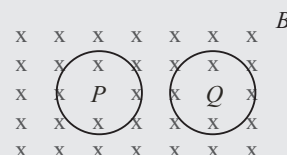
- The flux linked to a coil changes with time according to the equation $\phi = at^2 + bt + c$. Then SI unit of a and b are
 (a) volt and weber/sec (b) volt/sec and volt
 (c) volt.sec and weber (d) weber/sec and volt
- When a magnet is moved towards a coil, the emf induced in the coil does not depend on
 (a) the number of turns of the coil
 (b) the motion of the magnet
 (c) the magnetic moment of the magnet
 (d) the resistance of the coil
- A circular loop of radius R , carrying current I , lies in x - y plane with its centre at origin. The total magnetic flux through x - y plane is
 (a) directly proportional to I
 (b) directly proportional to R
 (c) directly proportional to R^2
 (d) zero
- An infinitely long cylindrical conducting rod is kept along $+z$ direction. A constant magnetic field is also present in $+z$ direction. Then current induced will be
 (a) 0
 (b) along $+z$ direction
 (c) clockwise as seen from $+z$
 (d) anticlockwise as seen from $+z$
- A coil of area 100 cm^2 has 500 turns. Magnetic field of 0.1 weber/m^2 is perpendicular to the coil. The field is reduced to zero in 0.1 second. The induced e.m.f. in the coil is
 (a) 1 V (b) 5 V
 (c) 50 V (d) 0

- A short bar magnet is moved with constant velocity along the axis of a short coil. The magnet enters into the coil and then leaves it. The variation of induced emf (ϵ) with time (t) is best represented as



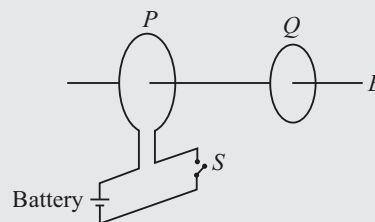
- Two different loops are concentric and lie in the same plane. The current in the outer loop is clockwise and increasing with time. The induced current in the inner loop is-
 (a) clockwise
 (b) 0
 (c) counter clockwise
 (d) in a direction that depends on the ratio of the loop radii

- P and Q are two circular thin coils of same radius and subjected to the same rate of change of flux. If coil P is made of copper and Q is made of iron, then the wrong statement is-



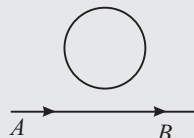
- emf induced in the two coils is the same
 - the induced current in P is more than that in Q
 - the induced current in P and Q are in the same direction
 - the induced currents are the same in both the coils
- Two identical circular loops of metal wire are lying on a table without touching each other. Loop-A carries a current which increases with time. In response, the loop-B
 (a) remains stationary
 (b) is attracted by the loop-A
 (c) is repelled by the loop-A
 (d) rotates about its CM, with CM fixed

- As shown in the figure, P and Q are two coaxial conducting loops separated by some distance. When the switch S is closed, a clockwise current I_P flows in P (as seen from E) and an induced current I_{Q1} flows in Q . The switch remains closed for a long time. When S is opened, a current I_{Q2} flows in Q . Then the directions of I_{Q1} and I_{Q2} (as seen from E) are:



- (a) clockwise and anti-clockwise respectively
- (b) both clockwise
- (c) both anti-clockwise
- (d) anti-clockwise and clockwise respectively

11. In the arrangement shown in given figure current from A to B is increasing in magnitude. Induced current in the loop will

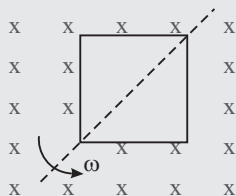


- (a) have clockwise direction
- (b) have anticlockwise direction
- (c) be zero
- (d) oscillate between clockwise and anticlockwise

12. A circular coil of n turns and radius r is placed in uniform magnetic field B . Initially the plane of the coil is perpendicular to the field. Now the coil is rotated by 90° . If its resistance is R then quantity of charge passing through the coil is-

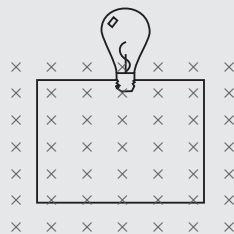
- (a) zero
- (b) $\frac{\pi n^2 r B}{R}$
- (c) $\frac{\pi n r^2 B}{R}$
- (d) $\frac{2\pi n r^2 B}{R}$

13. A square loop of edge a having n turns is rotated with a uniform angular velocity ω about one of its diagonals which is kept fixed in a horizontal position (figure). A uniform magnetic field B exists in the vertical direction. The maximum emf induced in the coil will be



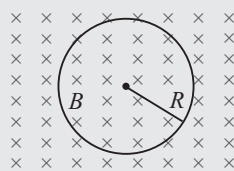
- (a) $NBa^2\omega$
- (b) $2NBa^2\omega$
- (c) NBa^2
- (d) None

14. A square wire loop of 10 cm side lies at right angles to a uniform magnetic field of 20T. A 10 V light bulb is in a series with the loop as shown in the fig. The magnetic field is decreasing steadily to zero over a time interval Δt . The bulb will shine with full brightness if Δt is equal to



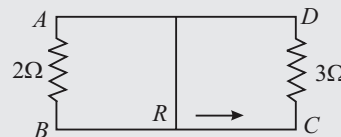
- (a) 20 ms
- (b) 0.02 ms
- (c) 2 ms
- (d) 0.2 ms

15. a conducting loop of radius R is present in a uniform magnetic field B perpendicular the plane of the ring. If radius R varies as a function of time ' t ', as $R = R_0 + t$. The e.m.f induced in the loop is



- (a) $2\pi(R_0 + t)B$ clockwise
- (b) $\pi(R_0 + t)B$ clockwise
- (c) $2\pi(R_0 + t)B$ anticlockwise
- (d) zero

16. A rectangular loop with a sliding connector of length 10 cm is situated in uniform magnetic field perpendicular to plane of loop. The magnetic induction is 0.1 tesla and resistance of connector (R) is 1 ohm. The sides AB and CD have resistances 2 ohm and 3 ohm respectively. Find the current in the connector during its motion with constant velocity of 1 m/s



- (a) $\frac{1}{110}$ A
- (b) $\frac{1}{220}$ A
- (c) $\frac{1}{55}$ A
- (d) $\frac{1}{440}$ A

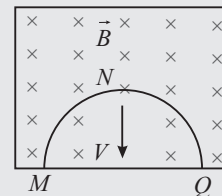
17. An electron is moving in a circular orbit of radius R with an angular acceleration α . At the centre of the orbit is kept a conducting loop of radius r ($r \ll R$). The e.m.f induced in the smaller loop due to the motion of the electron is

- (a) zero
- (b) $\frac{\mu_0 e r^2}{4R} \alpha$
- (c) $\frac{\mu_0 e r^2}{4\pi R} \alpha$
- (d) none of these

18. A metal rod moves at a constant velocity in a direction perpendicular to its length. A constant, uniform magnetic field exists in space in a direction perpendicular to the rod as well as its velocity. Select the correct statement (s) from the following

- (a) The entire rod is at the same electric potential
- (b) There is an electric field in the rod
- (c) The electric potential is highest at the centre of the rod and decreases towards its ends.
- (d) The electric potential is lowest at the centre of the rod, and increases towards its ends.

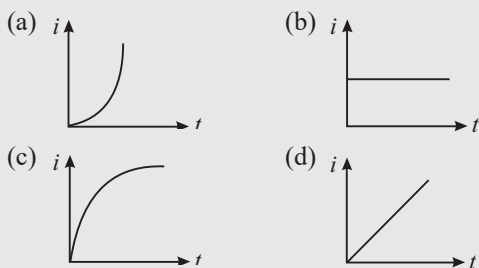
19. A thin semicircular conducting ring of radius R is falling with its plane vertical in horizontal magnetic induction B (Figure). At the position MNQ the speed of the ring is V . The potential difference developed across the ring is-



- (a) Zero
 (b) $BV\pi R^2/2$ and M is at higher potential
 (c) πRBV and Q is at higher potential
 (d) $2RBV$ and Q is at higher potential
20. A long metal bar of 30 cm length is aligned along a north south line and moves eastward at a speed of 10 ms^{-1} . A uniform magnetic field of 4.0 T points vertically downwards. If the south end of the bar has a potential of 0 V, the induced potential at the north end of the bar is

- (a) +12 V
 (b) -12 V
 (c) 0 V
 (d) cannot be determined since there is not closed circuit
21. A thin wire of length 2m is perpendicular to the xy plane. It is moved with velocity $\vec{v} = (2\hat{i} + 3\hat{j} + \hat{k}) \text{ m/s}$ through a region of magnetic induction $B = (\hat{i} + 2\hat{j}) \text{ Wb/m}^2$. Then potential difference induced between the ends of the wire is

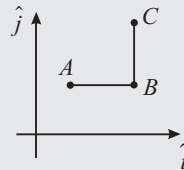
- (a) 2 volts
 (b) 4 volts
 (c) 0 volts
 (d) none of these
22. The figure shows an isosceles triangle wire frame with apex angle equal to $\pi/2$. The frame starts entering into the region of uniform magnetic field B with constant velocity v at $t = 0$. The longest side of the frame is perpendicular to the direction of velocity. If i is the instantaneous current through the frame then choose the alternative showing the correct variation of i with time.



23. A rod moves along a U shaped wire at a constant speed v under the action of a constant force F . The circuit is in a uniform magnetic field perpendicular to the plane. Calculate F if the rate of heat generation in the circuit is H .

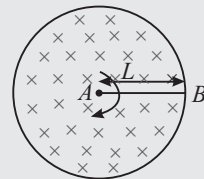
- (a) $F = Hv$
 (b) $F = \frac{H}{v}$
 (c) $F = \frac{H}{v}$
 (d) $F = \sqrt{Hv}$

24. There is a uniform magnetic field B directed in negative z direction. A conductor ABC has length $AB = l_1$, parallel to the x -axis, and length $BC = l_2$, parallel to the y -axis. ABC moves in the xy plane with velocity $v_x\hat{i} + v_y\hat{j}$. The potential difference between A and C is proportional to



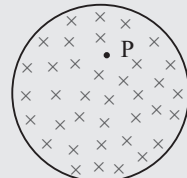
- (a) $v_x l_1 + v_y l_2$
 (b) $v_x l_2 + v_y l_1$
 (c) $v_y l_1 - v_x l_2$
 (d) $v_x l_1 - v_y l_2$
25. A short-circuited coil is placed in a time varying magnetic field. Electrical power is dissipated due to the current induced in the coil. If the number of turns were to be quadrupled and the coil radius halved, the electrical power dissipated would be-
- (a) Halved
 (b) The same
 (c) Doubled
 (d) Quadrupled

26. A copper rod AB of length L , pivoted at one end A , rotates at constant angular velocity ω , at right angles to a uniform magnetic field of induction B . The emf developed between the mid point of the rod and end B is

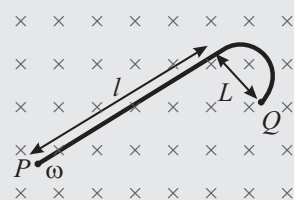


- (a) $\frac{B\omega l^2}{4}$
 (b) $\frac{B\omega l^2}{2}$
 (c) $\frac{3B\omega l^2}{4}$
 (d) $\frac{3B\omega l^2}{8}$

27. Figure shows a uniform magnetic field B confined to a cylindrical volume and is increasing at a constant rate. The instantaneous acceleration experienced by an electron placed at P is

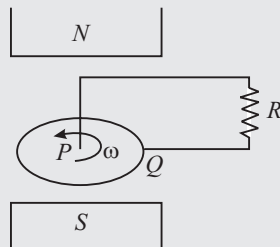


- (a) zero
 (b) towards right
 (c) towards left
 (d) upwards
28. When a 'J' shaped conducting rod is rotating in its own plane with constant angular velocity ω , about its end P , in a uniform magnetic field $\vec{B}$ directed normally into the plane of paper then magnitude of emf induced across it will be

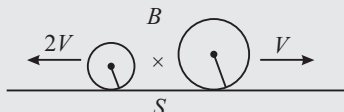


- (a) $B\omega\sqrt{L^2 + l^2}$
 (b) $\frac{1}{2}B\omega L^2$
 (c) $\frac{1}{2}B\omega(L^2 + l^2)$
 (d) $\frac{1}{2}B\omega l^2$

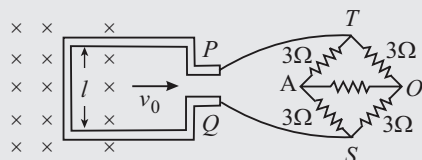
29. A metal disc rotates freely, between the poles of a magnet in the direction indicated. Brushes P and Q make contact with the edge of the disc and the metal axle. Current through R is



- (a) directed downward in the shown figure
 (b) directed upward in the shown figure
 (c) zero, because the emf induced in one side of the disc is opposed by the emf induced in the other side.
 (d) zero, because no emf is induced in the disc
30. Two conducting rings of radii r and $2r$ move in opposite directions with velocities $2v$ and v respectively on a conducting surface S . There is a uniform magnetic field B perpendicular to the plane of the rings. The potential difference between the highest points of the two rings is-



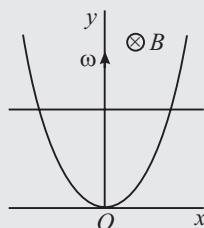
- (a) zero
 (b) $2rvB$
 (c) $4rvB$
 (d) $8rvB$
31. A square metal wire loop of side $l = 10$ cm and resistance 1 ohm is moved with a constant velocity v_0 in a uniform magnetic field induction $B = 2$ Wb/m² as shown in figure. The magnetic field lines are perpendicular to the plane of the loop. The loop is connected to a network of resistance each of value 3 ohm.



The resistance of the lead wires QS and PQ are negligible. What should be the speed of the loop so as to have a steady current of 1 milliampere in the wire PT ?

- (a) 1 cm/s
 (b) 2 cm/s
 (c) 4 cm/s
 (d) None of these

32. A wire bent as a parabola $y = ax^2$ is located in a uniform magnetic field of induction $\vec{B}$, the vector $\vec{B}$ being perpendicular to the xy plane. At the moment $t = 0$ a connector starts sliding in y direction from the parabola apex with a constant acceleration ω



(Figure). Find the emf of electromagnetic induction in the loop thus formed as a function of y .

- (a) $yB\sqrt{\frac{8\omega}{a}}$
 (b) $yB\sqrt{\frac{4\omega}{a}}$
 (c) $yB\sqrt{\frac{2\omega}{a}}$
 (d) None of these

33. A solenoid of inductance 50 mH and resistance 10Ω is connected to battery of 6V. The time elapsed before the current acquires half of its steady-state value will be nearly equal to-

- (a) 3.01 s
 (b) 3.02 s
 (c) 3.03 s
 (d) 3.5 ms

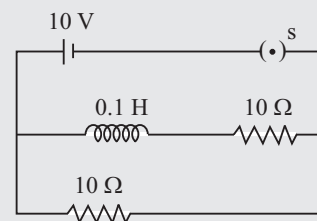
34. An inductor coil stores 32 J of magnetic field energy and dissipates energy as heat at a rate of 320 W, when a current of 4A is passed through it. The time constant of the circuit when this coil is joined across an ideal battery will be -

- (a) 0.2 s
 (b) 0.3 s
 (c) 0.4 s
 (d) 0.5 s

35. For L - R circuit, the time constant is equal to

- (a) twice the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
 (b) ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
 (c) half the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance
 (d) square of the ratio of the energy stored in the magnetic field to the rate of dissipation of energy in the resistance

36. In the adjoining circuit, initially the switch S is open. The switch ' S ' is closed at $t = 0$. The difference between the maximum and minimum current that can flow through the cell is



- (a) 2 A
 (b) 3 A
 (c) 1 A
 (d) nothing can be concluded

37. A current of 2A is increasing at a rate of 4 A/s through a coil of inductance 2H. The energy stored in the inductor per unit time is

(a) 2 J/s (b) 1 J/s
(c) 16 J/s (d) 4 J/s

38. The current in a coil of self-inductance 2.0 henry is increasing according to $I = 2 \sin t^2$ A. Find the amount of energy spent during the period when the current changes from 0 to 2 A.

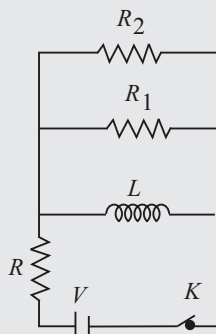
(a) 2J (b) 3J
(c) 4J (d) None of these

39. A solenoid having resistance 50Ω and inductance 80 H is connected to a 200 V battery. How long will it take for the magnetic energy to reach 1/4 of its maximum value?

(a) 1.1s (b) 2.07s
(c) 4s (d) None of these

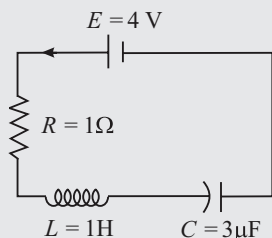
40. In the circuit shown, initially switch K is closed for long time. At time $t = 0$, switch K is opened. The total heat produced in resistor R_1 after opening the switch K is-

(a) $\frac{1}{2} \frac{LV^2}{R^2}$
(b) $\frac{1}{2} \left(\frac{R_1}{R_1 + R_2} \right) \frac{LV^2}{R^2}$
(c) $\frac{1}{2} \left(\frac{R_2}{R_1 + R_2} \right) \frac{LV^2}{R^2}$
(d) 0



41. The current in the given circuit is increasing at a rate 4 A/s. The charge on the capacitor at an instant when the current in the circuit is 2 A will be:

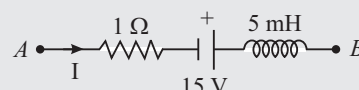
(a) $4\mu\text{C}$
(b) $5\mu\text{C}$
(c) $6\mu\text{C}$
(d) none of these



42. A long solenoid of N turns has a self inductance L and area of cross section A . When a current i flows through the solenoid, the magnetic field inside it has magnitude B . The current i is equal to:

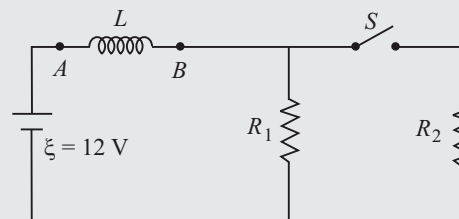
(a) BAN/L (b) $BANL$
(c) BN/AL (d) B/ANL

43. The network shown in the figure is part of a circuit. If at a certain instant, the current I is 5A and it is decreasing at a rate of 10^3 As^{-1} then $V_B - V_A$ equals



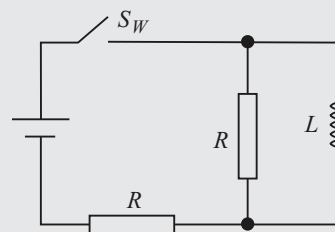
(a) 20 V (b) 15 V
(c) 10 V (d) 5 V

44. In the circuit shown, the switch S has been kept closed for a long time. The instant after the switch is opened, what is the voltage across the inductor (V_L) and which labeled point (A or B) of the inductor is at a higher potential? Take $R_1 = 4.0 \Omega$, $R_2 = 8.0 \Omega$, and $L = 2.5 \text{ H}$.



(a) $V_L = 12 \text{ V}$; Point A is at higher potential
(b) $V_L = 12 \text{ V}$; Point B is at higher potential
(c) $V_L = 6\text{V}$; Point A is at higher potential
(d) $V_L = 6\text{V}$; Point B is at higher potential

45. Find the time dependence of the current flowing through the inductance L of the circuit shown in figure. After the switch S_w is shorted at the moment $t = 0$.



(a) $\frac{\xi}{R} \left(1 - e^{-\frac{tR}{L}} \right)$ (b) $\frac{\xi}{R} \left(1 + e^{-\frac{tR}{L}} \right)$
(c) $\frac{\xi}{R} \left(1 - e^{-\frac{tR}{2L}} \right)$ (d) None of these

46. A small square loop of side l is placed inside a large square loop of side L ($L \gg l$). The loops are co-planar and their centres coincide. The mutual inductance of the system is proportional to:

(a) l/L (b) l^2/L
(c) L/l (d) L^2/l

47. In a car spark-coil, when the current in the primary is reduced from 4.0 A to zero in $10\mu\text{s}$, an emf of 40,000 V is induced in the secondary. Find the mutual

inductance M of the primary and the secondary windings of the spark coil.

- (a) 0.2 H (b) 0.1 H
(c) 0.3 H (d) None

48. Two circular coils can be arranged in any of the three situations shown in the figure. Their mutual inductance will be:



- (a) maximum in situation (a)
(b) maximum in situation (b)
(c) maximum in situation (c)
(d) the same in all situations

49. A coil of wire having inductance and resistance has a conducting ring placed coaxially within it. The coil is connected to a battery at time $t = 0$, so that a time dependent current $I_1(t)$ starts flowing through the coil.

If $I_2(t)$ is current induced in the ring, and $B(t)$ is the magnetic field at the axis of the coil due to $I_1(t)$, then as a function of time ($t > 0$), the product $I_2(t)B(t)$

- (a) increases with time
(b) decreases with time
(c) does not vary with time
(d) passes through a maximum

50. An ideal inductor (having zero initial current), a resistor and an ideal battery are connected in series at time $t = 0$. At any time t , the resistor dissipates energy at the rate P_R and the inductor stores energy at the rate P_L .

- (a) $P_R < P_L$ for all times t .
(b) For certain initial period of time $P_R < P_L$
(c) In the beginning $P_R > P_L$
(d) Both P_R and P_L increase continuously

Worksheet 2

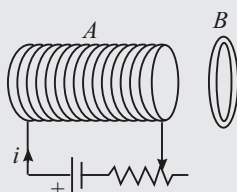
1. An aluminium ring B faces an electromagnet A . The current i through A can be altered.

(a) If i increases, A will repel B .

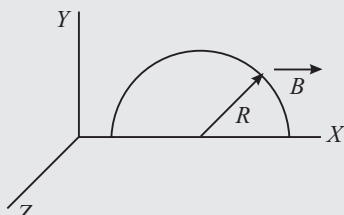
(b) If i increases, A will attract B .

(c) If i decreases, A will attract B .

(d) Repulsion or attraction depends on the sense of current in A



2. A semicircular conducting ring of radius R is placed in the xy plane, as shown in the figure. A uniform magnetic field is set up along the x -axis. No net emf, will be induced between the two ends of the ring if



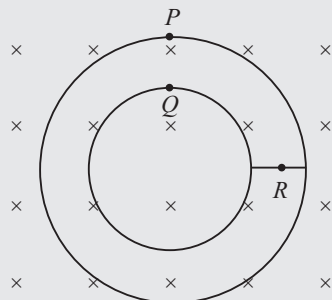
(a) it moves along the x -axis

(b) it moves along the y -axis

(c) it moves along the z -axis

(d) it remains stationary

3. Figure shows plane figure made of a conducting wire located in a magnetic field along the inward normal to the plane of the figure. The magnetic field starts diminishing. Then the induced current



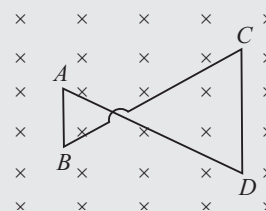
(a) at point P is clockwise

(b) at point Q is anticlockwise

(c) at point Q is clockwise

(d) at point R is zero

4. A conducting wire frame is placed in a magnetic field which is directed into the paper. The magnetic field is increasing at a constant rate. The directions of induced currents in wires AB and CD are



(a) B to A and D to C

(b) A to B and C to D

(c) A to B and D to C

(d) B to A and C to D

5. A bar magnet is moved along the axis of a copper ring placed far away from the magnet. Looking from the side of the magnet, an anticlockwise current is found to be induced in the ring. Which of the following may be true?

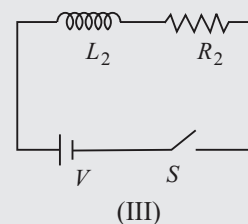
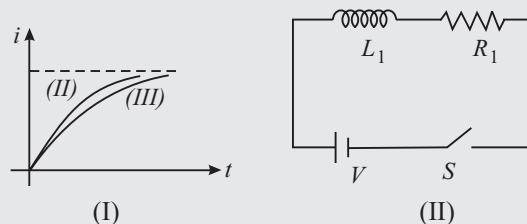
(a) The south pole faces the ring and the magnet moves towards it.

(b) The north pole faces the ring and the magnet moves towards it.

(c) The south pole faces the ring and the magnet moves away from it.

(d) The north pole faces the ring and the magnet moves away from it.

6. Current growth in two L - R circuits (II) and (III) are as shown in the graphs of figure (I). Let L_1 , L_2 , R_1 and R_2 be the corresponding values of the inductances and resistances in the two circuits. Then



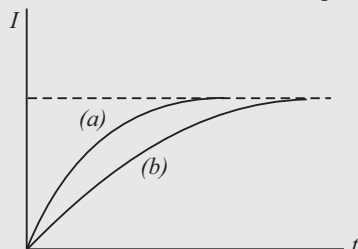
(a) $R_1 > R_2$

(b) $R_1 = R_2$

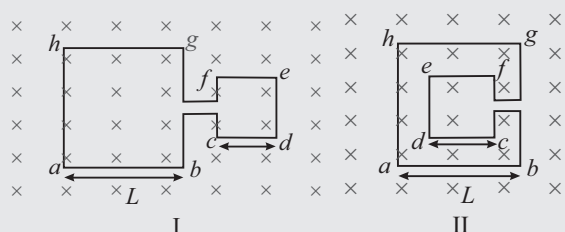
(c) $L_1 > L_2$

(d) $L_1 < L_2$

7. A circuit consisting of a constant e.m.f. 'E', a self induction 'L' and a resistance 'R' is closed at $t = 0$. The relation between the current I in the circuit and time t is as shown by curve 'a' in the figure. When one or more of parameters E , R and L are changed, the curve 'b' is obtained. The steady state current is same in both the cases. Then it is possible that:

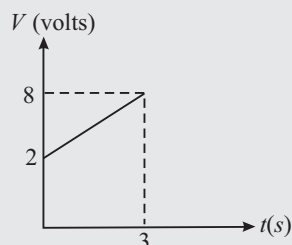


- (a) E & R are kept constant and L is increased.
 (b) E & R are kept constant and L is decreased
 (c) E & R are both halved and L is kept constant
 (d) E & L are kept constant and R is decreased
8. The adjoining figure shows two different arrangements in which two square wire frames of same resistance are placed in a uniform constantly decreasing magnetic field B .



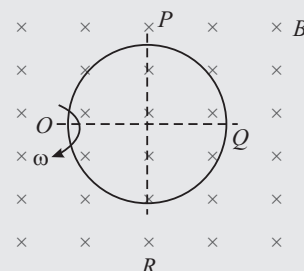
- (a) The value of magnetic flux can be given by -
 Case I: $\Phi = (L^2 + l^2)B$; Case II: $\Phi = (L^2 - l^2)B$
 (b) The direction of induced current in the case I is from b to a and from d to c
 (c) The direction of induced current in the case II is from b to a and from f to e
 (d) If I_1 and I_2 are the magnitudes of induced current in the cases I and II, respectively, then $I_1 > I_2$

9. A circuit element is placed in a closed box. At time $t = 0$, constant current generator supplying a current of 1 amp, is connected across the box. Potential difference across the box varies according to graph shown in figure. The element in the box is:



- (a) resistance of 2Ω (b) battery of emf 6V
 (c) inductance of 2H (d) capacitance of 0.5F

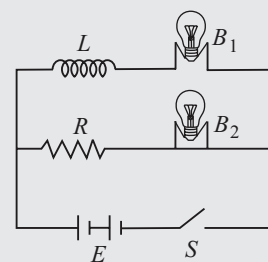
10. A conducting ring of radius a is rotated about a point O on its periphery as shown in the figure in a plane perpendicular to uniform magnetic field B which exists everywhere. The angular velocity is ω . Which of the followings are true regarding the potentials due to induced emf



- (a) $V_P = V_R > V_O$ (b) $V_Q - V_P = V_P - V_O$
 (c) $V_Q - V_O = 2B\omega a^2$ (d) $V_P - V_R = 2B\omega a^2$
11. Two different coils have self inductance 8mH and 2mH. The current in the two coils are increased at the same constant rate. The current in the second coil is also increased at the same constant rate. At a certain instant of time, the power given to the two coils is the same. At that time the current, the induced voltage and the energy stored in the first coil are I_1 , V_1 and W_1 respectively. Corresponding values for the second coil at the same instant are I_2 , V_2 and W_2 respectively. Then:

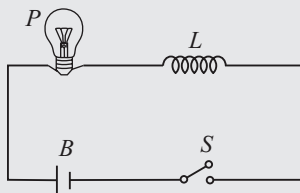
- (a) $\frac{I_1}{I_2} = \frac{1}{4}$ (b) $\frac{I_1}{I_2} = 4$
 (c) $\frac{W_2}{W_1} = 4$ (d) $\frac{V_2}{V_1} = \frac{1}{4}$

12. An inductor L , a resistance R and two identical bulbs B_1 and B_2 are connected to a battery through a switch S as shown in the figure. The resistance of coil having inductance L is also R . Which of the following statement gives the correct description of the happenings when the switch S is closed?

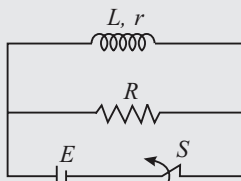


- (a) The bulb B_2 lights up earlier than B_1 and finally both the bulbs shine equally bright.
 (b) B_1 light up earlier and finally both the bulbs acquire equal brightness.
 (c) B_2 lights up earlier and finally B_1 shines brighter than B_2 .
 (d) B_1 and B_2 light up together with equal brightness all the time.

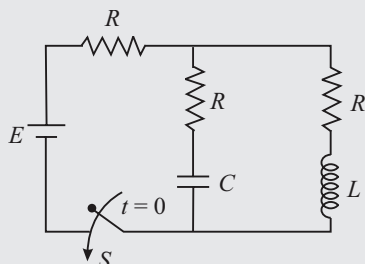
13. In figure, a lamp P is in series with an inductor. In two different experiments observations are recorded with an air core inductor and then with an iron core inserted inside the same inductor. When the switch S is closed



- The rise of current is slower if the inductor has an iron core.
 - The rise of current is faster if the inductor has an iron core.
 - The time constant is higher if the inductor has an iron core
 - The lamp will not glow for an iron core inductor
14. A solenoid of inductance L and resistance r is connected in parallel to a resistance R and a battery of emf E . Initially the switch is closed for long time and at $t = 0$, switch S is opened. Then



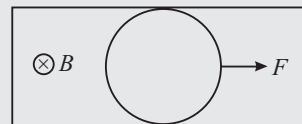
- Current through solenoid at any time t , after opening the switch is $\frac{E}{r} e^{-\frac{(R+r)}{L}t}$
 - Induced emf across solenoid at time $t = 0$ is $\frac{E(R+r)}{r}$
 - Amount of heat generated in solenoid is $\frac{E^2 L}{2r(r+R)}$
 - The direction of current in R changes immediately after operation of the switch
15. In the circuit shown in figure, switch S is closed at $t = 0$. Then:



- after a long time interval potential difference across capacitor and inductor will be equal.
- after a long time interval charge on capacitor will be $E C$.

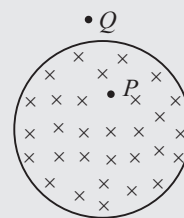
- after a long time interval current in the inductor will be E/R .
- after a long time interval current through battery will be same as the initial current through it.

16. Two long parallel conducting horizontal rails are connected by a conducting wire at one end. A uniform magnetic field B (directed vertically downwards) exists in the region of space. A light uniform ring of diameter d , which is practically equal to separation between the rails, is placed over the rails as shown in figure. The resistance of ring is λ per unit length.



- There is no current anywhere.
 - The force required to pull the ring with uniform velocity v is zero.
 - The force needed to pull the ring with uniform velocity v is $\frac{4B^2 v d}{\pi \lambda}$
 - The current in the conducting wire at the end of the rails is $\frac{4Bv}{\pi \lambda}$
17. Which of the following statements are not true
- Lenz's law is in accordance with the law of conservation of energy
 - According to the Lenz's law the induced current must produce a magnetic field that decreases the already existing field.
 - According to the Lenz's law if a magnet is pushed towards a stationary coil not connected to any source, it will face repulsion from the coil.
 - Lenz's law is not obeyed in phenomenon of mutual inductance.

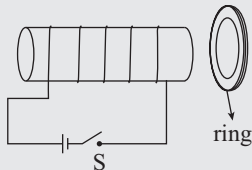
18. The diagram shows the cross section of an ideal solenoid and the magnetic field existing inside it. The current in the solenoid is increased.



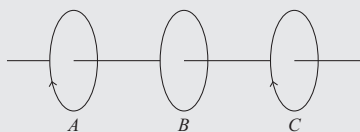
- There is an electric field at Q but no magnetic field
- There is a magnetic field at P but no electric field
- The electric field at P and Q may be of same magnitude
- A charge released at Q will accelerate.

Worksheet 3

1. When the switch S in the shown figure is closed a current flows in the solenoid. The direction of current is represented by the curling right hand fingers when you try to hold the solenoid in your right hand with the thumb pointing to the left. What is the direction of induced current in the ring

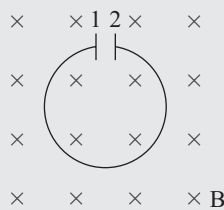


- (a) just after the switch is closed?
(b) a long time after the switch is closed?
(c) just after the switch is opened after remaining closed for a long time?
2. Three identical coils A , B and C are kept with their planes parallel and axes coinciding. Coils A and C carry equal currents as shown. Separation between A and B is same as the separation between B and C . The coil B is held fixed



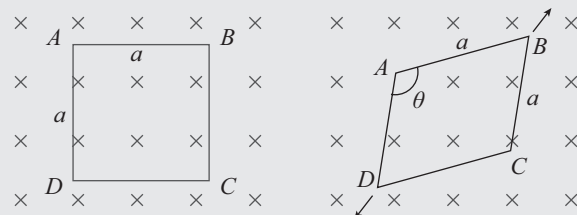
- (a) Coils A and C are moved with same speed towards the coil B . Is there any induced emf in B . If yes, What is the direction of induced current in B when seen from the side of coil A ?
(b) Coil C is held fixed and A is pushed towards B . Is there an induced emf in B .

3. A circular ring of radius r has a capacitor of capacitance C . It is placed in a uniform magnetic field (B) directed perpendicular to the figure. The resistance of the ring is R . The magnetic field changes at a rate $\frac{dB}{dt} = -k$. When k is



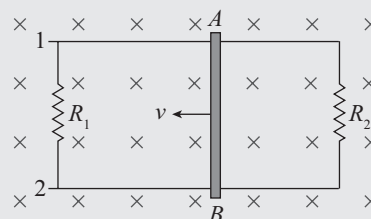
a positive constant. Find the maximum charge on the capacitor. Which plate is positive 1 or 2? Neglect the self inductance of loop.

4. A square loop of side length a is kept in a uniform magnetic field (B). The field is perpendicular to the plane of the loop. The corners B and D of the loop are pulled outward, as shown, and the loop changes. Its shape take place in a time interval Δt . Resistance of the loop is R .



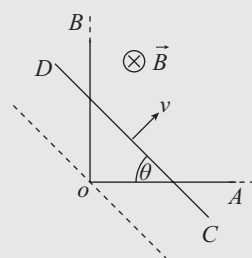
- (a) Find the average induced current in the loop.
(b) What is the direction of the induced current.

5. Two parallel conducting rails, 1 and 2, are at separation l . Two resistance R_1 and R_2 are connected to the rails as shown. A sliding bar AB is made to slide with a constant velocity v on the rails. A uniform magnetic field B exists perpendicular to the plane of the figure



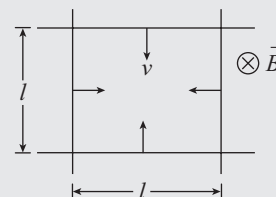
- (a) Find the power dissipated in the circuit.
(b) Find the force applied by the external agent to keep the bar AB moving with constant velocity.
(c) Will the power dissipated in the circuit increase or decrease if the resistor R gets burnt?

6. OA and OB are two perpendicular conducting rods. Another conducting rod CD slides on the two perpendicular rods with a constant velocity v , keeping angle θ fixed at 45° . Initial position of the rod CD is shown in dotted line.



Resistance per unit length is λ for each of the three rods. A uniform magnetic field B exists perpendicular to the plane of the figure. Find the induced current in the sliding rod as a function of time.

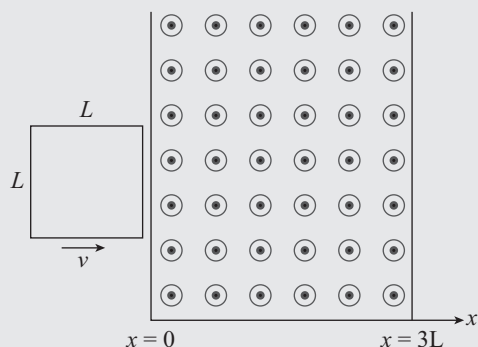
7. The four rods shown in the figure can slide over one another. At any instant they form a square. Each of them is moving with velocity v perpendicular to itself. Resistance of each rod is λ per unit length. At $t = 0$



the square formed by the rods has a side length l . A uniform magnetic field B exists perpendicular to the plane of the rods. Find the following as a function of time

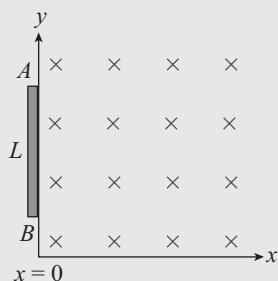
- emf induced in the square.
- Current induced in the square with direction.
- Force applied by the external agent on each rod to keep it moving uniformly.

8. Uniform magnetic field $\vec{B} = B_0 \hat{k}$ exists in the region $0 < x < 3L$. A square loop of side length L lies in the xy plane. The loop is moving with constant velocity in x direction. At $t = 0$ the front side of the loop is at $x = 0$. Plot the variation of the following quantities with time



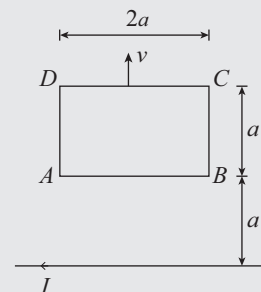
- Flux linked with the loop.
- Emf induced in the loop (Take anticlockwise as positive).
- External force needed to keep the loop moving with constant velocity.

9. In the region $x > 0$, there exists a magnetic field given by $\vec{B} = B_0(1 + ax^2)\hat{k}$. Here B_0 and a are positive constants. A straight conducting bar AB has length L and is lying on the y axis. It begins to move at time $t = 0$ with a constant acceleration $a\hat{i}$. Find the induced emf in the bar as a function of time.

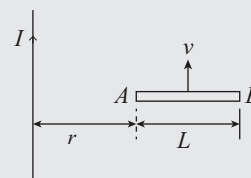


10. A rectangular loop $ABCD$ is moving in the magnetic field of a straight long wire carrying a current I . At the instant shown, the sides AB and CD are parallel to the long wire and the loop is moving with velocity v perpendicular to the side AB . Find the current in

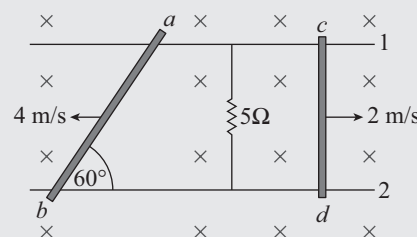
the loop at the instant shown. The resistance of the loop is R .



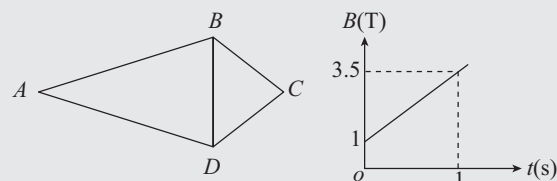
11. A long straight wire carries current I . A conducting bar AB has length L and is held perpendicular to the long wire as shown in the figure. The bar is moved with a velocity v parallel to the long wire. Find the emf induced in the conducting bar.



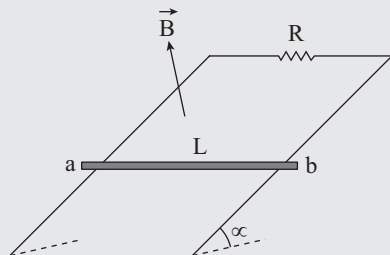
12. Two parallel rails, 1 and 2, have negligible resistance and are $l = 10$ cm apart. They are connected by a 5Ω resistor. Two metal rods ab and cd can slide on the rails. The rods have resistance 10Ω and 15Ω , respectively. The rods are moved with constant velocities of 4 m/s and 2 m/s parallel to the rails, as shown. A uniform magnetic field $B = 10$ mT exists in the entire space perpendicular to the figure. Find the current in the 5Ω resistor.



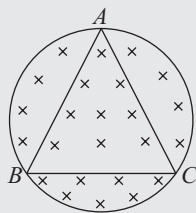
13. A uniform wire of resistance per unit length $2\Omega\text{m}^{-1}$ is used to make a two loop circuit as shown. $AB = AD = 2$ m, $BC = BD = CD = 1$ m. A uniform magnetic field exists perpendicular to the plane of the figure and its magnitude changes with the time as shown in the graph. Find the current in branch BD .



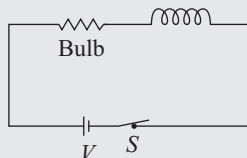
14. Two parallel, conducting rails are held inclined at an angle α to the horizontal at a separation L . The upper ends of the rails are connected using a resistance R . A conducting rod (ab) of mass m is free to slide on the rails. There is a uniform magnetic field perpendicular to the plane of the two rods. The rod is released and it begins to slide on the rails.



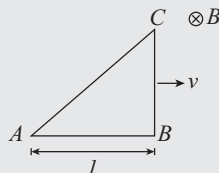
- What is the direction of induced current in the rod ab ?
 - Find the terminal speed of the rod.
 - What is the value of current in the rod when the terminal speed has been reached?
 - After the terminal speed has been reached, at what rate is energy dissipated in the resistance? At what rate is the gravitational potential energy of the rod decreasing?
15. Answer the last question if the direction of the magnetic field is vertically upward.
16. A circular region of radius r has a uniform magnetic field (B) directed perpendicular to the plane of the figure. ABC is an equilateral triangle made of a wire having resistance R . The magnetic field is increased at a constant rate $\frac{dB}{dt} = \alpha$. Find the reading of an ideal voltmeter connected across B and C . The wire ABC is uniform



17. An inductor is connected to a bulb in series and the combination is connected to a battery through a switch. The bulb is glowing at its rated power. A student fears that if the switch is suddenly opened a large emf will be induced in the inductor and the bulb will fuse. Is he right?

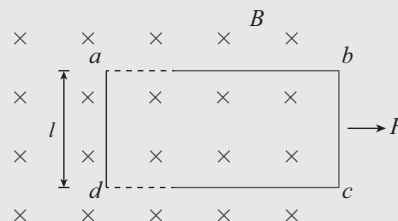


18. ABC is a wire frame in the shape of right angled triangle. Side lengths AB and BC are equal to l . Resistance of the three arms are



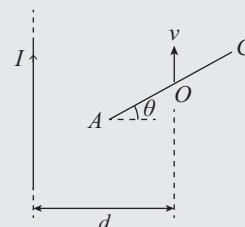
$R_{AB} = R_{BC} = R$ and $R_{AC} = 2R$. The wire frame translates with velocity v in a region having uniform magnetic field perpendicular to the plane of the figure, in a direction parallel to AB . Find the reading of an ideal voltmeter connected between A and B .

19. $abcd$ is conducting rectangular frame. It has a sides ad and bc each of length l and the other two sides are very long. The resistance of the frame is R and its mass is M . Initially the frame lies completely inside a uniform magnetic field B , with side bc at the edge of the field region. At time $t = 0$ a constant force F begins to pull the frame out of the magnetic field region.

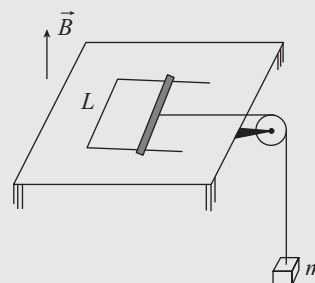


- Find the acceleration of the frame when its velocity is v .
- Show that after some time the frame will move with a constant velocity (v_0) till the complete frame moves out of the field region.
- Write the velocity of the frame as a function of time as long as some of its part is inside the field region.

20. A long straight wire carries a current I . A conducting rod AC is placed near the wire as shown. Length of the rod AC is $2a$ and its centre O is at distance d from the long wire. The rod AC makes an angle θ with the normal to the wire and is made to translate with a constant velocity v parallel to the wire. Find the emf induced in the rod AC .



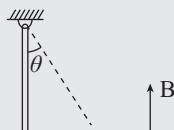
21. A pair of parallel horizontal conducting rails of negligible resistance shorted at one end is fixed on a table. The distance between the rails is L . A conducting massless rod of resistance R can slide on the rails frictionlessly. The rod is tied to a massless string which passes over a pulley fixed to the edge of the table. A mass m tied to the other end



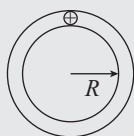
of the string hangs vertically. A constant magnetic field B exists perpendicular to the table. If the system is released from rest, calculate

- The terminal velocity attained by the rod and
- The acceleration of the mass at the instant when the velocity of the rod is half the terminal velocity.

22. A rectangular conducting loop is tied to a non conducting bar. The system oscillates like a pendulum with the angular displacement varying as $\theta = \theta_0 \sin(\omega t)$. When the pendulum is vertical the normal to the surface of the loop points rightward. There is a uniform magnetic field (B) in vertically upward direction. Write the induced emf in the rectangular loop as a function of time. Area of the loop is A .

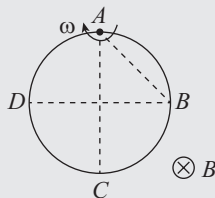


23. A small particle having mass m and charge q is kept inside a horizontal circular groove of radius R . The particle just fits inside the groove and can move frictionlessly inside it. A uniform magnetic field (B) is suddenly switched on inside the circular region of radius R . The field is directed perpendicular to the plane of the groove.



- Find the angular momentum (about the centre) of the charged particle after the field is switched on.
- Find the magnetic dipole moment associated with the circular motion of the charge.

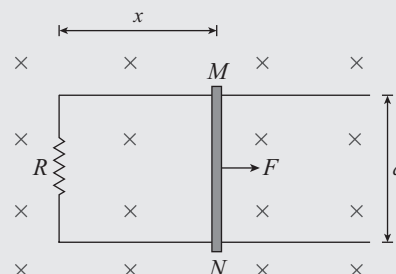
24. A conducting circular ring of radius a is being rotated with angular speed ω about point A on its circumference. The ring rotates in its own plane and there exists a uniform magnetic field (B) perpendicular to the plane of the ring. Find



- The emf induced along the diameter AC .
- The emf induced along the chord AB .

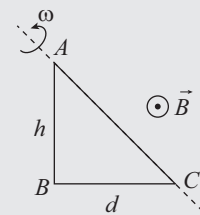
25. An ideal solenoid has radius r and n number of turns on its unit length. The current through the solenoid is increased at a rate b As/s. Find the emf induced along a semicircular arc of radius $\frac{r}{2}$ inside the solenoid. The plane of the arc is perpendicular to the axis of the solenoid and its centre of curvature is on the axis of solenoid.

26. Two long parallel horizontal rails, a distance d apart and each having a resistance λ per unit length, are joined at one end by a resistance R . A perfectly conducting rod MN of mass m is free to slide along the rails without friction. There is a uniform magnetic field B normal to the plane of the figure. A force F is applied on the rod MN to move it such that a constant current (I) flows through R .



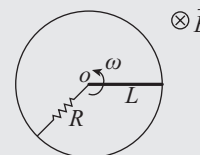
- Find the force F as function of distance x .
- What fraction of the power delivered by F gets dissipated in the resistance of the circuit.

27. A loop is made of a conducting wire. The loop has a shape of a right angled triangle and there are N turns in it. It can rotate about its longest side AC . Initially the plane of the loop is perpendicular to a uniform magnetic field B . The loop begins to rotate with constant angular velocity ω .

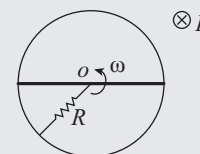


- Plot the variation of induced emf in the loop as a function of time.
- How will your plot change if the loop is rotated about the side AB in place of side AC ?

28. A rod of length L and resistance r rotates about an axis passing through its end O . Its other end slides on a perfectly conducting circular ring. A resistance R is connected between the end O of the rod and the circumference of the ring. The resistance does not interfere with the rotating rod. There is a uniform magnetic field B perpendicular to the plane of the figure and the angular speed of the rod is ω . Find the current in resistance R .



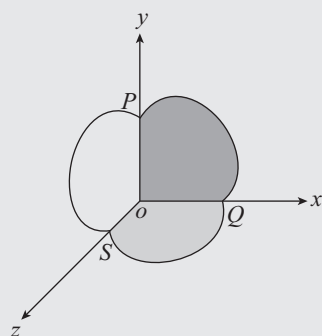
29. In the last question the rod is replaced with another rod of length $2L$ as shown. It rotates about its centre O and its resistance is $2r$. Everything else is same as in the last question. Find current in R .



30. A thermocol vessel contains 0.5 kg of distilled water at 30°C. A metal coil of area $5 \times 10^{-3} \text{ m}^2$, number of turns 100, mass 0.06 kg and resistance 1.6Ω is lying horizontally at the bottom of the vessel. A uniform time-varying magnetic field is set up at $t = 0$ and it is directed perpendicular to the plane of the coil. The field is increased from zero to 0.8 T at a constant rate between 0 and 0.2 s and then decreased to zero at the same rate between 0.2 to 0.4 s. The cycle is repeated 12000 times.

- (a) Draw graphs of the current in the coil and the power dissipated in the coil as a function of time for the first two cycles.
- (b) Assume that no heat is lost to the vessel or the surroundings and determine the final temperature of water. Specific heat of metal = $500 \text{ J kg}^{-1} \text{ K}^{-1}$ and specific heat of water = $4200 \text{ J kg}^{-1} \text{ K}^{-1}$. Neglect inductance of the coil.

31. Three identical wires are bent into semi-circular arcs of radius R . These arcs are connected with each other to form a closed mesh such that one of them lies in xy plane, one in yz plane and the other in xz plane (see figure). In the entire space



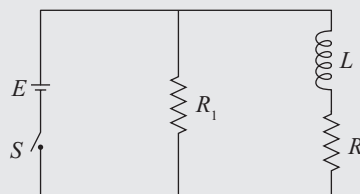
there is a uniform magnetic field $\vec{B} = B_0(\hat{i} + \hat{j})$ with its magnitude increasing at a constant rate of α . Calculate the emf induced in the mesh.

32. An inductor of inductance 2mH is connected across a charged capacitor of capacitance $5 \mu\text{F}$ and the resulting LC circuit is oscillating at its natural frequency. Q is the instantaneous charge on the capacitor and I is the current in the circuit. The maximum value of Q is $200 \mu\text{C}$.

- (a) What is the value of $\left| \frac{dI}{dt} \right|$, when $Q = 100 \mu\text{C}$?
- (b) What is the value of I , when $Q = 100 \mu\text{C}$?
- (c) Find the maximum value of I .
- (d) When I is equal to half its maximum value, what is the value of $|Q|$?

33. A solenoid has an inductance of 10H and a resistance of 2Ω . It is connected to a 10V battery at $t = 0$. How long will it take for the magnetic energy to reach $\frac{1}{4}$ of its maximum value?

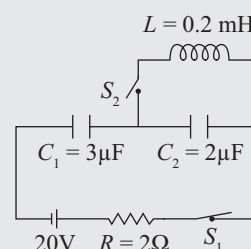
34. An inductor of inductance $L = 400 \text{ mH}$ and resistors of resistances $R_1 = 2 \Omega$ and $R_2 = 2 \Omega$ are connected to a battery of emf $E = 12 \text{ V}$ as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at time $t = 0$.



- (a) What is potential drop across L as a function of time?
- (b) After the steady state is reached, the switch is opened. What is the direction and magnitude of current through R_1 as a function of time?

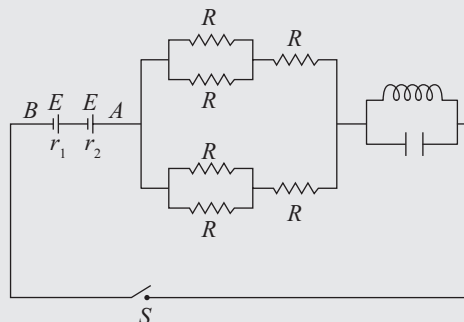
35. The switch S_1 shown in the figure is closed for a long time. At $t = 0$, it is opened and the switch S_2 is closed.

- (a) Write the variation of charge on the capacitance C_2 with time.



- (b) At what time the energy in the inductor becomes one third of that in C_2 for the second time after S_2 is closed?

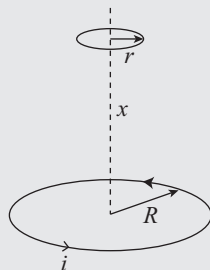
36. In the circuit shown in the figure, A and B are two cells of same emf E but different internal resistances r_1 and r_2 ($r_2 > r_1$), respectively. Find the value of resistance R such that the potential difference across the terminals of cell A is zero a long time after the switch S is closed.



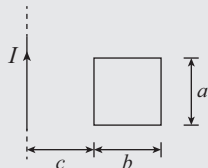
37. A solenoid has inductance L and resistance R . It is connected to a cell at $t = 0$.

- (a) Find the ratio of rate of energy storage in the magnetic field to the rate of heat dissipation, as a function of time.
- (b) At what time value of the above ratio is 1?

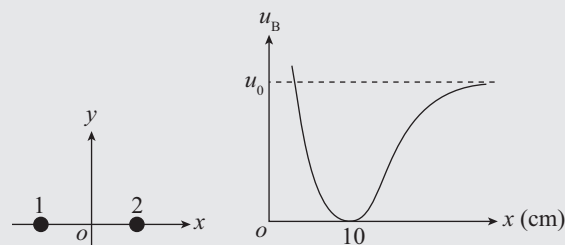
38. The figure shows two parallel and co axial loops. The smaller loop (radius r) is above the larger loop (radius R) at a distance $x \gg R$. The magnetic field due to a current i in the larger loop can be assumed to be a constant over the entire area of the smaller loop.



- (a) Find the mutual inductance of the pair of loops.
- (b) Find the emf induced and the direction of induced current in the smaller loop if x is increased at a constant rate $\frac{dx}{dt} = v$. Express your answer in terms of x .
39. Figure shows a rectangular loop near a long wire calculate their mutual inductance.



40. The first figure shows the cross sections of two long parallel current carrying wires. The ratio $\frac{i_1}{i_2}$ of the current in the wire 1 and wire 2 is $\frac{1}{4}$. Wire 1 is fixed but 2 can be moved along the positive side of the x axis. The second figure shows the variation of magnetic field energy density at the origin (O) with the x coordinate of wire 2. Find i_1 and i_2 . It is given that $u_0 = 2\text{ nJ/m}^3$.



Answer Sheet

Your Turn

1. $\frac{BS}{2}$
2. Zero, Zero
3. 20V, 2A
4. There is no induced current
5. $\varepsilon = 6.28 \text{ mV}$, $i_1 > i_2$
6. None is true
7. No
8. $\frac{BS}{R}$
9. $\sqrt{3} Blv$
10. (a) $\frac{Blv}{R}$ (b) $\frac{B^2 l^2 v}{R}$
11. $\frac{Bvl}{R}$
12. $\frac{1}{2} B\omega R^2$
13. $2B\omega r^2$
14. (a) $4 \times 10^{-4} \text{ V/m}$ (b) $6.25 \times 10^{-4} \text{ V/m}$
15. 6V
16. $\text{M}^1 \text{L}^2 \text{T}^{-2} \text{A}^{-2}$
17. 0.02 H
18. (a) 0 (b) -0.02 V
19. 6.28 mH
20. (a) 1A (b) 4A
21. (a) 0.1 ms (b) 0.0693 ms
22. 0.27A
23. (a) $R_2 > R_1$ (b) $L_2 > L_1$
24. (a) $0.2 \frac{V^2 L}{R}$ (b) $0.4 \frac{V^2}{R}$ (c) $0.23 \frac{V^2}{R}$
25. $(\ln 2) \frac{L}{R}$
26. $\frac{1}{8} \text{ mJ}$
27. (a) $6.28 \times 10^{-4} \text{ s}$ (b) 1A (c) $q = 10^{-4} \cos(10^4 t) \text{ C}$
28. CV_0
29. (a) $\frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$ (b) $\frac{V_0 L_1}{2(L_1 + L_2)}$
32. 0.1 H
33. 0.05V
34. 2M
35. $\frac{\mu_0 a^2}{2^{5/2} \cdot R}$

Worksheet 1

1. (b) 2. (d) 3. (d) 4. (a) 5. (b) 6. (b) 7. (c) 8. (d) 9. (c)
10. (d) 11. (a) 12. (c) 13. (a) 14. (a) 15. (c) 16. (b) 17. (b) 18. (b)
19. (d) 20. (a) 21. (a) 22. (d) 23. (b) 24. (c) 25. (b) 26. (d) 27. (b)
28. (c) 29. (a) 30. (d) 31. (b) 32. (a) 33. (d) 34. (a) 35. (a) 36. (c)
37. (c) 38. (c) 39. (a) 40. (c) 41. (c) 42. (a) 43. (b) 44. (b) 45. (c)
46. (b) 47. (b) 48. (a) 49. (d) 50. (b)

Worksheet 2

1. (a, c) 2. (a, b, c, d) 3. (a, c, d) 4. (a) 5. (b, c) 6. (b, d)
7. (a, c) 8. (a, b, c, d) 9. (d) 10. (a, b, c) 11. (a, c, d) 12. (a)
13. (a, c) 14. (a, b, c, d) 15. (d) 16. (c, d) 17. (a, c) 18. (a, c, d)

Worksheet 3

1. (a) clockwise as seen by an observer located on the side of the solenoid. (b) Zero current
(c) Anticlockwise
2. (a) yes, clockwise (b) yes 3. $\pi r^2 kC$, plate 1 is positive 4. (a). $\frac{Ba^2}{\Delta t R}(1 - \sin^2 \theta)$ (b) Clockwise
5. (a) $B^2 l^2 v^2 \left(\frac{R_1 + R_2}{R_1 R_2} \right)$ (b) $B^2 l^2 v \left(\frac{R_1 + R_2}{R_1 R_2} \right)$ (c) Decreases 6. $\frac{Bv}{(\sqrt{2} + 1)\lambda}$
7. (a) $4Bv(l - 2vt)$ (b) $\frac{Bv}{\lambda}$ (c) $\frac{B^2 v}{\lambda}(l - 2vt)$ 9. $B_0 \alpha L \left(1 + \frac{a\alpha^2 t^4}{4} \right) t$
10. $\frac{\mu_0 I v}{2\pi R}$ 11. $\frac{\mu_0 I v}{2\pi} \ln \left(1 + \frac{L}{r} \right)$ 12. $\frac{8}{55} \text{ mA}$ 13. 0.02 A
14. (a) a to b (b) $\frac{mgR \sin \alpha}{B^2 L^2}$ (c) $\frac{mg \sin \alpha}{BL}$ (d) $\frac{m^2 g^2 R \sin^2 \alpha}{B^2 L^2}$
15. (a) a to b (b) $\frac{mgR \tan \alpha}{B^2 L^2}$ (c). $\frac{mg \tan \alpha}{BL}$ (d) $\frac{m^2 g^2 \tan^2 \alpha \cdot R}{B^2 L^2}$
16. Zero 17. No 18. Zero
19. (a) $\frac{F}{M} - \frac{B^2 l^2 v}{mR}$ (b) $\frac{RF}{B^2 l^2}$ (c) $v_0 \left(1 - e^{\frac{-Ft}{mv_0}} \right)$ 20. $\frac{\mu_0 I v}{2\pi} \ln \left(\frac{d + a \cos \theta}{d - a \cos \theta} \right)$
21. (a) $\frac{mgR}{B^2 L^2}$ (b) $\frac{g}{2}$ 22. $\varepsilon = -BA\omega \theta_0 \cos(\omega t) \cdot \cos[\theta_0 \sin \omega t]$
23. (a) $\frac{R^2 qB}{2}$ (b) $\frac{q^2 R^2 B}{4m}$ 24. (a) $2B\omega a^2$ (b) $B\omega a^2$ 25. $\frac{\pi}{8} \mu_0 n b r^2$
26. (a) $BI d + \frac{2\lambda m I^2}{B^2 d^2}(R + 2\lambda x)$ (b) $\left[1 + \frac{2\lambda m I(R + 2\lambda x)}{B^3 d^3} \right]^{-1}$ 28. $\frac{BL^2 \omega}{2(R + r)}$
29. $\frac{BL^2 \omega}{r + 2R}$ 30. (b) 5.6°C 31. $\frac{R^2(2 + \pi)\alpha}{\sqrt{2}}$
32. (a) 10^4 As^{-1} (b) 0 (c) 2A (d) $1.73 \times 10^{-4} \text{ C}$ 33. 3.465 s
34. (a) $12e^{-5t}$ (b) $6e^{-10t}$ clockwise 35. (a) $Q = 24 \cos(50000t) \mu\text{C}$ (b) $52 \mu\text{s}$
36. $\frac{4}{3}(r_2 - r_1)$ 37. (a) $\frac{e^{-t/\tau}}{1 - e^{-t/\tau}}$ (b) $(\ln 2)\tau$ 38. (a) $\frac{\pi \mu_0 R^2 r^2}{2x^3}$ (b) $\frac{3}{2} \frac{\pi \mu_0 R^2 r^2 i}{x^4}$
39. $\frac{\mu_0 a}{2\pi} \ln \left(\frac{b}{c} + 1 \right)$ 40. $i_1 = 8.86 \mu\text{A}$, $i_2 = 35.44 \mu\text{A}$

Alternating Current Circuits

“Just as certain as death, Westinghouse will kill a customer within six months after he puts in a (alternating current) system of any size.”

– Thomas Edison

1. INTRODUCTION

A current (or emf) which keeps changing direction at regular intervals is known as an alternating current (or emf). ‘AC’ is an abbreviation for *alternating current*, however we mostly use it as an abbreviation for ‘alternating’. When we say *AC voltage* we mean *alternating voltage* (and not alternating current voltage!)

We have learnt in the last chapter that a sinusoidal alternating emf is developed in a coil if it is rotated with a constant angular velocity in a uniform magnetic field. Such an emf drives a sinusoidal alternating current. In this chapter, our main target is to learn about the response of a series LCR circuit (a circuit having an inductor, a capacitor and a resistor in series) when it is connected to an alternating emf source. This is important as the power supply that we receive in our houses and factories are all alternating.

In this chapter, we will be mostly using the term *voltage* for *alternating potential difference*.

Before we start discussing the alternating current and circuits, in the next section, we will quickly recapitulate some essential mathematical results.

2. ESSENTIAL MATHEMATICAL CONCEPTS TO DEAL WITH SINUSOIDAL QUANTITIES

We need to learn how to calculate average value of a continuously varying quantity and also how to add two or more sinusoidal quantities having same frequency.

2.1 Average of a quantity

y is a physical quantity that varies with some other quantity x and the dependence is expressed as

$$y = f(x)$$

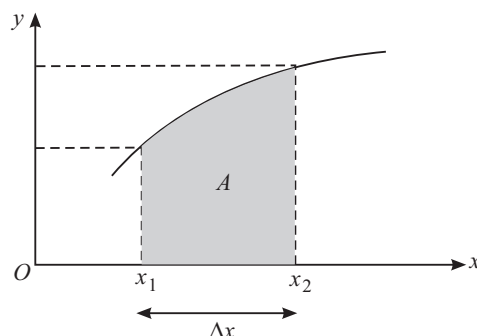
The average value of y as x changes from x_1 to x_2 is defined as

$$\langle y \rangle = \frac{\int_{x_1}^{x_2} y dx}{x_2 - x_1} \quad (1)$$

The average value of y is usually denoted as $\langle y \rangle$ or y_{av} or $\bar{y}$.

The integral $\int_{x_1}^{x_2} y dx$ is equal to the area under the graph (see figure). The average value of y is

$$\langle y \rangle = \frac{\text{Area}(A)}{\Delta x} \quad (2)$$

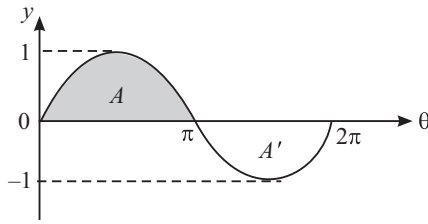


Example 1 Average of $\sin \theta$ and $\cos \theta$

- Find the average value of $\sin \theta$ when θ changes from 0 to π
- Find the average value of $\sin \theta$ when θ changes from 0 to 2π
- Find the average value of $\cos \theta$ when θ changes from $\pi/2$ to $3\pi/2$

Solution

(a) Let $y = \sin\theta$



$$\langle y \rangle = \frac{\int_0^\pi y d\theta}{\pi - 0}$$

$$= \frac{1}{\pi} \int_0^\pi \sin \theta d\theta = \frac{1}{\pi} [-\cos \theta]_0^\pi = \frac{2}{\pi}$$

Remember that the shaded area in the graph is

$$A = \int_0^\pi \sin \theta d\theta = 2$$

$$\text{Average value is } \frac{A}{\pi} = \frac{2}{\pi}$$

$$(b) \langle y \rangle = \frac{\int_0^{2\pi} \sin \theta d\theta}{2\pi} = \frac{1}{2\pi} [-\cos \theta]_0^{2\pi} = 0$$

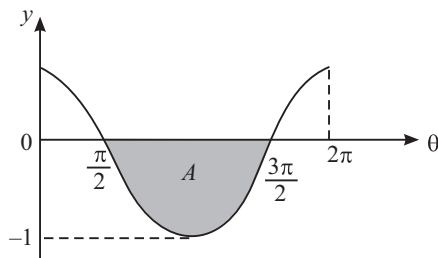
One can directly see that

$$\langle y \rangle = \frac{\text{area}(A) + \text{area}(A')}{2\pi} = 0$$

Mathematics (calculating by integration) treats area A' as negative. Magnitude of area A is definitely equal to A' .

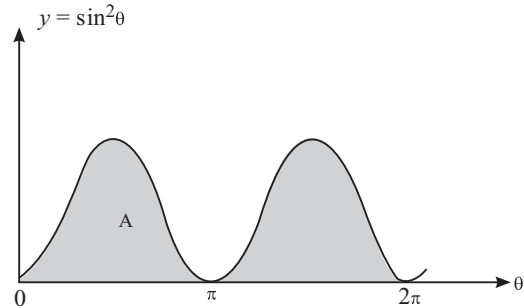
(c) $y = \cos\theta$

$$\langle y \rangle = \frac{\text{area}(A)}{\frac{3\pi}{2} - \frac{\pi}{2}} = -\frac{2}{\pi}$$



The area A in graph of $\sin\theta$ and in the graph of $\cos\theta$ are equal in magnitude.

Example 2 Average of $\sin^2\theta$



$$y = \sin^2\theta$$

$$\langle y \rangle = \frac{\int_0^{2\pi} \sin^2 \theta d\theta}{2\pi}$$

$$= \frac{1}{4\pi} \int_0^{2\pi} (1 - \cos 2\theta) d\theta$$

$$= \frac{1}{4\pi} \left[2\pi - \frac{1}{2} (\sin 2\theta)_0^{2\pi} \right] = \frac{1}{2}$$

If we average $\sin^2\theta$ for θ changing from 0 to π , we still get the same result. Obviously, this is true because

$$[\langle y \rangle]_0^\pi = \frac{A}{\pi} \text{ and } [\langle y \rangle]_0^{2\pi} = \frac{2A}{2\pi}$$

The average value of $\cos^2\theta$ is also $\frac{1}{2}$ for the interval 0 to π or 0 to 2π .

2.2 Understanding sinusoidal quantities

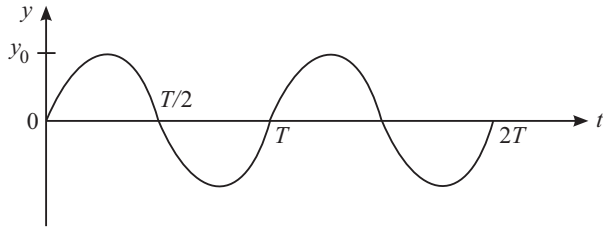
Consider a quantity y that changes sinusoidally with time (t).

$$y = y_0 \sin(\omega t) \quad [y_0 \text{ and } \omega \text{ are constants}]$$

y_0 is the peak value of y , often known as amplitude. The term $\phi = \omega t$ is known as *phase angle* or simply the *phase*. If we know the phase angle we know the *state* of y during its course of sinusoidal oscillations. For example, when ϕ is equal to 0, 2π , 4π we know that $y = 0$ and is increasing. When $\phi = \pi$, 3π , 5π y is zero but decreasing. When $\phi = \pi/2$, $5\pi/2$, y is at positive extreme.

At $t = 0$, value of y is zero (and is increasing). At $t = \frac{2\pi}{\omega}$, value of the phase angle ϕ becomes 2π and y completes one cycle. This time interval $T = \frac{2\pi}{\omega}$, needed for y to complete one cycle is known as *time period*. In an interval of 1 second, the function completes $\frac{1}{T} = \frac{\omega}{2\pi}$ cycles and this is known as frequency (f).

$$f = \frac{1}{T} = \frac{\omega}{2\pi} \quad (2)$$



At time T , the phase angle is $\phi = 2\pi$.

The constant ω is known as *angular frequency* or *circular frequency* and its unit is rad s^{-1} .

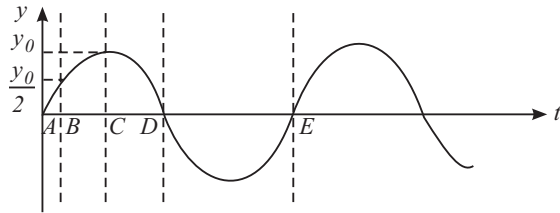
If an observer takes time $t = 0$ at an instant when y is not zero, then $y = y_0 \sin(\omega t)$ is not the appropriate representation. The observer must write

$$y = y_0 \sin(\omega t + \delta)$$

We emphasise that variation of y has not changed at all, it is just that one has chosen time to be zero when y is not zero.

Consider the sinusoidal variation of y shown in the figure below. If one puts on his stop-watch when the state of quantity y is at A , then the correct equation for representing the variation of y is $y = y_0 \sin \omega t$. If one starts counting time at B , then the correct equation will be $y = y_0 \sin(\omega t + \frac{\pi}{3})$. If you put $t = 0$ in this equation, you get $y = \frac{y_0}{2}$. Moreover, y is increasing with time as seen in the graph. If a person prefers taking $t = 0$ at C , then the correct equation will be

$$y = y_0 \sin\left(\omega t + \frac{\pi}{2}\right) = y_0 \cos \omega t.$$



Similarly, correct equation if you set $t = 0$ at D is

$$y = y_0 \sin(\omega t + \pi)$$

If $t = 0$ is set at E , the equation will be

$$y = y_0 \sin(\omega t + 2\pi) = y_0 \sin(\omega t)$$

Advancing the phase by 2π implies that one cycle has been completed and we are back to the initial state.

The quantity δ is known as *phase constant* or *initial phase* of the sinusoidally varying quantity y .

Example 3 Recognising initial phase

A quantity x changes sinusoidally with frequency 50Hz. Its amplitude is 2 unit. At time $t = 0$, value of x is 1 and it is decreasing with time. Write the equation showing variation of x with time.

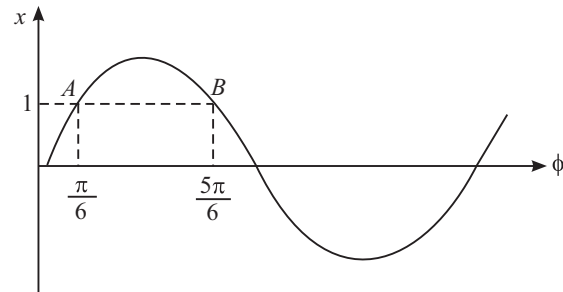
Solution

Concepts

(i) $x = x_0 \sin(\omega t + \delta)$

δ can be calculated as we know that value of x at $t = 0$

(ii) Key phrase in the question is 'decreasing with time'. We must choose δ so that x is decreasing.



The state represented by B is appropriate choice as x is decreasing.

$$\omega = 2\pi f = 2 \times 3.14 \times 50 = 314 \text{ rad s}^{-1}$$

$$\text{and } x_0 = 2$$

$$\therefore x = 2 \sin(314 t + \delta)$$

$$\text{At } t = 0, x = 1$$

$$\Rightarrow 1 = 2 \sin(0 + \delta)$$

$$\text{or, } \sin \delta = \frac{1}{2}$$

$$\text{or, } \delta = \frac{\pi}{6}, \frac{5\pi}{6}$$

Since x is decreasing, at $t = 0$, therefore, we choose

$$\delta = \frac{5\pi}{6}$$

$$\therefore x = 2 \sin\left(314t + \frac{5\pi}{6}\right)$$

Example 4 Ahead in phase

A quantity y_1 varies with time t as $y_1 = y_0 \sin(100 t)$. Another quantity y_2 also has the same amplitude and frequency but is ahead in phase (with respect to y_1) by $\frac{\pi}{2}$.

(a) Write the expression of y_2 .

(b) Plot the variation of y_1 and y_2 with time.

Solution

Concepts

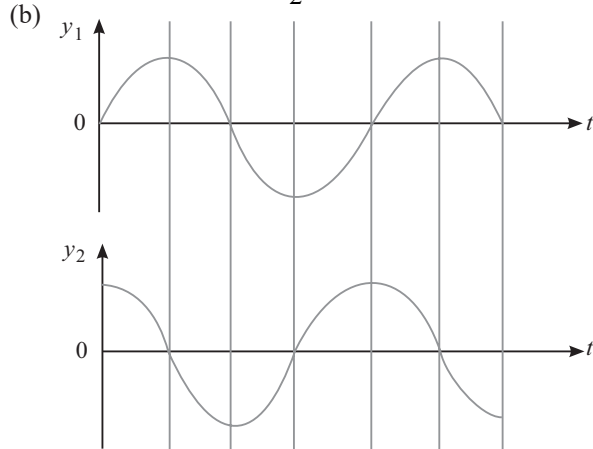
y_2 is $\frac{\pi}{2}$ ahead in phase with respect to y_1 , means the mathematical value of the phase angle for y_2 is always

$\frac{\pi}{2}$ more than that of y_1 .

6.4 Magnetism and Electromagnetic Induction

(a) $y_1 = y_0 \sin(100t)$

$y_2 = y_0 \sin(100t + \frac{\pi}{2}) = y_0 \cos(100t)$



y_2 is at peak at $t = 0$. y_1 will be at peak after some time. We say that y_2 is leading in phase.

Note: we may also say that y_1 lags in phase by $\frac{\pi}{2}$ with respect to y_2 .

2.3 Addition of sinusoidal quantities having same frequency

Consider two quantities y_1 and y_2 changing sinusoidally with time. Both have same frequency (i.e., same ω) but have a constant phase difference δ between them.

$$\left. \begin{aligned} y_1 &= a_1 \sin(\omega t) \\ y_2 &= a_2 \sin(\omega t + \delta) \end{aligned} \right\} \quad (i)$$

We wish to find $y = y_1 + y_2$

$$\begin{aligned} y &= a_1 \sin(\omega t) + a_2 \sin(\omega t + \delta) \\ &= a_1 \sin(\omega t) + (a_2 \cos \delta) \sin(\omega t) + (a_2 \sin \delta) \cos(\omega t) \\ &= (a_1 + a_2 \cos \delta) \sin(\omega t) + (a_2 \sin \delta) \cos(\omega t) \end{aligned}$$

or, $y = C \sin(\omega t) + D \cos(\omega t) \quad (ii)$

where

$$\left. \begin{aligned} C &= a_1 + a_2 \cos \delta \text{ and } \\ D &= a_2 \sin \delta \end{aligned} \right\} \quad (iii)$$

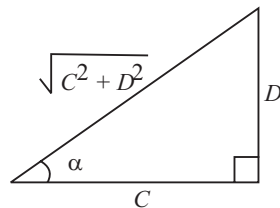
Equation (ii) can be written as

$$y = \sqrt{C^2 + D^2} \left[\frac{C}{\sqrt{C^2 + D^2}} \sin(\omega t) + \frac{D}{\sqrt{C^2 + D^2}} \cos(\omega t) \right]$$

Consider a right angled triangle as shown in the figure.

$$\sin \alpha = \frac{D}{\sqrt{C^2 + D^2}} ;$$

$$\cos \alpha = \frac{C}{\sqrt{C^2 + D^2}}$$



Also let $\sqrt{C^2 + D^2} = A$

Now, $y = A [\cos \alpha \sin(\omega t) + \sin \alpha \cos(\omega t)]$

or, $y = A \sin(\omega t + \alpha) \quad (iv)$

where $\alpha = \tan^{-1} \frac{D}{C} = \tan^{-1} \left(\frac{a_2 \sin \delta}{a_1 + a_2 \cos \delta} \right) \quad (v)$

and $A = \sqrt{C^2 + D^2} = \sqrt{(a_1 + a_2 \cos \delta)^2 + (a_2 \sin \delta)^2}$

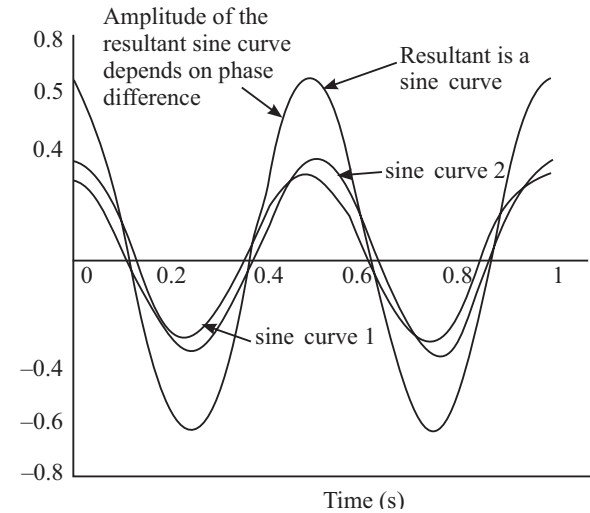
or, $A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \delta} \quad (3)$

Let us interpret the result.

(a) $y_1 = a_1 \sin \omega t, y_2 = a_2 \sin(\omega t + \delta)$

$y = y_1 + y_2 = A \sin(\omega t + \alpha)$

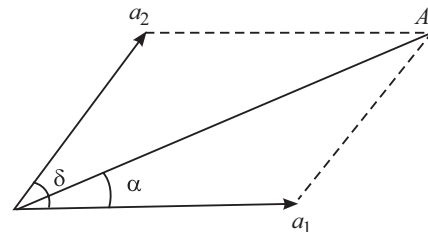
The resultant of two sinusoidally varying quantities is also sinusoidal and has the same frequency as the original quantities. This is remarkable! Take one sine curve and place over another sine curve. Both the curves have same frequency. Sum of the two curves is a sine curve even if you shift one curve over another.



(b) The amplitude to the resultant is given by

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \delta}$$

Looking at this result, we can frame a simple rule to get the resultant amplitude. Assume two vectors of magnitudes a_1 and a_2 inclined at δ . Their resultant gives the resultant amplitude A .



Also, angle α is given by the angle that $\vec{A}$ makes with $\vec{a}_1$ in the vector diagram.

2.4 Phasors

A sinusoidally varying quantity can be expressed using a rotating vector which is usually known as a phasor.

Consider a vector OA of length a . At time $t = 0$ it is along x axis and rotates in anticlockwise sense with a constant angular velocity ω .

At time t , it is in position OB . Now, drop a perpendicular from the tip of the rotating vector on y axis. C is the foot of the perpendicular. Obviously, $OC (= y)$ is

$$y = a \sin(\omega t)$$

It means, when a vector of length a rotates with angular speed ω (in anticlockwise sense), its projection on the y axis represents a sinusoidally varying quantity having angular frequency ω .

If initial position of the vector is not along the x axis, the equation representing y will be

$$y = a \sin(\omega t + \delta)$$

The angle made by the rotating vector, with the positive x direction, at a given instant is the phase angle for the sinusoidal quantity y .

Now, again consider two quantities having a constant phase difference δ between them.

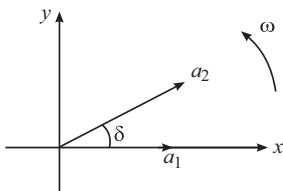
$$y_1 = a_1 \sin(\omega t) \text{ and}$$

$$y_2 = a_2 \sin(\omega t + \delta)$$

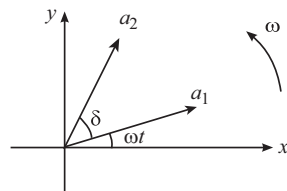
The figure given below shows the corresponding phasors. Note that the two phasors rotate with same angular velocity ω while maintaining a constant angle δ between them. The resultant of y_1 and y_2 is

$$y = y_1 + y_2$$

Instantaneous value of y is the sum of instantaneous values of y_1 and y_2 . We understand that y_1 is the y component of the phasor of length a_1 and y_2 is the y component of the phasor of length a_2 . The instantaneous value of y is



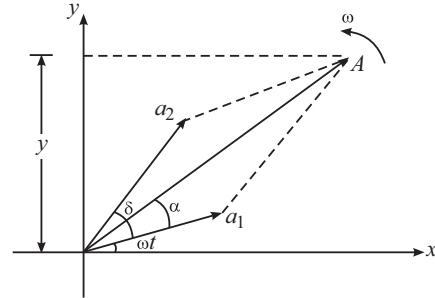
Phasors at $t = 0$



Phasors at time t

$$y = a_{1y} + a_{2y}$$

To get sum of y components of two vectors, we can also find the y component of their resultant. It means we first add vectors $\vec{a}_1$ and $\vec{a}_2$ and find their resultant $\vec{A}$. y -component of $\vec{A}$ gives us the instantaneous value of y . This result is exactly same as what we had derived in the section 2.3. A phasor having length A and phase angle $(\omega t + \alpha)$ represents the quantity y .



Add $\vec{a}_1$ and $\vec{a}_2$. Their resultant represents the resultant phasor

$$y = A \sin(\omega t + \alpha)$$

where

$$A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \delta} \text{ and } \tan \alpha = \frac{a_2 \sin \delta}{a_1 + a_2 \cos \delta}$$

Example 5 If $y_1 = 3 \sin(100t)$ and $y_2 = 4 \cos(100t)$, then find the expression for $y = y_1 + y_2$. Is it right to say that y is ahead in phase compared to y_2 ? What is the phase difference between y and y_1 ?

Solution

Concepts

$$y_2 = 4 \cos(100t) = 4 \sin(100t + \frac{\pi}{2})$$

It is recommended to express all terms as sine functions when trying to compare phases.

$$y_1 = 3 \sin(100t) ; y_2 = 4 \sin(100t + \frac{\pi}{2})$$

Resultant is a sine function with $\omega = 100$. The amplitude and phase constant of the resultant can be obtained by adding two vectors of length 3 and 4 at an angle of $\frac{\pi}{2}$.

$$A = \sqrt{3^2 + 4^2} = 5$$

$$\text{and } \tan \alpha = \frac{4}{3}$$

$$\Rightarrow \alpha = \tan^{-1}\left(\frac{4}{3}\right) \simeq 53^\circ$$

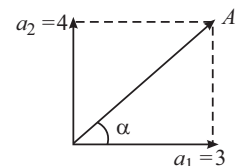
$$\therefore y = 5 \sin(100t + \tan^{-1}\frac{4}{3})$$

$$\text{Phase angle of } y \text{ is } \phi_1 = 100t + \tan^{-1}\left(\frac{4}{3}\right) = 100t + 53^\circ$$

$$\text{Phase angle of } y_2 \text{ is } \phi_2 = 100t + 90^\circ$$

$$\therefore y \text{ lags in phase compared to } y_2.$$

$$\text{And } y \text{ is ahead in phase by } \tan^{-1}\frac{4}{3} = 53^\circ, \text{ compared to } y_1.$$



In Short

- (i) If $y = f(x)$, then average value of y between $x = x_1$ to $x = x_2$ is

$$\langle y \rangle = \frac{\int_{x_1}^{x_2} y dx}{x_2 - x_1}$$

- (ii) If y is a sine function of x , then the average value of y over one full cycle is zero. However, the average is not zero for a half cycle.

$$y = y_0 \sin x$$

$$\langle y \rangle_{\text{Full cycle}} = y_0 \langle \sin x \rangle_{\text{Full cycle}} = 0$$

$$\langle y \rangle_{\text{Half cycle}} = y_0 \langle \sin x \rangle_{\text{Half cycle}} = \frac{2y_0}{\pi}$$

$$(iii) \langle \sin^2 x \rangle_{\text{Half cycle}} = \langle \sin^2 x \rangle_{\text{Full cycle}} = \frac{1}{2}$$

- (iv) $y = a \sin (\omega t + \delta)$

a is known as amplitude, $\phi = (\omega t + \delta)$ is known as phase angle, δ is known as initial phase (or phase constant)

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Here, f is frequency and T is known as time period.

- (v) $y_1 = a_1 \sin \omega t$,
 $y_2 = a_2 \sin (\omega t + \delta)$

The sum of y_1 and y_2 is also a sine function of angular frequency ω .

$$y = y_1 + y_2 = A \sin (\omega t + \alpha)$$

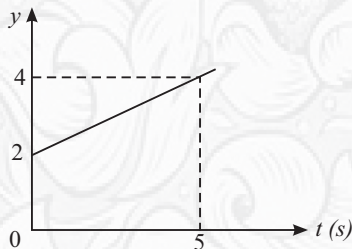
Here, A can be obtained as resultant of two vectors of length a_1 and a_2 inclined at δ . Angle α is the angle made by the resultant vector with the vector a_1 .

- (vi) Sinusoidally varying quantities are often represented by rotating vectors, known as phasors.



YOUR TURN

- Q.1** A quantity y varies with time t as shown. Find its average value in the interval 0 to 5s.



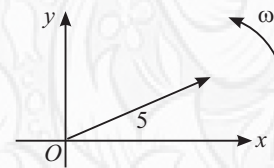
Q.2 $y = 1 - \sin (100\pi t)$

Find the average value of y in the interval $t = 0$ to $t = 0.02$ s.

Q.3 $y_1 = 5 \sin (200t)$ and $y_2 = 5 \sin (200t + \frac{2\pi}{3})$

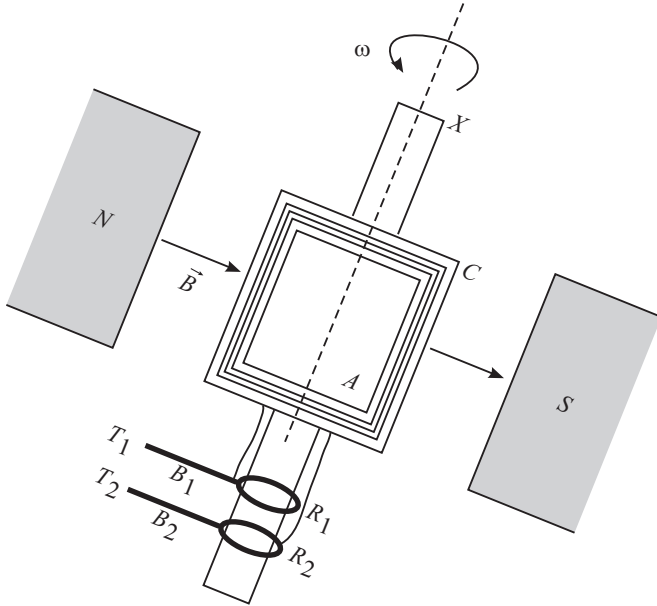
- (a) Find $y = y_1 + y_2$

- (b) Figure shows the phasor corresponding to y_1 . Draw a phasor corresponding to y .



3. AC GENERATOR

A coil (C) is wound over a soft iron core. This assembly is known as armature (A) of the generator. It is mounted on an axle (X) and can rotate freely between the poles of a magnet.



A : armature, C : coil. R_1 and R_2 : slip rings, B_1 and B_2 : carbon brushes, T_1 and T_2 terminals

Let the coil have N turns and its area be A . It is being rotated with angular speed ω in a magnetic field (B) created by the magnet. For simplicity, assume that the plane of the coil is perpendicular to the magnetic field at $t = 0$. Flux through each turn of the coil in this position is $BA \cos 0^\circ = BA$. At time t , the coil rotates by $\theta = \omega t$ and angle between the area vector ($\vec{A}$) and field ($\vec{B}$) becomes ωt . Flux through single turn is

$$\phi = BA \cos \omega t$$

Using Faraday's law, the induced emf in the coil is

$$\varepsilon = -N \frac{d\phi}{dt} = NBA\omega \sin \omega t$$

$$\text{or, } \varepsilon = \varepsilon_0 \sin \omega t$$

$$\text{where } \varepsilon_0 = NBA\omega \quad \dots(4)$$

The emf varies sinusoidally with time with angular frequency ω and peak value ε_0 .

The ends of the coil are connected to two slip rings (R_1 and R_2) fixed on the axle. These rings rotate with the armature assembly. Two carbon brushes (B_1 and B_2) are kept fixed while they touch the two rings. They are spring loaded to keep them pressed against the rings. These brushes act as the terminals (T_1 and T_2) of the generator. Any external circuit is connected to these terminals. The alternating emf developed in the coil drives an alternating current in the circuit.

4. AVERAGE OF AC CURRENT AND VOLTAGE

An alternating current can be expressed as

$$i = i_0 \sin (\omega t + \phi)$$

Average value of this current over a complete cycle is

$$\langle i \rangle = i_0 \langle \sin(\omega t + \phi) \rangle = 0$$

As we know that the average of a sine function over a full cycle is zero.

However, if we find the average of the current over half cycle we get

$$\langle i \rangle_{\text{Half cycle}} = i_0 \langle \sin(\omega t + \phi) \rangle_{\text{Half cycle}}$$

$$\text{or, } \langle i \rangle_{\text{Half cycle}} = \frac{2i_0}{\pi} \quad \dots(5)$$

[see example 1]

Similarly, if an alternating voltage is written as

$$v = v_0 \sin (\omega t + \phi) \text{ then,}$$

$$\langle v \rangle_{\text{Full cycle}} = 0$$

$$\langle v \rangle_{\text{Half cycle}} = \frac{2v_0}{\pi} \quad (6)$$

The average current expressed by equation (5) is not our favourite. It does not give the correct idea of power consumption in a circuit. A different kind of average, known as rms current, is a lot better. It gives us the exact idea of energy consumption in a circuit.

5. RMS CURRENT/VOLTAGE

The rms value of current i is obtained by finding the average (mean) of i^2 and taking the square root of the mean.

$$\text{Let } i = i_0 \sin (\omega t + \phi)$$

$$i^2 = i_0^2 \sin^2 (\omega t + \phi)$$

Mean of i^2 is

$$\langle i^2 \rangle = i_0^2 \langle \sin^2(\omega t + \phi) \rangle = \frac{i_0^2}{2} \quad (7)$$

[$\because$ Average of square of sine function in a complete cycle is $\frac{1}{2}$ as demonstrated in example 2]

The quantity $\langle i^2 \rangle$ is known as the *mean square current*. The square root of the mean square current is known as the *root mean square current* (rms current).

$$i_{\text{rms}} = \sqrt{\langle i^2 \rangle} = \frac{i_0}{\sqrt{2}} \quad (8)$$

Though we have taken the average over one cycle, the result remains valid if one takes average over a long period of time.

If we have an alternating voltage $v = v_0 \sin (\omega t + \phi)$, its rms value is

$$v_{\text{rms}} = \frac{v_0}{\sqrt{2}} \quad (9)$$

Significance of rms current and rms voltage

Consider an alternating current $i = i_0 \sin(\omega t + \phi)$ through a resistance R . The instantaneous power dissipated in the resistance is

$$P = i^2 R$$

Heat dissipated in a small interval dt is

$$dH = P dt = i^2 R dt$$

Total heat dissipated in one time period is

$$H = \int_0^T i^2 R dt = R \int_0^T i^2 dt$$

Average power (i.e., average heat dissipated per unit time) is

$$\langle P \rangle = \frac{H}{T} = R \frac{\int_0^T i^2 dt}{T}$$

$$\text{or, } \langle P \rangle = i_{\text{rms}}^2 \cdot R \quad (10)$$

Note that $\frac{\int_0^T i^2 dt}{T} = \langle i^2 \rangle = i_{\text{rms}}^2$

Therefore, if we pass a constant current i_{rms} through a resistance R , it will produce the same amount of heat in a time period as that dissipated when an alternating current is passed through it.

Similarly, we can show that a constant voltage v_{rms} applied across a resistance produces the same thermal energy as produced by a voltage $v = v_0 \sin(\omega t + \phi)$.

When talking about alternating current and alternating voltage, we generally mention their rms values. A current of 5A means an alternating current having a peak value of $i_0 = 5\sqrt{2}$ A. The instruments used to measure alternating currents and voltages are designed to directly measure the rms values.

Example 6 The household power supply

The household power supply in India is at 220V, 50Hz. Write the variation of supply voltage with time.

Solution**Concepts**

220V is the rms value of voltage.

$$v_0 = \sqrt{2} v_{\text{rms}}$$

Peak voltage, $v_0 = \sqrt{2} \times 220 = 311\text{V}$

Angular frequency, $\omega = 2\pi f = 2 \times 3.14 \times 50 = 314 \text{ rad s}^{-1}$

$$v = v_0 \sin(\omega t + \phi)$$

$$\text{or, } v = (311) \sin(314t + \phi) \text{ V}$$

ϕ is phase constant and its value depends on the state of the voltage when we take $t = 0$.

Example 7 AC superimposed over a DC

An alternating current $i_2 \sin \omega t$ has been added to a DC current i_1 and the resultant current can be written as $i = i_1 + i_2 \sin \omega t$

Find the rms value of i .

Solution**Concepts**

(i) A common mistake is to write

$$\begin{aligned} i_{\text{rms}} &= i_1 + (i_2 \sin \omega t)_{\text{rms}} \\ &= i_1 + \frac{i_2}{\sqrt{2}} \end{aligned}$$

(ii) For finding the i_{rms} , you must follow the steps – square, mean, root. Find i^2 , calculate its mean and then take root.

$$i^2 = (i_1 + i_2 \sin \omega t)^2 = i_1^2 + i_2^2 \sin^2 \omega t + 2i_1 i_2 \sin \omega t$$

Mean of i^2 is

$$\begin{aligned} \langle i^2 \rangle &= \langle i_1^2 \rangle + \langle i_2^2 \sin^2 \omega t \rangle + \langle 2i_1 i_2 \sin \omega t \rangle \\ &= i_1^2 + i_2^2 \langle \sin^2 \omega t \rangle + 2i_1 i_2 \langle \sin \omega t \rangle \\ &= i_1^2 + \frac{i_2^2}{2} + 0 \end{aligned}$$

[$\because$ Mean of $\sin^2 \omega t$ over a cycle is $\frac{1}{2}$ and the mean of $\sin \omega t$ over a cycle is zero]

rms current is

$$i_{\text{rms}} = \sqrt{\langle i^2 \rangle} = \sqrt{i_1^2 + \frac{i_2^2}{2}}$$

In Short

(i) The average value of sinusoidal AC current $i = i_0 \sin(\omega t + \phi)$ and voltage $v = v_0 \sin(\omega t + \phi)$ is zero for a full cycle. However, for half cycle the averages are $\frac{2i_0}{\pi}$ and $\frac{2v_0}{\pi}$ respectively.

(ii) Most useful kind of average for alternating current and voltage is their rms (root mean square) value. The rms current gives the exact idea of energy consumption in a circuit

(iii) Rms value of i is calculated by finding the average of i^2 and taking its root.

$$i_{\text{rms}} = \sqrt{\langle i^2 \rangle}$$

(iv) For $i = i_0 \sin(\omega t + \phi)$, $i_{\text{rms}} = \frac{i_0}{\sqrt{2}}$

$$\text{And for } v = v_0 \sin(\omega t + \phi), v_{\text{rms}} = \frac{v_0}{\sqrt{2}}$$



YOUR TURN

Q.4 The armature of an AC generator makes 50 rotations per seconds. What is the frequency of the alternating voltage generated by it?

Q.5 The peak voltage of an alternating current is 5A and its frequency is 100Hz. Find its rms value. How long will the current take to reach its peak value starting from zero?

Q.6 A current $i = 141 \sin(50t)$ A flows through a resistance $R = 1\Omega$. Find the average power dissipated in the resistance, averaged over a long period of time.

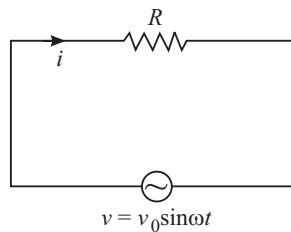
Q.7 Which is more dangerous – 220V DC supply or a 220V AC supply?

6. AC CIRCUIT WITH ONE ELEMENT

We will first consider AC circuit having only one element – a resistance, a capacitor or an inductor – connected to an alternating source.

6.1 Only resistor

Consider a circuit having a resistance (R) attached to an alternating source. Such a circuit is called a *purely resistive circuit*. Take a note of the symbol used to represent an AC source.



Let the voltage of the source be $v = v_0 \sin \omega t$.

The instantaneous current in the circuit must be

$$i = \frac{v}{R} = \frac{v_0}{R} \sin \omega t$$

or, $i = i_0 \sin \omega t$

where $i_0 = \frac{v_0}{R}$

Following points must be noted.

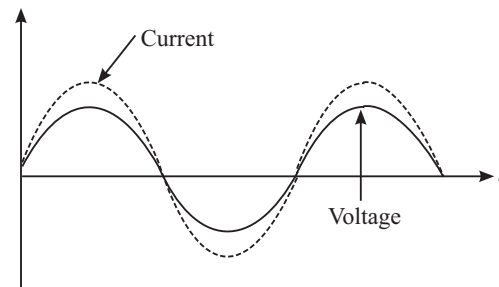
(a) Peak current is equal to peak voltage divided by R .

$$(b) i_0 = \frac{v_0}{R} \Rightarrow \frac{i_0}{\sqrt{2}} = \frac{v_0}{\sqrt{2}R}$$

$$\text{or } I = \frac{V}{R}$$

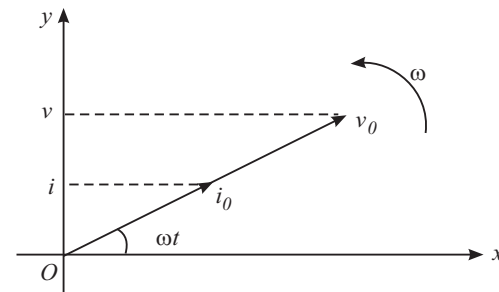
Here, capital letters (I and V) represent the rms values. Henceforth, we will always use capital letters (I and V) to represent rms values and the small letters (i and v) to represent the instantaneous values.

(c) The phase angles of the current and voltage are same. This implies that the current and voltage rise and fall together.



Variation of current and voltage which are in phase

(d) The phasors representing the current and voltage are as shown. They have lengths i_0 and v_0 and are always in same phase (i.e., make same angle with x direction)



i and v are instantaneous values of current and voltage

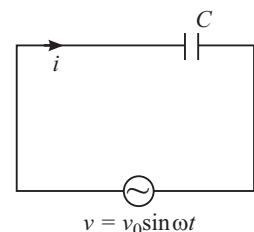
6.2 Only capacitor

Consider a capacitor connected to an AC source generating an emf

$$v = v_0 \sin \omega t \quad (i)$$

Such a circuit is often termed as *purely capacitive circuit*.

When a capacitor is connected to a DC battery, there is no current in the circuit once the capacitor gets fully charged. But in an AC circuit, the current exists as long as the source is there. As the voltage



6.10 Magnetism and Electromagnetic Induction

across the capacitor plates changes, charge on the capacitor plate also changes. It means charge will keep flowing to the plates or out of them. We always have a current in the connecting wires.

Suppose the charge on the capacitor is q and current in the circuit is i at time t . The current (i) is always same as the rate at which charge (q) on the plates changes with time.

$$\therefore i = \frac{dq}{dt}$$

Using Kirchhoff's loop law,

$$\frac{q}{C} = v_0 \sin \omega t$$

$$\text{or, } \frac{dq}{dt} = v_0 C \omega \cos \omega t$$

$$\text{or, } i = i_0 \cos \omega t$$

$$\text{where } i_0 = v_0 \omega C = \frac{v_0}{1/\omega C}$$

The following points are worth our attention:

$$(a) \text{ Peak current } i_0 = \frac{v_0}{1/\omega C} \quad (iii)$$

$\frac{1}{\omega C}$ is denoted by X_c and is known as *capacitive reactance*. Therefore,

$$i_0 = \frac{v_0}{X_c} \quad (11)$$

$$\text{Also, } \frac{i_0}{\sqrt{2}} = \frac{v_0/\sqrt{2}}{X_c} \Rightarrow I = \frac{V}{X_c}$$

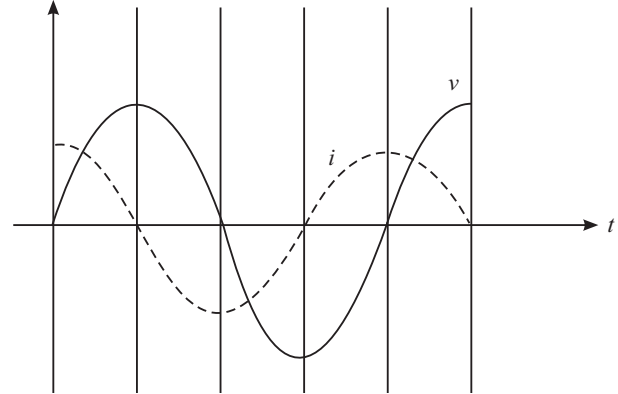
Here, we have used capital letters for rms values of current and voltage.

(b) $X_c = \frac{1}{\omega C}$ has dimensions of resistance. Its unit is ohm. It plays the role of effective resistance as far as calculation of current is concerned. Its value depends on the capacitance as well as the frequency of the source. For high frequency source, X_c is small and the current is large. For small frequency source, the capacitive reactance is high and the current is small.

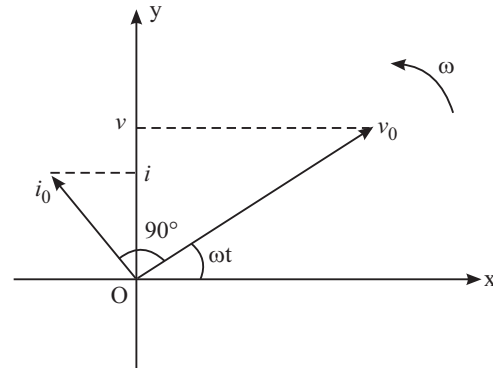
$$(c) v = v_0 \sin \omega t \text{ and } i = i_0 \cos \omega t = i_0 \sin \left(\omega t + \frac{\pi}{2} \right)$$

The current in the circuit leads the voltage by $\frac{\pi}{2}$. When the voltage is zero, the instantaneous current is at its maximum. When the voltage is maximum, the current is zero. There are instances when the voltage is positive but the current is negative. At these moments, the capacitor is discharging and giving energy to the source. Same is true when the voltage is negative and the current is positive. *On an average, a capacitor does not consume energy in an AC circuit.*

(d) The current phasor is ahead of the voltage phasor by $\pi/2$.



Current leads in phase by 90° . When both v and i are positive (or both negative), the source is supplying energy and the capacitor is gaining it. When v and i have opposite signs, the capacitor is losing energy and the source is gaining it.



Current phasor is 90° ahead of the voltage phasor. v and i are instantaneous values

6.3 Only inductor

The figure shows a pure inductor connected to an AC source supplying voltage

$$v = v_0 \sin \omega t$$

Such a circuit is known as a purely inductive circuit.

The induced emf in the inductor is $-L \frac{di}{dt}$, therefore using the Kirchhoff's loop law gives

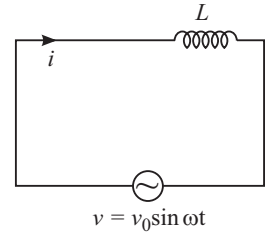
$$v_0 \sin \omega t - L \frac{di}{dt} = 0$$

$$\text{or, } \frac{di}{dt} = \frac{v_0}{L} \sin \omega t$$

$$\text{or, } \int di = \frac{v_0}{L} \int \sin \omega t dt$$

$$\text{or, } i = -\frac{v_0}{\omega L} \cos \omega t + K$$

where K is constant of integration.



(i)

The source voltage is sinusoidal and its average value is zero. Such a voltage always produces sinusoidal current. The average current must be zero. This will be possible only if value of the constant k in the equation (i) is zero.

$$\therefore i = -\frac{v_0}{\omega L} \cos \omega t$$

$$\text{or, } i = \frac{v_0}{X_L} \sin\left(\omega t - \frac{\pi}{2}\right)$$

where $X_L = \omega L$

X_L is known as *inductive reactance*.

We can write the expression of the current as

$$i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$$

where the peak current $i_0 = \frac{v_0}{X_L}$

Following are the important conclusions:

(a) The inductive reactance $X_L = \omega L$ has unit of ohm. It is the effective resistance offered by the inductor for finding the current. X_L is high for a high frequency source and small for a low frequency source.

(b) Peak current $i_0 = \frac{v_0}{X_L}$

$$\Rightarrow \frac{i_0}{\sqrt{2}} = \frac{v_0 / \sqrt{2}}{X_L}$$

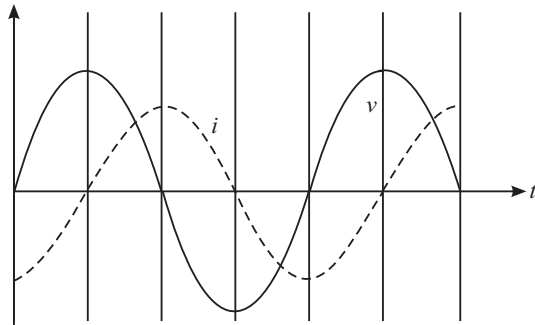
$$\text{or, } I = \frac{V}{X_L}$$

This is relation between rms current and rms voltage.

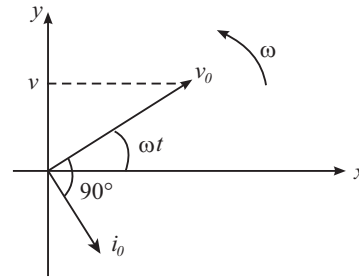
(c) $v = v_0 \sin \omega t$ and $i = i_0 \sin\left(\omega t - \frac{\pi}{2}\right)$

The current in the circuit lags the voltage by $\pi/2$.

(d) The following figures show the relationship between the phase of current and voltage.



Current lags in phase by 90° . When both v and i are positive (or both negative), the source supplies energy which gets stored in the inductor. When v and i are of opposite sign, the inductor loses energy.



The current phasor is 90° behind the voltage phasor

Example 8 An AC source supplying a voltage $v = 100 \sin(200t)$ is connected to a capacitor having capacitance $C = 100 \mu\text{F}$. Find

- the capacitive reactance
- the peak current
- the rms current
- the phase difference between the current in the circuit and the source voltage

Solution

(a) $\omega = 200$

$$\therefore X_C = \frac{1}{\omega C} = \frac{1}{200 \times 100 \times 10^{-6}} = 50 \Omega$$

$$(b) i_0 = \frac{v_0}{X_C} = \frac{100}{50} = 2 \text{ A}$$

$$(c) I = \frac{i_0}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \text{ A}$$

(d) In a purely capacitive circuit the current leads the voltage by 90°

Example 9 A pure inductance of 1.0 H is connected across a 110V , 70 Hz source. Find

- the reactance
- the current
- the value of instantaneous current, when the voltage is at its peak

Solution

$$(a) X_L = \omega L = (2\pi f) L$$

$$= 2 \times \frac{22}{7} \times 70 \times 1 = 440 \Omega$$

(b) When a question says “find current”, we must report the rms current.

$$I = \frac{V}{X_L} = \frac{110\text{V}}{440\Omega} = \frac{1}{4} \text{ A}$$

The voltage ($= 110\text{V}$) given in the question is rms voltage.

(c) Phase difference between current and voltage is 90° , current is zero when the voltage is at its peak.



YOUR TURN

Q.8 What is the reactance of a capacitor when it is connected to a DC source?

Q.9 An AC source is connected to a capacitor. Will the rms current increase or decrease when a dielectric slab is inserted between the plates of the capacitor?

Q.10 A bulb is designed to operate at 24V DC supply. It is connected to an AC source and glows at normal brightness. What is peak voltage of the source?

Q.11 Plot the variation of the reactance of a capacitor and the reactance of an inductor with change in the frequency of the source.

Q.12 A 50Hz source is connected to a $10\ \mu\text{F}$ capacitor and the current is I_1 . When the same source is connected to a 10mH inductor, the current is I_2 . Which one is higher – I_1 or I_2 ?

Q.13 A capacitor has reactance of 50Ω for a source of 50Hz frequency. The source voltage is $v = 10 \sin(100\pi t - \frac{\pi}{3})$. Write the instantaneous value of current at time $t = 0$.

7. SERIES AC CIRCUITS

Now we will discuss circuits in which an AC source has been connected to a series combination of a resistance, a capacitor and an inductor. In general, we will see that there is a phase difference between the current and the applied voltage. The peak current (i_0) and the peak voltage (v_0) are related as

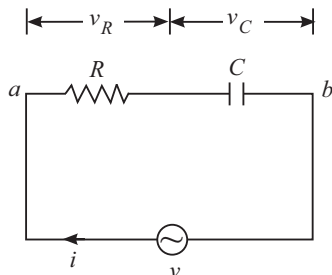
$$i_0 = \frac{v_0}{Z} \quad (14)$$

Z is known as *impedance* of the circuit. In a purely resistive circuit, the impedance is equal to R and in a purely capacitive circuit, the impedance is equal to X_C .

7.1 RC circuit

Consider an alternating source connected across a series combination of a resistor and a capacitor. Let the current in the circuit be

$$i = i_0 \sin \omega t$$



The instantaneous voltage across R will be

$$v_R = v_{R0} \sin \omega t \quad [v_{R0} = \text{peak voltage across } R]$$

$$\text{or, } v_R = (i_0 R) \sin \omega t \quad (i)$$

Recall that the current through a resistor and the voltage across it are always in the same phase.

The instantaneous voltage across the capacitor will be

$$v_C = v_{C0} \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$\text{or, } v_C = (i_0 X_C) \sin \left(\omega t - \frac{\pi}{2} \right) \quad (ii)$$

Here, $v_{C0} = i_0 X_C$ is the peak voltage across the capacitor.

The voltage across a capacitor lags in phase by $\frac{\pi}{2}$ compared to the current through it.

The instantaneous voltage across a and b (see figure) is sum of instantaneous voltage across R and C .

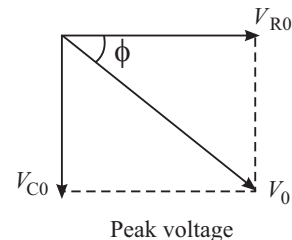
$$v = v_R + v_C$$

$$\text{or, } v = v_{R0} \sin \omega t + v_{C0} \sin \left(\omega t - \frac{\pi}{2} \right) \quad (iii)$$

As learnt in section 2.3 and 2.4, the above expression can be written as

$$v = v_0 \sin (\omega t - \phi) \quad (iv)$$

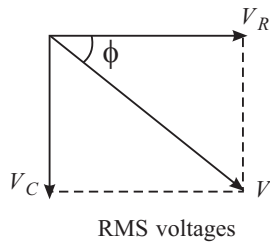
where v_0 and ϕ can be obtained by adding two vectors of magnitudes v_{R0} and v_{C0} inclined at 90° .



$$v_0 = \sqrt{v_{R0}^2 + v_{C0}^2} \quad (v)$$

Dividing by $\sqrt{2}$ on both sides, we get

$$\frac{v_0}{\sqrt{2}} = \sqrt{\left(\frac{v_{R0}}{\sqrt{2}} \right)^2 + \left(\frac{v_{C0}}{\sqrt{2}} \right)^2}$$



$$\text{or, } V = \sqrt{V_R^2 + V_C^2} \quad (15)$$

Here V is rms value of source voltage, V_R is rms value of voltage across R and V_C is rms value of the voltage across C .

Value of angle ϕ can be written as

$$\tan \phi = \frac{V_C}{V_R} = \frac{V_C}{V_R} \quad (16)$$

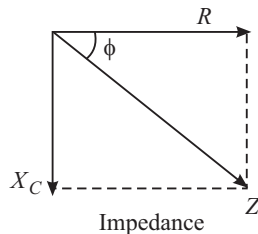
The voltage (i.e., the source voltage) lags behind the current in phase by an angle ϕ . In a purely capacitive circuit, the phase difference is 90° . Here, it is an acute angle.

Equation (v) can be written as

$$i_0 Z = \sqrt{(i_0 R)^2 + (i_0 X_C)^2}$$

$$\text{or, } Z = \sqrt{R^2 + X_C^2} \quad (17)$$

The impedance (Z) is the effective resistance offered by the combination of R and C .



The above discussion may appear a bit confusing to you. If needed, go through the sections 2.3 and 2.4 again. The basic thing to be understood is that the voltages across R and C both are changing sinusoidally with a phase difference of 90° between them. The peak value of the resultant voltage is not $(v_{R0} + v_{C0})$ because both v_R and v_C are not at their peak values simultaneously. The peak value of v is determined by the vector method of adding sinusoidal quantities.

Below we present the working method for solution of a series AC circuit. This follows directly from the above discussions.

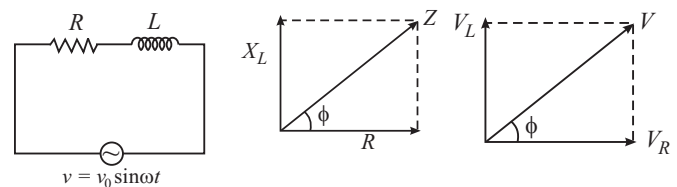
Represent the resistance by a vector of magnitude R drawn along the x -axis. Draw a vector of magnitude X_C , 90° behind of R (i.e., pointing in negative y direction). If there is an inductor in the circuit, draw a vector of magnitude X_L

90° ahead (pointing in the positive y direction) of R . Carry out vector addition of these to get the impedance (Z) of the circuit. The angle made by Z vector with the x -axis gives the phase difference between the current and the voltage. If Z lies above the x -axis, then the voltage leads in phase. When Z lies below the x -axis the current leads in phase. The peak current can be written as $i_0 = \frac{V_0}{Z}$.

In sections 7.2 and 7.3, we are illustrating this procedure.

7.2 LR circuit

Consider a series grouping of an inductor and a resistor connected to an AC voltage source. The voltage of the source is given by $v = v_0 \sin \omega t$.



To get the impedance (Z) of the circuit, draw a vector of magnitude R along the x -axis. Draw another vector of magnitude $X_L = \omega L$ along positive y -axis (i.e. 90° ahead of R). The impedance of the circuit is represented by the magnitude of the resultant of these two vectors.

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + (\omega L)^2} \quad (18)$$

The current leads in phase by an angle ϕ given as

$$\tan \phi = \frac{X_L}{R} = \frac{\omega L}{R} \quad (19)$$

$$\text{Peak current is } i_0 = \frac{V_0}{Z} = \frac{V_0}{\sqrt{R^2 + (\omega L)^2}}$$

$$\text{Rms current is } I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (\omega L)^2}}$$

Here, $V = \frac{V_0}{\sqrt{2}}$ is the rms voltage of the source.

The instantaneous current in the circuit is given by

$$i = i_0 \sin(\omega t - \phi) \quad [\text{voltage leads in phase}]$$

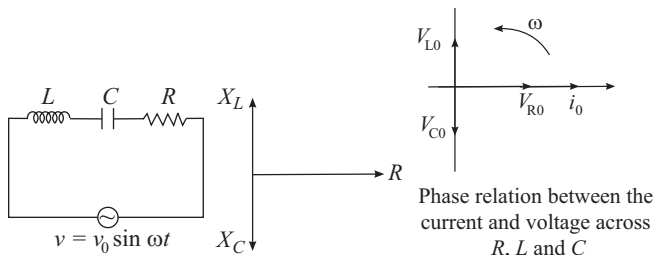
Also note that if V_R and V_L are rms voltages across R and L respectively, then the rms value of the source voltage is given by

$$V = \sqrt{V_R^2 + V_L^2}$$

7.3 LCR circuit

Consider a series combination of an inductor, a capacitor and a resistor connected to an AC source. The source voltage is $v = v_0 \sin \omega t$.

6.14 Magnetism and Electromagnetic Induction



The above figure shows the vector diagram to find the impedance of the circuit. Following three situations are possible:

(1) $X_L > X_C$

In this case, the vector sum of X_L and X_C is $X_L - X_C$ in the direction of X_L . This is the net reactance of the circuit. The impedance is

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad (20)$$

The voltage leads in phase by an angle ϕ given by

$$\tan \phi = \frac{X_L - X_C}{R} \quad (21)$$

The expression for current in the circuit will be

$$i = i_0 \sin(\omega t - \phi)$$

$$\text{where } i_0 = \frac{v_0}{Z}$$

(2) $X_C > X_L$

Reactance of the circuit is

$$X = X_C - X_L$$

Impedance is

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

The expression for Z is essentially same as given by equation (20).

In this case, the current leads in phase by an angle ϕ which is given by

$$\tan \phi = \frac{X_C - X_L}{R} \quad (22)$$

(3) $X_C = X_L$

In this case, the net reactance of the circuit is zero. The circuit behaves like a purely resistive circuit with $Z = R$. In our vector diagram, Z is along the x direction and $\phi = 0$. It means that the current and the voltage are in same phase.

$$\text{The peak current is } i_0 = \frac{v_0}{R}.$$

The rms current is $I = \frac{V}{R}$. Here, $V = \frac{v_0}{\sqrt{2}}$ is the rms voltage of the source.

In a LCR circuit this condition arises when

$$X_C = X_L \Rightarrow \frac{1}{\omega C} = \omega L$$

$$\Rightarrow \omega = \frac{1}{\sqrt{LC}} \quad (23)$$

The circuit is said to be in *resonance* under this condition. We will discuss resonance a little later.

Example 10 A 0.21H inductor and a 12Ω resistance are connected in series to a 220V, 50Hz AC source. Find the current in the circuit. Find the phase difference between the current and the applied voltage.

Solution

Concepts

- In an LR circuit, voltage leads in phase.
- Draw a vector of length R (along x direction) and draw another vector of length $X_L = \omega L$ directed along y direction. Resultant of R and X_L is impedance (Z).
- 220V is rms voltage. The question says 'find current'. It means we should report rms current. $I = \frac{V}{Z}$.
- The phase difference (ϕ) is the angle between the Z vector and R vector.

$$X_L = \omega L = 2\pi f \cdot L$$

$$= 2 \times 3.14 \times 50 \times 0.21 = 65.9\Omega$$

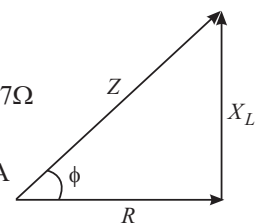
$$Z = \sqrt{R^2 + X_L^2}$$

$$= \sqrt{12^2 + (65.9)^2} = 67\Omega$$

$$\therefore I = \frac{V}{Z} = \frac{220}{67} = 3.28 \text{ A}$$

$$\text{Also, } \tan \phi = \frac{X_L}{R} = \frac{65.9}{12} = 5.5$$

$$\phi = \tan^{-1}(5.5) \simeq 80^\circ$$



Example 11 AC circuit with no resistance

A capacitance of 10μF and an inductor of 1H are connected in series to a 220V, 50Hz source. Find the

- impedance of the circuit
- current in the circuit
- phase difference between the current and the voltage

Solution
Concepts

- (i) There is no resistance. Circuit only has reactance. The resultant reactance is $X = X_C - X_L$ or $X = X_L - X_C$ depending on whether X_C is larger or X_L . The impedance
- $$Z = \sqrt{R^2 + (X_C - X_L)^2} = |X_C - X_L|$$
- (ii) If $X_C > X_L$, the current leads in phase and if $X_L > X_C$, the current lags.

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2 \times 3.14 \times 50 \times 10 \times 10^{-6}} = 318.5 \Omega$$

$$X_L = \omega L = 2\pi f L = 2 \times 3.14 \times 50 \times 1 = 314 \Omega$$

$$\therefore X_C > X_L$$

$$(a) Z = X_C - X_L = 4.5 \Omega$$

$$(b) I = \frac{V}{Z} = \frac{220}{4.5} = 48.9 \text{ A}$$

$$(c) \text{ Current leads in phase by } 90^\circ$$

Example 12 A 100V AC source of frequency 500 Hz is connected to an LCR circuit having $L = 7 \mu\text{H}$, $C = 70 \mu\text{F}$ and $R = 10 \Omega$; all connected in series. Find the voltage across the resistance. Does the current lead in phase?

Solution
Concepts

- (i) $Z = \sqrt{R^2 + (X_L - X_C)^2}$
 (ii) $V_R = IR$
 (iii) Remember, "Voltage across R" means the rms voltage across R

$$X_L = \omega L = 2\pi f L = 2 \times \frac{22}{7} \times 500 \times 7 \times 10^{-3} = 22 \Omega$$

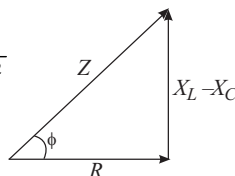
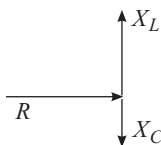
$$X_C = \frac{1}{\omega C}$$

$$= \frac{1}{2 \times \frac{22}{7} \times 500 \times 70 \times 10^{-6}} = 4.5 \Omega$$

$$X = X_L - X_C = 22 - 4.5 = 17.5 \Omega$$

$$Z = \sqrt{R^2 + X^2} = \sqrt{10^2 + (17.5)^2} = 20.2 \Omega$$

$$\therefore I = \frac{V}{Z} = \frac{100}{20.2} = 4.95 \text{ A}$$

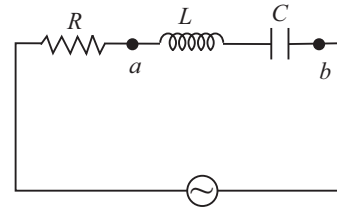


$$\therefore V_R = I.R. = 4.95 \times 10 = 49.5 \text{ V}$$

Since $X_L > X_C$, the voltage leads in phase.

Note: The voltage across the capacitor and the inductor can be written as $V_C = I X_C$, $V_L = I X_L$

Example 13 In the LCR circuit shown, the voltage across R, L and C are 40V, 70V and 40V, respectively.



(a) Find the source voltage

(b) If $R = 10 \Omega$, Find X_C .

(c) Find the voltage measured between the points a and b.

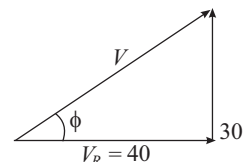
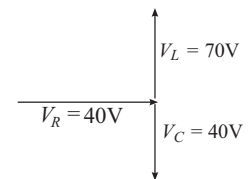
Solution
Concepts

- (i) Given voltages are the rms values.
 (ii) The three voltages are having phase difference between them. Their resultant will be obtained by vector method.
 (iii) The voltage across L is 90° ahead of the current and the voltage across C is 90° behind the current in phase. This means that the phase difference between these two voltages is 180° .

- (a) The voltage across L is 90° ahead of the voltage across R and the voltage across C is 90° behind the v_R .

To get the source voltage, we need to add three vectors as shown.

$$V = \sqrt{(V_R)^2 + (V_L - V_C)^2} = \sqrt{40^2 + 30^2} = 50 \text{ V}$$



$$(b) V_R = IR \Rightarrow I = \frac{40 \text{ V}}{10 \Omega} = 4 \text{ A}$$

$$\text{and } I X_C = V_C \Rightarrow X_C = \frac{40 \text{ V}}{4 \text{ A}} = 10 \Omega$$

$$(c) \text{ Voltage across the points a and b is } V_L - V_C = 30 \text{ V}$$

In Short

- (i) The voltage across a resistor is always in phase with the current through it. Peak current and peak voltage are related as $i_0 = \frac{V_0}{R}$. Rms current and rms voltage are related as $I = \frac{V}{R}$.
- (ii) In an AC circuit, capacitive reactance is $X_C = \frac{1}{\omega C}$ and an inductive reactance is $X_L = \omega L$.
- (iii) The effective reactance in a circuit having both C and L is $X = (X_C - X_L)$ or $(X_L - X_C)$ depending on whether X_C is larger or X_L .
- (iv) In a RC circuit or in an LCR circuit having $X_C > X_L$, the current leads in phase.
- (v) In an LR circuit or in an LCR circuit having $X_L > X_C$, the voltage leads in phase.
- (vi) The phase difference between the current and voltage is $\phi = \tan^{-1} \left(\frac{X}{R} \right)$.
- (vii) The voltage across a capacitor lags the current through it by 90° and the voltage across an inductor always leads the current by 90° .
- (viii) The impedance of a series LCR circuit is $Z = \sqrt{R^2 + (X_L - X_C)^2}$
when there is no capacitor, $Z = \sqrt{R^2 + X_L^2}$
when there is no inductor, $Z = \sqrt{R^2 + X_C^2}$
when there is no resistor, $Z = |X_L - X_C|$
when there are no capacitor and inductor or when $X_C = X_L$, $Z = R$. In this case, the current and voltage are in same phase.
- (ix) $i_0 = \frac{V_0}{Z}$ and $I = \frac{V}{Z}$



YOUR TURN

Q.14 A coil having a resistance $R = 50\Omega$ and an inductance of $L = 0.5H$ is connected to an AC source of 110V, 50Hz. Find the current in the circuit

Q.15 A pure inductor is connected across a 50V, 50Hz source and the peak current is 2A. Now the same inductor is connected across a 50V, 100Hz source. Find the peak current.

Q.16 A 100Ω resistance is connected in series to a 4H inductor. The voltage across the resistor varies as

$v_R = (20.0V) \sin(10^3 t)$. Here t is in second.

(a) Find the reactance in the circuit.

(b) Find the time dependence of voltage across the inductor.

Q.17 In a series LCR circuit, $L = 1H$, $C = 10^{-6}F$, $R = 10\Omega$. Find the frequency of the source for which the current in the circuit will be in phase with the source voltage.

8. POWER IN AC CIRCUITS

In general, there is a phase difference between the source voltage and the current in an AC circuit. The two can be expressed as

$$v = v_0 \sin \omega t \quad \text{and} \quad i = i_0 \sin (\omega t + \phi)$$

The work done by the source during the time interval t to $t + dt$ is

$$dW = v i dt = v_0 i_0 \sin \omega t \sin (\omega t + \phi) dt$$

$$= [(v_0 i_0 \cos \phi) \sin^2 \omega t + (v_0 i_0 \sin \phi) \sin \omega t \cos \omega t] dt$$

Work done by the source in one complete cycle is

$$W = \int_0^T dW$$

$$= (v_0 i_0 \cos \phi) \int_0^T \sin^2 \omega t \cdot dt + \left(\frac{1}{2} v_0 i_0 \sin \phi \right) \int_0^T \sin(2\omega t) dt$$

The average power delivered by the source is,

$$P = \frac{W}{T} = (v_0 i_0 \cos \phi) \frac{1}{T} \int_0^T \sin^2 \omega t \cdot dt +$$

$$\left(\frac{1}{2} v_0 i_0 \sin \phi \right) \frac{1}{T} \int_0^T \sin(2\omega t) dt$$

$$\text{or, } P = (v_0 i_0 \cos \phi) \left(\frac{1}{2} \right) + 0$$

Because $\frac{1}{T} \int_0^T \sin^2 \omega t \cdot dt$ is average of $\sin^2 \omega t$ over a cycle and it is $\frac{1}{2}$ and $\frac{1}{T} \int_0^T \sin(2\omega t) dt$ is average of $\sin(2\omega t)$ that is zero. One can prove these results by performing the integrations after putting $\omega = \frac{2\pi}{T}$.

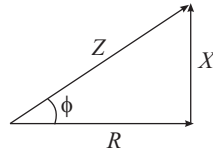
Hence, average power supplied by the source to the circuit is

$$P = \frac{v_0}{\sqrt{2}} \frac{i_0}{\sqrt{2}} \cos \phi = VI \cos \phi \quad \dots(24)$$

Following points are worth noting.

- (a) $\cos \phi$ is known as power factor of the circuit. It is equal to

$$\cos \phi = \frac{R}{Z}$$



- (b) $P = VI \cos \phi = VI \frac{R}{Z} = \left(\frac{V}{Z}\right) IR = I^2 R \quad \dots(25)$

$I^2 R$ is power consumed by the resistance in the circuit. In a way, we have proved that the power supplied by the source is consumed by the resistance only. *A capacitor or an inductor does not consume power in an AC circuit.*

- (c) When $X = 0$, $\phi = 0$ and $\cos \phi = 1$.

This is the ideal value of power factor.

- (d) For purely reactive circuits (circuits having zero resistance and only capacitance or/and inductance) $\phi = \pm \frac{\pi}{2}$. Therefore, $\cos \phi = 0$. Such circuits will not consume energy.
- (e) When a question asks for calculating power in an AC circuit, it is asking to calculate the average power given by equation (24). Over a long period of time, the average is still given by the same equation.

Example 14 Ideal value of power factor

Why the ideal value of power factor is considered to be 1.0? A factory deploying large motors has a poor power factor. What can you suggest to improve its power factor?

Solution

Suppose the supply voltage is V . For example, the transformer supplying power to your house supplies at $V = 220$ V. Now, you wish to consume a fixed power P by lighting some bulbs, fan, water pump, etc.

$$P = VI \cos \phi$$

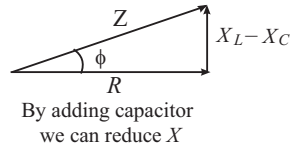
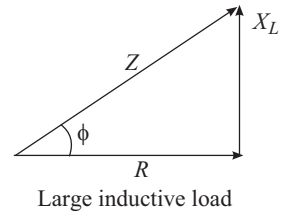
The current that you draw from the transformer is

$$I = \frac{P}{V \cos \phi}$$

If the power factor is low, say 0.4, you will draw a high current for consuming a fixed power P . On the other hand, you will draw half the current, for consuming same power P , if the power factor is 0.8.

The lesser the current, smaller will be the energy loss in transmission lines. The current will be smallest for $\cos \phi = 1.0$.

In a factory, which has large number of motors, the inductive load is high (due to coils in the armature of the motors). The phase difference ϕ between the voltage and the current is high and $\cos \phi$ is low. To improve the power factor, we need to add capacitor banks in the circuit. This does not lead to additional power consumption (a capacitor does not consume power) but decreases X . This decreases ϕ and increases $\cos \phi$.



Example 15 Wattless current

In an AC circuit, can we have negligible power consumption but a significant current?

Solution

A circuit having negligible resistance but some reactance (X) will have current given by

$$I = \frac{V}{X}$$

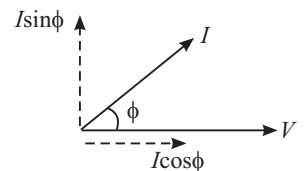
But power consumption in the circuit will be negligible. A capacitor and an inductor does not consume power.

Putting it differently, we can say that $\phi \rightarrow \frac{\pi}{2}$ and $\cos \phi \rightarrow 0$.

Such a current, which does not result in energy consumption is often termed as *wattless current*.

Note: $P = V(I \cos \phi)$

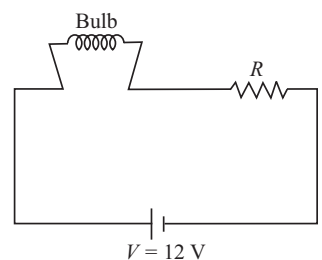
The component of the current phasor that is perpendicular to the voltage phasor ($= I \sin \phi$) is termed as wattless component of the current.



9. CHOKE COIL

Suppose you have a DC source of 12V and a bulb designed to operate at 6V. You cannot connect the bulb directly to the source as it will get damaged. The only thing that you can do is to put an additional resistance (R) in series with the bulb.

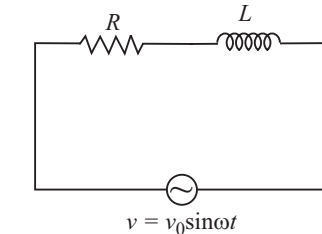
By selecting a suitable value of R , we can ensure that the potential difference across our bulb remains 6V. But this arrangement leads to unwanted energy loss in the resistance R .



6.18 Magnetism and Electromagnetic Induction

We avoid unwanted energy loss in AC circuits, while regulating voltage/current, by using an inductor coil in place of a resistance. Such an inductor coil having small resistance (ideally zero) is known as a choke coil. It is commonly seen in tubelights used in houses.

Household power supply is 220V, 50Hz. If we connect a tubelight directly to this supply it will get damaged. To reduce the current (and hence voltage across the tubelight), we connect a high inductance, low resistance choke coil in



A choke reduces the current without any additional energy loss

series. The figure shows the tubelight as a resistance and the choke coil as an inductance. [In practice, the inductor coil will also have a small resistance, r].

Impedance of the circuit is $Z = \sqrt{R^2 + (\omega L)^2}$

Current in the circuit is $I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (\omega L)^2}}$

The potential difference across R is $V_R = IR = V \frac{R}{\sqrt{R^2 + (\omega L)^2}}$

Thus, we have successfully reduced the potential difference across the resistance by increasing the impedance. But this does not result in any extra energy loss as the inductor does not consume power.



YOUR TURN

Q.18 Regulator of a fan consists of variable resistance. The resistance can be set at five distinct values controlling the speed of the fan. Does our energy bill reduce when we operate the fan at low speed?

Q.19 A fan regulator has a variable capacitance. When you turn the knob, the capacitance value changes. This lets you control the current and thereby, the speed of the fan.

Is this more effective regulator than the one using variable resistance?

Q.20 A bulb is rated 55W and is designed to operate at 110V DC supply. The bulb is to be connected across a 220V, 50 Hz AC supply. Find the inductance of the choke coil that must be connected in series, neglect the resistance of the coil.

10. RESONANCE IN SERIES LCR CIRCUIT

We studied in the last chapter that a current can persist forever in an ideal LC circuit having no resistance. The energy keeps shifting between the magnetic field energy in the inductor and the electric field energy in the capacitor. The natural frequency of oscillation is

$$\omega_0 = \frac{1}{\sqrt{LC}} \Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}} \quad (i)$$

However, in practice a circuit will always have some resistance and the energy gets dissipated quickly. The oscillations die. To maintain the current, we need to pump in energy on a continuous basis. This can be done by adding an AC source to the circuit. When the frequency of the source is equal to the natural frequency of the system given by (i), resonance happens. Under such conditions the current

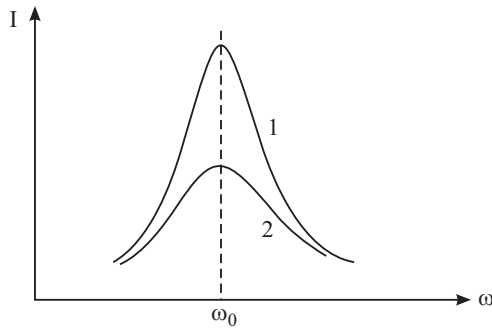
in the circuit reaches a maximum. This is what we observed in section 7.3.

In a series LCR circuit, if we have option of changing the frequency of the source, keeping everything else fixed, the impedance becomes least when the frequency of the source is given by equation (i). In this case $X_C = X_L$ and the impedance is $Z = R$. The current is maximum. Rms value of the current is

$$I_m = \frac{V}{R}$$

The current and voltage are in same phase when the circuit is in resonance.

The figure shown below gives the qualitative behaviour of the current in an LCR circuit when frequency of the applied source is varied. The current falls as frequency is increased above ω_0 or is decreased below ω_0 .



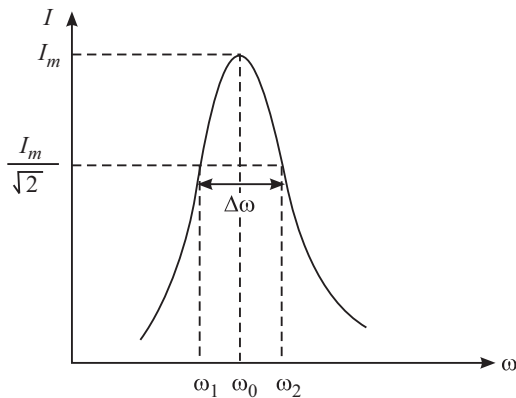
Resonance frequency is ω_0 . The graph 1 is for a circuit having lower resistance (R_1) and the graph 2 is for a circuit having higher resistance (R_2)

Sharpness of resonance

Consider a series LCR circuit in resonance. If a small percentage change in frequency of the source causes a big drop in the current (as shown in curve 1 of the above figure) in the circuit, the resonance is said to be sharp. The resonance curve depicted as 2 is relatively less sharp. Here the same percentage change in frequency will not cause much drop in the current.

Consider the resonance curve shown. The current is maximum (I_m) at the resonance angular frequency ω_0 . There are two frequencies ω_1 and ω_2 for which the current in the circuit is $\frac{I_m}{\sqrt{2}}$. These frequencies are known as *half power frequencies*.

Obviously, the power consumed ($P = I^2 R$) in the circuit at these frequencies is half the power consumed at resonance.



$\Delta\omega = \omega_2 - \omega_1$ is known as the *band width* of the resonance. let us calculate this.

The current in the circuit for source frequency ω is

$$I = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

Desired frequencies for $I = \frac{I_m}{\sqrt{2}}$ can be obtained as

$$\frac{I_m}{\sqrt{2}} = \frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$\text{But } I_m = \frac{V}{R}$$

$$\therefore \frac{1}{\sqrt{2}R} = \frac{1}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

$$\text{or, } 2R^2 = R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2$$

$$\text{or, } \omega L - \frac{1}{\omega C} = \pm R \quad \dots(ii)$$

When we proceed with + sign, the value of ω that we get represents ω_2 (the upper half power frequency) and when we solve the above equation for ω taking $-R$ on right side, the value of ω that we obtain represents ω_1 (the lower half power frequency). First we proceed with + sign.

$$\omega L - \frac{1}{\omega C} = R$$

$$\text{or, } (LC) \omega^2 - (RC) \omega - 1 = 0$$

$$\therefore \omega = \frac{RC \pm \sqrt{R^2 C^2 + 4LC}}{2LC}$$

$$\Rightarrow \omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{4L}\right)^2 + \frac{1}{LC}} \quad (iii)$$

Again starting with equation (ii) with $-R$ on the right side and solving the quadratic equation for ω gives

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{4L}\right)^2 + \frac{1}{LC}} \quad (iv)$$

$$\therefore \omega_2 - \omega_1 = \frac{R}{L}$$

$$\text{or, } \Delta\omega = \frac{R}{L} \quad \dots(26)$$

A dimensionless number known as quality factor (Q) is measure of sharpness of a resonance. It is defined as

$$Q = \frac{\omega_0}{\Delta\omega} \quad \dots(27)$$

Higher value of Q means a sharper resonance.

Using equation (26), the quality factor can be expressed as

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\sqrt{LC}} \frac{L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}} \quad \dots(28)$$

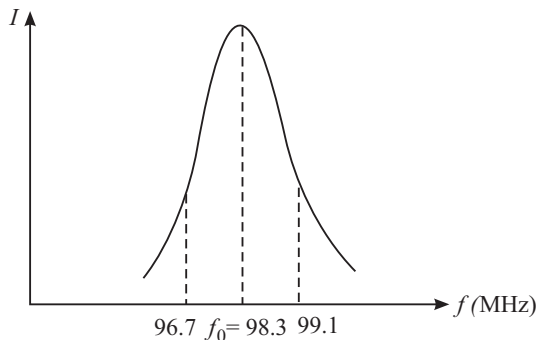
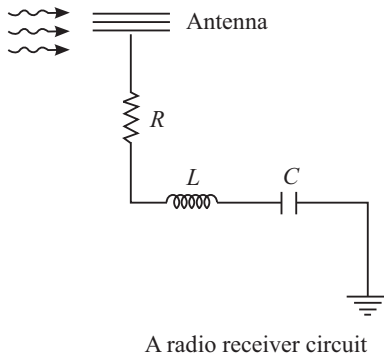
Significance of a sharp resonance

Consider that your city has three FM radio stations broadcasting at frequencies 96.7 MHz, 98.3 MHz and 99.1 MHz. You wish to listen to the music being played by 98.3 MHz. Your receiver set has a series LCR circuit and you can change its capacitance by turning a knob or pushing a button. The capacitor is adjusted to have a value such that

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = 98.3 \text{ MHz}$$

[In practice, the value of f_0 can be slightly larger or smaller than the exact value 98.3 MHz.]

The antenna of the receiver set receives all three radio waves and generates small emfs corresponding to each of them. These emfs are fed to the series LCR circuit. For emf corresponding to the frequency 98.3 MHz, resonance takes place. The current corresponding to this frequency is large. There is current in the circuit for other emfs (frequencies 96.7 and 99.1 MHz) also. But these currents are relatively smaller. If the resonance is not sharp, the current corresponding to these two frequencies can be reasonably high and in the further processing it will be difficult to suppress these frequencies. Most likely, we will hear all three stations simultaneously! But, if the resonance is sharp, the current corresponding to 96.7 and 99.1 MHz will be very small. They can be filtered out in further circuit and 98.3 MHz can be amplified.



A small change in frequency causes the current to drop sharply from its peak value at resonance. Such sharp resonance helps in picking up one frequency out of many

Therefore, a sharp resonance can be used to filter out one frequency from many others.

Example 16 Many formulae for Q

In a series LCR circuit, X_{L0} and X_{C0} represent the inductive and capacitive reactance at the resonance frequency. V_{C0} , V_{R0} and V_{L0} represent the voltage across C , R and L during resonance. Show that the quality factor of the resonance is equal to

- | | |
|-----------------------------|-----------------------------|
| (a) $\frac{X_{L0}}{R}$ | (b) $\frac{X_{C0}}{R}$ |
| (c) $\frac{V_{C0}}{V_{R0}}$ | (d) $\frac{V_{L0}}{V_{R0}}$ |

Solution

Concepts

- (i) From equation (28) $Q = \frac{\omega_0 L}{R}$
- (ii) In resonance $X_{C0} = X_{L0}$
- (iii) In resonance $V_{C0} = V_{L0}$

- (a) $Q = \frac{\omega_0 L}{R} = \frac{X_{L0}}{R}$
- (b) In resonance, $X_{L0} = X_{C0}$
 $\therefore Q = \frac{X_{C0}}{R}$
- (c) $Q = \frac{X_{L0}}{R} = \frac{IX_{L0}}{IR}$ [I = current during resonance]
 or, $Q = \frac{V_{L0}}{V_{R0}}$
- (d) In resonance, $V_{C0} = V_{L0}$
 $\therefore Q = \frac{V_{C0}}{V_{R0}}$

Example 17 An FM antenna is connected to a circuit having inductance $L = 10^{-6} \text{ H}$, a capacitance $C = 10^{-12} \text{ F}$ and a resistance $R = 100 \Omega$.

- (a) For what angular frequency ω_0 of the incoming wave will the circuit be in tune – that is, for what ω_0 will the current in the circuit be maximum?
- (b) Find the quality factor Q for the resonance.
- (c) Assume that the radio signal induces a sinusoidally changing emf of amplitude 10^{-4} V in the circuit. Find the current amplitude at this “in tune” frequency.

Solution
Concepts

- (i) Circuit will be in tune for resonance frequency of the given circuit.
- (ii) The current amplitude will be $i_0 = \frac{v_0}{R}$.

$$(a) \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-6} \times 10^{-12}}} = 10^9 \text{ rad s}^{-1}$$

$$(b) Q = \frac{\omega_0 L}{R} = \frac{10^9 \times 10^{-6}}{100} = 10$$

$$(c) i_0 = \frac{v_0}{R} = \frac{10^{-4}}{100} = 10^{-6} \text{ A}$$


YOUR TURN

Q.21 An LCR series circuit has a resonance frequency of 10^6 Hz and its frequency bandwidth is 10^3 Hz . Another circuit has a resonance frequency of 10^7 Hz and a frequency bandwidth of 10^3 Hz . Which has a sharper resonance curve?

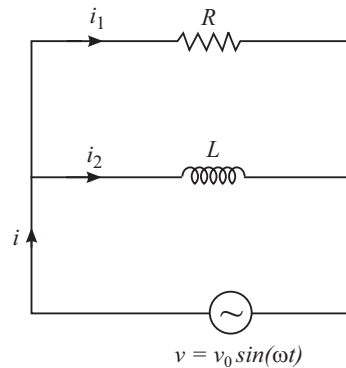
Q.22 A series LCR circuit has $R = 10 \Omega$ and $L = 0.1 \text{ mH}$. Find the frequency bandwidth of the resonance in Hz .

11. PARALLEL CIRCUIT

Consider a parallel LR circuit shown in the figure. The source voltage is

$$v = v_0 \sin(\omega t)$$

In a series circuit, the current is same for all elements. In a parallel circuit, it is the voltage that is same across all elements.



The current in R will be in phase with the applied voltage. Therefore, the current i_1 can be written as

$$i_1 = i_{10} \sin \omega t = \frac{v_0}{R} \sin \omega t.$$

The current in L will be 90° behind in phase compared to the voltage.

$$\therefore i_2 = i_{20} \sin \left(\omega t - \frac{\pi}{2} \right) = \frac{v_0}{X_L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

The current through the source is

$$i = i_1 + i_2$$

$$\text{or, } i = i_{10} \sin \omega t + i_{20} \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$\text{or, } i = i_0 \sin(\omega t - \phi)$$

$$\text{where } i_0 = \sqrt{i_{10}^2 + i_{20}^2}$$

$$\text{or, } i_0 = \sqrt{\frac{v_0^2}{R^2} + \frac{v_0^2}{X_L^2}}$$

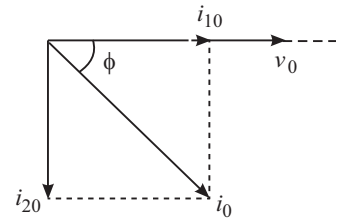
$$\text{or, } i_0 = v_0 \sqrt{\frac{1}{R^2} + \frac{1}{X_L^2}}$$

$$\text{or, } i_0 = \frac{v_0}{Z} \text{ where}$$

$$\frac{1}{Z} = \sqrt{\frac{1}{R^2} + \frac{1}{X_L^2}}$$

In general, when R , L and C are in parallel

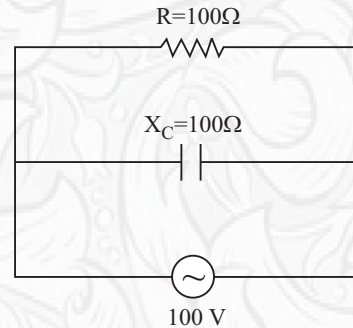
$$\frac{1}{Z} = \sqrt{\frac{1}{R^2} + \left(\frac{1}{X_L} - \frac{1}{X_C} \right)^2}$$





YOUR TURN

Q.23 In the circuit shown, find the current through the source. What is phase difference between the current and the voltage?



12. MEASUREMENT OF AC CURRENT AND VOLTAGE

In a DC ammeter or voltmeter, a coil is free to rotate in a magnetic field. When a current is passed, the coil experiences a torque and deflects. If an AC current is passed, through the coil, the magnetic torque will reverse its direction quickly as the direction of the current changes. Due to its inertia and friction, the coil will practically remain unmoved as the direction of torque changes at a rapid frequency.

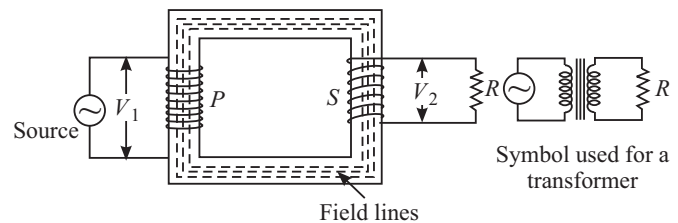
To be able to measure an AC current, we need to use a property so that the deflection of the moving part depends on i^2 . This will ensure that deflection remains independent of the direction of i . Hot-wire instruments are designed on this principle and their scale is graduated to directly give us the value of rms current. Voltmeters can also be designed to directly read v_{rms} .

AC current can also be measured by first rectifying it (i.e., converting it into dc) and then measuring it.

13. TRANSFORMER

A transformer is a device that can be used to *step-up* (i.e. increase) or to *step-down* (i.e. decrease) an AC voltage. Your household power supply is at 220V. If you wish to power a 440V electric motor for some purpose, you will have to use a transformer to step up the voltage from 220V to 440V.

A transformer works on the principle of mutual induction. The figure given below shows the design of a simple transformer. Two coils, having different number of turns, are wound separately on a *laminated, silicon steel core*. One of the coils is connected to the source of alternating voltage. This coil (P) is known as the *primary coil*. The other coil is known as the *secondary coil*. The induced emf in this coil is our output voltage.



The source drives an alternating current in the primary coil. This current produces a magnetic field which gets multiplied due to the ferromagnetic core. The entire core gets strongly magnetised and there is not much error in assuming that every magnetic field line that passes through the primary coil, also passes through the secondary coil. In other words, the magnetic flux (ϕ) through a single turn of the secondary coil is nearly same as the flux through a single turn of the primary. Since the current is alternating, the magnetic field that it produces is also alternating and therefore ϕ is changing with time. This induces an emf in the coils. Let the number of turns in the primary coil be N_1 . The emf induced in this coil is

$$|\varepsilon_1| = \left| N_1 \frac{d\phi}{dt} \right| \quad \dots(i)$$

The primary coil is made of thick copper wire and its resistance can be neglected. The circuit is essentially an inductance connected to an AC source. The magnitude of emf induced in the coil is always same as the source emf. It means the rms value of the source voltage (V_1) is same as rms value of induced voltage.

A typical transformer has large number of turns in its coils. The inductance of the primary coil is high. Its inductive reactance is also high. The current in the primary circuit is very low.

The emf induced in the secondary coil is

$$|\varepsilon_2| = \left| N_2 \frac{d\phi}{dt} \right| \quad \dots(ii)$$

where N_2 is number of turns in the secondary coil.

This induced emf acts as the output voltage of the transformer. From (i) and (ii)

$$\left| \frac{\mathcal{E}_2}{\mathcal{E}_1} \right| = \frac{N_2}{N_1}$$

or, $\frac{V_2}{V_1} = \frac{N_2}{N_1}$ (29)

Here, V_1 and V_2 are rms values of the output voltage and the input source voltage. The equations (i), (ii) and (29) are valid for any values of currents in the two coils.

Assume that a load resistance R is connected to the terminals of the secondary coil. A current i_2 flows in the secondary circuit. This current will induce an additional emf and hence a current i_1 flows in the primary coil. Original current in the primary coil was negligible. Therefore, i_1 is the net current in the coil. It can be shown that *this current is in phase with the source voltage*. Therefore, the power supplied by the source is

$$P_{\text{input}} = V_1 I_1$$

Here, I_1 is rms value of current (i_1) in the primary. The output power is

$$P_{\text{output}} = V_2 I_2$$

Here, V_2 is rms value of emf in the secondary coil and I_2 is the rms current in the load resistance. The current (I_2) and the voltage (V_2) in the *output resistance* are in same phase.

If our transformer is 100 percent efficient, i.e., there is no energy loss, then

$$P_{\text{output}} = P_{\text{input}}$$

$$V_2 I_2 = V_1 I_1$$

or, $\frac{I_2}{I_1} = \frac{V_1}{V_2} = \frac{N_1}{N_2}$... (30)

It can be shown that the output current is 180° out of phase with the input current.

A transformer having $N_2 > N_1$ will give the output voltage (V_2) larger than the input voltage (V_1). Such a transformer is known as a step up transformer. *There is no gain in terms of energy*. The secondary current is less than the primary current in this case (equation 30).

A transformer having $N_2 < N_1$ will have an output voltage less than the input source voltage. Such a transformer is known as a step down transformer. The current in the secondary (I_2) is higher than that in the primary ($= I_1$)

Efficiency

Equation (30) has been written assuming that the transformer is 100 percent efficient. It means that the entire power delivered by the source gets transferred to the load resistance. Practically, this is never true. There is always some loss of energy. The primary cause of energy loss are:

- $I^2 R$ loss:** The two coils will have some resistance and they will dissipate heat when a current flows through them. This loss is minimised by using thick copper wires for the coils.
- Eddy current loss:** Eddy current induced in the core causes heating of the core and energy loss. The silicon-steel used in the core has high resistivity and thereby, the strength of eddy current is low. Moreover, use of laminated thin sheets to make a core reduces the possible paths for eddy current.
- Hysteresis loss:** Hysteresis in the core causes heating. This loss is minimised by using special silicon steel for making the core. The hysteresis loop of the material has smaller area.
- Flux leakage:** We have assumed that the flux linked with one turn of the primary is same as that linked with one turn of the secondary. In practice, this may not be true. However, by winding the two coils one over another (with insulation in between) can greatly reduce the flux leakage.

The efficiency (η) of a transformer is defined as

$$\eta = \frac{P_{\text{output}}}{P_{\text{input}}} = \frac{I_2 V_2}{I_1 V_1} \quad (31)$$

$$\text{or, } \frac{I_2}{I_1} = \eta \frac{V_1}{V_2} = \eta \frac{N_1}{N_2} \quad (32)$$

Modern engineering design can easily give us transformer having efficiency of 0.99.

Use of transformer in power transmission

We use electricity in our house at 220V. A higher voltage is deemed to be unsafe and dangerous. Suppose a power plant is located hundreds of kilometers from a city. If the power is transmitted at 220V from the power plant to the city, current in the transmission lines will be high. This will cause a lot of energy to be wasted in the transmission wires as $I^2 R$ loss. A transformer helps in reducing such loss. The voltage at the power plant is stepped up (say to 66 kV) and transmitted to the city. The same amount of power gets transmitted at a much less current in the wire. The heat loss is low. This power (at 66 kV) is received at a substation (usually on outskirts of the city as 66kV is too dangerous) and a step down transformer reduces the voltage to 440V. The transmission lines running inside the city are typically carrying power at 440 V. Just before supplying to a house (or a group of houses), the voltage is again stepped down to 220V.

In nutshell, we try to transmit power at maximum possible voltage keeping safety in perspective. This reduces energy loss. At the same time, we wish to use the power at a safe low voltage. All this is possible due to transformers. It

6.24 Magnetism and Electromagnetic Induction

would not be wrong to say that the wide use of AC, instead of DC, is due to existence of transformers.

Example 18 A bulb in the primary circuit of a transformer

There is a bulb connected to the primary circuit of a transformer. When the secondary circuit is open (i.e. not connected to a load), the bulb is dim. The bulb becomes brighter after a load resistance is connected to the secondary of the transformer. Explain.

Solution

The inductive reactance of the primary coil is high and the current in the primary circuit is low. The bulb is dim.

When a load is connected to the secondary coil, a current flows in the secondary coil. Due to mutual inductance, this

induces an additional emf in the primary coil. This causes the current in the primary to surge and the bulb becomes brighter.

Example 19 A transformer has 1000 turns in its primary coil. It is used to step up a source voltage of 110V to 220V. How many turns must be there in the secondary coil?

Solution

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} \Rightarrow \frac{220}{110} = \frac{N_2}{1000}$$

$$\Rightarrow N_2 = 2000 \text{ turns}$$

In Short

- (i) Only resistance consumes power in an AC circuit.
- (ii) Power supplied by an AC source has an average value of $P = VI \cos \phi$.
Here, ϕ is phase difference between the current and the voltage and $\cos \phi$ is known as power factor.
- (iii) Power supplied by the source is same as the power dissipated in the resistance in the circuit. For a series circuit, $P = VI \cos \phi = I^2 R$.
- (iv) $I \sin \phi$ is known as wattless component of current.
- (v) A choke coil is basically an inductor with as small resistance as possible. It is used to regulate current in an AC circuit without wastage of energy.
- (vi) In a series LCR circuit, the current is maximum when the source has an angular frequency given by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The frequency is said to be the resonance frequency of a circuit. When a circuit is in resonance, the current is in phase with the voltage and impedance is $Z = R$.

- (vii) Current in a resonant circuit is $I = \frac{V}{R}$.
- (viii) A resonance is said to be sharp if the current in the circuit drops sharply when the source frequency is changed a little from the resonance frequency.

- (ix) Sharpness of a resonance is measured by quality factor, defined as

$$Q = \frac{\omega_0}{\Delta \omega} = \frac{\omega_0 L}{R}$$

Here, ω_0 = resonance angular frequency

$$\Delta \omega = \text{resonance (angular) bandwidth} = \frac{R}{L}$$

- (x) A sharp resonance can be used to filter out one frequency among many.
- (xi) In parallel branches, the voltage across each branch is the same. The currents in parallel branches may differ in phase.
- (xii) A transformer is used to step up or step down an AC voltage
- (xiii) The governing equation for a transformer is

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

Here, V_2 and V_1 are output and input voltage and N_2 and N_1 are the number of turns in the secondary and the primary coils, respectively.

- (xiv) The major sources of energy loss in a transformer are heating of coil, heating of the core due to hysteresis and eddy current and flux leakage.



YOUR TURN

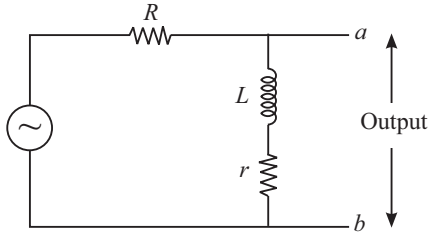
Q.24 A transformer has 100 turns in its primary and 300 turns in its secondary. The primary coil is connected to a 60V DC supply. Find the voltage across the secondary.

Q.25 A transformer has two coils having 1000 and 2000 turns. You wish to step down a 440 V supply. Which coil will be used as primary? What will be the output voltage?

Miscellaneous Examples

Example 20 A high pass filter

A high pass filter is a circuit which filters out low frequency AC. The figure shows an inductor coil having inductance L and resistance r connected in series to a resistance R . The output is received across the terminals a and b of the inductor coil. The source voltage is V .



- (a) Is it right to say that the output voltage increases with increasing frequency of the source? What is the limiting value of the output voltage as the frequency of the source tends to infinity? What is the value when the frequency of the source tends to zero?
- (b) Find the frequency at which $\frac{v_{20}}{v_{10}} = \frac{1}{2}$ where v_{20} and v_{10} are peak values of the output voltage and the input (source) voltage.

Solution

Concepts

With increasing ω , the inductive reactance ($X_L = \omega L$) increases.

- (a) For high ω , X_L is high. The voltage across the inductor coil will be high. For low ω , X_L is small and voltage across the inductor coil is low. It means for high frequency, we get higher output and for low frequency, we get lower output. If a mixture of frequencies is fed (i.e., there are multiplier sources), we receive significant output for high frequencies only. Hence the name high pass filter.

For $\omega \rightarrow \infty$, $X_L \rightarrow \infty$,

$V_{\text{output}} = V_{\text{source}} = V$

when $\omega \rightarrow 0$, $X_L \rightarrow 0$

$V_{\text{output}} \rightarrow 0$

(b) $Z = \sqrt{(R+r)^2 + X_L^2}$

Peak current, $i_0 = \frac{v_{10}}{\sqrt{(R+r)^2 + X_L^2}}$

Impedance of the inductor coil is

$Z_{\text{coil}} = \sqrt{r^2 + X_L^2}$

$$\therefore v_{20} = i_0 Z_{\text{coil}} = \frac{v_{10} \sqrt{r^2 + X_L^2}}{\sqrt{(R+r)^2 + X_L^2}}$$

for $\frac{v_{20}}{v_{10}} = \frac{1}{2}$, we must have

$$\frac{\sqrt{r^2 + X_L^2}}{\sqrt{(R+r)^2 + X_L^2}} = \frac{1}{2}$$

or, $4r^2 + 4X_L^2 = (R+r)^2 + X_L^2$

or, $X_L = \sqrt{\frac{(R+r)^2 - 4r^2}{3}}$

or, $\omega = \frac{1}{L} \sqrt{\frac{(R+r)^2 - 4r^2}{3}}$

Example 21 Variation of phase angle with ω

In an LCR circuit, the applied voltage and the current in the circuit are expressed as $v = v_0 \sin \omega t$, $i = i_0 \sin(\omega t + \phi)$. Plot a graph showing variation of ϕ with ω .

Solution

Concepts

(i) $\tan \phi = \frac{X_C - X_L}{R} = \frac{1}{\omega CR} - \frac{\omega L}{R}$

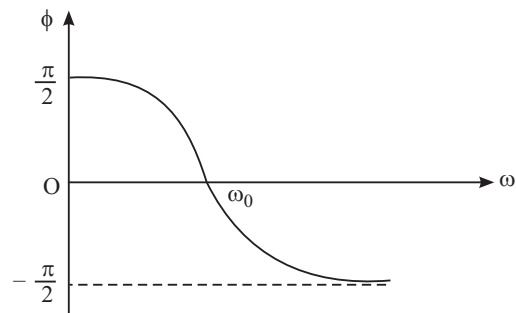
(ii) As $\omega \rightarrow 0$, $\tan \phi \rightarrow \infty$

$$\Rightarrow \phi \rightarrow \frac{\pi}{2}$$

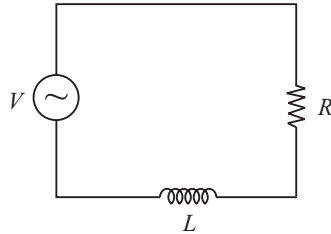
(iii) As $\omega \rightarrow \infty$, $\tan \phi \rightarrow -\infty$; $\Rightarrow \phi \rightarrow -\frac{\pi}{2}$

(iv) For $\omega_0 = \frac{1}{\sqrt{LC}}$, $\phi = 0$

The graph is as shown below.



Example 22 A R - L series circuit has been connected to an AC voltage source. The voltage across R is V_R and that across L is V_L . Plot the variation of V_R and V_L with the angular frequency (ω) of the source.

**Solution****Concepts**

$$(i) I = \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (\omega L)^2}}$$

$$(ii) V_R = IR \text{ and } V_L = IX_L$$

$$V_R = \frac{VR}{\sqrt{R^2 + (\omega L)^2}}$$

For $\omega \rightarrow 0$, $V_R \rightarrow V$

For $\omega \rightarrow \infty$, $V_R \rightarrow 0$

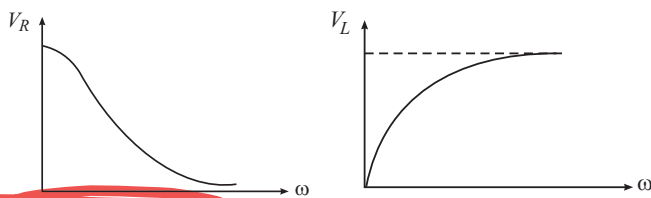
In effect, if we take output across R , then the circuit acts as a *low pass filter*. The output voltage is high for low frequency (ω).

$$V_L = \frac{VL\omega}{\sqrt{R^2 + (\omega L)^2}} = \frac{VL}{\sqrt{\left(\frac{R}{\omega}\right)^2 + L^2}}$$

For $\omega \rightarrow 0$, $V_L \rightarrow 0$

For $\omega \rightarrow \infty$, $V_L \rightarrow V$

If we take output across L , the circuit acts like a high pass filter.



Example 23 In a series LCR circuit, the current in the circuit is same for two angular frequencies, ω_1 and ω_2 , of the source. Find the resonance angular frequency for the circuit.

Solution**Concepts**

$$(i) \text{ The resonance angular frequency is } \omega_0 = \frac{1}{\sqrt{LC}}.$$

(ii) We can have same current in the circuit when the value of reactance happens to be same for two frequencies.

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

The impedance of the circuit can be same for two frequencies when

$$\omega_1 L - \frac{1}{\omega_1 C} = \frac{1}{\omega_2 C} - \omega_2 L \quad [\omega_1 > \omega_2]$$

$$\text{or, } (\omega_1 + \omega_2) L = \frac{\omega_1 + \omega_2}{\omega_1 \omega_2 C}$$

$$\text{or, } \omega_1 \omega_2 = \frac{1}{LC}$$

$$\text{or, } \omega_1 \omega_2 = \omega_0^2 \quad [\omega_0 = \text{resonance angular frequency}]$$

$$\text{or, } \omega_0 = \sqrt{\omega_1 \omega_2}$$

Example 24 Rotating coil

A coil has an area A and N number of turns. Its ends are shorted and the coil is rotated with constant angular speed ω in a uniform magnetic field. The self inductance of the coil is L and its resistance is R . Calculate the mechanical power needed to keep the coil rotating uniformly. Assume no friction.

Solution**Concepts**

- The mechanical work done in rotating the coil (against the magnetic torque) is dissipated as heat in the coil itself.
- The entire circuit is an LR circuit with sinusoidally varying emf.

The peak emf induced is $v_0 = NAB\omega$ [Refer to section 3 in this chapter]

The angular frequency of the alternating emf is ω .

$$\therefore Z = \sqrt{R^2 + (\omega L)^2}$$

$$I = \frac{V}{Z} = \frac{v_0 / \sqrt{2}}{Z} = \frac{NAB\omega}{\sqrt{2}\sqrt{R^2 + (\omega L)^2}}$$

Power dissipated in the resistance is

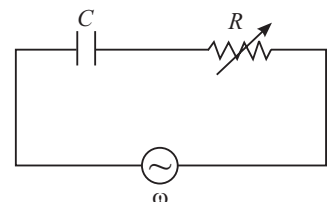
$$P = I^2 R = \frac{N^2 A^2 B^2 \omega^2 R}{2(R^2 + L^2 \omega^2)}$$

This is the necessary power needed to keep the coil rotating.

Example 25 Maximum power

An alternating source has a peak voltage v_0 . It is connected across a service combination of a capacitor and a variable resistor R .

Find the maximum power that can be dissipated in



the resistance. Express your answer in terms of capacitive reactance X_C and v_0 .

Solution

Concepts

(i) Power dissipated in R is $P = I^2 R$.

(ii) P is maximum when $\frac{dP}{dR} = 0$.

$$\text{Current } I = \frac{V}{\sqrt{X_C^2 + R^2}} \quad [V = \frac{v_0}{\sqrt{2}} = \text{rms voltage}]$$

$$P = I^2 R = \frac{V^2 R}{X_C^2 + R^2} \quad \dots(i)$$

$$P \text{ is maximum when } \frac{dP}{dR} = 0$$

[Note that X_C and the source are fixed. Only R is to be varied]

$$\text{or, } V^2 \left[\frac{(X_C^2 + R^2) \cdot 1 - R(2R)}{(X_C^2 + R^2)^2} \right] = 0$$

$$\Rightarrow R^2 = X_C^2 \Rightarrow R = X_C$$

$$\therefore P_{\max} = \frac{V^2 X_C}{2X_C^2} \quad [\text{Putting } R = X_C \text{ in (i)}]$$

$$= \frac{V^2}{2X_C} = \frac{v_0^2}{4X_C} \quad [\because V = \frac{v_0}{\sqrt{2}}]$$

Example 26 A capacitor in a square loop

A square loop of side length a has a capacitor of capacitance C , as shown. In the plane of the loop, there are two parallel wires. The wires are long and are at a separation $3a$. The wires are parallel to the sides of the square loop. The two wires carry alternating current given by $I = i_0 \sin \omega t$. The positive direction of currents in the two wires have been indicated by arrows in the figure. Neglect the self inductance of the loop.

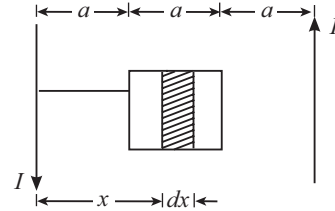
(a) Find the peak current in the loop.

(b) Write an expression for variation of current in the loop.

Solution

(a) Consider a strip of width dx in the loop. The magnetic field at the location of the strip is

$$B = \frac{\mu_0 I}{2\pi x} + \frac{\mu_0 I}{2\pi(3a-x)} \odot$$



[Direction of $\vec{B}$ is $\odot$ for indicated directions of the current in the two wires]

Flux through the strip is

$$d\phi = B (adx) = \frac{\mu_0 I a}{2\pi} \left[\frac{dx}{x} + \frac{dx}{3a-x} \right]$$

Flux through the square loop is

$$\phi = \frac{\mu_0 I a}{2\pi} \left[\int_a^{2a} \frac{dx}{x} + \int_a^{2a} \frac{dx}{3a-x} \right]$$

$$= \frac{\mu_0 I a}{2\pi} [\ln 2 + \ln 2] = \frac{\mu_0 I a \ln 2}{\pi}$$

$$= \frac{\mu_0 i_0 a \ln 2}{\pi} \sin \omega t$$

Emf induced in the loop is $\varepsilon = - \frac{d\phi}{dt}$

$$\text{or, } \varepsilon = - \frac{\mu_0 i_0 a (\ln 2) \omega}{\pi} \cos(\omega t)$$

This is an alternating emf with peak voltage

$$\varepsilon_0 = \frac{\mu_0 i_0 a (\ln 2) \omega}{\pi}$$

Peak current in the loop is

$$\therefore \frac{\varepsilon_0}{X_C} = \omega C \varepsilon_0 = \frac{\mu_0 i_0 a C \omega^2 (\ln 2)}{\pi} \quad (i)$$

$$(b) \quad \varepsilon = -\varepsilon_0 \cos \omega t = \varepsilon_0 \sin(\omega t - \frac{\pi}{2})$$

The loop has capacitive load only. Current in it will lead the emf by 90° .

$$\therefore i = (i_0)_{\text{loop}} \sin \omega t$$

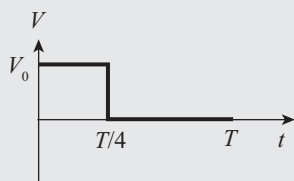
Here $(i_0)_{\text{loop}}$ is the peak current in the loop given by (i).

Worksheet 1

1. The peak value of an alternating emf E given by $E = E_0 \cos \omega t$ is 10V and frequency is 50 Hz. At time $t = (1/600)$ sec, the instantaneous value of emf is

(a) 10V (b) $5\sqrt{3}$ V
(c) 5V (d) 1V

2. One cycle of a periodic voltage V has been shown in the figure. The rms value of the voltage is



(a) $\frac{V_0}{8}$ (b) $\frac{V_0}{2}$
(c) V_0 (d) $\frac{V_0}{4}$

3. The ratio of rms value and peak value of current $i = 3 + 4 \sin(\omega t + \pi/3)$ is

(a) 5 (b) $\frac{\sqrt{17}}{7}$
(c) $\frac{5}{\sqrt{2}}$ (d) $\frac{7}{\sqrt{2}}$

4. Find the peak value of an AC emf given by $\epsilon = 3 \sin(314t) + 4 \cos(314t)$ where, ϵ is in volts.

(a) 5V (b) 7V
(c) 10V (d) None of the above

5. An AC voltage is given by $E = E_0 \sin\left(\frac{2\pi t}{T}\right)$. Then the mean value of voltage calculated over a time interval of $T/2$ seconds

(a) is always zero (b) is never zero
(c) is $(2E_0/\pi)$ always (d) may be zero

6. An AC voltage of $V = 220\sqrt{2} \sin\left(100\pi t + \frac{\pi}{2}\right)$ is applied across a DC voltmeter, its reading will be

(a) $220\sqrt{2}$ V (b) $\sqrt{2}$ V
(c) 220 V (d) zero

7. A coil of inductance 5.0 mH and negligible resistance is connected to an alternating voltage $V = 10 \sin(100t)$. The peak current in the circuit will be

(a) 2A (b) 1A
(c) 10A (d) 20A

8. The potential difference V across and the current I flowing through an instrument in an AC circuit are given by

$$V = (5 \cos \omega t) \text{ V}; \quad I = (2 \sin \omega t) \text{ A}$$

The power dissipated in the instrument is.

(a) zero (b) 5W
(c) 10W (d) 2.5W

9. A coil has an inductance of $\frac{22}{\pi}$ H and is joined in series with a resistance of 220Ω . When an alternating voltage of 220 V at 50 Hz is applied to it, then the wattless component of the rms current in the circuit is

(a) 5A (b) 0.5A
(c) 0.7A (d) 7A

10. An AC circuit has a source of frequency $f = 50$ Hz and a capacitor of capacitance $C = 100 \mu\text{F}$. If the peak value of the current in the circuit is 1.57A, the expression for the instantaneous voltage across the capacitor can be

(a) $50 \sin(100\pi t)$
(b) $50\sqrt{2} \sin(100\pi t - \pi/2)$
(c) $25 \cos(100\pi t - \pi/2)$
(d) None of the above

11. A good choke coil is connected to an AC source. The power factor is

(a) nearly zero (b) exactly zero
(c) nearly one (d) exactly one

12. In an AC circuit, the resistance value is kept unchanged. The percentage increase in the impedance of an AC circuit, when its power factor changes from 0.866 to 0.5 is

(a) 73.9% (b) 86.6%
(c) 90.8% (d) 66.6%

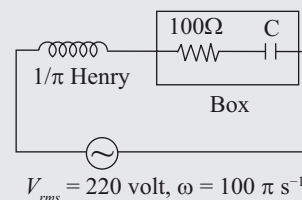
13. In a series LCR AC circuit, the voltage between the ends of the resistance R is V_R , that between the ends of capacitor is $V_C = 2V_R$ and that between the ends of inductor is $V_L = 3V_R$. The voltage of the source in terms of V_R will be

(a) $\sqrt{2} V_R$ (b) V_R
(c) $V_R/\sqrt{2}$ (d) $5V_R$

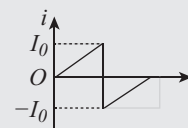
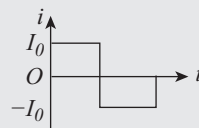
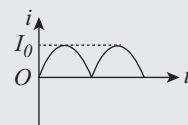
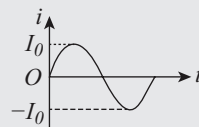
14. A long solenoid connected to a 12V DC source passes a steady current of 2A. When the solenoid

is connected to a source of 12V rms at 50Hz, the current flowing is 1A rms. The inductance of the solenoid is

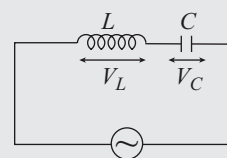
- (a) 0.044H (b) 0.033H
(c) 0.055H (d) None of the above
15. In a series LCR circuit $L = 1\text{H}$, $C = 6.25\text{ }\mu\text{F}$ and $R = 1\text{ ohm}$. The quality factor is
(a) 400 (b) 200
(c) 125 (d) 25
16. An AC source of emf $E = 200 \sin(100t)$ is connected to a choke coil of inductance 1 H and resistance 100Ω . Find the average power consumed.
(a) 400W (b) 200W
(c) 100W (d) None of the above
17. The self inductance of a choke coil is 10 mH. When it is connected with a 10V DC source, then the loss of power is 20W. When it is connected with 10 volt AC source, loss of power is 10W. The frequency of AC source will be nearly
(a) 50 Hz (b) 60 Hz
(c) 80 Hz (d) 100 Hz
18. When an AC source of emf $e = E_0 \sin(100t)$ is connected across a circuit, the phase difference between the emf e and the current i in the circuit is observed to be $\pi/4$, as shown in the diagram. If the circuit consists possibly only of R - C or R - L or L - C in series, find the relationship between the two elements.
(a) $R = 1\text{ k}\Omega$, $C = \mu\text{F}$
(b) $R = 1\text{ k}\Omega$, $C = 1\text{ }\mu\text{F}$
(c) $R = 1\text{ k}\Omega$, $L = 10\text{H}$
(d) $R = 1\text{ k}\Omega$, $L = 1\text{H}$
19. When a resistance R is connected in series with an element A , the electric current is found to be lagging behind the voltage by angle θ_1 . When the same resistance is connected in series with element B , current leads voltage by θ_2 . When R , A , B are connected in series, the current now leads voltage by θ . Assume same AC source is used in all cases, then
(a) $\theta = \theta_2 - \theta_1$
(b) $\tan \theta = \tan \theta_2 - \tan \theta_1$
(c) $\theta = \frac{\theta_1 + \theta_2}{2}$
(d) None of the above
20. In the circuit shown in the figure, current is 2.2 ampere. The power factor of the box is



- (a) $\frac{1}{\sqrt{2}}$ (b) 1
(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{2}$
21. Power factor of an L - R series circuit is 0.6 and that of a C - R series circuit is 0.5. If the element (L , C , and R) of the two circuits are joined in series, the power factor of this circuit is found to be 1. The ratio of the resistance in the L - R circuit to the resistance in the C - R circuit is
(a) 6/5 (b) 5/6
(c) $\frac{4}{3\sqrt{3}}$ (d) $\frac{3\sqrt{3}}{4}$
22. I_1 , I_2 , I_3 and I_4 are the respective rms values of the time varying currents as shown in the four cases I, II, III and IV. Identify the correct relations. The curves are sinusoidal.

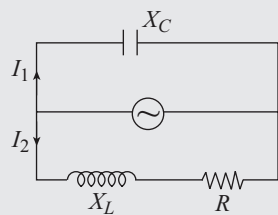


- (a) $I_1 = I_2 = I_3 = I_4$ (b) $I_3 > I_1 = I_2 > I_4$
(c) $I_3 > I_4 > I_2 = I_1$ (d) $I_3 > I_2 > I_1 > I_4$
23. The current I , potential difference V_L across the inductor and potential difference V_C across the capacitor in circuit as shown are best represented as



- (a) (b)
(c) (d)

24. In the shown AC circuit phase difference between currents I_1 and I_2 is



(a) $\frac{\pi}{2} - \tan^{-1} \frac{X_L}{R}$

(b) $\tan^{-1} \frac{X_L - X_C}{R}$

(c) $\frac{\pi}{2} + \tan^{-1} \frac{X_L}{R}$

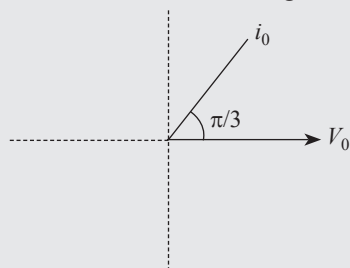
(d) $\tan^{-1} \frac{X_L - X_C}{R} + \frac{\pi}{2}$

Worksheet 2

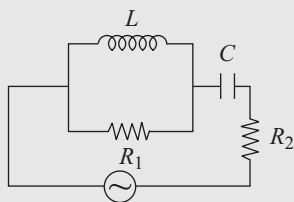
1. The voltage of an AC source varies with time according to the equation,

$V = 100 \sin 100\pi t \cos 100\pi t$ where t is in second and V is in volt. Then

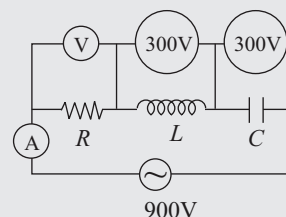
- the frequency of the source is 100 Hz
 - the peak voltage of the source is $(100/\sqrt{2})$ V
 - the peak voltage of the source is 50 V
 - the frequency of the source is 50 Hz
2. For an LCR series circuit, phasors of current i and applied voltage $v = v_0 \sin \omega t$ are shown in diagram at $t = 0$. Which of the following is/are correct?



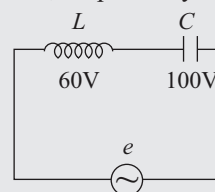
- At $t = \frac{\pi}{2\omega}$, instantaneous power supplied by source is negative
 - From $0 < t < \frac{2\pi}{3\omega}$, average power supplied by source is positive
 - At $t = \frac{5\pi}{6\omega}$, instantaneous power supplied by source is negative
 - If ω is increased slightly, angle between the two phasors decreases
3. For the given circuit



- The phase difference between I_L and I_{R_1} is 0°
 - The phase difference between V_C and V_{R_2} is 90°
 - The phase difference between I_L and I_{R_1} is 180°
 - The phase difference between V_C and V_{R_2} is 180°
4. In the series circuit shown in the figure, the voltmeters and the ammeter are ideal which can directly read rms values.

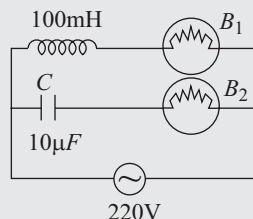


- The reading of the voltmeter across R is 300V
 - The reading of the voltmeter across R is 900V
 - The reading of ammeter will not change even if the resistance in the circuit is doubled
 - The reading of voltmeter connected across R will not change even if the resistance in the circuit is doubled
5. In the given figure, the voltages across L and C are 60 V and 100 V, respectively.



- The source voltage is 160V
 - The source voltage is 40V
 - The average power delivered by the source is zero
 - None of the above
6. An inductor 20×10^{-3} H, a capacitor $100 \mu\text{F}$ and a resistor 50Ω are connected in series across a source of emf $V = 10 \sin 314t$.
- Energy dissipated in the circuit in 20 minutes is more than 900 J
 - Energy dissipated in the circuit in 20 minutes is 628 J
 - If resistance is removed from the circuit and the value of inductance is doubled, then the variation of current with time in the new circuit is $i = 0.52 \cos 314t$
 - If resistance is removed from the circuit and the value of inductance is doubled, then the variation of current with time in the new circuit is $i = -0.52 \cos 314t$

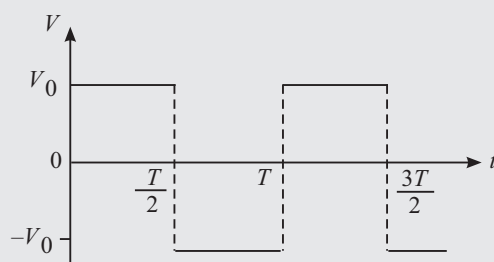
7. Two identical bulbs B_1 and B_2 are connected to an AC source. B_1 is connected in series with a coil of 100 mH and B_2 with a capacitor of $10\text{ }\mu\text{F}$ as shown in the figure. The brightness of B_1 and B_2 will



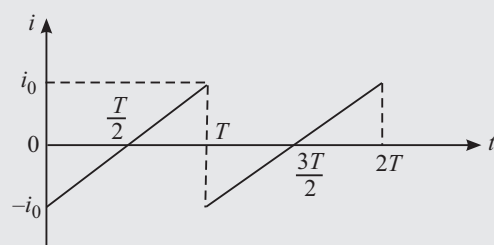
- (a) be same
 (b) be higher for B_1
 (c) depend on the frequency of the source
 (d) be higher in B_2
8. An LCR series circuit is in resonance with an AC source. The sharpness of the resonance is to be improved by reducing its bandwidth to half. A suitable way will be to
- (a) double the value of inductance
 (b) increase the inductance four fold
 (c) increase the capacitance four fold
 (d) half the resistance
9. An applied voltage signal consists of a superposition of a DC voltage and an alternating sinusoidal voltage of high frequency. The circuit has a capacitor and an inductor connected in series.
- (a) The voltage across the capacitor will be almost constant
 (b) The voltage across the inductor will be nearly sinusoidal
 (c) The voltage across the capacitor will be sinusoidal
 (d) There will be a sinusoidal current in the circuit
10. A choke coil is in series with a lamp. The lamp is shining brightly.
- (a) If an iron core is inserted inside the coil, the lamp becomes brighter if the power supply is DC
 (b) If an iron core is inserted inside the coil, the lamp becomes brighter if the power supply is AC
 (c) If an iron core is inserted inside the coil, the lamp gives same amount of light if the power supply is DC
 (d) If an iron core is inserted inside the coil the lamp becomes dimmer if the power supply is AC

Worksheet 3

1. Voltage in a circuit varies with time according to the graph shown. Find the rms value of the voltage.



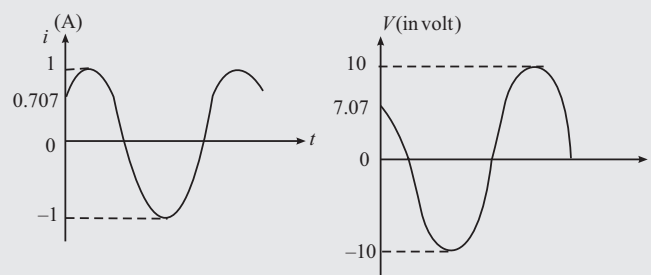
2. A direct current I_0 is superimposed on an alternating current $i = i_0 \sin \omega t$ flowing in a wire. What is the effective value of current in the wire?
3. Variation of current in a circuit is as shown in the graph. Find



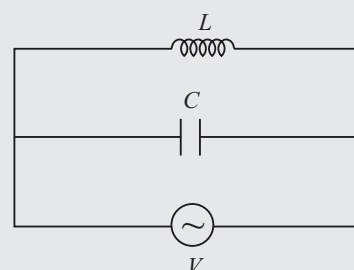
- (a) the average current over half cycle and
(b) the rms current over full cycle
4. When a 100V DC is applied across a coil, a current of 1A flows through it. When 100V, 50Hz AC source is connected across the same coil, a current of 0.5 A flows. Calculate the resistance and the inductance of the coil.
5. A capacitor and an inductor are connected in series to a 100V, 50Hz AC source. There is no resistance and the current is found to be very large. A 100Ω resistance is also connected in series with the capacitor and the inductor.
- (a) Find the current in the circuit.
(b) What is phase difference between the current and the applied voltage?
(c) If the inductor is removed, leaving R and C , the current in the circuit is observed to be 0.5A. Find the value of capacitance.
6. An LCR series circuit is powered by an AC source of variable frequency. Plot the variation of the impedance of the circuit with the source frequency.
7. A series LCR circuit is connected to a 200V, 300 rad s^{-1} source. The resistance in the circuit is 100Ω . When the capacitor is removed from the circuit, the

current in the circuit lags the source voltage by 60° . When only the inductance is removed, the current leads the voltage by 60° . Calculate the current and power dissipated in the original LCR circuit.

8. A series LR circuit is connected to a source which gives a voltage $v = 60 \sin (200t)$ V. The values of inductance and resistance are $L = 200\text{mH}$ and $R = 30\Omega$.
- (a) Find the variation of current with time.
(b) Find the average value of energy stored in the inductor, over a long period of time.
9. In an AC circuit having a source of angular frequency $\omega = 100 \text{ rad s}^{-1}$, the variation of current (i) and the source voltage (v) are as shown in the graphs.

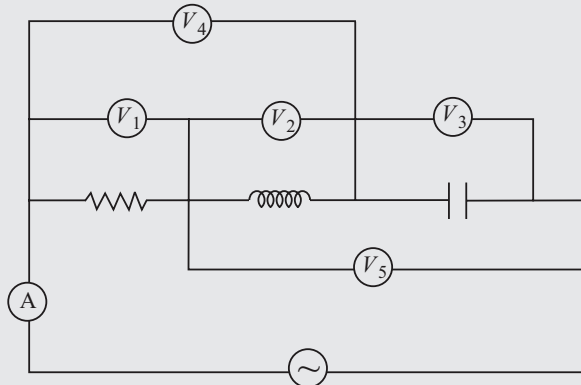


- (a) Write the variation of the current (i) and the voltage (v) with time.
(b) Write impedance of the circuit.
(c) Find the power consumption in the circuit.
10. For the circuit shown in the figure, the current in the inductance is 0.6A and that in the capacitor branch is 0.8 A. Find the current through the source.



11. The power factor of a series RL circuit is 0.5 when the applied voltage is $V = 100 \sin (100t)$. The resistance in the circuit is $R = 200\Omega$.
- (a) Find the value of inductance in the circuit.
(b) A capacitor is added to improve the power factor of the circuit. Find the capacitance, if the power factor becomes 1.0.

12. In the AC circuit shown in the figure, all the voltmeters and the ammeter are ideal and can directly measure rms values. Reading of the ammeter (A) is 2A. The voltmeters V_1 , V_3 and V_4 read 100V, 120V and 140V respectively.



- Find readings of the voltmeters V_2 and V_5 .
 - Find the source voltage.
 - Find R , X_L and X_C .
13. An LCR series circuit is resonant at 1000 kHz. The circuit has a quality factor of 100. Find the lower and upper half power frequencies.

14. A 750 Hz, 20V source is connected to a resistance $R = 100\Omega$, an inductance $L = 180 \text{ mH}$ and a capacitance $C = 10\mu\text{F}$ all in series. Assume that the heat dissipated in the resistor is retained by it. Find the time needed for the temperature of the resistor to rise by 10°C . It is given that the thermal capacity of the resistor is $2 \text{ J } ^\circ\text{C}^{-1}$.

15. A village has a demand of 220 kW of electric power at 220V. It is situated at a distance of 20km from a power plant. The plant generates electricity at 440V which is stepped up before transmission. The resistance of the two wire line used for transmission is 1.0Ω per km of wire. At the outskirts of the village, there is a substation which uses 4400–220V step down transformer to step down the voltage.

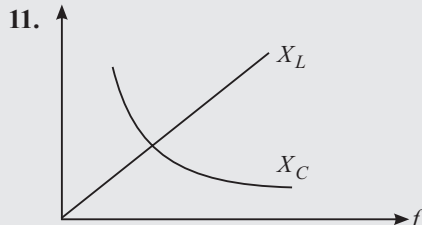
- Estimate the power loss in the transmission line in form of heat.
- How much power the plant must produce?
- What is turn ratio (number of turns in the primary to that in secondary) for the transformer being used at the power plant?

Assume power factor to be 1.0.

Answer Sheet

Your Turn

- | | | | |
|--------------|----------|--|-------------|
| 1. 3 | 2. 1 | 3. (a) $y = 5 \sin(200t + \frac{2\pi}{3})$ | 4. 50 Hz |
| 5. 2.5 ms | 6. 10 kW | 7. AC supply | 8. Infinity |
| 9. Increases | 10. 34 V | | |
| 11. | | 12. $I_2 > I_1$ | 13. 0.1 A |
| | | | 14. 0.67 A |



- | | | | |
|---|--------------------------------|--|---------------------------|
| 15. 1 A | 16. (a) $4 \times 10^3 \Omega$ | (b) $v_L = (800V) \sin\left(10^3 t + \frac{\pi}{2}\right)$ | 17. 160 Hz |
| 19. Yes | 20. 1.21 H | 21. Second circuit | 22. 1.59×10^5 Hz |
| 23. $I = \sqrt{2}$ A, current leads in phase by 45° | 24. Zero | | |
| 25. 2000 turns coil will be used as primary. Output = 220 V | | | |

Worksheet 1

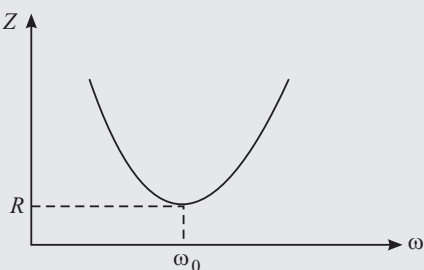
- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (b) | 3. (b) | 4. (a) | 5. (d) | 6. (d) | 7. (d) | 8. (a) | 9. (b) |
| 10. (a) | 11. (a) | 12. (a) | 13. (a) | 14. (b) | 15. (a) | 16. (c) | 17. (c) | 18. (a) |
| 19. (b) | 20. (a) | 21. (d) | 22. (b) | 23. (d) | 24. (c) | | | |

Worksheet 2

- | | | | | | |
|-----------|--------------|--------------|------------|-----------|-----------|
| 1. (a, c) | 2. (b, c, d) | 3. (b) | 4. (b, d) | 5. (b, c) | 6. (a, c) |
| 7. (c) | 8. (d) | 9. (a, b, d) | 10. (c, d) | | |

Worksheet 3

- | | | | |
|--|-------------------------------------|------------------------|---|
| 1. $v_{\text{rms}} = v_0$ | 2. $\sqrt{I_0^2 + \frac{i_0^2}{2}}$ | 3. (a) $\frac{i_0}{2}$ | (b) $i_{\text{rms}} = \frac{i_0}{\sqrt{3}}$ |
| 4. $R = 100 \Omega$, $L = 0.55 \text{ H}$ | | 5. (a) 1 A | (b) 0 (c) $18.4 \mu\text{F}$ |
| 6. Z | | 7. 2 A, 400 W | |



8. (a) $i = 1.2 \sin (200 t - \tan^{-1} \frac{4}{3})$ (b) 0.072 J

9. (a) $i = 1 \sin \left(100t + \frac{\pi}{4} \right)$; $v = 10 \sin \left(100t + \frac{3\pi}{4} \right)$ (b) $Z = 10\Omega$ (c) Zero

10. 0.2 A

11. (a) 3.464 (b) 28.9 μF

12. (a) $V_2 = 98V$, $V_5 = 22V$ (b) 102.4V (c) $R = 50\Omega$, $X_L = 49\Omega$, $X_C = 60\Omega$

13. 995 kHz, 1005 kHz

14. 345 s

15. (a) 100kW (b) 320kW (c) $\frac{11}{160}$

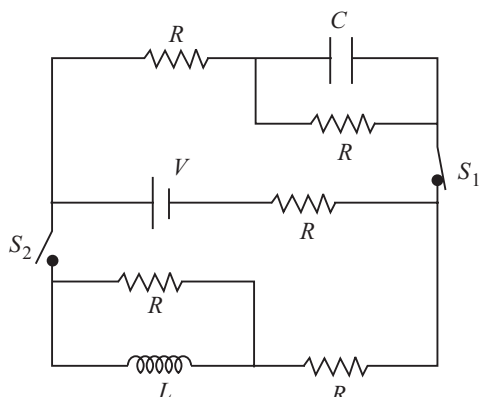
Miscellaneous Problems on Chapters 5 and 6

MATCH THE COLUMNS

- (1) Match the quantities given in column-I with their units in column-II.

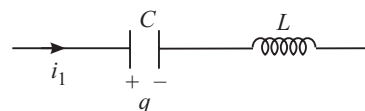
Column-I		Column-II	
(A)	Flux of magnetic field	(P)	$\text{Kg s}^{-2} \text{A}^{-1}$
(B)	Rate of change of flux	(Q)	Volt
(C)	Magnetic field	(R)	Coulomb-Ohm
(D)	Potential energy	(S)	Ampere-Weber

- (2) In the circuit shown, the switch S_1 is closed for a long time and S_2 is open. S_2 is closed at time $t = 0$ and later after a long time at $t = T$, S_1 is opened.



Column-I		Column-II	
(A)	Just after $t = 0$, the emf developed in the inductor is $\frac{V}{x}$. Value of x is	(P)	2
(B)	Just after $t = 0$, the current through the cell is $\frac{V}{xR}$. The value of x is	(Q)	4
(C)	Just before $t = T$, the current in the inductor is $\frac{V}{xR}$. The value of x is	(R)	$\frac{5}{2}$
(D)	Just after $t = T$, the current through the cell is $\frac{V}{xR}$. The value of x is	(S)	$\frac{15}{7}$

3. The figure shows a part of an electric circuit. The value of current at an instant is $i = 1\text{A}$ and is increasing at a rate of 20A/s .

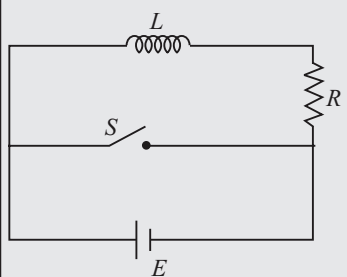
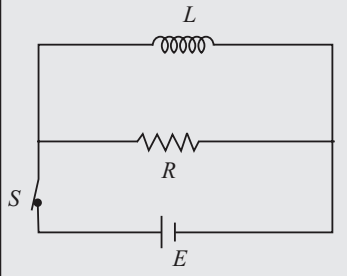
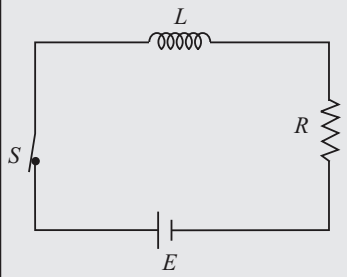


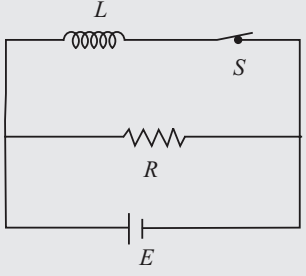
The charge on the capacitor is $q = 1\mu\text{C}$. The capacitor is an ideal parallel plate capacitor with plate separation, $d = 2\text{ mm}$ and the inductor is an ideal solenoid with volume 0.25m^3 and inductance 1mH . Match the entries in column-I with those in column-II if energy stored in the inductor is 1000 times the energy in the capacitor.

7.2 Magnetism and Electromagnetic Induction

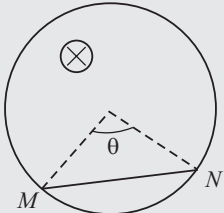
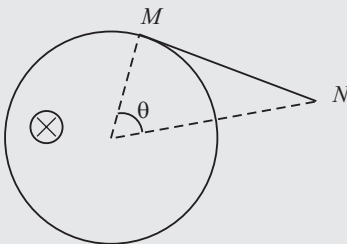
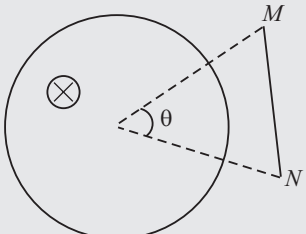
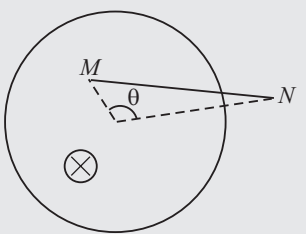
Column-I		Column-II	
(A)	$\frac{\text{Voltage across capacitor}}{\text{Voltage across inductor}}$	(P)	Greater than 1500
(B)	$\frac{\text{Energy density inside inductor}}{\text{Energy density inside capacitor}}$	(Q)	50
(C)	$\frac{\text{Rate of growth of electric field energy}}{\text{Rate of growth of magnetic field energy}}$	(R)	Less than 10
(D)	$\frac{\text{Electric field between capacitor plate (in V/m)}}{\text{Magnetic field inside solenoid (in } \mu\text{T)}}$	(S)	1

4. Column-I shows some circuits. In circuit (A) the switch (S) is initially open and is closed at $t = 0$. In all other circuits the switch (S) is initially closed (for a long time) and is opened at $t = 0$. Match the entries in the two columns.

Column-I		Column-II	
(A)		(P)	Induced emf in the inductor will be greater than E immediately after the operation of the switch
(B)		(Q)	Induced emf will be less than E
(C)		(R)	Finally, energy stored in the inductor is zero

(D)		(S)	Induced emf will be equal to less than or greater than E
-----	--	-----	--

5. Column-I shows a cylindrical region of radius r where a downward magnetic field B exists. B is increasing at a rate $\frac{dB}{dt}$. A conducting rod MN is placed as shown in the figure. Column-II has statements regarding the induced emf in the rod. Match the two columns.

Column-I		Column-II	
(A)		(P)	Induced emf in the rod is $\frac{1}{2} \theta r^2 \frac{dB}{dt}$
(B)		(Q)	Induced emf in the rod is less than $\frac{1}{2} \theta r^2 \frac{dB}{dt}$
(C)		(R)	End M is positive with respect to N
(D)		(S)	Induced emf in the rod is zero

6. The graphs given in cloumn-I represents the variation of current (i) and voltage (v) in a series ac circuit. Column-II gives some properties of the circuit. Match the two columns.

Column-I		Column-II	
(A)		(P)	Power factor is $\frac{1}{\sqrt{2}}$
(B)		(Q)	Power factor is zero
(C)		(R)	RC circuit
(D)		(S)	LCR Circuit
		(T)	LC circuit with $X_C > X_L$

7. The Column-I shows some ac circuits and the column-II shows the dependence of current on the source frequency. Match the two columns.

Column-I		Column-II	
(A)		(P)	
(B)		(Q)	
(C)		(R)	
(D)		(S)	

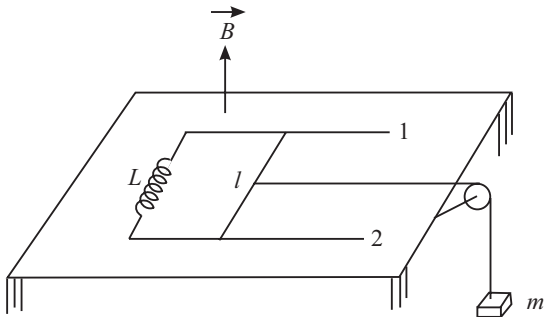
8. An inductor coil has a finite resistance. It is connected to an ac source. How can the quantities mentioned in column-I change if a capacitor is also connected in series with the coils. Choose your response from column-II.

Column-I		Column-II	
(A)	the power factor of the circuit	(P)	may increase
(B)	the power dissipated in the circuit	(Q)	may decrease
(C)	the voltage across the coil	(R)	may remain unchanged
(D)	the current in the circuit		

PASSAGE BASED PROBLEMS

Passage 1

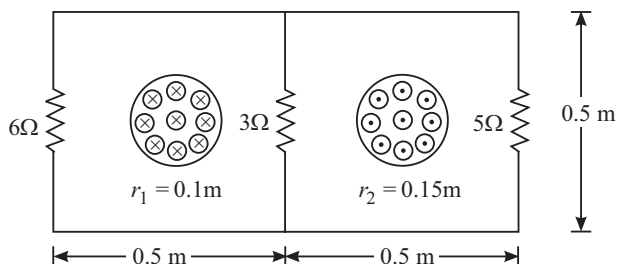
Two fixed parallel conducting rails – 1 and 2 – have their ends connected by an inductor coil of inductance L . Separation between the rails is l . A conducting rod of mass m can slide on the rails. The system is on a horizontal surface. A light non conducting string connects the rod to a mass m which is suspended vertically (see figure). A uniform vertical magnetic field (B) exists in the region. The system is released from rest. There is no friction and no resistance.



- The current in the rod when it has moved a distance x is
 - $\frac{Blx}{L}$
 - $\frac{BIL}{x}$
 - $\frac{2Blx}{L}$
 - $\frac{2BIL}{x}$
- The speed of the rod when it has moved a distance x is
 - $\sqrt{2gx - \frac{B^2 l^2 x^2}{2mL}}$
 - $\sqrt{2gx - \frac{B^2 l^2 x^2}{mL}}$
 - $\sqrt{gx - \frac{B^2 l^2 x^2}{2mL}}$
 - $\sqrt{\frac{gx}{2} + \frac{B^2 l^2 x^2}{2mL}}$

Passage 2

The circular regions shown in the figure are cross-sections of two long solenoids which have their lengths perpendicular to the plane of the figure. The direction of magnetic field inside the solenoids is as shown. The field is increasing at a rate of 100 T s^{-1} inside the two solenoids.



- The current induced in the 3Ω resistance is

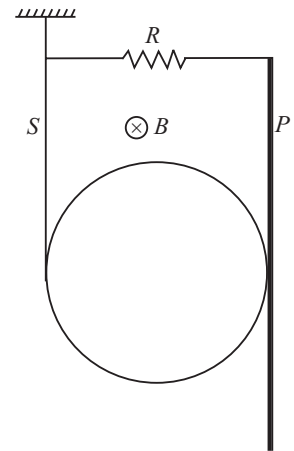
- 62 mA
- 860 mA
- 200 mA
- 922 mA

- The current in 6Ω resistance is

- 62 mA
- 860 mA
- 200 mA
- 922 mA

Passage 3

A conducting light string S is wound on the rim of a metal ring of radius r and mass m . The free end of the string is fixed to the ceiling. A vertical, long, smooth and perfectly conducting plane P always touches the ring. The plane P and the string S are connected by a resistance R . There is a uniform horizontal magnetic field (B) perpendicular to the plane of the ring, in the entire space. The ceiling is insulating and the resistance of the string and the ring is negligible. The ring is released from rest.



- The acceleration of the centre of the ring when its velocity is v is

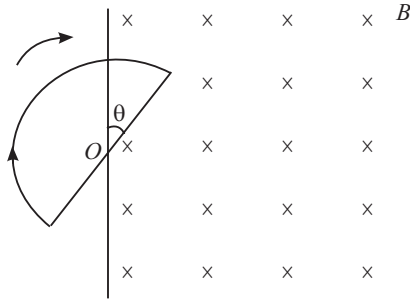
- $\frac{g}{2} - \frac{2B^2 r^2 v}{mR}$
- $g - \frac{4B^2 r^2 v}{mR}$
- $2g - \frac{B^2 r^2 v}{2mR}$
- $g - \frac{2B^2 r^2 v}{mR}$

- The terminal velocity of the centre of the ring is

- $\frac{mgR}{B^2 r^2}$
- $\frac{mgR}{2B^2 r^2}$
- $\frac{2mgR}{B^2 r^2}$
- $\frac{mgR}{4B^2 r^2}$

Passage 4

A wire loop enclosing a semicircle of radius R is located on the boundary of a uniform magnetic field B . The diameter coincides with the boundary and the loop is outside the boundary at $t = 0$. It begins to rotate with a constant angular acceleration α about an axis through O , perpendicular to the plane of the figure. The loop rotates clockwise. The arrow on the loop indicates the positive direction of the induced current.

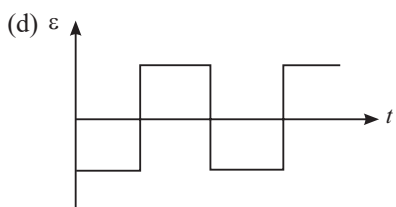
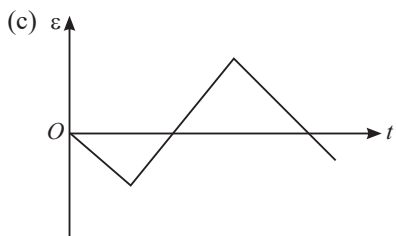
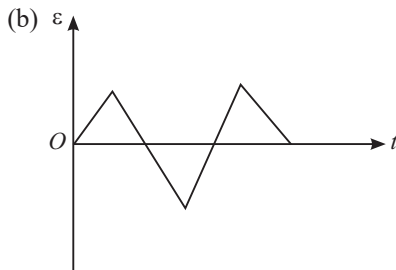
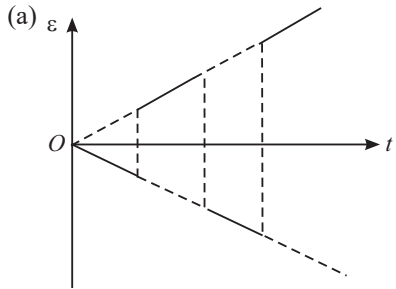


7. If n is number of half revolutions that the loop performs, then the induced emf in the loop can be expressed as

(a) $\varepsilon = (-1)^n BR^2\alpha t$ (b) $\varepsilon = (-1)^n \frac{1}{2} BR^2\alpha t$

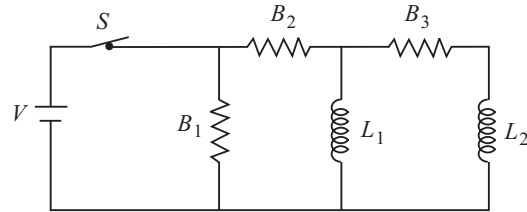
(c) $\varepsilon = \left(-\frac{1}{2}\right)^n BR^2\alpha t$ (d) $\varepsilon = \frac{1}{2} BR^2\alpha t$

8. The variation of emf in the loop can be best represented by



Passage 5

Three identical bulbs, B_1 , B_2 and B_3 are connected in a circuit as shown. The two inductors are also identical. The switch (S) is held closed for a long time.



9. Which of the followings is true?

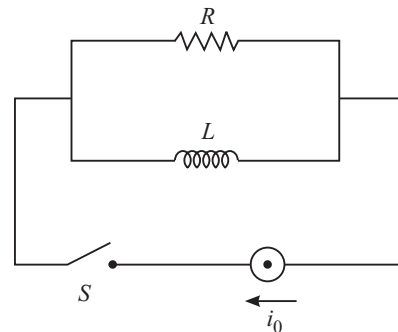
- (a) All the bulbs are glowing with same brightness
 (b) B_1 and B_2 are glowing with same brightness
 (c) B_2 and B_3 are having same brightness
 (d) Two of the bulbs are not glowing

10. Which of the following is true for the instant just after the switch is opened?

- (a) All three bulbs glow with same brightness
 (b) B_1 and B_2 glow with the same brightness
 (c) B_1 stops glowing and the other two have same brightness
 (d) Two of the bulbs do not glow

Passage 6

A parallel combination of a resistance (R) and a pure inductance (L) has been connected to a source through a switch (S). The source supplies a constant current i_0 . The switch is closed at $t = 0$.



11. The maximum current in R can be

- (a) Zero (b) i_0
 (c) $\frac{i_0}{2}$ (d) $\frac{i_0}{4}$

12. The current through the inductor changes with time as

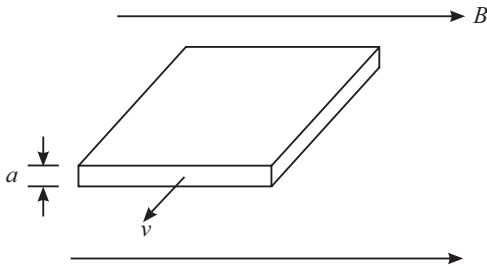
- (a) $i = i_0 e^{-\frac{2Rt}{L}}$ (b) $i = 2i_0 \left(1 - e^{-\frac{2Rt}{L}}\right)$
 (c) $i = i_0 \left(1 - e^{-\frac{Rt}{L}}\right)$ (d) $i = i_0 e^{-\frac{2Rt}{L}}$

13. Heat dissipated in the resistance after the switch is closed is

- (a) Li_0^2 (b) $\frac{1}{4} Li_0^2$
(c) $2 Li_0^2$ (d) $\frac{1}{2} Li_0^2$

Passage 7

A rectangular bar has large length and width but small thickness a . It is moved with a constant velocity v directed perpendicular to one of its edges. A uniform magnetic field B exists in the region parallel to the bar and perpendicular to its velocity.



14. The surface charge density (magnitude) on the upper and lower faces of the bar, if the material of the bar is a perfect conductor is

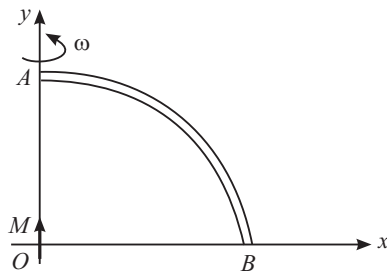
- (a) $\epsilon_0 vB$ (b) $\frac{1}{2} \epsilon_0 vB$
(c) $\epsilon_0^2 vB$ (d) $\frac{\epsilon_0 v}{B}$

15. The potential difference between the upper and lower face is

- (a) $\frac{vBa}{2}$ (b) $2vBa$
(c) vBa (d) zero

Passage 8

A quarter circular conducting ring of radius R is kept with its centre of curvature at the origin of a co-ordinate system. A small magnetic dipole is placed at the origin with its dipole moment $\vec{M}$ directed along the y -axis. The ring is made to rotate about y -axis with a constant angular velocity ω , in the magnetic field of the dipole. An emf ϵ , is induced between the ends A and B of the ring.



16. Which of the following are correct regarding the emf induced in the ring?

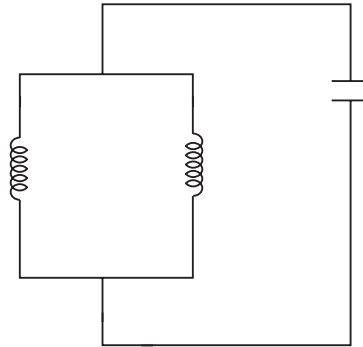
- (a) The end A is positive
(b) The end B is positive
(c) Polarity changes after every half revolution
(d) There is no emf

17. Magnitude of the induced emf is

- (a) $\frac{\mu_0 M \omega}{2\pi R}$ (b) $\frac{\mu_0 M \omega}{\pi R}$
(c) $\frac{\mu_0 M \omega}{4\pi R}$ (d) $\frac{\mu_0 M \omega}{3\pi R}$

Passage 9

The circuit shown has two coils each of inductance $L = 1.0\text{H}$ and a capacitor of capacitance $1.0\mu\text{F}$. At the instant the charge on the capacitor is zero, currents of 1.0A and 2.0A are flowing through the coils in the same direction.



18. The currents in the two inductors when the charge on the capacitor becomes maximum for the first time are

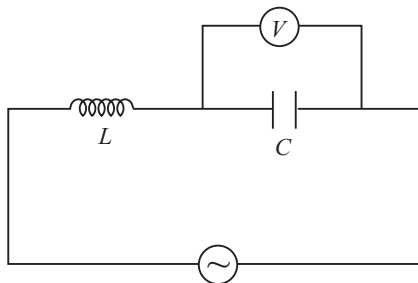
- (a) 1A and $\frac{1}{2}\text{A}$ in opposite directions
(b) 1A and 1A in same direction
(c) 2A and 1A in opposite directions
(d) $\frac{1}{2}\text{A}$ and $\frac{1}{2}\text{A}$ in opposite directions

19. The maximum charge on the capacitor is

- (a) $\frac{1}{\sqrt{2}}\text{mC}$ (b) $\sqrt{2}\text{mC}$
(c) $\frac{3}{\sqrt{2}}\text{mC}$ (d) $\sqrt{3}\text{mC}$

Passage 10

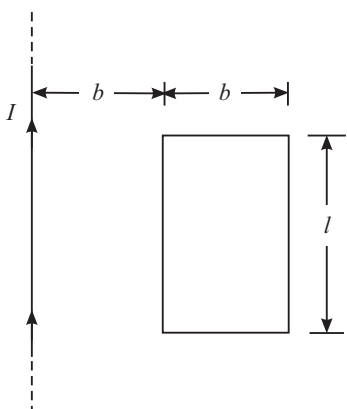
An inductance of 1.0 H is connected in series to a capacitor and an alternating voltage source of 220V, 50Hz is connected across this series combination. An ideal voltmeter, capable of measuring rms voltage is connected across the capacitor [Take $\pi^2 = 10$].



20. For what values of the capacitance, the reading of the voltmeter will be 220 V?
- (a) zero, $20\mu F$ (b) $10\mu F$, $20\mu F$
 (c) $20mF$, $40\mu F$ (d) $10\mu F$, $20\mu F$, $40\mu F$
21. Which of the following value of the capacitance will you never want to use in the circuit?
- (a) $20\mu F$ (b) $10\mu F$
 (c) $40\mu F$ (d) $22\mu F$

Passage 11

A rectangular loop of width b , length l , mass m and resistance R is kept at a distance b from a long straight current carrying wire having current I . The loop and the wire are in the same plane. The current in the wire is suddenly switched off.



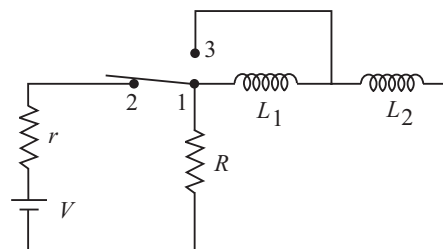
22. Which of the following is true?
- (a) The loop moves to right away from the wire
 (b) The loop moves to left towards the wire
 (c) The loop does not move at all
 (d) We cannot predict the motion of the loop as the event is sudden

23. The speed acquired by the loop is

- (a) $\frac{\mu_0^2 (\ln 2) l^2 I^2}{2\pi^2 b m R}$ (b) $\frac{\mu_0^2 (\ln 2) l^2 I^2}{4\pi^2 b m R}$
 (c) $\frac{\mu_0^2 (\ln 2) l^2 I^2}{\pi^2 b m R}$ (d) $\frac{\mu_0^2 (\ln 2) l^2 I^2}{16\pi^2 b m R}$

Passage 12

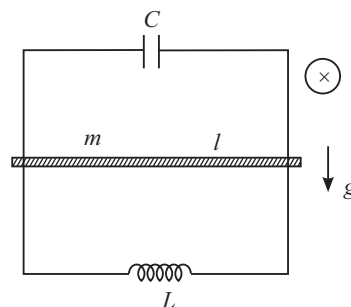
In the circuit shown in the figure, the two-way switch is in position 2 and the circuit is in steady state. Now the switch is shifted to position 3 at $t = 0$.



24. Current in L_1 is
- (a) decreasing (b) increasing
 (c) constant at $\frac{V}{r}$ (d) zero
25. Current through the switch as a function of time is
- (a) $\frac{V}{r} (1 - e^{-\frac{Rt}{L_2}})$ (b) $\frac{V}{r} e^{-\frac{Rt}{L_2}}$
 (c) $\frac{V}{r} (1 + e^{-\frac{Rt}{L_2}})$ (d) $\frac{V}{r} (e^{-\frac{Rt}{L_2}} + e^{-\frac{Rt}{L_2}})$

Passage 13

A conductor of mass m and length l can slide freely on a pair of long smooth, vertical rails. A magnetic field B exists in the region in direction perpendicular to the plane of the rails. The rails are connected at the top end by a capacitor of capacitance C and at the bottom end by an inductor of inductance L as shown. The acceleration due to gravity is g . The conductor is released from the top from rest at $t = 0$.



26. When the conductor falls through x , its acceleration is a . The current in the conductor at this instant is

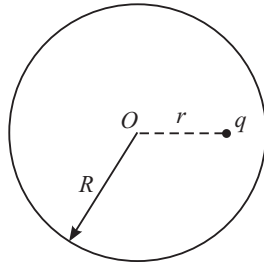
(a) $BlCa + \frac{Bl}{L}x$ (b) $2BlCa + \frac{Blx}{L}$
 (c) $BlCa + \frac{Blx}{2L}$ (d) $BlCa + \frac{2Blx}{L}$

27. The motion of the conductor is

- (a) Continuously accelerated
 (b) Initially accelerated and then uniform
 (c) Oscillatory but not SHM
 (d) SHM, with the mean position at $x = \frac{mgL}{B^2 l^2}$

Passage 14

Consider a circular region of radius R . A particle of mass m and charge q is kept at a distance $r (< R)$ from the centre of the circle. A magnetic field of magnitude B_0 is suddenly switched on. The field is directed into the plane of the paper and is present uniformly inside the circular region only.

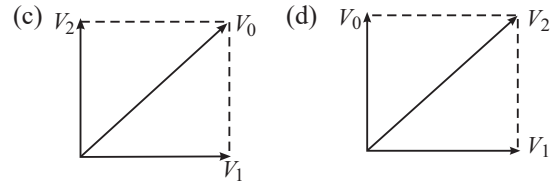
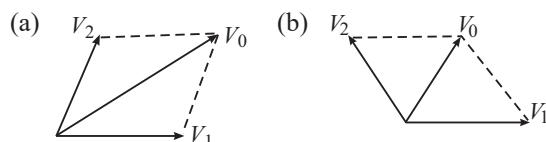


28. The particle will
- (a) Begin to move on a straight line and cross the circular region.
 (b) Begin to move on a circular path
 (c) Not move at all
 (d) Move on a parabolic path
29. For how long will the particle stay inside the field region
- (a) forever (b) $\frac{2\pi M}{qB_0}$
 (c) $\frac{\pi M}{qB_0}$ (d) $\frac{\sqrt{3}}{2} \frac{\pi M}{qB_0} \frac{r}{R}$

Passage 15

A coil having some finite resistance and inductance is connected in series to a resistor having resistance R . This combination is connected across an ac source of voltage V_0 . The ideal ac voltmeters connected across the resistor and the coil read V_1 and V_2 volts, respectively.

30. Which of the following is the best diagram representing the relationship between the phasors representing V_0 , V_1 and V_2 ?



31. The power dissipated in the coil is

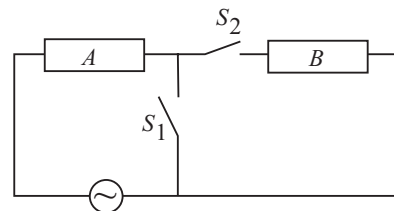
(a) $\frac{V_0^2 + V_1^2 - V_2^2}{R}$
 (b) $\frac{V_0^2 - V_1^2 - V_2^2}{R}$
 (c) $\frac{V_0^2 - V_1^2 - V_2^2}{2R}$
 (d) $\frac{V_0^2 + V_1^2 - V_2^2}{2R}$

32. The power dissipated in the circuit is

(a) $\frac{V_0^2 + V_1^2 + V_2^2}{R}$ (b) $\frac{V_0^2 + V_1^2 + V_2^2}{2R}$
 (c) $\frac{V_0^2 - V_1^2 - V_2^2}{R}$ (d) $\frac{V_0^2 - V_1^2 - V_2^2}{2R}$

Passage 16

The boxes A and B shown in the circuit have series combination of L , C and R inside them. Each box has two elements. (It means none of the boxes has only L or only C or only R or all three L , C and R). The source voltage is 100 V. When only S_1 is closed (and S_2 is open), the current through the source is 2A and the current leads in phase by 90° . When S_2 is closed and S_1 opened, the current through the source is 1A and the power dissipated in the circuit is 80W. The current lags in phase with respect to the source voltage.



33. The value of resistance in the box A is

(a) 0 (b) 10Ω
 (c) 100Ω (d) 50Ω

34. The value of resistance in the box B is

(a) 0 (b) 80Ω
 (c) 40Ω (d) 50Ω

35. The impedance of the box B is nearly

- (a) 100Ω (b) 120Ω
(c) 200Ω (d) 136Ω

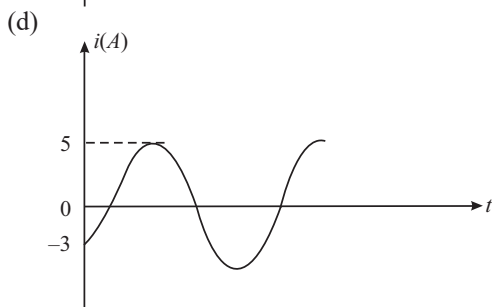
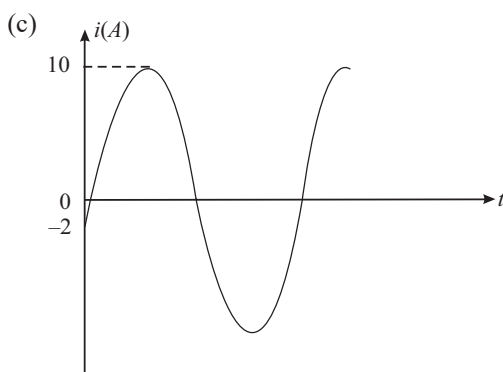
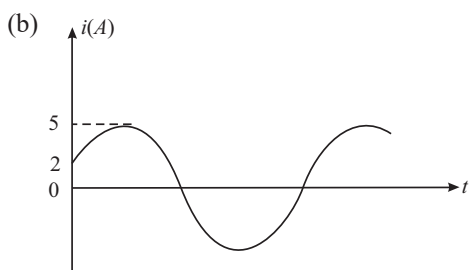
Passage 17

The figure shows four currents at a junction in an electric circuit. Current i_1 is a DC current having a steady value of 5A in the indicated direction. i_2 and i_3 are alternating currents given by

$$i_2 = 4 \sin(10\pi t), \quad i_3 = 3 \cos(10\pi t)$$

While writing the equations for i_2 and i_3 , the positive direction has been considered in the directions of respective arrows in the diagram. Current i_4 is also taken positive in the direction of the indicated arrow.

36. The variation of i_4 is correctly represented as

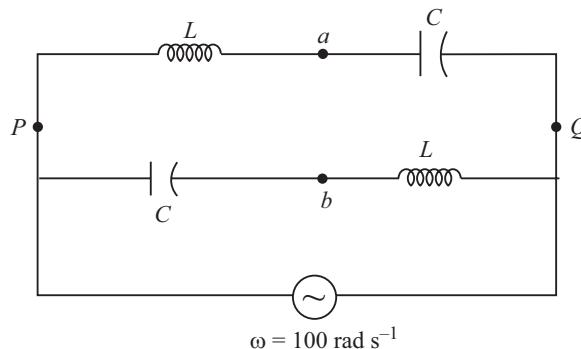


37. The rms value of i_4 is

- (a) 3.12A (b) 9.12A
(c) 6.12A (d) 8.0A

Passage 18

In the circuit shown, each of the two inductors has an inductance $L = 3.0\text{H}$. Capacitance of each capacitor is $C = 25\mu\text{F}$. The source has a peak voltage of 100V and its angular frequency is $\omega = 100 \text{ rad s}^{-1}$.



38. The ratio of voltage across the two inductors is

- (a) 2 : 1 (b) 3 : 5
(c) 4 : 1 (d) 1 : 1

39. The maximum value of instantaneous potential difference between the points a and b will be

- (a) 140V (b) 120V
(c) 100V (d) 200V

Passage 19

A uniform electric field is established inside a long cylindrical straight rod along its axis. The energy density of this electric field inside the rod is $u_E = 2 \times 10^{-17} \text{ Jm}^{-3}$.

The magnetic field energy density is found to be non-uniform inside the rod. Its value at a distance $r = 2\text{cm}$ from the axis, at a point inside the rod, is $u_B = 0.8 \text{ Jm}^{-3}$.

For the material of the rod, assume $\epsilon_r = 1$, $\mu_r = 1$.

40. The current density in the rod is

- (a) uniform
(b) non-uniform
(c) will depend on the radius of the rod
(d) varies linearly with distance from the axis of the rod

41. The conductivity of the material of the rod is

- (a) $2.0 \times 10^6 (\Omega\text{m})^{-1}$
(b) $2.0 \times 10^7 (\Omega\text{m})^{-1}$
(c) $3.7 \times 10^7 (\Omega\text{m})^{-1}$
(d) $8.0 \times 10^7 (\Omega\text{m})^{-1}$

Answer Sheet

Match the Columns

1. $(A) \rightarrow R; (B) \rightarrow Q; (C) \rightarrow P; (D) \rightarrow S$
2. $(A) \rightarrow Q; (B) \rightarrow P; (C) \rightarrow R; (D) \rightarrow S$
3. $(A) \rightarrow Q; (B) \rightarrow P; (C) \rightarrow Q; (D) \rightarrow R$
4. $(A) \rightarrow S, R; (B) \rightarrow R, S; (C) \rightarrow P, R; (D) \rightarrow P, R$
5. $(A) \rightarrow Q, R; (B) \rightarrow P, R; (C) \rightarrow P, R; (D) \rightarrow Q, R$
6. $(A) \rightarrow P, S; (B) \rightarrow P, R, S; (C) \rightarrow Q, T; (D) \rightarrow Q$
7. $(A) \rightarrow S; (B) \rightarrow R; (C) \rightarrow Q; (D) \rightarrow P$
8. $(A) \rightarrow P, Q, R; (B) \rightarrow P, Q, R; (C) \rightarrow P, Q, R; (D) \rightarrow P, Q, R$

Passage Based Problems

- | | | | | | | | | |
|---------|---------|---------|-------------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (c) | 3. (d) | 4. (a) | 5. (a) | 6. (d) | 7. (b) | 8. (a) | 9. (b) |
| 10. (b) | 11. (b) | 12. (c) | 13. (d) | 14. (a) | 15. (c) | 16. (b) | 17. (a) | 18. (d) |
| 19. (c) | 20. (a) | 21. (b) | 22. (b) | 23. (d) | 24. (c) | 25. (a) | 26. (a) | 27. (d) |
| 28. (b) | 29. (a) | 30. (a) | 31. (d) | 32. (b) | 33. (a) | 34. (b) | 35. (d) | 36. (a) |
| 37. (c) | 38. (d) | 39. (b) | 40. (a,c,d) | 41. (c) | | | | |

Electromagnetic Waves

“It requires a much higher degree of imagination to understand the electromagnetic field than to understand invisible angelsI speak of E and B fields and wave my arms and you may imagine that I can see them....I cannot really make a picture that is even nearly like the true waves.”

— Richard Feynman

1. INTRODUCTION

Twenty-first century is based almost entirely on the physics of electromagnetic waves. The television that you see, the mobile phone that you use to chat endlessly, the internet which consumes lot of your time, the microwave oven that you use for cooking and many more devices of today's world use electromagnetic waves for their functioning.

The visible light, infrared, ultraviolet, X-rays, radio waves, etc., all have similar characteristics and are part of a larger universe known as electromagnetic waves. In this chapter, we will try to understand the Maxwell's idea of displacement current and how it helped his understanding of electromagnetic waves. We will limit ourselves mostly to qualitative discussion of the characteristics of electromagnetic waves.

2. AMPERE'S LAW AND DISPLACEMENT CURRENT

Faraday's law of electromagnetic induction established that a time changing magnetic field produces an electric field. The next logical question was whether a time varying electric field generates a magnetic field. James Clerk Maxwell discovered that the answer to this question is –yes. He discovered this result during his attempt to remove an inconsistency in the Ampere's circuital law.

Inconsistency in Ampere's law

Consider a parallel plate capacitor during its charging phase. Instantaneous charge on the capacitor plate is q and it is increasing. The current in the wires is

$$i = \frac{dq}{dt}$$

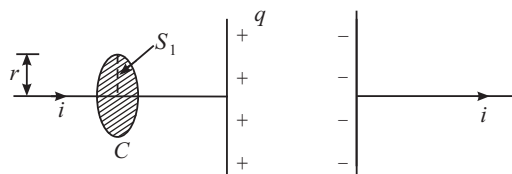


Fig (a)

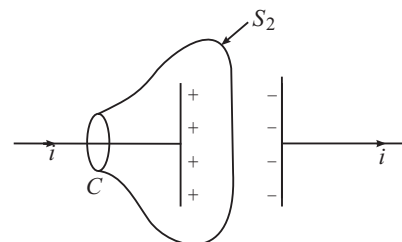


Fig (b)

$\oint_C \vec{B} \cdot d\vec{l}$ is equal to $\mu_0 i$, where i is the current piercing through S_1 or S_2

Consider a circular loop (C) of radius r with the current carrying wire along the axis of the loop. The current (i) sets up a symmetrical magnetic field (B) on the loop C . We can write, using the Ampere's law

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 i \quad \dots(i)$$

Here, i is the current piercing through the surface S_1 of the current loop (see figure (a)).

Now, we will again consider the line integral of $\vec{B}$ on the closed loop C but will consider surface S_2 (see figure (b)) for writing the value of current in the Ampere's law. The surface (S_2) is a pot-shaped surface with its mouth as the loop (C). The surface passes between the parallel plates of the capacitor. No current pierces through this surface.

$$\therefore \oint_C \vec{B} \cdot d\vec{l} = 0 \quad \dots(ii)$$

8.2 Magnetism and Electromagnetic Induction

The equations (i) and (ii) give two different values of the line integral of $\vec{B}$ on the same loop (C) when we consider two different surfaces for deciding the current on the right side of equation.

Maxwell introduced the idea of displacement current to correct the Ampere's circuital law and give it a more general form.

Displacement current

Maxwell corrected the Ampere's law as

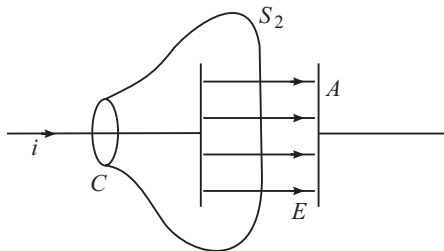
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i + i_d) \quad \dots(1)$$

Here i is the usual conduction current and i_d is the displacement current. The quantity i_d is not a current in its usual sense but is a quantity having the dimensions of the current. It is defined as

$$i_d = \epsilon_0 \frac{d\phi_E}{dt} \quad \dots(2)$$

Here, ϕ_E is the flux of electric field over the surface considered in the Ampere's law.

Once again, consider the surface S_2 for application of Ampere's law over the closed loop (C). There is an electric field in the space between the capacitor plates.



Electric flux through S_2 is $\phi_E = EA$

$$E = \frac{q}{A\epsilon_0} \quad [A = \text{plate area}]$$

The flux of electric field through the surface S_2 is

$$\phi_E = AE = \frac{q}{\epsilon_0}$$

[Number of field lines passing through an area A held parallel to the plates will be same as the number of lines passing through S_2].

Since the capacitor is getting charged, q is increasing and therefore ϕ_E is also increasing

$$\frac{d\phi_E}{dt} = \frac{1}{\epsilon_0} \frac{dq}{dt}$$

$$\text{or, } \epsilon_0 \frac{d\phi_E}{dt} = i \quad \left[\because \frac{dq}{dt} = i = \text{conduction current} \right]$$

In our discussion, we see that

$$i_d = i \quad \dots(\text{iii})$$

No more inconsistency in Ampere's law

Ampere's law in its corrected form is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i + i_d)$$

When we consider surface S_1 (in our previous discussion), there is no electric field. Therefore, $i_d = 0$. We write

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 i$$

i is the current in the wire. When we consider surface S_2 , there is no conduction current through it. But there is a displacement current through it. The flux of electric field linked to it is changing with time. Equation (iii) tells us that the displacement current is equal to i .

$$\begin{aligned} \therefore \oint_C \vec{B} \cdot d\vec{l} &= \mu_0 (i + i_d) \\ &= \mu_0 (0 + i) = \mu_0 i \end{aligned}$$

Now, considering any surface (S_1 , S_2 or any other) will give the same result.

Significance of displacement current

The Ampere's law, in its new form tells us that a magnetic field has one more source, apart from the conventional electric current. This source is time-changing electric field. If there is no conduction current in space but there is time changing E , it will induce a magnetic field (B). One can detect magnetic field between the plates of a capacitor where there is no conventional current but only displacement current.

With the introduction of the displacement current, the laws of electricity and magnetism have become more symmetrical. Now, we say that a time-changing B produces E and a time-changing E produces B .

This symmetry is the cause of electromagnetic waves. In the sections to follow, we will have qualitative discussion on characteristics of electromagnetic waves.

3. MAXWELL'S EQUATIONS

The whole subject of electricity and magnetism can be described mathematically by four equations. These equations are collectively known as Maxwell's equations. We have studied them all in simple form. They are:

$$\oint \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} \quad [\text{Gauss's law for electricity}]$$

$$\oint \vec{B} \cdot d\vec{S} = 0 \quad [\text{Gauss's law for magnetism}]$$

$$\oint \vec{E} \cdot d\vec{l} = - \frac{d\phi_B}{dt} \quad [\text{Faraday's law of electromagnetic induction}]$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i + \epsilon_0 \frac{d\phi_E}{dt} \right) \quad [\text{Ampere's law}]$$

In vacuum, where there is no charge and no conduction current, the Ampere's law is reduced to

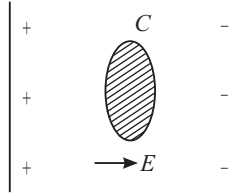
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

Example 1 An ideal parallel plate capacitor is connected to a battery. What is the magnetic field between the capacitor plates after it is completely charged?

Solution

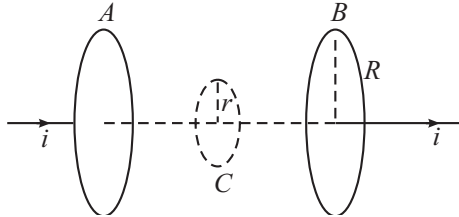
The magnetic field is created either by a conduction current or a displacement current. Both are zero inside a fully charged capacitor.

If we consider a closed loop C inside the capacitor, there is no current piercing through it. Since the electric field is constant, there is no change in ϕ_E . It means that the displacement current is also zero. Magnetic field inside a fully charged capacitor is zero.



Example 2 Magnetic field inside capacitor

A parallel plate capacitor has circular plates of radius R . The capacitor is getting charged and the instantaneous current is i . Consider a circular loop of radius r between the plates. The plane of the loop is parallel to the plane of the plates. (a) Find the value of line integral $\oint \vec{B} \cdot d\vec{l}$ over the closed loop C . (b) Find the magnitude of the magnetic field on the circumference of loop C .



Solution

Concepts

- (i) There is no conduction current through the surface of the loop C . But there is a displacement current through it. When the capacitor is charging, the displacement current through the surface of the loop is non-zero.

$$(ii) i_d = \epsilon_0 \frac{d\phi_E}{dt}$$

$$(a) \text{ Electric field between the plates } E = \frac{q}{A\epsilon_0} = \frac{q}{\pi R^2 \epsilon_0}$$

Electric flux through the surface of the loop is

$$\phi_E = E \cdot \pi r^2 = \frac{qr^2}{R^2 \epsilon_0}$$

$$\frac{d\phi_E}{dt} = \frac{r^2}{R^2 \epsilon_0} \frac{dq}{dt}$$

$$\text{or, } i_d = \epsilon_0 \frac{d\phi_E}{dt} = \frac{r^2}{R^2} i$$

Applying the Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i + i_d)$$

$$\text{or, } \oint \vec{B} \cdot d\vec{l} = \mu_0 \left(0 + \frac{r^2}{R^2} i \right) = \frac{\mu_0 r^2}{R^2} i \quad \dots(i)$$

$$(b) \oint_C \vec{B} \cdot d\vec{l} = B \cdot 2\pi r$$

Here, B is magnetic field at every point on the circumference of the loop. The symmetrical position of the loop allows us to assume this.

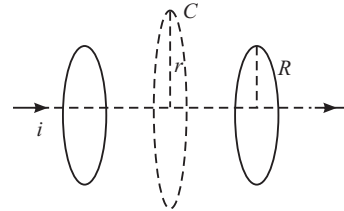
$\therefore$ From (i)

$$B \cdot 2\pi r = \frac{\mu_0 r^2}{R^2} i$$

$$\text{or, } B = \frac{\mu_0 i}{2\pi R^2} \cdot r$$

Note: (i) At $r = R$, $B = \frac{\mu_0 i}{2\pi R}$

- (ii) If we assume that the electric field is uniform only inside the plates and there is no field outside, then for $r > R$



$$\phi_E = E \cdot \pi R^2 = \frac{q}{\pi R^2 \epsilon_0} \pi R^2$$

$$\text{or, } \phi_E = \frac{q}{\epsilon_0}$$

$$\text{or, } i_d = \epsilon_0 \frac{d\phi_E}{dt} = \frac{dq}{dt}$$

$$\text{or, } i_d = i$$

$$\therefore B \cdot 2\pi r = \mu_0 i$$

$$\Rightarrow B = \frac{\mu_0 i}{2\pi r}$$

- (iii) The above results show that for writing the magnetic field we can simply consider the charging capacitor as a current carrying wire of cross-sectional radius R . The displacement current ($i_d = i$) can be assumed to be uniformly distributed over a cross-section of radius R .

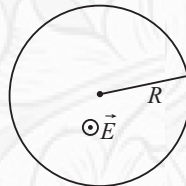


YOUR TURN

Q.1 What is the dimension of quantity $\epsilon_0 \frac{d\phi_E}{dt}$, where ϕ_E is electric flux?

Q.2 A capacitor is in series with a 10Ω resistance. The combination is connected to a 10V cell. Find the displacement current immediately after the circuit is completed.

Q.3 The figure shows a circular region of radius R . A uniform electric flux is directed out of the plane of the page. The electric flux through the region is given by $\phi_E = at$, where a is a constant and t is time in second.



- What is the unit of constant a ?
- Write the magnetic field at a point at a radial distance $r < R$.
- Write the magnetic field at a point at a radial distance $r > R$.

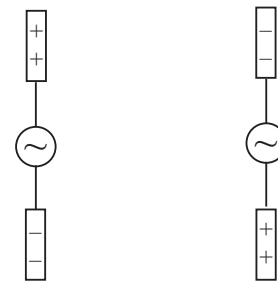
4. SOURCE OF ELECTROMAGNETIC WAVES

We know that a stationary charge produces an electric field and a steady current creates a magnetic field. Maxwell's theory says that an accelerated charge radiates electromagnetic waves. To put a qualitative argument, consider an oscillating charge. An oscillating charge is accelerated. This oscillating charge will produce an oscillating electric field in its surrounding space. This time changing electric field will produce a time changing magnetic field, which in turn will produce a time changing electric field, and so on. The disturbance is in the form of electric field and magnetic field. This disturbance propagates as the two fields keep regenerating one another. Propagation of linked electric and magnetic fields through the space is known as electromagnetic wave. The oscillating charge is the source of energy. Frequency of the oscillating charge is the frequency of the electromagnetic wave.

In an ac circuit, the electrons in the wire oscillate at the frequency of the source. Can we generate light from the oscillating electrons? The answer is no. Visible light has frequency of the order of 10^{14} Hz and the frequency even in modern day electronic circuits is not that high. Therefore, it has not been possible to generate light using an ac circuit. However, smaller frequency (high wavelength) electromagnetic radiation can always be produced using this.

Hertz was successful in generating electromagnetic waves of long wavelengths (in radio wave region) and thereby testing the validity of Maxwell's theory. Marconi, in Italy, followed Hertz work and was successful in generating powerful radio waves and transmitting them over long distances. This marked the beginning of the era of communication using electromagnetic waves.

A common way to produce electromagnetic waves is to apply a sinusoidal voltage source to an antenna. The effect is to produce an oscillating electric dipole. This antenna generates an electromagnetic wave— usually a radio wave.



The oscillating charges on the antenna produce electromagnetic waves.

5. CHARACTERISTICS OF ELECTROMAGNETIC WAVES

The electric and magnetic fields in an electromagnetic wave are perpendicular to each other, and to the direction of propagation of the wave. The variation of electric field and magnetic field in a wave propagating in x direction can be expressed as

$$\left. \begin{aligned} E &= E_y = E_0 \sin \omega \left(t - \frac{x}{c} \right) \\ \text{and } B &= B_z = B_0 \sin \omega \left(t - \frac{x}{c} \right) \end{aligned} \right\} \quad (3)$$

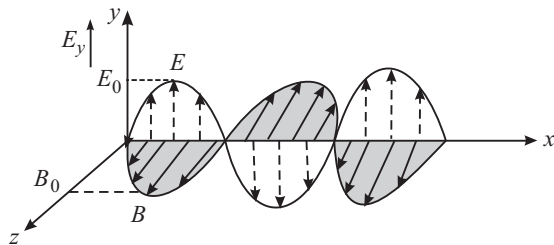
These two equations can be shown to satisfy the Faraday's law as well as the Ampere's law (the last two Maxwell's equations). The above two equations satisfy the Maxwell's equation under conditions

$$E_0 = cB_0 \quad \dots(4)$$

$$\text{and } c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \quad \dots(5)$$

In a way, the equations given in (3) are solutions to the Maxwell's equation under conditions (4) and (5).

The equation given in (3) represents a wave travelling in x direction with speed c . The electric field remains along y direction and oscillates between $-E_0$ and $+E_0$. The magnetic field remains along z direction and changes its value between $-B_0$ and $+B_0$. The values E_0 and B_0 are related by equation (4). The speed of wave is given by equation (5). Putting the values of ϵ_0 and μ_0 yields a value of $c = 2.9979 \times 10^8 \text{ ms}^{-1}$. Experimenters had already measured the speed of light to be near this value. This provided the proof that light is an electromagnetic wave as described by Maxwell. Speed of all electromagnetic waves in vacuum is a universal constant given by the equation (5).



The wave is propagating in x direction. The electric field oscillates in y direction and the magnetic field oscillates in z direction. Actually, this is a linearly polarised wave.

In a medium, the speed of an electromagnetic wave is given by

$$v = \frac{1}{\sqrt{\epsilon \mu}} \quad \dots(6)$$

Here, ϵ = Permittivity of the medium = $\epsilon_r \epsilon_0$ and
 μ = Permeability of the medium = $\mu_r \mu_0$

No material medium is required for propagation of an electromagnetic wave. They are self-sustaining electric and magnetic oscillations which can travel indefinitely in space.

Energy density

The electromagnetic wave is made up of electric and magnetic fields. These fields contain energy. Therefore, an electromagnetic wave contains energy.

For a plane electromagnetic wave, the electric and magnetic fields are expressed by equation (3). In a small volume (dV) of space, the energy in an electromagnetic wave is sum of energies in electric and magnetic fields.

$$\begin{aligned} dU &= dU_E + dU_B \\ &= \frac{1}{2} \epsilon_0 E^2 \cdot dV + \frac{1}{2\mu_0} B^2 \cdot dV \end{aligned}$$

The energy density is

$$\begin{aligned} u &= \frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 \\ &= \frac{1}{2} \epsilon_0 E_0^2 \sin^2 \omega \left(t - \frac{x}{c} \right) + \frac{1}{2\mu_0} B_0^2 \sin^2 \omega \left(t - \frac{x}{c} \right) \end{aligned}$$

The average of the above expression is

$$\begin{aligned} \langle u \rangle &= \frac{1}{2} \epsilon_0 E_0^2 \left\langle \sin^2 \omega \left(t - \frac{x}{c} \right) \right\rangle \\ &\quad + \frac{1}{2\mu_0} B_0^2 \left\langle \sin^2 \omega \left(t - \frac{x}{c} \right) \right\rangle \\ &= \frac{1}{4} \epsilon_0 E_0^2 + \frac{1}{4\mu_0} B_0^2 \quad \dots(i) \end{aligned}$$

The first term in the above expression is the average value of electric field energy density and the second term is the average magnetic field energy density. It is easy to show that both these energy densities are same.

$$\frac{1}{4\mu_0} B_0^2 = \frac{1}{4} \frac{E_0^2}{c^2 \mu_0} \quad [\text{using (4)}]$$

$$= \frac{1}{4} \epsilon_0 E_0^2 \quad [\text{using (5)}]$$

Therefore, an electromagnetic wave contains equal amount of energies in electric and magnetic fields. We can write equation (i) as

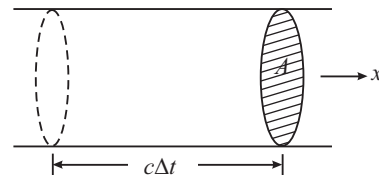
$$\langle u \rangle = \frac{1}{4} \epsilon_0 E_0^2 + \frac{1}{4} \epsilon_0 E_0^2$$

$$\text{or, } \langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 + \frac{1}{2\mu_0} B_0^2 \quad \dots(7)$$

Intensity

Intensity of a wave is defined as amount of energy crossing in unit time through an unit area held normal to the direction of the wave propagation.

Consider an electromagnetic wave propagating in x direction. Think of a cylindrical volume of cross-sectional area A and length $c\Delta t$, as shown. The amount of energy contained in the volume of the cylinder will get transmitted through the cross-sectional area A in time interval Δt .



Energy contained in the cylinder is

$$U = \langle u \rangle (Ac\Delta t) = \frac{1}{2} \epsilon_0 E_0^2 (Ac\Delta t)$$

8.6 Magnetism and Electromagnetic Induction

This energy crosses the area A in time Δt . Therefore, the intensity of the wave is

$$I = \frac{U}{A \Delta t}$$

or, $I = \frac{1}{2} \epsilon_0 E_0^2 c \quad \dots(8)$

Momentum

When an electromagnetic wave is incident on a surface, it exerts force on it. A portion of wave having energy U carries a linear momentum given by

$$P = \frac{U}{c} \quad \dots(9)$$

We will discuss more on this in modern physics.

Example 3 The electric field in a plane electromagnetic wave is given by $E_z = 60 \sin(500x + 1.5 \times 10^{11} t)$ V/m.

- Write wavelength and frequency of the wave.
- Write the expression of the magnetic field.

Solution

Concepts

- In a wave equation $E = E_0 \sin(kx + \omega t)$,
 $k = \frac{2\pi}{\lambda}$ and $\omega = 2\pi f$
- $B_0 = \frac{E_0}{c}$
- The magnetic field oscillates along y direction.

$$(a) \quad \frac{2\pi}{\lambda} = 500 \quad \Rightarrow \quad \lambda = \frac{2\pi}{500} \text{ m} = 1.26 \text{ cm}$$

$$2\pi f = 1.5 \times 10^{11}$$

$$\Rightarrow f = \frac{1.5 \times 10^{11}}{2\pi} = 23.9 \times 10^9 \text{ Hz}$$

$$(b) \quad B_0 = \frac{E_0}{c} = \frac{60}{3 \times 10^8} = 2 \times 10^{-7} \text{ T}$$

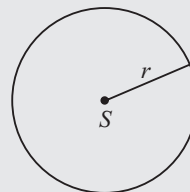
$$\therefore B = 2 \times 10^{-7} \sin(500x + 1.5 \times 10^{11} t) \text{ T}$$

Example 4 Electric and magnetic fields due to a bulb
 A small bulb has a 100W rating and its efficiency to produce light is 4%. Find the peak value of electric and magnetic fields in the light emitted by the bulb, at a distance of 1.0 m from it.

Solution

Concepts

- We can treat the small bulb like a point source.
- The bulb radiates uniformly in all directions. Imagine a spherical surface of radius r with the bulb at its centre. The intensity of light on the spherical surface will be



The electromagnetic wave travels radially from the source S and is incident normally on a spherical surface

$$I = \frac{P}{4\pi r^2}$$

where P = light energy emitted

- Once we know I , we can find E_0 using equation (8).

Intensity of light at a distance of 1m from the bulb is

$$I = \frac{4\% \text{ of } 100\text{W}}{4\pi(1)^2} = \frac{1}{\pi} = 0.32 \text{ Wm}^{-2}$$

Using equation (8), we can write

$$E_0^2 = \frac{2I}{\epsilon_0 c}$$

$$\text{or, } E_0^2 = \frac{2 \times 0.32}{8.85 \times 10^{-12} \times 3 \times 10^8} = 241$$

$$\text{or, } E_0 = 15.5 \text{ V/m}$$

$$\text{And } B_0 = \frac{E_0}{c} = 5.2 \times 10^{-8} \text{ T}$$

Note:

- Though the energy contained in electric field is same as the energy contained in magnetic field, the strength of magnetic field is very weak.
- The energy emitted by a point source goes on spreading over larger and larger surface and its intensity falls off as $\frac{1}{r^2}$. On the other hand, intensity of a plane wave remains constant and there is no spreading in its energy.



YOUR TURN

Q.4 A plane electromagnetic wave is propagating in x direction. The amplitude of the electric field in the wave is $E_0 = 10 \text{ Vm}^{-1}$. Find

- the amplitude of the magnetic field,
- the intensity of the wave.

Q.5 Which physical property is same for an ultraviolet ray of wavelength 200 nm, radiowaves of wavelength 100 m and X-rays of wavelength 1 Å?

Q.6 In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of $1 \times 10^{14} \text{ Hz}$. The amplitude of the electric field is 50 Vm^{-1} .

- Find wavelength of the wave.
- Find the energy density in the wave.

6. ELECTROMAGNETIC SPECTRUM

Electromagnetic waves have widely varying frequencies. Those having frequencies in the range $4.3 \times 10^{14} \text{ Hz}$ to $7.7 \times 10^{14} \text{ Hz}$ (i.e., wavelengths in the range of nearly 380 to 700 nm) produce sensation of vision and are usually called as light. The electromagnetic radiations having higher frequencies than the visible light are ultraviolet rays, X-rays and gamma rays. Those having frequencies less than that of visible light are infrared, microwaves and radiowaves. The classification is mainly based on how the waves are produced and/or detected. There is no sharp distinction between the frequencies (or wavelengths) of the waves. We will give brief description of different types of electromagnetic waves in order of decreasing wavelengths.

Radio waves

The radio waves can be produced by oscillating charges in an electric circuit. A common way is to set up oscillation of charges in an antenna using an ac source. Their usual frequency range is 500 kHz to about 1000 MHz (i.e., wavelength range of a fraction of a metre to over few kilometre). The amplitude modulated (AM) radio functions in the range 530 kHz to 1710 kHz. Such radio stations were ubiquitous, few decades back. Higher frequencies up to 54 MHz are used for short wave bands. All India radio still broadcasts in short wave band. The TV waves range from 54 MHz to 890 MHz and, the now common, FM (frequency modulated) radio band has a range of 88 MHz to 108 MHz. The mobile phones operate in ultra high frequency range. A typical range can be 1710–1860 MHz.

Microwaves

These are electromagnetic radiations having frequency in the range of 300 MHz to 300 GHz (i.e., wavelength in the range of 1 mm to 1 m). High power microwave sources use special vacuum tubes (known as magnetron, klystron,

gyrotron). Low power sources use devices like Gunn diodes and IMPATT diodes.

Microwaves have shorter wavelength than radiowaves. This causes less diffraction when they encounter buildings, trees, hills etc.. This makes them suitable for radars. They require smaller antenna compared to the radiowaves. Microwaves are also employed in microwave ovens.

Infrared waves

These waves have wavelength greater than the red end of the visible spectrum and have wavelength range of 700 nm to 1 mm (frequency range of $3 \times 10^{11} \text{ Hz}$ to $4.3 \times 10^{14} \text{ Hz}$).

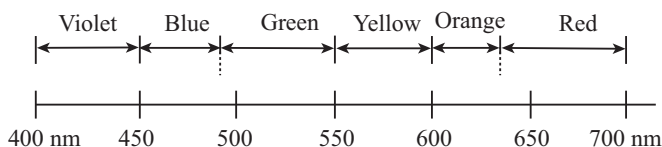
Infrared waves are sometimes called as *heat waves*. They are readily absorbed by H_2O , CO_2 , NH_3 molecules. Water molecules are present in many of the daily life materials and get heated by absorbing any infrared radiation falling on them.

Our Earth receives energy from sunlight. It radiates predominantly in infrared region. This radiated energy is unable to escape the atmosphere in totality. The H_2O , CO_2 , NH_3 molecules (known as greenhouse gases) in the atmosphere absorb a significant part of the energy radiated by the Earth. This helps the Earth in maintaining its average temperature. With rising pollution and increasing quantity of carbon dioxide in the atmosphere, more heat will be retained than necessary. This will lead to global warming.

Infrared detectors are used by satellites to gather photographs for military purpose as well as for other purposes like knowing the area under cotton farming (or any other crop). TV remotes also use infrared radiation to send signal to the television set.

Visible light

The electromagnetic radiation in the wavelength range of 380 nm to 700 nm helps us to see the world. This is commonly known as light.



Approximate wavelength of different colours in the visible light spectrum

Ultraviolet rays (UV rays)

These are electromagnetic waves having wavelengths shorter than the visible violet light. Their wavelength range is 0.6 nm to 400 nm. Ultraviolet radiation is produced by very hot bodies and some special lamps.

The Sun sends a lot of ultraviolet radiation to us. But fortunately, most of it is absorbed by the ozone layer in the atmosphere. A large exposure of UV rays is harmful for us. UV radiation is effectively absorbed by simple glass. Hence, one cannot get sunburn through a glass window. Welders wear special glass goggles and glass face mask to protect them from the UV rays produced in the welding arc.

Due to their short wavelengths, they cannot be diffracted easily and can be focussed into very narrow beams. This makes UV rays suitable for LASIK eye surgery.

UV lamps are used to kill germs in water purifiers.

X-rays

X-rays are produced when high speed electrons are bombarded on a metal target. They are short wavelength radiations with wavelengths in the range of 10^{-4} nm to 10 nm.

X-rays have high penetration power. They are used in detecting bone fractures. They are also used extensively in security checks at airports and other places.

Excessive exposure to X-ray is dangerous.

Gamma-rays

This high frequency radiation is produced in nuclear reactions and radioactivity. They are commonly used to destroy cancer cells. They have very high frequency and short wavelengths. Their wavelength ranges from 10^{-10} m to 10^{-14} m.

This interaction of an electromagnetic wave with matter takes place via the electric and magnetic fields. These fields set in oscillation the charges present in the matter. The detailed interaction depends on the wavelength of the wave. The phenomenon of scattering or absorption of wave depends on the wavelength. The infrared rays (having wavelength higher than the visible light), are capable of inducing vibrations in atoms and molecules of a substance. This vibration increases the internal energy, and hence temperature, of the substance.

In Short

- (i) Ampere's law, as corrected by Maxwell is

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i + i_d)$$

i = usual conduction current

$$i_d = \epsilon_0 \frac{d\phi_E}{dt} = \text{displacement current}$$

- (ii) The significance of the displacement current is that it tells us that a time changing E produces a B .

- (iii) Time changing electric and magnetic fields can keep on regenerating one another and the disturbance in the form of linked E and B can travel in space. This forms electromagnetic wave. Speed of electromagnetic wave in vacuum is given by

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \simeq 3 \times 10^8 \text{ m/s}$$

- (iv) An electromagnetic wave, is produced by an accelerated charge. A charge oscillating harmonically with frequency f produces waves of frequency f .

- (v) In an electromagnetic wave, the electric and the magnetic field oscillate in mutually perpendicular directions which are perpendicular to the direction of the wave propagation. The wave propagates in the direction of $\vec{E} \times \vec{B}$.

- (vi) $\lambda f = c$

λ = wavelength of the wave

f = frequency of the wave

- (vii) $B_0 = \frac{E_0}{c}$

B_0 and E_0 are the peak values of the electric and magnetic fields in an electromagnetic wave

- (viii) In an electromagnetic wave:

Energy density of electric field = Energy density of magnetic field

- (ix) The average value of total energy density is

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2$$

- (x) Intensity of an electromagnetic wave is

$$I = \langle u \rangle \cdot c = \frac{1}{2} \epsilon_0 E^2 \cdot c$$

- (xi) The electromagnetic waves in decreasing order of wavelengths are:

Radio waves, Microwaves, Infrared waves, Visible light, Ultraviolet rays, X-rays and Gamma-rays

Miscellaneous Examples

Example 5 A parallel plate capacitor has plate area A and plate separation d . It is connected to a battery of emf V and internal resistance R , at time $t = 0$. Consider a plane surface of area $\frac{A}{2}$, parallel to the plates and located symmetrically

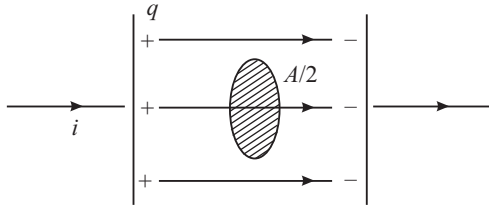
between them. Find the displacement current through the surface as a function of time.

Solution

Concepts

- (i) The charge on the capacitor plate changes with time as $q = CV(1 - e^{-t/RC})$
- (ii) $i_d = \epsilon_0 \frac{d\phi_E}{dt}$

Charge on the capacitor plate varies with time as
 $q = CV(1 - e^{-t/RC})$



Charging current is

$$i = \frac{dq}{dt} = \frac{V}{R} e^{-t/RC}$$

The electric field between the plates is $E = \frac{q}{A\epsilon_0}$

Flux through area $A/2$ is $\phi_E = E \frac{A}{2} = \frac{q}{2\epsilon_0}$

The displacement current through the area $A/2$ is

$$i_d = \epsilon_0 \frac{d\phi_E}{dt} = \epsilon_0 \frac{1}{2\epsilon_0} \frac{dq}{dt} = \frac{1}{2} i$$

$$\text{or, } i_d = \frac{V}{2R} e^{-t/RC} = \frac{V}{2R} e^{-\frac{td}{\epsilon_0 AR}} \quad [\because C = \frac{\epsilon_0 A}{d}]$$

Note: The displacement current behaves like an usual current that is uniformly distributed over area A . Therefore, the displacement current through area $A/2$ is $\frac{i}{2}$.

[i = displacement current through area A = Conduction current in the wire]

Example 6 Displacement current for alternating source connected to capacitor

A capacitor has circular plates of radius $R = 16$ cm and has a capacitance $C = 200$ pF. It is connected to a 200 V ac supply having an angular frequency of $\omega = 100$ rad s^{-1} .

- (a) Find the rms value of displacement current.
- (b) Find the amplitude of the magnetic field B at a point located symmetrically between the plates at a distance of 3.0 cm from the axis.

Solution

Concepts

- (i) Displacement current is always equal to the conduction current charging the capacitor.
- (ii) Rms value of displacement current will be same as rms value of the charging current.
- (iii) The displacement current is distributed uniformly over the cross section of the plate. One can treat the capacitor as a current carrying wire of radius R . (Refer to example 2)

$$(a) X_c = \frac{1}{\omega C} = \frac{1}{100 \times 200 \times 10^{-12}} = 5 \times 10^7 \Omega$$

Rms value of the charging current is

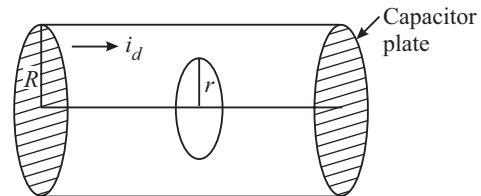
$$I = \frac{V}{X_c} = \frac{200}{5 \times 10^7} = 4 \times 10^{-6} \text{ A}$$

Instantaneous value of displacement current = instantaneous value of conduction current

$\therefore$ Rms value of displacement current = $I = 4 \times 10^{-6}$ A

(b) Magnetic field at a distance r from the axis is

$$B = \left(\frac{\mu_0 i}{2} \right) r$$



$$\text{or, } B = \frac{\mu_0 i}{2\pi R^2} r$$

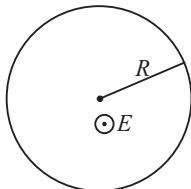
B is maximum when i is maximum.

$$\therefore B_0 = \frac{\mu_0 i}{2\pi R^2} r \quad [i_0 = \text{peak current} = \sqrt{2} I]$$

$$\text{or, } B_0 = \frac{2 \times 10^{-7} \times \sqrt{2} \times 4 \times 10^{-6}}{(0.16)^2} \times 3 = 1.32 \times 10^{-10} \text{ T}$$

Example 7 Non-uniform electric field

In the figure shown, an electric field is directed out of the page within a circular region of radius R . The magnitude of the field is $E = a \left(1 - \frac{r}{R}\right)t$, where t is time, r is radial distance ($r \leq R$) and a is a constant.



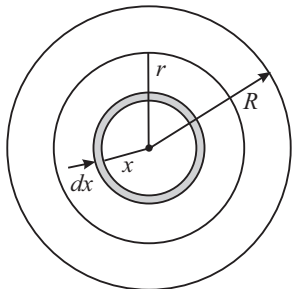
- Find the magnitude of the induced magnetic field at a radial distance $r (< R)$.
- Find the induced magnetic field at a radial distance $r (> R)$.
- At what radial distance is the induced magnetic field maximum?

Solution**Concepts**

(i) Time changing E induces B

$$(ii) \oint \vec{B} \cdot d\vec{l} = \mu_0 (i + i_d)$$

- (a) Consider a circle of radius $r (< R)$.



To calculate the flux of electric field through the circle of radius r , we divide the region into multiple thin rings. Flux through a ring of radius x and width dx is

$$d\phi_E = 2\pi x dx \cdot E = (2\pi x dx) a \left(1 - \frac{x}{R}\right)t$$

$\therefore$ Flux through circle of radius r is

$$\begin{aligned} \phi_E &= 2\pi a t \left[\int_0^r x dx - \frac{1}{R} \int_0^r x^2 dx \right] \\ &= 2\pi a t \left[\frac{r^2}{2} - \frac{r^3}{3R} \right] \end{aligned} \quad \dots(i)$$

$$\therefore \frac{d\phi_E}{dt} = 2\pi a r^2 \left[\frac{1}{2} - \frac{r}{3R} \right]$$

Displacement current through area of the circle is

$$i_d = \epsilon_0 \frac{d\phi_E}{dt} = 2\pi \epsilon_0 a r^2 \left(\frac{1}{2} - \frac{r}{3R} \right) \quad \dots(ii)$$

From symmetry, we can assume that the induced magnetic field is tangential to the circle and has same strength at all points on its circumference.

$$\oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi r$$

Using Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i + i_d) \quad [i = 0]$$

$$\text{or, } B \cdot 2\pi r = 2\pi \mu_0 \epsilon_0 a r^2 \left(\frac{1}{2} - \frac{r}{3R} \right)$$

$$\text{or, } B = \mu_0 \epsilon_0 a r \left(\frac{1}{2} - \frac{r}{3R} \right) \quad \dots(iii)$$

- (b) The displacement current through the circular region of radius R can be obtained by putting $r = R$ in (ii).

$$i_d = \frac{1}{3} \pi \epsilon_0 a R^2$$

Applying the Ampere's law over the circle of radius $r (> R)$,

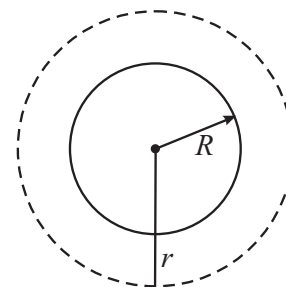
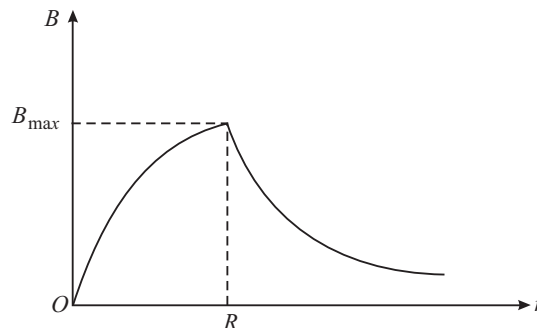
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i_d$$

$$\text{or, } B \cdot 2\pi r = \mu_0 \frac{1}{3} \pi \epsilon_0 a R^2$$

$$\text{or, } B = \frac{\mu_0 \epsilon_0 a R^2}{6r} \quad \dots(iv)$$

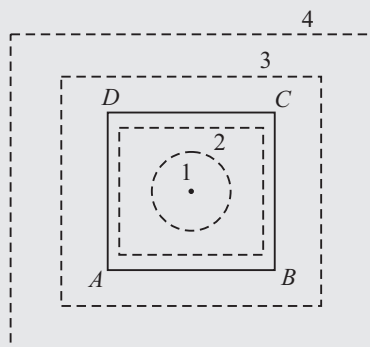
- (c) The induced B is maximum at $r = R$. It can be obtained by putting $r = R$ in (iii) or (iv)

$$B_{\max} = \frac{\mu_0 \epsilon_0 a R}{6}$$



Worksheet 1

- At which of the following points can we detect an electric field?
 - A point near a straight wire carrying a steady current of 5.0A
 - A point inside a solenoid carrying a steady current of 5.0A
 - A point near a wire carrying an alternating current of 5.0A
 - None of the above
- The figure is a view of a plate ($ABCD$) of a parallel plate capacitor from within the capacitor. The dashed lines show four closed paths along which we evaluate $\oint \vec{B} \cdot d\vec{l}$. Rank the paths according to the magnitude of the integral along the paths, when the capacitor is getting discharged, greatest first.



- $4 = 3 > 2 > 1$
 - $4 > 3 > 2 > 1$
 - $2 > 1 > 3 > 4$
 - $2 > 3 > 1 > 4$
- Which of the following has longest wavelength?
 - Gamma ray
 - X-ray
 - Visible light
 - Radio waves
 - An electromagnetic wave has wavelength of $2\text{\AA}$. Which kind of wave is it likely to be?
 - Radio wave
 - Microwave
 - X-ray
 - Ultraviolet ray
 - Which of the following is known as heat waves?
 - Infrared wave
 - Ultraviolet ray
 - X-ray
 - Microwave

- Which of the following will you use for a radar system?
 - A wave of wavelength 2m
 - A wave of wavelength 1mm
 - A wave of wavelength 10m
 - None of the above
- Which of the following is the chief cause of global warming?
 - Increased absorption, of the radiation emitted by the Earth, by the atmosphere
 - Hole in ozone layer
 - Decreasing green house gases in the atmosphere
 - Increased radioactivity inside the Earth
- Energy contained in a small volume of space through which an electromagnetic wave passes, oscillates with
 - the frequency of the wave
 - half the frequency of the wave
 - double the frequency of the wave
 - zero frequency
- The sunlight reaching the Earth has peak electric field of 900 V/m. The peak magnetic field in this light is
 - $3\mu\text{T}$
 - $9\mu\text{T}$
 - $0.3\mu\text{T}$
 - $4\mu\text{T}$
- A charged particle is oscillating with a frequency of 1 GHz. The wavelength of the electromagnetic wave produced by it is
 - 3 mm
 - $1\text{\AA}$
 - $3\text{\AA}$
 - 30 cm
- In a series LCR circuit we have $R = 10\Omega$, $L = 1\text{H}$ and $C = 1\mu\text{F}$. The circuit is in resonance. The electromagnetic wave generated by such a circuit will have wavelength
 - $1.89 \times 10^6 \text{ m}$
 - $1.4 \times 10^{-6} \text{ m}$
 - $2 \mu\text{m}$
 - 4 mm
- Which of the following quantities have no association with an electromagnetic wave?
 - Momentum
 - Electric field
 - Energy
 - Angular momentum

Worksheet 2

- In which of the following cases can we expect a magnetic field in the space between the plates of a parallel plate capacitor?
 - A capacitor is connected to an ac voltage source
 - A capacitor is connected to a dc source and the circuit is in steady state.
 - Immediately after a capacitor is connected to a dc source.
 - A charged capacitor is getting discharged through a resistance.
- An electromagnetic wave in space can be described by its electric field as $E = E_0 \sin(kx - \omega t)$. Which of the following is/are independent of the frequency of the wave?
 - k
 - ω
 - $\frac{\omega}{k}$
 - ωk
- E_{rms} and B_{rms} represent the rms values of the electric and magnetic field in an electromagnetic wave of intensity I . Which of the following relations are correct?
 - $I = \frac{E_{\text{rms}}^2}{c\mu_0}$
 - $I = \frac{cB_{\text{rms}}^2}{\mu_0}$
 - $cB_{\text{rms}} = E_{\text{rms}}$
 - None of the above
- Which of the following are incorrect?
 - The ozone layer in the atmosphere absorbs the infrared radiation coming from the Sun
 - The average surface temperature of the Earth, in absence of the atmosphere would have been higher than present value
 - X-ray coming from the Sun is mostly absorbed by the atmosphere
 - The ultraviolet rays can kill germs
- Which of the following are incorrectly matched?
 - X ray – 1000 Å
 - Gamma ray – 1000 Å
 - Micro wave – 10^{-6} m
 - Television – 500 MHz

Worksheet 3

1. The magnetic field in a plane electromagnetic wave is given by

$$B = (10\mu\text{T}) \sin [(3 \times 10^{15} \text{ s}^{-1}) \left(1 - \frac{x}{c}\right)]$$

- (a) In which part of the electromagnetic spectrum does this wave lie?
- (b) Find the peak value of electric field in the wave.
- (c) Find the average energy density in the wave.
2. Find the amplitude of the electric field in a parallel beam of light of intensity 8.0 Wm^{-2} .

3. A laser beam is operating at 10 mW. Find the amount of energy contained in 10cm length of the beam.

4. A series *LCR* circuit has $R = 10\Omega$, $X_c = 40\Omega$ and $X_L = 40\Omega$ for a source frequency of 100Hz. The voltage of the source is 50V. Find the rms value of displacement current in the capacitor.

5. In the previous question, how will the displacement current change if the frequency of the voltage source is increased?

Answer Sheet

Your Turn

1. [A]

2. 1A

3. (a) $\frac{Vm}{s}$ (b) $\left(\frac{\mu_0 \epsilon_0 a}{2\pi R^2}\right)r$ (c) $\left(\frac{\mu_0 \epsilon_0 a}{2\pi r}\right)$

4. (a) $3.3 \times 10^{-7} \text{ T}$ (b) 0.133 Wm^{-2}

6. (a) 0.03 mm (b) $1.1 \times 10^{-8} \text{ Jm}^{-3}$

5. Speed of wave is same for all. $c = 3 \times 10^8 \text{ ms}^{-1}$

Worksheet 1

1. (c)

2. (a)

3. (d)

4. (c)

5. (a)

6. (b)

7. (a)

8. (c)

9. (a)

10. (d)

11. (a)

12. (d)

Worksheet 2

1. (a, c, d)

2. (c)

3. (a, b, c)

4. (a, b)

Worksheet 3

1. (a) Visible light (b) 3000 V/m (c) $3.98 \times 10^{-5} \text{ Jm}^{-3}$

2. 77.6 Vm^{-1}

3. $3.33 \times 10^{-12} \text{ J}$

4. 5A

5. Decrease

Previous Years' JEE Questions

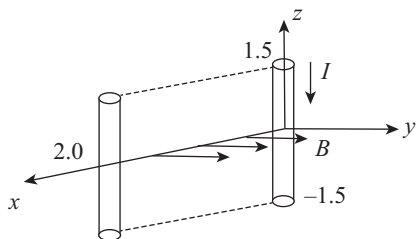
MAGNETIC FIELD

AIEEE/JEE Main Questions

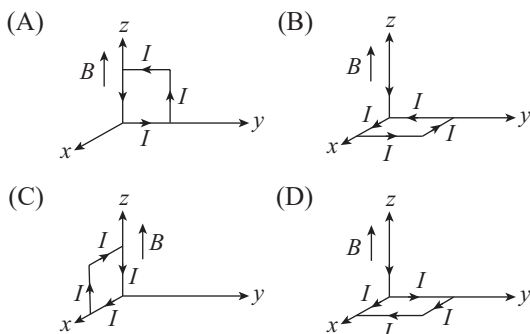
1. A uniform electric field and a uniform magnetic field are produced, pointing in the same direction. An electron is projected with its velocity pointing in the same direction. [AIEEE 2005]
 - (a) The electron will turn to its right
 - (b) The electron will turn to its left
 - (c) The electron's velocity will increase in magnitude
 - (d) The electron's velocity will decrease in magnitude
2. A charged particle of mass m and charge q travels on a circular path of radius r that is perpendicular to a magnetic field B . The time taken by the particle to complete one revolution is [AIEEE 2005]
 - (a) $\frac{2\pi qB}{m}$
 - (b) $\frac{2\pi m}{qB}$
 - (c) $\frac{2\pi mq}{B}$
 - (d) $\frac{2\pi q^2 B}{m}$
3. In a region, steady and uniform electric and magnetic fields are present. These two fields are parallel to each other. A charged particle is released from rest in this region. The path of the particle will be a [AIEEE 2006]
 - (a) circle
 - (b) helix
 - (c) straight line
 - (d) ellipse
4. A charged particle with charge q enters a region of constant, uniform and mutually orthogonal fields $\vec{E}$ and $\vec{B}$ with a velocity $\vec{v}$ perpendicular to both $\vec{E}$ and $\vec{B}$, and comes out without any change in magnitude or direction of $\vec{v}$. Then [AIEEE 2007]
 - (a) $\vec{v} = \vec{B} \times \vec{E} / E^2$
 - (b) $\vec{v} = \vec{E} \times \vec{B} / B^2$
 - (c) $\vec{v} = \vec{B} \times \vec{E} / B^2$
 - (d) $\vec{v} = \vec{E} \times \vec{B} / E^2$
5. A charged particle moves through a magnetic field perpendicular to its direction. Then [AIEEE 2007]
 - (a) kinetic energy changes but the momentum is constant
 - (b) the momentum changes but the kinetic energy is constant
 - (c) both momentum and kinetic energy of the particle are not constant
 - (d) both momentum and kinetic energy of the particle are constant
6. Proton, deuteron and alpha particle of same kinetic energy are moving in circular trajectories in a constant magnetic field. The radii of the paths of proton, deuteron and alpha particle are respectively r_p , r_d and r_α . Which one of the following relations is correct ? [AIEEE 2012]
 - (a) $r_\alpha = r_d > r_p$
 - (b) $r_\alpha = r_p = r_d$
 - (c) $r_\alpha = r_p < r_d$
 - (d) $r_\alpha > r_d > r_p$
7. Two short bar magnets of length 1 cm each have magnetic moments 1.20 Am^2 and 1.00 Am^2 , respectively. They are placed on a horizontal table parallel to each other with their N poles pointing towards the South. They have a common magnetic equator and are separated by a distance of 20.0 cm. The value of the resultant horizontal magnetic induction at the mid-point O of the line joining their centres is close to (Horizontal component of Earth's magnetic induction is $3.6 \times 10^{-5} \text{ Wb m}^{-2}$) [AIEEE 2013]
 - (a) $3.6 \times 10^{-5} \text{ Wb m}^{-2}$
 - (b) $2.56 \times 10^{-4} \text{ Wb m}^{-2}$
 - (c) $3.50 \times 10^{-4} \text{ Wb m}^{-2}$
 - (d) $5.80 \times 10^{-4} \text{ Wb m}^{-2}$

8. A conductor lies along the z -axis at $-1.5 \leq z < 1.5 \text{ m}$ and carries a fixed current of 10.0 A in $(-\hat{a}_z)$ direction (see figure). For a field $\vec{B} = 3.0 \times 10^{-4} e^{-0.2x} \hat{a}_y \text{ T}$, find the power required to move the conductor at a constant speed to $x = 2.0 \text{ m}$, $y = 0 \text{ m}$ in $5 \times 10^{-3} \text{ s}$. Assume parallel motion along the x -axis.

[JEE Main 2014]



- (a) 1.57 W (b) 2.97 W
(c) 14.85 W (d) 29.7 W
9. A rectangular loop of sides 10 cm and 5 cm carrying a current I of 12 A is placed in different orientations as shown in the figures below: [JEE Main 2015]



If there is a uniform magnetic field of 0.3 T in the positive z direction, in which orientations the loop would be in (i) stable equilibrium and (ii) unstable equilibrium?

- (a) (B) and (C), respectively
(b) (A) and (B), respectively
(c) (A) and (C), respectively
(d) (B) and (D), respectively
10. An electron, a proton and an alpha particle having same kinetic energy are moving in circular orbits of radii r_e, r_p, r_α respectively in a uniform magnetic field B . The relation between r_e, r_p, r_α is

[JEE Main 2018]

- (a) $r_e > r_p = r_\alpha$ (b) $r_e < r_p = r_\alpha$
(c) $r_e < r_p < r_\alpha$ (d) $r_e < r_\alpha < r_p$

IIT JEE / JEE Advanced Questions

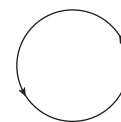
1. In a moving coil galvanometer, torque on the coil can be expressed as $\tau = ki$, where i is current through the wire and k is constant. The rectangular coil of the galvanometer having number of turns N , area A and moment of inertia I is placed in magnetic field B .

[IIT-JEE 2005]

- (a) Find k in terms of given parameters N, I, A and B
(b) Find the torsion constant of the spring, if a current i_0 produces a deflection of $\pi/2$ in the coil.
(c) Find the maximum angle through which the coil is deflected, if charge Q is passed through the coil almost instantaneously. (Ignore the damping in mechanical oscillations)
2. The following field line does not represent

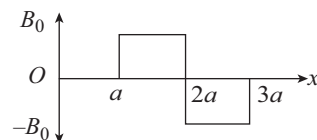
[IIT-JEE 2006]

- (a) induced electric field
(b) magnetostatic field
(c) gravitational field of a mass at rest
(d) electrostatic field



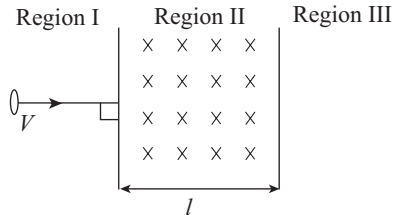
3. A magnetic field $\vec{B} = B_0 \hat{j}$ exists in the region $a < x < 2a$ and $\vec{B} = -B_0 \hat{j}$ in the region $2a < x < 3a$, where B_0 is a positive constant. A positive point charge moving with a velocity $\vec{v} = v_0 \hat{i}$, where v_0 is a positive constant, enters the magnetic field at $x = a$. The trajectory of the charge in this region can be like

[IIT-JEE 2007]

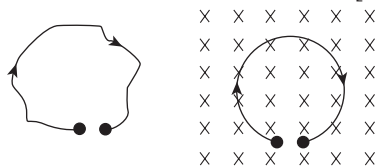


- (a)
- (b)
- (c)
- (d)

4. A particle of mass m and charge q , moving with velocity V enters Region II normal to the boundary as shown in the figure. Region II has a uniform magnetic field B perpendicular to the plane of the paper. The length of the Region II is l . Choose the correct choice (s) [IIT-JEE 2008]

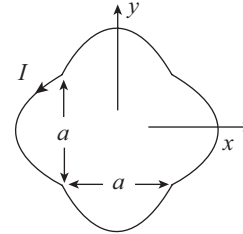


- (a) The particle enters Region III only if its velocity $V > \frac{q\ell B}{m}$
 (b) The particle enters Region III only if its velocity $V < \frac{q\ell B}{m}$
 (c) Path length of the particle in Region II is maximum when velocity $V = \frac{q\ell B}{m}$
 (d) Time spent in Region II is same for any velocity V as long as the particle returns to Region I
5. A thin flexible wire of length L is connected to two adjacent fixed points and carries a current I in the clockwise direction, as shown in the figure. When the system is put in a uniform magnetic field of strength B going into the plane of the paper, the wire takes the shape of a circle. The tension in the wire is [IIT-JEE 2010]



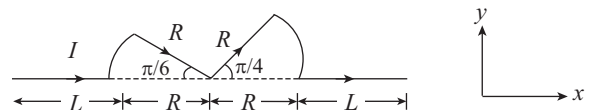
- (a) IBL (b) $\frac{IBL}{\pi}$
 (c) $\frac{IBL}{2\pi}$ (d) $\frac{IBL}{4\pi}$
6. An electron and a proton are moving on straight parallel paths with same velocity. They enter a semi-infinite region of uniform magnetic field perpendicular to the velocity. Which of the following statement(s) is/are true? [IIT-JEE 2011]
- (a) They will never come out of the magnetic field region
 (b) They will come out travelling along parallel paths
 (c) They will come out at the same time
 (d) They will come out at different times

7. A loop carrying current I lies in the $x - y$ plane as shown in the figure. The unit vector $\hat{k}$ is coming out of the plane of the paper. The magnetic moment of the current loop is [IIT-JEE 2012]



- (a) $\left(\frac{\pi}{2} + 1\right)a^2 I \hat{k}$ (b) $\left(\frac{\pi}{2} + 1\right)a^2 I \hat{k}$
 (c) $-\left(\frac{\pi}{2} + 1\right)a^2 I \hat{k}$ (d) $(2\pi + 1)a^2 I \hat{k}$
8. Consider the motion of a positive point charge in a region where there are simultaneous uniform electric and magnetic fields $\vec{E} = E_0 \hat{j}$ and $\vec{B} = B_0 \hat{j}$. At time $t = 0$, this charge has velocity $\vec{v}$ in the $x - y$ plane, making an angle θ with the x -axis. Which of the following option(s) is (are) correct for time $t > 0$? [IIT-JEE 2012]
- (a) If $\theta = 0^\circ$, the charge moves in a circular path in the $x - z$ plane
 (b) If $\theta = 0^\circ$, the charge undergoes helical motion with constant pitch along the y -axis
 (c) If $\theta = 10^\circ$, the charge undergoes helical motion with its pitch increasing with time, along the y -axis
 (d) If $\theta = 90^\circ$, the charge undergoes linear but accelerated motion along the y -axis

9. A conductor (shown in the figure) carrying constant current I is kept in the $x - y$ plane in a uniform magnetic field $\vec{B}$. If F is the magnitude of the total magnetic force acting on the conductor, then the correct statement(s) is (are) [JEE (Advanced) 2015]



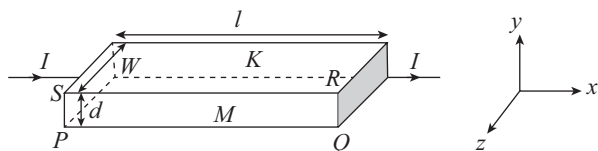
- (a) If B is along $\hat{z}$, $F \propto (L + R)$
 (b) If B is along $\hat{x}$, $F = 0$
 (c) If B is along $\hat{y}$, $F \propto (L + R)$
 (d) If B is along $\hat{z}$, $F = 0$

9.4 Magnetism and Electromagnetic Induction

Q.No. 10 to 11 are based on following passage :

In a thin rectangular metallic strip, a constant current I flows along the positive x -direction, as shown in the figure. The length, width and thickness of the strip are l , w and d , respectively. A uniform magnetic field $\vec{B}$ is applied on the strip along the positive y -direction. Due to this, the charge carriers experience a net deflection along the z -direction. This results in accumulation of charge carriers on the surface $PQRS$ and appearance of equal and opposite charges on the face opposite to $PQRS$. A potential difference along the z -direction is thus developed. Charge accumulation continues until the magnetic force is balanced by the electric force. The current is assumed to be uniformly distributed on the cross section of the strip and carried by electrons.

[JEE (Advanced) 2015]



10. Consider two different metallic strips (1 and 2) of the same material. Their lengths are the same, widths are w_1 and w_2 and thicknesses are d_1 and d_2 , respectively. Two points K and M are symmetrically located on the opposite faces parallel to the x - y plane (see figure). V_1 and V_2 are the potential differences between K and M in strips 1 and 2, respectively. Then, for a given current I flowing through them in a given magnetic field strength B , the correct statement(s) is (are)

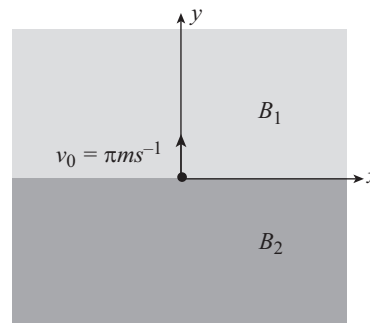
- If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = 2V_1$
- If $w_1 = w_2$ and $d_1 = 2d_2$, then $V_2 = V_1$
- If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = 2V_1$
- If $w_1 = 2w_2$ and $d_1 = d_2$, then $V_2 = V_1$

11. Consider two different metallic strips (1 and 2) of same dimensions (length l , width w and thickness d) with carrier densities n_1 and n_2 , respectively. Strip 1 is placed in magnetic field B_1 and strip 2 is placed in magnetic field B_2 , both along positive y -directions. Then V_1 and V_2 are the potential differences developed between K and M in strips 1 and 2, respectively. Assuming that the current I is the same for both the strips, the correct option(s) is (are)

- If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = 2V_1$
- If $B_1 = B_2$ and $n_1 = 2n_2$, then $V_2 = V_1$
- If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = 0.5V_1$
- If $B_1 = 2B_2$ and $n_1 = n_2$, then $V_2 = V_1$

12. In the xy -plane, the region $y > 0$ has a uniform magnetic field $B_1\hat{k}$ and the region $y < 0$ has another uniform magnetic field $B_2\hat{k}$. A positively charged particle is projected from the origin along the positive

y -axis with speed $v_0 = \pi ms^{-1}$ at $t = 0$, as shown in the figure. Neglect gravity in this problem. Let $t = T$ be the time when the particle crosses the x -axis from below for the first time. If $B_2 = 4B_1$, the average speed of the particle, in ms^{-1} , along the x -axis in the time interval T is _____. [JEE Advanced 2018]



13. A moving coil galvanometer has 50 turns and each turn has an area $2 \times 10^{-4} \text{ m}^2$. The magnetic field produced by the magnet inside the galvanometer is 0.02 T . The torsional constant of the suspension wire is $10^{-4} \text{ Nm rad}^{-1}$. When a current flows through the galvanometer, a full scale deflection occurs if the coil rotates by 0.2 rad . The resistance of the coil of the galvanometer is 50Ω . This galvanometer is to be converted into an ammeter capable of measuring current in the range $0 - 1.0 \text{ A}$. For this purpose, a shunt resistance is to be added in parallel to the galvanometer. The value of this shunt resistance, in ohms, is _____. [JEE Advanced 2018]

MAGNETIC EFFECT OF CURRENT

AIEEE / JEE Main Questions

1. Two concentric coils each of radius equal to $2\pi \text{ cm}$ are placed at right angles to each other. 3 ampere and 4 ampere are the currents flowing in each coil respectively. The magnetic induction in Weber/m^2 at the centre of the coils will be $(\mu_0 = 4\pi \times 10^{-7} \text{ Wb/Am})$

[AIEEE 2005]

- 5×10^{-5}
- 7×10^{-5}
- 12×10^{-5}
- 10^{-5}

2. A long solenoid has 200 turns per cm and carries a current i . The magnetic field at its centre is $6.28 \times 10^{-2} \text{ Weber/m}^2$. Another long solenoid has 100 turns per cm and it carries a current $i/3$. The value of the magnetic field at its centre is

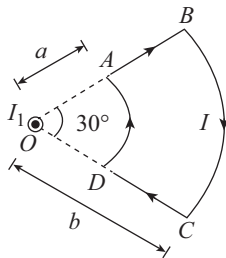
[AIEEE 2006]

- $1.05 \times 10^{-4} \text{ w/m}^2$
- $1.05 \times 10^{-2} \text{ w/m}^2$
- $1.05 \times 10^{-5} \text{ w/m}^2$
- $1.05 \times 10^{-3} \text{ w/m}^2$

3. A long straight wire of radius a carries a steady current i . The current is uniformly distributed across its cross section. The ratio of the magnetic field at $a/2$ and $2a$ is- [AIEEE 2007]
- (a) $1/2$ (b) $1/4$
(c) 4 (d) 1
4. A current I flows along the length of an infinitely long, straight, thin-walled pipe. Then [AIEEE 2007]
- (a) the magnetic field at all points inside the pipe is the same, but not zero
(b) the magnetic field is zero only on the axis of the pipe
(c) the magnetic field is different at different points inside the pipe
(d) the magnetic field at any point inside the pipe is zero
5. A horizontal overhead power line is at a height of 4m from the ground and carries a current of 100A from east to west. The magnetic field directly below it on the ground is ($\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$) [AIEEE 2008]
- (a) $2.5 \times 10^{-7} \text{ T}$ southward
(b) $5 \times 10^{-6} \text{ T}$ northward
(c) $5 \times 10^{-6} \text{ T}$ southward
(d) $2.5 \times 10^{-7} \text{ T}$ northward

Question numbers 6 and 7 are based on the following paragraph:

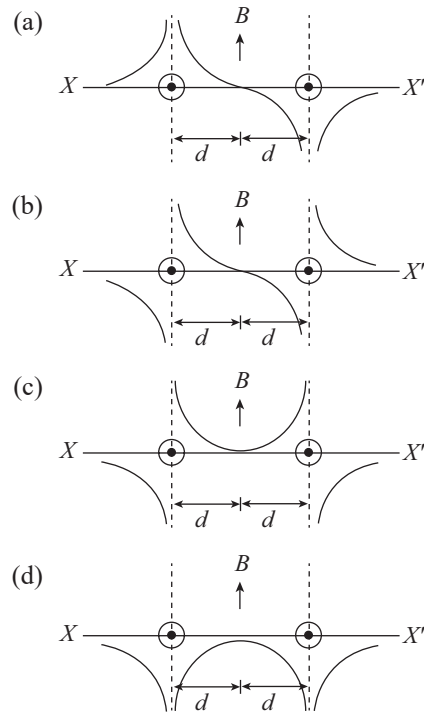
A current loop $ABCD$ is held fixed on the plane of the paper as shown in the figure. The arcs BC (radius = b) and DA (radius = a) of the loop are joined by two straight wires AB and CD . A steady current I is flowing in the loop. Angle made by AB and CD at the origin O is 30° . Another straight thin wire with steady current I_1 flowing out of the plane of the paper is kept at the origin.



6. The magnitude of the magnetic field (B) due to the loop $ABCD$ at the origin (O) is [AIEEE 2009]
- (a) zero (b) $\frac{\mu_0 I(b-a)}{24ab}$
(c) $\frac{\mu_0 I}{4\pi} \left[\frac{b-a}{ab} \right]$ (d) $\frac{\mu_0 I}{4\pi} \left[2(b-a) + \frac{\pi}{3}(a+b) \right]$
7. Due to the presence of the current I_1 at the origin
- (a) The forces on AB and DC are zero
(b) The forces on AD and BC are zero

- (c) The magnitude of the net force on the loop is given by $\frac{I_1 I}{4\pi} \mu_0 \left[2(b-a) + \frac{\pi}{3}(a+b) \right]$
- (d) The magnitude of the net force on the loop is given by $\frac{\mu_0 I I_1}{24ab} (b-a)$

8. Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper as shown. The variation of the magnetic field B along the line XX' is given by [AIEEE 2010]

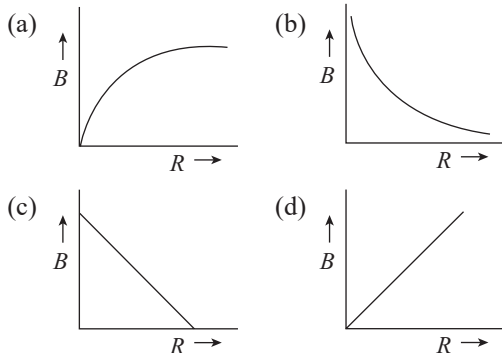


9. A current I flows in an infinitely long wire with cross section in the form of a semicircular ring of radius R . The magnitude of the magnetic induction along its axis is [AIEEE 2011]

- (a) $\frac{\mu_0 I}{2\pi R}$ (b) $\frac{\mu_0 I}{4\pi R}$
(c) $\frac{\mu_0 I}{\pi^2 R}$ (d) $\frac{\mu_0 I}{2\pi^2 R}$

10. A charge Q is uniformly distributed over the surface of non-conducting disc of radius R . The disc rotates about an axis perpendicular to its plane and passing through its centre with an angular velocity ω . As a result of this rotation, a magnetic field of induction B is obtained at the centre of the disc. If we keep both the amount of charge placed on the disc and its angular velocity to be constant and vary the radius of the disc then the variation of the magnetic induction

at the centre of the disc will be represented by the figure [AIEEE 2012]



11. Two long current carrying thin wires, both with current I , are held by insulating threads of length L and are in equilibrium as shown in the figure, with threads making an angle ' θ ' with the vertical. If wires have mass λ per unit length then the value of I is (g = gravitational acceleration)

[JEE Main 2015]

- (a) $\sqrt{\frac{\pi\lambda g L}{\mu_0}} \tan \theta$ (b) $\sin \theta \sqrt{\frac{\pi\lambda g L}{\mu_0 \cos \theta}}$
 (c) $2 \sin \theta \sqrt{\frac{\pi\lambda g L}{\mu_0 \cos \theta}}$ (d) $2 \sqrt{\frac{\pi g L}{\mu_0}} \tan \theta$

12. Two coaxial solenoids of different radii carry current I in the same direction. Let $\vec{F}_1$ be the magnetic force on the inner solenoid due to the outer one and $\vec{F}_2$ be the magnetic force on the outer solenoid due to the inner one. Then [JEE Main 2015]

- (a) $\vec{F}_1$ is radially outwards and $\vec{F}_2 = 0$
 (b) $\vec{F}_1 = \vec{F}_2 = 0$
 (c) $\vec{F}_1$ is radially inwards and $\vec{F}_2$ is radially outwards
 (d) $\vec{F}_1$ is radially inwards and $\vec{F}_2 = 0$

13. Two identical wires A and B , each of length ' l ' carry the same current I . Wire A is bent into a circle of radius R and wire B is bent to form a square of side ' a '. If B_A and B_B are the values of magnetic field at the centres of the circle and square respectively, then the ratio $\frac{B_A}{B_B}$ is [JEE Main 2016]

- (a) $\frac{\pi^2}{8}$ (b) $\frac{\pi^2}{16\sqrt{2}}$
 (c) $\frac{\pi^2}{16}$ (d) $\frac{\pi^2}{8\sqrt{2}}$

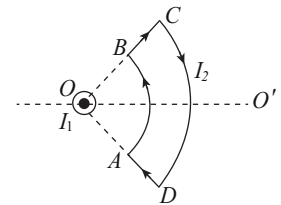
14. The dipole moment of a circular loop carrying a current I , is m and the magnetic field at the centre of the loop is B_1 . When the dipole moment is doubled by keeping the current constant, the magnetic field at the centre of the loop is B_2 . The ratio $\frac{B_1}{B_2}$ is

[JEE Main 2018]

- (a) 2 (b) $\sqrt{3}$
 (c) $\sqrt{2}$ (d) $\frac{1}{\sqrt{2}}$

IIT JEE / JEE Advanced Questions

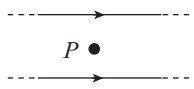
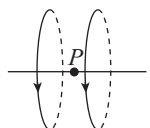
1. Current I_1 is flowing out from the plane of paper. A steady state current I_2 is flowing in the loop $ABCD$ as shown.



[IIT-JEE 2006]

- (a) The net force on the loop is zero
 (b) The net torque on the loop is zero
 (c) As seen from O , the loop will rotate clockwise along OO' axis
 (d) As seen from O , the loop will rotate in anticlockwise direction along OO' axis.

2. Two wires each carrying a steady current I are shown in four configurations in Column-I. Some of the resulting effects are described in Column-II. Match the statements in Column-I with the statements in Column-II and indicate your answer by darkening appropriate bubbles in the 4×4 matrix given in the ORS. [IIT-JEE 2007]

Column-I		Column-II	
(A)	Point P is situated midway between the wires 	(p)	The magnetic fields (B) at P due to the currents in the wires are in the same direction
(B)	Point P is situated at the midpoint of the line joining the centres of the circular wires, which have same radii 	(q)	The magnetic fields (B) at P due to the currents in the wires are in opposite directions

(C)	Point P is situated at the midpoint of the line joining the centres of the circular wires, which have same radii	(r)	There is no magnetic field at P
(D)	Point P is situated at the common centre of the wires	(s)	The wires repel each other

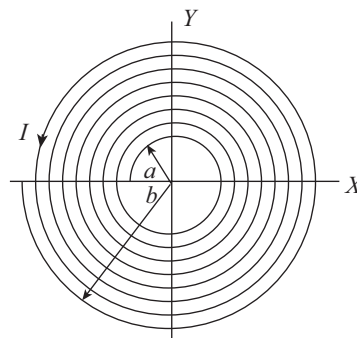
3. Six point charges, each of the same magnitude q , are arranged in different manners as shown in Column II. In each case, a point M and a line PQ passing through M are shown. Let E be the electric field and V be the electric potential at M (potential at infinity is zero) due to the given charge distribution when it is at rest. Now, the whole system is set into rotation with a constant angular velocity about the line PQ . Let B be the magnetic field at M and μ be the magnetic moment of the system in this condition. Assume each rotating charge to be equivalent to a steady current.

[IIT-JEE 2009]

Column-I	Column-II
(A) $E = 0$	(p) Charges are at the corners of a regular hexagon. M is at the centre of the hexagon. PQ is perpendicular to the plane of the hexagon.
(B) $V \neq 0$	(q) Charges are on a line perpendicular to PQ at equal intervals. M is the mid-point between the two innermost charges.
(C) $B = 0$	(r) Charges are placed on two coplanar insulating rings at equal intervals. M is the common centre of the rings. PQ is perpendicular to the plane of the rings.
(D) $\mu \neq 0$	(s) Charges are placed at the corners of a rectangle of sides a and $2a$ and at the mid-points of the longer sides. M is at the centre of the rectangle. PQ is parallel to the longer sides.

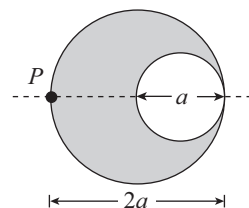
(t)	Charges are placed on two coplanar, identical insulating rings at equal intervals. M is the mid-point between the centres of the rings. PQ is perpendicular to the line joining the centres and coplanar to the rings.
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4. A long insulated copper wire is closely wound as a spiral of ' N ' turn. The spiral has inner radius ' a ' and outer radius ' b '. The spiral lies in the X - Y plane and a steady current ' I ' flows through the wire. The Z -component of the magnetic field at the centre of the spiral is given by [IIT-JEE 2011]



- (a) $\frac{\mu_0 N I}{2(b-a)} \ln\left(\frac{b}{a}\right)$ (b) $\frac{\mu_0 N I}{2(b-a)} \ln\left(\frac{b+a}{b-a}\right)$
 (c) $\frac{\mu_0 N I}{2b} \ln\left(\frac{b}{a}\right)$ (d) $\frac{\mu_0 N I}{2b} \ln\left(\frac{b+a}{b-a}\right)$

5. A cylindrical cavity of diameter a exists inside a cylinder of diameter $2a$ as shown in the figure. Both the cylinder and the cavity are infinitely long. A uniform current density J flows along the length.



the magnitude of the magnetic field at the point P is given by $\frac{N}{12} \mu_0 a J$, then the value of N is

[IIT-JEE 2012]

6. A steady current I flows along an infinitely long hollow cylindrical conductor of radius R . This cylinder is placed coaxially inside an infinite solenoid of radius $2R$. The solenoid has n turns per unit length and carries a steady current I . Consider a point P at a distance r from the common axis. The correct statement(s) is (are) [JEE Advanced 2013]

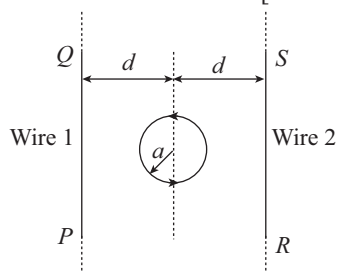
- (a) In the region $0 < r < R$, the magnetic field is non-zero

- (b) In the region $R < r < 2R$, the magnetic field is along the common axis.
- (c) In the region $R < r < 2R$, the magnetic field is tangential to the circle of radius r , centered on the axis
- (d) In the region $r > 2R$, the magnetic field is non-zero

Q.No. 7 and 8 are based on following passage :

The figure shows a circular loop of radius a with two long parallel wires (numbered 1 and 2) all in the plane of the paper. The distance of each wire from the centre of the loop is d . The loop and the wires are carrying the same current I . The current in the loop is in the counterclockwise direction if seen from above.

[JEE Advanced 2014]



7. When $d \approx a$ but wires are not touching the loop, it is found that the net magnetic field on the axis of the loop is zero at a height h above the loop. In that case
- (a) current in wire 1 and wire 2 is in the direction PQ and RS , respectively and $h \approx a$
- (b) current in wire 1 and wire 2 is in the direction PQ and SR , respectively and $h \approx a$
- (c) current in wire 1 and wire 2 is in the direction PQ and SR , respectively and $h \approx 1.2a$
- (d) current in wire 1 and wire 2 is in the direction PQ and RS , respectively and $h \approx 1.2a$
8. Consider $d \gg a$, and the loop is rotated about its diameter parallel to the wires by 30° from the position shown in the figure. If the current in the wires are in the opposite directions, the torque on the loop at its new position will be (assume that the net field due to the wires is constant over the loop)

- (a) $\frac{\mu_0 I^2 a^2}{d}$ (b) $\frac{\mu_0 I^2 a^2}{2d}$
- (c) $\frac{\sqrt{3}\mu_0 I^2 a^2}{d}$ (d) $\frac{\sqrt{3}\mu_0 I^2 a^2}{2d}$

9. Two infinitely long straight wires lie in the xy -plane along the lines $x = \pm R$. The wire located at $x = +R$ carries a constant current I_1 and the wire located at $x = -R$ carries a constant current I_2 . A circular loop of radius R is suspended with its centre at $(0, 0, \sqrt{3}R)$

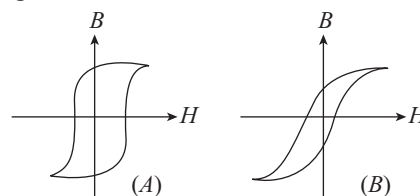
and in a plane parallel to the xy -plane. This loop carries a constant current I in the clockwise direction as seen from above the loop. The current in the wire is taken to be positive if it is in the $+\hat{j}$ direction. Which of the following statements regarding the magnetic field $\vec{B}$ is (are) true?

[JEE Advanced 2018]

- (a) If $I_1 = I_2$, then $\vec{B}$ cannot be equal to zero at the origin $(0, 0, 0)$
- (b) If $I_1 > 0$ and $I_2 < 0$, then $\vec{B}$ can be equal to zero at the origin $(0, 0, 0)$
- (c) If $I_1 < 0$ and $I_2 > 0$, then $\vec{B}$ can be equal to zero at the origin $(0, 0, 0)$
- (d) If $I_1 = I_2$, then the z -component of the magnetic field at the centre of the loop is $\left(-\frac{\mu_0 I}{2R}\right)$

MAGNETIC PROPERTIES OF MATTER

- Needles N_1 , N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will [AIEEE 2006]
 - attract N_1 and N_2 strongly but repel N_3
 - attract N_1 strongly, N_2 weakly and repel N_3 weakly
 - attract N_1 strongly, but repel N_2 and N_3 weakly
 - attract all three of them
- Relative permittivity and permeability of a material are ϵ_r and μ_r respectively. Which of the following values of these quantities are allowed for a diamagnetic material ? [AIEEE 2008]
 - $\epsilon_r = 0.5$, $\mu_r = 1.5$
 - $\epsilon_r = 1.5$, $\mu_r = 0.5$
 - $\epsilon_r = 0.5$, $\mu_r = 0.5$
 - $\epsilon_r = 1.5$, $\mu_r = 1.5$
- The coercivity of a small magnet where the ferromagnet gets demagnetised is $3 \times 10^3 \text{ Am}^{-1}$. The current requires to be passed in a solenoid of length 10 cm and number of turns 100, so that the magnet gets demagnetised when inside the solenoid, is [JEE Main 2014]
 - 30 mA
 - 60 mA
 - 3 A
 - 6 A
- Hysteresis loops for two magnetic materials A and B are given below



These materials are used to make magnets for electric generators, transformer core and electromagnet core. Then it is proper to use [JEE Main 2017]

- (a) A for electric generators and transformers
- (b) A for electromagnets and B for electric generators
- (c) A for transformers and B for electric generators
- (d) B for electromagnets and transformers

ELECTROMAGNETIC INDUCTION

AIEEE / JEE Main Questions

1. The flux linked with a coil at any instant t is given by $\phi = 10t^2 - 50t + 250$. The induced emf at $t = 3s$ is [AIEEE 2006]

- (a) 190 V
- (b) -190 V
- (c) -10 V
- (d) 10 V

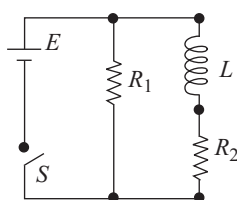
2. An ideal coil of $10H$ is connected in series with a resistance of 5Ω and a battery of $5V$. Two seconds after the connection is made, the current flowing in ampere in the circuit is [AIEEE 2007]

- (a) $(1 - e^{-1})$
- (b) $(1 - e)$
- (c) e
- (d) e^{-1}

3. Two coaxial solenoids are made by winding a thin insulated wire over a pipe of cross-sectional area $A = 10 \text{ cm}^2$ and length $= 20 \text{ cm}$. If one of the solenoids has 300 turns and the other 400 turns, their mutual inductance is ($\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1}$) [AIEEE 2008]

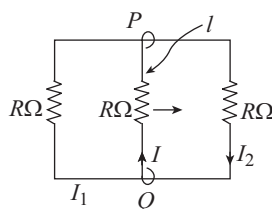
- (a) $2.4\pi \times 10^{-5} \text{ H}$
- (b) $4.8\pi \times 10^{-4} \text{ H}$
- (c) $4.8\pi \times 10^{-5} \text{ H}$
- (d) $2.4\pi \times 10^{-4} \text{ H}$

4. An inductor of inductance $L = 400 \text{ mH}$ and resistors of resistances $R_1 = 2\Omega$ and $R_2 = 2\Omega$ are connected to a battery of emf $12V$ as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at $t = 0$. The potential drop across L as a function of time is [AIEEE 2009]



- (a) $6e^{-5t} \text{ V}$
- (b) $\frac{12}{t} e^{-3t} \text{ V}$
- (c) $6\left(1 - e^{-\frac{t}{0.2}}\right) \text{ V}$
- (d) $12e^{-5t} \text{ V}$

5. A rectangular loop has a sliding connector PQ of length l and resistance $R\Omega$ and it is moving with a speed v as shown. The set-up is placed in a uniform magnetic field



going into the plane of the paper. The three currents I_1 , I_2 and I are [AIEEE 2010]

- (a) $I_1 = I_2 = \frac{Blv}{6R}, I = \frac{Blv}{3R}$
- (b) $I_1 = -I_2 = \frac{Blv}{R}, I = \frac{2Blv}{R}$
- (c) $I_1 = I_2 = \frac{Blv}{3R}, I = \frac{2Blv}{3R}$
- (d) $I_1 = I_2 = I = \frac{Blv}{R}$

6. A boat is moving due east in a region where the Earth's magnetic field is $5.0 \times 10^{-5} \text{ NA}^{-1}\text{m}^{-1}$ due north and horizontal. The boat carries a vertical aerial $2m$ long. If the speed of the boat is 1.50 ms^{-1} , the magnitude of the induced emf in the wire of aerial is [AIEEE 2011]

- (a) 0.50 mV
- (b) 0.15 mV
- (c) 1 mV
- (d) 0.75 mV

7. A fully charged capacitor C with initial charge q_0 is connected to a coil of self inductance L at $t = 0$. The time at which the energy is stored equally between the electric and the magnetic fields is [AIEEE 2011]

- (a) $2\pi\sqrt{LC}$
- (b) $\sqrt{LC}$
- (c) $\pi\sqrt{LC}$
- (d) $\frac{\pi}{4}\sqrt{LC}$

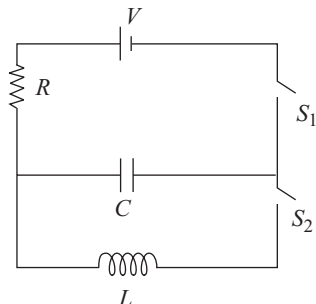
8. A coil is suspended in a uniform magnetic field, with the plane of the coil parallel to the magnetic lines of force. When a current is passed through the coil, it starts oscillating; it is very difficult to stop. But if an aluminium plate is placed near to the coil, it stops. This is due to [AIEEE 2012]

- (a) electromagnetic induction in the aluminium plate giving rise to electromagnetic damping
- (b) development of air current when the plate is placed
- (c) induction of electrical charge on the plate
- (d) shielding of magnetic lines of force as aluminium is a paramagnetic material

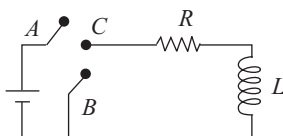
9. A circular loop of radius 0.3 cm lies parallel to a much bigger circular loop of radius 20 cm . The centre of the small loop is on the axis of the bigger loop. The distance between their centres is 15 cm . If a current of $2.0A$ flows through the smaller loop, then the flux linked with bigger loop is [JEE Main 2013]

- (a) $6 \times 10^{-11} \text{ Wb}$
- (b) $3.3 \times 10^{-11} \text{ Wb}$
- (c) $6.6 \times 10^{-9} \text{ Wb}$
- (d) $9.1 \times 10^{-11} \text{ Wb}$

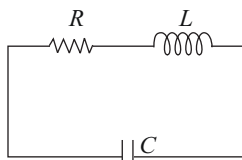
10. In an LCR circuit as shown below, both switches are open initially. Now switch S_1 is closed, S_2 is kept open. (q is charge on the capacitor and $\tau = RC$ is capacitive time constant). Which of the following statements is correct? [JEE Main 2013]



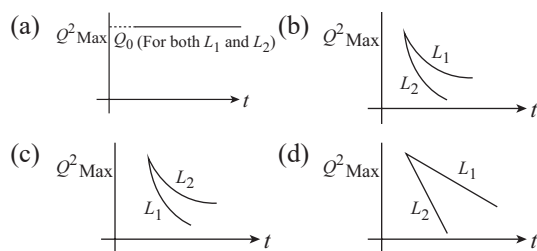
- (a) At $t = \tau, q = CV/2$
 (b) At $t = 2\tau, q = CV(1 - e^{-2})$
 (c) At $t = \frac{\tau}{2}, q = CV(1 - e^{-1})$
 (d) Work done by the battery is half of the energy dissipated in the resistor
11. In the circuit shown here, the point 'C' is kept connected to point 'A' till the current flowing through the circuit becomes constant. Afterward, suddenly, point 'C' is disconnected from point 'A' and connected to point 'B' at time $t = 0$. Ratio of the voltage across resistance and the inductor at $t = L/R$ will be equal to [JEE Main 2014]
- (a) $\frac{e}{1-e}$ (b) 1
 (c) -1 (d) $\frac{1-e}{e}$



12. An LCR circuit is equivalent to a damped pendulum. In an LCR circuit, the capacitor is charged to Q_0 and then connected to the L and R as shown. If a student plots graphs of the square of maximum charge ($Q_{\max}^2$) on the capacitor with time (t) for two different values L_1 and L_2 ($L_1 > L_2$) of L then which of the following represents this graph correctly? (plots are schematic and not drawn to scale)

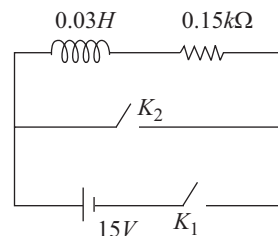


[JEE Main 2015]

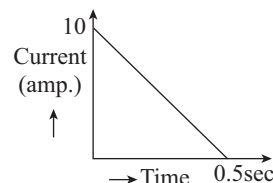


13. An inductor ($L = 0.03\text{H}$) and a resistor ($R = 0.15\text{k}\Omega$) are connected in series to a battery of 15V emf in a circuit shown below. The key K_1 has been kept closed for a long time. Then at $t = 0$, K_1 is opened and key K_2 is closed simultaneously. At $t = 1\text{ms}$, the current in the circuit will be ($e^5 \approx 150$)

[JEE Main 2015]



- (a) 0.67 mA (b) 100 mA
 (c) 67 mA (d) 6.7 mA
14. In a coil of resistance 100Ω , a current is induced by changing the magnetic flux through it as shown in the figure. The magnitude of change in flux through the coil is [JEE Main 2017]

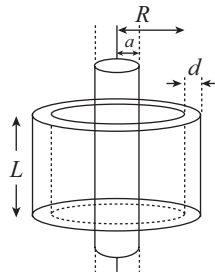


- (a) 200 Wb (b) 225 Wb
 (c) 250 Wb (d) 275 Wb

IIT JEE / JEE Advanced Questions

1. An infinitely long cylinder is kept parallel to a uniform magnetic field B directed along positive z axis. The direction of induced current as seen from the z axis will be [IIT-JEE 2005]
- (a) zero
 (b) anticlockwise of the +ve z axis
 (c) clockwise of the +ve z axis
 (d) along the magnetic field

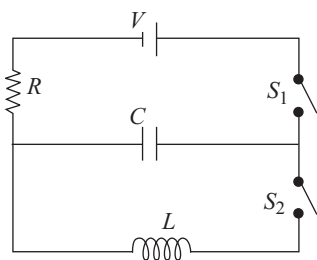
2. A long solenoid of radius a and number of turns per unit length n is enclosed by cylindrical shell of radius R , thickness d ($d \ll R$) and length L . A variable current $i = i_0 \sin \omega t$ flows through the coil. If the resistivity of the material of cylindrical shell is ρ , find the induced current in the shell.



[IIT-JEE 2005]

Paragraph for questions 3 to 5

In the given circuit, initially S_1 is closed at $t = 0$ and S_2 is kept open. The maximum charge on the capacitor subsequently is Q_0 . After a long time, the switch S_1 is opened and S_2 is closed. $\tau = RC$



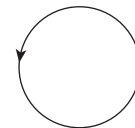
3. Which of the following is correct?
- Charge $Q = \frac{Q_0}{2}$ in time $t = \tau$
 - Charge $Q = Q_0(1 - e^{-2})$ in time $t = 2\tau$
 - Work done by battery $= \frac{1}{2}$ energy dissipated in resistor
 - Charge $Q = Q_0(1 - e^{-1})$ in time $t = 2\tau$
4. Take the instant when S_1 is opened and S_2 is closed at $t = 0$. The charge Q on capacitor at any time can hold a value given by
- $Q = Q_0 \cos\left(\frac{\pi}{2} - \omega t\right)$
 - $Q = Q_0 \cos\left(\frac{\pi}{2} + \omega t\right)$
 - $Q = -LC \frac{d^2 Q}{dt^2}$
 - $Q = \frac{1}{\sqrt{LC}} \frac{d^2 Q}{dt^2}$
5. After S_2 is closed, which one is correct ?
- The current always flows in one direction
 - The energy is exchanged between the capacitor and the inductor

- The energy initially is magnetic
- The value of a maximum instantaneous current

$$I = V \sqrt{\frac{L}{C}}$$

6. A field line is shown in the figure. This field cannot represent [IIT-JEE 2006]

- magnetostatic field
- electrostatic field
- induced electric field
- gravitational field

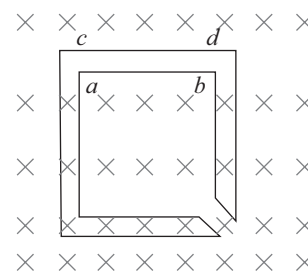


7. Match the following (one item of Column-I may match more than one item in Column-II)

Column-I		Column-II	
(A)	Dielectric ring uniformly charged	(p)	Constant electrostatic field out of system
(B)	Dielectric ring uniformly charged rotating with angular velocity ω	(q)	Magnetic field is present
(C)	Constant current in ring i	(r)	Electric field (induced)
(D)	A current $I = I_0 \cos \omega t$ in a ring	(s)	Magnetic dipole moment

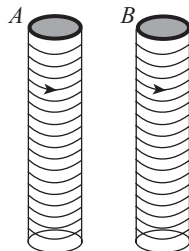
8. The figure shows certain wire segments joined together to form a coplanar loop. The loop is placed in a perpendicular magnetic field in the direction going into the plane of the figure. The magnitude of the field increases with time. I_1 and I_2 are the currents in the segments ab and cd . Then

[IIT-JEE 2009]



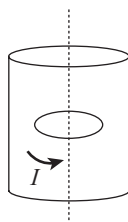
- $I_1 > I_2$
- $I_1 < I_2$
- I_1 is in the direction ba and I_2 is in the direction cd
- I_1 is in the direction ab and I_2 is in the direction dc

9. Two metallic rings A and B , identical in shape, size but having different resistivities ρ_A and ρ_B , are kept on top of two identical solenoids as shown in the figure. When current I is switched on in both the solenoids in identical manner, the rings A and B jump to heights h_A and h_B , respectively, with $h_A > h_B$. The possible relation(s) between their resistivities and their masses m_A and m_B is (are)



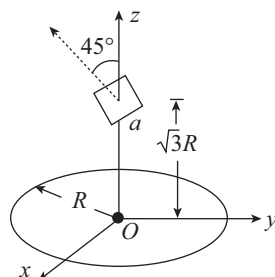
[IIT-JEE 2009]

- (a) $\rho_A > \rho_B$ and $m_A = m_B$
 (b) $\rho_A < \rho_B$ and $m_A = m_B$
 (c) $\rho_A > \rho_B$ and $m_A > m_B$
 (d) $\rho_A < \rho_B$ and $m_A < m_B$
10. A long circular tube of length 10 m and radius 0.3 m carries a current I along its curved surface as shown. A wire-loop of resistance 0.005Ω and of radius 0.1 m is placed inside the tube with its axis coinciding with the axis of the tube. The current varies as $I = I_0 \sin(300t)$ where I_0 is constant. If the magnetic moment of the loop is $N\mu_0 I_0 \cos(300t)$, then 'N' is.....



[IIT-JEE 2011]

11. A circular wire loop of radius R is placed in the x - y plane centered at the origin O . A square loop of side a ($a \ll R$) having two turns is placed with its centre at $z = \sqrt{3}R$ along the axis of the circular wire loop, as shown in figure. The plane of the square loop makes an angle of 45° with respect to the z -axis. If the mutual inductance between the loops is given by $\frac{\mu_0 a^2}{2^{p/2} R}$, then the value of p is



[IIT-JEE 2012]

Q.12 and 13 are based on the following passage

A point charge Q is moving in a circular orbit of radius R in the x - y plane with an angular velocity ω .

This can be considered as equivalent to a loop carrying a steady current $\frac{Q\omega}{2\pi}$.

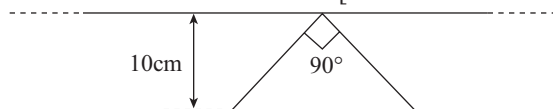
A uniform magnetic field along the positive z -axis is now switched on, which increases at a constant rate from 0 to B in one second. Assume that the radius of the orbit remains constant. The application of the magnetic field induces an emf in the orbit. The induced emf is defined as the work

done by an induced electric field in moving an unit positive charge around a closed loop. It is known that for an orbiting charge, the magnetic dipole moment is proportional to the angular momentum with a proportionality constant γ .

[IIT-JEE 2013]

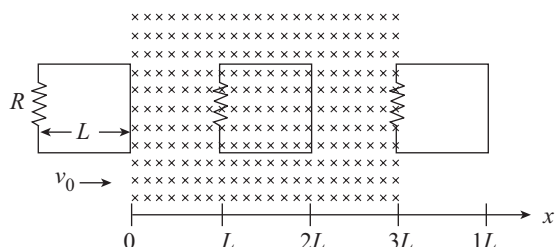
12. The magnitude of the induced electric field in the orbit at any instant of time during the time interval of the magnetic field change is
- (a) $\frac{BR}{4}$ (b) $\frac{BR}{2}$
 (c) BR (d) $2BR$
13. The change in the magnetic dipole moment associated with the orbit, at the end of the time interval of the magnetic field change is
- (a) $-\gamma BQR^2$ (b) $-\gamma \frac{BQR^2}{2}$
 (c) $\gamma \frac{BQR^2}{2}$ (d) γBQR^2
14. A conducting loop in the shape of a right angled isosceles triangle of height 10 cm is kept such that the 90° vertex is very close to an infinitely long conducting wire (see the figure). The wire is electrically insulated from the loop. The hypotenuse of the triangle is parallel to the wire. The current in the triangular loop is in counterclockwise direction and increased at a constant rate of 10 A s^{-1} . Which of the following statement(s) is (are) true?

[JEE Advanced 2016]



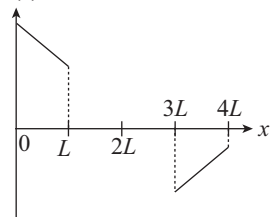
- (a) The magnitude of induced emf in the wire is $\left(\frac{\mu_0}{\pi}\right) V$
 (b) If the loop is rotated at a constant angular speed about the wire, an additional emf of $\left(\frac{\mu_0}{\pi}\right) V$ is induced in the wire
 (c) The induced current in the wire is in opposite direction to the current along the hypotenuse
 (d) There is a repulsive force between the wire and the loop
15. A rigid wire loop of square shape having side of length L and resistance R is moving along the $x =$ axis with a constant velocity v_0 in the plane of the paper. At $t = 0$, the right edge of the loop enters a region of length $3L$ where there is a uniform magnetic field B_0 into the plane of the paper, as shown in the figure. For sufficiently large v_0 , the loop eventually crosses the region. Let x be the location of the right edge of the loop. Let $v(x)$, $I(x)$ and $F(x)$ represent the velocity

of the loop, current in the loop, and force on the loop, respectively, as a function of x . Counter-clockwise current is taken as positive. [JEE Advanced 2016]

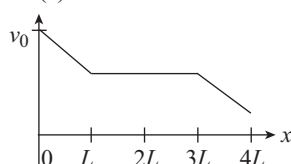


Which in the following schematic plot(s) is(are) correct? (Ignore gravity)

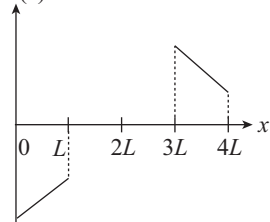
(a) $I(x)$



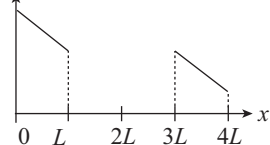
(b) $v(x)$



(c) $F(x)$



(d) $F(x)$

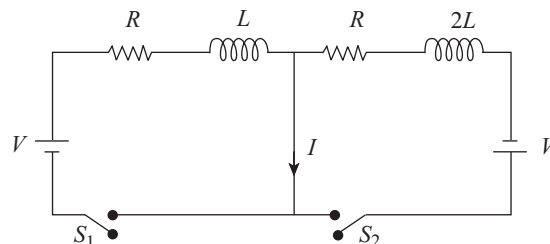


16. Two inductors L_1 , (inductance 1 mH, internal resistance 3Ω) and L_2 (inductance 2 mH, internal resistance 4Ω), and a resistor R (resistance 12Ω) are all connected in parallel across a 5V battery. The circuit is switched on at time $t = 0$. The ratio of the maximum to the minimum current ($I_{\max}/I_{\min}$) drawn from the battery is.....

[JEE Advanced 2016]

17. In the figure below, the switches S_1 and S_2 are closed simultaneously at $t = 0$ and a current starts to flow in the circuit. Both the batteries have the same

magnitude of the electromotive force (emf) and the polarities are as indicated in the figure. Ignore mutual inductance between the inductors. The current I in the middle wire reaches its maximum magnitude $I_{\max}$ at time $t = \tau$. Which of the following statements is (are) true? [JEE Advanced 2018]



- (a) $I_{\max} = \frac{V}{2R}$ (b) $I_{\max} = \frac{V}{4R}$
 (c) $\tau = \frac{L}{R} \ln 2$ (d) $\tau = \frac{2L}{R} \ln 2$

AC CIRCUITS

AIEEE / JEE Main Questions

- The self-inductance of the motor of an electric fan is 10 H. In order to impart maximum power at 50 Hz, it should be connected to a capacitance of nearly [AIEEE 2005]

(a) $4 \mu\text{F}$ (b) $8 \mu\text{F}$
 (c) $1 \mu\text{F}$ (d) $2 \mu\text{F}$
- A circuit has a resistance of 12Ω and an impedance of 15Ω . The power factor of circuit will be [AIEEE 2005]

(a) 0.8 (b) 0.4
 (c) 1.25 (d) 0.125
- The phase difference between an alternating current and emf is $\pi/2$. Which of the following cannot be the constituent of the circuit? [AIEEE 2005]

(a) C alone (b) R, L
 (c) L, C (d) L alone
- In a series resonant LCR circuit, the voltage across R is 100 V and $R = 1 \text{ k}\Omega$ with $C = 2 \mu\text{F}$. The resonant frequency ω is 200 rad s^{-1} . At resonance, the voltage across L is [AIEEE 2006]

(a) $4 \times 10^{-3} \text{ V}$ (b) $2.5 \times 10^{-2} \text{ V}$
 (c) 40 V (d) 250 V
- In an AC generator, a coil with N turns, all of the same area A and total resistance R , rotates with frequency ω in a magnetic field B . The maximum value of emf generated in the coil is [AIEEE 2006]

(a) $NABR\omega$ (b) NAB
 (c) $NABR$ (d) $NAB\omega$

9.14 Magnetism and Electromagnetic Induction

6. In an AC circuit, the voltage applied is $E = E_0 \sin \omega t$. The resulting current in the circuit is $I = I_0 \sin \left(\omega t - \frac{\pi}{2} \right)$.

The power consumption in the circuit is given by

[AIEEE 2007]

- (a) $P = \frac{E_0 I_0}{\sqrt{2}}$ (b) $P = \text{zero}$
 (c) $P = \frac{E_0 I_0}{2}$ (d) $P = \sqrt{2} E_0 I_0$

7. In a series LCR circuit $R = 200 \Omega$ and the voltage and the frequency of the main supply is 220 V and 50 Hz respectively. On taking out the capacitance from the circuit, the current lags behind the voltage by 30° . On taking out the inductor from the circuit, the current leads the voltage by 30° . The power dissipated in the LCR circuit is [AIEEE 2010]

- (a) 242 W (b) 305 W
 (c) 210 W (d) Zero W

8. An arc lamp requires a direct current of 10 A at 80 V to function. If it is connected to a 220 V (rms), 50 Hz AC supply, the series inductor needed for it to work is close to [JEE Main 2016]

- (a) 80 H (b) 0.08 H
 (c) 0.044 H (d) 0.065 H

9. For an RLC circuit driven with voltage of amplitude v_m and frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ the current exhibits resonance. The quality factor, Q is given by [JEE Main 2018]

- (a) $\frac{\omega_0 L}{R}$ (b) $\frac{\omega_0 R}{L}$
 (c) $\frac{R}{(\omega_0 C)}$ (d) $\frac{CR}{\omega_0}$

IIT JEE / JEE Advanced Questions

1. An AC voltage source of variable angular frequency ω and fixed amplitude v_0 is connected in series with a capacitance C and an electric bulb of resistance R (inductance zero). When ω is increased

[IIT-JEE 2010]

- (a) the bulb glows dimmer
 (b) the bulb glows brighter
 (c) total impedance of the circuit is unchanged
 (d) total impedance of the circuit increases
2. You are given many resistances, capacitors and inductors. These are connected to a variable DC voltage source (the first two circuits) or an AC voltage source of 50 Hz frequency (the next three circuits) in

different ways as shown in Column-II. When a current I (steady state for DC or rms for AC) flows through the circuit, the corresponding voltage V_1 and V_2 . (indicated in circuits) are related as shown in Column-I. Match the two columns. [IIT-JEE 2010]

Column-I		Column-II	
(A)	$I \neq 0$, V_1 is proportional to I	(p)	
(B)	$I \neq 0$, $V_2 > V_1$	(q)	
(C)	$V_1 = 0$, $V_2 = V$	(r)	
(D)	$I \neq 0$, V_2 is proportional to I	(s)	
		(t)	

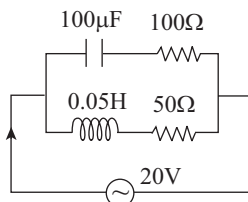
3. A series R-C circuit is connected to AC voltage source. Consider two cases; (A) when C is without a dielectric medium and (B) when C is filled with dielectric of constant 4. The current I_R through the resistor and voltage V_C across the capacitor are compared in the two cases. Which of the following is/are true? [IIT-JEE 2011]

- (a) $I_R^A > I_R^B$ (b) $I_R^A < I_R^B$
 (c) $V_C^A > V_C^B$ (d) $V_C^A < V_C^B$

4. A series R-C combination is connected to an AC voltage of angular frequency $\omega = 500 \text{ rad s}^{-1}$. If the impedance of the R-C circuit is $R\sqrt{1.25}$, the time constant (in millisecond) of the circuit is.....

[IIT-JEE 2011]

5. In the given circuit, the AC source has $\omega = 100 \text{ rad s}^{-1}$. Considering the inductor and capacitor to be ideal, the correct choice (s) is(are) [IIT-JEE 2012]



- (a) The current through the circuit, I is 0.3 A
 (b) The current through the circuit, I is $0.3\sqrt{2}$ A
 (c) The voltage across 100Ω resistor = $10\sqrt{2}$ V
 (d) The voltage across 50Ω resistor = 10 V

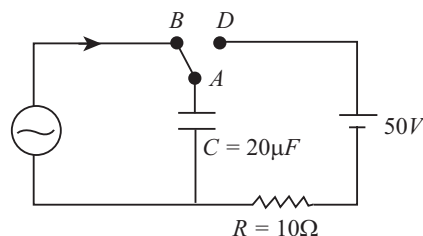
Q.No. 6 and 7 are based on following passage:

A thermal power plant produces electric power of 600 kW at 4000 V which is to be transported to a place 20 km away from the power plant for consumers usage. It can be transported either directly with a cable of large current carrying capacity or by using a combination of step-up and step-down transformers at the two ends. The drawback of the direct transmission is the large energy dissipation. In the method using transformers, the dissipation is much smaller. In this method, a step-up transformer is used at the plant side so that the current is reduced to a smaller value. At the consumers end, a step-down transformer is used to supply power to the consumers at the specified lower voltage. It is reasonable to assume that the power cable is purely resistive and the transformers are ideal with a power factor unity. All the currents and voltages mentioned are rms values.

[JEE Advanced 2013]

6. In the method using the transformers, assume that the ratio of the number of turns in the primary to that in the secondary in the step-up transformers is 1:10. If the power to the consumers has to be supplied at 200 V, the ratio of the number of turns in the primary to that in the secondary in the step-down transformers is
- (a) 200 : 1 (b) 150 : 1
 (c) 100 : 1 (d) 50 : 1
7. If the direct transmission method with a cable of resistance $0.4 \Omega \text{ km}^{-1}$ is used, the power dissipation (in %) during transmission is
- (a) 20 (b) 30
 (c) 40 (d) 50
8. At time $t = 0$, terminal A in the circuit shown in the figure is connected to B by a key and an

alternating current $I(t) = I_0 \cos(\omega t)$, with $I_0 = 1 \text{ A}$ and $\omega = 500 \text{ rad s}^{-1}$ starts flowing in it with the initial direction shown in the figure. At $t = \frac{7\pi}{6\omega}$, the key is switched from B to D . Now onwards only A and D are connected. A total charge Q flows from the battery to charge the capacitor fully. If $C = 20 \mu\text{F}$, $R = 10 \Omega$ and the battery is ideal with *emf* of 50 V, identify the correct statement(s). [JEE Advanced 2014]



- (a) Magnitude of the maximum charge on the capacitor before $t = \frac{7\pi}{6\omega}$ is $1 \times 10^{-3} \text{ C}$
 (b) The current in the left part of the circuit just before $t = \frac{7\pi}{6\omega}$ is clockwise
 (c) Immediately after A is connected to D , the current in R is 10 A
 (d) $Q = 2 \times 10^{-3} \text{ C}$

ELECTROMAGNETIC WAVES

JEE Main Questions

1. The magnetic field in a travelling electromagnetic wave has a peak value of 20 nT. The peak value of electric field strength is [JEE Main 2013]
- (a) 3 V m^{-1} (c) 6 V m^{-1}
 (b) 9 V m^{-1} (d) 12 V m^{-1}
2. During the propagation of electromagnetic waves in a medium [JEE Main 2014]
- (a) Electric energy density is double of the magnetic energy density
 (b) Electric energy density is half of the magnetic energy density
 (c) Electric energy density is equal to the magnetic energy density
 (d) Both electric and magnetic energy densities are zero

3. Match Column-I (electromagnetic wave type) with Column-II (Its association / application) and select the correct option from the choice given below the column:
[JEE Main 2014]

Column-I		Column-II	
(A)	Infrared wave	(i)	To treat muscular strain
(B)	Radio waves	(ii)	For broadcasting
(C)	X-rays	(iii)	To detect fracture of bones
(D)	Ultraviolet rays	(iv)	Absorbed by the ozone layer of the atmosphere

- (A) (B) (C) (D)
 (a) (iv) (iii) (ii) (i)
 (b) (i) (ii) (iv) (iii)
 (c) (iii) (ii) (i) (iv)
 (d) (i) (ii) (iii) (iv)

4. An EM wave from air enters a medium. The electric fields are $\vec{E}_1 = E_{01} \hat{x} \cos \left[2\pi \nu \left(\frac{z}{c} - t \right) \right]$ in air and $\vec{E}_2 = E_{02} \hat{x} \cos [k(2z - ct)]$ in medium, where the wave number k and frequency ν refer to their values in air. The medium is non-magnetic. If ϵ_{r1} and ϵ_{r2} refer to relative permittivities of air and medium respectively, which of the following options is correct?
[JEE Main 2018]

- (a) $\frac{\epsilon_{r1}}{\epsilon_{r2}} = 4$ (b) $\frac{\epsilon_{r1}}{\epsilon_{r2}} = 2$
 (c) $\frac{\epsilon_{r1}}{\epsilon_{r2}} = \frac{1}{4}$ (d) $\frac{\epsilon_{r1}}{\epsilon_{r2}} = \frac{1}{2}$

Answer Sheet

MAGNETIC FIELD

AIEEE/JEE Main Questions

1. (d) 2. (b) 3. (c) 4. (b) 5. (b) 6. (c) 7. (b) 8. (b) 9. (d)
10. (b)

IIT JEE / JEE Advanced Questions

1. (a) $k = NAB$ (b) $C = \frac{2Ni_0AB}{\pi}$ (c) $\theta_{\max} = Q\sqrt{\frac{NAB\pi}{2Ii}}$
2. (c,d) 3. (a) 4. (a,c,d) 5. (c) 6. (b,d) 7. (b) 8. (c,d) 9. (a,b,c)
10. (a,d) 11. (a,c) 12. 2ms^{-1} 13. 5.56

MAGNETIC EFFECT OF CURRENT

AIEEE / JEE Main Questions

1. (a) 2. (b) 3. (d) 4. (d) 5. (c) 6. (b) 7. (b) 8. (b) 9. (c)
10. (b) 11. (c) 12. (b) 13. (d) 14. (c)

IIT JEE / JEE Advanced Questions

1. (a,c) 2. $A \rightarrow q, r; B \rightarrow p; C \rightarrow q, r; D \rightarrow q$
3. $A \rightarrow p, r, s; B \rightarrow r, s; C \rightarrow p, q, t; D \rightarrow r, s$
4. (a) 5. (5) 6. (a,d) 7. (c) 8. (b) 9. (a,b,d)
10. (b) 11. (c) 12. (b) 13. (d) 14. (c)

MAGNETIC PROPERTIES OF MATTER

1. (b) 2. (b) 3. (c) 4. (d)

ELECTROMAGNETIC INDUCTION

AIEEE / JEE Main Questions

1. (c) 2. (a) 3. (d) 4. (d) 5. (c) 6. (b) 7. (d) 8. (a)
9. (d) 10. (b) 11. (c) 12. (b) 13. (a) 14. (c)

IIT JEE / JEE Advanced Questions

1. (a) 2. $i = \frac{\varepsilon}{R} = \frac{\mu_0 L d n a^2 i_o \omega \cos \omega t}{2\rho R}$ 3. (b) 4. (c) 5. (b) 6. (b,d)
7. $A \rightarrow p; B \rightarrow p, q, s; C \rightarrow q, s; D \rightarrow q, r, s$
8. (d) 9. (b,d) 10. (6) 11. (7) 12. (b) 13. (c) 14. (a,d) 15. (a,b) 16. (8)
17. (b,d)

AC CIRCUITS

AIEEE / JEE Main Questions

1. (c) 2. (a) 3. (b) 4. (d) 5. (d) 6. (b) 7. (a) 8. (d) 9. (a)

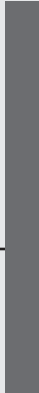
IIT / JEE Advanced Questions

1. (b) 2. $A \rightarrow r, s, t; B \rightarrow q, r, s, t; C \rightarrow p, q; D \rightarrow q, r, s, t$
3. (b,c) 4. (4) 5. (a,c) 6. (a) 7. (b) 8. (c,d)

ELECTROMAGNETIC WAVES

AIEEE / JEE Main Questions

1. (b) 2. (c) 3. (d) 4. (c)



Solutions

1. Magnetic Field
2. Magnetic Effect of Current
3. Miscellaneous Problems on Chapters 1 and 2
4. Magnetic Properties of Matter and Earth's Magnetism
5. Electromagnetic Induction
6. Alternating-Current Circuits
7. Miscellaneous Problems on Chapters 5 and 6
8. Electromagnetic Waves
9. Previous Years' JEE Questions

Chapter 1 Magnetic Field



Your Turn

3. Magnetic length (= distance between the poles)

$$= 0.84 \times 10 \text{ cm} = 8.4 \text{ cm} = 8.4 \times 10^{-2} \text{ m}$$

$$\therefore M = m l_m = 12 \times 8.4 \times 10^{-2}$$

$$= 1.008 \text{ A-m}^2$$

4. Using the expression $U = -\vec{M} \cdot \vec{B}$, one can quickly see that unit of M can be written as J/T.

5. Using equation (4) we can write,

$$B = \frac{\mu_0}{4\pi} \frac{2Mx}{(x^2 - d^2)^2}$$

$$= 10^{-7} \times \frac{2 \times 1.0 \times 0.1}{[(0.1)^2 - (0.05)^2]^2} \quad [\because 2d = 10 \text{ cm}]$$

$$= 0.36 \times 10^{-4} \text{ T.}$$

6. $B = \frac{\mu_0}{4\pi} \frac{M}{(x^2 + d^2)^{3/2}}$ [equation (6)]

$$= 10^{-7} \times \frac{1.0}{[(0.1)^2 + (0.05)^2]^{3/2}}$$

$$= \frac{10^{-4}}{(1.25)^{3/2}} = \frac{10^{-4}}{1.4} = 0.71 \times 10^{-4}$$

$$= 7.1 \times 10^{-5} \text{ T}$$

8. $F = qvB \sin 180^\circ = 0$

There is no force when $\vec{v}$ is parallel or antiparallel to $\vec{B}$.

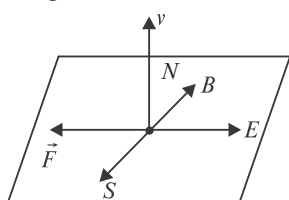
9. Maximum force is experienced by the charge when its velocity is perpendicular to the direction of the field.

$$F_{\max} = qvB \sin 90^\circ = 1 \times 10^{-6} \times 1 \times 10^6 \times 10^{-3} \times 1$$

$$= 10^{-3} \text{ N.}$$

$$\therefore a_{\max} = \frac{F_{\max}}{m} = \frac{10^{-3} \text{ N}}{10^{-6} \text{ kg}} = 10^3 \text{ ms}^{-2}$$

10. The vector $\vec{v} \times \vec{B}$ points in west direction. Force on the positive charge is towards west and that on the negative charge is towards east.



Direction of force on positive charge.

$$11. \vec{F} = -q \vec{v} \times \vec{B} = -q (v_0 \hat{i} + v_0 \hat{j}) \times (B_0 \hat{k})$$

$$= -qv_0 B_0 (\hat{i} \times \hat{k}) - qv_0 B_0 (\hat{j} \times \hat{k})$$

$$= qv_0 B_0 \hat{j} - qv_0 B_0 \hat{i}$$

$$12. F = qvB \Rightarrow [B] = \frac{[F]}{[qv]}$$

$$\Rightarrow [B] = \frac{[M^1 L^1 T^{-2}]}{[A^1 T^1 L^1 T^{-1}]} = [M^1 L^0 T^{-2} A^{-1}]$$

13. Alpha particles are doubly ionised He ions. They have two protons and two neutrons.

$$\text{Mass of } \alpha \text{ particle, } M = 4m_p = 4 \times 1.67 \times 10^{-27} \text{ kg.}$$

$$\text{Charge on } \alpha \text{ particle, } q = +2e = 3.2 \times 10^{-19} \text{ C}$$

$$\text{Radius of circular path, } R = \frac{mv}{qB}$$

$$\therefore R = \frac{4 \times 1.67 \times 10^{-27} \times 10 \times 10^3}{3.2 \times 10^{-19} \times 1.0} = 2.1 \times 10^{-4} \text{ m}$$

$$\text{Frequency of revolution, } f = \frac{qB}{2\pi m}$$

$$\therefore f = \frac{3.2 \times 10^{-19} \times 1.0}{2 \times 3.14 \times 4 \times 1.67 \times 10^{-27}} = 7.6 \times 10^6 \text{ Hz}$$

14. Kinetic energy of the particle after being accelerated through a potential difference of V is $K = qV$.

$$\therefore \text{Momentum, } P = \sqrt{2mK} = \sqrt{2mqV}$$

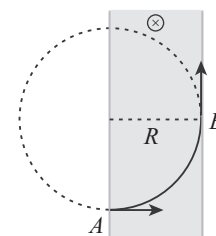
$$\therefore R = \frac{p}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}}$$

15. Radius of the circular path is

$$R = \frac{mv}{qB} = \frac{1.6 \times 10^{-27} \times 4 \times 10^6}{1.6 \times 10^{-19} \times 0.4}$$

$$= 0.1 \text{ m} = 10 \text{ cm.}$$

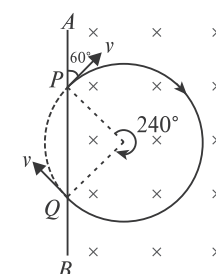
The width of the field region is just less than 10 cm. The shaded region in the diagram shows the field region. The particle enters into the field at A and leaves it at B. It gets deflected by 90° .



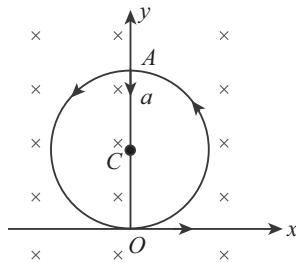
16. Negatively charged particle will rotate clockwise and its path will be as shown in the figure.

The particle rotates by an angle of 240° in the field region.

$$t = \frac{2}{3} T = \frac{2}{3} \frac{2\pi m}{qB} = \frac{4\pi m}{3qB}$$



18. The circular path of the particle will be xy plane as shown in figure. Its acceleration is towards the centre of the circle. It is in negative y -direction when the particle is at A . Time needed to travel from O to A is



$$t = \frac{T}{2} = \frac{\pi m}{qB}$$

19. A neutron is a neutral particle. It moves undeflected in a magnetic field. Path shown as D must correspond to neutrons.

Force on an electron is $-e(\vec{v} \times \vec{B})$. The electrons must deflect towards right as represented by path A. Radius of curvature of the path $R = \frac{mv}{qB}$ will be small due to very small mass of the electron.

The positively charged particle – protons and alpha particles will deflect to left. For alpha particle $\frac{m}{q}$ is larger than that for a proton. In fact, mass of an alpha particle is nearly 4 times that of a proton and charge on it is only twice that of a proton. The ratio $\frac{m}{q}$ for alpha is twice that for a proton.

$$\therefore R_\alpha = 2R_p$$

$\therefore C$ represents the track followed by the α particles and B represents the path of protons.

20. $K = \frac{1}{2}mv^2$

$$\Rightarrow v = \sqrt{\frac{2K}{m}} = \sqrt{\frac{2 \times 90 \times 1.6 \times 10^{-19}}{9.0 \times 10^{-31}}}$$

$$= 5.66 \times 10^6 \text{ m/s.}$$

$$v_{11} = v \cos \theta = 5.66 \times 10^6 \times \cos 53^\circ$$

$$= 5.66 \times 10^6 \times \frac{3}{5} = 3.4 \times 10^6 \text{ m/s.}$$

$$v_1 = v \sin \theta = 5.66 \times 10^6 \times \sin 53^\circ$$

$$= 5.66 \times 10^6 \times \frac{4}{5} = 4.5 \times 10^6 \text{ m/s.}$$

Radius of the path,

$$R = \frac{mv_1}{qB} = \frac{9 \times 10^{-31} \times 4.5 \times 10^6}{1.6 \times 10^{-19} \times 3.14 \times 10^{-4}}$$

$$= 8.06 \times 10^{-2} \text{ m} = 8.06 \text{ cm}$$

$$\text{Time period of motion is } T = \frac{2\pi m}{qB}$$

$$\text{or, } T = \frac{2 \times 3.14 \times 9 \times 10^{-31}}{1.6 \times 10^{-19} \times 3.14 \times 10^{-4}} \\ = 1.13 \times 10^{-7} \text{ s}$$

$$\text{Pitch, } P = Tv_{11} = 1.13 \times 10^{-7} \times 3.4 \times 10^6 \\ = 3.8 \times 10^{-1} \text{ m} \\ = 38 \text{ cm}$$

21. Magnetic force is

$$\vec{F}_B = q(\vec{v} \times \vec{B}) = q(100\hat{i} \times 4 \times 10^{-3}\hat{j}) = 0.4q\hat{k}$$

The electric force must be equal and opposite to the magnetic force. Hence the electric field is along z -direction.

$$q\vec{E} = (0.4q)\hat{k}$$

$$\Rightarrow \vec{E} = (0.4 \text{ V/m})\hat{k}$$

22. (a) Frequency of circular motion = 10^6 Hz. In every round, a deuteron enters the gap twice. Hence the required answer is 2×10^6 .

$$(b) v = \omega R_0 = 2\pi f R_0 \\ = 2 \times 3.14 \times 10^6 \times 0.8 = 5.0 \times 10^6 \text{ m/s.}$$

$$23. \frac{m_2}{m_1} = \frac{R_2}{R_1} = \frac{1.4}{1.2} = \frac{7}{6}$$

If one isotope of carbon is ^{12}C , the other must be ^{14}C .

$$24. \vec{F} = I \vec{L} \times \vec{B} = ILB_0 [\hat{i} \times (2\hat{j} + \hat{k})] \\ = ILB_0 [2\hat{k} - \hat{j}]$$

26. Let current divide at A as I_1 into $APQRB$ and I_2 into path AB .

F_{APQRB} = Force on AB carrying current I .

$\therefore$ Force on pentagon is

F = Force on AB having current I_1 + Force on AB having current I_2

$$= \text{Force on } AB \text{ having current } I = IaB$$

Direction is normal to AB (upward in the figure)

27. (a) Force on AB is directed along $\vec{AB} \times \vec{B}$, i.e. perpendicularly into the plane of the figure. Magnitude of the force is

$$F_{AB} = IaB \sin 90^\circ = IaB$$

- (b) Current is parallel to $\vec{B}$. Force is zero.

- (c) Force on a closed current loop in a uniform magnetic field is zero.

28. Current $I = \frac{V}{R}$

Magnetic force on the wire PQ is $F = IdB = \frac{VdB}{R}$.

This force is horizontally perpendicular to the wire PQ . Friction force on the wire can have a maximum value of μmg . The wire can move if

$$F = \mu mg$$

$$\text{or, } \frac{VdB}{R} = \mu mg \Rightarrow \frac{VdB}{Rmg} = \mu$$

when $R = 20\Omega$, the wire just moves

$$\therefore \mu = \frac{10 \times 0.1 \times 0.5}{20 \times 0.02 \times 10} = \frac{1}{8}$$

Actually, decreasing R , increases the current I and the magnetic force increases. When the magnetic force increases beyond the friction force, the wire moves.

29. Current, $I = 2q(f) = 2q \frac{\omega}{2\pi} = \frac{q\omega}{\pi}$

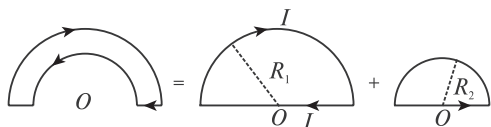
$$\therefore M = \pi l^2 \cdot I = q\omega l^2$$

$$\text{Moment of inertia} = 2ml^2$$

$$\text{Angular momentum} = 2ml^2 \cdot \omega$$

$$\therefore \text{Required ratio} = \frac{M}{L} = \frac{q}{2m}$$

30. The given loop may be considered as the superposition of two loops, shown in the figure given below.



Dipole moment of the semicircular loop of radius R_1 is

$$M_1 = \frac{\pi R_1^2}{2} I \otimes$$

Dipole moment of the semicircular loop of radius R_2 is

$$M_2 = \frac{\pi R_2^2}{2} I \odot$$

$\therefore$ Net dipole moment of the given loop is

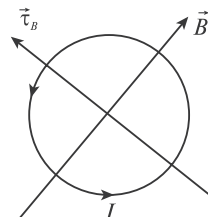
$$M = \frac{\pi R_1^2}{2} I - \frac{\pi R_2^2}{2} I = \frac{\pi}{2} I (R_1^2 - R_2^2) \otimes$$

31. The loop can be thought as a combination of two semicircular parts in xy and yz planes.

$$\vec{M} = \frac{\pi R^2 I}{2} (-\hat{i}) + \frac{\pi R^2 I}{2} (-\hat{k}).$$

32. Magnetic dipole moment of the disc is parallel to $\vec{B}$. Torque acting on it is zero

33. The dipole moment vector is directed perpendicular to the plane of the figure. It is out of the plane of the figure.



$$\tau = MB \sin 90^\circ = I\pi R^2 B$$

Direction of $\vec{\tau}$ is perpendicular to $\vec{B}$ in the plane of the figure.

34. $I = \frac{C}{NAB} \theta \Rightarrow C = \frac{INAB}{\theta}$

$$C = \frac{(10 \times 10^{-3}) \times (100) \times (2.0 \times 10^{-4}) (0.01)}{0.05}$$

$$= 4.0 \times 10^{-5} \text{ Nm rad}^{-1}$$



Worksheet 1

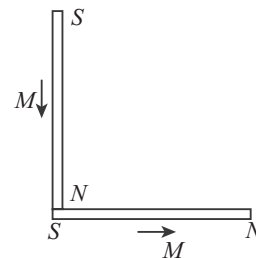
1. Magnetic field lines are closed loops. Inside a magnet, the field lines are directed from the south pole to the north pole.

2. Use equation (5)

$$B = \frac{\mu_0}{4\pi} \frac{2M}{x^3} = 10^{-7} \times \frac{2 \times 1.2}{(0.1)^3} = 2.4 \times 10^{-4} \text{ T}$$

3. The resultant dipole moment is the vector sum of the dipole moments of the two magnets.

$$M_{\text{net}} = \sqrt{M^2 + M^2} = \sqrt{2} M = \sqrt{2} ml$$



4. Freely suspended magnet will orient itself parallel to the direction of the field. Work needed to rotate it from this position to a new position where it makes an angle θ with the field is

$$W = \int_0^\theta \tau d\theta = MB \int_0^\theta \sin \theta d\theta = MB (1 - \cos \theta)$$

5.6 Magnetism and Electromagnetic Induction

5. If m is the pole strength of the magnet, force acting on its pole will be

$$F = mB$$

$$\text{or, } 6 \times 10^{-4} = m \times 2 \times 10^{-5}$$

$$\Rightarrow m = 30 \text{ Am}$$

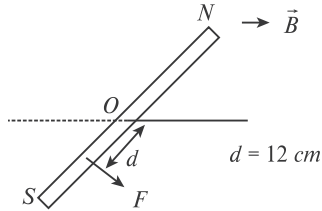
If l is magnetic length, then dipole moment is

$$ml = M \Rightarrow l = \frac{3}{30} = \frac{1}{10} \text{ m} = 0.1 \text{ m}$$

$$\text{Geometric length is } l_g = \frac{l}{0.84} = \frac{0.1}{0.84} = 0.12 \text{ m}$$

6. Magnetic torque

$$\begin{aligned} \tau_B &= MB \sin\theta = (24 \times 0.25)(0.25)\sin\theta \\ &= 1.5 \times \sin 30^\circ = 0.75 \text{ N-m} \end{aligned}$$



Force F is applied at 12cm from pivot (O). Its torque is

$$\tau_F = F(0.12)$$

For equilibrium, $F(0.12) = 0.75$

$$\Rightarrow F = 6.25 \text{ N}$$

7. $F = qvB \sin\theta = 1 \times 10 \times 0.5 \times \frac{1}{2} = 2.5 \text{ N}$
8. Path is helical when the velocity makes an angle θ with the direction of the field.
9. Kinetic energy acquired by a charge q after being accelerated through a potential difference of V is $K = qV$.

For proton $K_p = eV$

For alpha particle, $K_\alpha = 2eV = 2K_p$

$$R_p = \frac{\sqrt{2mK_p}}{eB}$$

$$R_\alpha = \frac{\sqrt{2(4m)(2K_p)}}{2eB}$$

[Mass of μ is nearly 4 times the mass of a proton]

$$\therefore \frac{R_p}{R_\alpha} = \frac{1}{\sqrt{2}}$$

10. Electric force on the electron is in positive y -direction. For the magnetic force to be in negative y -direction, the magnetic force must be in negative z -direction.

$$\vec{F}_r = -e(\vec{v} \times \vec{B}) = -e[v\hat{i} \times B(-\hat{k})] = evB(-\hat{j})$$

11. Both magnetic and electric force are proportional to charge. Hence doubling the charge will double them and net force will remain zero.

$$12. R = \frac{\sqrt{2mK}}{eB} \Rightarrow R \propto \frac{\sqrt{K}}{B}$$

$$13. \text{ For deuteron } R = \frac{2mv}{qB} \quad [\text{Mass of deuteron} = 2m]$$

$$\text{For proton } R = \frac{(m)v_\perp}{qB}$$

where v_0 is velocity component which is perpendicular to $\vec{B}$. For the two radii to be same, we must have $v_\perp = 2v$

$\therefore$ Speed of the proton is certainly greater than $2v$ (it has a component of velocity along $\vec{B}$ also).

$$\therefore KE_p > \frac{1}{2}m(2v)^2 = \left[\frac{1}{2}(2m)v^2\right] 2$$

$$\therefore KE_p > (50 \text{ keV})2 = 100 \text{ keV.}$$

14. Refer to example 8. The ion which has the smallest radius of circular path, gets deflected most

$$R_H = \frac{\sqrt{2mK}}{eB}$$

$$R_{He} = \frac{\sqrt{2(4m)K}}{eB}$$

$$R_0 = \frac{\sqrt{2(16m)K}}{2eB}$$

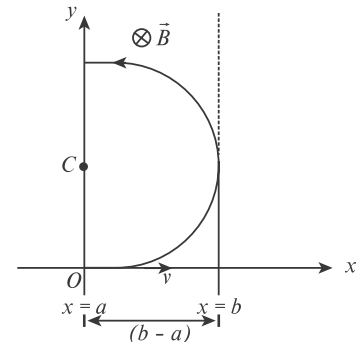
$$\therefore R_0 = R_{He} > R_H$$

$\therefore O^{++}$ and He^+ are deflected equally and H^+ is deflected most.

16. The diagram shows the situation when the particle is just not able to cross into the region $x > b$. In this case, radius of circular path is just less than $(b-a)$. For the particle to enter into the region $x > b$, we must have

$$\text{Radius} > (b-a)$$

$$\frac{mv}{qB} > (b-a) \Rightarrow v > \frac{(b-a)qB}{m}$$



17. Assume that only magnetic field ($\vec{B} = B_0 \hat{i}$) exists.

The velocity given to the particle is $\vec{v} = v_0 \hat{j}$. The particle will move in a circle in yz plane.

Now, imagine that the electric field ($\vec{E} = E_0 \hat{i}$) also exists. The particle experiences force in x direction due to this field. This force will not change the motion in yz plane.

when speed is $2v_0$

$$(2v_0)^2 = v_x^2 + v_y^2 + v_z^2$$

$$\text{But } v_y^2 + v_z^2 = v_0^2$$

$$\therefore 4v_0^2 = v_x^2 + v_0^2$$

$$\Rightarrow v_x = \sqrt{3} v_0$$

$$\text{Now, acceleration in } x \text{ direction is } a_x = \frac{qE_0}{m}$$

$$\therefore v_x = a_x t$$

$$\Rightarrow \sqrt{3} v_0 = \frac{qE_0}{m} t \Rightarrow t = \frac{\sqrt{3} v_0 m}{qE_0}$$

18. In electric field, the force normal to the direction of velocity is eE .

$$\therefore \frac{mv_0^2}{r_1} = eE \Rightarrow r_1 = \frac{mv_0^2}{eE}$$

$$\text{In magnetic field, } r_2 = \frac{mv_0}{eB}$$

$$\therefore \frac{r_1}{r_2} = \frac{Bv_0}{E}$$

19. Time period is same for both the particles.

$$\therefore a = 1.$$

$$R_1 = \frac{mv \sin 30^\circ}{qB}; \quad R_2 = \frac{mv \sin 60^\circ}{qB}$$

$$\Rightarrow b = \frac{R_1}{R_2} = \frac{1}{\sqrt{3}}$$

$$P_1 = T v \cos 30^\circ; \quad P_2 = T v \cos 60^\circ$$

$$\Rightarrow c = \frac{P_1}{P_2} = \sqrt{3}$$

$$\therefore abc = 1$$

20. The positive ions gather speed in positive x -direction and the negative ions gather speed in negative x -direction due to the electric field. Magnetic force on both will be in negative y -direction.

21. Radius $\propto$ momentum.

$$R_A > R_B \Rightarrow m_A v_A > m_B v_B$$

22. Particle is deflected by 60° . This means that it is rotated by 60° on its circular path.

$$t = \frac{T}{6} = \frac{1}{6} \cdot \frac{2\pi m}{qB} = \frac{\pi m}{3qB}$$

23. Speed of the particle at (x_0, y_0) is

$$v = \sqrt{4^2 + 3^2} = 5$$

Gain in kinetic energy is due to work done by the electric force.

$$\therefore (qE_0)(x_0) = \frac{1}{2} m (5)^2$$

$$\text{or, } x_0 = \frac{25}{2\alpha E_0}$$

24. Assume that only magnetic field exists. The path will be a helix with the particle advancing in negative y -direction with constant speed v . Time period and pitch are

$$T = \frac{2\pi m}{qB}; \quad p = vt$$

When an electric field is present in y -direction, the particle experiences an electric force in y -direction. Due to this force it retards in $-y$ -direction, stops at some point and then starts moving in positive y -direction. It can pass through origin, if its displacement is zero in y -direction after a time $t = nT$, where $n = 1, 2, 3, \dots$

$$y = u_y t + \frac{1}{2} a_y t^2$$

$$0 = -vt + \frac{1}{2} \left(\frac{qE}{m} \right) t^2$$

$$\text{or, } t = \frac{2mv}{qE}$$

$$\text{or, } n \cdot \frac{2\pi m}{qB} = \frac{2mv}{qE}$$

$$\text{or, } n = \frac{vB}{\pi E}$$

25. Magnetic force is always perpendicular to velocity. It does no work.

$$26. \vec{F}_1 = (-e) \vec{v}_1 \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$\text{or, } -2\hat{j} = -2e [\hat{i} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})]$$

$$\text{or, } \hat{j} = e [B_y \hat{k} - B_z \hat{j}]$$

$$\text{or, } B_y = 0 \text{ and } B_z = -\frac{1}{e}$$

$$\text{and } \vec{F}_2 = (-e) \vec{v}_2 \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$2\hat{i} = -2e [\hat{j} \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})]$$

$$\text{or, } \hat{i} = B_x e \hat{k} - B_z e \hat{i} \quad [\because B_y = 0]$$

$$\text{or, } \hat{i} = B_x e \hat{k} + \hat{i} \quad [\because B_z = -\frac{1}{e}]$$

$$\therefore B_x = 0$$

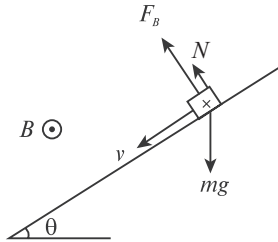
$$\therefore \vec{B} = -\left(\frac{1}{e}\right) \hat{k}$$

When the particle is projected along z -direction, it will experience no force.

27. Particles will move in helical paths, revolving in opposite sense. Time period $\left(T = \frac{2\pi m}{qB}\right)$ for the positive particle is T and for the negative particle it is $2T$. Pitch for the positive particle is P and for the negative particle is $2P$ ($\because$ pitch = $v \cos \theta T$)
The particles will meet after the positive particle has completed two turns and the negative particle has completed one turn.

$$\begin{aligned} \text{Required distance} &= 2P = 2 \cdot \frac{2\pi M}{QB} \cdot v \cos \theta \\ &= \frac{4\pi M v \cos \theta}{Q \cdot B} \end{aligned}$$

28. Speed after time t is $v = g \sin \theta \cdot t$



For the block to lose contact, $N = 0$

$$\begin{aligned} \Rightarrow F_B &= mg \cos \theta \\ \Rightarrow qvB &= mg \cos \theta \\ \Rightarrow qg \sin \theta \cdot t \cdot B &= mg \cos \theta \\ \Rightarrow t &= \frac{m \cot \theta}{qB} \end{aligned}$$

29. When released from rest, the proton experiences electric force only.

$$ma_0 = eE \Rightarrow E = \frac{ma_0}{e} \text{ (west)}$$

When projected towards north, its acceleration becomes $3a_0$ towards west. The magnetic force must be

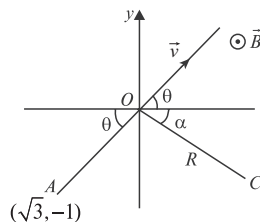
$$F_B = 2ma_0 \text{ towards west}$$

$$\text{or, } ev_0 B = 2ma_0 \Rightarrow B = \frac{2ma_0}{ev_0}$$

Direction of $\vec{B}$ must be downward for the force to be in west direction.

30. Higher the strength of field, smaller is the radius of the circular path.

31. The initial force on the particle is along a direction perpendicular to the line AO as shown.



$$\text{Obviously, } \tan \theta = \frac{\sqrt{3}}{1}$$

$$\Rightarrow \theta = 60^\circ$$

$$\therefore \alpha = 30^\circ$$

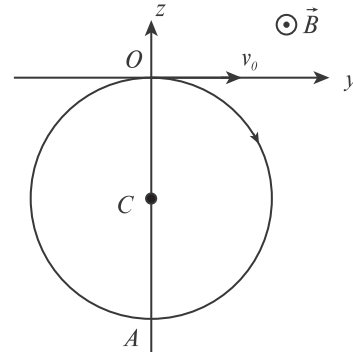
$$\text{Radius } R = \frac{mv}{qB} = \frac{1 \times 1}{1 \times 1} = 1 \text{ m.}$$

$\therefore$ Co-ordinates of the centre C are

$$x = R \cos \alpha = 1 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

$$y = -R \sin \alpha = -\frac{1}{2}$$

32. Particle moves in a helical path. Let's first consider the motion in yz plane. In time $t = \frac{\pi}{B_0 \alpha}$, the particle will complete half circle and reach A .



$\therefore$ y co-ordinate = 0

$$z \text{ co-ordinate} = 2R = \frac{2mv_0}{qB_0} = \frac{2v_0}{\alpha B_0}$$

In x direction, the particle advances with constant speed v_0 .

$$\therefore x \text{ co-ordinate} = v_0 t = \frac{v_0 \pi}{B_0 \alpha}$$

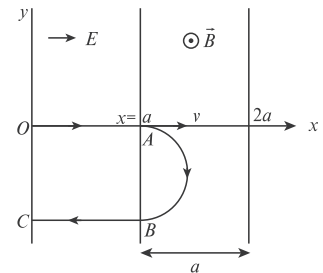
33. Let $a = 0.167 \text{ m}$. The electric field accelerates the charge from O to A . The speed acquired by the proton is given as

$$\frac{1}{2} mv^2 = qEa$$

$$\Rightarrow v = \sqrt{\frac{2qEa}{m}}$$

The proton is deflected back if the radius of circular path in the magnetic field is less than a . [If $R > a$, the particle will exit out of the magnetic field from the other side].

$$\therefore \frac{mv}{qB} < a$$



$$\therefore \frac{mv}{qa} < B$$

$$\text{or, } \frac{m}{qa} \sqrt{\frac{2qEa}{m}} < B$$

$$\text{or, } \sqrt{\frac{2Em}{qa}} < B$$

Substituting the values gives $B > 5\sqrt{2}$ mT.

34. The situation is as shown in figure.

$$\tan\theta = \frac{AC}{AB} = \frac{4}{3}$$

$$\theta = 53^\circ$$

$$BA = BD = 3\text{m}$$

[Tangent from same point]

$$\therefore MD = BD \sin\theta$$

$$= 3\left(\frac{4}{5}\right) = \frac{12}{5}$$

$$DN = R\cos\theta = R\left(\frac{3}{5}\right)$$

$$\therefore R + \frac{3R}{5} = \frac{12}{5} \Rightarrow R = 1.5\text{m.}$$

$$\text{or, } \frac{mv}{qB} = 1.5 \Rightarrow \frac{1 \times v}{1 \times 2} = 1.5$$

$$\Rightarrow v = 3 \text{ m/s}$$

35. In equilibrium $kx_0 = mg$
when the magnetic field is switched on, the equilibrium is established when stretch in the spring is $2x_0$.

$$\therefore \text{Magnetic force} = kx_0 = mg \quad (\downarrow)$$

$$BIL = mg \Rightarrow B = \frac{mg}{LT} = \frac{mgR}{L\varepsilon}$$

Current in the rod is from left to right. The field must be outward for the magnetic force to be downward.

36. For $x = 2$,

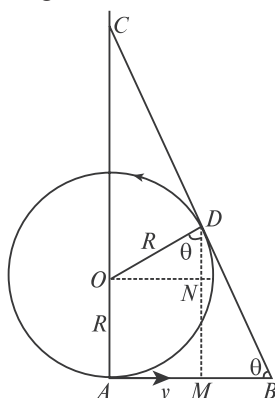
$$y = \sqrt{2(2)} = 2\text{m}$$

$$\therefore \text{Length } AB = 4\text{m.}$$

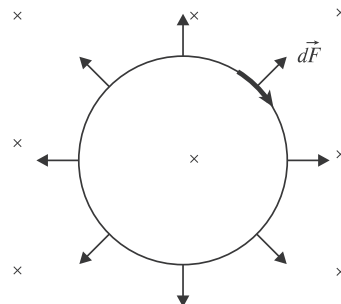
The curved wire can be replaced with a straight wire AB having current from A to B .

$$\therefore F = ILB = 2 \times 4 \times 4$$

$$= 32\text{N (Direction is along positive } x)$$



37. Net force on the loop = 0

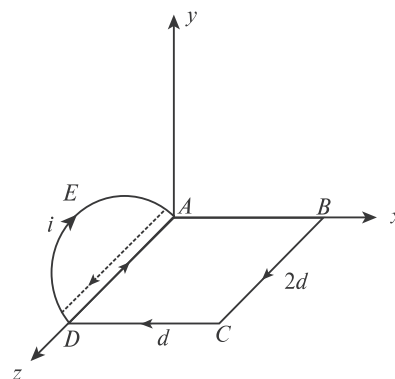


$\therefore$ Coil will not move.

Every small element of the coil is pulled radially outward by the magnetic force. Hence it expands.

38. We assume two separate current loops in xz plane and yz plane. The current in branch AD are equal and opposite and they will make no difference to our calculation. For semicircular part

$$\vec{M}_1 = i \frac{\pi d^2}{2} (-\hat{i})$$

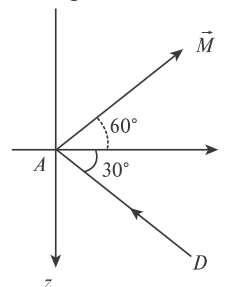


For rectangular part

$$\vec{M}_2 = i 2d^2 (-\hat{j})$$

$$\therefore \vec{M} = id^2 \left(-\frac{\pi}{2}\hat{i} - 2\hat{j} \right)$$

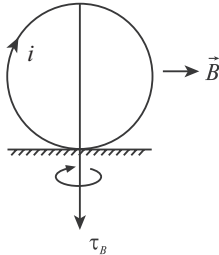
39. $M = iA = 10(0.1)^2 = 0.1 \text{ Am}^2$. Direction of the magnetic moment vector is normal to the plane of the loop. The direction of $\vec{M}$ can be shown perpendicular to side DA in xz plane.



$$\begin{aligned}\therefore \vec{M} &= 0.1 \cos 60^\circ \hat{i} - 0.1 \sin 60^\circ \hat{k} \\ &= 0.05 (\hat{i} - \sqrt{3} \hat{k}).\end{aligned}$$

40. $\vec{M}$ is into the plane of the figure. The torque $\vec{\tau}_B (= \vec{M} \times \vec{B})$ is directed vertically down. The torque will try to rotate the ring about its vertical diameter. Moment of inertia of the ring about this axis is

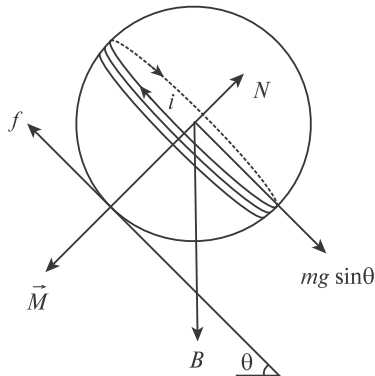
$$\begin{aligned}I &= \frac{1}{2} mR^2 \\ \therefore \alpha &= \frac{\tau}{I} = \frac{MB}{\frac{1}{2} mR^2} = \frac{2i\pi R^2 B}{mR^2} \\ &= \frac{2i\pi B}{m} = \frac{2 \times 4 \times \pi \times 10}{2} = 40\pi \text{ rad s}^{-2}\end{aligned}$$



41. For equilibrium, the friction must balance $mg \sin \theta$.

$$\therefore f = mg \sin \theta \quad (i)$$

Friction produces a torque which must be balanced by the magnetic torque. The magnetic dipole moment ($\vec{M}$) is perpendicular to the plane of the coil, i.e., perpendicular to the incline plane. The direction of current (i) must be as shown in figure.



$$\begin{aligned}M &= NiA = Ni\pi R^2 \\ \tau_B &= MB \sin \theta = Ni\pi R^2 B \sin \theta\end{aligned}$$

For equilibrium, $Ni\pi R^2 B \sin \theta = fR$

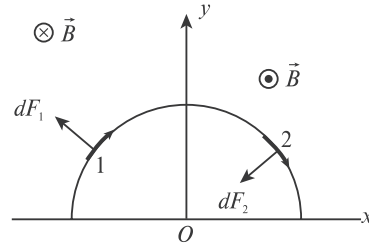
$$\text{or, } Ni\pi R^2 B \sin \theta = mg \sin \theta \cdot R$$

$$\text{or, } B = \frac{mg}{N\pi iR} = \frac{mg}{3\pi iR}$$

Note

If the direction of i is opposite to the direction shown, the dipole moment will be in the direction of N and direction of τ_B will support the torque due to friction.

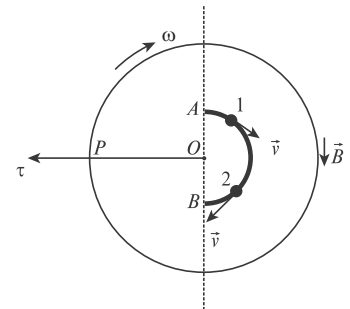
$$42. \vec{B} = \frac{B_0}{2R} x (\hat{k})$$



For $x > 0$, the field is in positive x direction and for $x < 0$, the field is in negative x direction. Direction of forces on two elements 1 and 2, at x -coordinates $-x$ and $+x$ are as shown in figure. Magnitude of forces on these two identical elements will be same as magnitude of $\vec{B}$ at their locations is same.

The y components of the forces cancel out. The x -components add and the resultant is in negative x -direction.

43. Consider a semicircular arc on the disc as shown in dark colour (see figure). Consider two small identical charged elements 1 and 2 on this ring. Charge 1 experiences force in a direction into the plane of the figure ($q\vec{v} \times \vec{B}$) and the charge 2 experiences force in a direction out of the plane of the figure. The forces are equal and opposite and they form torque. Similar torque is due to forces on other pair of charges at symmetrical locations with respect to the centre O . The torque tries to rotate the disc about the line OP .



44. The magnetic dipole moment is normal to the plane of the coil. It makes 30° angle with $\vec{B}$

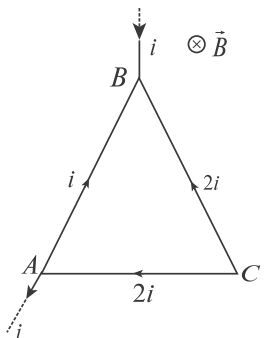
$$\therefore \tau = 2nIAB \sin 30^\circ = nIAB$$

45. According to the Kirchhoff's law, the current in side BC and CA will be $2i$.

Force on BCA is same as force on straight wire BA carrying current $2i$ from B to A .

Therefore, force on the triangle will be same as force on straight wire BA carrying current i from B to A .

$$F = i l B$$



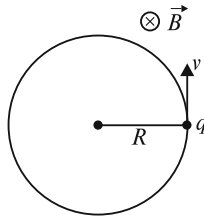


Worksheet 2

$$1. R = \frac{mv}{qB}; T = \frac{2\pi m}{qB}$$

Rate of sweeping the area is

$$\frac{dA}{dt} = \frac{\pi R^2}{T} = \frac{\pi \left(\frac{mv}{qB} \right)^2}{\frac{2\pi m}{qB}} = \frac{1}{2} \frac{mv^2}{qB}$$



$$2. R = \frac{mv \sin \theta}{qB}$$

$$P = \frac{2\pi m}{qB} v \cos \theta$$

when $R = P$

$$\sin \theta = 2\pi \cos \theta \Rightarrow \tan \theta = 6.28$$

$$\Rightarrow \theta = 81^\circ$$

when $\theta < 81^\circ$, $v_{||}$ is higher and the pitch becomes greater than the radius.

when $\theta > 81^\circ$, $v_{||}$ is smaller and the pitch becomes smaller.

3. When only electric field is there, the proton can deflect in positive x direction only if $\vec{E}$ is directed in positive x direction.

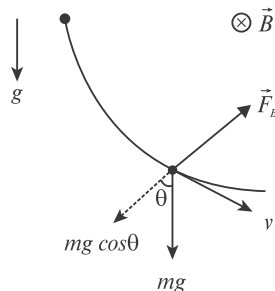
when only $\vec{B}$ is there, the electron may experience force in positive x -direction if velocity is in positive y -direction and $\vec{B}$ is in positive z -direction. The same may be true with $\vec{v}$ along negative y direction and $\vec{B}$ in negative z direction. In fact, the same can be true with electron's velocity in z -direction and $\vec{B}$ in $-y$ -direction.

4. For finding the force one can replace the conductor with a straight wire AE .

Length $AE = \lambda$

[If you put $x = \lambda$, the angle $\frac{2\pi}{\lambda}x$ becomes 2π]

5. The particle starts moving due to gravity. Initially, the magnetic force deflects it. Subsequently, the deflecting normal force is resultant of the magnetic force and a component of the weight of the particle.



When the speed of the particle becomes high, the magnetic force will turn the particle up. As the particle climbs, its speed will begin to decrease. The speed will not increase continuously.

$$6. \vec{F} = q \vec{v} \times \vec{B} = q (x\hat{i} + y\hat{j}) \times (y\hat{i} + x\hat{j}) \\ = q (x^2 - y^2) \hat{k} \\ |F| = |q(x^2 - y^2)| \\ = q(x^2 - y^2) \text{ if } x > y$$

$\vec{F}$ is along positive z direction if $x > y$

7. Change in kinetic energy = Work done by the electric force

$$\frac{1}{2} m (2v)^2 - \frac{1}{2} m (v^2) = qE(2a)$$

$$\text{or, } E = \frac{3}{4} \frac{mv^2}{qa}$$

Rate of work done by electric force at P = power of the electric force

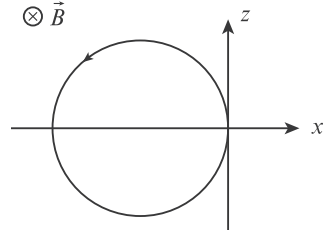
$$= \vec{F}_e \cdot \vec{v} = qEv = \frac{3}{4} \frac{mv^3}{a}$$

At point Q ,

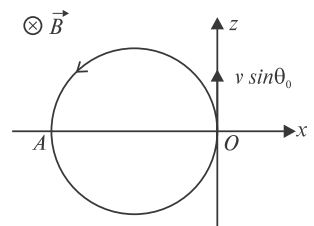
$$\vec{F}_e \cdot \vec{v}_Q = 0. \quad [\because \vec{F}_e \perp \vec{v}_Q]$$

Magnetic force does no work.

9. The proton goes in circular path due to its velocity component along z direction. Due to its velocity component along y direction, it keeps advancing in y direction and follows a helical path. The y co-ordinate will be proportional to the time of motion ($y = v_y t$).



10. Path is helical. The circular path in yz plane is as shown in figure. The particle advances uniformly in x direction with velocity $v \cos \theta_0$.



In time $t = \frac{T_0}{2}$, the

particle advances a distance equal to half the pitch in positive x direction. In yz plane, at $t = \frac{T_0}{2}$, the particle

is at A . Hence its y -coordinate is zero and z -coordinate is $-2R_0$. Statement (A) is true.

S.12 Magnetism and Electromagnetic Induction

One can verify that at $t = \frac{3}{2} T_0$, the particle will again

be at A , in yz plane. Its z co-ordinate is not correct in option (B). Therefore, this is an incorrect statement.

Obviously, maximum distance of the particle from the x axis is $2R_0$.



Worksheet 3

1. Force on the particle is perpendicular to $\vec{B}$. It means that the acceleration is also perpendicular to $\vec{B}$.

$$\therefore \vec{a} \cdot \vec{B} = 0$$

$$\text{or, } x(0.03) + 4(0.02) + 2(-0.1) = 0$$

$$\text{or, } x = 4$$

2. Let $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$

$$\text{We know that } \vec{F} = q (\vec{v} \times \vec{B})$$

$$\text{or, } -5\sqrt{2} \times 10^{-3} \hat{k}$$

$$= (10 \times 10^{-6}) \left[\frac{10^6}{\sqrt{2}} \hat{i} + \frac{10^6}{\sqrt{2}} \hat{j} \right] \times (B_x \hat{i} + B_y \hat{j} + B_z \hat{k})$$

$$\text{or, } -5\sqrt{2} \times 10^{-3} \hat{k}$$

$$= \frac{10}{\sqrt{2}} [B_z \hat{i} - B_z \hat{j} + (B_y - B_x) \hat{k}]$$

$$\therefore B_z = 0 \text{ and } B_y - B_x = -10^{-3} \text{ T} \quad (\text{i})$$

In the second case

$$F_2 \hat{j} = (10 \times 10^{-6}) (10^6 \hat{k}) \times (B_x \hat{i} + B_y \hat{j})$$

$$\text{or, } F_2 \hat{j} = 10B_x \hat{j} - 10B_y \hat{i}$$

$$\Rightarrow B_y = 0 \text{ and } 10 B_x = F_2 \quad (\text{ii})$$

$$\text{From (i) } B_y - B_x = -10^{-3}$$

$$\text{or, } B_x = 10^{-3} \text{ T}$$

$$\text{From (ii) } F_2 = 10^{-2} \text{ N.}$$

3. Charge on the first ion = $+e$

Its kinetic energy after acceleration is $k = e\Delta V$

$\therefore$ Radius of its circular path is

$$R = \frac{\sqrt{2mk}}{eB} = \frac{\sqrt{2me\Delta V}}{eB} = \frac{1}{B} \sqrt{\frac{2m\Delta V}{e}}$$

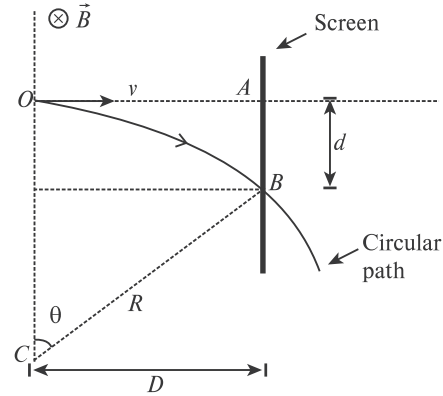
For the second particle, charge is $+2e$. Radius for it is

$$\therefore 2R = \frac{1}{B} \sqrt{\frac{2M\Delta V}{2e}}$$

$$\text{or, } \frac{2}{B} \sqrt{\frac{2m\Delta V}{e}} = \frac{1}{B} \sqrt{\frac{2M\Delta V}{2e}}$$

$$\Rightarrow M = 8m.$$

4. (a) The electron beam hits the screen at point B . The question is asking us to calculate the distance $AB = d$.



C is the centre of the circular path of the particle.

$$\begin{aligned} R &= \frac{mv}{eB} \\ &= \frac{\sqrt{2mk}}{eB} = \frac{\sqrt{2meV}}{eB} \\ &= \frac{1}{B} \sqrt{\frac{2mV}{e}} \end{aligned}$$

The deflection angle θ is given by $\sin\theta = \frac{D}{R}$

$$\begin{aligned} \therefore d &= R - R\cos\theta \\ &= R \left[1 - \sqrt{1 - \frac{D^2}{R^2}} \right] \\ &= R \left[1 - \left(1 - \frac{D^2}{R^2} \right)^{\frac{1}{2}} \right] \\ &= R \left[1 - \left(1 - \frac{D^2}{2R^2} \right) \right] \end{aligned}$$

Here, we have expanded using binomial theorem and neglected all higher order terms since $\frac{D}{R} \ll 1$.

$$\therefore d \simeq \frac{D^2}{2R^2} = \frac{D^2 B}{2} \sqrt{\frac{e}{2mV}}$$

- (b) For $V = 750 \text{ V}$, $D = 0.5 \text{ m}$ and $B = 0.5 \mu\text{T}$, we get

$$\begin{aligned} d &= \frac{0.5^2 \times 0.5 \times 10^{-6}}{2} \sqrt{\frac{1.6 \times 10^{-19}}{2 \times 9.1 \times 10^{-31} \times 750}} \\ &= 0.067 \text{ m} = 6.7 \text{ cm} \end{aligned}$$

5. Refer to example 8. The angular deviation caused due to magnetic force is given by

$$\sin\theta = \frac{\Delta x}{R} = \frac{\Delta x q B}{mv} \quad \left[R = \frac{mv}{qB} \right]$$

If $\Delta x \ll R$, Then $\sin\theta$ is small. We can write

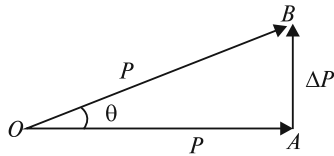
$$\sin\theta \simeq \theta$$

$$\therefore \theta = \frac{\Delta x q B}{mv}$$

Change in momentum of the particle can be written as

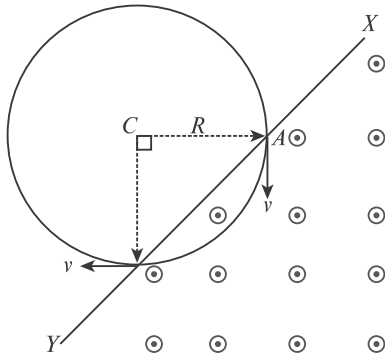
$$\Delta P = P \cdot \theta = mv\theta = \Delta x \cdot qB$$

This is the required impulse.



In the diagram shown $\overline{OA}$ represents the initial momentum and $\overline{OB}$ represents the final momentum after a deflection of θ . The magnitude of momentum does not change only its direction changes. For small angle θ we can write $\Delta P = P\theta$.

6. Path of the particle is as shown in the figure. The particle completes quarter of a circle inside the magnetic field.



- $\therefore$ Distance travelled is

$$S = \frac{2\pi R}{4} = \frac{\pi}{2} \frac{mv}{qB}$$

7. Velocity of the particle is inclined to $\vec{B}$. Therefore, it will move in a helical path. The pitch of the path will be

$$p = TV_{||} = \left(\frac{2\pi m}{qB} \right) v \cos\theta$$

The particle will touch the line GS when it completes full turns. Therefore, it will hit the spot S only if

$$GS = np \text{ with } n = 1, 2, 3, \dots$$

$$\text{or, } n \cdot \left(\frac{2\pi m}{qB} \right) v \cos\theta = (GS)$$

$$\begin{aligned} \Rightarrow B &= n \cdot \frac{2\pi \cos\theta}{q(GS)} (mv) \\ &= n \cdot \frac{2\pi \cos\theta}{(GS)} \frac{\sqrt{2mk}}{q} \quad [\because mv = p = \sqrt{2mk}] \end{aligned}$$

B is smallest when $n = 1$

$$\begin{aligned} \therefore B_{\min} &= \frac{2\pi \cos 60^\circ \sqrt{2 \times 9.1 \times 10^{-31} \times 2 \times 10^3 \times 1.6 \times 10^{-19}}}{0.1 \times 1.6 \times 10^{-19}} \\ &= 15\pi \times 10^{-4} = 4.71 \times 10^{-3} \text{ T.} \end{aligned}$$

Particle can hit the spot S if $B = 2 B_{\min}$. In this case it makes two full turn on the helical path to hit the spot S . Particle can similarly hit S for all values of $n = 1, 2, 3, 4, \dots$

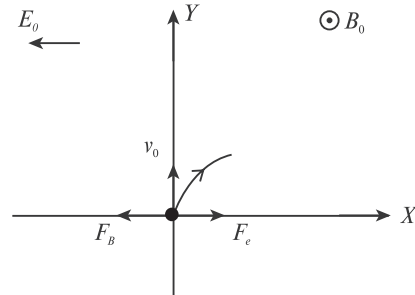
$\therefore$ Next higher value of B that is appropriate is

$$B = 2 B_{\min} = 9.42 \times 10^{-3} \text{ T}$$

8. $\vec{F}_e = -q(-E_0 \hat{i}) = qE_0 \hat{i}$

$$\vec{F}_e = -(v_0 \hat{j} \times B_0 \hat{k}) = -qv_0 B_0 \hat{i}$$

$$= -\frac{qE_0}{2} \hat{i} \quad [\because v_0 = \frac{E_0}{2B_0}]$$



Resultant force on the charge is $\vec{F} = \frac{qE_0}{2} \hat{i}$.

Velocity of the particle is v_0 along y -direction. The force is perpendicular to the velocity.

$$\therefore \frac{mv_0^2}{R} = \frac{qE_0}{2}$$

$[R = \text{radius of curvature of the path}]$

$$\text{or, } R = \frac{2mv_0^2}{qE_0}$$

9. (a) Refer to example 8.

$$\sin\theta = \frac{b}{R} \Rightarrow \sin 30^\circ = \frac{bqB_0}{mv_0}$$

$$\text{or } b = \frac{mv_0}{2qB_0}$$

- (b) when the width of the region containing the field is higher than the radius of the circular path, the particle will trace a semicircle and come out of

the field region. $2.6b$ is definitely greater than the radius $\left(\frac{qB_0}{mv_0}\right)$.

10. (a) The path of the particle is a helix with increasing pitch. The axis of the helix is parallel to the y -axis. The particle has a circular motion in a plane perpendicular to the direction of the magnetic field (i.e., xz plane). The electric force in y direction accelerates the particle in y -direction and its pitch goes on increasing.

The particle crosses the y axis at regular intervals

$$T = \frac{2\pi m}{qB_0}$$

$\therefore$ Particle will cross the y -axis n^{th} time at a time

$$t = nT = \frac{2\pi nm}{qB_0} = \frac{2\pi n}{\sigma B_0}$$

[Specific charge $\sigma = \frac{q}{m}$]

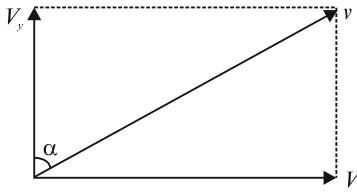
y co-ordinate at this time is

$$\begin{aligned} y &= \frac{1}{2} a_y t^2 = \frac{1}{2} \left(\frac{qE_0}{m} \right) \left(\frac{2\pi n}{\sigma B_0} \right)^2 \\ &= \frac{2\pi^2 n^2 E_0}{\sigma B_0^2} \end{aligned}$$

- (b) y component of velocity at time t is

$$v_y = a_y t = \left(\frac{qE_0}{m} \right) \left(\frac{2\pi n}{\sigma B_0} \right) = 2\pi n \left(\frac{E_0}{B_0} \right)$$

The angle between the particle's velocity vector and the y -axis at this instant is perpendicular to B .



$$\begin{aligned} \alpha &= \tan^{-1} \left(\frac{v_0}{v_y} \right) \\ &= \tan^{-1} \left(\frac{v_0 B}{2\pi n E} \right) \end{aligned}$$

v_0 is component of velocity in xz plane that is perpendicular to B .

11. Velocity component of the particle parallel to $\vec{B}$ is $u_z = u$.

Velocity component in xy plane (i.e., perpendicular to $\vec{B}$) is also u . This velocity makes the particle move in circle. The electric force on the particle is directed in negative z direction.

The velocity component in xy plane remains constant (its direction keeps changing in xy plane as the particle rotates). But the z component of velocity changes due to the electric force on the particle.

$$az = -\frac{qE_0}{m}$$

[Electric field is in negative z direction]

Speed will be smallest when u_z becomes zero.

$$\Rightarrow v_z = u_z + a_z t = 0$$

$$\text{or, } u - \frac{qE_0}{m} t = 0 \Rightarrow t = \frac{mu}{qE_0}$$

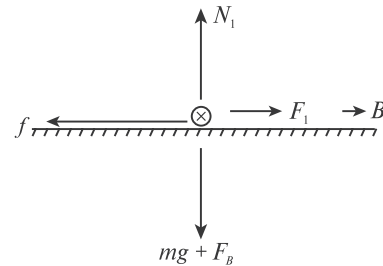
When $u_z = 0$, the particle only has its motion in xy plane. Its speed is smallest and its value is u .

12. Electric force does work on the particle to change its kinetic energy. Work done by the electric force is maximum when the particle is at its maximum x -coordinate. At this instant its kinetic energy is maximum.

$$\begin{aligned} \therefore \frac{1}{2} mv_{\text{max}}^2 &= qE_0 d \\ \Rightarrow v_{\text{max}} &= \sqrt{\frac{2qE_0 d}{m}} \end{aligned}$$

The key observation is that the magnetic force performs no work.

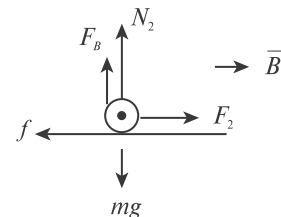
13. The diagram shows the cross section of the wire with the direction of current into the plane of the figure, Magnetic force is downward and this increases the normal force. This increases the friction force. For the wire to just start moving



$$F_1 = \mu N_1 = \mu (mg + F_B)$$

$$\text{or, } F_1 = \mu (mg + \mu B)$$

When the direction of the current is reversed, the direction of the magnetic force also reverses. For the wire to just start moving



$$F_2 = \mu N_2 = \mu (mg - ILB) \quad (\text{ii})$$

Eliminating μ between (i) and (ii) gives

$$mg = BIl \left(\frac{F_1 + F_2}{F_1 - F_2} \right)$$

14. The diagram shows cross section of the wire. Current in the wire is I into the plane of the figure. The normal force (N), weight (mg) and the magnetic force (F_B) acting on the wire are as shown in the figure. The wire moves without acceleration if

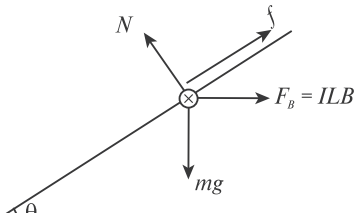
$$F_B \cos \theta = mg \sin \theta$$

$$\Rightarrow ILB \cos \theta = mg \sin \theta$$

$$\Rightarrow B = \frac{mg}{IL} \tan \theta$$

$$\text{or, } B = \frac{50 \times 10^{-3} \times 10}{2.5 \times 0.5} \times \frac{3}{4} = 0.3 \text{ T}$$

15. When B is small, the component of F_B up the incline is smaller than $mg \sin \theta$ and the bar has a tendency to slide down. The friction is directed up the plane. F_B is smallest (i.e., B is smallest) when the friction is at its maximum value.



$$N = mg \cos \theta + ILB \sin \theta$$

For equilibrium

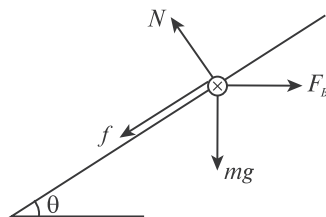
$$\mu N + F_B \cos \theta = mg \sin \theta$$

$$\text{or, } \mu(mg \cos \theta + ILB \sin \theta) + ILB \cos \theta = mg \sin \theta$$

$$\text{or, } B = \frac{mg(\sin \theta - \mu \cos \theta)}{IL(\cos \theta + \mu \sin \theta)}$$

This is the smallest value of B for which equilibrium is possible. If B decreases, F_B will decrease and the wire will slide down the incline.

When B is high ($\Rightarrow F_B$ is high) the wire has a tendency to move up. Friction gets directed down the incline to prevent the wire



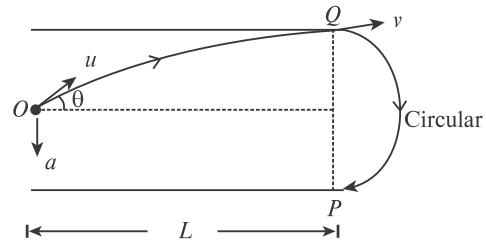
from moving. Maximum B for which the equilibrium is possible is obtained by assuming the friction to be at its maximum. Doing similar calculations as above gives

$$B_{\max} = \frac{mg(\sin \theta + \mu \cos \theta)}{IL(\cos \theta - \mu \sin \theta)}$$

16. Acceleration in the electric field is

$$a = \frac{qE}{m} = \frac{(10^{-6})(10^{-3})}{(10^{-10})}$$

$$\text{or, } a = 10 \text{ m/s}^2$$



Motion of the charge from O to Q is on a parabolic path just like a projectile. The particle will emerge out of the plate region, grazing the plate at Q if

$$L = \frac{\text{Range}}{2}$$

$$\text{or, } 0.17 = \frac{u^2 \sin 2\theta}{2a}$$

$$\Rightarrow 0.17 = \frac{u^2 \sin 60^\circ}{2 \times 10}$$

$$\text{or, } u = 1.98 \text{ m/s}$$

Velocity of the particle at Q is

$$v = u \cos \theta = 1.98 \times \cos 30^\circ$$

$$\text{or, } v = 1.7 \text{ m/s}$$

The particle will re enter the plate region at P after describing a semicircle in the region having magnetic field.

$$\Rightarrow QP = 2 \frac{mv}{qB} \quad \Rightarrow 0.1 = 2 \frac{10^{-10} \times 1.7}{10^{-6} \times B}$$

$$\Rightarrow B = 3.4 \text{ mT.}$$

17. Acceleration in y -direction is $a = \frac{qE}{m}$

Time needed to reach $N =$ time needed to get displaced by $\frac{d}{2}$ in y -direction.

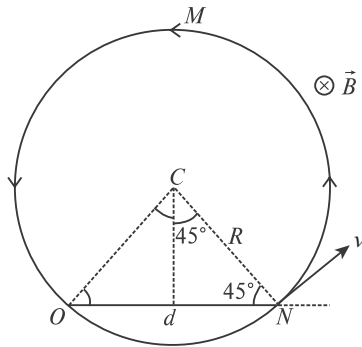
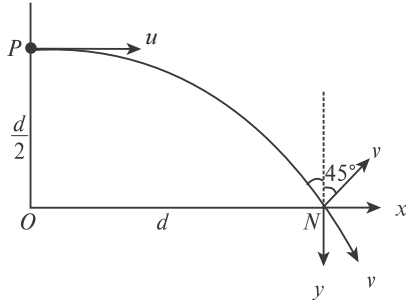
This can be calculated as

$$y = \frac{1}{2} a_y t^2 \quad \Rightarrow \quad \frac{d}{2} = \frac{1}{2} \frac{qE}{m} t^2$$

$$\Rightarrow t = \sqrt{\frac{md}{qE}}$$

The initial speed of the particle is

$$u = \frac{d}{t} = \sqrt{\frac{qEd}{m}}$$



Just before hitting the surface at N, the velocity components are

$$v_x = u \quad \text{and} \quad v_y = a_y t = \frac{qE}{m} \sqrt{\frac{md}{qE}} = \sqrt{\frac{qEd}{m}}$$

The angle made by the velocity with the surface is

$$\tan \theta = \frac{v_y}{v_x} = 1$$

$$\Rightarrow \theta = 45^\circ$$

$$\text{Speed } v = \sqrt{v_x^2 + v_y^2} = \sqrt{\frac{2qEd}{m}}$$

Since the collision is elastic, the particle rebounds with same speed making 45° with the surface as shown in figure. Now, it moves in a circular path along NMO to hit the point O. Radius of the path can be calculated as

$$R = \frac{mv}{qB} = \sqrt{\frac{2qEd}{m}} \cdot \frac{1}{B}$$

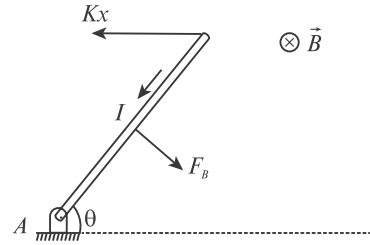
From the figure,

$$R = \sqrt{2} \frac{d}{2} = \frac{d}{\sqrt{2}}$$

$$\text{or, } \sqrt{\frac{2mEd}{q}} \cdot \frac{1}{B} = \frac{d}{\sqrt{2}}$$

$$\text{or, } B = 2 \sqrt{\frac{mE}{qd}}$$

18. (a) Magnetic force on every segment of equal length will be same. Therefore, we can consider the magnetic force to be acting at the centre of the rod. Direction of magnetic force is as shown in figure. It produces a clockwise torque about A. The spring must produce an equal anticlockwise torque about A. It means that the spring must be stretched.



- (b) Net torque about A will be zero if

$$IlB \frac{l}{2} = kx l \sin 30^\circ$$

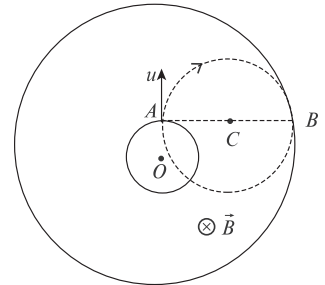
$$\text{or, } x = \frac{IlB}{k}$$

$\therefore$ Energy stored in the spring is

$$U = \frac{1}{2} kx^2 = \frac{1}{2} k \left(\frac{IlB}{k} \right)^2$$

$$\text{or, } U = \frac{I^2 l^2 B^2}{2k}$$

19. The question is all about understanding the position of the centre of the circular path taken by the electron. Electron starts moving from point A in the radial direction OA. Centre of the circular path is on a line perpendicular to initial velocity (i.e., perpendicular to OA). For the electron to not hit the outer cylinder, the diameter of its circular path shall be less than the geometrical length AB.

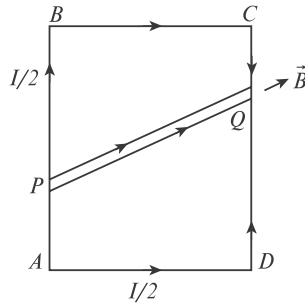


$$\therefore 2 \frac{mu}{eB} < AB$$

$$\text{or, } \frac{2mu}{eB} < \sqrt{OB^2 - OA^2} = \sqrt{b^2 - a^2}$$

$$\text{or, } \frac{2mu}{e\sqrt{b^2 - a^2}} < B.$$

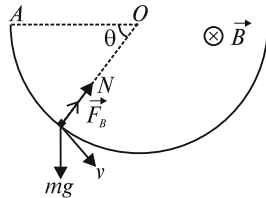
20. Current will divide equally in two identical parallel paths $PBCQ$ and $PADQ$. Force on $PBCQ$ can be written by replacing it with a straight wire PQ . Similarly, force on $PADQ$ can also be written by replacing it with a straight wire PQ . But wire PQ is parallel to $\vec{B}$. Hence magnetic force is zero.



21. Magnetic force does not perform work. Therefore, energy of the block remains conserved.

Let v be the speed of the block when it reaches B .

$$\frac{1}{2}mv^2 = mgR \sin \theta \Rightarrow v = \sqrt{2gR \sin \theta}$$



The magnetic force is radial and has magnitude

$$F_B = QvB$$

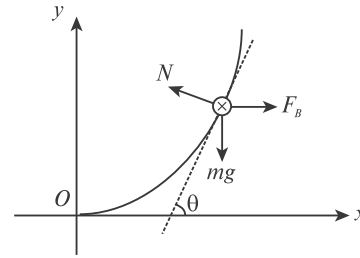
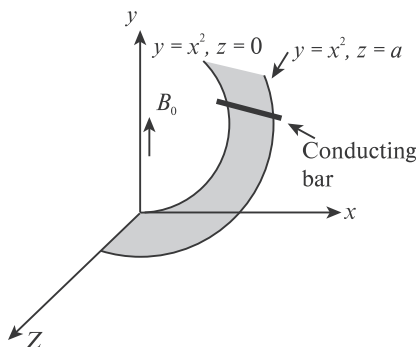
Writing equation for centripetal force gives

$$N + F_B - mg \sin \theta = \frac{mv^2}{R}$$

$$\text{or, } N + Q\sqrt{2gR \sin \theta} B - mg \sin \theta = 2 mg \sin \theta$$

$$\text{or, } N = 3 mg \sin \theta - QB\sqrt{2gR \sin \theta}$$

22. The situation is as shown in the figure below, In the second figure, we have drawn one parabolic rail and cross section of the conducting bar carrying current i .



Current in the conductor is going into the figure.

Slope of the parabola is given by $\frac{dy}{dx}$.

$$\therefore \tan \theta = \frac{dy}{dx} = 2x$$

If the equilibrium happens at $x = x_0$, then

$$F_B \cos \theta = mg \sin \theta \Rightarrow iaB_0 = mg \tan \theta$$

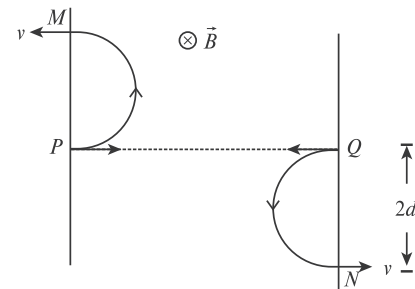
$$\Rightarrow iaB_0 = mg (2x_0)$$

$$\text{or, } x_0 = \frac{iaB_0}{2mg}$$

$$\text{or, } \sqrt{y_0} = \frac{iaB_0}{2mg} \Rightarrow y_0 = \left(\frac{iaB_0}{2mg} \right)^2$$

23. The two particles trace semicircular paths inside the field region and come out at points M and N . Separation between the particles at this instant is

$$r = MN = 4\sqrt{2} d$$

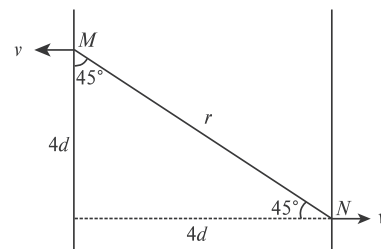


Velocity of B relative to A in a direction perpendicular to MN is

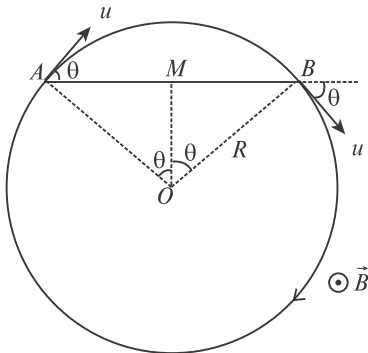
$$\frac{v}{\sqrt{2}} + \frac{v}{\sqrt{2}} = \sqrt{2} v$$

$\therefore$ Relative angular speed is

$$\omega = \frac{\sqrt{2}v}{r} = \frac{\sqrt{2}v}{4\sqrt{2}d} = \frac{v}{4d}$$



24. The circular path of the particle is shown in figure. AB is a chord of the circular path and $\angle AOB = 2\theta$.

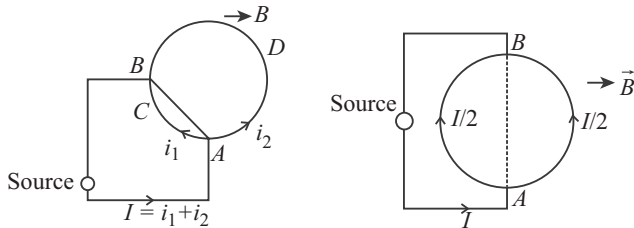


$$\frac{MB}{R} = \sin\theta$$

$$\text{or, } \frac{LqB_0}{2mu} = \sin\theta$$

$$\text{or, } B_0 = \frac{2mu}{qL} \sin\theta$$

25. If the battery is connected as shown in the first figure, the force on segment ACB is same as force on straight wire AB carrying current i_1 . Similarly, force on the segment ADB is same as force on a straight wire AB carrying current i_2 . The total force is equal to force on a straight wire AB carrying a current



$$I = i_1 + i_2.$$

Obviously, force will be maximum when AB is of maximum length and perpendicular to $\vec{B}$. The second figure shows the situation. Maximum force is

$$F = \frac{I}{2}(2a)B + \frac{I}{2}(2a)B = 2IaB \otimes$$

26. Magnetic force on the semicircular ring = force on a straight wire BA carrying same current $I = I(2R)B_0$. This force is vertically upward and acts at the centre. When torque due to this force exceeds torque due to weight (about A), the ring rotates about A and the circuit gets broken at P. Limiting value of I can be obtained as

$$(2IRB_0)R = mgR$$

$$\text{or, } I = \frac{mg}{2RB_0}$$

27. Kinetic Energy of the electrons in the beam = $e\Delta V$.

If v is velocity of electrons in the beam, then

$$\frac{1}{2}mv^2 = e\Delta V \Rightarrow v = \sqrt{\frac{2e\Delta V}{m}} \quad (i)$$

Since θ is small, the velocity component parallel to $\vec{B}$ is $v_{||} = v\cos\theta \simeq v$

$$\text{Time to travel distance } d \text{ is } t = \frac{d}{v} \quad (ii)$$

The electrons entering the field at an angle, move in helical path.

$$\text{The time period is } T = \frac{2\pi m}{eB}$$

The time period of helical motion is independent of angle θ or speed. All the electrons reach O, if time period of helical path is same as time given by (ii). The electrons can also reach O if time t given by (ii) is two times, or three times the time period T .

$\therefore$ For electrons to hit O, we must have

$$\frac{d}{v} = n \cdot T; \quad n = 1, 2, 3, \dots$$

$$\text{or, } d = n \cdot \frac{2\pi m}{eB} \cdot \sqrt{\frac{2e\Delta V}{m}}$$

For smallest d , $n = 1$

$$\therefore d_{\min} = \frac{2\pi}{B} \sqrt{\frac{2m\Delta V}{e}}$$

28. Area of the loop is A = Area of two squares + Area of two quarter circles

$$\therefore A = 2 \times (0.2)^2 + 2 \times \frac{\pi(0.1)^2}{4}$$

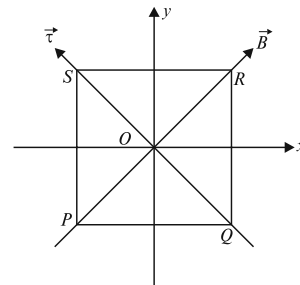
$$= \left(\frac{16+\pi}{2}\right) \times 10^{-2} \text{ m}^2$$

$\therefore$ Magnetic dipole moment

$$\vec{M} = i\vec{A} = 5 \left(\frac{16+\pi}{2}\right) \times 10^{-2} (\hat{k})$$

$$\text{Torque is } \vec{\tau} = \vec{M} \times \vec{B} = 6.7 (\hat{j} - \hat{i}) \text{ Nm}$$

29. (a) Magnetic Moment $\vec{M} = I_0 L^2 \hat{k}$



$$\text{Magnetic field } \vec{B} = \frac{B}{\sqrt{2}} (\hat{i} + \hat{j})$$

$$\vec{\tau} = \vec{M} \times \vec{B}$$

$$= \frac{I_0 L^2 B}{\sqrt{2}} (\hat{j} - \hat{i})$$

$$|\vec{\tau}| = I_0 L^2 B$$

- (b) The torque is along QS . Moment of inertia about QS is $I = \frac{1}{2} I_z$ where I_z is moment of inertia about z -axis.

$$\text{or, } I = \frac{1}{2} \left[4 \times \left(\frac{ML^2}{12} + \frac{ML^2}{4} \right) \right] = \frac{2}{3} ML^2$$

$$\text{Angular acceleration, } \alpha = \frac{\tau}{I} = \frac{3I_0 B}{2M}$$

$$\therefore \Delta\theta = \frac{1}{2} \alpha (\Delta t)^2 = \frac{3I_0 B}{4M} (\Delta t)^2$$

30. You should attempt this question after studying Bohr's theory of H-atom.

- (a) According to the Bohr's theory

$$mvR = \frac{h}{2\pi} \Rightarrow v = \frac{h}{2\pi mR}$$

$$\text{Time period of revolution, } T = \frac{2\pi R}{v}$$

Frequency of revolution,

$$f = \frac{1}{T} = \frac{v}{2\pi R} = \frac{h}{4\pi^2 mR^2}$$

$$\text{Current, } i = ef = \frac{eh}{4\pi^2 mR^2}$$

$$\therefore \text{Magnetic dipole moment, } M = i\pi R^2 = \frac{eh}{4\pi m}$$

Direction of $\vec{M}$ is perpendicular to the plane of the orbit.

$$(b) \vec{\tau} = \vec{M} \times \vec{B} \Rightarrow \tau = MB \sin 30^\circ$$

$$\text{or, } \tau = \frac{ehB}{8\pi m}$$

Direction of $\vec{\tau}$ is perpendicular to both $\hat{n}$ and $\vec{B}$.

31. Let the direction of current in PQ be from P to Q , and its magnitude be I .

Magnetic moment of the loop is

$$\vec{\tau}_B = \vec{M} \times \vec{B} = (-Iab\hat{k}) \times (3\hat{i} + 4\hat{k})B_0$$

$$= -3IabB_0\hat{j}$$

Torque due to weight is

$$\vec{\tau}_g = \frac{mga}{2}\hat{j}$$

[curl your right hand fingers in the sense in which the weight of the loop likes to rotate it and the thumb gives the direction of the torque]

Since $\vec{\tau}_B$ and $\vec{\tau}_g$ are oppositely directed, equilibrium is possible.

$\therefore$ Direction of the current from P to Q is correct

$$\vec{F} = I(-b\hat{j}) \times (3\hat{i} - 4\hat{k})B_0 = IbB_0(3\hat{k} - 4\hat{i})$$

For equilibrium

$$\tau_B = \tau_g$$

$$\text{or, } 3IabB_0 = \frac{mga}{2}$$

$$\text{or, } I = \frac{mg}{6bB_0}$$

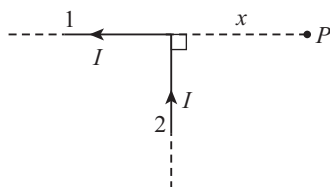
Chapter 2 Magnetic Effect of Current



Your Turn

- Field at P due to straight wire AB as well as due to BC is directed into the plane of the figure.
- Field at P due to the straight segment 1 is zero. Field at P due to the straight segment 2 (a semi-infinite wire) is

$$B = \frac{\mu_0 I}{4\pi r} \otimes$$



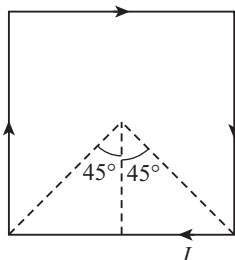
This is the resultant field at P .

- Field due to each side is directed into the plane of the figure. Field due to any one side at the centre is

$$B = \frac{\mu_0 I}{4\pi \frac{a}{2}} (\sin 45^\circ + \sin 45^\circ)$$

$$= \frac{2\sqrt{2}\mu_0 I}{4\pi a} \otimes$$

contribution of each side is the same. The resultant field at the centre is

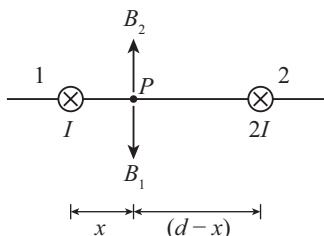


$$B_{\text{centre}} = 4B = \frac{2\sqrt{2}\mu_0 I}{\pi a} \otimes$$

- Current will divide equally in the parallel paths ABC and ADC . Fields due to each side has same magnitude (at point O). Direction of field due to AB and BC is into the plane of the figure and that due to AD and DC is out of the plane of the figure. No field is contributed by the wire carrying the current I to A and that going out at C . The point O lies on the extension of these wires.

Therefore, resultant field at O is zero.

- Consider a point P between the two wires. The magnetic field due to the two wires are in opposite directions at P . For the resultant field



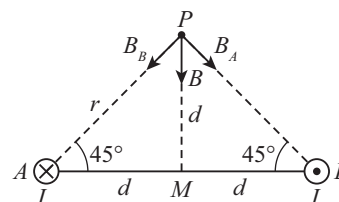
to be zero at P , the two fields must be of same magnitude.

$$B_1 = B_2$$

$$\frac{\mu_0 I}{4\pi x} = \frac{\mu_0 (2I)}{4\pi (d-x)}$$

$$\Rightarrow d - x = 2x \Rightarrow x = \frac{d}{3}$$

- From the geometry of the figure, $\angle APB = 90^\circ$. The magnetic field at P due to the wire at A is



$$B_A = \frac{\mu_0 I}{4\pi r} = \frac{\mu_0 I}{4\pi (\sqrt{2}d)}$$

Direction of this field is perpendicular to AP , along PB . Field due to the wire at B is along PA . [To visualise this you may draw circular field lines around the wire].

$$B_B = B_A = \frac{\mu_0 I}{4\pi (\sqrt{2}d)}$$

The resultant of B_A and B_B is along PM and its magnitude is

$$B = \sqrt{B_A^2 + B_B^2} = \frac{\mu_0 I}{4\pi d}$$

- Let the radius of the coil be r . Then

$$n 2\pi r = L \Rightarrow r = \frac{L}{2\pi n}$$

$$\text{Field at the centre is } B = n \frac{\mu_0 I}{2r}$$

$$\text{or, } B = n \frac{\mu_0 I}{2 \left(\frac{L}{2\pi n} \right)} = \frac{\mu_0 \pi n^2 I}{L}$$

- Field at O due to the straight part joining the circle at A is zero. $B_1 = 0$

Field at O due to the circular arc is

$$B_2 = \frac{\mu_0 I}{2R} \cdot \frac{3}{4} = \frac{3\mu_0 I}{8R} \odot$$

Field at O due to the straight part joining the circle at B is

$$B_3 = \frac{\mu_0 I}{4\pi R} \odot \quad [\text{Semi infinite wire}]$$

$$\therefore \text{Resultant field } B = \frac{\mu_0 I}{4R} \left(\frac{3}{2} + \frac{1}{\pi} \right) \odot$$

9. There is no field at O due to the two straight segments.

Field due to the semicircle in yz plane is

$$\vec{B}_1 = \left(\frac{\mu_0 I}{4R} \right) \hat{j} = \left(\frac{\pi \times 10^{-7} \times 5}{0.1} \right) \hat{j} = (5\pi \times 10^{-6} T) \hat{j}$$

Field due to the semicircle in xy plane is

$$\begin{aligned} \vec{B}_2 &= -\left(\frac{\mu_0 I}{4r} \right) \hat{k} = -\left(\frac{\pi \times 10^{-7} \times 5}{0.05} \right) \hat{k} \\ &= -(10\pi \times 10^{-6} T) \hat{k} \end{aligned}$$

$$\therefore \vec{B} = 5\pi \times 10^{-6} (\hat{j} - 2\hat{k}) T.$$

10. Field due to all straight segments is zero (at O).
Field due to the circular arcs are:

$$B_{AB} = \frac{\mu_0 I}{2(3r)} \frac{1}{12} = \frac{\mu_0 I}{72r} \otimes$$

$$B_{CD} = \frac{\mu_0 I}{2(2r)} \frac{1}{12} = \frac{\mu_0 I}{48r} \odot$$

$$B_{EF} = \frac{\mu_0 I}{4r} \otimes$$

$$\begin{aligned} \text{Resultant field is } B &= \left(\frac{\mu_0 I}{72r} + \frac{\mu_0 I}{4r} - \frac{\mu_0 I}{48r} \right) \otimes \\ &= \frac{35}{144} \frac{\mu_0 I}{r} \otimes \end{aligned}$$

11. Current divides into two parts at A and combines at B . Refer to example 7. The currents in two semicircular parts will produce zero field at O . The two straight pieces produce equal and opposite fields at O . Therefore, the net magnetic field at O is zero.

12. Given $\frac{\mu_0 I}{2R} = 5 \times 10^{-5} T$ (i)

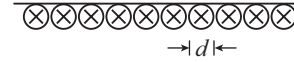
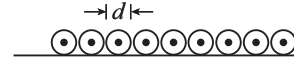
Field on the axis is given by equation (12)

$$\begin{aligned} B &= \frac{\mu_0}{2} \frac{R^2 I}{(R^2 + x^2)^{3/2}} = \frac{\mu_0 I}{2R} \frac{1}{\left(1 + \frac{x^2}{R^2} \right)^{3/2}} \\ &= (5 \times 10^{-5}) \frac{1}{\left[1 + \left(\frac{4}{3} \right)^2 \right]^{3/2}} = (5 \times 10^{-5}) \frac{3^3}{5^3} \\ &= 1.08 \times 10^{-5} T \end{aligned}$$

13. Number of turns in 1m length of the solenoid is

$$n = \frac{1\text{m}}{1\text{mm}} = 1000$$

$$\begin{aligned} \therefore B &= \mu_0 n I = 4\pi \times 10^{-7} \times 1000 \times 1 \\ &= 4\pi \times 10^{-4} T \end{aligned}$$



14. The field is uniform around the midpoint on the axis. Both the poles of the magnet experience equal and opposite forces. Resultant force is zero.

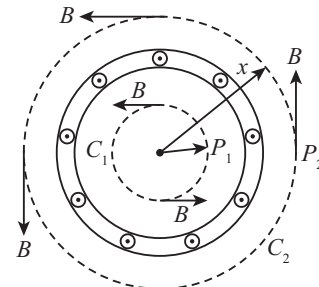
15. Every point on the circle is not a symmetrical location relative to the wire. We cannot assume the field to be of same strength and tangential direction at all points.

$$\oint \vec{B} \cdot d\vec{l} = 0 \text{ is absolutely right but } \oint \vec{B} \cdot d\vec{l} \neq B \cdot 2\pi r.$$

16. $\vec{B}$ is field due to all source currents. In the given question it is the resultant field due to I_1 and I_2 both.

17. From Ampere's law, both the line integrals have the same value equal to $\mu_0 I$.

18. The figure shows the cross sectional view. Current is directed out of the plane of the figure.



Consider an Amperian loop C_1 – a circle passing through P_1 . Field will be tangential and will have same strength at all points on this circle.

$$\therefore \oint_{C_1} \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$\text{or, } B 2\pi x = \mu_0 (0) \quad [\text{No current pierces through the circle } C_1]$$

$$\Rightarrow B = 0.$$

At all points inside the hollow tube the field is zero. Similarly, consider a circular loop C_2 through point P_2 .

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$\Rightarrow B \cdot 2\pi x = \mu_0 i$$

$$\text{or, } B = \frac{\mu_0 I}{2\pi x}$$

19. Refer to equation 19.

Field at A due to the upper sheet is

$$B_{1A} = \frac{\mu_0 j a}{2} (\rightarrow)$$

Field at A due to the lower sheet is

$$B_{2A} = \frac{\mu_0 j a}{2} (\leftarrow)$$

Therefore, field at A is zero.

Similarly, field at D is zero.

Fields due to the two sheets are in the same direction at C .

$$\begin{aligned} \therefore B_C &= \frac{\mu_0 j a}{2} (\leftarrow) + \frac{\mu_0 j a}{2} (\leftarrow) \\ &= \mu_0 j a (\leftarrow) \end{aligned}$$

20. (a) $\oint \vec{B} \cdot d\vec{l} = \mu_0 N I = 4\pi \times 10^{-7} \times 3000 \times 5$
 $= 6\pi \times 10^{-3} \text{ T.m.}$

$$\begin{aligned} \text{(b) } B &= \mu_0 n I = 4\pi \times 10^{-7} \times \frac{3000}{2} \times 5 \\ &= 3\pi \times 10^{-3} \text{ T} \end{aligned}$$

21. Given: $\frac{\mu_0}{2\pi} \frac{I_1 I_2}{d} = 10^{-5}$

$$\text{or, } 2 \times 10^{-7} \times \frac{10(I_2)}{0.1} = 10^{-5}$$

$$\text{or, } I_2 = 2A.$$

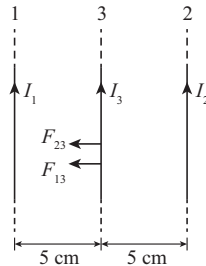
Force on unit length of I_3 is

$$F = F_{13} + F_{23}$$

[I_1 attracts I_3 and I_2 repels I_3]

$$= \frac{\mu_0}{2\pi} \frac{I_1 I_3}{0.05} + \frac{\mu_0}{2\pi} \frac{I_2 I_3}{0.05}$$

$$= \frac{2 \times 10^{-7}}{0.05} \times (10+2)2 = 9.6 \times 10^{-5} \text{ N/m}$$



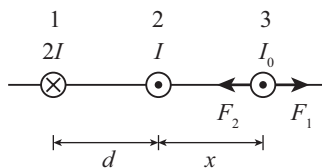
22. The third wire shall be placed nearer to the wire carrying current I . Let the third wire be at a distance x from the second wire having current I . (Note that if we place the third wire between the two wires, it will experience both the force in the same direction. Also, the direction of current in the third wire will not affect our answer.)

For equilibrium of the third wire

$$F_1 = F_2$$

$$\text{or, } \frac{\mu_0}{2\pi} \frac{(2I)(I_0)}{(d+x)} L = \frac{\mu_0}{2\pi} \frac{II_0}{x} L$$

$$\text{or, } 2x = d + x \Rightarrow x = d$$



Worksheet 1

1. Electric field is not produced because the wire contains even distribution of negative and positive charges. A finite piece of wire is neutral.

3. Field due to each of the straight sections is

$$B_1 = \frac{\mu_0 i}{4\pi r} \odot$$

$$\text{Field due to the circular part is } B_2 = \frac{\mu_0 i}{2r} \otimes$$

$$\begin{aligned} \therefore \text{Resultant field is } B &= \left(\frac{\mu_0 i}{2r} - \frac{2 \cdot \mu_0 i}{4\pi r} \right) \otimes \\ &= \frac{\mu_0 2i}{4\pi r} (\pi - 1) \otimes \end{aligned}$$

4. Frequency of revolution $f = \frac{1}{2 \times 10^{-6}} = \frac{10^6}{2} \text{ HZ}$

charge on an alpha particle is $q = +2e$

$$\begin{aligned} \therefore \text{current } i &= qf = 2 \times 1.6 \times 10^{-19} \times \frac{10^6}{2} \\ &= 1.6 \times 10^{-13} \text{ A} \end{aligned}$$

$$\therefore B = \frac{\mu_0 i}{2r} = \frac{\mu_0 \times 1.6 \times 10^{-13}}{2 \times 0.8} = 10^{-13} \mu_0$$

5. $B = \mu_0 n i = 4\pi \times 10^{-7} \times (10 \times 100) \times 5$
 [Number of turns per meter is 10×100]

$$\therefore B = 2\pi \times 10^{-3} \text{ T.}$$

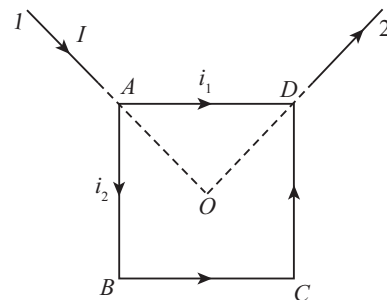
6. The maximum magnetic field is on the surface of the wire.

$$B_{\max} = \frac{\mu_0 I}{2\pi r} \quad [r = \text{radius}]$$

$$\text{or, } 5 \times 10^{-3} = \frac{2 \times 10^{-7} \times i}{0.8 \times 10^{-3}}$$

$$\text{or, } i = 20A$$

7. The field due to the wires bringing in and taking out the current (the wires 1 and 2), from the square frame, is zero in figure a, b and d.



In figure (a), the current I divides into two parallel paths in the inverse ratio of the resistances.

$$\therefore \frac{i_1}{i_2} = \frac{3}{1} \left[\frac{\text{resistance of path } AD}{\text{resistance of path } ABCD} = \frac{1}{3} \right]$$

$$\therefore i_1 = \frac{3I}{4}, i_2 = \frac{I}{4}$$

Field due to AB , BC and CD are equal and in same direction. If field due to each of the three arms is B_0 , then the resultant field due to the three sides is $3B_0 \odot$.

Field due to AD is also $3B_0 \otimes$ [$\because i_1 = 3i_2$]
Both fields cancel out.

Therefore resultant field is zero at the centre.

In all four diagrams, the field at the centre due to the square loop is zero.

In figure (c) the field due to the input and output wires add to produce a resultant field at the centre.

$$8. \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + x^2)^{3/2}} = \frac{1}{8} \frac{\mu_0 I}{2R}$$

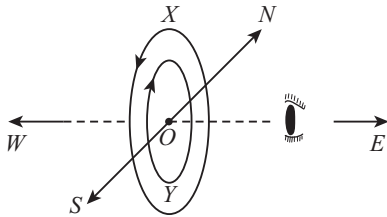
$$\text{or, } \left(\frac{R}{\sqrt{R^2 + x^2}} \right)^3 = \frac{1}{8}$$

$$\text{or, } 2R = \sqrt{R^2 + x^2} \Rightarrow 4R^2 = R^2 + x^2$$

$$\text{or } x = R\sqrt{3}$$

9. Field due to X

$$B_x = \frac{N_1 \mu_0 I_1}{2r_1} (\rightarrow)$$



Field due to Y

$$B_y = \frac{N_2 \mu_0 I_2}{2r_2} (\leftarrow)$$

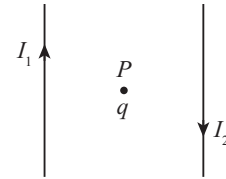
$$\therefore B = \frac{\mu_0}{2} \left[\frac{N_2 I_2}{r_2} - \frac{N_1 I_1}{r_1} \right] (\leftarrow)$$

$$= 2\pi \times 10^{-7} \left[\frac{25 \times 18}{0.1} - \frac{20 \times 16}{0.16} \right]$$

$$= 5\pi \times 10^{-4} (\leftarrow)$$

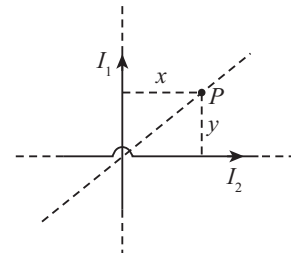
i.e., $5\pi \times 10^{-4} T$ directed towards West.

10. Field due to I_1 and I_2 will be normal to the plane of the paper, at a point like P . When charge q is projected normal to the paper it will experience no force.



11. Field at P can be zero because the field due to I_1 is directed into the plane, while the field due to I_2 is directed out of the plane of the figure.

For zero field at $P(x, y)$



$$\frac{\mu_0 I_1}{2\pi x} = \frac{\mu_0 I_2}{2\pi y} \Rightarrow y = \left(\frac{I_2}{I_1} \right) x$$

This is an equation of a straight line.

12. Let the length of the wire be L .

When bent into a circle of one turn, the radius is given by $R = \left(\frac{L}{2\pi} \right)$

$$\therefore B_0 = \frac{\mu_0 I}{2R} = \frac{\pi \mu_0 I}{L}$$

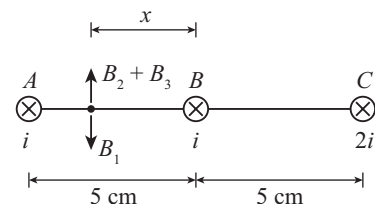
When bent into a circular coil having n turns, the radius is given by

$$r = \frac{L}{2\pi n}$$

$$\therefore B = \frac{n \mu_0 I}{2r} = \frac{\pi n^2 \mu_0 I}{L} = n^2 B_0$$

13. At a point lying between A and B , the fields created by B and C are upward and that created by A is downward. At a point that is closer to A (as compared to B) there is a possibility that the field can be zero. Let this point be at a distance x from B .

B_1 , B_2 and B_3 are fields due to A , B and C respectively. For zero field.



$$B_1 = B_2 + B_3$$

$$\Rightarrow \frac{\mu_0 i}{2\pi(5-x)} = \frac{\mu_0 i}{2\pi x} + \frac{\mu_0(2i)}{2\pi(x+5)}$$

$$\text{or, } \frac{1}{5-x} = \frac{1}{x} + \frac{2}{x+5}$$

Solving this gives $x = 3.2\text{cm}$

You are advised to argue that the magnetic field cannot be zero at a point between B and C.

$$14. F_{AC} = \frac{\mu_0(10)(10)}{2\pi(0.1)} (0.15) (\leftarrow)$$

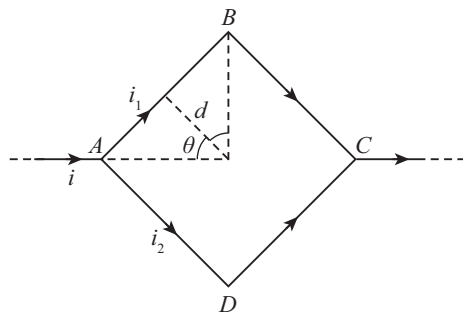
$$F_{BC} = \frac{\mu_0(20)(10)}{2\pi(0.1)} (0.15) (\rightarrow)$$

$$\therefore \text{Resultant force is } F = \frac{\mu_0}{2\pi} \frac{10 \times 10 \times 0.15}{0.1}$$

$$= 2 \times 10^{-7} \times 1.5 \times 100 = 3 \times 10^{-5} \text{ N } (\rightarrow)$$

15. The current will divide into two parallel paths in inverse ratio of the resistances

$$\frac{i_1}{i_2} = \frac{2r}{r} = \frac{2}{1}$$



$$\text{And } i_1 + i_2 = i$$

$$\therefore i_1 = \frac{2i}{3}, i_2 = \frac{i}{3}$$

$$\text{In figure, } d = \frac{a}{2} \text{ and } \theta = 45^\circ$$

$$\therefore B_{AB} = \frac{\mu_0 i_1}{4\pi \frac{a}{2}} (\sin 45^\circ + \sin 45^\circ) = \frac{\mu_0 i_1}{2\pi a} \frac{2}{\sqrt{2}} \otimes$$

$$B_{BC} = \frac{\mu_0 i_1}{2\pi a} \frac{2}{\sqrt{2}} \otimes$$

$$\text{Similarly, } B_{AD} = \frac{\mu_0 i_2}{2\pi a} \frac{2}{\sqrt{2}} \odot$$

$$B_{DC} = \frac{\mu_0 i_2}{2\pi a} \frac{2}{\sqrt{2}} \odot$$

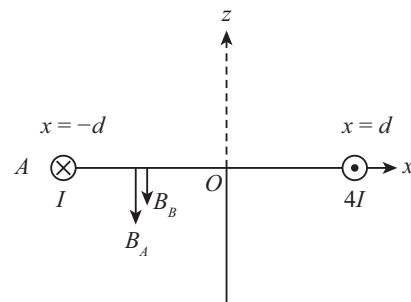
$$\therefore B = \frac{\sqrt{2}\mu_0}{\pi a} [i_1 - i_2] = \frac{\sqrt{3}}{3} \frac{\mu_0 i}{\pi a} \otimes$$

16. Each ring will create field along its own axis. Therefore, there are three fields of equal magnitude in mutually perpendicular directions.

If B_0 is field due to one ring then the resultant is

$$B = \sqrt{B_0^2 + B_0^2 + B_0^2} = \sqrt{3} B_0 = \sqrt{3} \frac{\mu_0 I}{2R}$$

17. At all points on the x -axis, between $x = -d$ and $x = d$, the magnetic field due to both the wires is in negative z direction.



Field is nowhere zero between the wires.

Field will be too high near the wires and at some point between them it will form a minima

18. Draw circular amperian loops through 1, 2 and 3 with their axis along the axis of the wire.

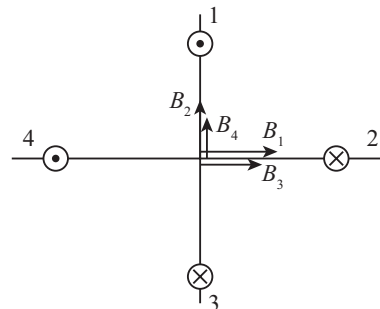
In each of the three circles there is symmetry at all points on the circumference. Using the Ampere's law in each case we will write

$$B2\pi r = \mu_0 I \Rightarrow B = \frac{\mu_0 I}{2\pi r}$$

There will be no difference in value of B in three cases.

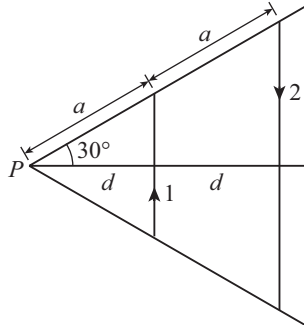
19. The directions of the four fields have been shown in the figure. Their resultant is

$$\sqrt{(2B)^2 + (2B)^2} = 2\sqrt{2} B$$



20. $d = a \cos 30^\circ = \frac{\sqrt{3}a}{2}$

Field at P due to wire 1 is



$$B_1 = \frac{\mu_0 I}{4\pi d} (\sin 30^\circ + \sin 30^\circ) \odot$$

or, $B_1 = \frac{\mu_0 I}{4\pi d} \odot$

Similarly, field due to wire 2 is

$$B_2 = \frac{\mu_0 I}{4\pi(2d)} (\sin 30^\circ + \sin 30^\circ) \otimes$$

$$= \left(\frac{\mu_0 I}{4\pi d} \right) \left(\frac{1}{2} \right) \otimes$$

$\therefore$ Resultant field at P is

$$B = \frac{\mu_0 I}{4\pi d} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \dots \right) \odot$$

$$= \frac{\mu_0 I}{4\pi d} \ln 2 = \frac{\mu_0 I}{4\pi \frac{\sqrt{3}a}{2}} \ln 2 = \frac{\mu_0 I}{4\pi \sqrt{3}a} (\ln 4) \odot$$

$$[\because \ln(1 + 1) = 1 - \frac{1}{2} + \frac{1}{3} \dots]$$

21. Both wires will produce field which are mutually perpendicular.

22. The straight wire pointing towards O produces no field at O .

$$\text{Field due to semicircle is } \vec{B}_1 = \frac{\mu_0 I}{4R} (-\hat{i})$$

$$\text{Field due to quarter circle is } \vec{B}_2 = \frac{\mu_0 I}{8R} (-\hat{k})$$

$$\text{Field due to straight wire } \vec{B}_3 = \frac{\mu_0 I}{4\pi R} (-\hat{k})$$

$$\therefore \vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 = -\frac{\mu_0 I}{4R} \hat{i} - \frac{\mu_0 I}{8\pi R} (\pi + 2) \hat{k}$$

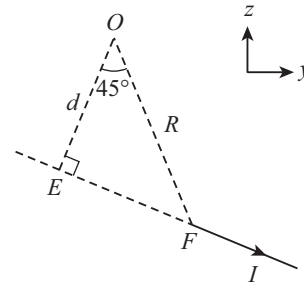
23. Field due to straight wire in xy plane is

$$\vec{B}_1 = \frac{\mu_0 I}{4\pi R} (-\hat{k})$$

$$\text{Field due to } \left(\frac{3}{4} \right) \text{th circle is } \vec{B}_2 = \frac{3}{4} \frac{\mu_0 I}{2R} (-\hat{i})$$

Field due to straight wire in yz plane is calculated as given below.

$\vec{B}_3 = (\text{Field due to a semi infinite wire starting from } E) - (\text{Field due to straight wire } EF)$



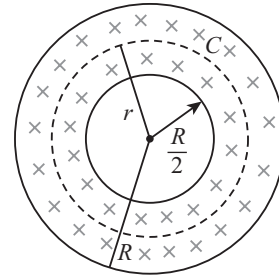
$$= \left[\frac{\mu_0 I}{4\pi d} - \frac{\mu_0 I}{4\pi d} (0 + \sin 45^\circ) \right] (\hat{i})$$

$$= \frac{\mu_0 I}{4\pi d} \left(1 - \frac{1}{\sqrt{2}} \right) \hat{i} = \frac{\mu_0 I}{4\pi \frac{R}{\sqrt{2}}} \left(1 - \frac{1}{\sqrt{2}} \right) \hat{i}$$

$$\therefore \vec{B} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3$$

$$= \left[\frac{\mu_0 I}{4\pi R} (\sqrt{2} - 1) - \frac{3\mu_0 I}{8R} \right] \hat{i} - \frac{\mu_0 I}{4\pi R} \hat{k}$$

24. Cross section is shown. It is easy to understand that the field inside the cavity is zero and the field at an outside point is



$$B = \frac{\mu_0 I}{2\pi r} \text{ where } I \text{ is the current.}$$

$$\Rightarrow B \propto \frac{1}{r}$$

For writing the field at a point at a distance r from the axis where $\frac{R}{2} < r < R$, imagine a circular Amperian loop C . By symmetry, field must be tangential and of same magnitude at all points on the circle.

$$\oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi r$$

The current passing through the loop C is

$$i = j \pi \left(r^2 - \frac{R^2}{4} \right)$$

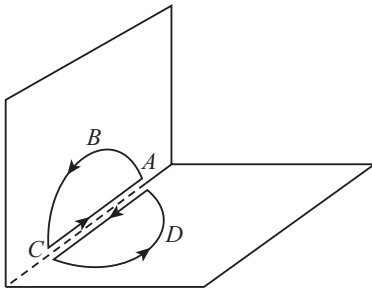
$$\therefore B \cdot 2\pi r = \mu_0 j \pi \left(r^2 - \frac{R^2}{4} \right)$$

$$\text{or, } B = \frac{\mu_0 j}{2} \left(r - \frac{R^2}{4r} \right)$$

Therefore, variation of B with r (for $\frac{R}{2} < r < R$) is non-linear. Non of the graph has this characteristic.

$$25. \oint_{ABC} \vec{B} \cdot d\vec{l} = \mu_0 (i_1 + i_3)$$

$$\oint_{CDA} \vec{B} \cdot d\vec{l} = \mu_0 (i_2 - i_3)$$



The desired integral can be obtained by adding the above two integrals. In the first integral we took a segment from C to A and in the second integral we took a segment A to C . When we add, the contribution of integrals along CA sums to zero.

$$\therefore \oint \vec{B} \cdot d\vec{l} = \mu_0 (i_1 + i_2)$$

26. Go through the solution of question 23 in worksheet 3, for detailed expressions.

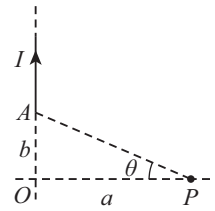
However, to recognize the correct graph, you need not write the equations.

For $r < R_1$, the field is contributed by the current in inner cylinder only and we need to recall that it will increase linearly with r . For $R_1 < r < R_2$, there is no contribution from outer current and field due to the inner cylinder falls off inversely with r .

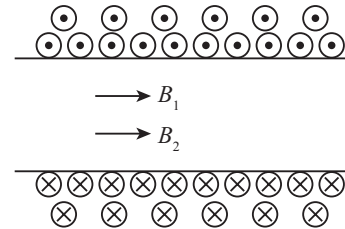
For $R_2 < r < R_3$, the field is contributed by the inner cylinder as well as the part of the current in the outer cylinder that lies in the range R_2 to r . The field continues to fall but the equation representing the variation of B has changed. Therefore, the curve in the graph must change. For $r > R_3$, net current through a circular Amperian loop is zero. Hence the field falls to zero.

27. Field at P = (field due to a semi infinite wire) – (field due to the segment OA)

$$\begin{aligned} \text{or } B &= \frac{\mu_0 I}{4\pi a} - \frac{\mu_0 I}{4\pi a} (\sin 0^\circ + \sin \theta) \\ &= \frac{\mu_0 I}{4\pi a} (1 - \sin \theta) \\ &= \frac{\mu_0 I}{4\pi a} \left(1 - \frac{b}{\sqrt{a^2 + b^2}} \right) \end{aligned}$$



28. The figure shows the two layers



$$\begin{aligned} B &= B_1 + B_2 \\ &= \mu_0 n_1 I + \mu_0 n_2 I \\ &= 4\pi \times 10^{-7} \left(\frac{350}{0.4} + \frac{250}{0.4} \right) \times 4 \\ &= 7.54 \times 10^{-3} \text{ T} \end{aligned}$$



Worksheet 2

1. Each of the rails (and even arm AB) sets a field in upward direction. The conductor experiences a force towards right ($I\vec{L} \times \vec{B}$).

$$3. \vec{B}_{BC} = \frac{\mu_0 i}{4\pi \frac{l}{2}} (\sin 0^\circ + \sin 45^\circ) \hat{j}$$

$$= \frac{\mu_0 I}{2\sqrt{2}\pi l} \hat{j}$$

$\vec{B}_{DE}$ will be in x direction, therefore option (B) is wrong.

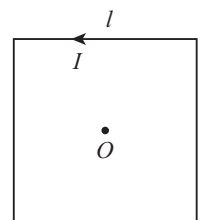
If we have a plane square loop of side length l , the field at the centre is

$$B_0 = \frac{\mu_0 I}{4\pi \frac{l}{2}} (\sin 45^\circ + \sin 45^\circ) \times 4$$

$$= \frac{\sqrt{2}\mu_0 I}{\pi l}$$

Field due to each half loop will be half the above value

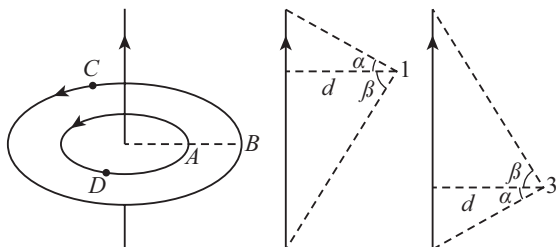
$$\therefore \vec{B} = B_0 \hat{i} + B_0 \hat{j} = \frac{\sqrt{2}\mu_0 I}{\pi l} (\hat{i} + \hat{j})$$



4. In the figure shown in option (A), the two parallel paths between X and Y are not identical. Current in them are not same. Field at the centre will not be zero.

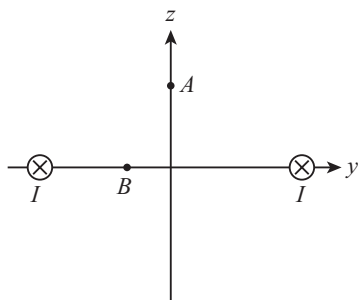
In all other cases, the currents in the two parallel paths produce opposing fields at the centre.

5.



Field at the points like A and B are in same direction but have different magnitudes. Fields at the points like A and D have same magnitude but different directions. Field at points like 1 and 3 will be same.

6. The figure shown here is an easier depiction of the situation given in the problem. One can easily show that the magnetic field at a point like A will be along y direction and the field at a point like B will be along z direction.



9. For finding the force, the circular part ADC can be replaced with a straight conductor AC . Similarly, the part ABC can also be replaced with a straight conductor AC .

$$\therefore F = BI_1 l + BI_2 l = BIl$$

where I_1 = current in part ADC

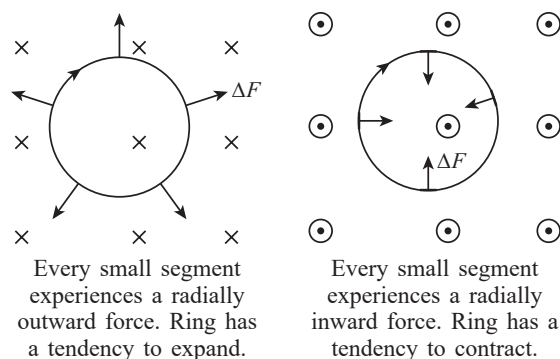
I_2 = current in part ABC

$$I_1 + I_2 = I$$

We cannot say that the field due to the loop is zero at all outside points.

10. Inside an ideal solenoid the magnetic field is uniform (B) parallel to its axis. A current loop will experience no net force in a uniform field.

When the ring is held with its plane normal to the axis, two situations can arise as shown below.



Every small segment experiences a radially outward force. Ring has a tendency to expand.

Every small segment experiences a radially inward force. Ring has a tendency to contract.

$$11. \oint \vec{B} \cdot d\vec{l} = 0$$

The above expression shall be true for any closed path inside the field.

12. The cylinder is just like an ideal solenoid. Field inside an ideal solenoid is $B = \mu_0 ni$

$$\text{Here, } ni = \lambda$$

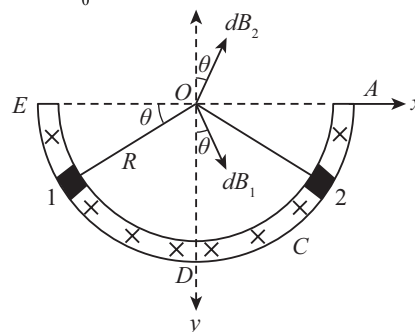
$$\therefore \text{Field inside the cylinder is } B = \mu_0 \lambda.$$

Field outside the cylinder is zero.

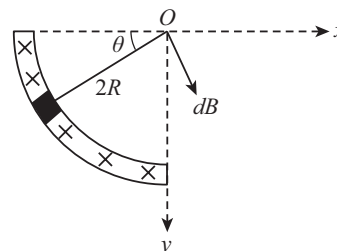
13. Consider long thin wires 1 and 2, at symmetrical locations with respect to O . Field due to 1 and 2 cancel out in y direction but add in x direction.

$$\therefore B_1 \text{ is along the extreme radius } OA.$$

$$B_1 = 2 \int_0^{\pi/2} dB \sin \theta \quad (i)$$



When the sheet is bent to produce a quarter circle section, radius will be double. Since current has been halved, the value of current through an element identical to the one considered above will be same. Since the thin wire is now at $2R$ distance from O , value of dB will be half of that in equation (i)



$$B_x = \int_0^{\pi/2} \frac{dB}{2} \sin \theta$$

$$\text{or, } B_x = \frac{B_1}{4}$$

From symmetry of the situation, B_x and B_y cannot be different

$$\therefore B_y = B_x = \frac{B_1}{4}$$

$$\therefore B = \frac{\sqrt{2}B_1}{4} = \frac{B_1}{2\sqrt{2}}$$

Obviously, B makes 45° with the x direction.

14. Every small current element experiences a radial magnetic force. The ring has a tendency to expand (or contract if the direction of field is reversed). It acquires tension. The magnetic force on one half ACB will be same as the force on a straight wire AB .

$$\begin{aligned} F_B &= IB(2r) \\ &= 100 \times 0.2 \times (2 \times 0.5) \\ &= 20\text{N}. \end{aligned}$$

This force on the half ring is balanced by the tension force at its two ends A and B .

$$\therefore 2T = 20 \Rightarrow T = 10\text{N}$$

PQR is one third of the circle.

It experiences two tension forces inclined at 60°

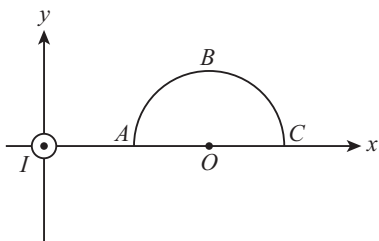
$\therefore$ Resultant force on the one third part due to the remaining two third part is

$$\sqrt{T^2 + T^2 + 2TT \cos 60^\circ} = \sqrt{3}T = 10\sqrt{3}\text{ N}.$$

15. Consider a closed amperian loop $ABCOA$. No current is passing through the area of the loop

$$\therefore \oint \vec{B} \cdot d\vec{l} = 0$$

$$\text{or, } \int_{ABC} \vec{B} \cdot d\vec{l} + \int_C^A \vec{B} \cdot d\vec{l} = 0$$



$$\text{or, } \int_{ABC} \vec{B} \cdot d\vec{l} = - \int_C^A \vec{B} \cdot d\vec{l}$$

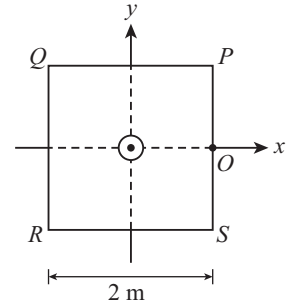
But the magnetic field is perpendicular to the x axis at all point on it.

$$\therefore \int_C^A \vec{B} \cdot d\vec{l} = 0$$

$$\therefore \int_{ABC} \vec{B} \cdot d\vec{l} = 0$$

Now, consider a square shaped Amperian loop $PQRS$. O is the centre of the given circle.

$$\oint_{RSPQR} \vec{B} \cdot d\vec{l} = \mu_0 I$$



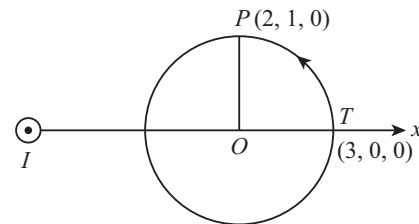
$$\text{or, } \int_R^S \vec{B} \cdot d\vec{l} + \int_S^P \vec{B} \cdot d\vec{l} + \int_P^Q \vec{B} \cdot d\vec{l} + \int_Q^R \vec{B} \cdot d\vec{l} = \mu_0 I$$

All 4 integrals must be equal.

$$\therefore \int_R^S \vec{B} \cdot d\vec{l} = \frac{\mu_0 I}{4}$$

$$\therefore \int_O^P \vec{B} \cdot d\vec{l} = \frac{1}{2} \frac{\mu_0 I}{4} = \frac{\mu_0 I}{8}$$

For the given circle.



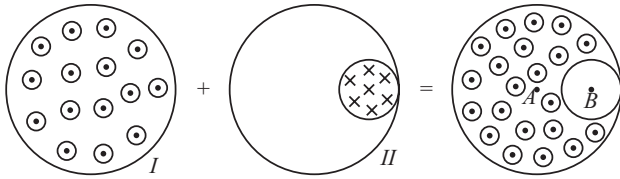
$$\oint_{OTP} \vec{B} \cdot d\vec{l} = 0$$

$$\int_O^P \vec{B} \cdot d\vec{l} + \int_P^T \vec{B} \cdot d\vec{l} + \int_T^O \vec{B} \cdot d\vec{l} = 0$$

$$\text{or, } 0 + \int_T^P \vec{B} \cdot d\vec{l} = - \int_P^O \vec{B} \cdot d\vec{l} = \int_O^P \vec{B} \cdot d\vec{l} = \frac{\mu_0 I}{8}$$

$$16. \text{ Current density } j = \frac{I}{\pi R^2 - \frac{\pi R^2}{4}} = \frac{4I}{3\pi R^2}$$

We can assume that the entire cross section has a uniform current density j directed out of the plane of the paper and the circle of radius $\frac{R}{2}$ has an additional current density j directed into the paper.



Field at A = Field due to current shown in I + Field due to current in II

$$B_A = 0 + \frac{\mu_0 I}{2\pi \frac{R}{2}}$$

Where i = current in cross section $\frac{R}{2} = j \frac{\pi R^2}{4}$

$$= \frac{4I}{3\pi R^2} \cdot \frac{\pi R^2}{4} = \frac{I}{3}$$

$$\therefore B_A = \frac{\mu_0 I / 3}{\pi R} = \frac{\mu_0 I}{3\pi R}$$

Similarly, $B_B = B_I + B_{II}$

$$= \frac{\mu_0 j}{2} \left(\frac{R}{2} \right) + 0.$$

$$= \frac{\mu_0 R}{4} \frac{4I}{3\pi R^2} = \frac{\mu_0 I}{3\pi R}$$



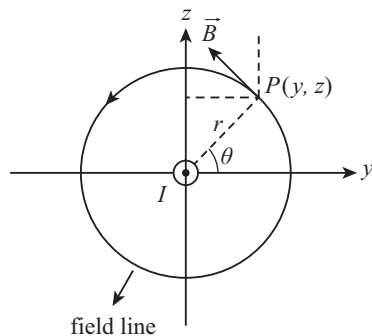
Worksheet 3

- The field lines are circles in yz plane (or a plane parallel to it).

Distance of the point (x, y, z) from the current carrying wire is

$$r = \sqrt{y^2 + z^2}$$

$\therefore$ strength of the field at the given point is



$$B = \frac{\mu_0 I}{4\pi \sqrt{y^2 + z^2}}$$

In vector form,

$$\vec{B} = -(B \sin \theta) \hat{j} + (B \cos \theta) \hat{k}$$

$$= \frac{\mu_0 I}{4\pi \sqrt{y^2 + z^2}} \left[-\frac{z}{\sqrt{y^2 + z^2}} \hat{j} + \frac{y}{\sqrt{y^2 + z^2}} \hat{k} \right]$$

$$= \frac{\mu_0 I}{4\pi (y^2 + z^2)} (-z\hat{j} + y\hat{k})$$

- The two currents create fields in opposite directions at O . The field is zero when the fields created by the two currents have same magnitude. According to the graph this happens when

$$I_2 = 1.0 \text{ A}$$

$$\therefore \frac{\mu_0 I_2}{2\pi(2R)} = \frac{\mu_0 I_1}{2R}$$

$$\text{or, } I_1 = \frac{1}{2\pi} = 0.16 \text{ A}$$

- Field at O is zero due to segments AB and DE .

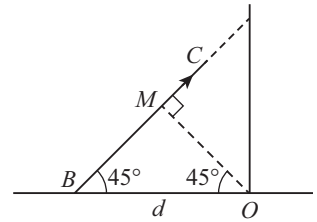
$$\text{Distance } OM = \frac{d}{\sqrt{2}}$$

Field at O due to BC is

$$B_1 = \frac{\mu_0 I}{4\pi \frac{d}{\sqrt{2}}} (\sin 45^\circ + \sin 90^\circ) \otimes$$

$$= \frac{\mu_0 I}{4\pi d} (1 + \sqrt{2}) \otimes$$

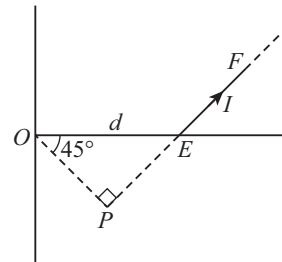
$$\text{Distance } OP = \frac{d}{\sqrt{2}}$$



[OP is perpendicular to the extended line FE]

Field due to wire EF can be written as field due to a semi infinite wire PE minus the field due to segment PE .

$$\therefore B_2 = B_{PF} - B_{PE} = \frac{\mu_0 I}{4\pi \frac{d}{\sqrt{2}}} - \frac{\mu_0 I}{4\pi \frac{d}{\sqrt{2}}} [\sin 0^\circ + \sin 45^\circ]$$



$$= \frac{\mu_0 I}{4\pi d} (\sqrt{2} - 1) \odot$$

∴ Resultant field at O is

$$B = B_1 + B_2 = \frac{\mu_0 I}{4\pi d} (1 + \sqrt{2}) \otimes + \frac{\mu_0 I}{4\pi d} (\sqrt{2} - 1) \odot$$

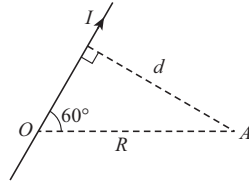
$$= \frac{\mu_0 I}{2\pi d} \otimes$$

4. (a) Points on the circumference are not at equal perpendicular distance from the wire.
 (b) Ampere's law tells us that

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

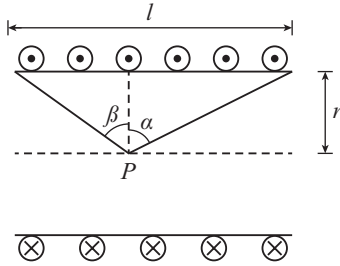
(c) $d = R \sin 60^\circ = \frac{\sqrt{3}}{2} R$

$$\therefore B_A = \frac{\mu_0 I}{2\pi d} = \frac{\mu_0 I}{\sqrt{3}\pi R}$$



5. From equation 15

$$B = \frac{\mu_0 n I}{2} (\sin \alpha + \sin \beta)$$



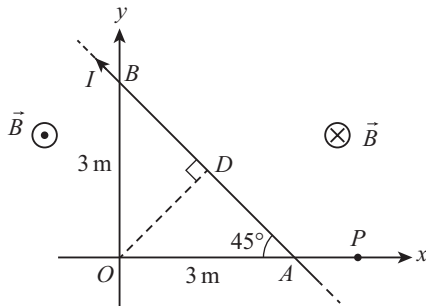
when $r \gg l$, both α and β will be small angles.

$$\sin \alpha = \tan \alpha = \frac{l}{2r}$$

Since point P is at the centre, $\alpha = \beta$

$$\therefore B \simeq \frac{\mu_0 n I}{2} \left(\frac{l}{2r} + \frac{l}{2r} \right) = \frac{\mu_0 n I l}{2r}$$

6. The current carrying wire is as shown in the figure.



- (a) To the left of the wire the field is directed out of the plane of the figure.

Distance of the point O from the wire is

$$d = OD = \frac{3}{\sqrt{2}} \text{ m.}$$

$$\therefore \vec{B} = \left(\frac{\mu_0 I}{2\pi d} \right) \hat{k} = \left(2 \times 10^{-7} \times \frac{10}{\frac{3}{\sqrt{2}}} \right) \hat{k}$$

$$= (9.4 \times 10^{-7} \text{ T}) \hat{k} = (0.94 \mu\text{T}) \hat{k}$$

- (b) Point P has co-ordinates $(6, 0, 0)$.

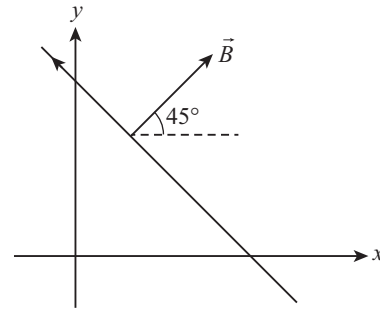
Distance $AP = 3 \text{ m}$.

Again the perpendicular distance of the point P from the wire is $d = \frac{3}{\sqrt{2}} \text{ m}$.

$$\vec{B} = - (9.4 \times 10^{-7} \text{ T}) \hat{k} = - (0.94 \mu\text{T}) \hat{k}$$

Field at P is directed into the plane of the figure.

- (c) Point $Q(1, 2, 2)$ is exactly above the wire (in the figure) at a distance of 2 m. To understand this, note that the point $(1, 2, 0)$ lies on the given line. The direction of the field at Q is along the tangent on a circle around the wire. This direction is parallel to the xy plane and perpendicular to the wire. The figure on the right shows the direction.

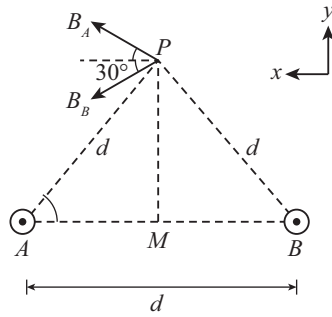


$$B = \frac{\mu_0 I}{2\pi d} = 2 \times 10^{-7} \times \frac{10}{2} = 10^{-6} \text{ T.}$$

$$\therefore \vec{B} = 10^{-6} \left(\frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{j} \right) \text{ T}$$

$$= \frac{1}{\sqrt{2}} (\hat{i} + \hat{j}) \mu\text{T.}$$

7. (a) The situation is shown in the figure. A and B are cross sections of the two wires carrying current out of the plane of the figure.



Field due to the wire A is B_A , directed perpendicular to AP. Field at P, due to the wire B is B_B , directed perpendicular to BP.

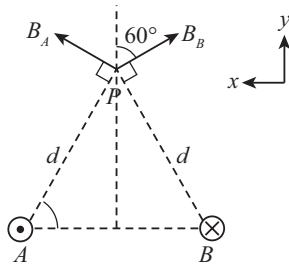
$$B_A = B_B = \frac{\mu_0 I}{2\pi d}$$

Resultant of the two fields is along x direction (see figure)

$$B = B_A \cos 30^\circ + B_B \cos 30^\circ = \frac{\sqrt{3}\mu_0 I}{2\pi d}$$

$$= \sqrt{3} \times 2 \times 10^{-7} \times \frac{10}{0.1} = 2\sqrt{3} \times 10^{-5} \text{ T}$$

- (b) Currents in the two wires are directed as shown. Direction of B_B reverses when the current in the wire B is reversed.



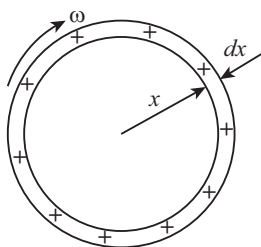
Resultant field is along y direction

$$B = B_A \cos 60^\circ + B_B \cos 60^\circ$$

$$= \frac{\mu_0 I}{2\pi d} = 2 \times 10^{-7} \times \frac{10}{0.1}$$

$$= 2 \times 10^{-5} \text{ T.}$$

8. Consider a ring of radius x and width dx . Charge on the ring is $dq = \sigma \cdot 2\pi x dx$.



$$\text{Frequency of rotation is } f = \frac{\omega}{2\pi}.$$

Current associated with the thin ring of radius x is

$$dI = (dq)f = \sigma \omega x dx$$

Magnetic field at the centre due to this current is

$$dB = \frac{\mu_0 (dI)}{2x} = \frac{\mu_0 \sigma \omega}{2} dx$$

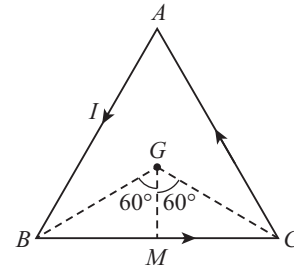
Adding contributions made by all such rings, gives

$$B = \int dB = \frac{\mu_0 \sigma \omega}{2} \int_a^b dx = \frac{\mu_0 \sigma \omega}{2} (b - a)$$

The field is normal to the plane of the ring.

9. Distance of the centroid (G) from the wire BC is

$$GM = \frac{L}{2} \cot 60^\circ = \frac{L}{2\sqrt{3}}$$



Magnetic field at G due to BC is

$$B = \frac{\mu_0 I}{4\pi \left(\frac{L}{2\sqrt{3}} \right)} (\sin 60^\circ + \sin 60^\circ)$$

$$= \frac{3\mu_0 I}{2\pi L} \odot$$

All three sides contribute same field in same direction. Hence, the resultant field is

$$B_0 = 3B = \frac{9\mu_0 I}{2\pi L} \odot$$

10. Field at a distance x on the axis of C_1 is

$$\vec{B}_1 = \frac{\mu_0 I_1}{2} \frac{R^2}{(R^2 + x^2)^{3/2}} (\hat{i})$$

The field is directed along the axis. This field can be treated to be nearly same at all points in the loop C_2 because it has a small size.

Magnetic dipole of the loop C_2 is $\vec{M} = (\pi r^2 I_2) \hat{k}$

$\therefore$ Torque on the loop due to field $\vec{B}_1$ is

$$\vec{\tau} = \vec{M} \times \vec{B}$$

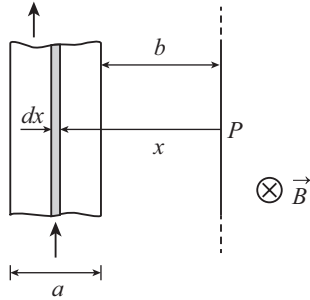
$$\vec{\tau} = \pi r^2 I_2 = \frac{\mu_0 I_1}{2} \frac{R^2}{(R^2 + x^2)^{3/2}} = \frac{\mu_0 \pi I_1 I_2 R^2 r^2}{2(R^2 + x^2)^{3/2}}$$

11. Consider a thin strip of width dx as shown. Current in it is

$$dI_0 = \frac{I_0}{a} dx$$

Field due to this small current at a point P is

$$dB_0 = \frac{\mu_0(dI_0)}{2\pi x} = \frac{\mu_0 I_0}{2\pi a} \frac{dx}{x}$$



$\therefore$ Field at P due to the entire strip is

$$B_0 = \frac{\mu_0 I_0}{2\pi a} \int_b^{b+a} \frac{dx}{x} = \frac{\mu_0 I_0}{2\pi a} \ln \left(\frac{b+a}{b} \right) \otimes$$

This is the field at every point on the wire carrying current I .

$\therefore$ Force on length L of the wire is

$$F = ILB_0 = \frac{\mu_0 I_0 IL}{2\pi a} \ln \left(1 + \frac{a}{b} \right)$$

The force is directed towards left perpendicular to the wire.

12. (a) Field at the origin is obtained by adding the contributions made by the two semicircles.

$$\vec{B} = \frac{\mu_0 I}{4R} (-\hat{i}) + \frac{\mu_0 I}{4R} (\hat{j}) = \frac{\mu_0 I}{4R} (-\hat{i} + \hat{j})$$

Force on the charge is

$$\begin{aligned} \vec{F} &= q \vec{v} \times \vec{B} = q (-v_0 \hat{i}) \times \frac{\mu_0 I}{4R} (-\hat{i} + \hat{j}) \\ &= - \frac{\mu_0 q v_0 I}{4R} \hat{k} \end{aligned}$$

(b) Force on each of the semicircles can be calculated by replacing them with straight wire KM .

$$\therefore \text{Force on the loop} = \vec{F}_{KM} + \vec{F}_{KM}$$

$$= [BI(2R)(-\hat{k}) \times B_0 \hat{j}] \times 2$$

$$= (4BIR) \hat{i}$$

13. Consider a segment of length dx in the copper wire at D . The magnetic field at D due to the current I_2 is

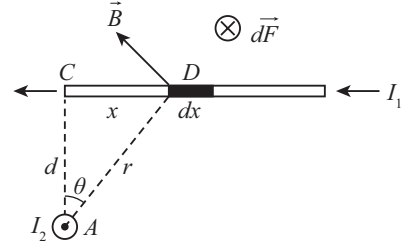
$$B = \frac{\mu_0 I_2}{2\pi r} = \frac{\mu_0 I_2}{2\pi \sqrt{d^2 + x^2}}$$

The field is perpendicular to AD as shown.

Force on the segment is

$$dF = I_1 dx \cdot B \cdot \sin \theta$$

$$= \left(\frac{\mu_0 I_1 I_2}{2\pi \sqrt{d^2 + x^2}} dx \right) \left(\frac{x}{\sqrt{d^2 + x^2}} \right)$$



This force is directed perpendicular to the plane of the figure.

- $\therefore$ Forces on every segment of the copper wire is directed into the plane of the figure. Therefore, the resultant force is

$$\begin{aligned} F &= \int dF \\ &= \frac{\mu_0 I_1 I_2}{2\pi} \int_{x=0}^L \frac{xdx}{d^2 + x^2} = \frac{\mu_0 I_1 I_2}{4\pi} \int_0^L \frac{2xdx}{d^2 + x^2} \\ &= \frac{\mu_0 I_1 I_2}{4\pi} [\ln(d^2 + x^2)]_0^L = \frac{\mu_0 I_1 I_2}{4\pi} \ln \left(\frac{d^2 + x^2}{d^2} \right) \end{aligned}$$

14. (a) Field at R due to both the wires are directed ($\downarrow$) perpendicular to the x axis.

$\therefore$ Net field at R is

$$\begin{aligned} B &= B_P + B_Q \\ &= \frac{\mu_0}{2\pi} \left(\frac{2.5}{5} \right) + \frac{\mu_0}{2\pi} \left(\frac{I}{2} \right) = \frac{\mu_0}{4\pi} (1 + I) \end{aligned}$$

Force on the electron, $F = eBv \sin 90^\circ$

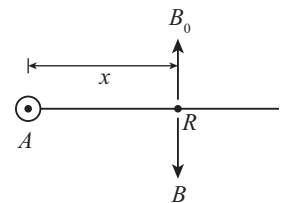
$$\text{or, } 3.2 \times 10^{-20} = 1.6 \times 10^{-19} (10^{-7}) (1 + I) (4 \times 10^5)$$

or, $I = 4\text{A}$

(b) Field at R due to the wires at P and Q is

$$B = \frac{\mu_0}{4\pi} (I + 1) = \frac{5\mu_0}{4\pi}$$

To neutralise this field one can place a wire at A having current I_0 directed out of the plane of the figure. This current produces a field B_0 that is directed opposite to B .



$$\frac{\mu_0 I_0}{2\pi x} = \frac{5\mu_0}{4\pi}$$

It is given that $I_0 = 2.5$

$$\therefore x = 1\text{m}$$

The third wire can also be placed at C having current in a direction into the plane of the paper.

15. Field due to segments BC and EF are zero.

Field due to AB and FA are directed into the plane of the figure. Their resultant is

$$B_1 = 2 \left[\frac{\mu_0}{4\pi} \frac{I}{2a} (\sin 0^\circ + \sin 45^\circ) \right] = \frac{\mu_0 I}{4\sqrt{2}\pi a} \otimes$$

Field due to CD and DE are directed out of the plane of the figure. Their resultant is

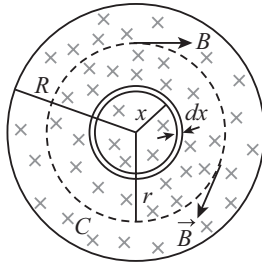
$$B_2 = 2 \left[\frac{\mu_0}{4\pi} \frac{I}{a} (\sin 0^\circ + \sin 45^\circ) \right] = \frac{\mu_0 I}{2\sqrt{2}\pi a} \odot$$

$\therefore$ Resultant field at P is

$$B = B_2 - B_1 = \frac{\mu_0 I}{4\sqrt{2}\pi a} \odot$$

16. (a) The figure shows the cross section of the conductor. The current density is directed into the plane of the figure.

A circle of radius r (represented by C) is our Amperian loop. The magnetic field will have same strength at all the points on this circle and will be tangential.



$$\oint_C \vec{B} \cdot d\vec{t} = B \cdot 2\pi r$$

From Ampere's law $B \cdot 2\pi r = \mu_0 i$ (i)

Where i is the total current piercing through the circle of radius r . For finding this current, consider a circular ring of radius x and width dx . Current through the width dx of the ring is

$$di = j \cdot 2\pi x \, dx = 2\pi a x^2 \, dx$$

Total current through the circle of radius r is

$$i = 2\pi a \int_0^r x^2 \, dx = \frac{2\pi a r^3}{3} \quad \text{(ii)}$$

Putting this in (i) gives

$$B \cdot 2\pi r = \mu_0 \frac{2\pi a r^3}{3} \Rightarrow B = \frac{\mu_0 a r^2}{3}$$

- (b) The total current in the wire can be found by changing the upper limit of integration to R in equation (ii).

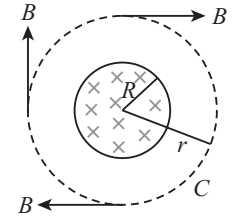
$$I = \frac{2\pi a R^3}{3}$$

Again using the Ampere's law for the circular loop (C) shown in the figure:

$$B \cdot 2\pi r = \mu_0 I$$

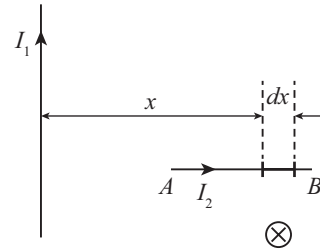
$$\text{or, } B \cdot 2\pi r = \mu_0 \frac{2\pi a R^3}{3}$$

$$\text{or, } B = \frac{\mu_0}{3} \frac{a R^3}{r}$$



17. Consider a segment of length dx on the side AB , at a distance x from the straight wire. Field due to I_1

$$\text{at this location is } B_1 = \frac{\mu_0 I_1}{2\pi x} \otimes$$



Force on the segment is $dF = (I_2 dx)(B_1)$

$$= \frac{\mu_0 I_1 I_2}{2\pi} \frac{dx}{x}$$

This force is perpendicular to AB parallel to I_1 . Total force on the side is

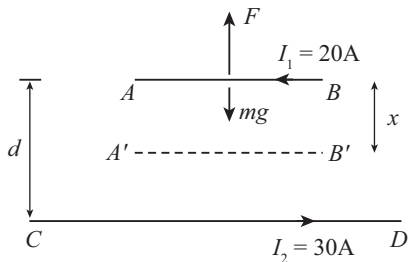
$$F = \int dF = \frac{\mu_0 I_1 I_2}{2\pi} \int_a^{2a} \frac{dx}{x} = \frac{\mu_0 I_1 I_2}{2\pi} \ln(2).$$

Forces on the sides AB and CD act along the same line. They do not produce torque. Similarly, the forces on AD and BC act along the same line and produce no torque.

18. Consider a length L of the wire AB . Let the mass of the segment be m . In equilibrium position its weight is balanced by the repulsion due to current I_2 .

$$\therefore \frac{\mu_0 I_1 I_2 L}{2\pi d} = mg \quad \text{(i)}$$

When the wire AB is in a position x below its equilibrium position, the magnetic repulsion is higher than mg . This creates a restoring force pushing the wire back to its equilibrium.



$$m \frac{d^2 x}{dt^2} = mg - \frac{\mu_0 I_1 I_2}{2\pi d(d-x)} L$$

$$= mg - \frac{\mu_0 I_1 I_2 L}{2\pi d} \left(1 - \frac{x}{d}\right)^{-1}$$

For $\frac{x}{d} \ll 1$, $\left(1 - \frac{x}{d}\right)^{-1} \simeq 1 + \frac{x}{d}$

$$\therefore m \frac{d^2 x}{dt^2} = mg - \frac{\mu_0 I_1 I_2 L}{2\pi d} - \frac{\mu_0 I_1 I_2 L}{2\pi d} \frac{x}{d}$$

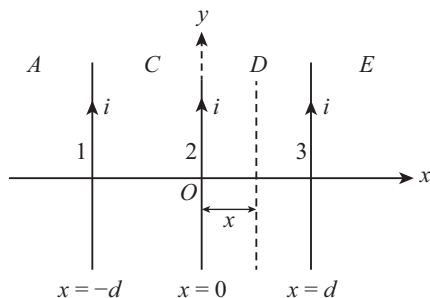
using (i)

$$m \frac{d^2 x}{dt^2} = -\left(\frac{mg}{d}\right)x$$

$$\Rightarrow \frac{d^2 x}{dt^2} = -\left(\frac{g}{d}\right)x \quad [\text{SHM}]$$

Time period is $T = 2\pi \sqrt{\frac{d}{g}} = 2\pi \sqrt{\frac{0.01}{9.8}} \simeq 0.2\text{s}$

19. The field cannot be zero in the region A. field contributed by all the three wires will be directed out of the plane of the paper in the region A. For similar reason, the field cannot be zero in the region E. It can be zero at points in the region C and D.



Let the field be zero on the dotted line shown in the figure.

$$B_1 + B_2 = B_3$$

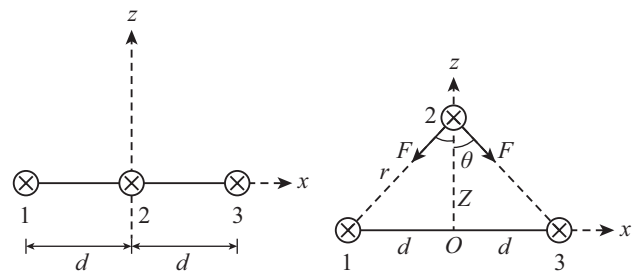
$$\text{or, } \frac{\mu_0}{2\pi} \frac{i}{(d+x)} + \frac{\mu_0}{2\pi} \frac{i}{x} = \frac{\mu_0}{2\pi} \frac{i}{(d-x)}$$

$$\text{or, } \frac{i}{d+x} + \frac{1}{x} = \frac{1}{d-x}$$

$$\Rightarrow x = \pm \frac{d}{\sqrt{3}}$$

The lines $x = + \frac{d}{\sqrt{3}}$ and $x = - \frac{d}{\sqrt{3}}$ are two lines having zero field.

20. The figure shows the cross sections of the wires in the xz plane. If wire 2 is displaced along z direction, it feels the attraction of the other two wires. The resultant force is in negative z direction. This is the restoring force causing oscillations. Resultant force on a length L of wire 2 is.



$$F_0 = 2 F \cos \theta = 2 \left(\frac{\mu_0 i^2 L}{2\pi r} \right) \left(\frac{z}{r} \right)$$

$$= \frac{\mu_0}{\pi} \frac{i^2 L z}{(d^2 + z^2)}$$

If $z \ll d$, then $d^2 + z^2 \simeq d^2$.

$$F_0 \simeq \frac{\mu_0}{\pi} \frac{i^2 L z}{d^2}$$

For the motion of wire 2 we can write

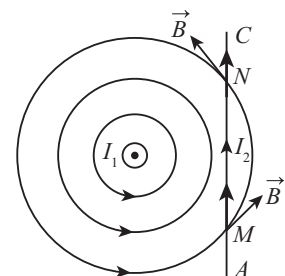
$$\lambda L \frac{d^2 z}{dt^2} = - \left(\frac{\mu_0 i^2 L}{\pi d^2} \right) z$$

$$\text{or, } \frac{d^2 z}{dt^2} = - \left(\frac{\mu_0 i^2}{\pi \lambda d^2} \right) z$$

Hence the motion is SHM.

$$f = \frac{1}{2\pi} \sqrt{\frac{\mu_0 i^2}{\pi \lambda d^2}} = \frac{i}{2\pi d} \sqrt{\frac{\mu_0}{\pi \lambda}}$$

21. The figure shows the circular field lines around I_1 . If we consider a small element in the wire AC at M, it will experience force in the direction into the plane of the paper. An identical element at N



will experience same force but in a direction out of the plane of the paper. The net force on AC will be zero.

22. Take hint from question 8 in this worksheet.

23. (a) Field at P due to circular arc is

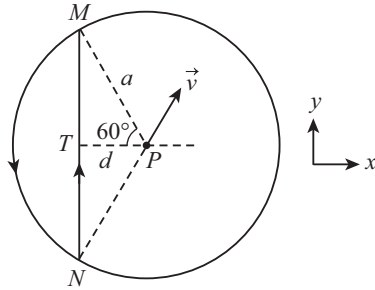
$$\vec{B}_1 = \frac{1}{3} \frac{\mu_0 I}{2a} \hat{k} = \frac{\mu_0 I}{6a} \hat{k}$$

Field at P due to the straight wire NM is

$$\vec{B}_2 = \frac{\mu_0 I}{4\pi d} (\sin 60^\circ + \sin 60^\circ)(-\hat{k})$$

$$\text{Here } d = PT = a \cos 60^\circ = \frac{a}{2}$$

$$\begin{aligned} \therefore \vec{B}_2 &= \frac{\mu_0 I}{4\pi \frac{a}{2}} \sqrt{3} (-\hat{k}) \\ &= \frac{\sqrt{3}\mu_0 I}{2\pi a} (-\hat{k}) \end{aligned}$$



$\therefore$ Resultant field at P is

$$\begin{aligned} \vec{B} &= \vec{B}_1 + \vec{B}_2 = \frac{\mu_0 I}{2a} \left(\frac{\sqrt{3}}{\pi} - \frac{1}{3} \right) (-\hat{k}) \\ &= -\frac{0.11\mu_0 I}{a} \hat{k} \end{aligned}$$

Velocity of the particle is

$$\begin{aligned} \vec{v} &= (v \cos 60^\circ) \hat{i} + (v \sin 60^\circ) \hat{j} \\ &= \frac{v}{2} \hat{i} + \frac{\sqrt{3}}{2} v \hat{j} \end{aligned}$$

Force on the particle is $\vec{F} = Q (\vec{v} \times \vec{B})$

$$\text{or, } \vec{F} = \frac{0.11\mu_0 IQv}{2a} (\hat{j} - \sqrt{3}\hat{i})$$

$$\therefore \text{Acceleration, } \vec{a} = \frac{\vec{F}}{m} = \frac{0.11\mu_0 IQv}{2am} (\hat{j} - \sqrt{3}\hat{i})$$

(b) Magnetic dipole moment of the loop is

$$\vec{M} = (IA) \hat{k}, \text{ where the area (A) is given by}$$

$A = \text{Area of sector } PMN - \text{area of triangle } PMN$

$$= \frac{1}{3} \pi a^2 - \frac{1}{2} (2 \cdot a \sin 60^\circ) (a \cos 60^\circ) \simeq 0.6a^2$$

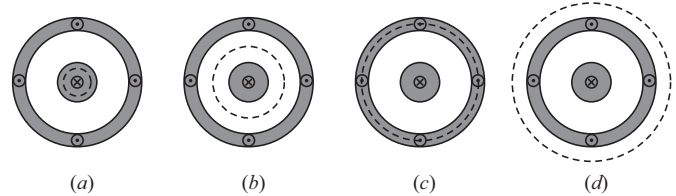
$$\therefore \vec{M} = (0.6 Ia^2) \hat{k}$$

$$\therefore \vec{\tau} = \vec{M} \times \vec{B} = 0.6 Ia^2 B (\hat{k} \times \hat{i}) = 0.6 Ia^2 B \hat{j}$$

24. For solving each part of the problem, we will consider a circular Amperian loop. The figures given below show the loops for the four parts of the problem. By symmetry, the magnetic field at each point of the circular loop must be of same magnitude and tangential in direction. The line integral of B can be written as

$$\oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi x$$

This is true for all four circular loops.



The dotted circle is the Amperian loop. Its radius is x .

(a) current enclosed by the circular loop is

$$i = \frac{I}{\pi a^2} \cdot \pi x^2 = \frac{Ix^2}{a^2}$$

Using Ampere's law gives, $\oint \vec{B} \cdot d\vec{l} = \mu_0 i$

$$\Rightarrow B \cdot 2\pi x = \frac{\mu_0 Ix^2}{a^2}$$

$$\Rightarrow B = \frac{\mu_0 Ix}{2\pi a^2}$$

(b) Current enclosed by the loop is I

$$\therefore B \cdot 2\pi x = \mu_0 I$$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi x}$$

(c) Current density in the outer shell is,

$$j = \frac{I}{\pi(c^2 - b^2)}$$

The area of cross section of the outer shell that falls inside our Amperian loop is $\pi(x^2 - b^2)$.

$\therefore$ Current in outer shell, inside our Amperian loop is

$$j \pi (x^2 - b^2) = \frac{I(x^2 - b^2)}{(c^2 - b^2)}$$

This is in opposite direction of the current I in the inner wire.

$\therefore$ Net current inside the circle is

$$i = I - \frac{I(x^2 - b^2)}{(c^2 - b^2)} = \frac{I(c^2 - x^2)}{(c^2 - b^2)}$$

$$\therefore B \cdot 2\pi x = \frac{\mu_0 I(c^2 - x^2)}{(c^2 - b^2)}$$

$$\text{or, } B = \frac{\mu_0 I(c^2 - x^2)}{2\pi x(c^2 - b^2)}$$

(d) Net current enclosed by the Amperian loop is zero.

$\therefore B = 0$.

25. (a) The straight, radial parts like CD do not produce any field at O .

$$B_{\text{inner arcs}} = 4 \times \left[\frac{\mu_0 I \frac{1}{8}}{2r_1} \right] = \frac{\mu_0 I}{4r_1} \odot$$

[$\because$ Each arc is $\frac{1}{8}$ of a complete circle and there are 4 such arcs]

$$\text{Similarly, } B_{\text{outer arcs}} = 4 \times \left[\frac{\mu_0 I \frac{1}{8}}{2r_2} \right] = \frac{\mu_0 I}{4r_2} \odot$$

$$\therefore B = \frac{\mu_0 I}{4} \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \odot$$

Substituting the values gives $B = 6.54 \times 10^{-5} \text{ T } \odot$

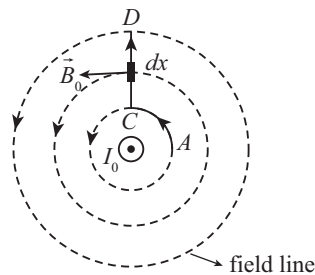
(b) The field due to the current in the loop is directed outward at O . The straight wire carries current in the same direction. Therefore, it experiences no force.

The magnetic field lines due to the straight wire are circular (see figure). An arc like AC falls on one of these circular field lines. Every segment on AC has current in the direction of the field. It will experience no force.

$$\therefore F_{AC} = 0$$

To find force on CD , consider a small segment of length dx on the arm CD . Magnetic field (due to I_0) at the location of this element is

$B_0 = \frac{\mu_0 I_0}{2\pi x}$ where x is the radial distance of the element from the centre.



This field is perpendicular to $I \overrightarrow{dx}$. Therefore, the force on the element is

$$dF = B_0 (I dx) \sin 90^\circ = \frac{\mu_0 I_0 I}{2\pi} \left(\frac{dx}{x} \right)$$

The force is directed along $I \overrightarrow{dx} \times \vec{B}_0$. This direction is out of the plane of the paper.

$$\begin{aligned} \therefore F_{CD} &= \frac{\mu_0 I_0 I}{2\pi} \int_{r_1}^{r_2} \frac{dx}{x} = \frac{\mu_0 I_0 I}{2\pi} \ln \left(\frac{r_2}{r_1} \right) \\ &= \frac{\mu_0 I_0 I}{2\pi} \ln (1.5) \odot \end{aligned}$$

Substituting the values gives $F_{CD} = 8.2 \times 10^{-6} \text{ N } \odot$

26. $\vec{B}_1$ = Field due to current in the entire cross section. (i.e., due to $I_1 = j \pi a^2$)

$\vec{B}_2$ = Field due to current directed outward in the cavity region (i.e., due to current $I_2 = j \pi \left(\frac{a}{2} \right)^2$)

Resultant field at any point is $\vec{B} = \vec{B}_1 + \vec{B}_2$

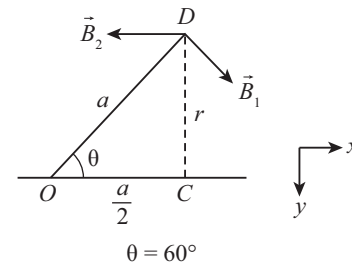
(a) For point A:

$$\begin{aligned} \vec{B}_A &= \vec{B}_{1A} + \vec{B}_{2A} = \left(\frac{\mu_0 I_1}{2\pi a} \right) \hat{j} - \frac{\mu_0 I_2}{2\pi \left(\frac{3a}{2} \right)} \hat{j} \\ &= \frac{\mu_0}{2\pi a} \left[j \pi a^2 - \frac{2}{3} j \pi \frac{a^2}{4} \right] \hat{j} \end{aligned}$$

[Please avoid getting confused between the current density j and the unit vector $\hat{j}$]

$$\vec{B}_A = \left(\frac{5}{12} \mu_0 j a \right) \hat{j}$$

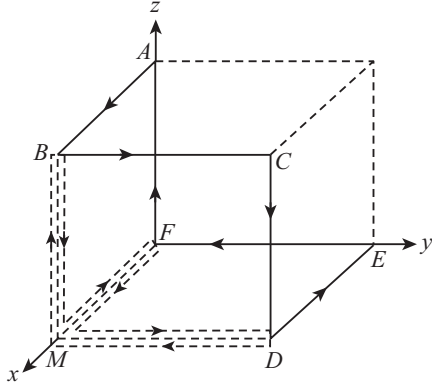
- (b) At D , the field due to I_1 is tangential, i.e., perpendicular to OD . Field due to I_2 is perpendicular to CD .



Therefore, y component of the field at $D = y$ component of $\vec{B}_1$

$$\begin{aligned} \therefore B_{yD} &= B_1 \cos \theta = B_1 \cos 60^\circ = \frac{B_1}{2} \\ &= \frac{1}{2} \left(\frac{\mu_0 j a}{2} \right) = \frac{\mu_0 j a}{4} \end{aligned}$$

27. In figure (1) the field, at the centre must be along y direction. This follows from the symmetry.



Consider two currents along BM – from B to M and another from M to B . Similarly, consider equal and opposite currents along MF and MD .

This gives us three complete square loops,

Field at the centre due to the loop $ABMFA$ is $B_0 \hat{j}$.

Field at the centre due to the loop $FMDEF$ is $B_0 \hat{k}$. Field at the centre due to the loop $BCDMB$ is $-B_0 \hat{i}$

$$\therefore \vec{B} = -B_0 \hat{i} + B_0 \hat{j} + B_0 \hat{k}$$

$$28. V = \frac{1}{4\pi\epsilon_0} \sum \frac{\Delta q}{r} \quad \dots(i)$$

Here r is distance of a small charge (Δq) from the origin. When this Δq charge rotates with angular speed ω , the current associated with its motion is $\Delta i = \Delta q \frac{\omega}{2\pi}$. This current produces a magnetic field at O given by

$$\Delta B = \frac{\mu_0 \Delta i}{2r} = \frac{\mu_0 \omega \Delta q}{4\pi r}$$

The current is in xy plane and the field (ΔB) is along z direction. The resultant field is obtained by adding fields due to all small charges.

$$\begin{aligned} B &= \sum \Delta B = \frac{\mu_0 \omega}{4\pi} \sum \frac{\Delta q}{r} \\ &= \frac{\mu_0 \omega}{4\pi} 4\pi\epsilon_0 V = \mu_0 \epsilon_0 \omega V. \end{aligned}$$

Chapter 3 Miscellaneous Problems on Chapters 1 and 2



Match the Columns

1. (A) If the g and B are in same direction and the velocity of the particle is also in the same direction, the particle will experience only gravitational force and move in a straight line with acceleration.
If a particle is moving horizontally, we can apply a horizontal $\vec{B}$ perpendicular to its velocity so that the magnetic force can balance $m\vec{g}$. Therefore, it can move unaccelerated on a straight line.
When $\vec{B}$ and $\vec{g}$ are parallel, we can have a helical path with changing pitch.
- (B) Uniform $\vec{E}$ and uniform $\vec{g}$ means that the force on the particle is constant. In such case, the path can be a straight line or a parabola. When force is zero, the particle can have uniform motion.
- (C) The electric field due to a point charge is radial. A particle can move along a straight line with variable speed. It can also move in a circle like an electron in a hydrogen atom.
- (D) The particle can move in the directions of the fields with an acceleration $a = \frac{qE}{m}$.
It can move in a helical path with variable pitch.

2. (A) $B = \frac{\mu_0 I}{2\pi x} \Rightarrow B \propto \frac{1}{x}$

(B) $B = \frac{\mu_0}{2} \frac{IR^2}{(R^2 + x^2)^{3/2}}$

The field is maximum at $x = 0$ and decreases with x .

(C) $B = B_1 \cos \theta + B_2 \cos \theta$

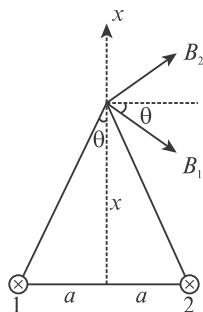
$$= 2 \cdot \frac{\mu_0 I}{2\pi \sqrt{a^2 + x^2}} \cdot \frac{x}{\sqrt{a^2 + x^2}}$$

$$= \frac{\mu_0 Ix}{\pi(a^2 + x^2)}$$

At $x = 0$, $B = 0$

when $x \rightarrow \infty$, $B \rightarrow 0$

Appropriate graph is (P)

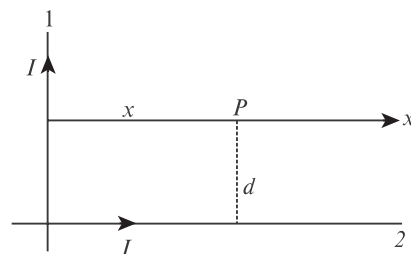


- (D) Field at P due to wire 1 is

$$B_1 = \frac{\mu_0 I}{2\pi x} \otimes$$

Field at P due to wire 2 is

$$B_2 = \frac{\mu_0 I}{2\pi d} \odot$$



Resultant field is zero at $x = d$

for $x < d$, the field is $(B_1 - B_2)$ directed $\otimes$. For $x > d$, B_2 is larger in magnitude and the field is

$$B = \frac{\mu_0 I}{2\pi} \left(\frac{1}{d} - \frac{1}{x} \right) \odot$$

For $x \rightarrow \infty$; $B = \frac{\mu_0 I}{2\pi d} \odot$

Graph (R) is appropriate.

3. (A) Field at O due to two straight segments is

$$B_1 = \frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{4\pi r} = \frac{\mu_0 I}{2\pi r} \odot$$

Field due to the circular part is greater than that due to a semicircular arc but less than that due to a complete circle.

$$\therefore \frac{\mu_0 I}{4\pi r} < B_2 < \frac{\mu_0 I}{2\pi r} \odot$$

$\therefore$ Resultant field is

$$\left(\frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{4r} \right) < B < \left(\frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{2r} \right)$$

or, $\frac{\mu_0 I}{2r} \left(\frac{1}{\pi} + \frac{1}{2} \right) < B < \frac{\mu_0 I}{2r} \left(\frac{1}{\pi} + 1 \right)$

or, $0.4 \frac{\mu_0 I}{r} < B < 0.65 \frac{\mu_0 I}{r}$

(B) Due to straight wires $B_1 = \frac{\mu_0 I}{2\pi r} = 0.15 \frac{\mu_0 I}{r}$

Due to circular arc $B_2 < \frac{\mu_0 I}{4r} = 0.25 \frac{\mu_0 I}{r}$

$\therefore 0.15 \frac{\mu_0 I}{r} < B < 0.45 \frac{\mu_0 I}{r}$

(C) Due to straight wire $\vec{B}_1 = \frac{\mu_0 I}{4\pi r} (-\hat{k})$

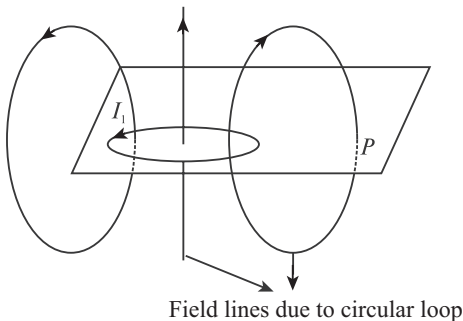
Due to half circle $\vec{B}_2 = \frac{\mu_0 I}{4r} (-\hat{i})$

$\therefore B = \frac{\mu_0 I}{4r} \sqrt{\frac{1}{\pi^2} + 1} = \frac{\mu_0 I}{r} \frac{\sqrt{1.09}}{4} = 0.26 \frac{\mu_0 I}{r}$

(D) $B = \frac{\mu_0 I}{2\pi r} + \frac{\mu_0 I}{4r} = \frac{\mu_0 I}{r} \left[\frac{1}{2\pi} + \frac{1}{4} \right]$
 $= 0.4 \frac{\mu_0 I}{r}$

4. (A) Fields due to the straight wire are circular. The circular loop lies on one such field line. Therefore, the loop does not experience any force.

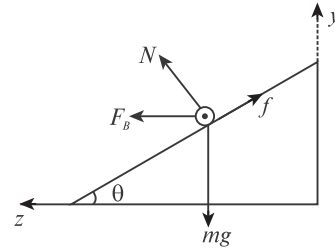
The circular loop will produce a field (at P) that is parallel to the straight wire. To visualise this, you can draw few field lines due to the circular loop.



Due to the straight wire, field is in the plane of the circle, at P . Therefore, the two fields are perpendicular.

- (C) Force due to I_1 on arm AB is repulsive and force due to I_1 on arm CD is attractive. But force on AB is higher. Therefore, the result is repulsion.
- (D) The magnetic fields (at P) due to the two loops are along their respective axes. The fields are perpendicular. One loop will experience torque in the field of the other loop.

5. (A) The figure shows the cross-section of the wire. The current is out of the plane of the figure in positive x direction.

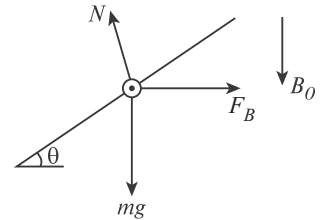


When magnetic field is along positive y direction, force on the conductor is in z direction.

The conductor cannot be in equilibrium. The angle θ is larger than angle of repose. The component of F_B down the incline further helps the conductor to move. Moreover, F_B has a component along the normal force N and this decreases the value of N also. This weakens the friction.

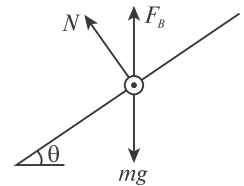
- (B) When $\vec{B} = -B_0 \hat{j}$,

the magnetic force is directed as shown. It F_B is too large, its component up the incline can cause the conductor to go up. If F_B is weak, the component of mg will cause the conductor to slide down. For particular values of F_B , the conductor can stay in equilibrium also.



- (C) When $\vec{B} = B_0 \hat{k}$, F_B is directed along mg . Since θ is greater than angle of repose, the conductor will move down. Friction will be up the plane.

- (D) When $\vec{B} = -B_0 \hat{k}$, F_B is directed vertically up. Component of F_B along the incline is up. This is directed opposite to the component of mg along the incline. Depending on the strength of F_B , the conductor can move up or move down or can stay in equilibrium.



6. (A) Field at P due to current i is $\vec{B}_1 = \frac{\mu_0 i}{4\pi d} (-\hat{j})$

Field due to $\frac{i}{3}$ in x direction is $\vec{B}_2 = \frac{\mu_0 i / 3}{4\pi d} (-\hat{j})$

Fields due to $i/3$ current in y and negative y directions cancel out.

$\therefore B = -\frac{\mu_0}{4\pi d} \frac{4i}{3} \hat{j}$

- (B) Field due to i is

$$\vec{B}_1 = \frac{\mu_0 i}{4\pi d} (-\hat{j})$$

Field due to $i/2$ in first quadrant is

$$B_2 = \frac{\mu_0 i / 2}{4\pi d} \text{ (direction as shown)}$$

Similarly, the field due to $i/2$ in fourth quadrant is B_3 as shown.

$$\begin{aligned} \therefore \vec{B} &= -(B_1 + \sqrt{2}B_2)\hat{j} \\ &= -\left(\frac{\mu_0 i}{4\pi d} + \frac{\sqrt{2}\mu_0 i / 2}{4\pi d}\right)\hat{j} \\ &= -\frac{\mu_0 i}{4\pi d} \left(1 + \frac{1}{\sqrt{2}}\right)\hat{j} \\ |B| &< \frac{\mu_0 i}{2\pi d} \end{aligned}$$

Similar workings can be done for option (C) and (D).

7. (A) Aluminium has higher resistivity than copper.

$$I = \frac{k\theta}{NAB}$$

$$\text{Current sensitivity is } \frac{\theta}{I} = \frac{NAB}{k}$$

Current sensitivity has no relation to the resistance

$$\text{of coil } \frac{\theta}{IR_g} = \frac{NAB}{kR_g}$$

With increase in coil resistance R_g , the voltage sensitivity drops.

$\therefore$ Range to measure voltage increases.

- (B) When N is doubled, the current sensitivity doubles. But the voltage sensitivity does not change since R_g is proportional to length of the wire and it also doubles when N is doubled. Since the current sensitivity has increased, the range to measure the current drops.

- (C) Coil area $A = lb$ [l = length, b = width]. Doubling l means A doubles.

Length of the wire used in making the coil increases (but is not doubled).

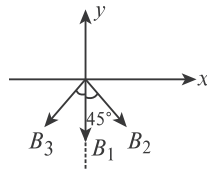
Current sensitivity increases. Range to measure current decreases.

$\therefore$ A has doubled but R_g has less than doubled, therefore the voltage sensitivity also increases.

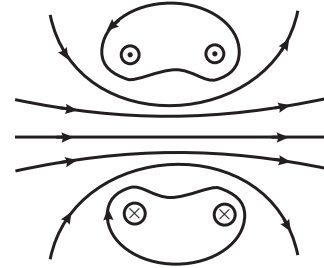
- (D) k is increased.

Both the current sensitivity and voltage sensitivity decrease.

$\therefore$ Range to measure voltage increases.

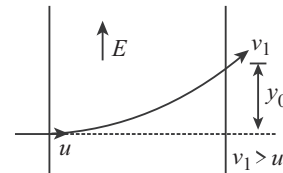


8. The last figure is field lines due to an arrangement like Helmholtz coils.

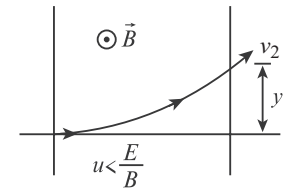


The field tends to get uniform between the coils.

9. In the first experiment, the particle follows a curved path as shown. Work done by the electric force ($= qEy_0$) is gain in KE of the particle.



In the second experiment, when $u = \frac{E}{B}$, the particle moves undeviated. Displacement $y = 0$. Work done by electric force = 0. There is no change in speed.

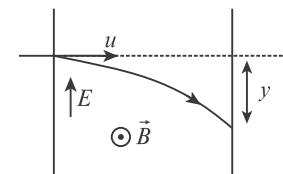


when $u < \frac{E}{B}$, the initial electric force is stronger. The

particle deflects upwards.

Electric force does positive work. Speed $v_2 > u$. But $y < y_0$. Work done by electric force is less than that in experiment 1. Therefore, $v_2 < v_1$.

When $u > \frac{E}{B}$, the particle deflects downwards. Work done by electric force is negative. Therefore, $v_2 < v_1$. When direction of B is reversed, the particle will deflect upwards and $y > y_0$. Work done by the electric force will be larger than that in the first experiment. Therefore, $v_2 > v_1$



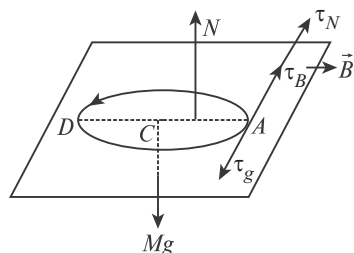
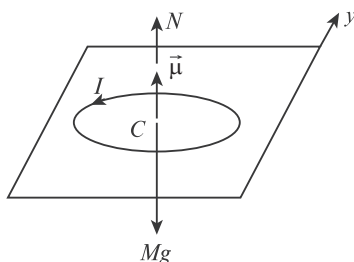


Passage Based Problems

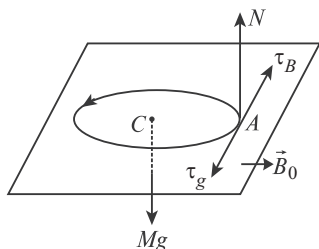
Passage 1

- Initially, when there is no magnetic field, the normal force and the weight of the loop acts at its centre. When a magnetic field is applied, a magnetic torque acts on the loop (magnetic force on the loop is zero). Direction of this torque is in y direction. This torque tries to rotate the loop about a tangential axis passing through point A (see figure). In other words, this torque tries to bring the ring in standing position at A . [Make sure that you understand the direction of torque and its effect.] Now, torque of normal force (N) also supports the magnetic torque. The torque due to Mg opposes the effort of magnetic and normal torque to topple the ring. When B is increased, the magnetic torque increases and the ring begins to lose contact with the floor at end D . The effective normal force moves from C towards A . Torque due to N (about A) decreases and the equilibrium is maintained.

$$\tau_B + \tau_N = \tau_g$$



Magnetic torque on the current loop tries to topple it about A . Point of action of N moves to right.



In extreme case, point of action of N moves to point A .

When $B = B_0$, the ring touches the surface only at the point A . $\tau_N = 0$. Now,

$$\tau_B = \tau_g$$

If B is increased further, the magnetic torque becomes larger than τ_g and the ring begins to topple.

- Value of B_0 can be calculated as

$$\tau_B = \tau_g$$

$$\Rightarrow I(\pi R^2) B_0 = Mg \cdot R$$

$$\text{or, } B_0 = \frac{Mg}{\pi IR}$$

when $B = 2B_0$, The magnetic torque will be

$$\tau_B = (2B_0)(I \cdot \pi R^2) = 2Mg \cdot R$$

The normal force acts at point A . Net torque on the loop is $\tau = \tau_B - \tau_g = MgR$.

Moment of inertia of the loop about the rotation axis is $\frac{3}{2}MR^2$.

$$\therefore \frac{3}{2}MR^2 \cdot \alpha = MgR \Rightarrow \alpha = \frac{2}{3} \frac{g}{R}$$

- Immediately after release, the acceleration of the CoM of the ring is

$$a_{CM} = R\alpha = \frac{2g}{3} (\uparrow)$$

Using $F_{\text{ext}} = Ma_{\text{cm}}$, we get

$$N - Mg = \frac{2Mg}{3} \Rightarrow N = \frac{5Mg}{3}$$

Passage 2

- In equilibrium, magnetic torque = spring torque

$$\text{or, } Ni_0BA = C \frac{\pi}{2} \Rightarrow C = \frac{2NBi_0A}{\pi}$$

- Charge is passed in a small time interval. The coil receives a magnetic torque for a small duration and gains angular speed (ω).

Change in angular momentum = Angular impulse

$$I\omega = NBA \int_0^{\Delta t} i dt$$

$$\text{or, } I\omega = NBAQ \Rightarrow \omega = \frac{NBAQ}{I}$$

The coil has a KE after the charge is passed through it. The maximum deflection occurs when this entire KE gets transferred to the spring as its potential energy.

$$\frac{1}{2} C \theta_0^2 = \frac{1}{2} I \omega^2$$

$$\Rightarrow \theta_0 = \sqrt{\frac{I}{C}} \cdot \omega = \sqrt{\frac{I \pi}{2 N B i_0 A}} \frac{N B A Q}{I}$$

$$\text{or, } \theta_0 = Q \sqrt{\frac{\pi N B A}{2 I i_0}}$$

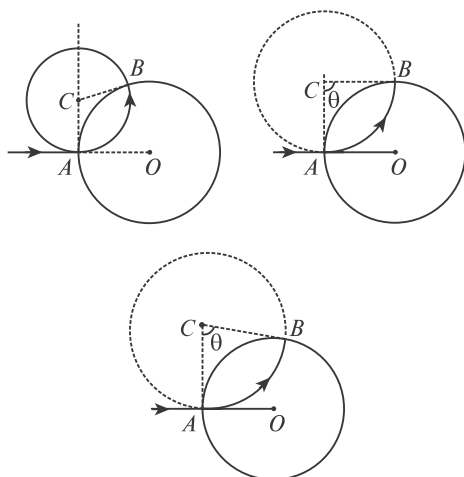
Passage 3

6. The diagrams given below show the path of a particle. The first diagram shows the path of a particle with relatively smaller speed. It moves in a circle of smaller radius. Centre of the circle is at C (on a line perpendicular to the initial velocity direction). Angle $\angle ACB = \theta$ is an obtuse angle. The particle needs time t to move from A to B which is given by

$$t = \frac{\theta}{2\pi} \cdot T$$

where $T = \frac{2\pi m}{qB}$ is time period of circular motion.

In the second figure, the particle has a higher speed. It moves on a circle of larger radius. Angle θ is smaller. Time needed to cross the circular field region is also smaller.



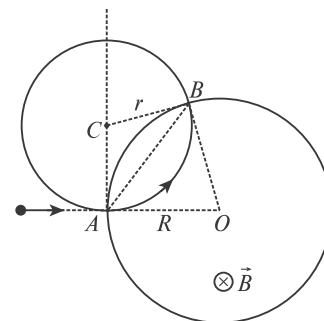
In the third diagram, the particle has much higher speed. It moves on a circular path of much larger radius. Angle θ is even smaller and the time spent in the field region is also smaller.

Therefore, it is easy to see that higher the speed, lesser is the time spent in the field region.

7. According to the question, the particle must spend $t = \frac{T}{3}$ time in the field region. It must turn through an angle of 120° in the field region.

Radius of the circular path is $r = \frac{mv}{qB}$

Particle enters the field region at A and leaves it at B . Centre of the circular path is C . $\angle ACB = 120^\circ$



Also,

$$\angle CAO = \angle CBO = 90^\circ$$

$$\therefore \angle AOB = 60^\circ$$

ΔAOB is equilateral. Therefore, $AB = R$

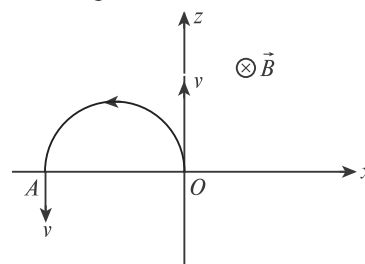
$$\text{In } \Delta ACB : AB = 2r \cos 30^\circ$$

$$\Rightarrow R = 2r \frac{\sqrt{3}}{2} \Rightarrow r = \frac{R}{\sqrt{3}}$$

$$\text{or, } \frac{mv}{qB} = \frac{R}{\sqrt{3}} \Rightarrow v = \frac{qBR}{\sqrt{3}m}$$

Passage 4

8. The particle is given a velocity in positive z direction. The magnetic field that it encounters is B in y direction. The particle goes in a circle in xz plane. It crosses the $z = 0$ plane at a point A such that



$$OA = 2R = \frac{2mv}{qB}$$

9. After crossing the xy plane at A , the particle moves below $z = 0$ and encounters a field B in x direction. The centre of the circular path now lies above point A in the figure shown. The particle describes a semicircle and rises to a point B at a height $2R$ above A . There it again crosses the $z = 0$ plane. The y co-ordinate of point B will be

$$y = -2R = -\frac{2mv}{qB}$$

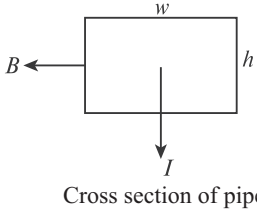
$\therefore$ co-ordinates of the point B are

$$\left(\frac{-2mv}{qB}, \frac{-2mv}{qB} \right)$$

Passage 5

10. $T = \frac{2\pi m}{qB_0}$ is time period of circular motion in the magnetic field. The particle enters the field region and completes a circular path. The moment it completes the circle, the direction of field reverses. Now, the particle describes the circle in opposite sense. The path looks like 8.
11. In this case, the field changes direction everytime the particle completes a semicircle. Once a semicircle is completed, the sense of rotation changes. The resulting path is a series of semicircles with their centres on opposite sides.

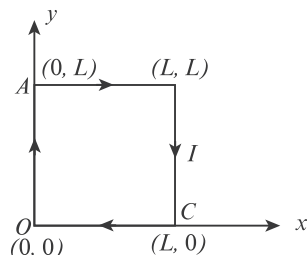
Passage 6

12. Current is directed downward in the figure. Direction of magnetic force on the liquid is along $I(\vec{h} \times \vec{B})$. In the figure shown, it is directed into the figure along the axis of the pipe.
- 
13. $F_B = IhB$
 $= (j\Delta L w)(hB)$
 $\therefore \text{Pressure} = \frac{F_B}{hw} = j\Delta LB$

Passage 7

14. When the springs are relaxed, the magnetic force alone balances the weight. Magnetic force is directly proportional to current. This makes the device work effectively.
15. $W = ILB$
 Maximum current, for full scale deflection in the ammeter is
 $I_0 = 10\text{A}$
 $\therefore W_{\max} = I_0 LB = 10 \times 0.5 \times 0.5 = 2.5\text{N}$

Passage 8

16. Tangent to field lines must give the direction of field. For example, at a point $(0, y)$, $B_y = 0$. The field is along x direction only. Tangent on the field line at $(0, y)$ must be along x direction.
17. The situation is shown in the figure below. The current is clockwise so that dipole moment is in negative z direction. Force on side OA can be calculated as follows.
- 

Consider a small length dy of the wire at co-ordinate y . Force on this segment is

$$d\vec{F} = I(dy \hat{j}) \times \left[\left(\frac{B_0 y}{L} \right) \hat{i} + \left(\frac{B_0 x}{L} \right) \hat{j} \right]$$

$$= \frac{IB_0}{L} y dy (-\hat{k})$$

$$\therefore \text{Force on wire OA is } \vec{F}_1 = \frac{IB_0}{L} \int_0^L y dy (-\hat{k})$$

$$= -\frac{1}{2} B_0 LI (\hat{k})$$

Similarly, we can calculate forces on AB , BC and CO . They are

$$\vec{F}_2 = \frac{1}{2} B_0 IL \hat{k}$$

$$\vec{F}_3 = \frac{1}{2} B_0 IL \hat{k}$$

$$\vec{F}_4 = -\frac{1}{2} B_0 IL \hat{k}$$

Torques due to $\vec{F}_1$ and $\vec{F}_3$ about x axis cancel out. There is no torque due to $\vec{F}_4$. The force $\vec{F}_2$ produces torque about x axis.

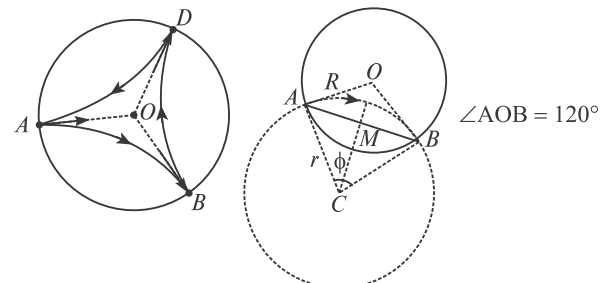
$$\tau_x = L.F_2 = \frac{1}{2} B_0 IL^2$$

18. Forces $\vec{F}_2$ and $\vec{F}_4$ have equal and opposite torques about y axis. $\vec{F}_1$ has no torque. Only $\vec{F}_3$ produces torque about y axis.

$$|\tau_y| = |LF_3| = \frac{1}{2} B_0 IL^2$$

Passage 9

- 19.



The particle starts from A , hits the cylinder at B , then at D before it returns to A . It hits the surface normally. Obviously, the path from A to B must be same as the path from B to D and from D to A .

$$\angle AOB = 120^\circ$$

The circular path of the particle from A to B has its centre at C .

$$\angle OAC = \angle OBC = 90^\circ$$

$$\therefore \phi = 60^\circ$$

$$AB = 2R \sin 60^\circ = 2r \sin 30^\circ$$

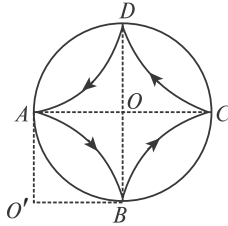
$$\Rightarrow r = \sqrt{3}R$$

$$\text{or, } \frac{mv_0}{qB} = \sqrt{3}R \Rightarrow v_0 = \frac{\sqrt{3}RqB}{m}$$

20. Particle takes the path $ABCD A$. When moving on curve AB , the centre of the circular path is at O' . Particle rotates by 90° when moving from A to B .

$$t_{AB} = \frac{T}{4} = \frac{\pi m}{2qB}$$

$$\therefore \text{Total travel time} = 4t_{AB} = \frac{2\pi m}{qB}$$

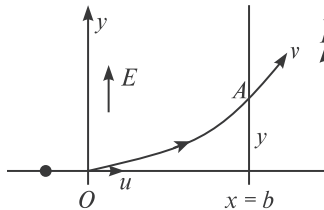


Passage 10

21. Speed (u) of the particle after being accelerated through a potential difference V is given by $\frac{1}{2}mu^2 = eV$

$$\Rightarrow u = \sqrt{\frac{2eV}{m}} \quad \dots(i)$$

In the region of electric field, acceleration is $a_y = \frac{eE}{m}$



Time needed to travel a distance b along x direction is

$$t = \frac{b}{u}$$

y - co-ordinate of point A will be

$$y = \frac{1}{2}a_y t^2 = \frac{1}{2} \left(\frac{eE}{m} \right) \left(\frac{b}{u} \right)^2$$

$$\text{or, } y = \frac{Eb^2}{4V} \quad [\text{Substituting for } u \text{ from (i)}]$$

22. When the particle exits the electric field, y component of its velocity is

$$v_y = 0 + a_y t = \frac{eE}{m} \cdot \frac{b}{u}$$

This is velocity component parallel to $\vec{B}$. The pitch is

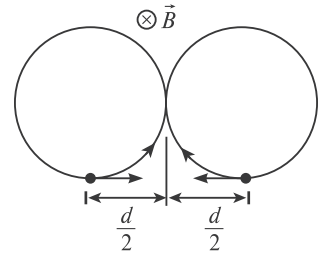
$$\begin{aligned} p &= v_y T = \left(\frac{eE b}{m u} \right) \left(\frac{2\pi m}{eB} \right) \\ &= \frac{2\pi b E}{Bu} = \frac{\pi b E}{B} \sqrt{\frac{2m}{eV}} \end{aligned}$$

Passage 11

23. Radius of circular paths of the two particles can be written as

$$R_1 = \frac{mv}{qB}$$

$$R_2 = \frac{2mv}{2qB} = R_1$$



The time period of circular motion is also same for the two particles. The two particles will just miss each other if their circular paths have radius just less than $\frac{d}{2}$, as shown.

$$\frac{mv}{qB} < \frac{d}{2} \Rightarrow v < \frac{qBd}{2m}$$

$$\therefore v_0 = \frac{qBd}{2m}$$

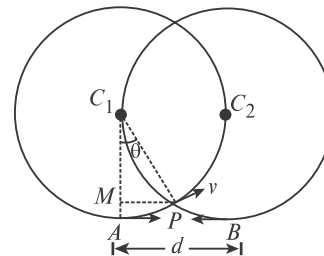
24. When $v = 2v_0$, the radius of circular paths doubles and the particles collide at point P . C_1 and C_2 are centres of their circular paths. Due to symmetry, we can judge that the particles collide at P such that $MP = \frac{d}{2}$.

$$\therefore \sin \theta = \frac{d/2}{R} = \frac{d/2}{d} = \frac{1}{2}$$

[Radius for speed v_0 is $\frac{d}{2}$. Hence, radius for speed $2v_0$ is d]

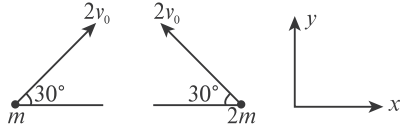
$$\text{or, } \theta = \pi/6$$

Time for the particles to collide = time needed to rotate by an angle of $\pi/6$



$$\therefore t = \frac{1}{12} \frac{2\pi m}{qB} = \frac{\pi m}{6qB}$$

25. At the time of collision the velocities of the two particles are as indicated. Let the velocity components in x and y directions, after collision, be v_x and v_y . Using conservation of momentum.



$$3mv_x = m(2v_0)\cos 30^\circ - 2m(2v_0)\cos 30^\circ$$

$$\text{or, } v_x = -\frac{v_0}{\sqrt{3}}$$

$$\text{And, } 3mv_y = m(2v_0)\sin 30^\circ + 2m(2v_0)\sin 30^\circ$$

$$\text{or, } v_y = v_0$$

$$\therefore \text{Speed of the combined mass, } v = \sqrt{v_x^2 + v_y^2} = \frac{2}{\sqrt{3}}v_0$$

$$\therefore \text{Radius of circular path } r = \frac{(3m)v}{qB} = \frac{(3m)\left(\frac{2}{\sqrt{3}}\frac{qBd}{2m}\right)}{qB}$$

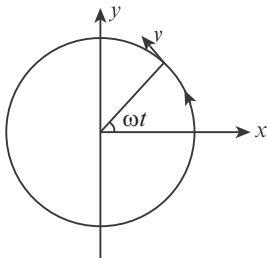
$$\text{or, } r = \sqrt{3}d$$

Passage 12

26. The bead rotates in xy plane with angular speed ω . Its velocity component in xy plane is $v = \omega a$. Its velocity at time t is

$$\vec{v} = -a\omega \sin(\omega t)\hat{i} + a\omega \cos(\omega t)\hat{j} + v_z\hat{k}$$

$$\text{Magnetic force } \vec{F} = q\vec{v} \times \vec{B}$$



$$\text{or, } \vec{F} = [-a\omega \sin(\omega t)\hat{i} + a\omega \cos(\omega t)\hat{j} + v_z\hat{k}] \times (B_0\hat{j})$$

z -component of the force is

$$F_z = -qa\omega B_0 \sin(\omega t)$$

$$a_z = -\frac{qa\omega B_0}{m} \sin(\omega t)$$

$$27. \frac{dv_z}{dt} = -\frac{qa\omega B_0}{m} \sin(\omega t)$$

$$\int_0^{v_z} dv_z = -\frac{qa\omega B_0}{m} \int_0^t \sin(\omega t) dt$$

$$\text{or, } v_z = -\frac{qaB_0}{m} (1 - \cos \omega t)$$

$$\text{or, } \frac{dz}{dt} = -\frac{qaB_0}{m} (1 - \cos \omega t)$$

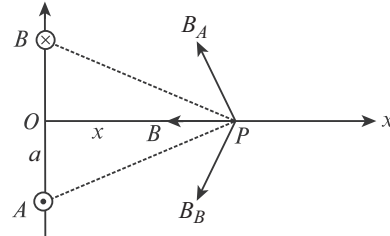
$$\text{or, } \int_a^0 dz = -\frac{qaB_0}{m} \int_0^T (1 - \cos \omega t) dt$$

$$\text{or, } a = \frac{qaB_0}{m} \left[T - \frac{1}{\omega} \sin \omega T \right]$$

$$\text{or, } B_0 = \frac{m\omega}{q(\omega T - \sin \omega T)}$$

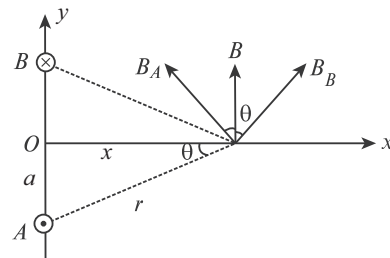
Passage 13

28. If the wire B carries a current I_0 in negative z direction, the fields due to the two wires at a point on the x axis are directed as shown and their resultant is along negative x direction. A current along x axis will experience no force due to this field.



29. Fields due to the wires A and B , at a point on the x axis, are directed as shown. The resultant field is along y direction and has magnitude

$$B = B_A \cos \theta + B_B \cos \theta = 2 \cdot \frac{\mu_0 I_0}{2\pi r} \cos \theta$$

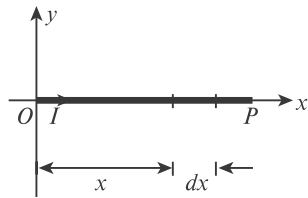


$$= \frac{\mu_0 I_0}{\pi r} \cdot \frac{x}{r} = \frac{\mu_0 I_0 x}{\pi(a^2 + x^2)}$$

A segment of length dx on the wire along x axis will experience a force

$$\vec{dF} = (I dx \hat{i}) \times \left(\frac{\mu_0 I_0 x}{\pi(a^2 + x^2)} \hat{j} \right)$$

$$= \frac{\mu_0 I I_0}{\pi(a^2 + x^2)} x dx \hat{k}$$



∴ Net force on the wire OP is

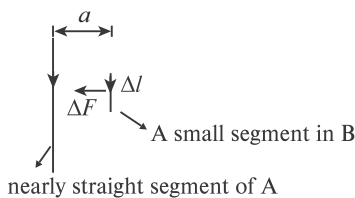
$$\begin{aligned} F &= \int_0^L \frac{\mu_0 I_0}{\pi(a^2 + x^2)} x dx \\ &= \frac{\mu_0 I_0}{2\pi} \int_0^L \frac{2x dx}{a^2 + x^2} \\ &= \frac{\mu_0 I_0}{2\pi} [\ln(a^2 + x^2)]_0^L \\ &= \frac{\mu_0 I_0}{2\pi} \ln \left(\frac{a^2 + L^2}{a^2} \right) \end{aligned}$$

The force is in z direction.

Note: $\int \frac{2x dx}{a^2 + x^2}$ can be easily evaluated by substituting $a^2 + x^2 = t$.

Passage 14

30. A small element of length Δl on the circumference of the ring B sees a magnetic field due to a long straight section of A. [We all are on the surface of the Earth and it appears as a large flat plate. Similarly, an observer close to the circumference of a large ring will see a long straight wire]. The attractive force felt by the small segment in B is



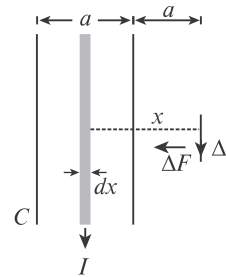
$$\Delta F = \frac{\mu_0 I^2}{2\pi a} \Delta l$$

Total force experienced by the ring B is

$$F_1 = \sum \Delta F = \frac{\mu_0 I^2}{2\pi a} \sum \Delta l$$

$$\text{or, } F_1 = \frac{\mu_0 I^2}{2\pi a} 2\pi R = \frac{\mu_0 I^2 R}{2a}$$

31. A small element in B will see a force due to a long strip in C having width a . Consider a thin wire in C having width dx . Current in this wire is $dI = \frac{Id x}{a}$.



Field at the location of the small segment of B, due to this current dI is

$$dB = \frac{\mu_0 dI}{2\pi x} = \frac{\mu_0 I}{2\pi a} \frac{dx}{x}$$

∴ Field due to C is

$$B = \frac{\mu_0 I}{2\pi a} \int_a^{2a} \frac{dx}{x} = \frac{\mu_0 I}{2\pi a} \ln(2)$$

Force on the segment of length Δl is

$$\Delta F = \left[\frac{\mu_0 I}{2\pi a} \ln 2 \right] (I \Delta l) = \frac{\mu_0 I^2}{2\pi a} \ln(2) \Delta l$$

∴ Force on B is

$$\begin{aligned} F_2 &= \sum \Delta F = \frac{\mu_0 I^2 (\ln 2)}{2\pi a} \sum \Delta l \\ &= \frac{\mu_0 I^2 (\ln 2)}{2\pi a} \cdot 2\pi R = \frac{\mu_0 I^2 R (\ln 2)}{2a} \end{aligned}$$

Passage 15

$$\begin{aligned} 32. \quad U &= -\vec{M} \cdot \vec{B} = -(\pi b^2 I_2) \frac{\mu_0}{2} \frac{I_1 a^2}{(a^2 + x^2)^{3/2}} \cos 0 \\ &= -\frac{\pi \mu_0}{2} \frac{I_1 I_2 a^2 b^2}{(a^2 + x^2)^{3/2}} \end{aligned}$$

$$33. \quad F = -\frac{dU}{dx} = \frac{3\pi \mu_0}{2} \frac{I_1 I_2 a^2 b^2 x}{(a^2 + x^2)^{5/2}}$$

Passage 16

34. When S is closed, a current (I) flows into the wire. It experiences an upward magnetic force. This accelerates the wire. Its acceleration is given by

$$a = \frac{ILB - mg}{m} \quad \dots(i)$$

The magnetic force is maintained till the wire moves $h_0 = 0.025\text{m}$. Its speed at the instant it leaves the liquid is

$$v = \sqrt{2ah_0} \quad \dots(ii)$$

After this, the contact is broken and there is no magnetic force. Wire retards with acceleration g . It rises further by $h - h_0 = 0.35 - 0.025 = 0.325$ m

$$\therefore v^2 = 2g(h - h_0) = 2 \times 9.8 \times 0.325 = 6.37$$

$$\text{or, } v = 2.52 \text{ ms}^{-1}$$

35. Putting this into (ii) gives

$$a = \frac{v^2}{2h_0} = \frac{6.37}{2 \times 0.025} = 127.4 \text{ m/s}^2$$

Putting this into (i)

$$\text{or, } \frac{ILB}{m} - g = 127.4$$

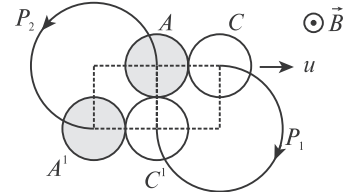
$$\text{or, } \frac{I \times 0.15 \times 6.5 \times 10^{-3}}{5.4 \times 10^{-5}} = 137.2$$

$$\text{or, } I = 7.6 \text{ A}$$

Passage 17

36. A hits C and transfers its entire velocity to C (elastic head on collision of equal masses). A comes to rest and C begins to move with velocity u . The charge in C experiences magnetic force and its centre moves on circular path P_1 . Radius of circular path is

$$R = \frac{mu}{qB} = \frac{mu}{q \left(\frac{mu}{2qr} \right)} = 2r$$



The disc C rotates by 270° and hits back A . Time needed for this is

$$t_0 = \frac{3}{4} T = \frac{3}{4} \times \frac{2\pi m}{qB} = \frac{3\pi m}{2q \left(\frac{mu}{2rq} \right)} = \frac{3r}{u}$$

37. Now, A acquires velocity u travelling in y direction. The negative charge experiences a magnetic force and the centre of A traces a circle P_2 as shown. It will further take time t_0 to hit C .

$\therefore$ Overall time after which A again hits C is $2t_0$

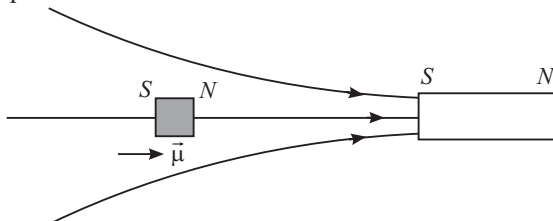
$$2t_0 = \frac{6r}{u}$$

Chapter 4 Magnetic Properties of Matter and Earth's Magnetism



Your Turn

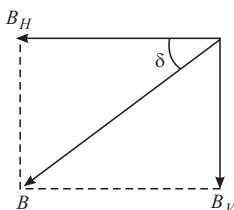
2. Magnetic field is towards right at the location of the cubes. A magnetic dipole moment is induced in the cubes (due to alignment of atomic dipoles) directed parallel to the field.



Cube 2 is located in higher field. It will have better alignment of dipoles. It has higher dipole moment.

The two cubes get attracted by the magnet.

4. $I = \frac{\mu}{V} = \frac{1}{1 \times 10^{-6}} = 10^6 \text{ Am}^{-1}$
5. (a) χ is negative. The material is diamagnetic.
 (b) $\mu_r = 1 + \chi = 1 - 0.00002 = 0.99998$
 (c) $B = \mu_r B_0 = 0.99998 \times 1 \text{ mT} = 0.99998 \text{ mT}$
6. $\mu_r = \frac{\mu}{\mu_0} = \frac{0.252 \text{ TmA}^{-1}}{4\pi \times 10^{-7} \text{ TmA}^{-1}} = 2 \times 10^5$
 $\chi = \mu_r - 1 \approx 2 \times 10^5$
7. (a) $B_0 = \mu_0 n i \Rightarrow H = \frac{B_0}{\mu_0} = n i$
 $\therefore 1000 \text{ A/m} = (1000 \text{ m}^{-1}) i$
 $\Rightarrow i = 1 \text{ A}$
 (b) $I = \chi H \Rightarrow \chi = \frac{I}{H} = \frac{0.1}{1000} = 10^{-4}$
 A small positive value of χ means that the material is paramagnetic.
8. $\frac{B_H}{B} = \cos \delta \Rightarrow \frac{20 \mu\text{T}}{B} = \cos 37^\circ$
 $\Rightarrow B = \frac{20 \times 5}{4} \mu\text{T} = 25 \mu\text{T}$
9. The tension and weight of the magnet are two forces which pass through its centre and produce no torque. The magnet experiences torque only due to the magnetic field of the Earth. Therefore,



the magnet aligns in the direction of the field. This direction is in magnetic meridian plane making an angle δ with the horizontal such that

$$\tan \delta = \frac{B_V}{B_H} = \frac{18.75}{25} = \frac{3}{4}$$

$$\delta \approx 37^\circ$$



Worksheet 1

1. Induced magnetic moment is opposite to the field of the magnet. The dipole moment of the specimen and that of the magnet are oppositely directed. There is always a repulsion.
11. $H = ni = (500 \text{ m}^{-1}) (0.5 \text{ A}) = 250 \text{ Am}^{-1}$
 $\chi = \mu_r - 1 \approx 1000$
 Magnetisation of the rod is
 $I = \chi H = 1000 \times 250 = 2.5 \times 10^5 \text{ Am}^{-1}$
 $\therefore$ Magnetic moment of the rod is $M = IV$
 $= (2.5 \times 10^5 \text{ Am}^{-1}) (10^{-4} \text{ m}^3)$
 $= 25 \text{ Am}^2$
12. $B = \mu_0 \mu_r H$
 $\therefore \mu_r = \frac{B}{\mu_0 H} = \frac{8\pi}{4\pi \times 10^{-7} \times 2 \times 10^3} = 10^4$
13. $B_H = B \cos 60^\circ = 0.36 \times \frac{1}{2} = 0.18 \text{ G}$
 $B_V = B \sin 60^\circ = 0.36 \times \frac{\sqrt{3}}{2} = 0.18 \sqrt{3} \text{ G}$
14. The horizontal component (B_H) of the field is in magnetic meridian. Angle between B_H and the true north direction is the declination angle (θ).
 $\tau_B = MB_H \sin \theta$
 or, $1.2 \times 10^{-3} = 60 \times 40 \times 10^{-6} \times \sin \theta$
 $\Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$



Worksheet 2

1. The domain which has dipole moment in the direction of external field will grow. Others will decrease in size.
2. $I = \chi H$
 There are two points on the loop where $I = 0$ but $H \neq 0$. It means that susceptibility is zero. There are points where I is positive but H is negative implying that the susceptibility is negative. Also there are points where $H = 0$ but $I \neq 0$, meaning that χ is infinity.

4. We know $I \propto \frac{B_0}{T}$

When the ratio of the external field and the temperature does not change, the magnetisation will not change.

5. (a) The B_H is perpendicular to the needle. The needle is not allowed to rotate because of torque due to this field. But B_V will align the needle parallel to itself. The needle will be vertical.
 (b) The needle is in magnetic meridian. It will point in the direction of the resultant field. It will be inclined at dip angle to the horizontal.
 (c) Effective magnetic field is only B_V .
6. The effective horizontal component of the field in the given plane is $B'_H = B \cos \beta$

$$\text{Now, } B'_H = B_H \cos \alpha$$

$$\text{Therefore, } B_H = \frac{B \cos \beta}{\cos \alpha}$$

$$\text{Also } B_V = B \sin \beta$$

$$\tan \delta = \frac{B_V}{B_H} = \frac{\sin \beta \cos \alpha}{\cos \beta}$$



Worksheet 3

1. There may be minor differences in the strength of individual atomic dipoles of two different materials. But if all the dipole of a paramagnet are aligned strictly parallel to one another it will acquire a large magnetisation of the same order as a ferromagnet. But of course, such saturation will require very high magnetic fields.
2. $\mu_r = 1 + \chi = 1.000068$
 Magnetic field after insertion of tungsten is
 $B = \mu_r B_0$ [B_0 = original field]
 $\therefore$ Percentage change = $\frac{B - B_0}{B_0} \times 100 = \frac{\mu_r - 1}{1} \times 100$
 $= 0.0068\%$
3. Magnetisation is dipole moment per unit volume. It is decided by the degree of alignment of the atomic dipoles. By cutting the magnet I will not change.
4. Mass of the cubical domain = density $\times$ volume
 $= (7.9 \text{ g/cm}^3) \times (10^{-4} \text{ cm})^3 = 7.9 \times 10^{-12} \text{ g}$
 55g iron contains 1 mol = 6.02×10^{23} atoms
 $\therefore$ Number of atoms inside the domain
 $= \frac{7.9 \times 10^{-12}}{55} \times 6.02 \times 10^{23}$
 $= 8.65 \times 10^{10}$ atoms

Maximum possible dipole moment is obtained when all the dipole are in perfect alignment.

$$\therefore \mu_{\max} = (8.65 \times 10^{10}) \times (9.27 \times 10^{-24})$$

$$= 8.0 \times 10^{-13} \text{ Am}^2$$

5. $\chi = 1 \times 10^{-5}$

Material is paramagnetic.
 Using Curie's law $\chi \propto \frac{1}{T}$

$$\frac{\chi_2}{\chi_1} = \frac{T_1}{T_2} \Rightarrow \frac{1.5 \times 10^{-5}}{1 \times 10^{-5}} = \frac{300}{T_2}$$

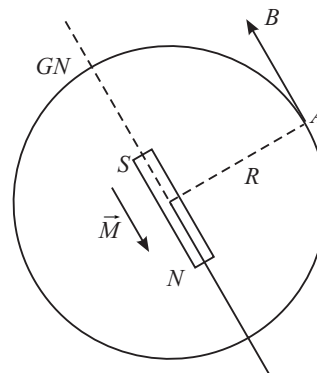
$$\Rightarrow T_2 = 200 \text{ K}$$

6. The magnetising field intensity must be $2 \times 10^4 \text{ Am}^{-1}$ to demagnetise the sample.

$$H = ni$$

$$\text{or, } 2 \times 10^4 = n \times 5 \Rightarrow n = 4000/\text{m}$$

7. A is a point on the magnetic equator. Field at A is antiparallel to the dipole moment and its magnitude is given by



$$B = \frac{\mu_0}{4\pi} \frac{M}{R^3}$$

$$\text{or, } M = \frac{4 \times 10^{-5} \times (6.4 \times 10^6)^3}{\mu_0 / 4\pi}$$

$$= 4 \times 10^{-5} \times (6.4 \times 10^6)^3 \times 10^7$$

$$\simeq 1.05 \times 10^{23} \text{ Am}^2$$

8. Geomagnetic poles are points on the axis of the dipole. The field at a pole will be

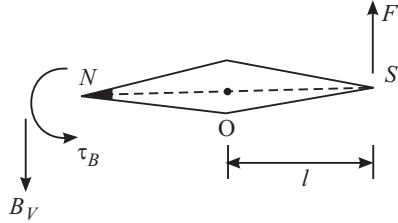
$$B_1 = \frac{\mu_0}{4\pi} \frac{2M}{R^3} \quad (\text{i})$$

Field on the equator is

$$B_2 = \frac{\mu_0}{4\pi} \frac{M}{R^3} \quad (\text{ii})$$

$$\text{From (i) and (ii) } B_2 = \frac{1}{2} B_1 = 30 \mu\text{T}$$

9. In magnetic meridian plane $\frac{B_V}{B_H} = \tan \delta$
 or, $\frac{B_V}{B_H} = \tan 45^\circ$
 $\Rightarrow B_V = B_H = 18 \mu\text{T}$



In the horizontal position of the needle, the magnetic torque acts only due to B_V .

$$\begin{aligned}\tau_B &= MB_V \sin 90^\circ \\ &= 0.16 \times 18 \times 10^{-6} \times 1 \\ &= 2.88 \times 10^{-6} \text{ N-m}\end{aligned}$$

Torque due to the applied force must balance this.

$$\begin{aligned}\therefore Fl &= 2.88 \times 10^{-6} \text{ N-m} \\ \text{or, } F \times (0.05 \text{ m}) &= 2.88 \times 10^{-6} \text{ N-m} \\ \text{or, } F &= 57.6 \times 10^{-6} \text{ N}\end{aligned}$$

10. Current $I = q \cdot f = \sigma \cdot 2\pi r l \cdot \frac{\omega}{2\pi} = \sigma r l \omega$

The rotating rod is like an ideal solenoid

$$H = ni$$

Here ni can be replaced with I .

$$\begin{aligned}\therefore H &= I \\ \therefore B &= \mu_0 \mu_r H = \mu_0 (1 + \chi) I \\ &= \mu_0 (1 + \chi) \sigma r l \omega\end{aligned}$$

Chapter 5 Electromagnetic Induction



Your Turn

1. Initially $\vec{B}$ is parallel to $\vec{S}$. When the surface is tilted by 60° , the $\vec{S}$ tilts by 60° . Now, the flux is

$$\phi = BS \cos 60^\circ = \frac{BS}{2}$$

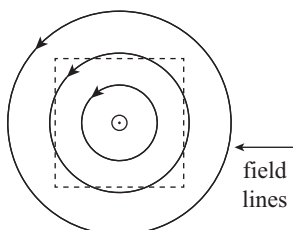
2. Flux linked with the loop is $\phi = BS$.

Flux does not change when the loop moves in the way described in the problem. Hence, there is no induced emf.

$$\begin{aligned} 3. \quad \varepsilon &= \left| N \frac{d\phi}{dt} \right| \\ &= N \frac{d(SB)}{dt} = NS \frac{dB}{dt} \\ &= 100 \times 0.1 \times \left(\frac{1T}{0.5s} \right) = 20 \text{ volt} \end{aligned}$$

$$\text{Induced current, } i = \frac{\varepsilon}{R} = \frac{20}{10} = 2A$$

4. The magnetic field produced by the current in the straight wire can be represented by circular field lines as shown. No field line pierces through the square loop. There is no flux through the square loop. There is no induced emf/current.



5. Induced emf does not depend on the material of the conducting ring. Flux for both the rings is

$$\phi = B \cdot S$$

The magnitude of the induced emf in both the rings is

$$\begin{aligned} \varepsilon &= \left| \frac{d\phi}{dt} \right| = S \left| \frac{dB}{dt} \right| \\ &= \pi (0.1)^2 \times 0.2 = 6.28 \times 10^{-3} \text{ V} \end{aligned}$$

Resistance of copper ring will be smaller and hence it will have higher current.

Note: The magnitude of induced emf is not dependent on the fact whether the flux is increasing or decreasing. It is 'the rate of change of the flux' that is important.

6. When the magnet is above the ring it is approaching the ring. The flux linked with the ring changes and a current is induced in it. Cause of induction can be stated as 'the magnet approaching the ring'. This has to be opposed. The magnetic dipole of the ring repels the magnet. Thus, $a_1 < g$.

After the magnet has crossed the ring, there is still a induced current in the ring. This time the reason is the magnet is moving away from the ring. The dipole formed in the ring opposes this. The magnet gets attracted (Upward). Again its acceleration is less than g . $a_2 < g$.

[If you are unable to understand, read example 3 again]

7. Change in flux is necessary for induction. Nonuniform field simply means that field varies with position. Electromagnetic induction happens when field varies with time.

8. Refer to example 6.

Charge flow does not depend on the time needed to rotate the coil, and

$$\therefore \Delta q = \frac{\Delta \phi}{R} = \frac{BS}{R}$$

[Initial flux = BS , final flux = 0. Change is BS]

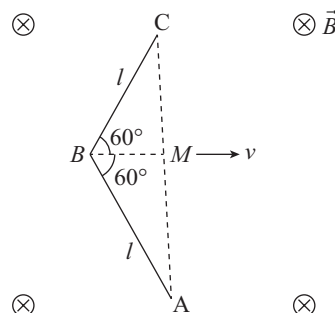
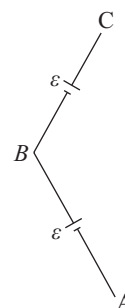
9. Emf in the arm BC is $\varepsilon = Blv \sin 60^\circ$
 $= \frac{\sqrt{3}}{2} Blv$

Both the arms have equal emfs and the resultant emf between A and C is $2e = \sqrt{3} Blv$.

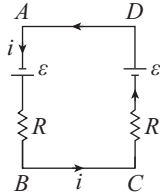
Alternatively, you can replace the bent wire with a straight wire AC .

$$AC = 2l \sin 60^\circ = \sqrt{3} l$$

$$\begin{aligned} \therefore \varepsilon_{AC} &= B v (\sqrt{3} l) \\ &= \sqrt{3} Blv \end{aligned}$$



10. (a) emf induced in each bar is $\varepsilon = Blv$. For CD , the end D is positive and for AB the end B is positive. The equivalent electric circuit is as shown. The current in the loop is



$$i = \frac{2\varepsilon}{2R} = \frac{\varepsilon}{R} = \frac{Blv}{R}$$

[Net emf in the loop is 2ε]

(b) Magnetic force on CD is towards left. An equal force to the right must be applied by the external agent.

$$F = ilB = \frac{B^2 l^2 v}{R}$$

Magnetic force on AB is towards right ($\vec{I} \times \vec{B}$) and same force must be applied by the external agent or it towards left.

11. $\varepsilon = Bv$ ($AB \sin 60^\circ$) = Bvl

$$\therefore i = \frac{Bvl}{R}$$

12. Refer to example 10.

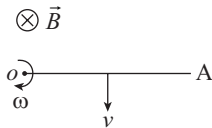
A disc is a wheel with infinite number of spokes.

If we go on increasing the number of spokes in example 10, will the emf induced between the centre and the circumference change? Obviously, no.

13. The semicircular wire can be replaced with a straight wire OA .

$$\varepsilon = \frac{1}{2} B\omega(2r)^2 = 2B\omega r^2$$

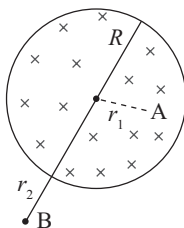
Consider a small segment on straight wire OA . Mark the direction of its velocity ($\vec{v}$). The direction indicated by $\vec{v} \times \vec{B}$ gives the positive end. End A is at higher potential.



14. Figure shows that circular cross section of the solenoid. The magnetic field is uniform inside the solenoid

$$B = \mu_0 ni = 4\pi \times 10^{-7} \times 1000 \times i = 4\pi \times 10^{-4} i$$

$$\begin{aligned} \frac{dB}{dt} &= 4\pi \times 10^{-4} \frac{di}{dt} \\ &= 4\pi \times 10^{-4} \times \frac{200}{\pi} = 0.08 \text{ T/s} \end{aligned}$$



- (a) The induced electric field at point A is given by equation (8)

$$E = \frac{r_1}{2} \frac{dB}{dt} = \frac{1 \times 10^{-2}}{2} \times 0.08 = 4 \times 10^{-4} \text{ V/m.}$$

- (b) At point B , the induced electric field is given by equation (9)

$$\begin{aligned} E &= \left(\frac{R^2}{2} \frac{dB}{dt} \right) \frac{1}{r_2} \\ &= \frac{(2.5 \times 10^{-2})^2}{2} \times 0.08 \times \frac{1}{0.04} = 6.25 \times 10^{-4} \text{ V/m} \end{aligned}$$

16. $\varepsilon = L \left| \frac{di}{dt} \right|$

$$\text{or, } [L] = \frac{[\varepsilon]}{\left[\frac{di}{dt} \right]} = \left[\frac{\text{J/C}}{\text{A/s}} \right]$$

[$J \rightarrow$ energy, $C \rightarrow$ Coulomb]

$$\text{or, } [L] = \left[\frac{ML^2T^{-2} / AT}{A/T} \right] = [M^1L^2T^{-2}A^{-2}]$$

17. $L = \frac{\phi}{i} = \frac{0.04}{2} = 0.02 \text{ H}$

18. (a) When i is constant there is no emf in the inductor. It is like a conducting wire.

$$V_a - 1 \times 2 + 2 = V_b \quad \begin{array}{c} a \xrightarrow{\quad 1\Omega \quad} 2V \quad b \\ i = 2A \end{array}$$

- (b) When i is decreasing, the emf induced in the inductor can be represented as shown.

$$\varepsilon = \left| L \frac{di}{dt} \right| \quad \begin{array}{c} a \xrightarrow{\quad 1\Omega \quad} 2V \quad b \\ i = 2A \quad \varepsilon \end{array}$$

$$= 0.2 \times 1 = 0.2 \text{ V}$$

$$\therefore V_a - 1 \times 2 + 0.2 + 2 = V_b$$

$$\text{or, } V_a - V_b = -0.2 \text{ Volt}$$

19. $L = \mu_0 \pi r^2 n^2 l = \mu_0 (\pi r^2 l) n^2$

$$\text{or, } L = \mu_0 V n^2 \text{ [V = Volume of the solenoid]}$$

$$= 4\pi \times 10^{-7} \times 0.02 \times (500)^2$$

$$= 6.28 \times 10^{-3} \text{ H} = 6.28 \text{ mH}$$

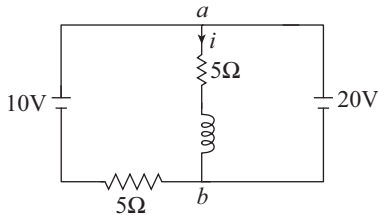
20. (a) When the switch is open, current in the left loop is

$$i_0 = \frac{10V}{10\Omega} = 1A.$$

After closing the switch (S), the current in L cannot change immediately. Therefore, current in L at $t = 0^+$ is $i_0 = 1A$.

(b) After a long time the circuit again attains steady state and the current in branch ab is

$$i = \frac{20V}{5\Omega} = 4A.$$



21. (a) $i_0 = \frac{V}{R} = \frac{10}{100} = 0.1A$

when $V_R = 6.3V$, $i = \frac{6.3}{100} = 0.063A$

The question is basically asking us to find the time in which the current becomes 63% of the final maximum value.

$$\therefore t = \tau = \frac{L}{R} = \frac{0.01}{100} = 0.0001s = 0.1 \text{ ms}$$

(b) When the emf induced in the inductor is 5V

$$V_R + 5 = 10 \Rightarrow V_R = 5 \text{ Volt}$$

$$\Rightarrow iR = 5$$

$$\Rightarrow i = 0.05A$$

$$\text{or, } i = \frac{i_0}{2} \Rightarrow i_0 e^{-t/\tau} = \frac{i_0}{2}$$

$$\text{or, } \frac{t}{\tau} = \ln 2 \Rightarrow t = (\ln 2)\tau$$

$$= 0.693 \times 0.1 = 0.0693 \text{ ms}$$

22. Initial current $i_0 = \frac{10V}{5\Omega} = 2A$

After the switch is moved to b , the battery gets disconnected and the current begins to decay

$$\tau = \frac{L}{R} = \frac{0.2}{5} = 0.04 \text{ s}$$

$$i = i_0 e^{-t/\tau} = 2 e^{-\frac{0.08}{0.04}} = 2e^{-2}$$

$$\text{or, } i = 0.27 \text{ A.}$$

23. (a) In both cases, the final steady state current is same.

$$\therefore \frac{V}{R_1} = \frac{2V}{R_2} \Rightarrow \frac{R_1}{R_2} = \frac{1}{2}$$

(b) Looking at the graphs we can say that the time needed for the current to grow to 50% of its peak value for the second circuit is double that of the first circuit

$$\frac{i_0}{2} = i_0 (1 - e^{-t/\tau}) \Rightarrow e^{-t/\tau} = \frac{1}{2}$$

$$\therefore e^{-t_0/\tau_1} = e^{-2t_0/\tau_2} = \frac{1}{2}$$

$$\Rightarrow \tau_1 = \frac{\tau_2}{2} \Rightarrow \frac{L_1}{R_1} = \frac{L_2}{2R_2}$$

$$\Rightarrow \frac{L_1}{L_2} = \frac{R_1}{2R_2} = \frac{1}{4}$$

24. Current in the circuit at $t = \frac{L}{R}$ is

$$i = i_0 (1 - e^{-1}) = 0.63i_0 = 0.63 \frac{V}{R}$$

(a) $U = \frac{1}{2} Li^2 = \frac{1}{2} L \left(0.63 \frac{V}{R} \right)^2 = 0.2 L \frac{V^2}{R}$

(b) $P_R = i^2 R = 0.4 \frac{V^2}{R}$

(c) $P_L = \frac{dU}{dt} = \frac{1}{2} L \left(2i \frac{di}{dt} \right) = Li \frac{di}{dt}$

$$i = i_0 (1 - e^{-t/\tau})$$

$$\therefore \frac{di}{dt} = \frac{i_0}{\tau} e^{-t/\tau} = \frac{V/R}{L/R} e^{-t/\tau} = \frac{V}{L} e^{-t/\tau}$$

$$\text{At } t = \frac{L}{R}$$

$$\frac{di}{dt} = \frac{V}{L} e^{-1} = 0.37 \frac{V}{L}$$

$$\therefore P_L = Li \frac{di}{dt} = L \left(0.63 \frac{V}{R} \right) \left(0.37 \frac{V}{L} \right) = 0.23 \frac{V^2}{R}$$

25. $U_B = \frac{B^2}{2\mu_0}$

u_B becomes $\frac{1}{4}$ of its original value if B becomes $\frac{1}{2}$ of its initial value. This will happen when the current becomes half.

$$\frac{I}{2} = I (1 - e^{-t/\tau})$$

$$\text{or, } e^{-t/\tau} = \frac{1}{2} \Rightarrow \frac{t}{\tau} = \ln 2$$

$$\text{or, } t = (\ln 2)\tau = (\ln 2) \frac{L}{R}$$

26. $f = \frac{1}{2\pi \sqrt{LC}}$

$$\text{or, } L = \frac{1}{4\pi^2 C f^2} = \frac{1}{4 \times 10 \times 10^{-6} \times 10^8} = \frac{1}{4} \times 10^{-3} \text{ H.}$$

$$\begin{aligned}\text{Energy } E &= \frac{1}{2} Li_0^2 = \frac{1}{2} \times \frac{1}{4} \times 10^{-3} \times 1^2 \\ &= \frac{1}{8} \times 10^{-3} \text{ J} \\ &= \frac{1}{8} \text{ mJ}\end{aligned}$$

27. (a) $\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 10 \times 10^{-6}}} = 10^4 \text{ rad s}^{-1}$.

or, $\frac{2\pi}{T} = 10^4 \Rightarrow T = \frac{2\pi}{10^4} = 6.28 \times 10^{-4} \text{ s}$.

(b) Maximum charge on the capacitor
 $q_0 = CV = 10 \times 10^{-6} \times 10 = 10^{-4} \text{ C}$
 $\frac{1}{2} Li_0^2 = \frac{q_0^2}{2C}$

$\Rightarrow i_0 = \frac{q_0}{\sqrt{LC}} = 10^{-4} \times 10^4 = 1 \text{ A}$

(c) At $t = 0$, charge on the capacitor is $+q_0$

$\therefore q = q_0 \cos \omega t$

or, $q = 10^{-4} \cos(10^4 t)$

28. There are only two elements in the circuit – L and C . The voltage across both of them is always same.

$\therefore$ Maximum voltage across on the capacitor $= V_0$

Maximum charge on the capacitor $= CV_0$

29. (a) Equivalent inductance is $L = L_1 + L_2$

$\therefore \omega = \frac{1}{\sqrt{(L_1 + L_2)C}} \Rightarrow f = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$

(b) Potential difference across the capacitor $= \frac{V_0}{2}$

Potential difference across the series combination of L_1 and L_2 will also be $\frac{V_0}{2}$

$V_{L1} = L_1 \frac{di}{dt}$ and $V_{L2} = L_2 \frac{di}{dt}$

$\Rightarrow \frac{V_{L1}}{V_{L2}} = \frac{L_1}{L_2}$

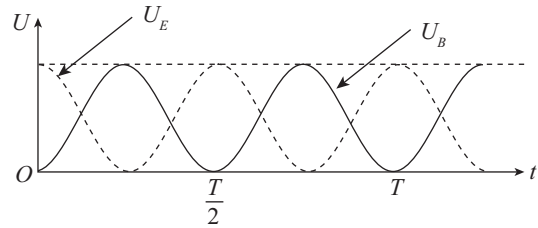
And $V_{L1} + V_{L2} = \frac{V_0}{2}$

Solving (i) and (ii) we get

$V_{L1} = \frac{V_0 L_1}{2(L_1 + L_2)}$

30. $U_E = \frac{q^2}{2C} = \frac{q_0^2}{2C} \cos^2 \omega t$

$U_B = \frac{1}{2} Li^2 = \frac{1}{2} L i_0^2 \sin^2 \omega t$



31. The current is maximum and entire energy is with the inductor. The current cannot become zero immediately. The emf induced in the inductor recharges the capacitor.

32. $|\varepsilon_2| = M \left| \frac{di_1}{dt} \right|$

or, $M = \frac{0.2V}{2A/s} = 0.1 \text{ H}$

33. $|\varepsilon_B| = M \left| \frac{di_A}{dt} \right| \Rightarrow M = \frac{0.1}{4} = 0.025 \text{ H}$

Value of M does not change when B is made the primary coil.

$\therefore |\varepsilon_A| = M \left| \frac{di_B}{dt} \right| = 0.025 \times 2 = 0.05 \text{ V}$

34. Mutual inductance is proportional to the number of turns in both the coils.

35. Consider a current i in the circular loop. The magnetic field on its axis at a distance R is

$B = \frac{\mu_0}{2} \frac{R^2 i}{(R^2 + R^2)^{3/2}} = \frac{\mu_0 i}{2^{5/2} R}$

Because the square loop is small in size, we can assume the field to be B over its entire area.

The magnetic flux linked to the square loop is

$\phi = a^2 \cdot B = \left(\frac{\mu_0 a^2}{2^{5/2} R} \right) i$

$\therefore M = \frac{\mu_0 a^2}{2^{5/2} R}$

(i) Worksheet 1

(ii) 1. $\phi = at^2 + bt + c$

$\frac{d\phi}{dt} = 2at + b$

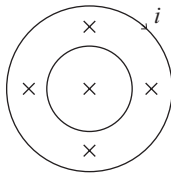
Unit of $\frac{d\phi}{dt}$ (induced emf) is volt.

$\therefore$ Unit of a is volt sec^{-1} and unit of b is volt.

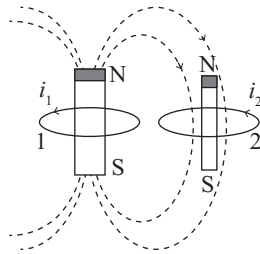
3. Magnetic field lines are closed loops. A field line that pierces out of the xy plane will also pierce into it at some point.
4. There is no motional emf and there is no induced electric field.
5. $\frac{dB}{dt} = \frac{0.1T}{0.1s} = 1\text{T s}^{-1}$

$$|\varepsilon| = NA \frac{dB}{dt} = 500 \times (100 \times 10^{-4}) \times 1 = 5\text{V}$$

6. The direction of emf will change as the flux first increases and then begins to decrease. Only the graph in (B) shows change in direction of emf.
7. The current in the outer loop produces a magnetic field directed into the plane of the figure. With increasing current the flux increases. The induced current must be anticlockwise which will create its own B in outward direction.



8. Induced emf will be same. The iron ring has higher resistance and current in it will be smaller.
9. Consider a current i_1 in coil 1 as shown. The field lines are as shown. The flux linked to coil 2 increases when i_1 is increased. Induced current (i_2) is such that it creates its own field in upward direction.

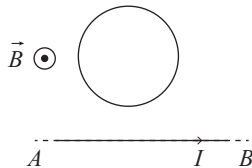


Now, replace the two coils with small magnets. They have their dipole moments in the same direction. the two magnets repel.

10. A clockwise current in P Produces a field directed leftward (on the axis). The induced current in Q must produce a rightward field so as to oppose the change in the flux. Current in Q is anticlockwise.

When the current is switched off the field due to P (towards left) decreases. Induced current in Q is clockwise so as to produce a field towards left.

11. Field due to AB is directed outward at the circular loop. When I increases, this field increases. Current induced in the loop must create inward field.



$\therefore$ Current must be clockwise.

12. Initial flux $\phi_1 = n \cdot \pi r^2 B$
Final flux $\phi_2 = 0$

$$\therefore \text{change in flux, } |\Delta\phi| = n\pi r^2 B$$

$$\text{Emf, } \varepsilon = \frac{\Delta\phi}{\Delta t}$$

$$\text{Current } i = \frac{\varepsilon}{R} = \frac{1}{R} \frac{\Delta\phi}{\Delta t}$$

$$\text{Charge flow } \Delta q = i\Delta t = \frac{1}{R} \Delta\phi = \frac{n\pi r^2 B}{R}$$

13. The area vector rotates with angular speed ω . If we assume that at $t = 0$, the situation is as shown in the figure, then total flux at $t = 0$ is

$$\phi = N B a^2 \cos 0^\circ$$

$$\text{At time } t, \phi = N B a^2 \cos(\omega t)$$

$$\therefore \varepsilon = -\frac{d\phi}{dt} = N B a^2 \omega \sin \omega t$$

$$\text{Maximum value of } \varepsilon \text{ is } N B a^2 \omega$$

14. The bulb will shine with full brightness if an emf of 10V is induced.

$$\varepsilon = A \frac{dB}{dt}$$

$$\text{or, } \frac{dB}{dt} = \frac{10\text{V}}{(0.1\text{m})^2} = 10^3$$

$$\text{or, } \frac{20T}{\Delta t} = 10^3$$

$$\Rightarrow \Delta t = 20 \text{ ms}$$

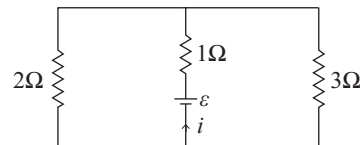
15. $\phi = B\pi R^2 = B\pi(R_0 + t)^2$

$$\therefore \varepsilon = \frac{d\phi}{dt} = 2B\pi(R_0 + t)$$

As flux is increasing, the induced current must be anticlockwise. So as to produce an outward field.

16. $\varepsilon = Blv = 0.1 \times 0.1 \times 1 = 0.01 \text{ V}$

The equivalent circuit is as shown. 2Ω and 3Ω in parallel have an equivalent of $\frac{2 \times 3}{2 + 3} = \frac{6}{5} = 1.2\Omega$



This 1.2Ω and 1Ω are in series.

$$\therefore R = 2.2\Omega$$

$$\therefore i = \frac{\varepsilon}{R} = \frac{0.01}{2.2} = \frac{1}{220} \text{ A}$$

17. Current due to motion of electron is

$$i = e \cdot f = e \frac{\omega}{2\pi} \quad [f = \text{Frequency of revolution}]$$

Magnetic field at the centre is

$$B = \frac{\mu_0 i}{2R} = \frac{\mu_0}{2R} \left(\frac{e\omega}{2\pi} \right) = \frac{\mu_0 e}{4\pi R} \omega$$

Flux linked with the loop of radius r is nearly equal to

$$\phi = B \cdot \pi r^2 = \frac{\mu_0 e r^2}{4R} \omega$$

$$\therefore |\epsilon| = \frac{d\phi}{dt} = \frac{\mu_0 e r^2}{4R} \frac{d\omega}{dt} = \frac{\mu_0 e r^2}{4R} \alpha$$

19. The ring can be replaced with a straight wire PQ .

$$\therefore \epsilon = Bv(2R)$$

Positive terminal is in the direction $\vec{V} \times \vec{B}$.

20. The positive terminal is indicated by the direction $\vec{V} \times \vec{B}$. $\vec{V}$ is toward east and $\vec{B}$ is downwards. Therefore, $\vec{V} \times \vec{B}$ is towards north. The north end is at higher potential.

21. The wire is along z direction. magnetic field is in xy plane. It is perpendicular to the wire. Emf is induced only because of velocity component in xy plane. The relevant velocity is $\vec{V}_0 = 2\hat{i} + 3\hat{j}$

In fact, the velocity component that is perpendicular to $\vec{B}$ is only important. That is why we write

$$\epsilon = lBv_0 \sin \theta \text{ when } \theta \text{ is angle between } \vec{V}_0 \text{ and } \vec{B}$$

$$\text{or, } \epsilon = l \left| (\vec{V}_0 \times \vec{B}) \right|$$

$$= 2 \text{ volt}$$

22. The figure shows the situation at time t .

$$AD = vt$$

$$EF = 2(AD \tan 45^\circ) = 2vt$$

Emf induced can be calculated as

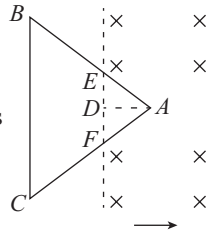
$$\phi = B \frac{1}{2} (AD)(EF) = Bv^2 t^2$$

$$\therefore |\epsilon| = \frac{d\phi}{dt} = 2Bv^2 t$$

Alternatively, one can think in terms of motional emf. Emf induced in EAF is same as emf induced in a straight wire EF .

$$\therefore \epsilon = Bvl = Bv(2vt) = 2v^2 Bt$$

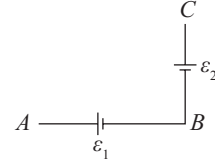
$\therefore$ graph of ϵ versus t is a straight line as shown in (D)



23. Mechanical power supplied by the external agent gets converted into heat.

$$\therefore H = Fv \quad \Rightarrow F = \frac{H}{v}$$

24. Emf induced is due to velocity component perpendicular to the length of the conductor.



$$\text{Emf induced in } AB \text{ is } \epsilon_{AB} = Bl_1 v_y \quad [\text{A positive}]$$

$$\text{Emf induced in } BC \text{ is } \epsilon_{CB} = Bl_2 v_x \quad [\text{C positive}]$$

$$\therefore \epsilon_{AC} = l_1 v_y - l_2 v_x$$

25. Power dissipated, $P = \frac{\epsilon^2}{R}$

Where ϵ = induced emf and R = resistance of the coil.

The emf becomes 4 times if number of turns is made 4 times.

The length of the wire used in making the coil becomes 4 times. Cross sectional area of the wire becomes $\frac{1}{4}$ times if radius is halved.

The resistance $R = \frac{\rho l}{A}$ becomes 16 times.

$$\therefore \text{Power dissipated, } P' = \frac{(4\epsilon)^2}{16R} = \frac{\epsilon^2}{R} = P$$

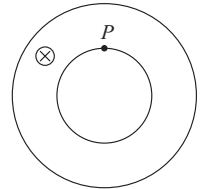
26. Emf between A and B is $\epsilon_1 = \frac{1}{2} B\omega L^2$

$$\text{Emf between } A \text{ and } C \text{ is } \epsilon_2 = \frac{1}{2} B\omega \left(\frac{L}{2} \right)^2 = \frac{1}{8} B\omega L^2$$

$$\therefore \text{Emf between } C \text{ and } B \text{ is } \epsilon_1 - \epsilon_2 = \frac{3}{8} B\omega L^2$$

27. Think of a loop passing through P .

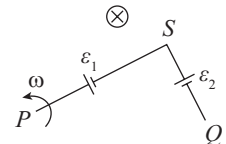
The induced current in this loop is anticlockwise when B is increased. The induced electric field will exert a tangentially rightward force on an electron at P



28. We can replace the curved part of the rod with a straight part (shown as SQ in the figure)

Emf induced in PS is

$$\epsilon_1 = \frac{1}{2} B\omega l^2$$

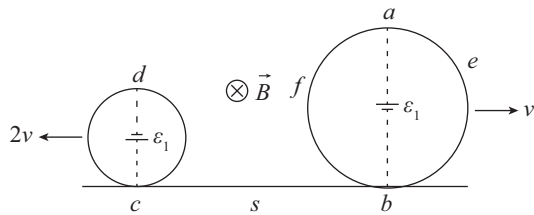


Emf induced in part SQ is $\varepsilon_2 = \frac{1}{2}B\omega L^2$

[Refer to example 33]

$\therefore$ Emf between P and Q is $\varepsilon_1 + \varepsilon_2 = \frac{1}{2}B\omega(l^2 + L^2)$

29. One can refer to example 10 and Your turn question 12.
30. Think of a ring as made of two semicircular parts such as aeb and afb . To find emf in a semicircular part we can replace it with a straight conductor ab . Emf induced in both semicircular parts will be $\varepsilon_1 = Bv(4r) = 4Bvr$

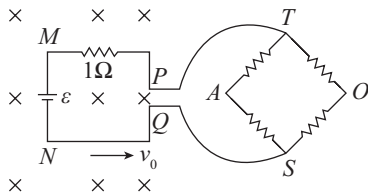


Similarly, emf induced in the two semicircular parts of the smaller ring will be $\varepsilon_2 = B(2v)(2r) = 4Bvr$

Note that the positive end of the emf is indicated by $\vec{v} \times \vec{B}$.

As one moves from d to c to b (along conducting surface S) to a the total emf is $\varepsilon_1 + \varepsilon_2 = 8Bvr$

31. The network of resistance is a balanced wheatstone bridge. The resistance between A and O will receive no current. It can be removed. The equivalent resistance is 3Ω .



Emf induced in the branch MN of the loop is

$$\varepsilon = Blv = 2 \times 0.1 \times v_0 = 0.2v_0$$

$$\therefore i = \frac{\varepsilon}{R}$$

$$\text{or, } 10^{-3} = \frac{0.2v_0}{3+1}$$

$$\text{or, } v_0 = 2 \times 10^{-2} \frac{m}{s}$$

32. Refer to example 11. The emf in the loop is the motional emf induced in length AC of the sliding connector.

$$PC = x = \sqrt{\frac{y}{a}}$$

$$\text{Length } AC = 2x = 2\sqrt{\frac{y}{a}}$$

Velocity of the connector after it gets displaced by y is given by

$$v^2 = 2\omega y$$

$$\text{or, } v = \sqrt{2\omega y}$$

$\therefore$ Emf induced in AC is $\varepsilon = Blv$

$$= B \left(2\sqrt{\frac{y}{a}} \right) (\sqrt{2\omega y})$$

$$= yB \sqrt{\frac{8\omega}{a}}$$

$$33. \tau = \frac{L}{R} = \frac{50 \times 10^{-3}}{10} = 5 \times 10^{-3} \text{ s} = 5 \text{ ms.}$$

The growth of current in LR circuit is given by

$$i = i_0 (1 - e^{-t/\tau})$$

$$\text{When } i = \frac{i_0}{2}, \quad \frac{1}{2} = 1 - e^{-t/\tau}$$

$$\text{or, } e^{-t/\tau} = \frac{1}{2} \quad \Rightarrow \quad \frac{t}{\tau} = \ln 2$$

$$\Rightarrow t = \tau (\ln 2) = 5 \times 0.693 \text{ ms} \approx 3.5 \text{ ms}$$

34. The inductor has some resistance as it dissipates energy as heat.

Value of inductance can be calculated as

$$\frac{1}{2} Li^2 = 32 \text{ J}$$

$$\text{or, } \frac{1}{2} L(4)^2 = 32 \quad \Rightarrow \quad L = 4 \text{ H}$$

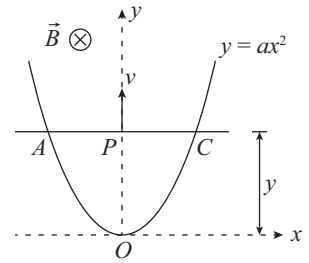
value of resistance can be calculated as

$$i^2 R = 320 \quad \Rightarrow \quad (4)^2 R = 320$$

$$\text{or, } R = 20\Omega$$

$$\therefore \tau = \frac{L}{R} = \frac{4}{20} = 0.2 \text{ s}$$

$$35. \frac{U_B}{P_R} = \frac{\frac{1}{2} Li^2}{Ri^2} = \frac{L}{2R} = \frac{\tau}{2}$$



36. Immediately after the switch is closed, there is no current in the branch having the inductor. There is current in the parallel branch having only 10Ω resistance. The current is $i_1 = \frac{10V}{10\Omega} = 1A$.

[This is the minimum current through the cell]

The current i_1 remains constant and the current (i_2) in the branch having the inductor begins to grow. After a long time the current i_2 is $\frac{10V}{10\Omega} = 1A$. Now the current through the cell is $2A$. This is the maximum current.

$$\therefore i_2 - i_1 = 1A$$

37. $U = \frac{1}{2}Li^2$

$$\Rightarrow \frac{dU}{dt} = Li \frac{di}{dt} = 2 \times 2 \times 4 = 16Js^{-1}$$

38. No matter how the current has changed, if current in an inductor is I , the energy stored in it is

$$U = \frac{1}{2}LI^2 = \frac{1}{2} \times 2 \times 2^2 = 4J$$

39. Magnetic energy is $U_B = \frac{1}{2}Li^2$

The maximum magnetic energy is $U_{Bo} = \frac{1}{2}Li_0^2$ where i_0 is the final steady state current.

The magnetic energy becomes $\frac{1}{4}$ of maximum when the current becomes i_0 .

$$\therefore \frac{i_0}{2} = i_0(1 - e^{-t/\tau})$$

$$\text{or, } e^{-t/\tau} = \frac{1}{2} \Rightarrow \frac{t}{\tau} = \ln 2$$

$$\text{or, } t = (\ln 2)\tau = 0.693 \times \frac{L}{R} = 0.693 \times \frac{80}{50} = 1.1 \text{ s}$$

40. With the switch closed, the inductor is like a plain wire and R_1 and R_2 are short circuited. Current through the inductor is $i_0 = \frac{V}{R}$.

After the switch is opened, the energy stored in the inductor gets dissipated in R_1 and R_2 .

Power dissipated in R_1 and R_2 are $\frac{\varepsilon^2}{R_1}$ and $\frac{\varepsilon^2}{R_2}$, respectively where ε is the emf induced in the inductor.

$$\frac{P_1}{P_2} = \frac{R_2}{R_1} \text{ (always)}$$

$\therefore$ Heat dissipated in the two resistors over any period of time will also be in the above ratio.

$$\therefore \frac{H_1}{H_2} = \frac{R_2}{R_1} \quad (i)$$

$$\text{Also total heat dissipated } H_1 + H_2 = \frac{1}{2}Li_0^2$$

$$\text{or, } H_1 + H_2 = \frac{1}{2}L\left(\frac{V}{R}\right)^2 \quad (ii)$$

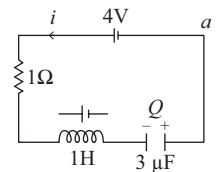
solving (i) and (ii)

$$H_1 = \left(\frac{R_2}{R_1 + R_2}\right) \left(\frac{1}{2}L\frac{V^2}{R^2}\right)$$

41. $V_a + 4 - iR - L\frac{di}{dt} + \frac{Q}{C} = V_a$

$$\text{or, } 4 - 2 \times 1 - 1 \times 4 + \frac{Q}{3\mu F} = 0$$

$$\text{or, } Q = 6\mu C$$



42. $L = \pi\mu_0 n^2 r^2 l$

$$= \mu_0 (\pi r^2) (nl)^2 \frac{1}{l}$$

$$\text{or, } L = \frac{\mu_0 AN^2}{l}$$

$$B = \mu_0 ni = \mu_0 \frac{N}{l} i$$

$$\text{or, } i = \frac{BL}{\mu_0 N} = \frac{BAN}{L}$$

43. Current I is decreasing. This means that emf induced in the inductor has its positive end towards right.

$$|\varepsilon| = L \frac{di}{dt} = (5mH) \times (10^3 As^{-1}) = 5 \text{ volt.}$$

$$V_A - 5 \times 1 + 15 + 5 = V_B$$

$$\text{or, } V_B - V_A = 15V$$

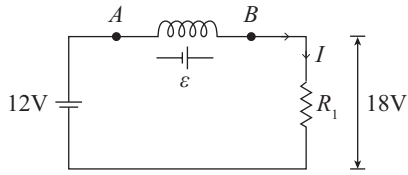
44. When the switch is closed, effective resistance is parallel combination of R_1 and R_2

$$R_0 = \frac{R_1 R_2}{R_1 + R_2} = \frac{4 \times 8}{4 + 8} = \frac{8}{3}\Omega$$

$$\text{Current through } L \text{ is } I = \frac{12V}{\frac{8}{3}\Omega} = \frac{9}{2}A$$

Immediately after the switch is opened, the current in the inductor remains $I = \frac{9}{2}A$ and the only resistance in the circuit is $R_1 = 4\Omega$.

Drop of potential in $R_1 = \frac{9}{2} \times 4 = 18V$



Immediately after S is opened

Obviously, induced emf in the inductor is 6V with B at higher potential.

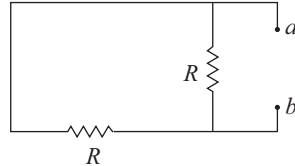
45. For a quick answer, refer to the note given at the end of example 20.

The effective resistance is obtained across the inductor after shorting the cell.

$R_0 = R$ and R in parallel

$$= \frac{R}{2}$$

$$\therefore \tau = \frac{L}{R_0} = \frac{2L}{R}$$



After a long time, the resistance connected in parallel to L will have no current as the inductor will be a zero resistance path.

Current through the inductor will be $i_0 = \frac{\mathcal{E}}{R}$

$$\therefore i = i_0 (1 - e^{-t/\tau}) = \frac{\mathcal{E}}{R} \left(1 - e^{-\frac{tR}{2L}} \right)$$

46. Magnetic field due to current in side AB , at the centre of the square is

$$B = \frac{\mu_0 I}{4\pi \frac{L}{2}} (\sin 45^\circ + \sin 45^\circ) \odot$$

$$= \frac{\mu_0 I}{\sqrt{2}\pi L} \odot$$

Field due to all four arms is

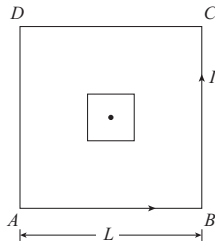
$$B_0 = 4B = \frac{2\sqrt{2}\mu_0 I}{\pi L}$$

This field can be treated as uniform over the entire area of the smaller square as it is very small in size.

$\therefore$ Flux linked with the smaller loop is

$$\phi = B \cdot A = \frac{2\sqrt{2}\mu_0 I}{\pi L} \cdot l^2 = \left(\frac{2\sqrt{2}\mu_0 l^2}{\pi L} \right) I$$

$$\therefore M = \frac{2\sqrt{2}\mu_0 l^2}{\pi L}$$



$$47. |\mathcal{E}| = \left| M \frac{di}{dt} \right|$$

given $\mathcal{E} = 40,000 V$ and

$$\frac{di}{dt} = \frac{4A}{10 \times 10^{-6} s} = 4 \times 10^5 A/s$$

$$\therefore M = \frac{40000}{4 \times 10^5} = 0.1 H$$

48. When a current I is given to one of the coils, maximum number of magnetic field lines will pass through the other coil in case (a).

49. Current I_1 changes with time as

$$I_1 = I_0 (1 - e^{-t/\tau})$$

Magnetic field at the centre due to this current will be proportional to the current itself.

$$\therefore B \propto (1 - e^{-t/\tau})$$

Flux linked with the ring

$$\phi = MI_1$$

(M = mutual inductance)

$\therefore$ Emf in the ring

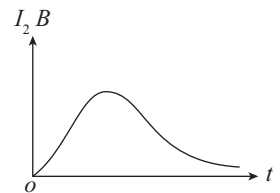
$$\mathcal{E} = \frac{d\phi}{dt} \propto \frac{dI_1}{dt}$$

or, $\mathcal{E} \propto e^{-t/\tau}$

$\therefore$ current in the ring $I_2 \propto e^{-t/\tau}$

The product $I_2 \cdot B \propto e^{-t/\tau} (1 - e^{-t/\tau})$

The variation of this quantity will be as shown in the graph. Note that at $t = 0$, the value of $I_2 B$ is zero. At $t = \infty$, the value is again zero.



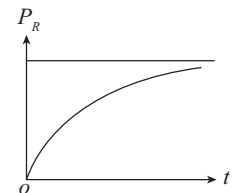
$$50. P_L = Li \frac{di}{dt} = Li_0 (1 - e^{-t/\tau}) i_0 e^{-t/\tau}$$

$$= Li_0^2 e^{-t/\tau} (1 - e^{-t/\tau})$$

$$P_R = i^2 R = i_0^2 (1 - e^{-t/\tau})^2 \cdot R$$

The variation of P_L will be similar to the graph drawn in the last solution.

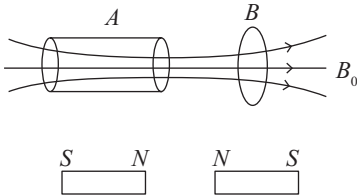
The variation of P_R is as shown here.





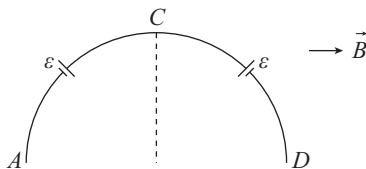
Worksheet 2

1. B_0 is field produced by the current in A . When the current in A increases, B_0 becomes stronger and the flux in ring B increases. The current induced in B is so as to oppose this increasing B_0 . Replacing A and B with equivalent magnets we can see that they will repel. Even if the direction of current in A is opposite so that it produces field (B_0) directed leftward, the two will repel (show this yourself).



Similarly, one can show that when the current in A decreases, the two attract one another.

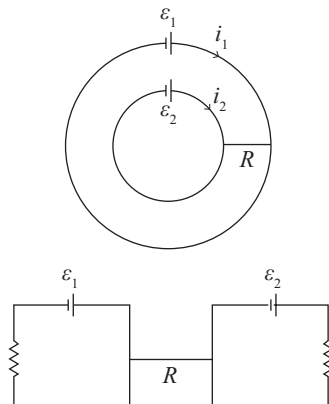
2. For emf to get induced between the ends, the three vectors $\vec{B}$, $\vec{v}$ and $\vec{L}$ shall be mutually perpendicular, or at least they should have components which are mutually perpendicular.



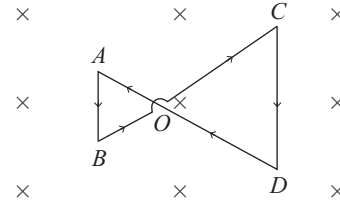
Velocity is outward

Movement of the semicircle in xy plane will produce no emf. When the ring moves along z direction, an emf is induced in the segment AC as well as in CD . But the net emf between A and D is still Zero.

3. The emfs induced in the two loops are as shown. They drive currents i_1 and i_2 as shown. There cannot be a current in path R . It is like two electric loops connected by a wire. The connecting wire cannot have a current.



4. Let's apply Faraday's Law. The arrows in the diagram show the assumed positive direction of current in the loop. The direction of area vector is indicated by the right hand thumb when the fingers are curled in the sense of the current. Area CDO is positive and area OAB is negative.

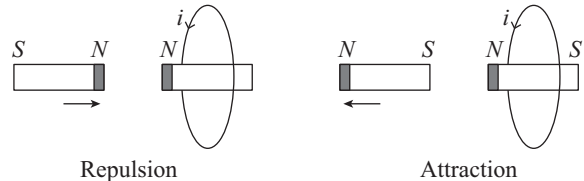


$$\begin{aligned}\therefore \phi &= B \cdot (\text{area } CDO) - B \cdot (\text{area } ABO) \\ &= B(A_1 - A_2)\end{aligned}$$

$$\therefore \varepsilon = -\frac{d\phi}{dt} = -(A_1 - A_2) \frac{dB}{dt}$$

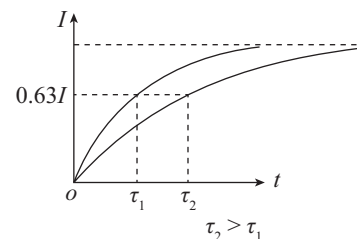
$\therefore$ emf is negative. The current flows in direction opposite to that indicated by the arrows.

5. From Lenz's Law, the approaching magnet must be repelled by the induced current and the receding magnet must be attracted.



6. The source voltage is same in both the circuits. Final steady state current is also same in both the circuits (see the graphs).

$$\therefore \frac{V}{R_1} = \frac{V}{R_2} \Rightarrow R_1 = R_2$$



The graph shows that $\tau_2 > \tau_1$

$$\Rightarrow \frac{L_2}{R_2} > \frac{L_1}{R_1} \Rightarrow L_2 > L_1$$

7. Steady state current remains same

$$\therefore \frac{E}{R} \text{ must remain constant}$$

The time constant has increased.

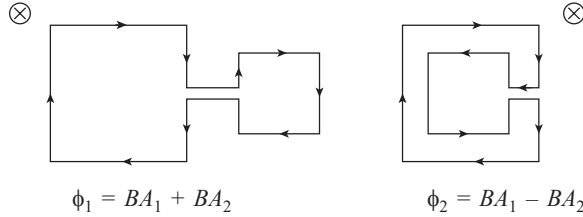
$\therefore \frac{L}{R}$ has increased.

The options (A) and (C) satisfy the above requirements.

8. The arrows show the assumed positive direction in the two loops.

A_1 = area of larger square = L^2 and

A_2 = area of the smaller square = l^2



In the first figure, as the right hand fingers are curled in the sense of the current the thumb points inwards for both the square loops. Both $\vec{A}_1$ and $\vec{A}_2$ are inwards. In the second figure the thumb is directed inward when fingers are curled in the sense of current in the outer square and it is directed outward when the fingers are curled in the sense of current in the inner square.

$$\varepsilon_1 = -\frac{d\phi_1}{dt} = -B(A_1 + A_2) \left(\frac{dB}{dt} \right) = a \quad \text{positive number}$$

$$\left[\therefore \frac{dB}{dt} \text{ is negative} \right]$$

$\therefore$ Current is in the direction indicated in the figure above

$$\varepsilon_2 = -\frac{d\phi_2}{dt} = -B(A_1 - A_2) \left(\frac{dB}{dt} \right) = a \quad \text{positive number}$$

$\therefore$ Current is in the direction indicated above.

9. current is constant $i = \frac{dq}{dt} = 1$

q is increasing at a constant rate and the potential difference is also increasing at a constant rate. This means that the element in the box is a capacitor.

10. Emf induced in quarter circle OP is same as emf induced in a straight wire OP .

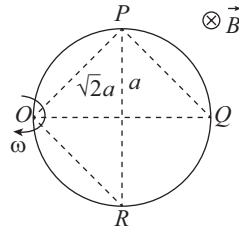
$$V_{po} = \frac{1}{2} B\omega(\sqrt{2}a)^2 = B\omega a^2$$

Similarly, $V_{Ro} = B\omega a^2$

$$V_{Qo} = \frac{1}{2} B\omega(2a)^2 = 2B\omega a^2$$

Note that $PQ \perp OP$ and refer to example 33.

$$V_{QP} = \frac{1}{2} B\omega(\sqrt{2}a)^2 = B\omega a^2$$



$$11. P = Li \frac{di}{dt}$$

$$P_1 = P_2$$

$$\Rightarrow L_1 I_1 \frac{dI_1}{dt} = L_2 I_2 \frac{dI_2}{dt}$$

$$\text{or, } 8I_1 = 2I_2$$

$$\left[\therefore \frac{dI_1}{dt} = \frac{dI_2}{dt} \right]$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{1}{4}$$

$$\frac{V_2}{V_1} = \frac{L_2 \frac{dI_2}{dt}}{L_1 \frac{dI_1}{dt}} = \frac{L_2}{L_1} = \frac{2}{8} = \frac{1}{4}$$

$$\frac{W_2}{W_1} = \frac{\frac{1}{2} L_2 I_2^2}{\frac{1}{2} L_1 I_1^2} = \left(\frac{L_2}{L_1} \right) \left(\frac{I_2}{I_1} \right)^2 = \frac{1}{4} \times \left(\frac{4}{1} \right)^2 = \frac{4}{1}$$

12. The two parallel paths are independent. Final current in both the paths will be same. Due to presence of inductance, the growth of current in B_1 gets delayed.

13. Inductance is higher for the iron core inductor.

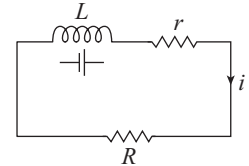
14. In steady state the current through the solenoid is $i_s = \frac{E}{r}$.

As the switch is opened, the initial current through the solenoid will remain i_0 .

After this the current decays with a time constant of

$$\tau = \frac{L}{R+r}$$

$$\therefore i = i_s e^{-t/\tau} = \frac{E}{r} e^{-\frac{(R+r)t}{L}}$$



Decaying current

At time $t = 0$, the current is i_0 and the emf induced in the inductor must be

$$\varepsilon = i_s(R+r) = \frac{E}{r}(R+r)$$

$$\text{Energy stored in the solenoid } U = \frac{1}{2} Li_s^2 = \frac{L}{2} \left(\frac{E}{r} \right)^2$$

This energy is dissipated as heat in r and R . The amount of heat dissipated in the two resistances will be in the ratio of resistances

$$\frac{H_1}{H_2} = \frac{r}{R} \quad (i)$$

$$\text{And } H_1 + H_2 = \frac{L}{2} \left(\frac{E}{r} \right)^2 \quad (ii)$$

Solving (i) and (ii) gives

$$H_1 = \left(\frac{r}{R+r} \right) \frac{L E^2}{2 r^2} = \frac{L E^2}{2r(R+r)}$$

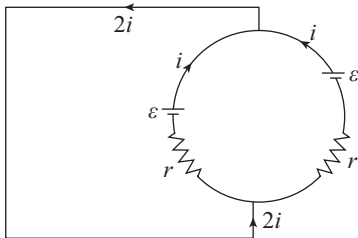
15. After a long time, there will be no current in the branch having capacitor. The inductor will behave like a plain wire.

$$\therefore \text{Current in the inductor } i = \frac{E}{2R}$$

Initially, the current flows only through the capacitor branch. The capacitor acts like a zero resistance. The initial current through the cell is $i_0 = \frac{E}{2R}$.

16. The emf induced in two semicircular parts is equal to $\varepsilon = Bvd$.

The equivalent electric circuit is as shown.



$$r = \lambda \pi r = \lambda \pi \frac{d}{2}$$

Using Kirchhoff's law we can write

$$\varepsilon = ir \Rightarrow i = \frac{\varepsilon}{r} = \frac{Bvd}{\lambda \pi \frac{d}{2}} = \frac{2Bv}{\lambda \pi}$$

Magnetic force on one semicircular part is $= idB$

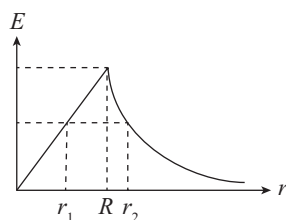
$$= \frac{2B^2vd}{\pi \lambda}$$

Total force on the ring is twice of this

$$F = \frac{4B^2vd}{\pi \lambda}$$

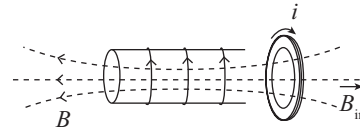
18. Then will be induced electric field at P as well as Q . For an ideal solenoid there is no magnetic field at Q .

The variation of induced electric field with distance (r) from the centre is as shown. The magnitude of E can be same at points at distances r_1 ($r_1 < R$) and r_2 ($r_2 > R$).



Worksheet 3

1. (a) Magnetic field created by the solenoid at the ring is towards left. When the switch is put on the flux through the ring increases and a clockwise (seen from the side of the solenoid) current is induced in it. This current (i) produces a field (B_{in}) directed to right, thereby weakening the flux



- (b) When current in the solenoid is constant there is no change in B . There is no current induced in the ring
(c) When the current in solenoid decreases, flux in the ring also decreases. The induced current is anticlockwise so as to create a field towards right.
2. Both coils produce field in same direction at the coil B .

$$3. \phi = \pi r^2 B; \quad \varepsilon = \left| \frac{d\phi}{dt} \right| = \left| \pi r^2 \frac{dB}{dt} \right| = \pi r^2 k$$

Since B is decreasing the induced current is clockwise so as to produce a magnetic field directed inward.

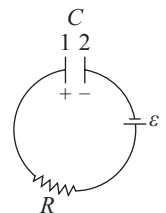
The equivalent circuit is as shown.

It is an RC circuit with a source of constant emf.

The final maximum change on the capacitor will be

$$Q_0 = CE = \pi r^2 kC$$

Plate 1 is positive.



4. (a) Area of the square $= a^2$

$$\text{Area of the rhombus} = a^2 \sin \theta$$

[Recall that area of parallelogram with $\vec{a}$ and $\vec{b}$ as adjacent sides is $|\vec{a} \times \vec{b}| = ab \sin \theta$]

$$\therefore \text{Change in area of the loop } \Delta A = a^2(1 - \sin \theta)$$

$$\text{Change in flux } \Delta \phi = B \Delta A = Ba^2(1 - \sin \theta)$$

$$\text{Induced emf has magnitude } \varepsilon = \frac{\Delta \phi}{\Delta t} = \frac{Ba^2(1 - \sin \theta)}{\Delta t}$$

$$\text{Induced current is } i = \frac{\varepsilon}{R} = \frac{Ba^2}{\Delta t R}(1 - \sin \theta)$$

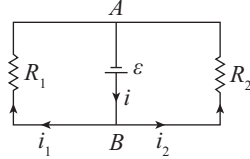
- (b) As the loop is deformed, its area decreases. The flux decreases. Induced current must be in a direction

so as to produce a magnetic field in the direction of the field so as to strengthen it. Therefore the direction of the current must be clockwise.

5. (a) Emf induced in the sliding bar $\varepsilon = Blv$. The equivalent electric circuit is as shown

$$p = p_1 + p_2 = \frac{\varepsilon^2}{R_1} + \frac{\varepsilon^2}{R_2}$$

$$= B^2 l^2 v^2 \left(\frac{R_1 + R_2}{R_1 R_2} \right)$$



(b) $i_1 = \frac{\varepsilon}{R_1}; \quad i_2 = \frac{\varepsilon}{R_2}$

$$i = i_1 + i_2 = Blv \left(\frac{R_1 + R_2}{R_1 R_2} \right)$$

External force on the bar = magnetic force on the bar

$$\text{or, } F = ilB = B^2 l^2 v \left(\frac{R_1 + R_2}{R_1 R_2} \right)$$

- (c) If the resistor R_1 is burnt, only R_2 remains. The power dissipation becomes $p = \frac{\varepsilon^2}{R_2}$

$\therefore$ Power dissipated in the circuit gets reduced

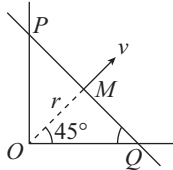
6. The figure shows the situation at time t .

$$r = OM = vt$$

$$OQ = OP = \sqrt{2}r = \sqrt{2}vt$$

Area of $\triangle OPQ$ is

$$S = \frac{1}{2}(\sqrt{2}vt)(\sqrt{2}vt) = v^2 t^2$$



Flux linked with area S is

$$\phi = B \cdot S = Bv^2 t^2$$

$$\text{Emf induced, } E = \left| \frac{d\phi}{dt} \right|$$

$$= 2Bv^2 t$$

Resistance of the loop, $R = \lambda[OP + OQ + PQ]$

$$= \lambda[\sqrt{2}r + \sqrt{2}r + 2r] = 2(\sqrt{2} + 1)r\lambda$$

$$= 2(\sqrt{2} + 1)v\lambda t$$

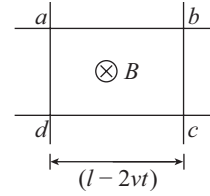
$$\therefore i = \frac{\varepsilon}{R} = \frac{2Bv^2 t}{2(\sqrt{2} + 1)v\lambda t} = \frac{Bv}{(\sqrt{2} + 1)\lambda}$$

7. (a) Situation at time t is as shown.

Area of the square is $S = (l - 2vt)^2$.

Flux linked with the square loop is

$$\phi = BS = B(l - 2vt)^2$$



$$\therefore \varepsilon = \left| \frac{d\phi}{dt} \right| = 4B(l - 2vt)v$$

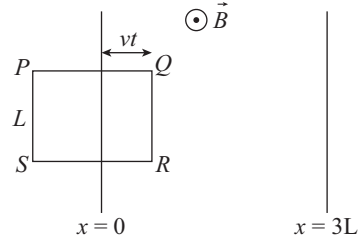
- (b) Resistance $R = \lambda \cdot 4(l - 2vt)$

$$\therefore i = \frac{\varepsilon}{R} = \frac{Bv}{\lambda}$$

Area is decreasing and therefore the flux is decreasing. The induced current is clockwise so as to produce an inward field and strengthen the magnetic field (B)

- (c) Magnetic force on a rod is $F_B = i(l - 2vt)B = \frac{B^2 v}{\lambda}(l - 2vt)$. The same force is needed in opposite direction by the external agent

8. (a) The loop will be completely inside the field region at time $t_1 = \frac{L}{v}$. In the interval $0 < t < t_1$, the flux linked with the loop is increasing linearly.

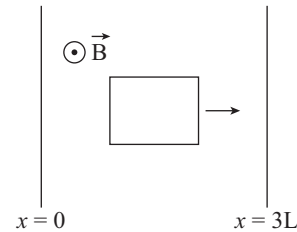


Flux increasing

$$\phi = B(L \cdot vt)$$

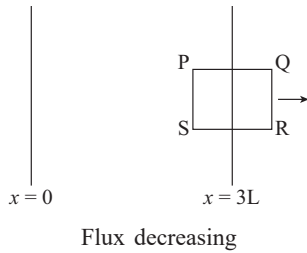
At $t = t_1$ the flux is $\phi = BL^2$

In the interval $t_1 < t < t_2$ the loop is completely inside the field region. Here $t_2 = \frac{3L}{v}$. In this interval the flux does not change

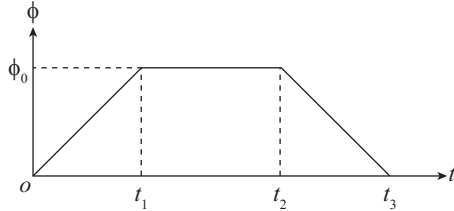


Flux constant

In the interval $t_2 < t < t_3$ the loop exits the field region. Here $t_3 = \frac{4L}{v}$. In this interval the flux decreases linearly to zero.



The variation of the flux is shown in the graph below.



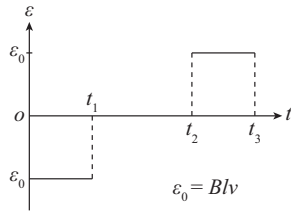
(b) Emf induced is $\varepsilon = \left| \frac{d\phi}{dt} \right|$

It means the induced emf in the loop can be obtained by slope of the $\phi - t$ graph. The direction of induced emf can be obtained using Lenz's Law.

Alternatively, one may think in terms of motional emf. In the interval $0 < t < t_1$, emf is induced in the side QR only with the end R positive [positive end is indicated by $\vec{v} \times \vec{B}$]. Therefore it is negative emf and its magnitude is Blv .

In the interval $t_1 < t < t_2$ the emf in the loop is zero.

In the interval $t_2 < t < t_3$, emf is induced only in side PS and end S is positive. This emf will drive an anticlockwise current. It is positive emf and its magnitude is Blv .



- (c) The magnetic force opposes the motion of the loop. External force must be in the direction of motion.

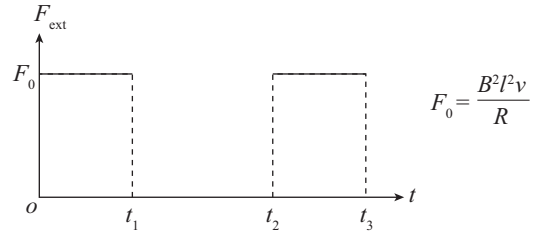
In the interval $0 < t < t_1$, the magnetic force acts on arm QR

$$F_{\text{ext}} = F_B = iLB = \frac{Blv}{R} LB = \frac{B^2 l^2 v}{R} = a \text{ constant}$$

In the interval $t_1 < t < t_2$, there is no current in the loop. No force is needed.

In the interval $t_2 < t < t_3$, the magnetic force acts on the arm PS

$$F_{\text{ext}} = F_0 = \frac{B^2 l^2 v}{R}$$



9. Velocity of the bar at time t is $v = \alpha t$
 x coordinate of the bar at time t is $x = \frac{1}{2} \alpha t^2$
 $\therefore$ Induced emf $\varepsilon = Bvl$

$$= B_0 \left[1 + a \left(\frac{1}{2} \alpha t^2 \right)^2 \right] (\alpha t) \cdot L$$

$$= B_0 \alpha L \left(1 + \frac{a \alpha^2 t^4}{4} \right) t$$

End A is positive

10. Field at the location of side AB is, $B_1 = \frac{\mu_0 I}{2\pi a}$

Emf induced in AB is, $\varepsilon_1 = B_1 v(2a) = \frac{\mu_0 I v}{\pi}$

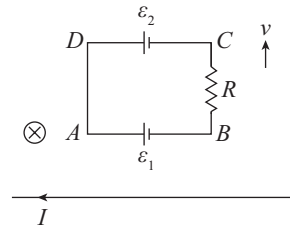
Field (B) due to current I is directed inwards. Positive terminal of the emf is indicated by the direction $\vec{v} \times \vec{B}$.

Similarly, field at the location of side CD is

$$B_2 = \frac{\mu_0 I}{2\pi(2a)}$$

Emf induced in CD is, $\varepsilon_2 = B_2 v(2a) = \frac{\mu_0 I v}{2\pi}$

Polarity of ε_2 is as shown.



Net emf in the loop is $\varepsilon = \varepsilon_1 - \varepsilon_2 = \frac{\mu_0 I v}{2\pi}$

$\therefore$ Current in the loop is $i = \frac{\varepsilon}{R} = \frac{\mu_0 I v}{2\pi R}$

11. Consider a segment of length dx in the conducting bar AB . Magnetic field at the location of the segment is

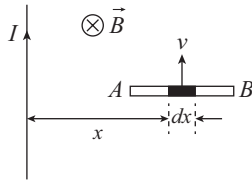
$$B = \frac{\mu_0 I}{2\pi x}$$

Emf induced in the small segment is

$$d\varepsilon = Bv(dx) = \frac{\mu_0 Iv}{2\pi} \frac{dx}{x}$$

$$\therefore \varepsilon = \int dE = \frac{\mu_0 Iv}{2\pi} \int_r^{r+L} \frac{dx}{x} = \frac{\mu_0 Iv}{2\pi} \ln\left(\frac{r+L}{r}\right)$$

$$\text{or, } \varepsilon = \frac{\mu_0 Iv}{2\pi} \ln\left(1 + \frac{L}{r}\right)$$



12. Emf induced in rod ab is $\varepsilon_1 = Bv_1 L \sin 60^\circ$

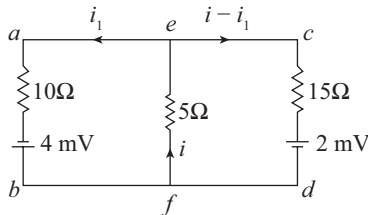
Here L = length of ab

$L \sin 60^\circ = l = 10 \text{ cm}$ = separation between the rails

$$\therefore \varepsilon_1 = Bv_1 l = 10 \times 10^{-3} \times 4 \times 0.1 = 4 \text{ mV}$$

$$\text{Emf induced in } cd \text{ is } \varepsilon_2 = Bv_2 l = 10 \times 10^{-3} \times 2 \times 0.1 = 2 \text{ mV}$$

The equivalent circuit is as shown. Current distribution is also shown. We will apply Kirchhoff's loop law for the two loops.



For loop $abfe$:

$$5i + 10i_1 = 4 \quad (i)$$

For loop $cdfe$:

$$15(i - i_1) + 5i = -2$$

$$\text{or, } 20i - 15i_1 = -2$$

$$\text{Solving (i) and (ii), gives } i = \frac{8}{55} \text{ mA}$$

[In equations (i) and (ii), the voltages in mV]

13. $\frac{dB}{dt}$ = slope of the graph = 2.5 Ts^{-1}

$$\text{Area of } \triangle ABD = \frac{1}{2} \times 1 \times \sqrt{2^2 - \left(\frac{1}{2}\right)^2} = \frac{\sqrt{15}}{4} \text{ m}^2$$

$$\text{Area of } \triangle BCD = \frac{\sqrt{3}}{4} \text{ m}^2$$

Emf induced in ABD is

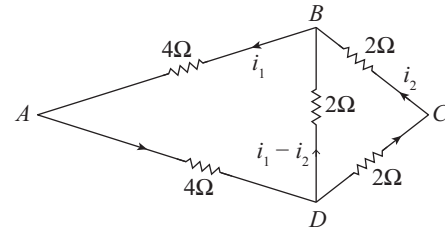
$$\varepsilon_1 = \left| \frac{d\phi}{dt} \right| = \frac{\sqrt{15}}{4} \left| \frac{dB}{dt} \right| = \frac{\sqrt{15}}{4} \times 2.5 = \frac{5\sqrt{15}}{8} \text{ V}$$

$$\text{Emf induced in } BCD \text{ is } \varepsilon_2 = \frac{\sqrt{3}}{4} \frac{dB}{dt} = \frac{5\sqrt{3}}{8} \text{ V}$$

Resistance of $AB = 4\Omega$ = resistance of AD

Resistance of $BC = 2\Omega$ = resistance of CD = resistance of BD

The diagram shows the current distribution. Note that we have not shown the emfs as we do not know the locations of the emfs. Only thing that we know is that there is an emf ε_1 in the loop ABD and there is an emf ε_2 in the loop BCD . The sense of the two emfs can be decided using the Lenz's Law. In both the loops the two emfs tend to drive anticlockwise currents.



In loop $BADB$:

$$\frac{5\sqrt{15}}{8} = 4i_1 + 4i_1 + 2(i_1 - i_2) \quad (i)$$

In loop $BDCB$:

$$\frac{5\sqrt{3}}{8} = 2i_2 + 2i_2 - 2(i_1 - i_2) \quad (ii)$$

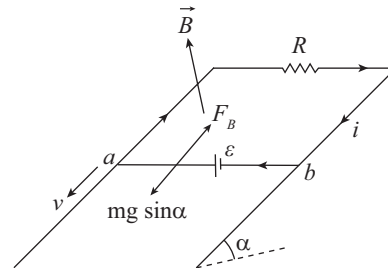
Solving (i) and (ii) gives

$$i_1 = 0.3 \text{ A}, i_2 = 0.28 \text{ A}, i_1 - i_2 = 0.02 \text{ A}$$

14. (a) From the concept of motional emf, the positive terminal of the induced emf in the rod is directed towards $\vec{v} \times \vec{B}$.

$\therefore$ End a of the rod is positive. Current in the loop is clockwise. Current in the rod is from a to b .

- (b) Induced emf $\varepsilon = BLv$



$$\text{Induced current } i = \frac{BLv}{R}$$

$$\text{Magnetic force, } F_B = iLB = \frac{B^2 L^2 v}{R}$$

When the rod has attained terminal speed (i.e., acceleration is zero)

$$F_B = mg \sin \alpha$$

$$\frac{B^2 L^2 v_0}{R} = mg \sin \alpha$$

$$\Rightarrow v_0 = \frac{mgR \sin \alpha}{B^2 L^2}$$

$$(c) \quad i = \frac{BLv_0}{R} = \frac{mg \sin \alpha}{BL}$$

$$(d) \quad \text{Power dissipated, } P = i^2 R = \frac{m^2 g^2 \sin^2 \alpha \cdot R}{B^2 L^2}$$

Gravitational potential energy also decreases at the same rate. It is the gravitational energy that is getting converted into heat in the resistance.

15. Direction of F_B becomes horizontal. At terminal speed

$$F_B \cos \alpha = mg \sin \alpha$$

$$\text{or, } \frac{B^2 L^2 v_0}{R} \cos \alpha = mg \sin \alpha$$

$$\text{or, } v_0 = \frac{mgR \tan \alpha}{B^2 L^2}$$

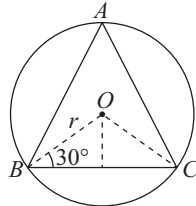
All other calculation steps are similar to the last question

16. Side length of the triangle is $a = 2r \cos 30^\circ = \sqrt{3}r$

Area of $\triangle ABC$ is

$$A = \frac{\sqrt{3}}{4} a^2$$

$$= \frac{3\sqrt{3}}{4} r^2$$



Emf induced in the triangle is

$$\varepsilon = A \frac{dB}{dt} = \frac{3\sqrt{3}}{4} r^2 \cdot \alpha$$

Due to symmetry, the emf induced in each side of the triangle is

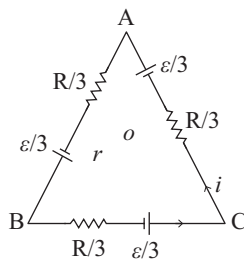
$$\frac{\varepsilon}{3} = \frac{\sqrt{3}}{4} r^2 \cdot \alpha$$

The equivalent electric circuit is as shown

$$i = \frac{\varepsilon}{R}$$

$$V_B - i \frac{R}{3} + \frac{\varepsilon}{3} = V_C$$

$$\text{or, } V_B - V_C = 0$$

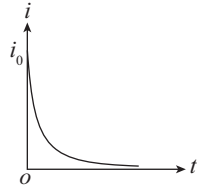


$$17. \quad \text{Induced emf } \varepsilon = L \frac{di}{dt}$$

The induced emf is large when $\frac{di}{dt}$ is large. As the switch is opened, it creates a gap in the circuit and this air gap acts as a large resistance.

The initial current, after opening the switch, is same as the current (i_0) before opening the switch. The current drops sharply as the time constant is small

($\tau = \frac{L}{R}$, R is large).



The current in the circuit is never larger than i_0 and the bulb is safe.

18. Emf induced in $AB = 0$

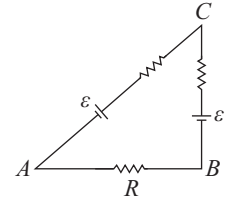
Emf induced in $BC = Blv$

Emf induced in $AC = Blv$

Not emf in the loop = 0

Current = 0

$$V_{AB} = 0$$



19. (a) Emf induced in side ab is $\varepsilon = Blv$

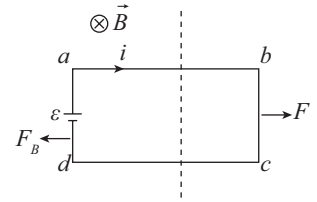
Current in the frame is $i = \frac{Blv}{R}$

Magnetic force on side ad is

$$F_B = i l B = \frac{B^2 l^2}{R} v$$

$$\therefore ma = F - F_B$$

$$\text{or, } a = \frac{F}{m} = \frac{B^2 l^2 v}{Rm}$$



- (b) When F_B becomes equal to F , the frame acquires its terminal velocity

$$\frac{B^2 l^2 v_0}{R} = F \quad \Rightarrow \quad v_0 = \frac{RF}{B^2 l^2} \quad (i)$$

- (c) At time t , $\frac{dv}{dt} = \frac{F}{m} - \frac{B^2 l^2}{mR} v$

$$\text{or, } \int_0^v \frac{dv}{RF - B^2 l^2 v} = \frac{1}{mR} \int_0^t dt$$

$$\text{or, } \left[\ln(RF - B^2 l^2 v) \right]_0^v = -\frac{B^2 l^2}{mR} t$$

$$\text{or, } \ln \left(\frac{RF - B^2 l^2 v}{RF} \right) = -\frac{Ft}{mv_0} \quad [\text{using (i)}]$$

$$\text{or, } 1 - \frac{v}{v_0} = e^{-\frac{Ft}{mv_0}}$$

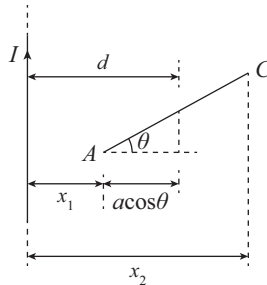
$$\text{or, } v = v_0 \left(1 - e^{-\frac{Ft}{mv_0}} \right)$$

20. Distance of end A of the rod from the wire is $x_1 = d - a \cos \theta$

Distance of the end B of the rod from the wire is $x_2 = d + a \cos \theta$

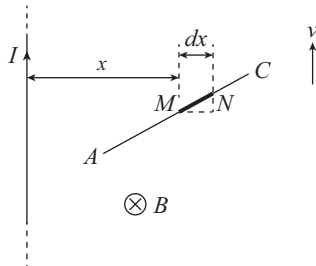
Magnetic field at a distance x from the wire

$$\text{is } B = \frac{\mu_0 I}{2\pi x} \otimes$$



Geometry of the situation

The emf developed in a small segment MN of the rod is $d\varepsilon = Bvdx = \frac{\mu_0 Iv}{2\pi} \frac{dx}{x}$

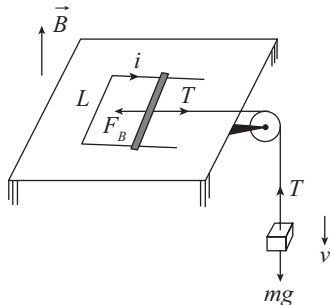


$\therefore$ Emf induced in the rod AC is

$$\varepsilon = \frac{\mu_0 Iv}{2\pi} \int_{x_1}^{x_2} \frac{dx}{x}$$

$$\text{or, } \varepsilon = \frac{\mu_0 Iv}{2\pi} \ln \left(\frac{x_2}{x_1} \right) = \frac{\mu_0 Iv}{2\pi} \ln \left(\frac{d + a \cos \theta}{d - a \cos \theta} \right)$$

21. Let the velocity of the block and the rod be v at some instant



Emf induced in the rod $\varepsilon = BLv$

Current in the loop, $i = \frac{BLv}{R}$

Magnetic force on the rod is

$$F_B = iLB = \frac{B^2 L^2 v}{R}$$

For the motion of the rod mass system we can write

$$mg - F_B = ma \quad (i)$$

- (a) When terminal velocity is attained the acceleration is zero.

$$\text{or, } F_B = mg$$

$$\text{or, } \frac{B^2 L^2 v_0}{R} = mg$$

$$\text{or, } v_0 = \frac{mgR}{B^2 L^2}$$

- (b) Use equation (i) with $v = \frac{v_0}{2}$ and find a .

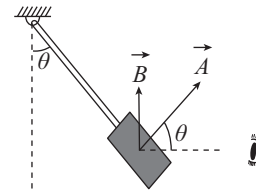
22. Position at time t is as shown flux through the loop is

$$\phi = BA \cos(90 - \theta) = BA \sin \theta$$

$$= BA \sin[\theta \cdot \sin \omega t]$$

Induced emf is

$$\varepsilon = -\frac{d\phi}{dt} = -BA \cdot \omega \cos[\theta \cdot \sin \omega t] \cdot \cos \omega t$$



Anticlockwise induced current is positive for the observer shown

For the shown direction of area vector $\vec{A}$, an anticlockwise induced current is positive. The emf is positive when it tends to drive an anticlockwise current, when seen from the right.

23. (a) Induced electric field on the circumference of the groove is $E = \frac{R}{2} \frac{dB}{dt}$

Force on charge due to the induced electric field is

$$F = Eq = \frac{Rq}{2} \frac{dB}{dt}$$

$$\text{Impulse of the force} = \int F dt = \frac{Rq}{2} \int dB = \frac{RqB}{2}$$

[$\because$ Change in magnetic field has magnitude of B]

$\therefore$ Momentum acquired by the particle is $P = \frac{RqB}{2}$

Angular momentum of the particle about the centre is

$$L = P \cdot R = \frac{R^2 q B}{2}$$

- (b) $M = \left(\frac{q}{2m} \right) L = \frac{q^2 R^2 B}{4m}$

$$24. \quad \varepsilon_{AC} = \frac{1}{2} B \omega l^2 = \frac{1}{2} B \omega (2a)^2 = 2B\omega a^2$$

$$\varepsilon_{AB} = \frac{1}{2} B \omega (\sqrt{2}a)^2 = B\omega a^2$$

25. Field inside the solenoid is $B = \mu_0 n i$. When i changes, B changes.

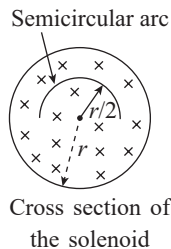
$$\frac{dB}{dt} = \mu_0 n \frac{di}{dt} = \mu_0 n b$$

The time changing B induces an electric field. The electric field at a distance $r/2$ from the axis of the solenoid is

$$\varepsilon = \frac{r/2}{2} \frac{dB}{dt} = \frac{r}{4} \mu_0 n b$$

The electric field is tangential along the arc. The emf induced along the arc is

$$\varepsilon = \int \vec{E} \cdot d\vec{l} = \frac{\mu_0 n b r}{4} \cdot \pi \left(\frac{r}{2} \right) = \frac{\pi \mu_0 n b r^2}{8}$$



26. (a) Induced emf $\varepsilon = Bdv$

$$\text{Induced current } I = \frac{Bvd}{R + 2\lambda x}$$

[Resistance of the two rails in the closed loop = $2(\lambda x)$]

$$\text{or, } v = \frac{I(R + 2\lambda x)}{B \cdot d} \quad (i)$$

The current carrying rod experiences a magnetic force that opposes its motion.

$$F_B = IdB = \frac{B^2 d^2 v}{(R + 2\lambda x)}$$

Acceleration of the rod is

$$a = \frac{dv}{dt} = \frac{2\lambda I}{B \cdot d} \frac{dx}{dt} = \frac{2\lambda I}{Bd} v$$

Using $F = ma$, we can write

$$F - F_B = ma$$

$$\text{or, } F = F_B + ma = \frac{B^2 d^2 v}{(R + 2\lambda x)} + \frac{2\lambda I m}{B \cdot d} v$$

$$= BId + \frac{2\lambda m I^2}{(Bd)^2} (R + 2\lambda x) \quad [\text{using (i)}]$$

- (b) Power delivered by the external force is

$$P = Fv = \left[BId + \frac{2\lambda m I^2}{(Bd)^2} (R + 2\lambda x) \right] \frac{I(R + \lambda x)}{B \cdot d}$$

$$= I^2 (R + 2\lambda x) + \frac{2\lambda m I^3}{B^3 d^3} (R + 2\lambda x)^2$$

Heat dissipated per unit time in the loop is

$$P_R = I^2 (R + 2\lambda x)$$

$$\therefore \frac{P_R}{P} = \left[1 + \frac{2\lambda m I (R + 2\lambda x)}{B^3 d^3} \right]^{-1}$$

You must think why this ratio is not 1

27. (a) At time $t = 0$, the area vector of the loop is parallel to $\vec{B}$. At time t the angle between the area vector and the magnetic field becomes

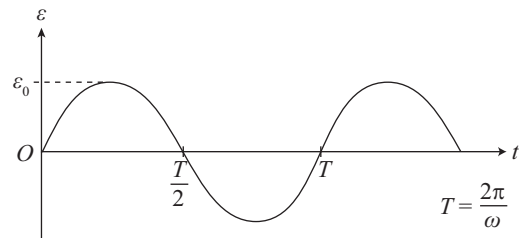
$$\theta = \omega t$$

$\therefore$ Total flux linked to the loop at time t is

$$\phi = N(dh)B \cdot \cos(\omega t)$$

$$\therefore \varepsilon = -\frac{d\phi}{dt} = NdhB\omega \sin \omega t$$

$$= \varepsilon_0 \sin \omega t \quad [\varepsilon_0 = N \cdot dhB\omega]$$

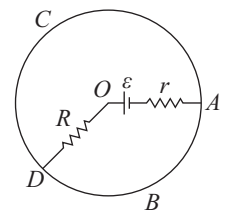


- (b) There is no change in answer.

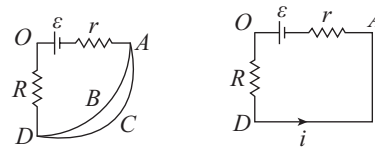
28. Induced emf in the rod is

$$\varepsilon = \frac{1}{2} BL^2 \omega$$

The effective electric circuit is as shown. Note that there are two parallel resistance free path between D and A .



$$i = \frac{\varepsilon}{R + r} = \frac{\frac{1}{2} BL^2 \omega}{R + r}$$



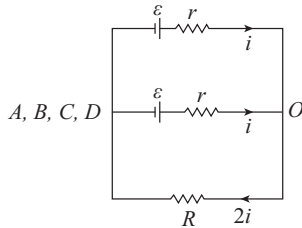
29. The induced emf in each half of the rod is

$$\varepsilon = \frac{1}{2}BL^2\omega$$

The polarity of the emfs is as shown. A simplified circuit is also shown. Using Kirchhoff's Law we can write

$$ir + 2iR = \varepsilon$$

$$\text{or, } i = \frac{\varepsilon}{r + 2R}$$



$$\therefore \text{Current in } R \text{ is } 2i = \frac{2\varepsilon}{r + 2R} = \frac{BL^2\omega}{r + 2R}$$

30. (a) Flux linked with the coil $\phi = NBA \cos 0^\circ = NBA$
Emf induced,

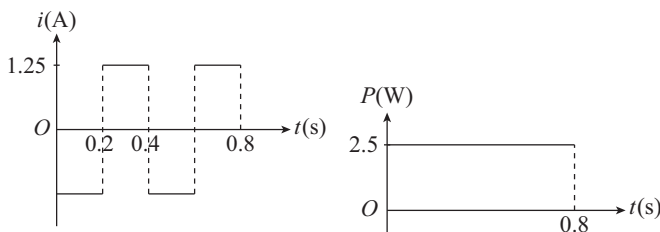
$$\varepsilon = -\frac{d\phi}{dt} = -NA \frac{dB}{dt} = -100 \times 5 \times 10^{-3} \times \frac{0.8}{0.2} = -2V$$

When B is decreased, the induced emf has same magnitude but opposite direction.

Induced current has magnitude equal to

$$i = \frac{\varepsilon}{R} = \frac{2}{1.6} = 1.25A$$

$$\text{Power dissipated} = i^2 R = (1.25)^2 \times 1.6 = 2.5 \text{ W}$$



- (b) Heat supplied to the water and the coil
 $= P_t = 2.5 \times 0.4 \times 12000 = 12000 \text{ J}$

From the principle of calorimetry

$$m_1 s_1 \Delta\theta + m_2 s_2 \Delta\theta = 12000$$

$$\text{or, } \Delta\theta = \frac{12000}{m_1 s_1 + m_2 s_2} = \frac{12000}{0.5 \times 4200 + 0.06 \times 500} = 5.6^\circ\text{C}$$

31. Consider the semicircular arc in xy plane. The centre of curvature is at C . The arc enclosed between the xy axes and the arc is

$$A = \frac{\pi R^2}{2} + \frac{1}{2}(\sqrt{2}R \cdot \sqrt{2}R) = R^2 \left(1 + \frac{\pi}{2}\right) [\text{Area of semicircle} + \text{area of } \triangle OPQ]$$

Magnetic flux through this area is $\phi_1 = \vec{A} \cdot \vec{B}$

$$= (A\hat{k}) \cdot B_0(\hat{i} + \hat{j}) = 0$$

Magnetic flux through the area in yz plane is

$$\phi_2 = R^2 \left(1 + \frac{\pi}{2}\right) \hat{i} \cdot B_0(\hat{i} + \hat{j}) = R^2 B_0 \left(1 + \frac{\pi}{2}\right)$$

Flux through the area in xz plane is

$$\phi_3 = R^2 \left(1 + \frac{\pi}{2}\right) \hat{j} \cdot B_0(\hat{i} + \hat{j}) = R^2 B_0 \left(1 + \frac{\pi}{2}\right)$$

$$\text{Total flux } \phi = \phi_1 + \phi_2 + \phi_3 = B_0 R^2 (2 + \pi)$$

Given $B = \sqrt{2}B_0$ [Magnitude of the field]

$$\frac{dB}{dt} = \sqrt{2} \frac{dB_0}{dt} = \alpha$$

$$\therefore \frac{d\phi}{dt} = R^2 (2 + \pi) \frac{dB_0}{dt} = \frac{R^2 (2 + \pi) \alpha}{\sqrt{2}}$$

32. (a) The potential difference across the capacitor is equal to the emf induced in the inductor, at any instant

$$\left| L \frac{dI}{dt} \right| = \frac{Q}{C}$$

$$\text{or, } \left| \frac{dI}{dt} \right| = \frac{Q}{LC} = \frac{100 \times 10^{-6}}{2 \times 10^{-3} \times 5 \times 10^{-6}} = 10^4 \text{ A/s}$$

- (b) When $Q = 200 \mu\text{C}$, the entire energy is with the capacitor. There is no energy in the inductor.

$$\therefore I = 0$$

$$(c) \frac{1}{2} L I_0^2 = \frac{Q_0^2}{2C} \Rightarrow I_0 = \frac{Q_0}{\sqrt{LC}}$$

$$\text{or, } I_0 = \frac{200 \times 10^{-6}}{\sqrt{2 \times 10^{-3} \times 5 \times 10^{-6}}} = 2A$$

- (d) Given $I = \frac{I_0}{2} = 1A$

Energy in the inductor

$$= \frac{1}{2} L I^2 = \frac{1}{2} \times 2 \times 10^{-3} \times 1^2 = 10^{-3} \text{ J}$$

Total energy in the circuit

$$= \frac{1}{2} LI_0^2 = \frac{1}{2} \times 2 \times 10^{-3} \times (2)^2 = 4 \times 10^{-3} \text{ J}$$

Energy with the capacitor when $I = 1 \text{ A}$ is

$$= 4 \times 10^{-3} - 10^{-3}$$

$$= 3 \times 10^{-3} \text{ J}$$

$$\frac{Q^2}{2C} = 3 \times 10^{-3}$$

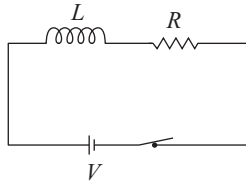
$$\text{or, } Q^2 = 2 \times 3 \times 10^{-3} \times 5 \times 10^{-6}$$

$$\text{or, } Q = \sqrt{3} \times 10^{-4} = 1.73 \times 10^{-4} \text{ C}$$

33. Magnetic energy stored in the solenoid is $U_B = \frac{1}{2} Li^2$. This is maximum when the current attains its final steady value $i_0 = \frac{V}{R}$.

The energy stored in the magnetic field is $\frac{1}{4}$ of the maximum value when

$$i = \frac{i_0}{2}$$



A solenoid has inductance as well as resistance. Both the properties can be shown separately in series.

The variation of current with time in the circuit is given by

$$i = i_0 (1 - e^{-t/\tau})$$

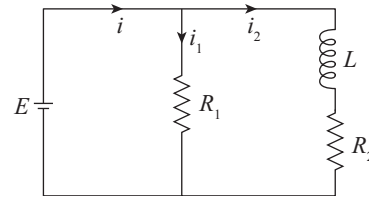
$$\text{or, } \frac{i_0}{2} = i_0 (1 - e^{-t/\tau})$$

$$\text{or, } 1 - e^{-t/\tau} = \frac{1}{2} \Rightarrow e^{-t/\tau} = \frac{1}{2}$$

$$\text{or, } \frac{t}{\tau} = \ln 2 \Rightarrow t = (\ln 2) \tau = (\ln 2) \frac{L}{R}$$

$$\therefore t = 0.693 \times \frac{10}{2} = 3.465 \text{ s}$$

34. (a) Resistance R_1 is connected in parallel to the series combination of R_2 and L . The two parallel paths are independent. The current in R_1 is constant $i_1 = \frac{E}{R_1}$.



The growth of current i_2 has nothing to do with i_1

The current (i_2) grows with a time constant

$$\tau = \frac{L}{R_2} = \frac{0.4}{2} = 0.25.$$

Steady state value of i_2 is

$$i_0 = \frac{E}{R_2} = \frac{12}{2} = 6 \text{ A}$$

After the switch is closed, the current i_2 changes as

$$i_2 = i_0 (1 - e^{-t/\tau}) = 6(1 - e^{-t/0.25}) = 6(1 - e^{-5t})$$

The potential drop across L at time t is

$$V_L = L \left| \frac{di_2}{dt} \right| = 0.4 \times 6 \times 5 e^{-5t}$$

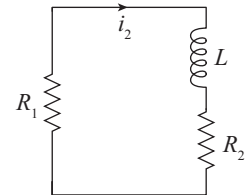
$$= 12 e^{-5t} \text{ volt.}$$

- (b) In steady state the current through L is $i_0 = 6 \text{ A}$. If the switch is opened, the battery gets disconnected and the current through the inductor begins to decay. The effective circuit is as shown. The time constant is

$$\tau = \frac{L}{R_1 + R_2} = 0.1 \text{ s}$$

$$i_2 = i_0 e^{-t/\tau} = 6 e^{-10t} \text{ A}$$

The direction of the decaying current is clockwise as shown.



35. With S_1 closed and S_2 open, C_1 and C_2 are in series. Their equivalent is

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{3 \times 2}{3 + 2} = 1.2 \mu\text{F}$$

In steady state, the charge on the capacitors is

$$Q_0 = CV = 1.2 \times 20 = 24 \mu\text{C}.$$

At $t = 0$, the charged capacitor C_2 is connected to the inductor and LC oscillation starts. The angular frequency of the oscillation is

$$\omega = \frac{1}{\sqrt{LC_2}} = \frac{1}{\sqrt{0.2 \times 10^{-3} \times 2 \times 10^{-6}}} = 50000 \text{ rad s}^{-1}$$

- (a) $Q = Q_0 \cos \omega t = 24 \cos(50000t) \mu\text{C}.$

- (b) The energy in the inductor is $\frac{1}{3}$ of that in C_2 . This means that

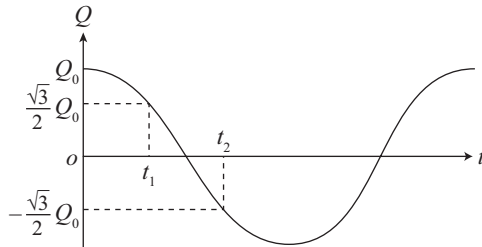
$$\frac{U_E}{3} + U_E = \frac{Q_0^2}{2C} \quad [U_E = \text{energy in the capacitor}]$$

$$\text{or, } U_E = \frac{3}{4} \frac{Q_0^2}{2C} \Rightarrow \frac{Q^2}{2C} = \frac{3}{4} \frac{Q_0^2}{2C}$$

$$\text{or, } Q = \frac{\sqrt{3}}{2} Q_0 \quad (i)$$

So the given situation arises when the magnitude of the charge on the capacitor is given by (i).

We are supposed to report time t_2 (see the graph).



$$\therefore Q = Q_0 \cos(50000t)$$

$$\text{or, } -\frac{\sqrt{3}}{2} Q_0 = Q_0 \cos(50000t)$$

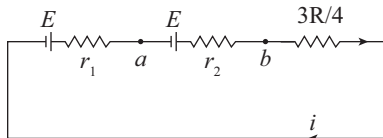
$$\text{or, } 50,000 t = \frac{5\pi}{6}$$

$$\text{or, } t = \frac{\pi}{6} \times 10^{-4} = 52 \mu\text{s}$$

36. In steady state there is no current through the capacitor and the inductor behaves like a plain conducting wire. The equivalent of all resistors having resistance R is

$$R_{eq} = \frac{3R}{4}$$

The effective circuit for finding the current is as shown.



$$i = \frac{2E}{r_1 + r_2 + \frac{3R}{4}}$$

a and b are terminals of cell A .

$$V_a - V_b = 0 \text{ if}$$

$$E = ir_2$$

$$\text{or, } E = \frac{2Er_2}{r_1 + r_2 + \frac{3R}{4}}$$

$$\text{or, } R = \frac{4}{3}(r_2 - r_1)$$

$$37. (a) i = i_0(1 - e^{-t/\tau})$$

$$\text{Energy stored in the magnetic field is } U_B = \frac{1}{2} Li^2$$

$$\text{Rate of change of } U_B \text{ is } P_B = \frac{dU_B}{dt} = Li \frac{di}{dt}$$

$$\text{Rate of heat dissipation is } P_R = i^2 R$$

$$\therefore \frac{P_B}{P_R} = \frac{L \frac{di}{dt}}{R i} = \frac{L \frac{i_0}{\tau} e^{-t/\tau}}{R i_0 (1 - e^{-t/\tau})}$$

$$\text{or } \frac{P_B}{P_R} = \frac{e^{-t/\tau}}{1 - e^{-t/\tau}} \quad \left[\because \tau = \frac{L}{R} \right]$$

- (b) The above ratio is 1 if

$$e^{-t/\tau} = 1 - e^{-t/\tau} \Rightarrow e^{-t/\tau} = \frac{1}{2}$$

$$\text{or, } \frac{t}{\tau} = \ln 2 \Rightarrow t = (\ln 2)\tau$$

38. (a) Magnetic field at the location of the smaller loop is due to the current in the bigger loop is

$$B = \frac{\mu_0 R^2 i}{2(R^2 + x^2)^{3/2}}$$

The flux associated with the smaller loop is

$$\phi = BA = \frac{\mu_0 R^2 i}{2(R^2 + x^2)^{3/2}} \cdot \pi r^2 = \frac{\pi \mu_0 R^2 r^2}{2(R^2 + x^2)^{3/2}} i$$

For $x \gg R$

$$\phi \simeq \frac{\pi \mu_0 R^2 r^2}{2x^3} i$$

$$\therefore M = \frac{\pi \mu_0 R^2 r^2}{2x^3}$$

- (b) Induced emf in the smaller coil is

$$\varepsilon = \left| \frac{d\phi}{dt} \right| = \left| \frac{\pi \mu_0 R^2 r^2 i}{2} \frac{d}{dt} (x^{-3}) \right| = \frac{3}{2} \frac{\pi \mu_0 R^2 r^2 i}{x^4}$$

Magnetic field at the smaller loop is upward for the direction of current shown in the bigger loop. When x increases, the flux decreases. The induced current will produce a field in upward direction to oppose the change in flux. Direction of induced current will be same as the current in the bigger loop.

39. Refer to example 50. Consider a current I in the straight wire, field due to this current at a distance r from the wire is

$$B = \frac{\mu_0 I}{2\pi r}$$

Flux through a strip of width dr is

$$d\phi = B \cdot a dr = \frac{\mu_0 I a}{2\pi} \frac{dr}{r}$$

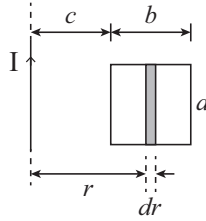
Flux through the rectangular loop is

$$\phi = \frac{\mu_0 I a}{2\pi} \int_c^{b+c} \frac{dr}{r} = \frac{\mu_0 I a}{2\pi} \ln\left(\frac{b}{c} + 1\right)$$

$$\therefore M = \frac{\mu_0 a}{2\pi} \ln\left(\frac{b}{c} + 1\right)$$

40. $u_B = 0$ for $x = 10$ cm. This means that the field at the origin is zero for $x = 10$ cm. The two wires create equal and opposite fields.

$$\frac{\mu_0 i_1}{2\pi d} = \frac{\mu_0 i_2}{2\pi x} \quad [d = \text{distance of wire 1 from the origin}]$$



$$\text{or, } d = x \left(\frac{i_1}{i_2} \right) = 10 \times \frac{1}{4} = 2.5 \text{ cm.}$$

When $x \rightarrow \infty$, the field at O is only due to i_1 . The energy density is

$$\frac{B_1^2}{2\mu_0} = u_0 \quad \Rightarrow \quad B_1^2 = 2\mu_0 \cdot u_0$$

$$\text{or, } \frac{\mu_0^2 i_1^2}{4\pi^2 d^2} = 2\mu_0 u_0 \quad \Rightarrow \quad i_1^2 = \frac{8u_0 \pi^2 d^2}{\mu_0}$$

$$\begin{aligned} \text{or, } i_1^2 &= \frac{8 \times 2 \times 10^{-9} \times \pi^2 \times (2.5 \times 10^{-2})^2}{4\pi \times 10^{-7}} \\ &= 78.5 \times 10^{-6} \text{ A} \end{aligned}$$

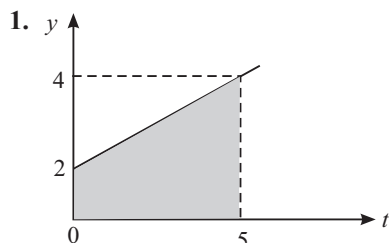
$$\text{or, } i_1 = 8.86 \times 10^{-6} \text{ A} = 8.86 \text{ } \mu\text{A}$$

$$i_2 = 4i_1 = 35.44 \text{ } \mu\text{A.}$$

Chapter 6 Alternating Current Circuits



Your Turn



$$\langle y \rangle = \frac{\int_0^5 y dt}{5} = \frac{\text{Area under the graph}}{5}$$

$$= \frac{\frac{1}{2} \times 5 \times (2+4)}{5} = 3$$

2. $\omega = 100\pi \Rightarrow \frac{2\pi}{T} = 100\pi$
 $\Rightarrow T = 0.02 \text{ s}$

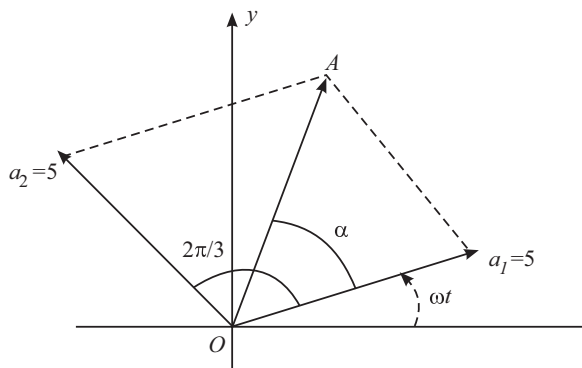
This is the time period of the sine term.

$$\langle y \rangle = 1 - \langle \sin(100\pi t) \rangle$$

$$= 1 - 0 = 1$$

Average of sine function in one cycle is zero. In interval $t = 0$ to $t = T$, the sine function completes one cycle.

3. The phasor representing y_2 is $\frac{2\pi}{3}$ ($= 120^\circ$) ahead of the one representing y_1 . The amplitude of the resultant is obtained by vector sum of $a_1 = 5$ and $a_2 = 5$, placed at 120° . The amplitude is



$$A = \sqrt{5^2 + 5^2 + 2 \cdot 5 \cdot 5 \cos 120^\circ} = 5$$

$$\alpha = \frac{\pi}{3} (= 60^\circ)$$

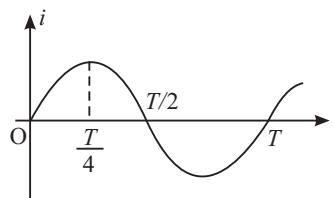
$$\therefore y = A \sin(\omega t + \alpha) = 5 \sin\left(\omega t + \frac{\pi}{3}\right)$$

Note that the phasor diagram itself tells you that phase of the resultant is α ahead of the phase of y_1 .

4. The frequency of rotation of the coil is the frequency of the voltage generated.

5. $i_{\text{rms}} = \frac{i_0}{\sqrt{2}} = \frac{5}{\sqrt{2}} = 3.5 \text{ A}$

Time period, $T = \frac{1}{f} = \frac{1}{100} = 0.01 \text{ s}$



The current takes one fourth of the time period to reach the peak value starting from zero. Hence, the required time is

$$t = \frac{T}{4} = \frac{0.01}{4} = 2.5 \times 10^{-3} \text{ s}$$

6. $i_{\text{rms}} = \frac{i_0}{\sqrt{2}} = \frac{141}{1.41} = 100 \text{ A}$

Power dissipated is, $P_{\text{av}} = i_{\text{rms}}^2 \cdot R = (100)^2 \times 1 = 1000 \text{ W}$.

7. When we say 220 V AC, we are referring to a voltage having rms value of 220V. Its peak value is $220\sqrt{2} = 311 \text{ V}$.

Obviously, a supply having a peak voltage of 311 V is more dangerous than the one having a constant voltage 220 V.

8. Frequency of a DC source is zero.

$$X_C = \frac{1}{\omega C} = \infty$$

9. Inserting the dielectric, increases the capacitance. This decreases the reactance X_C .

$\therefore$ Current increases.

10. The bulb will show normal brightness when the rms voltage (V) of the AC source is 24V.

or, $\frac{v_0}{\sqrt{2}} = 24 \Rightarrow v_0 = 34 \text{ V}$

11. $X_C = \frac{1}{\omega c} = \frac{1}{2\pi f C}$

$$\therefore X_C \propto \frac{1}{f}$$

And $X_L = \omega L = 2\pi fL$

$\Rightarrow X_L \propto f$.

12. $X_C = \frac{1}{\omega C} = \frac{1}{\omega(10 \times 10^{-6})}$

$I_1 = \frac{V}{X_C}$
 $= V \cdot \omega \times 10^{-5} = V(2\pi \times 50) \times 10^{-5} = \pi V \times 10^{-3}$

$X_L = \omega L = \omega(10 \times 10^{-3}) = 10^{-2}(2\pi \times 50) = \pi$

$I_2 = \frac{V}{X_L} = \frac{V}{\pi}$

clearly, $I_2 > I_1$

13. $i_0 = \frac{v_0}{X_C} = \frac{10}{50} = 0.2 \text{ A}$

Phase of current is $\frac{\pi}{2}$ ahead of the phase of the voltage.

$\therefore i = 0.2 \sin(100\pi t - \frac{\pi}{3} + \frac{\pi}{2}) = 0.2 \sin(100\pi t + \frac{\pi}{6})$

At $t = 0$,

$i = 0.2 \sin \frac{\pi}{6} = 0.1 \text{ A}$

14. $\omega = 2\pi f = 2 \times 3.14 \times 50 \simeq 314 \text{ rad s}^{-1}$

$Z = \sqrt{R^2 + X_L^2} = \sqrt{(50)^2 + (\omega L)^2}$
 $= \sqrt{2500 + (314 \times 0.5)^2} = 165 \Omega$

$\therefore \text{rms current} = \frac{110V}{165\Omega} = 0.67 \text{ A}$

15. $i_0 = \frac{v_0}{X_L} = \frac{v_0}{\omega L}$

when ω is doubled, X_L doubles and the peak current halves.

16. $i = \frac{v_R}{R} = \frac{20 \sin(10^3 t)}{100} = 0.2 \sin(10^3 t)$

The current in the circuit is in phase with v_R .

(a) $X_L = \omega L = 10^3 \times 4 = 4000 \Omega$

(b) $v_L = (i_0 X_L) \sin\left(\omega t + \frac{\pi}{2}\right)$

or, $v_L = (0.2 \times 4000) \sin\left(10^3 t + \frac{\pi}{2}\right)$
 $= 800 \sin\left(10^3 t + \frac{\pi}{2}\right)$

17. The current and the voltage will be in phase when

$X_L = X_C$

$\Rightarrow \omega = \frac{1}{\sqrt{LC}} \Rightarrow f = \frac{1}{2\pi\sqrt{LC}}$

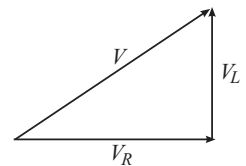
or, $f = \frac{1}{2 \times 3.14 \sqrt{1 \times 10^{-6}}} = 160 \text{ Hz}$

19. By changing the capacitance, we can change the impedance of the fan circuit and thereby control the current. This is a better method as a capacitor will not consume energy.

20. $V_L^2 + V_R^2 = V^2$

or, $V_L^2 + (110)^2 = (220)^2$

or, $V_L = 110\sqrt{3} \text{ V}$
 $= 190.5 \text{ V}$



The resistance of the bulb is

$R = \frac{(110)^2}{55} = 220 \Omega$

In the LR circuit, we need

$\frac{V_L}{V_R} = \frac{190.5}{110} \Rightarrow \frac{IX_L}{IR} = \frac{190.5}{110}$

or, $X_L = \frac{190.5}{110} \times 220 = 381 \Omega$

or, $\omega L = 381 \Rightarrow 2\pi f \cdot L = 381$

or, $L = \frac{381}{2 \times 3.14 \times 50} = 1.21 \text{ H}$

21. $Q = \frac{\omega_0}{\Delta\omega} = \frac{f_0}{\Delta f}$

$Q_1 = \frac{10^6}{10^3} = 1000$

$Q_2 = \frac{10^7}{10^3} = 10000$

$\therefore Q_2 > Q_1$

The second circuit has a sharper resonance.

22. $\Delta\omega = \frac{R}{L} = \frac{10}{0.1 \times 10^{-3}} = 10^5 \text{ rad s}^{-1}$

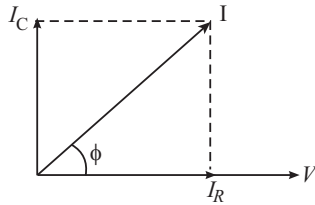
or, $\Delta f = \frac{10^5}{2 \times 3.14} = 1.59 \times 10^5 \text{ Hz}$

23. $I_R = \frac{100V}{100\Omega} = 1 \text{ A}$

$I_C = \frac{100V}{100\Omega} = 1 \text{ A}$

$\therefore I = \sqrt{I^2 + I^2} = \sqrt{2} \text{ A}$

$\tan \phi = \frac{I_C}{I_R} = \frac{1}{1} \Rightarrow \phi = \frac{\pi}{4}$



24. A transformer works with AC sources only. With a DC source, there is no change in flux and no emf is induced in the secondary.



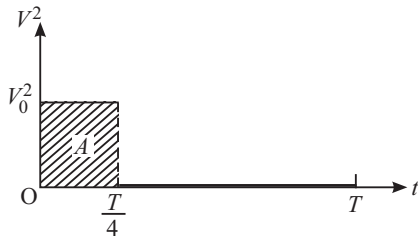
Worksheet 1

1. $E = 10 \cos(2\pi \times 50t)$

At $t = \frac{1}{600} \text{ s}$

$E = 10 \cos\left(100\pi \times \frac{1}{600}\right) = 10 \cos\left(\frac{\pi}{6}\right) = 5\sqrt{3} \text{ V}$

2. The graph of V^2 will be as shown



$\int_0^T V^2 dt = \text{area}(A) = V_0^2 \frac{T}{4}$

$\therefore \langle V^2 \rangle = \frac{\int_0^T V^2 dt}{T} = \frac{V_0^2}{4}$

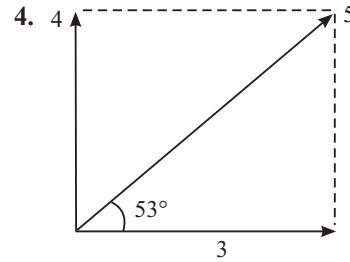
$\therefore V_{\text{rms}} = \sqrt{\langle V^2 \rangle} = \frac{V_0}{2}$

3. Refer to example 7.

The rms value is $I = \sqrt{3^2 + \frac{4^2}{2}} = \sqrt{17} \text{ A}$.

Peak value of the current is $i_0 = 3 + 4 = 7 \text{ A}$

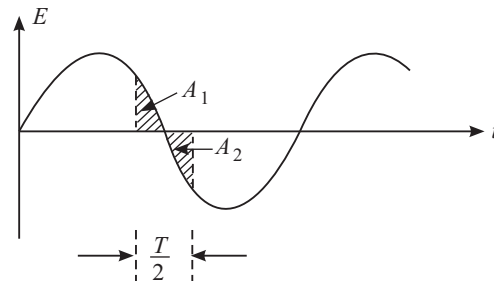
$\therefore \text{Required ratio} = \frac{\sqrt{17}}{7}$



$\varepsilon = 3 \sin(314t) + 4 \cos(314t)$
 $= 3 \sin(314t) + 4 \sin\left(314t + \frac{\pi}{2}\right)$
 $= 5 \sin(314t + 53^\circ)$

$\therefore \varepsilon_0 = 5$

5. The average value calculated over a time interval can be zero if the area (A_1) above the time axis is same as the area (A_2) below the time axis.



6. A DC voltmeter will always show zero value for a high frequency AC voltage. Due to inertia and friction the coil in the voltmeter will not be able to move quickly and keep changing direction so fast as the voltage.

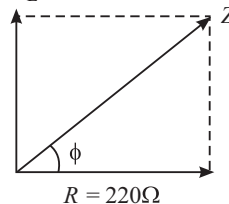
7. $X_L = \omega L = 100 \times 5 \times 10^{-3} = 0.5 \Omega$

$i_0 = \frac{v_0}{X_L} = \frac{10}{0.5} = 20 \text{ A}$

8. The phase difference between the current and voltage is 90° . Therefore, the power dissipated is zero.

9. Wattless current = $I \sin \phi$

$X_L = 220 \Omega$



where ϕ is phase difference between the current and the voltage.

$X_L = 2\pi f L = 2\pi \times 50 \times \frac{2 \cdot 2}{\pi} = 220 \Omega$

$$Z = \sqrt{R^2 + X_L^2} = 220\sqrt{2} \Omega$$

$$\therefore I = \frac{V}{Z} = \frac{220}{220\sqrt{2}} = \frac{1}{\sqrt{2}} \text{ A}$$

$$\text{Also } \tan \phi = \frac{X_L}{R} = 1 \Rightarrow \phi = 45^\circ$$

$$\therefore I \sin \phi = \frac{1}{2}$$

$$10. X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times 3.14 \times 50 \times 100 \times 10^{-6}} = \frac{100}{3.14} \Omega$$

$$v_0 = i_0 X_C = 1.57 \times \frac{100}{3.14} = 50 \text{ V}$$

Only option (a) has peak voltage of 50V.

The phase angle can be anything. Only thing is that the current must lead in phase by $\frac{\pi}{2}$ compared to the voltage across the capacitor.

11. A good choke coil has negligible resistance. The current lags the voltage by 90° and therefore, $\cos \phi = 0$.

$$12. \cos \phi = 0.866 = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \phi = 30^\circ$$

$$Z = R \sec 30^\circ = \frac{2R}{\sqrt{3}} = 1.15R$$

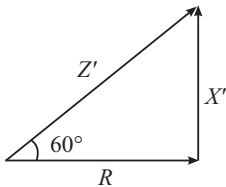
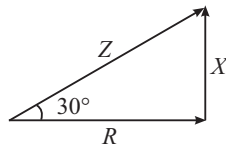
$$\text{Now, } \cos \phi' \text{ becomes } = \frac{1}{2}$$

$$\Rightarrow \phi' = 60^\circ$$

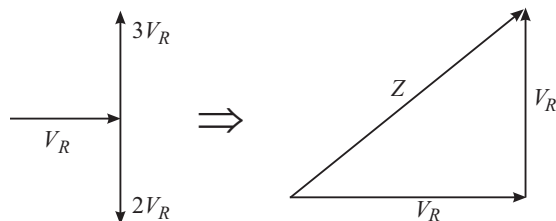
$$\therefore Z' = R \sec 60^\circ = 2R$$

$\therefore$ percentage change in impedance is

$$\frac{Z' - Z}{Z} \times 100 = \frac{2R - 1.15R}{1.15R} \times 100 = 73.9\%$$



13.



$$Z = \sqrt{2} V_R$$

14. When connected to a DC source, the inductance offers no resistance in steady state. Only resistance of the coil decides the value of current.

$$R = \frac{12V}{2A} = 6\Omega$$

When connected to the AC source, the impedance is

$$Z = \frac{12V}{1A} = 12\Omega$$

$$\text{or, } \sqrt{R^2 + X_L^2} = 12 \Rightarrow X_L = \sqrt{108} = 10.4\Omega$$

$$\text{or, } 2\pi fL = 10.4$$

$$\text{or, } L = \frac{10.4}{2 \times 3.14 \times 50} = 0.033 \text{ H}$$

$$15. Q = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{1} \sqrt{\frac{1}{6.25 \times 10^{-6}}} = 400$$

$$16. X_L = \omega L = 100 \times 1 = 100\Omega.$$

$$R = 100\Omega$$

$$\therefore Z = \sqrt{R^2 + X_L^2} = 100\sqrt{2} \Omega$$

$$\therefore I = \frac{V}{Z} = \frac{200/\sqrt{2}}{100\sqrt{2}} = 1A \quad \left[V_{\text{rms}} = \frac{200}{\sqrt{2}} \right]$$

$$P = I^2 R = 100W$$

$$17. \text{Power loss in DC circuit is } P = \frac{V^2}{R}$$

$$\Rightarrow R = \frac{V^2}{P} = \frac{10^2}{20} = 5\Omega$$

When connected to the AC source, power loss is given by

$$I^2 R = P$$

$$\text{or, } I^2(5) = 10 \Rightarrow I = \sqrt{2} \text{ A}$$

$$\Rightarrow Z = \frac{10V}{\sqrt{2}A} = 5\sqrt{2} \Omega$$

$$\text{or, } \sqrt{R^2 + X_L^2} = 5\sqrt{2}$$

$$\text{or, } 25 + X_L^2 = 50 \Rightarrow X_L = 5\Omega$$

$$\text{or, } 2\pi fL = 5 \Rightarrow 2 \times 3.14 \times f \times 10 \times 10^{-3} = 5$$

$$\text{or, } f = 79.6\Omega$$

18. The current leads in phase by $\frac{\pi}{4}$. Therefore, the circuit must be an RC circuit.

For $\phi = 45^\circ$

$$X_C = R$$

$$\Rightarrow \frac{1}{\omega C} = R \Rightarrow \frac{1}{100} = CR \quad (i)$$

Option (a) satisfies the above equation.

19. The element A is an inductor.

$$\tan \theta_1 = \frac{X_L}{R}$$

The element B is a capacitor.

$$\tan \theta_2 = \frac{X_C}{R}$$

With R , L and C in series

$$\tan \theta = \frac{X_C - X_L}{R} = \frac{X_C}{R} - \frac{X_L}{R}$$

$$\text{or, } \tan \theta = \tan \theta_2 - \tan \theta_1$$

$$20. I = \frac{V}{Z} \Rightarrow Z = \frac{220}{2.2} = 100\Omega$$

$$X_L = \omega L = (100\pi) \left(\frac{1}{\pi} \right) = 100\Omega$$

Because Z is same as R , the reactance of the circuit must be zero.

$$\therefore X_C = X_L = 100\Omega$$

$$\text{For the box, } \tan \phi = \frac{X_C}{R} = 1 \Rightarrow \phi = 45^\circ$$

$$\text{Power factor is } \cos \phi = \frac{1}{\sqrt{2}}$$

$$21. \cos \phi_1 = 0.6 \Rightarrow \tan \phi_1 = \frac{4}{3}$$

$$\text{or, } \frac{X_L}{R_1} = \frac{4}{3} \quad (i)$$

For CR circuit

$$\cos \phi_2 = 0.5 \Rightarrow \tan \phi_2 = \sqrt{3}$$

$$\text{or, } \frac{X_C}{R_2} = \sqrt{3} \quad (ii)$$

When all elements are joined in series, the power factor is 1.0.

This means $X_C = X_L$

From (i) and (ii)

$$\frac{X_C / R_2}{X_L / R_1} = \frac{\sqrt{3}}{4/3}$$

$$\text{or, } \frac{R_1}{R_2} = \frac{3\sqrt{3}}{4}$$

22. The graph of i^2 versus time for case (A) and (B) will be same. Therefore, $\langle i^2 \rangle$ will be equal in two cases.

$$\therefore I_1 = I_2$$

Just by observation, one can see that the area under i^2 graph will be maximum in case (C).

$$\therefore I_3 > I_1 = I_2 = I_4$$

23. V_L leads in phase by 90° relative to the current and V_C lags relative to the current.

24. I_1 is ahead in phase by 90° relative to the source voltage.

I_2 lags in phase relative to the source voltage by

$$\phi = \tan^{-1} \left(\frac{X_L}{R} \right)$$

$\therefore$ Phase difference between I_1 and I_2 is

$$\frac{\pi}{2} + \tan^{-1} \left(\frac{X_L}{R} \right)$$



Worksheet 2

$$1. \quad v = 100 \sin(100\pi t) \cos(100\pi t) \\ = 50 \sin(200\pi t)$$

$$\therefore v_0 = 50 \text{ V}$$

$$\omega = 200\pi \Rightarrow 2\pi f = 200\pi$$

$$\Rightarrow f = 100\text{Hz}$$

2. Instantaneous power

$$P = vi = (v_0 \sin \omega t) i_0 \sin \left(\omega t + \frac{\pi}{3} \right)$$

$$= v_0 i_0 \sin(\omega t) \sin \left(\omega t + \frac{\pi}{3} \right)$$

P will be positive as long as both $\sin \omega t$ and $\sin \left(\omega t + \frac{\pi}{3} \right)$ are positive or when both are negative.

When $\omega t + \frac{\pi}{3} < \pi$; P will be positive as both sine terms are positive.

$$\text{or, } \omega t < \frac{2\pi}{3}$$

$$\text{or, } t < \frac{2\pi}{3\omega}$$

$$\text{At } t = \frac{5\pi}{6\omega}; \omega t = \frac{5\pi}{6}$$

$\sin(\omega t)$ is positive but $\sin \left(\omega t + \frac{\pi}{3} \right)$ is negative.

$\therefore P$ is negative

If ω is increased slightly, X_L will increase and X_C will decrease.

Current is ahead in phase. This implies $X_C > X_L$ in the circuit. Therefore, increasing ω slightly will result in decrease in value of $X_C - X_L$. This will cause ϕ to decrease.

3. L and R_1 are in parallel. Voltage across them is same. The current through them must have a phase difference of 90° .

C and R_2 are in series voltages across them have a phase difference of 90° and current is same.

4. $V_L = V_C \Rightarrow X_L = X_C$

The circuit is in resonance. The voltage drop across R is equal to the source voltage in this case. Changing the value of R will change the current but $V_R = IR$ will remain same.

5. The voltage across L and C are at 180° phase difference.

$$\therefore \text{Source voltage} = 100 - 60 = 40\text{V}$$

No power is consumed unless there is some resistance in the circuit. The current in the circuit and the applied voltage have a phase difference of 90° .

6. $X_L = \omega L = 314 \times 20 \times 10^{-3} = 6.28\Omega$

$$X_C = \frac{1}{\omega C} = \frac{1}{314 \times 100 \times 10^{-6}} = 31.85\Omega$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2} = \sqrt{50^2 + (25.57)^2} = 56.2\Omega$$

$$\therefore I = \frac{V}{Z} = \frac{10/\sqrt{2}}{56.2} = 0.125\text{A}$$

$$\text{Power dissipated, } P = I^2 R = (0.125)^2 \times 50 = 0.78 \text{ watt}$$

$$\therefore \text{Heat dissipated in 20 min} = 0.78 \times 20 \times 60 = 936\text{J}$$

When L is doubled, $X_L = 6.28 \times 2 = 12.56\Omega$

With resistance is removed, the impedance is

$$Z = X_C - X_L = 31.85 - 12.56 = 19.3\Omega$$

$$\therefore i_0 = \frac{v_0}{Z} = \frac{10}{19.3} = 0.52\text{A}$$

Since $X_C > X_L$, the current leads in phase by 90°

$$\therefore i = 0.52 \sin \left(314t + \frac{\pi}{2} \right) = 0.52 \cos (314t)$$

7. The two parallel paths are independent. For high frequency source, X_L will be high and B_1 will be dim.

For low frequency source, X_C will be high and B_2 will be dim.

8. We shall not disturb the resonance frequency. For this, we will not change L and C . Reducing the resistance reduces the resonance bandwidth.

$$\Delta\omega = \frac{R}{L}$$

By making R half, the value of $\Delta\omega$ will become half.

9. For DC, impedance of L is negligible and that of C is very high. Therefore, the DC signal appears across C . For high frequency AC, X_L is high and X_C is negligible. The AC signal appears across L .

There is no current corresponding to the DC signal as the capacitor blocks the current. The circuit has current only corresponding to the AC signal.

10. Insertion of iron core increases the inductance. In AC circuit, this increases the impedance and the current falls. The bulb becomes less bright.

In DC circuit, after the transient phase, an inductor has no effect on the current value.



Worksheet 3

1. The rms value will be same for the interval 0 to $\frac{T}{2}$ or 0 to T .

$$\text{For } 0 < t < \frac{T}{2}, \quad v = v_0$$

$$\langle v^2 \rangle = \frac{\int_0^{\frac{T}{2}} v^2 dt}{\frac{T}{2}} = \frac{2v_0^2}{T} \int_0^{\frac{T}{2}} dt = v_0^2$$

$$\therefore v_{\text{rms}} = \sqrt{\langle v^2 \rangle} = v_0$$

2. Effective value refers to the rms value. The instantaneous current is

$$I = I_0 + i_0 \sin \omega t$$

$$\therefore I^2 = I_0^2 + i_0^2 \sin^2 \omega t + 2 I_0 i_0 \sin \omega t$$

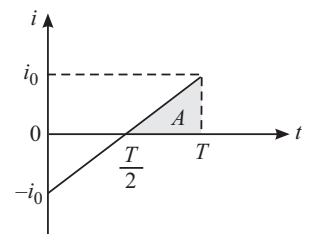
$$\langle I^2 \rangle = I_0^2 + i_0^2 \langle \sin^2 \omega t \rangle + 2 I_0 i_0 \langle \sin \omega t \rangle$$

$$= I_0^2 + \frac{i_0^2}{2} + 0$$

$$\therefore I_{\text{rms}} = \sqrt{\langle I^2 \rangle} = \sqrt{I_0^2 + \frac{i_0^2}{2}}$$

3. (a) $\langle i \rangle = \frac{\text{Area}(A)}{T/2}$

$$= \frac{\frac{1}{2} \cdot \frac{T}{2} \cdot i_0}{\frac{T}{2}} = \frac{i_0}{2}$$



- (b) Value of rms current will be same for the interval 0 to $\frac{T}{2}$ or 0 to T .

$$\text{For the interval } 0 < t < \frac{T}{2},$$

$$i = \left(\frac{2i_0}{T} \right) t - i_0 \quad [\text{Equation of the straight line shown in the graph}]$$

$$\therefore i^2 = \left(\frac{2i_0}{T} \right)^2 t^2 + i_0^2 - \left(\frac{4i_0^2}{T} \right) t$$

$$\langle i^2 \rangle = \frac{1}{T/2} \left[\left(\frac{2i_0}{T} \right)^2 \int_0^{T/2} t^2 dt + i_0^2 \int_0^{T/2} dt - \left(\frac{4i_0^2}{T} \right) \int_0^{T/2} t dt \right]$$

$$= \frac{2}{T} \left[\frac{4i_0^2}{T^2} \cdot \frac{T^3}{24} + i_0^2 \frac{T}{2} - \frac{4i_0^2}{T} \cdot \frac{T^2}{8} \right]$$

$$= i_0^2 \left[\frac{1}{3} + 1 - 1 \right] = \frac{i_0^2}{3}$$

$$\therefore i_{\text{rms}} = \frac{i_0}{\sqrt{3}}$$

4. For a DC source, the inductor offers no resistance in steady state.

$$1\text{A} = \frac{100\text{V}}{R} \Rightarrow R = 100\Omega$$

For AC source, the impedance is

$$Z = \frac{100\text{V}}{0.5\text{A}} = 200\Omega$$

$$\text{or, } \sqrt{X_L^2 + R^2} = 200$$

$$\text{or, } X_L^2 = (200)^2 - (100)^2 = 3 \times (100)^2$$

$$\text{or, } X_L = 100\sqrt{3}\Omega$$

$$\text{or, } 2\pi f L = 173 \Rightarrow L = \frac{173}{2 \times 3.14 \times 50}$$

$$\text{or, } L = 0.55\text{ H}$$

5. Originally $X_L = X_C$ and therefore, $z = 0$. This makes the current very large.

- (a) When R is added, the effective impedance becomes $R = 100\Omega$

$$\therefore I = \frac{100\text{V}}{100\Omega} = 1\text{A}$$

- (b) The circuit is in resonance and the phase difference between the current and the voltage is zero.

- (c) After the inductor is removed,

$$Z = \frac{V}{I} = \frac{100\text{V}}{0.5\text{A}} = 200\Omega$$

$$\text{or, } R^2 + X_C^2 = (200)^2$$

$$\text{or, } X_C^2 = 3(100)^2$$

$$\text{or, } X_C = 100\sqrt{3}\Omega$$

$$\text{or, } \frac{1}{2\pi f C} = 100\sqrt{3}$$

$$\Rightarrow C = \frac{1}{2 \times 3.14 \times 50 \times 100 \times 1.73}$$

$$= 18.4\mu\text{F}$$

$$6. Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

At $\omega_0 = \frac{1}{\sqrt{LC}}$, the impedance is minimum equal to R .

As $\omega < \omega_0$ or $\omega > \omega_0$, the impedance is higher.

7. When C is removed, the circuit becomes an LR circuit with

$$\tan \phi = \frac{X_L}{R} \Rightarrow X_L = R \tan \phi = 100\sqrt{3}\Omega$$

when L is removed, the circuit becomes a CR circuit with

$$\tan \phi = \frac{X_C}{R} \Rightarrow X_C = R \tan \phi = 100\sqrt{3}\Omega$$

when L , C and R all are present the circuit is in resonance as $X_L = X_R$

$$\therefore Z = R = 100\Omega$$

$$\therefore I = \frac{V}{Z} = \frac{200}{100} = 2\text{A}$$

$$P = VI \cos \phi = 200 \times 2 \times 1 = 400\text{W}$$

8. (a) $X_L = \omega L = 200 \times 200 \times 10^{-3} = 40\Omega$

$$\therefore Z = \sqrt{R^2 + X_L^2} = \sqrt{30^2 + 40^2} = 50\Omega$$

$$\therefore i_0 = \frac{v_0}{Z} = \frac{60}{50} = 1.2\text{A}$$

$$\tan \phi = \frac{X_L}{R} = \frac{4}{3}$$

$$\therefore i = 1.2 \sin \left(200t - \tan^{-1} \frac{4}{3} \right) \text{A}$$

Current lags voltage in an LR circuit

$$(b) U = \frac{1}{2} Li^2$$

$$= \frac{1}{2} \times 200 \times 10^{-3} \times (1.2)^2 \sin^2 \left(200t - \tan^{-1} \frac{4}{3} \right)$$

$$\langle U \rangle = 0.144 \left\langle \sin^2 \left(200t - \tan^{-1} \frac{4}{3} \right) \right\rangle$$

$$= 0.144 \times \frac{1}{2} = 0.072\text{J}$$

9. (a) Let $i = i_0 \sin(\omega t + \phi_1) = 1 \sin(100t + \phi_1)$

$$\text{at } t = 0, i = 0.707 = \frac{1}{\sqrt{2}}$$

$$\text{or, } \sin \phi_1 = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

Look at the graph. At $t = 0$, i is positive and increasing with time. This is possible only of

$$\phi_1 = \frac{\pi}{4}$$

$$\therefore i = 1 \sin \left(100t + \frac{\pi}{4} \right)$$

(i)

$$\text{Let } v = v_0 \sin(\omega t + \phi_2)$$

$$\text{or, } v = 10 \sin(100t + \phi_2)$$

$$\text{At } t = 0, v = 7.07 = \frac{10}{\sqrt{2}}$$

$$\therefore \phi_2 = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

v is decreasing at $t = 0$

$$\therefore \phi_2 = \frac{3\pi}{4}$$

$$\therefore v = 10 \sin \left(100t + \frac{3\pi}{4} \right) \quad (\text{ii})$$

$$(b) Z = \frac{v_0}{i_0} = \frac{10}{1} = 10\Omega$$

(c) Phase difference between the voltage and current is $\frac{\pi}{2}$ (look at (i) and (ii))

$$\therefore P = VI \cos \phi = 0.$$

10. Voltage is same across L and C . Current in C leads the voltage by 90° and the current in L lags the voltage by 90° . It means that the two currents in L and C have a phase difference of 180° between them. The resultant current through the source can be obtained by adding two vectors of length 0.8 and 0.6, placed at an angle of 180° . The resultant is 0.2 A.

11. (a) $\cos \phi = 0.5$

$$\Rightarrow \tan \phi = \sqrt{3}$$

$$\text{or, } \frac{X_L}{R} = \sqrt{3}$$

$$\text{or, } \omega L = 200\sqrt{3}$$

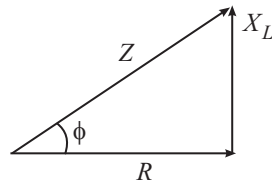
$$\Rightarrow L = 2\sqrt{3} \text{ H} = 3.46 \text{ H}$$

- (b) Power factor will become 1.0 if $\phi = 0^\circ$

This means $X_C = X_L$

$$\text{or, } \frac{1}{\omega C} = 200\sqrt{3}$$

$$\Rightarrow C = \frac{1}{2\sqrt{3} \times 10^4} = 28.9 \mu\text{F}$$



$$12. (a) V_1^2 + V_2^2 = V_4^2$$

$$\text{or, } V_2^2 = (140)^2 - (100)^2 \Rightarrow V_2 = 98\text{V}$$

The voltages across L and C are at 180° phase difference.

$$\therefore V_5 = V_3 - V_2 = 120 - 98 = 22\text{V}$$

$$(b) V = \sqrt{100^2 + 22^2} = 102.4\text{V}$$

$$(c) IR = V_1$$

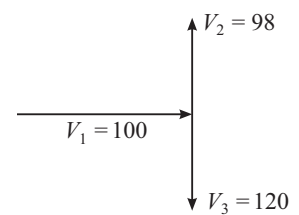
$$\text{or, } R = \frac{100}{2} = 50\Omega$$

$$IX_L = V_2$$

$$\text{or, } X_L = \frac{98}{2} = 49\Omega$$

$$IX_C = V_3$$

$$\text{or, } X_C = \frac{120}{2} = 60\Omega$$



$$13. Q = \frac{\omega_0}{\Delta\omega} = \frac{f_0}{\Delta f}$$

$$\Rightarrow \Delta f = \frac{1000 \times 10^3}{100} = 104 \text{ Hz} = 10 \text{ kHz}$$

$$\therefore \text{Lower half power frequency is } f_1 = f_0 - \frac{\Delta f}{2} = 995 \text{ kHz.}$$

$$\text{The upper half power frequency is } f_2 = f_0 + \frac{\Delta f}{2} = 1005 \text{ kHz.}$$

$$14. z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2} \quad \text{Here } \omega = 2\pi f$$

Substituting the values and calculating gives $z = 833\Omega$.

$$P = VI \cos \phi = (V) \left(\frac{V}{Z} \right) \left(\frac{R}{Z} \right) = \left(\frac{V}{Z} \right)^2 R$$

$$= \left(\frac{20}{833} \right)^2 \times 100 = 0.058 \text{ Js}^{-1}$$

Heat needed to raise the temperature of the resistor by 10°C is

$$Q = C\Delta\theta = 2 \times 10 = 20\text{J}$$

$$\therefore Pt = 20 \Rightarrow t = \frac{20}{0.058} = 345 \text{ s}$$

$$15. \text{Line resistance} = (1 \text{ ohm km}^{-1}) \times (40 \text{ km}) = 40\Omega$$

The power is transmitted at 4400 V.

$$\therefore \text{Current (rms value) in the line } I = \frac{P}{V}$$

$$\text{or, } I = \frac{220 \times 1000}{4400} = 50\text{A}$$

$$(a) P_{\text{Loss}} = I^2 R = (50)^2 \times 40 = 100,000\text{W} = 100\text{kW}$$

$$(b) \text{Power produced by the plant} = 100 + 220 = 320\text{kW}$$

$$(c) \text{Voltage drop in the transmission line} = IR = 50 \times 40 = 2000\text{V}$$

$$\therefore \text{Output of the transformer at the plant} = 4400 + 2000 = 6400\text{V}$$

$$\therefore \frac{N_1}{N_2} = \frac{440}{6400} = \frac{11}{160}$$

Chapter 7 Miscellaneous Problems on Chapter 5 and 6



Match the Columns

1. $\frac{\Delta\phi}{R} = q \Rightarrow \Delta\phi = q \cdot R \Rightarrow \text{Coulomb - Ohm}$

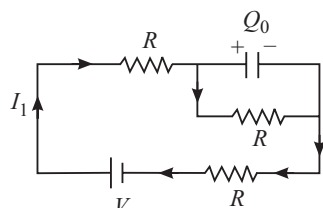
$\frac{d\phi}{dt} = \varepsilon \Rightarrow \text{volt}$

$BIL = F \Rightarrow B = \frac{F}{IL}$

$\Rightarrow \frac{\text{newton}}{\text{A-m}} = \frac{\text{kgms}^{-2}}{\text{A-m}} = \text{kg A}^{-1}\text{s}^{-2}$

Energy = $LI^2 = (LI) (I) = \text{Weber - Ampere}$

2. When only S_1 is closed, the current flows as shown.

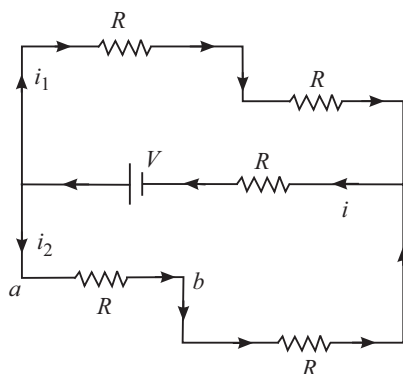


$I_1 = \frac{V}{3R}$

The potential difference across the capacitor is

$I_1 R = \frac{V}{3}$

Immediately after $t = 0$, there is no current in the inductor. The effective circuit is as shown in the second diagram. Due to symmetry $i_1 = i_2 = i_0$ (say)



Current through the cell is $i = i_1 + i_2 = 2i_0$.

Using Kirchhoff's law

$2i_0 R + i_0 R + i_0 R = V$

$\Rightarrow i_0 = \frac{V}{4R}$

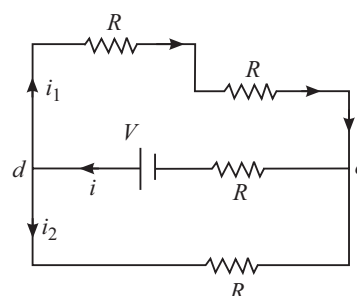
Emf across the inductor = $V_{ab} = i_0 R = \frac{V}{4}$

Change in current through the cell

$= 2i_0 - I_1 = \frac{V}{2R} - \frac{V}{3R} = \frac{V}{6R}$

Just before $t = T$, the effective current is as shown in the third diagram. The inductor is like a plain wire and resistance between a and b is short-circuited.

The resistances $2R$ and R are in parallel between d and e .



$\therefore \text{Their equivalent} = \frac{2R \times R}{2R + R} = \frac{2R}{3}$

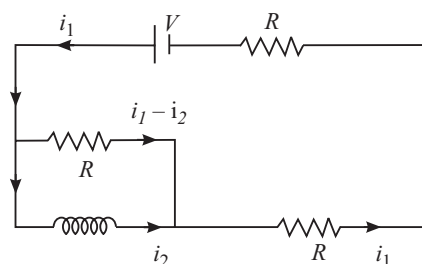
$\therefore \text{Current through the cell is } i = \frac{V}{\frac{2R}{3} + R} = \frac{3V}{5R}$

$i_2 R = i_1 (2R) \Rightarrow \frac{i_2}{i_1} = \frac{2}{1}$

And $i_2 + i_1 = i = \frac{3V}{5R}$

$\therefore i_2 = \frac{2V}{5R}$

Just after $t = T$ (when S_1 is opened), the current in inductor will not change. The diagram represents the situation just after $t = T$.



$Ri_1 + Ri_1 + R(i_1 - i_2) = V$

or, $3i_1 - \frac{2V}{5R} = \frac{V}{5R}$

or, $3i_1 = \frac{7V}{5R} \Rightarrow i_1 = \frac{7V}{15R}$

$$3. U_B = \frac{1}{2} L i^2 = \frac{1}{2} \times 10^{-3} \times 1^2 = 0.5 \times 10^{-3} \text{ J} \\ = 0.5 \text{ mJ}$$

$$\text{Energy density in magnetic field } U_B = \frac{0.5 \text{ mJ}}{0.25 \text{ m}^3} \\ = 2 \text{ mJ m}^{-3}$$

$$V_L = L \frac{di}{dt} = 1 \times 10^{-3} \times 20 = 20 \text{ mV}$$

For capacitor:

$$U_E = \frac{0.5 \text{ mJ}}{1000} = 0.5 \text{ } \mu\text{J}$$

$$\frac{q^2}{2C} = 0.5 \text{ } \mu\text{J}$$

$$\text{or, } \frac{(10^{-6})^2}{2C} = 0.5 \times 10^{-6} \Rightarrow C = 1 \text{ } \mu\text{F}$$

$$V_C = \frac{q}{C} = \frac{1 \text{ } \mu\text{C}}{1 \text{ } \mu\text{F}} = 1 \text{ V}$$

$$\text{Electric field between the plates } E = \frac{V_C}{d} = \frac{1 \text{ V}}{2 \text{ mm}} \\ = 500 \text{ V m}^{-1}$$

$$\therefore u_E = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \times 8.8 \times 10^{-12} \times (500)^2 = 1.1 \text{ } \mu\text{J/m}^3$$

$$\frac{V_C}{V_L} = \frac{1 \text{ V}}{20 \text{ mV}} = 50$$

$$\frac{u_B}{u_E} = \frac{2 \times 10^{-3}}{1.1 \times 10^{-6}} = 1.8 \times 10^3 = 1800$$

Rate of growth of electric field energy is given as

$$U_E = \frac{q^2}{2C}$$

$$\Rightarrow \frac{dU_E}{dt} = \frac{q}{C} \frac{dq}{dt} = \frac{qi}{C} = \frac{10^{-6} \times 1}{10^{-6}} = 1 \text{ J/s}$$

For magnetic field energy

$$U_B = \frac{1}{2} L i^2$$

$$\therefore \frac{dU_B}{dt} = L i \frac{di}{dt} = 10^{-3} \times 1 \times 20 = 20 \times 10^{-3} \text{ J/s}$$

$$\therefore \frac{dU_E / dt}{dU_B / dt} = \frac{1 \text{ J/s}}{20 \times 10^{-3} \text{ J/s}} = 50$$

$$\text{For the solenoid, } \frac{B^2}{2\mu_0} = u_B$$

$$\text{or, } B^2 = (2 \times 10^{-3}) \times 2 \times 4\pi \times 10^{-7}$$

$$\text{or, } B = 7.08 \times 10^{-5} \text{ T} = 70.8 \text{ } \mu\text{T}$$

$$\therefore \frac{E(\text{in V/m})}{B(\text{in } \mu\text{T})} = \frac{500}{70.8} = 7.06$$

4. In (A), when S is closed the current in L - R circuit begins to fall. Before closing the switch, $i = \frac{E}{R}$.

Immediately after closing the switch, $i = \frac{E}{R}$. It means induced emf is E . Thereafter, it decreases.

In (B), when the switch is closed, current in L can have any value. Only rate of change of current in L will be such that $L \frac{di}{dt} = E$.

When S is opened, initial current in R will be same as the current in L . This could have a value such that $iR > E$.

$\therefore$ Induced emf can be greater than E as well. It can obviously, be less than or equal to E also.

In (C), when the switch is opened, $\frac{di}{dt}$ is very large.

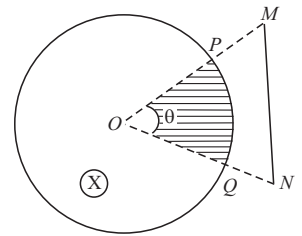
$\therefore$ Induced emf will be greater than E . The open switch is like a huge resistance. Therefore, despite large emf, the current is limited.

In (D) also, the situation is very similar to (C).

5. Consider the figure shown in (C). If we consider ΔOMN as a closed loop, then emf induced in it will be

$$\epsilon = \frac{d\phi}{dt} = (\text{area of sector } OPQ) \frac{dB}{dt} \\ = \frac{1}{2} r^2 \theta \frac{dB}{dt} \quad (i)$$

Now think in terms of induced electric field. The lines of induced electric field are circles centred at O . When one moves along OM the electric field is always perpendicular to the radial path OM (or, ON).



$$\therefore \int_O^M \vec{E} \cdot d\vec{l} = \int_O^N \vec{E} \cdot d\vec{l} = 0$$

$\therefore$ Emf induced in the ΔOMN is entirely in MN .

$$\therefore \epsilon_{\text{rod}} = \frac{1}{2} r^2 \theta \frac{dB}{dt}$$

Proceeding similarly, we see that same emf is induced in the rod in (B) also.

In (A) and (D), the flux through ΔOMN is less than $\frac{1}{2} r^2 \theta B$. Therefore, induced emf is less than that given by (i).

B is increasing. This means that flux through $\triangle OMN$ is increasing. The induced emf will try to drive an anticlockwise current in $\triangle OMN$. Therefore, end M is positive in all diagrams.

6. (A) Voltage leads in phase by 45°

$$\cos \phi = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Because the voltage leads, the circuit is either LR or LCR with $X_L > X_C$.

- (B) Current leads in phase by 45°

$$\therefore \cos \phi = \frac{1}{\sqrt{2}}$$

The circuit is either CR or LCR with $X_C > X_L$.

- (C) Current leads in phase by 90°

$$\therefore \cos \phi = 0$$

The circuit must be LC circuit with $X_C > X_L$.

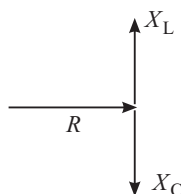
- (D) Voltage leads in phase by 90°

$$\therefore \cos \phi = 0.$$

The circuit is LC circuit with $X_L > X_C$.

8. If $X_C < X_L$, then ϕ will decrease. The power factor increases. Impedance decreases and hence the current drops. The power dissipated in the coil will decrease ($P = I^2 R$). The voltage across the coil will also decrease ($V_{\text{coil}} = I Z_{\text{coil}}$).

Similarly think of situations when $X_C = X_L$ and when $X_C > X_L$. You must also think about a situation when $X_C > 2X_L$. In this case, the impedance will increase.



Passage Based Problems

Passage 1

1. Let the speed of the rod at a time t after its release be v .

$$Blv - L \frac{di}{dt} = 0 \quad \Rightarrow \quad Bl v dt = L di$$

$$\text{or, } Bl dx = L di \quad [dx = v dt = \text{small displacement of the rod in interval } dt]$$

$$\text{or, } Bl \int_0^x dx = L \int_0^I di$$

$$\text{or, } Blx = LI \quad \Rightarrow \quad I = \frac{Blx}{L} \quad (\text{i})$$

2. For the system of the block and the rod, we can write

$$2ma = mg - F_B$$

$[F_B = \text{magnetic force on the rod}]$

$$\text{or, } 2mv \frac{dv}{dx} = mg - IlB$$

$$\text{or, } 2mv \frac{dv}{dx} = mg - \frac{B^2 l^2 x}{L}$$

$$\text{or, } 2v dv = \left(g - \frac{B^2 l^2}{mL} x \right) dx$$

$$\text{or, } 2 \int_0^v v dv = \int_0^x \left(g - \frac{B^2 l^2}{mL} x \right) dx$$

$$\text{or, } v^2 = gx - \frac{B^2 l^2 x^2}{2mL}$$

$$\text{or, } v = \sqrt{gx - \frac{B^2 l^2 x^2}{2mL}}$$

Passage 2

3. Using Faraday's law, we will write the emfs in the two loops. Emf in the left loop is

$$\epsilon_1 = \frac{d\phi}{dt} = \pi r_1^2 \frac{dB}{dt} = \pi (0.1)^2 \times 100 = \pi \text{ V}$$

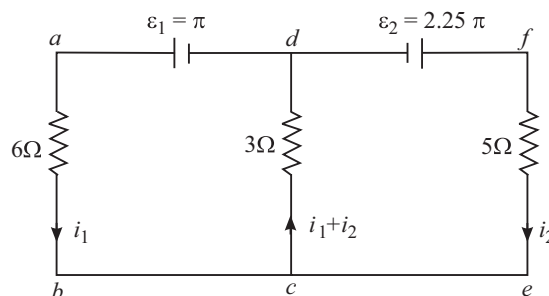
The field is increasing in inward direction. The induced emf will try to induce an anticlockwise current in the loop.

Emf in the right loop is

$$\epsilon_2 = \pi r_2^2 \frac{dB}{dt} = \pi (0.15)^2 \times 100 = 2.25\pi \text{ V}$$

The field is directed outward and is increasing. The induced emf will try to induce a clockwise current in the loop.

For the loop $abcd$, we can write



$$6i_1 + 3(i_1 + i_2) = \pi$$

$$\text{or, } 9i_1 + 3i_2 = \pi \quad (\text{i})$$

For the loop $cdfe$, we can write

$$3(i_1 + i_2) + 5i_2 = 2.25\pi$$

$$\text{or, } 3i_1 + 8i_2 = 2.25\pi \quad (\text{ii})$$

Solving (i) and (ii) gives

$$i_1 = 62 \text{ mA}; i_2 = 860 \text{ mA}$$

$$\text{Current in } 3\Omega \text{ resistance} = i_1 + i_2 = 922 \text{ mA}$$

Passage 3

6. Let the velocity of the ring be v at time t . Emf induced along semicircle DEF is

$$\varepsilon = Bv(2r) = 2Bvr$$

Same emf is induced in the semicircle DGF .

$\therefore$ Current in R is

$$i = \frac{\varepsilon}{R} = \frac{2Bvr}{R}$$

Current in two semicircular parts is

$$\frac{i}{2} = \frac{Bvr}{R}$$

Magnetic force on the semicircular part DEF is

$$= \frac{i}{2}(2r)B = irB,$$

Same force acts on the other half of the ring. The total magnetic force on the ring is

$$F_B = 2irB \\ = \frac{4B^2r^2v}{R} \text{ (upward)}$$

For the translational motion of the ring we can write

$$ma = mg - F_B - T$$

$$\text{or, } a = g - \frac{4B^2r^2v}{mR} - \frac{T}{m} \quad (i)$$

For rotational motion about the centre

$$(mr^2)\alpha = T \cdot r \quad [mr^2 = \text{moment of inertia} \\ \alpha = \text{angular acceleration}]$$

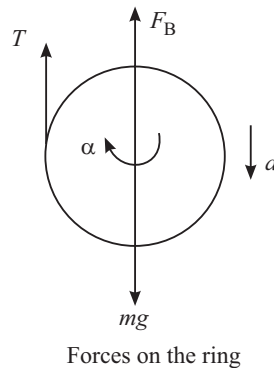
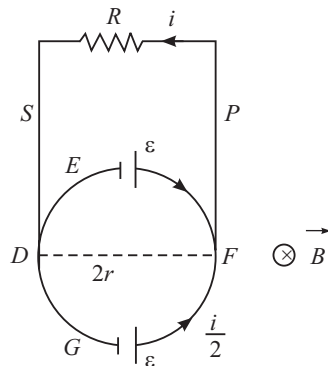
$$\text{or, } a = \frac{T}{m} \quad [\because r\alpha = a] \quad (ii)$$

Solving (i) and (ii) gives

$$a = \frac{g}{2} - \frac{2B^2r^2v}{mR}$$

When the terminal velocity is attained, $a = 0$

$$\Rightarrow v_0 = \frac{mgR}{4B^2r^2}$$



Passage 4

7. The semicircular loop rotates through an angle θ in time t given by

$$\theta = \frac{1}{2} \alpha t^2$$

$$\text{or, } t = \sqrt{\frac{2\theta}{\alpha}}$$

For $\theta = 0$ to π , 2π to 3π , 4π to 5π , etc., the flux linked with the loop increases and the induced current must be anticlockwise (i.e., negative) so as to weaken the magnetic field.

Similarly, one can see that for $\theta = \pi$ to 2π , 3π to 4π , 5π to 6π etc., the direction of induced current (or induced emf) is positive.

Time needed to rotate through an angle π is

$$t_1 = \sqrt{\frac{2\pi}{\alpha}}$$

Time when the ring completes a rotation of 2π is

$$t_2 = \sqrt{\frac{4\pi}{\alpha}}$$

$$\text{Time taken to rotate through } n\pi \text{ is } t_n = \sqrt{\frac{2n\pi}{\alpha}}$$

For interval 0 to t_1 the emf is negative, for t_1 to t_2 it is positive, for t_2 to t_3 it is again negative and so on. Area of the loop inside the field at time t is

$$A = \frac{1}{2} R^2 \theta = \frac{1}{2} R^2 \left(\frac{1}{2} \alpha t^2 \right) = \frac{1}{4} R^2 \alpha t^2$$

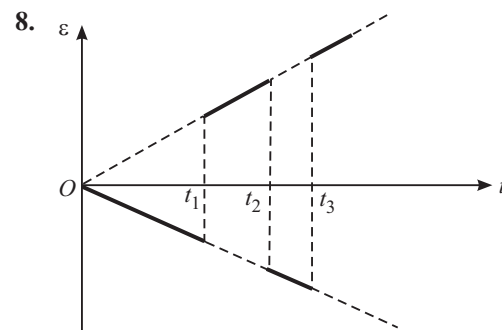
$$\therefore \phi = BA = \frac{1}{4} R^2 \alpha B t^2$$

$$\therefore \varepsilon = \left| \frac{d\phi}{dt} \right| = \frac{1}{2} R^2 \alpha B t$$

$\therefore$ Magnitude of ε is proportional to time

$$\therefore \varepsilon = (-1)^n \left(\frac{1}{2} R^2 \alpha B t \right)$$

$n = 1, 2, 3, \dots$ is number of half revolutions.

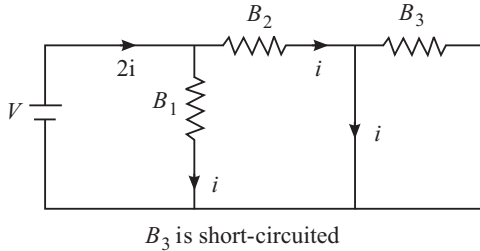


Note that time needed to complete successive half revolutions goes on decreasing

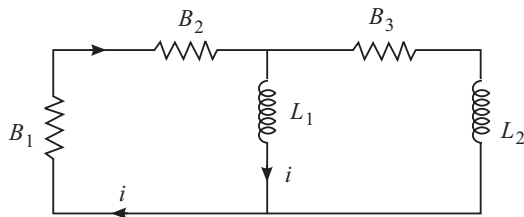
$$t_1 > (t_2 - t_1) > (t_3 - t_2)$$

Passage 5

9. Effective circuit in steady state is as shown.



10. After the switch is opened, the currents in the branches having inductors cannot change suddenly. Current in L_2 (and hence in B_3) will remain zero. Same current will circulate in B_1 and B_2 .

**Passage 6**

11. Immediately after the switch is closed, the entire current i_0 will pass through the resistance. The current in the inductor begins to increase and after a long time, the entire current will pass through the inductor.
12. Let current in L and R be i and $(i_0 - i)$, respectively at time t .

$$L \frac{di}{dt} = R(i_0 - i)$$

$$\text{or, } \int_0^i \frac{di}{i_0 - i} = \frac{R}{L} \int_0^t dt$$

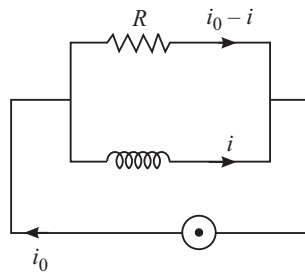
$$\ln \left(\frac{i_0 - i}{i_0} \right) = - \frac{Rt}{L}$$

$$\text{or, } i = i_0 \left(1 - e^{-\frac{Rt}{L}} \right)$$

13. Current in R changes with time as

$$i_1 = i_0 - i = i_0 e^{-\frac{Rt}{L}}$$

$$\begin{aligned} \therefore \text{Heat dissipated in } R &= \int_0^\infty i_1^2 R dt \\ &= i_0^2 R \int_0^\infty e^{-\frac{2Rt}{L}} dt \end{aligned}$$



$$= \frac{i_0^2 R}{-2R} \left[e^{-\frac{2Rt}{L}} \right]_0^\infty = \frac{1}{2} Li_0^2$$

Passage 7

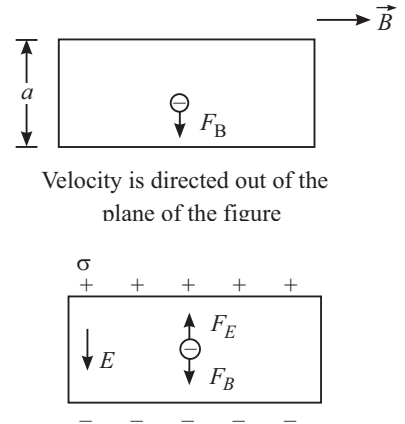
14. An electron in the material experiences a downward magnetic field $[\vec{F}_B = -e\vec{v} \times \vec{B}]$. The lower face becomes negative

and the upper face becomes positive. This produces an electric field inside the material. The accumulation of charges on the two faces stops when the electric force balances the magnetic force.

$$eE = evB$$

$$\Rightarrow E = vB$$

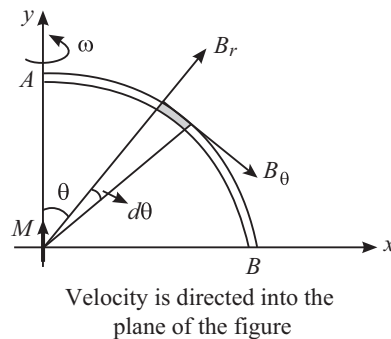
$$\text{or, } \frac{\sigma}{\epsilon_0} = vB \Rightarrow \sigma = \epsilon_0 vB$$



$$15. V = E \cdot a = vBa$$

Passage 8

17. The magnetic field due to the magnetic dipole at an angular position θ has two components



$$B_r = \frac{\mu_0}{4\pi} \frac{2M \cos \theta}{R^3}$$

$$B_\theta = \frac{\mu_0}{4\pi} \frac{M \sin \theta}{R^3}$$

B_θ is parallel to the length of a small element on the ring. B_r is perpendicular to the element and an emf gets induced due to this. Emf induced in the element of length $Rd\theta$ is

$$\begin{aligned}
 d\varepsilon &= (B_r) (Rd\theta) (\omega R) \\
 &= \left(\frac{\mu_0}{4\pi} \frac{2M \cos \theta}{R^3} \right) (Rd\theta) (\omega R) \\
 &= \frac{\mu_0}{2\pi} \frac{M\omega}{R} \cos \theta d\theta
 \end{aligned}$$

Emf induced between A and B is

$$\varepsilon = \frac{\mu_0}{2\pi} \frac{M\omega}{R} \int_0^{\pi/2} \cos \theta d\theta = \frac{\mu_0 M \omega}{2\pi R}$$

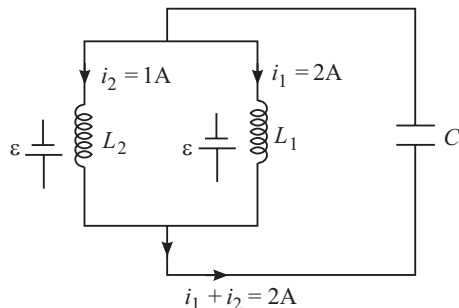
The positive side of the emf is indicated by $\vec{v} \times \vec{B}_r$.
The end B is positive.

Passage 9

18. The two inductors are in parallel. Therefore, induced emf in both of them must always be same.

$$\left| L_1 \frac{di_1}{dt} \right| = \left| L_2 \frac{di_2}{dt} \right|$$

$$\Rightarrow \left| \frac{di_1}{dt} \right| = \left| \frac{di_2}{dt} \right| \quad [\because L_1 = L_2]$$

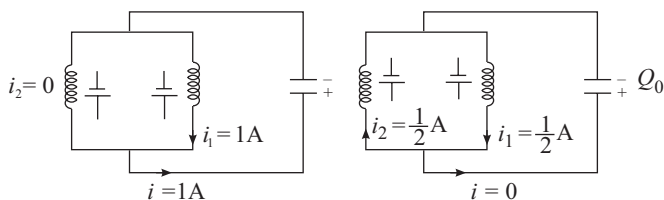


The current in the two inductors must change at the same rate. Initially the charging current for the capacitor is $i_1 + i_2 = 3A$.

After some time, the current in L_2 will reduce to zero and the current in L_1 will become 1A (i.e. in both the inductors the current will reduce by same amount 1A). The current in L_1 will further keep reducing and that in L_2 will begin to increase in opposite direction.

When $i_1 = \frac{1}{2}A$ and $i_2 = \frac{1}{2}A$ (in opposite direction),

there is no current to the capacitor. Till this time there was a charging current. At this instant, the charge on the capacitor is maximum.



Rate of change of i_1 and i_2 is always same

19. Energy conservation gives

$$\frac{Q_0^2}{2C} + \frac{1}{2}L\left(\frac{1}{2}\right)^2 + \frac{1}{2}L\left(\frac{1}{2}\right)^2 = \frac{1}{2}L(1)^2 + \frac{1}{2}L(2)^2$$

$$\text{or, } \frac{Q_0^2}{C} = \left(1 + 4 - \frac{1}{4} - \frac{1}{4}\right) \times 1$$

$$\text{or, } Q_0 = \sqrt{\frac{9}{2} \times 10^{-6}} = \frac{3}{\sqrt{2}} \text{ mC}$$

Passage 10

20. When $C = 0$, $X_C = \frac{1}{\omega C} = \infty$

Current in the circuit will be zero and $V_2 = 0$.

$\therefore$ The voltmeter reads the source voltage which is equal to 220 V. Also when $V_L = 440V$ and $V_C = 220V$, the voltage across the L - C combination will be 220V. This is because the two voltages have a phase difference of π .

$$V_L = 2V_C$$

$$\Rightarrow X_L = 2X_C \Rightarrow \omega L = \frac{2}{\omega C}$$

$$\Rightarrow C = \frac{2}{\omega^2 L} = \frac{2}{(2\pi f)^2 \times 1} = \frac{2}{4 \times \pi^2 \times 50^2 \times 1}$$

$$\text{or, } C = 20\mu F$$

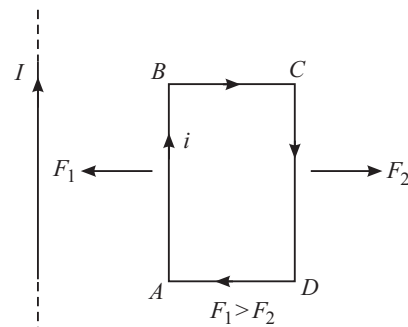
21. When $X_C = X_L \Rightarrow \frac{1}{\omega C} = \omega L \Rightarrow C = 10\mu F$,

the impedance in the circuit will become zero and the current will be very large.

Passage 11

22. Magnetic field due to the straight wire is directed into the plane of the paper ($\otimes$) at the location of the loop. When the current is switched off, the flux linked to the loop decreases. A clockwise current is induced in the loop to oppose the change in flux. The arm AB gets attracted by the wire and the arm CD is repelled. Because AB is nearer to the wire, it experiences a larger force.

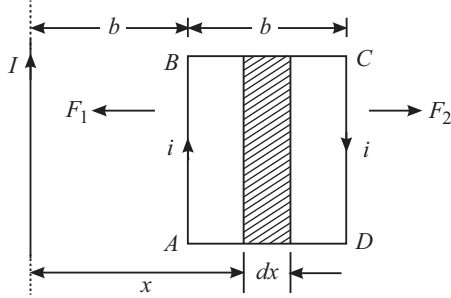
$\therefore$ The loop gets attracted.



23. Flux linked with the rectangular loop is

$$\phi = \int_b^{2b} \frac{\mu_0 I}{2\pi x} l dx = \frac{\mu_0 I l}{2\pi} \int_b^{2b} \frac{dx}{x}$$

$$= \frac{\mu_0 I l}{2\pi} \ln(2)$$



Emf induced in the loop is

$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0 \ln(2) l}{2\pi} \frac{dI}{dt}$$

Current induced in the loop is $i = \frac{\varepsilon}{R}$

$$= \frac{\mu_0 \ln(2) l}{2\pi R} \frac{dI}{dt}$$

Force on arm AB is $F_1 = \frac{\mu_0 I i}{2\pi b} l$

Force on arm CD is $F_2 = \frac{\mu_0 I i}{2\pi(2b)} l$

Force on the loop, $F = F_1 - F_2 = \frac{\mu_0 I i l}{4\pi b}$

Impulse of this force = $\int F dt = \int \frac{\mu_0 I l}{4\pi b} i dt$

$$= \int \frac{\mu_0 I l}{4\pi b} \frac{\mu_0 \ln(2) l}{2\pi R} dI \quad [\text{using (i)}]$$

$$= \frac{\mu_0^2 (\ln 2) l^2}{8\pi^2 b R} \int I dI = \frac{\mu_0^2 (\ln 2) l^2 I^2}{16\pi^2 b R}$$

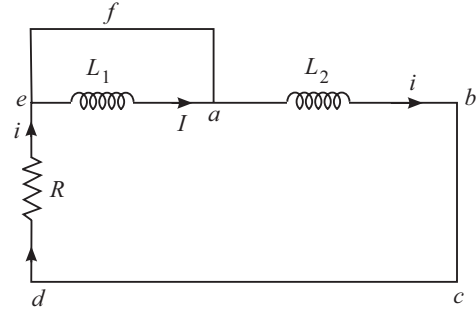
$$\therefore \text{Velocity acquired} = \frac{\text{Impulse}}{\text{Mass}} = \frac{\mu_0^2 (\ln 2) l^2 I^2}{16\pi^2 b m R}$$

Passage 12

24. With the switch at position 1, the current flows along L_1 and L_2 (there is no current in R as the inductors behave like plain wires and R is short-circuited). The current is $I = \frac{V}{r}$.

After the switch is shifted to position 3, the potential difference across L_1 becomes zero. This means that there is no induced emf in L_1 . This means that the current in L_1 does not change.

25. The current in L_2 decays with time constant $\frac{L_2}{R}$. Effective path is abcdefa.



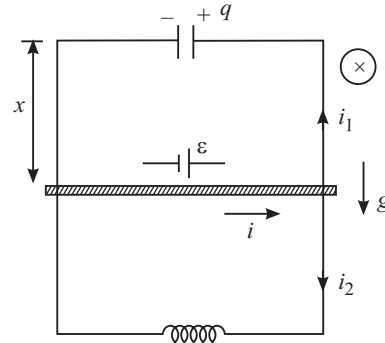
$$i = \frac{V}{r} e^{-\frac{Rt}{L_2}}$$

Current in the path fe is superposition of I and i .

$$\therefore \text{Current} = I - i = \frac{V}{r} \left[1 - e^{-\frac{Rt}{L_2}} \right]$$

Passage 13

(i) 26. Emf induced, $\varepsilon = Blv$



If q is charge on the capacitor

$$\frac{q}{C} = Blv$$

$$\text{or, } \frac{dq}{dt} = B l C \frac{dv}{dt}$$

$$\text{or, } i_1 = B l C a \quad (i)$$

$$\text{Also, } \varepsilon = L \frac{di_2}{dt}$$

$$\text{or, } Blv = L \frac{di_2}{dt}$$

$$\text{or, } \int_0^{i_2} di_2 = \frac{Bl}{L} \int_0^t v dt$$

$$\text{or, } i_2 = \frac{Blx}{L} \quad (ii)$$

$$\text{Current in the conductor is } i = i_1 + i_2 = B l C a + \frac{Blx}{L}$$

27. Force equation for the motion of the conductor is

$$ma = mg - ilB$$

$$\text{or, } a = g - \frac{B^2 l^2 C a}{m} - \frac{B^2 l^2 x}{mL}$$

$$\text{or, } a \left(1 + \frac{B^2 l^2 C}{m} \right) = - \frac{B^2 l^2}{mL} \left(x - \frac{mgL}{B^2 l^2} \right)$$

$$\text{or, } a = - \frac{B^2 l^2}{L(m + B^2 l^2 C)} \left(x - \frac{mgL}{B^2 l^2} \right)$$

$\therefore$ Motion is SHM with equilibrium at $x = \frac{mgL}{B^2 l^2}$

Passage 14

28. Induced electric field, $E = \frac{r}{2} \frac{dB}{dt}$

$$\text{Force on the charge, } F = qE = \frac{qr}{2} \frac{dB}{dt}$$

This force is tangential (perpendicular to the radius)

$$\therefore \text{Impulse } \int F dt = \frac{qr}{2} \int dB = \frac{qr}{2} B_0$$

$$[\int dB = \text{change in field} = B_0]$$

$$\text{Momentum of the particle } P = \frac{qrB_0}{2}$$

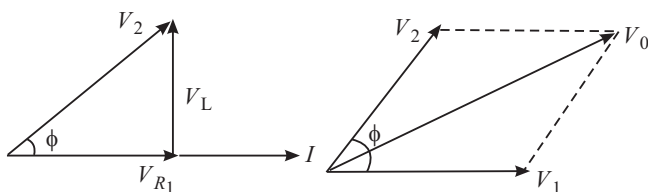
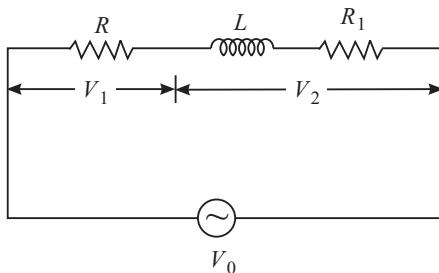
Now the particle begins to move in circle in the uniform magnetic field B_0 . The radius of the circular

$$\text{path is } r' = \frac{P}{qB_0} = \frac{r}{2}$$

29. The particle will keep moving in the circle forever.

Passage 15

30. L is the inductance and R_1 is the resistance of the coil. The voltage (V_2) across the coil is leading in phase to the current by some angle ϕ .



Therefore, the correct representation of V_1 and V_2 is given in option (A).

$$31. V_1^2 + V_2^2 + 2V_1V_2 \cos \phi = V_0^2$$

$$\text{or, } V_2 \cos \phi = \frac{V_0^2 - V_1^2 - V_2^2}{2V_1}$$

The power dissipated in the coil is

$$P_{\text{coil}} = I V_2 \cos \phi = \frac{V_1}{R} \frac{V_0^2 - V_1^2 - V_2^2}{2V_1}$$

$$\text{or, } P_{\text{coil}} = \frac{V_0^2 - V_1^2 - V_2^2}{2R}$$

$$32. P = P_{\text{coil}} + P_R = \frac{V_0^2 - V_1^2 - V_2^2}{2R} + \frac{V_1^2}{R}$$

$$\text{or, } P = \frac{V_0^2 + V_1^2 - V_2^2}{2R}$$

Passage 16

33. When only box A is connected to the source, the impedance is

$$Z_A = \frac{100V}{2A} = 50\Omega$$

Because the current leads in phase by 90° , the box A must have a capacitor and an inductor with $X_C > X_L$.

For the box A

$$Z_A = X_C - X_L = 50\Omega$$

There is no resistance in the box.

34. When both A and B are connected, the power dissipated is 80W.

$$\therefore I^2 R = 80 \quad \Rightarrow R = \frac{80}{1} = 80\Omega$$

This resistance is there in box B .

35. Overall impedance of the circuit is

$$Z = \frac{100V}{1A} = 100\Omega$$

The second element in the box B must be an inductor. Then only the voltage can lead the current in phase.

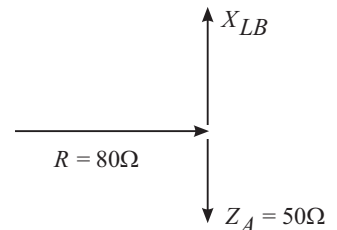
$$(100)^2 = (80)^2 + (X_{LB} - 50)^2$$

$$\text{or, } X_{LB} - 50 = 60$$

$$\text{or, } X_{LB} = 110\Omega$$

$\therefore$ Impedance of box B is

$$Z_B = \sqrt{80^2 + (110)^2} = 136\Omega$$



Passage 17

36. From Kirchhoff's junction law,

$$i_4 + i_3 = i_1 + i_2 \quad [\text{Take note of the positive direction of each current}]$$

$$\text{or, } i_4 = 5 + 4 \sin(10\pi t) - 3 \cos(10\pi t)$$

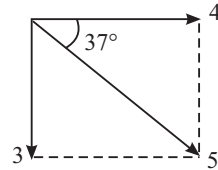
$$= 5 + 4 \sin(10\pi t) + 3 \sin(10\pi t - \frac{\pi}{2})$$

$$= 5 + 5 \sin(10\pi t - 37^\circ)$$

The value of i_4 oscillates between +10 and 0.

At $t = 0$, its value is 2A.

Therefore, graph given in (A) is correct.



- 37.
- $i_4^2 = 5^2 + 5^2 \sin^2(10\pi t - 37^\circ) + 50 \sin(10\pi t - 37^\circ)$

$$\langle i_4^2 \rangle = 25 + \frac{25}{2} + 0 = 37.5$$

$$\therefore \sqrt{\langle i_4^2 \rangle} = 6.12 \text{ A}$$

Passage 18

38. Current (
- I
-) in both the parallel paths will be same. Therefore, voltage across both the inductors is

$$V_L = IX_L$$

$\therefore$ The required ratio is 1 : 1.

39. The current in both the parallel paths will always be same and will have same phase.

The voltage across L will be 90° ahead of the current and the voltage across C will be 90° behind the current. Therefore, the voltage v_{pa} and v_{pb} will have a phase difference of 180° between them. When one is at positive maximum the other is at its negative maximum. This means that the maximum value of instantaneous potential difference between the points a and b will be

$$(v_{ab})_{\max} = v_{LO} + v_{CO} \quad (i)$$

where v_{LO} and v_{CO} are peak voltages across the inductor and the capacitor

$$X_L = \omega L = 300 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 25 \times 10^{-6}} = 400 \Omega$$

$$\text{Impedance of one branch is } Z = \sqrt{X_L^2 + X_C^2} = 500 \Omega$$

$$\text{Peak current in one branch is } i_0 = \frac{v_0}{Z} = \frac{100}{500} = \frac{1}{5} \text{ A}$$

$$\therefore v_{LO} = i_0 X_L = \frac{1}{5} \times 300 = 60 \text{ V}$$

$$v_{CO} = i_0 X_C = \frac{1}{5} \times 400 = 80 \text{ V}$$

$$\text{From (i) } (v_{ab})_{\max} = 60 + 80 = 140 \text{ V}$$

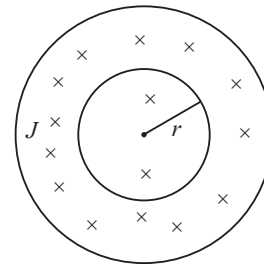
Passage 19

$$40. \frac{1}{2} \epsilon_0 E^2 = u_E \quad \Rightarrow \quad E = \sqrt{\frac{2u_E}{\epsilon_0}} \quad (i)$$

Due to the electric field, there will be a current in the rod. The current density is given by Ohm's law

$$j = \sigma E = \sigma \sqrt{\frac{2u_E}{\epsilon_0}} \quad (\text{uniform})$$

41. Magnetic field at a distance
- r
- from the axis is



Cross section of the rod. Current density is perpendicular to the plane of the paper

$$B = \left(\frac{\mu_0 j}{2} \right) r = \left(\mu_0 \sigma \sqrt{\frac{u_E}{2\epsilon_0}} \right) \cdot r$$

$$u_B = \frac{B^2}{2\mu_0}$$

$$\text{or, } u_B = \left(\frac{\sigma^2 u_E \mu_0}{4\epsilon_0} \right) r^2$$

$$\text{or, } \sigma = \frac{2}{r} \sqrt{\frac{\epsilon_0 u_B}{\mu_0 u_E}} = 3.7 \times 10^7 (\Omega \text{ m})^{-1}$$

Chapter 8 Electromagnetic Waves



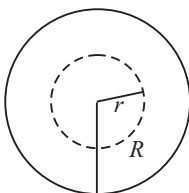
Your Turn

2. $i_d = \text{conduction current} = \frac{V}{R} = \frac{10}{10} = 1 \text{ A}$

3. (a) Unit of ϕ_E is $\frac{V}{m} \cdot m^2 = Vm$

$\therefore$ Unit of a is $\frac{Vm}{s}$

- (b) Consider a circular loop of radius r ($< R$) with its centre at the centre of the given region. Due to symmetry, we can assume the magnetic field to be tangential at all points on the loop having same magnitude B .



$$\oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi r$$

Electric flux through the surface of the loop is

$$\phi = \left(\frac{\pi r^2}{\pi R^2} \right) at$$

$$\text{or, } \frac{d\phi}{dt} = \frac{r^2}{R^2} \cdot a$$

$\therefore$ Using Ampere's law

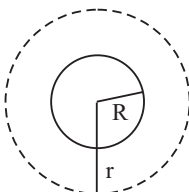
$$B \cdot 2\pi r = \mu_0 i_d$$

$$\text{or, } B \cdot 2\pi r = \mu_0 \epsilon_0 \frac{d\phi}{dt}$$

$$\text{or, } B \cdot 2\pi r = \mu_0 \epsilon_0 \frac{r^2 a}{R^2}$$

$$\text{or, } B = \left(\frac{\mu_0 \epsilon_0 a}{2\pi R^2} \right) r$$

- (c) Consider a loop of radius r ($> R$). Again due to symmetry, we can assume field at all points to be B tangential to the loop.



$$\oint \vec{B} \cdot d\vec{l} = B \cdot 2\pi r$$

$$\text{Using Ampere's law, } B \cdot 2\pi r = \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

$$\text{or, } B \cdot 2\pi r = \mu_0 \epsilon_0 \cdot a$$

$$\text{or, } B = \frac{\mu_0 \epsilon_0 a}{2\pi r}$$

4. (a) $B_0 = \frac{E_0}{c} = \frac{10}{3 \times 10^8} = 3.3 \times 10^{-7} \text{ T}$

(b) $I = \frac{1}{2} \epsilon_0 E_0^2 \cdot c$
 $= \frac{1}{2} \times 8.85 \times 10^{-12} \times 10^2 \times 3 \times 10^8$
 $= 0.133 \text{ W/m}^2$

6. (a) $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^{14}} = 3 \times 10^{-5} \text{ m}$

(b) $\langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 = \frac{1}{2} \times 8.85 \times 10^{-12} \times 50^2$
 $= 1.1 \times 10^{-8} \text{ Jm}^{-3}$



Worksheet 1

1. Oscillating charge in a wire carrying alternating current will produce an electromagnetic wave.

2. $\oint \vec{B} \cdot d\vec{l} = \mu_0 i_d$

Displacement current is maximum and equal for paths 4 and 3, followed by 2 and 1.

6. Microwaves are preferred for radar systems due to their short wavelengths.

7. Gases like water vapour, carbon dioxide, etc., absorb the radiation emitted by the Earth in infrared region (which would otherwise escape into space). Increased absorption causes global warming.

8. $dU = \left(\frac{1}{2} \epsilon_0 E^2 + \frac{1}{2\mu_0} B^2 \right) dV$ [dV = small volume]

$$= \epsilon_0 E^2 dV \quad \left[\because \frac{1}{2\mu_0} B^2 = \frac{1}{2} \epsilon_0 E^2 \right]$$

$$= \epsilon_0 E_0^2 \sin^2(\omega t - kx) \cdot dV$$

$$= \frac{1}{2} \epsilon_0 E_0^2 \cdot dV [1 - \cos^2(\omega t - kx)]$$

Angular frequency of oscillation = 2ω

9. $B_0 = \frac{E_0}{c} = \frac{900}{3 \times 10^8} = 3 \times 10^{-6} \text{ T}$

10. Frequency of wave = $1 \text{ GHz} = 10^9 \text{ Hz}$.

$$\therefore \lambda = \frac{c}{f} = \frac{3 \times 10^8}{10^9} = 0.3 \text{ m}$$

$$11. \omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 1 \times 10^{-6}}} = 10^3 \text{ rad s}^{-1}$$

$$\text{or, } f = \frac{10^3}{2 \times 3.14} = 159 \text{ Hz}$$

The electrons oscillate with this frequency. They produce electromagnetic waves of same frequency.

$$\lambda = \frac{3 \times 10^8}{159} = 1.89 \times 10^6 \text{ m}$$



Worksheet 2

1. When the charge on the capacitor plate is changing there is time changing E inside it. This will induce a magnetic field.

$$2. \frac{\omega}{k} = \frac{2\pi f}{2\pi/\lambda} = \lambda f = \text{speed of the wave (c)}$$

Speed of electromagnetic wave in space is an universal constant.

If frequency changes, ω changes and $k \left(= \frac{2\pi}{\lambda} \right)$ will also change.

$$3. I = \frac{1}{2} \epsilon_0 E_0^2 \cdot c = \frac{1}{2} \epsilon_0 \left(\frac{E_0}{\sqrt{2}} \right)^2 c = \epsilon_0 c E_{\text{rms}}^2$$

[E is varying sinusoidally. Its rms value is related to its peak value as $E_{\text{rms}} = \frac{E_0}{\sqrt{2}}$]

$$\therefore I = \frac{E_{\text{rms}}^2}{c\mu_0} \left[\because c^2 = \frac{1}{\epsilon_0\mu_0} \Rightarrow \epsilon_0 c = \frac{1}{c\mu_0} \right]$$

$$\text{Because } B_0 = \frac{E_0}{c}$$

$$\therefore B_{\text{rms}} = \frac{E_{\text{rms}}}{c}$$

$$\therefore I = \frac{(cB_{\text{rms}})^2}{c\mu_0} = \frac{cB_{\text{rms}}^2}{\mu_0}$$

4. In absence of the atmosphere, there would be no greenhouse effect. The Earth will become colder.



Worksheet 3

$$1. (a) \omega = 3 \times 10^{15} \text{ rad s}^{-1}$$

$$\text{or, } f = \frac{3 \times 10^{15}}{2\pi} \text{ Hz}$$

$$\therefore \lambda = \frac{c}{f} = \frac{3 \times 10^8 \times 2\pi}{2 \times 10^{15}} = 6.28 \times 10^{-7} \text{ m} \\ = 6280 \text{ \AA}$$

This is visible light.

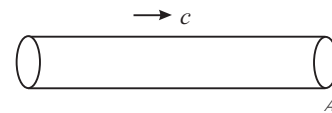
$$(b) E_0 = cB_0 = 3 \times 10^8 \times 10 \times 10^{-6} = 3000 \text{ Vm}^{-1}$$

$$(c) \langle u \rangle = \frac{1}{2} \epsilon_0 E_0^2 = \frac{1}{2} \times 8.85 \times 10^{-12} \times (3000)^2 \\ = 3.98 \times 10^{-15} \text{ Jm}^{-3}$$

$$2. I = \langle u \rangle c = \frac{1}{2} \epsilon_0 E_0^2 \cdot c$$

$$\therefore E_0 = \sqrt{\frac{2I}{\epsilon_0 c}} = \sqrt{\frac{2 \times 8}{8.85 \times 10^{-12} \times 3 \times 10^8}} = 77.6 \text{ Vm}^{-1}.$$

3. 10 mJ energy crosses through any cross-section in 1 second. It means that a cylinder of length $c = 3 \times 10^8 \text{ m}$ will contain 10 mJ energy inside it.



$\therefore$ Energy contained in 10 cm length will be

$$E = \frac{10 \times 10^{-3} \text{ J}}{(3 \times 10^8 \text{ m})} \times (0.1 \text{ m}) = 3.33 \times 10^{-12} \text{ J}$$

4. rms value of $i_d = \text{rms value of current in the circuit}$
 $= \frac{V}{R} = 5 \text{ A}.$
5. The circuit described in Q.4 is in resonance. When the frequency is increased, the current decreases.

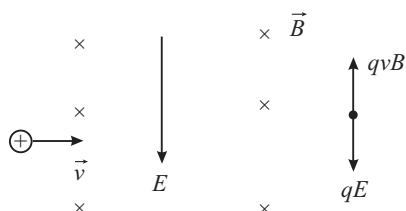
Chapter 9 Previous Years' Questions

MAGNETIC FIELD

ALIEEE/JEE Main Questions

1. Because velocity is parallel to $\vec{B}$, there will be no magnetic force on the electron. The electric force is opposite to motion and hence the electron slows down.
2. Time period of circular motion, $T = \frac{2\pi m}{qB}$
3. Only electric force will act and the particle will accelerate along the direction of the field.
4. Force on charge must be zero

$$q\vec{E} + q\vec{v} \times \vec{B} = 0 \Rightarrow \vec{E} = -\vec{v} \times \vec{B}$$



The diagram shows such a situation when the electric force and the magnetic force are equal and opposite. $\vec{E}$ is directed along $-\vec{v} \times \vec{B}$. One can easily see that $\vec{v}$ is directed along $\vec{E} \times \vec{B}$.

Magnitude of v must be $\frac{E}{B}$.

Therefore, option (b) is appropriate.

5. The particle gets deflected. Direction of its momentum keeps changing. But the speed of the particle does not change. Magnetic force on a moving charge never performs any work on it.

$$6. r = \frac{\sqrt{2mK}}{qB}$$

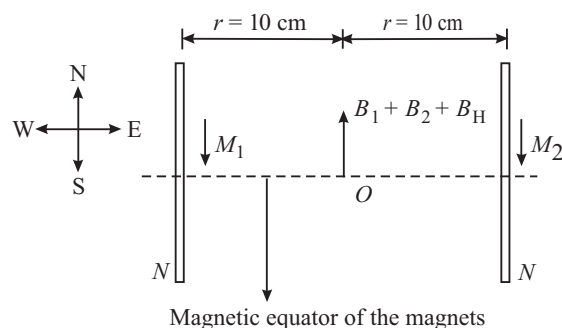
Kinetic energy (K) and magnetic field (B) are same for all.

$$\begin{aligned} \therefore r &\propto \frac{\sqrt{m}}{q} \\ r_p : r_d : r_\alpha &= \frac{\sqrt{m}}{e} : \frac{\sqrt{2m}}{e} : \frac{\sqrt{4m}}{2e} \\ &= 1 : \sqrt{2} : 1 \\ \therefore r_\alpha &= r_p < r_d \end{aligned}$$

Note: A deuteron has one proton and one neutron. An alpha particle has 2 protons and 2 neutrons.

7. The situation is as shown in the figure. The field at a point on the equator of a dipole is directed opposite

to its dipole moment. Therefore, the fields due to the two magnets and the horizontal component of the Earth's magnetic field, all are in the same direction (towards north). The resultant field is



$$\begin{aligned} B &= B_1 + B_2 + B_H \\ &= \frac{\mu_0 M_1}{4\pi r^3} + \frac{\mu_0 M_2}{4\pi r^3} + B_H \\ &= 10^{-7} \times \frac{1.2}{(0.1)^3} + 10^{-7} \times \frac{1.0}{(0.1)^3} + 3.6 \times 10^{-5} \\ &= 2.2 \times 10^{-4} + 0.36 \times 10^{-4} = 2.56 \times 10^{-4} \text{ T} \end{aligned}$$

$$8. \text{ Constant velocity of the conductor is } = \frac{2m}{5 \times 10^{-3} \text{ s}} = 400 \text{ ms}^{-1}$$

As the conductor moves along x direction, the magnitude of magnetic field changes through its direction remains fixed along the y direction.

The magnetic force on the conductor when its x co-ordinate is x is

$$\begin{aligned} \vec{F}_B &= I \vec{L} \times \vec{B} = 10 (-3\hat{k}) \times (3 \times 10^{-4} e^{-0.2x} \hat{j}) \\ &= (9 \times 10^{-3} e^{-0.2x}) \hat{i} \end{aligned}$$

Work done by the magnetic force on the conductor is

$$\begin{aligned} W_B &= \int_0^2 F_B dx = 9 \times 10^{-3} \int_0^2 e^{-0.2x} dx \\ &= \frac{9 \times 10^{-3}}{-0.2} [e^{-0.2x}]_0^2 = 4.5 \times 10^{-2} [1 - e^{-0.4}] \end{aligned}$$

Average power of the magnetic force, $P_B = \frac{W}{\Delta t}$

$$\begin{aligned} P_B &= \frac{4.5 \times 10^{-2}}{5 \times 10^{-3}} [1 - e^{-0.4}] \\ &= 9 (1 - e^{-0.4}) \end{aligned}$$

$$\text{Now, } e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

$$e^{-0.4} = 1 - \frac{0.4}{1} + \frac{(0.4)^2}{2} - \frac{(0.4)^3}{6} + \dots$$

$$\simeq 1 - 0.4 = 0.6$$

$$\therefore P_B = 9 \times 0.6 = 3.6 \text{ W}$$

The external agent must apply a force equal and opposite to the magnetic force to keep the velocity constant. Work done by external agent is negative and the power delivered is also negative.

$$P_{\text{ext}} = -3.6 \text{ W}$$

The question does not have options with negative sign and the nearest magnitude to our calculated value is 2.97W [Exact value of $e^{-0.4} = 1.67$ and this indeed gives $P = 2.97\text{W}$].

9. The loop is in stable equilibrium when the magnetic dipole moment is parallel to $\vec{B}$ [in option (b)] and the loop is in unstable equilibrium when the dipole moment is antiparallel to $\vec{B}$ [in option (d)]
10. This question is similar to question (6) above.

IIT JEE/JEE Advanced Questions

1. (a) The magnetic torque on the coil is
 $\tau = \mu B \cos 0^\circ = N A i B$
 $\therefore K = NAB$
- (b) In equilibrium, the spring torque balances the magnetic torque
or, $C\theta = NABi$
or, $C \cdot \frac{\pi}{2} = NAB i_0 \Rightarrow C = \frac{2NABi_0}{\pi} \dots(i)$
- (c) Magnetic torque $\tau = NAB i$

The torque gives a sharp angular impulse to the coil as the current persists for a small time.

$$\text{Angular impulse} = \int \tau dt = NAB \int i dt = NABQ$$

$$\text{Change in angular momentum} = NABQ$$

$$\text{or, } I\omega_0 = NABQ$$

$$\text{or, } \omega_0 = \frac{NABQ}{I}$$

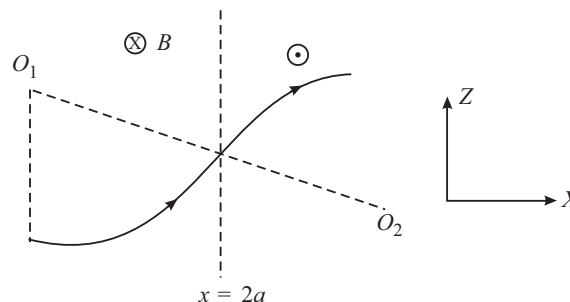
The coil acquires an angular speed ω when the sudden current is passed through it. As the coil rotates, the spring twists and the kinetic energy of the coil starts getting converted into the spring potential energy. The rotation stops when the entire kinetic energy gets converted into the spring potential energy. The potential energy stored in the spring when it is twisted by angle

$$\theta \text{ is } U = \frac{1}{2} C\theta^2.$$

$$\therefore \frac{1}{2} C\theta_{\text{max}}^2 = \frac{1}{2} I\omega_0^2$$

$$\begin{aligned} \text{or, } \theta_{\text{max}} &= \sqrt{\frac{I}{C}} \cdot \omega_0 = \sqrt{\frac{I}{C}} \cdot \frac{NABQ}{I} \\ &= \frac{NABQ}{\sqrt{I}} \sqrt{\frac{\pi}{2NABi_0}} \quad [\text{using (i)}] \\ &= Q \sqrt{\frac{NAB\pi}{2Ii_0}} \end{aligned}$$

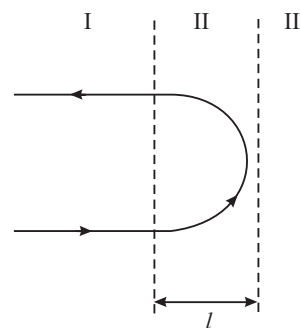
2. The magnetic field lines and the lines representing an induced electric field are closed loops.
3. In the diagram shown, the positive y direction is directed into the plane of the figure. Initially the particle moves on a circle with its centre at O_1 in the region $x = a$ to $x = 2a$. Then in the region $2a < x < 3a$ it moves on a circular path with its centre at O_2 .



4. The particle will follow a trajectory as shown in the figure if the radius of its circular path in region II is less than l .

$$\frac{mV}{qB} \leq l$$

$$\text{or, } V \leq \frac{qBl}{m}$$



In such a case the time spent in magnetic field = time needed to complete a semicircle

$$= \frac{\pi m}{qB} \quad (\text{independent of } V)$$

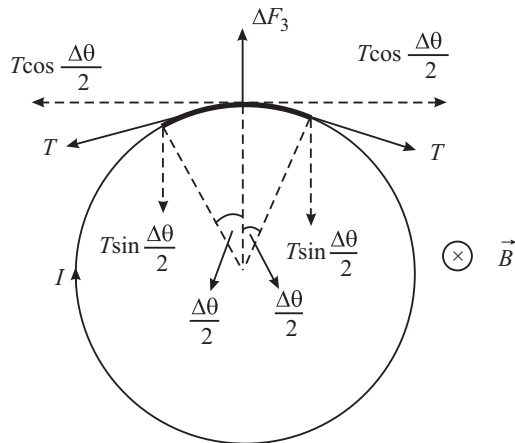
Path length inside the magnetic field is maximum when radius of the circular path is just equal to l . In this case,

$$V = \frac{qBl}{m}$$

The particle enters into region III if $V > \frac{qBl}{m}$

5. Consider a small segment on the circular wire subtending a small angle $\Delta\theta$ at the centre. Length of the segment is

$$\Delta l = R\Delta\theta$$



Magnetic force on the element is perpendicular to the length of the element. The force is in radial direction.

$$\Delta F_B = I\Delta l B = IRB\Delta\theta$$

Thus, force on the small segment gets balanced by the components of the tensions on its two ends (see figure)

$$2T \sin\left(\frac{\Delta\theta}{2}\right) = \Delta F_B$$

$$\text{or, } 2T \cdot \frac{\Delta\theta}{2} = IRB\Delta\theta \quad [\text{For small angle } \sin\theta \approx \theta]$$

$$\text{or, } T = IRB$$

$$= \frac{ILB}{2\pi} \quad \left[\because \text{Radius } R = \frac{L}{2\pi} \right]$$

6. The particles will take semicircular paths inside the magnetic field and exit the field region travelling in direction opposite to the initial direction of motion.

Since the two particles have different mass, they will have different time periods of circular motion

$\left(T = \frac{2\pi m}{qB}\right)$. They will spend different amount of time in the field region for completing their semicircular paths.

7. Area enclosed by the loop is

$A = 4$ (Area of a semicircle of radius $\frac{a}{2}$) + Area of a square of side a

$$= 4 \cdot \frac{1}{2} \pi \left(\frac{a}{2}\right)^2 + a^2 = \left(\frac{\pi}{2} + 1\right)a^2$$

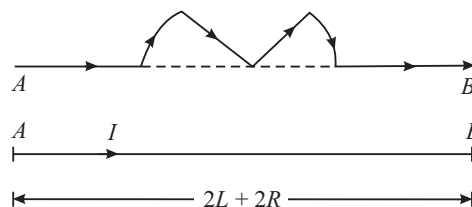
Curl your right-hand finger in the sense of the current. The thumb points outward in positive z direction.

$\therefore$ Magnetic dipole moment of the loop is

$$\vec{\mu} = \left(\frac{\pi}{2} + 1\right)a^2 I \hat{k}$$

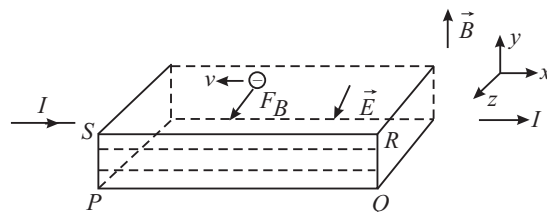
9. For calculation of the magnetic force, we can replace the entire wire with a straight wire AB having length $2L + 2R = 2(L + R)$

A magnetic force proportional to length (i.e., $2(L + R)$) acts when $\vec{B}$ is along y or z direction. There will be no force when $\vec{B}$ is along x direction.



10. An electron is moving in negative x direction inside the conductor. The magnetic field is along positive y direction. The magnetic force on a moving electron is $\vec{F}_B = -e(\vec{v} \times \vec{B}) = -evB(-\hat{i} \times \hat{j}) = evB\hat{k}$.

$\therefore$ Electrons move towards face $PQRS$ and the face becomes negatively charged. A potential difference develops between the two faces parallel to xy plane. Face $PQRS$ is at lower potential. Let the electric field directed from K to M be E . This field exerts a force eE on an electron in negative z direction.



No further charge accumulation will happen when

$$eE = evB \quad [v = \text{drift speed}]$$

$$\text{or, } E = vB \Rightarrow \frac{V}{w} = vB \quad \dots(i)$$

Here, V is potential difference between the two faces at separation w .

The current I is

$$I = (nev)(w \cdot d) \Rightarrow v = \frac{I}{newd} \quad \dots(ii)$$

when I is constant

for $w_1 = w_2$ and $d_1 = 2d_2$

$$v_1 = \frac{v_2}{2}$$

$\therefore$ From (i) $V_1 = w_1 v_1 B$

$$\text{and } V_2 = w_2 v_2 B = w_2 (2v_1) B = 2V_1$$

$\therefore$ option (a) is correct.

Check other options in similar way.

11. From (i) $V = wvB$

width w is same for both the strips

$$\therefore V \propto vB \quad \dots(i)$$

$$I = (nev)(wd)$$

$$\text{For the two strips } (n_1 ev_1)(wd) = (n_2 ev_2)(wd)$$

$$\Rightarrow n_1 v_1 = n_2 v_2 \quad \dots(ii)$$

In option (a):

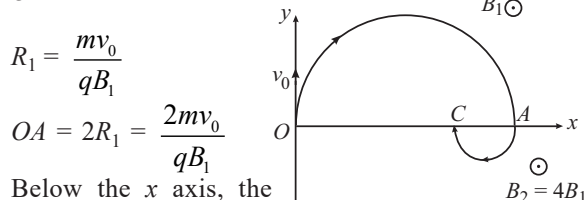
$$B_1 = B_2 \text{ and } n_1 = 2n_2$$

$$\therefore \text{From (ii) } 2v_1 = v_2$$

$$\therefore \text{From (i) } \frac{V_1}{V_2} = \frac{v_1}{v_2} = \frac{1}{2} \Rightarrow V_2 = 2V_1$$

Similarly, check other options.

12. The particle moves on circular path of radius R_1 from O to A .



$$R_1 = \frac{mv_0}{qB_1}$$

$$\therefore OA = 2R_1 = \frac{2mv_0}{qB_1}$$

Below the x axis, the field strength increases to $4B_1$ and the radius of the circular path becomes

$$R_2 = \frac{mv_0}{qB_2} = \frac{mv_0}{4qB_1} = \frac{R_1}{4}$$

$$\therefore AC = 2R_2 = \frac{R_1}{2} = \frac{mv_0}{2qB_1}$$

Time needed for this motion is

$$T = \frac{\pi m}{qB_1} + \frac{\pi m}{qB_2} = \frac{\pi m}{qB_1} + \frac{\pi m}{4qB_1} = \frac{5}{4} \frac{\pi m}{qB_1}$$

- $\therefore$ Question asks us to calculate average speed along x axis.

Distance travelled along x axis is $x = OA + AC$

$$= \frac{2mv_0}{2qB_1} + \frac{mv_0}{2qB_1}$$

$$= \frac{5mv_0}{qB_1}$$

$$\therefore \text{Average speed } \langle v \rangle = \frac{\frac{5mv_0}{qB_1}}{\frac{5}{4} \frac{\pi m}{qB_1}} = \frac{2v_0}{\pi} = 2 \text{ ms}^{-1}$$

13. In our discussion on a galvanometer, we considered a spring which opposes that rotation of the coil. In older versions, we had the coil suspended by a thread. The twisted thread exerts a counter torque just like a spring.

If i_g is the full scale deflection current, then

$$Ni_g AB = C\theta$$

$$\text{or, } i_g = \frac{C\theta}{NAB} = \frac{10^{-4} \times 0.2}{50 \times 2 \times 10^{-4} \times 0.02} = 0.1 \text{ A}$$

The value of shunt resistance is given by

$$R_s = \frac{i_g R_g}{i - i_g} = \frac{0.1 \times 50}{1 - 0.1}$$

$$\text{or, } R_s = \frac{50}{9} = 5.56 \Omega$$

MAGNETIC EFFECT OF CURRENT

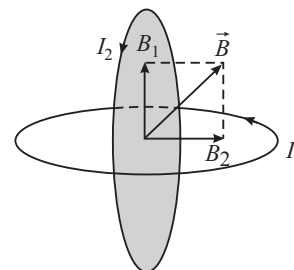
$$1. B_1 = \frac{\mu_0 I_1}{2R}$$

$$B_2 = \frac{\mu_0 I_2}{2R}$$

$$\therefore B = \sqrt{B_1^2 + B_2^2}$$

$$= \frac{\mu_0}{2R} \sqrt{I_1^2 + I_2^2}$$

$$= \frac{4\pi \times 10^{-7}}{2 \times 2\pi \times 10^{-2}} \sqrt{3^2 + 4^2} = 5 \times 10^{-5} \text{ T}$$



$$2. \mu_0 n_1 i_1 = B_1 \text{ and } \mu_0 n_2 i_2 = B_2$$

$$\therefore \frac{n_2 i_2}{n_1 i_1} = \frac{B_2}{B_1} \Rightarrow \frac{100 \times i / 3}{200 \times i} = \frac{B_2}{6.28 \times 10^{-2}}$$

$$\text{or, } B_2 = 1.05 \times 10^{-2} \text{ T}$$

3. The magnetic field at a distance r ($< a$) is given by

$$B = \frac{\mu_0 j}{2} r = \frac{\mu_0 i}{2\pi a^2} r$$

$$\text{At } r = \frac{a}{2}, \quad B_1 = \frac{\mu_0 i}{4\pi a}$$

Field at a distance r ($> a$) from the axis is given by

$$B = \frac{\mu_0 i}{2\pi r}$$

$$\therefore B_2 = \frac{\mu_0 i}{2\pi(2a)} = \frac{\mu_0 i}{4\pi a}$$

$$\therefore B_1 = B_2$$

$$5. B = \frac{\mu_0 i}{2\pi r} = 2 \times 10^{-7} \times \frac{100}{4} = 5 \times 10^{-6} \text{ T}$$

The direction of the field is given by the right hand rule and is southward.

6. Field due to circular arc BC is

$$B_1 = \frac{\mu_0 I}{2b} \cdot \frac{1}{12} = \frac{\mu_0 I}{24b} \otimes$$

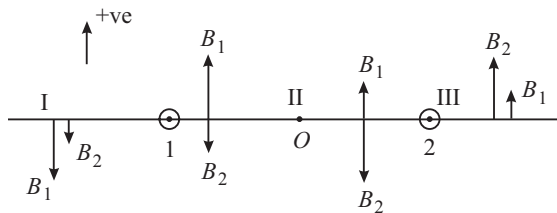
Field at O due to circular arc DA is

$$B_2 = \frac{\mu_0 I}{2a} \cdot \frac{1}{12} = \frac{\mu_0 I}{24a} \odot$$

Field at O due to AB and CD is zero. Therefore, the resultant field at O is

$$B = \frac{\mu_0 I}{24} \left[\frac{1}{a} - \frac{1}{b} \right] \odot = \frac{\mu_0 I(b-a)}{24ab} \odot$$

7. Field lines of the magnetic field produced by I , are circular. The arc AC and BC fall on two circular field lines. Any small element on these arcs carries current in the direction (or opposite) of the magnetic field due to I_1 . Force on these arcs are zero.
8. The diagram shows the directions of magnetic fields at various points due to the two currents.

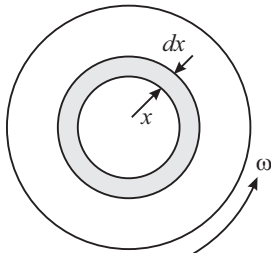


At mid-point O the two fields are equal and opposite and the resultant is zero. Close to wire 1, to the right of it, the field is very high and is directed upward ($B_1 > B_2$). The upward field has been considered as positive. In region I, the two fields are in negative direction. In region III, both the fields are in positive directions.

9. Refer to example 20 in chapter 2.

10. Surface charge density $\sigma = \frac{Q}{\pi R^2}$. Charge on a ring of radius x and thickness dx is

$$\begin{aligned} dQ &= \sigma \cdot 2\pi x dx = \frac{Q}{\pi R^2} \cdot 2\pi x dx \\ &= \frac{2Q}{R^2} x dx \end{aligned}$$



Current corresponding to this charge is

$$i = (dQ) f = \frac{2Q}{R^2} x dx \cdot \frac{\omega}{2\pi} = \frac{Q\omega}{\pi R^2} x dx$$

Field at the centre due to this current is

$$dB = \frac{\mu_0 i}{2x} = \frac{\mu_0 Q\omega}{2\pi R^2} dx$$

$\therefore$ Field due to the entire disc is

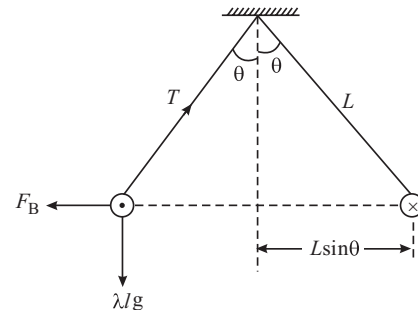
$$B = \frac{\mu_0 Q\omega}{2\pi R^2} \int_0^R dx = \frac{\mu_0 Q\omega}{2\pi R}$$

$$\therefore B \propto \frac{1}{R}$$

11. Magnetic repulsion force on length l of each wire is

$$F_B = \frac{\mu_0 I^2}{2\pi(2L \sin \theta)} l$$

Considering equilibrium of length l of one of the wires:



$$T \sin \theta = F_B$$

$$T \cos \theta = \lambda l g$$

$$\therefore \tan \theta = \frac{F_B}{\lambda l g} \Rightarrow \frac{\sin \theta}{\cos \theta} = \frac{\mu_0 I^2}{4\pi L \lambda g \sin \theta}$$

$$\therefore I = 2 \sin \theta \sqrt{\frac{\pi \lambda g L}{\mu_0 \cos \theta}}$$

13. Radius of circle, $r = \frac{l}{2\pi}$

$$\text{Field at the centre of the circle, } B_A = \frac{\mu_0 i}{2r} = \frac{\mu_0 \pi i}{l}$$

$$\text{Side length of the square, } a = \frac{l}{4}$$

Field at the centre of the square

$$\begin{aligned} B_B &= 4 \frac{\mu_0 i}{4\pi \left(\frac{a}{2}\right)} [\sin 45^\circ + \sin 45^\circ] = \frac{2\sqrt{2}\mu_0 i}{\pi a} \\ &= \frac{8\sqrt{2}\mu_0 i}{\pi l} \end{aligned}$$

$$\therefore \frac{B_A}{B_B} = \frac{\pi^2}{8\sqrt{2}}$$

14. Keeping the current constant, the dipole moment can be doubled by making the area of the loop two times. It means the radius is made $\sqrt{2}$ times.

$$B_1 = \frac{\mu_0 i}{2r} \text{ and } B_2 = \frac{\mu_0 i}{2\sqrt{2}r}$$

$$\therefore \frac{B_1}{B_2} = \sqrt{2}$$

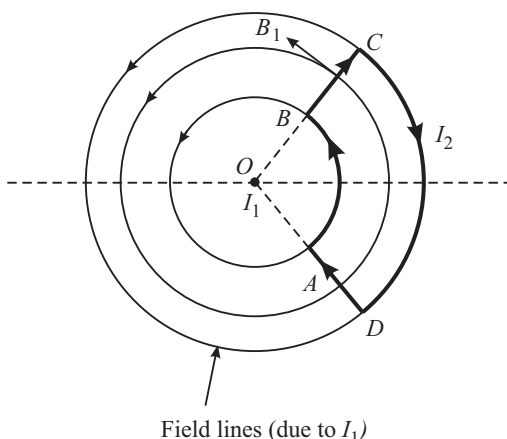
IIT JEE/JEE ADVANCED QUESTIONS

1. As explained in question 7 above (in JEE Main questions), there will be no force on the arc AB and CD .

Any small element on BC will experience a force

$$d\vec{F} = I_2 d\vec{l} \times \vec{B}_1$$

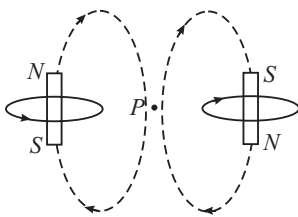
This force is directed out of the plane of the figure. Similarly, force on DA is directed into the plane of the figure.



Forces on BC and DA will rotate the loop clockwise when seen from O .

These two forces will be of same magnitude and the net force on the loop will be zero.

2. (A) For two parallel wires, the fields at P are oppositely directed and have same magnitude. Their resultant is zero.
- (B) Both the circular loops produce a field directed towards right, at P . The two parallel currents attract.
- (C) To have a better understanding let us replace the two loops with equivalent magnetic dipoles. We have shown one field line due to each dipole that passes through point P . Obviously, the two fields are oppositely directed. The resultant field at P is zero. The two magnetic dipoles attract one another as the north pole of one is closer to the south pole of the other.
- (D) Again replace the two loops with their equivalent magnetic dipoles to clearly understand the situation.



3. (p) The three positive charges are at 120° angular separation. They produce electric field of same strength at mutual angles of 120° . The resultant field at M is zero.

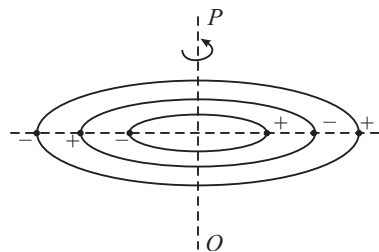
Similarly, the electric field due to the three negative charges has a zero resultant.

The potential at M is zero as sum of all the charges is zero and they all are at same separation. When the charge system rotates, the equivalent current is zero. ($\because$ total charge is zero). It means that $B = 0$ and $\mu = 0$.

- (q) It is easy to see that $E \neq 0$ and $V = 0$.

The rotating system is like three circular loops. Each loop has zero charge and there is no effective current.

$$\therefore B = 0; \mu = 0$$



- (r) Three + charges produce zero E .

Three - charges also produce zero E .

The negative potential due to the three negative charges is higher than the positive potential due to the positive charges.

- $\therefore$ At M we have a net negative potential.

When rotated with same frequency, the currents in the two circles are equal (but opposite).

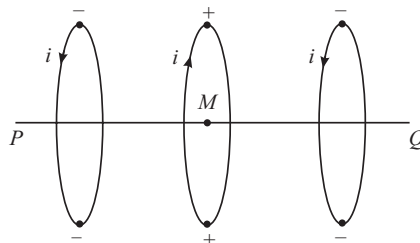
Magnetic field $\left(B = \frac{\mu_0 i}{2r} \right)$ due to the smaller circle has higher magnitude, therefore, there is a net B at M . There is of course a magnetic dipole moment as well.

- (s) It is easy to show that $V \neq 0$ and $E = 0$

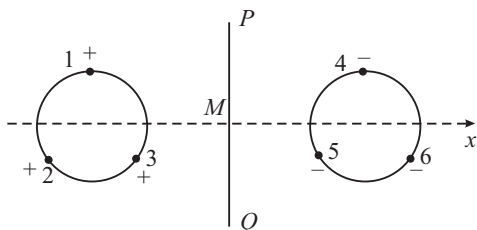
When rotated, there are three current loops.

There can be non-zero B at M .

There is a non-zero μ as well.



- (t) Charges 1 and 4 produce zero potential at M . Similarly, 2, 6, 3 and 5 produce zero potential. The electric field at M due to 1 and 4 is directed along x . Resultant field due to 2 and 6 is also in the same direction. Electric field due to 3 and 5 is also in the same direction.



$$\therefore E \neq 0$$

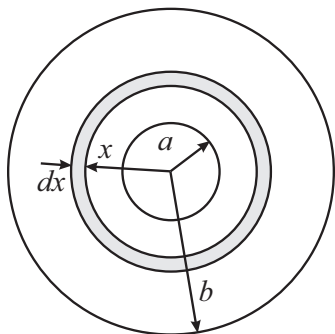
When rotated about PQ , the pair of charges 1 and 4 lie on a circle and the associated current is zero. Similarly the pairs 2 and 6 and 3 and 5 form loops with zero currents.

$$\therefore B = \mu = 0.$$

4. There are N turns in a width of $(b - a)$.

Consider a circular strip of width dx having radius x . Number of turns in the strip is

$$dN = \left(\frac{N}{b-a} \right) dx$$



Magnetic field due to the coil of radius x having dN turns, at its centre is

$$dB = \frac{\mu_0 I}{2\pi r} \cdot \left(\frac{N}{b-a} \right) dx$$

- $\therefore$ Field due to the entire spiral is

$$B = \int dB = \frac{\mu_0 NI}{2\pi(b-a)} \int_a^b \frac{dx}{x} = \frac{\mu_0 NI}{2\pi(b-a)} \ln \left(\frac{b}{a} \right)$$

5. Field at P can be calculated as

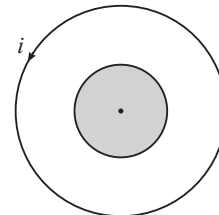
$$B = B_1 - B_2$$

where B_1 = Field due to a cylindrical wire without cavity

B_2 = Field due to a current inside the cavity region only

$$\therefore B = \frac{\mu_0 (J \cdot \pi a^2)}{2\pi a} - \frac{\mu_0 [J \cdot \pi (a/2)^2]}{2\pi \left(\frac{3a}{2} \right)} = \frac{5\mu_0 aJ}{12}$$

6. The diagram shows the cross-section of the arrangement. The shaded region has a current perpendicular to the plane of the figure. Field due to this is in tangential direction for any $r > 0$.



A current (i) in the solenoid coil produces a field that is directed normal to the plane of the figure. This field exists inside the outer circle ($r < 2R$). For $r > 2R$, there is no field due to the solenoid.

For the region $0 < r < R$, the magnetic field has a component in the plane of the figure and another component that is perpendicular to the figure.

In the region $r > 2R$, field is only because of the current in the cylinder.

7. Field due to the circular wire at a height h on the axis is

$$B_0 = \frac{\mu_0}{2} \frac{a^2 I}{(a^2 + h^2)^{3/2}}$$

Direction of this field is out of plane of the paper.

The two straight wires must produce field of equal magnitude in the downward direction.

The figure shows the cross-sections of the two wires. O is the centre of the circular loop and P is the point under consideration. Current in wire 1 is inward and that in wire 2 is outward.

B_1 is perpendicular to AP and B_2 is perpendicular to CP . Resultant of B_1 and B_2 is

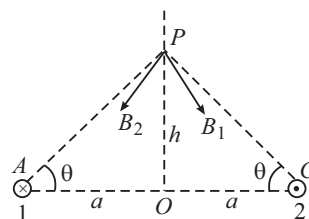
$$B_1 \cos\theta + B_2 \cos\theta = 2 \cdot \frac{\mu_0 I}{2\pi\sqrt{h^2 + a^2}} \cos\theta$$

$$= \frac{\mu_0 I}{\pi\sqrt{h^2 + a^2}} \frac{a}{\sqrt{h^2 + a^2}} = \frac{\mu_0 Ia}{\pi(h^2 + a^2)}$$

This field must have same magnitude as B_0

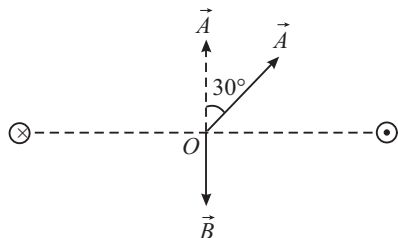
$$\therefore \frac{\mu_0 Ia}{\pi(h^2 + a^2)} = \frac{\mu_0}{2} \frac{a^2 I}{(a^2 + h^2)^{3/2}}$$

$$\text{or, } 4(a^2 + h^2) = \pi^2 a^2 \Rightarrow h \simeq 1.2a$$



8. Field at O due to the parallel wires is

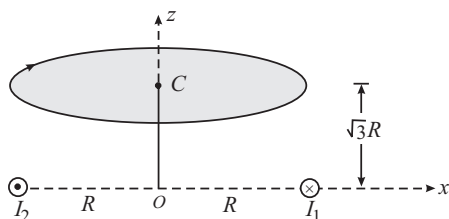
$$B = \frac{\mu_0 I}{2\pi d} + \frac{\mu_0 I}{2\pi d} = \frac{\mu_0 I}{\pi d}$$



The original area vector of the loop is opposite to $\vec{B}$. When the loop rotates by 30° , the area vector makes an angle of $180^\circ - 30^\circ$ with $\vec{B}$.

$$\begin{aligned} \therefore \tau &= \mu B \sin(180^\circ - 30^\circ) = I \cdot \pi a^2 \frac{\mu_0 I}{\pi d} \sin 30^\circ \\ &= \frac{\mu_0 I^2 a^2}{2d} \end{aligned}$$

9. (A) If I_1 and I_2 are in same direction, the magnetic field at O due to the straight wires is



This figure shows I_1 and I_2 in opposite directions

$$\begin{aligned} \vec{B}_1 - \vec{B}_2 &= \left(\frac{\mu_0 I_1}{2\pi R} - \frac{\mu_0 I_2}{2\pi R} \right) \hat{k} \\ &= 0 \quad (\text{For } I_1 = I_2) \end{aligned}$$

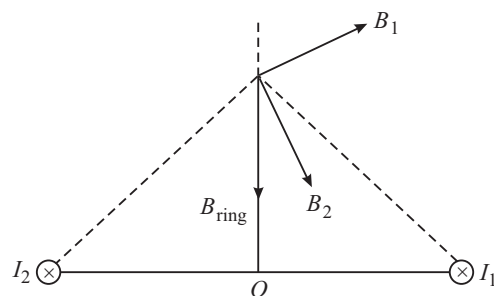
Field due to the ring is the net field at O in this case.

- (B) $I_1 > 0$ means it is in positive y direction. $I_2 < 0$ means it is in negative y directions. The above figure shows the situation where field at O due to the straight wires is

$$\vec{B}_1 + \vec{B}_2 = \frac{\mu_0}{2\pi R} (I_1 + I_2) \hat{k}$$

Field due to the ring is in negative z direction. Therefore, the resultant field at O can be zero for a certain value of I .

- (C) If $I_1 < 0$ and $I_2 > 0$, the field due to the straight wires will be in negative z direction. The field due to the ring is also in same direction. The resultant cannot be zero.
- (D) The situation is as shown. The z components of B_1 and B_2 cancel out.



$$\therefore B_z = B_{\text{ring}} = \frac{\mu_0 I}{2R}$$

Direction is in negative z direction.

ELECTROMAGNETIC INDUCTION

AIEEE/JEE Main Questions

1. $\varepsilon = - \frac{d\phi}{dt} = -20t + 50$

At $t = 3$, $\varepsilon = -20 \times 3 + 50 = -10\text{V}$

2. $\tau = \frac{L}{R} = \frac{10}{5} = 2 \text{ s}$

Current after a time constant is

$$i = \frac{V}{R} (1 - e^{-1}) = 1(1 - e^{-1})$$

3. $M = \pi \mu_0 n_1 n_2 r^2 l = \pi \mu_0 \frac{(n_1 l)(n_2 l)r^2}{l}$

$$= (\pi r^2) \frac{\mu_0 N_1 N_2}{l} = \frac{(10 \times 10^{-4})(4\pi \times 10^{-7})(300 \times 400)}{0.2}$$

$$= 2.4\pi \times 10^{-4} \text{ H}$$

4. The two parallel paths are independent. The current grows in the inductor with a time constant of

$$\tau = \frac{L}{R_2} = \frac{400\text{mH}}{2\Omega} = 200 \text{ ms} = 0.2\text{s}$$

$$\therefore i = i_0 (1 - e^{-t/\tau}) = \frac{E}{R_2} \left(1 - e^{-\frac{t}{0.2}} \right)$$

or, $i = 6 (1 - e^{-5t})$

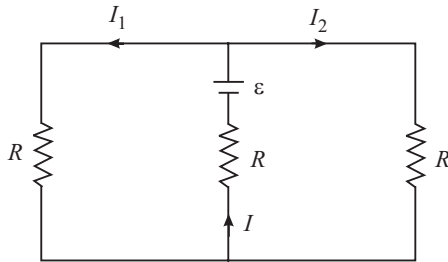
$$V_L = L \frac{di}{dt} = 400 \times 10^{-3} \times 6 \times 5 \times e^{-5t} = 12e^{-5t} \text{ V}$$

5. The effective electric circuit is as shown. Induced emf, $\varepsilon = Blv$

From symmetry, $I_1 = I_2$

And, $I = I_1 + I_2 = 2I_1$

Using Kirchoff's loop law



$$I_1 R + IR = \varepsilon$$

$$\text{or, } I_1 R + 2I_1 R = Blv \Rightarrow I_1 = \frac{Blv}{3R}$$

$$\therefore I = \frac{2Blv}{3R}$$

6. B , v and l are mutually perpendicular

$$\therefore \varepsilon = Blv = 5 \times 10^{-5} \times 1.5 \times 2 = 0.15 \text{ mV}$$

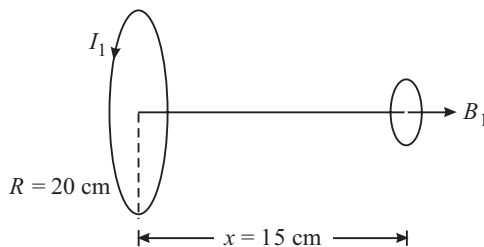
7. The energy stored in capacitor becomes half the original when both L and C have same amount of energy. This implies that the charge on the capacitor becomes $\frac{q_0}{\sqrt{2}}$.

$$q = q_0 \cos \omega t \Rightarrow \frac{q_0}{\sqrt{2}} = q_0 \cos \omega t$$

$$\Rightarrow \omega t = \frac{\pi}{4}$$

$$\Rightarrow t = \frac{\pi}{4} \sqrt{LC}$$

8. The eddy currents induced in the plate will oppose the cause of motion, i.e. the oscillations.
9. We will first find the mutual inductance for the pair of the coils. Consider the bigger coil as primary and let i_1 be the current given to it. The magnetic field produced by this current at a distance x on the axis is



$$B_1 = \frac{\mu_0}{2} \frac{R^2 i_1}{(R^2 + x^2)^{3/2}}$$

This field can be assumed to be uniform over the entire area of the smaller loop as its radius is too small ($r = 0.3 \text{ cm}$).

- $\therefore$ Flux linked with the smaller loop is

$$\phi_2 = B_1 \cdot \pi r^2 = \frac{\mu_0 \pi}{2} \frac{R^2 r^2}{(R^2 + x^2)^{3/2}} i$$

$$\therefore M = \frac{\mu_0 \pi R^2 r^2}{2(R^2 + x^2)^{3/2}}$$

Value of M does not change when the smaller loop is made primary. When a current $i_2 = 2.0 \text{ A}$ is given to the smaller loop the flux linked with the primary is

$$\begin{aligned} \phi_2 &= M i_2 = \frac{\mu_0 \pi R^2 r^2}{2(R^2 + x^2)^{3/2}} i_2 \\ &= \frac{4\pi \times 10^{-7} \times \pi \times (0.2)^2 \times (0.003)^2}{2[(0.2)^2 + (0.15)^2]^{3/2}} = 9.1 \times 10^{-11} \text{ Wb} \end{aligned}$$

10. Switch S_2 is kept open. It means that the inductor is never in the circuit. In the charging circuit of the capacitor the charge on the capacitor varies with time as $q = q_0 (1 - e^{-t/\tau}) = CV (1 - e^{-t/\tau})$.
11. When A and C are connected and the circuit is in steady state, the current through the RL combination is $i_0 = \frac{V}{R}$

When C is connected to B , the current begins to decay.

$$i = i_0 e^{-t/\tau}$$

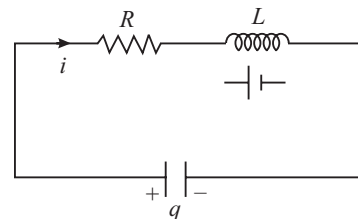
$$\frac{di}{dt} = \frac{i_0}{\tau} e^{-t/\tau}$$

$$\frac{V_R}{V_L} = \frac{iR}{-L \frac{di}{dt}} = \frac{R i_0 e^{-t/\tau}}{-L \frac{1}{\tau} e^{-t/\tau}} = -1 \quad \left[\because \tau = \frac{L}{R} \right]$$

The negative sign indicates that the two voltages are opposite. One is drop and the other is gain when you move around the loop.

12. The energy of the system will decrease exponentially with time. The energy decay happens slowly when the inertia is higher.

To throw more light, consider the circuit at any time t .



$$\frac{q}{C} = Ri + L \frac{di}{dt} \quad \therefore i = -\frac{dq}{dt}$$

$$\therefore \frac{q}{C} = -R \frac{dq}{dt} - L \frac{d^2 q}{dt^2}$$

$$\text{or, } L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = 0$$

The solution to this differential element is of the form

$$q = q_0 e^{-t/\tau} \cos(\omega t + \delta)$$

$$\text{Here } \tau = \frac{2L}{R} \text{ and } \omega = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

The amplitude of charge varies as

$$Q_{\max} = q_0 e^{-t/\tau}$$

$$\therefore Q_{\max}^2 = q_0^2 e^{-2t/\tau}$$

Larger τ means slower decay.

$$13. i_0 = \frac{15V}{0.15k\Omega} = \frac{1}{10} \text{ A}$$

$$\tau = \frac{L}{R} = \frac{0.03}{0.15 \times 10^3} = 2 \times 10^{-4} = 0.2 \text{ ms}$$

$t = 1\text{ms}$ is 5 times the time constant. In the decaying current circuit, the current at $t = 1\text{ms}$ is $i = i_0 e^{-t/\tau}$

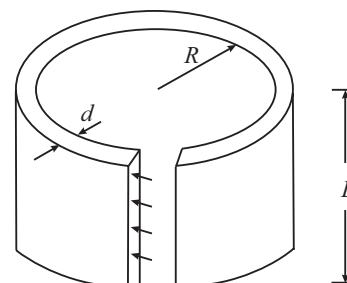
$$= 0.1 e^{-5} = \frac{0.1}{150} \text{ s} = 0.67 \text{ ms}$$

$$14. \text{ Total charge flow } \Delta Q = \int i dt = \text{area under the graph.}$$

$$\therefore \Delta Q = \frac{1}{2} \times 0.5 \times 10 = 2.5 \text{ C}$$

$$\text{we know that } \Delta Q = \frac{\Delta \phi}{R}$$

$$\therefore \Delta \phi = 100 \times 2.5 = 250 \text{ Wb}$$



The figure shows how the current is directed.
It flows through a cross section of Ld .

$$\therefore i = \frac{\varepsilon}{R} = - \frac{\mu_0 L d n a^2 i_0 \omega \cos \omega t}{2\rho R}$$

3. It is a charging RC circuit.

$$Q = Q_0 (1 - e^{-t/\tau})$$

Obviously, option (b) is correct.

$$4. \frac{Q}{C} - L \frac{di}{dt} = 0$$

$$\text{Put } i = -\frac{dQ}{dt}$$

$$L \frac{d^2 Q}{dt^2} = -\frac{Q}{C} \Rightarrow Q = -LC \frac{d^2 Q}{dt^2}$$

5. The energy is exchanged between the electric field energy in the capacitor and the magnetic field energy in the inductor.

6. Magnetic field and induced electric field are represented by closed lines.

7. A static charge produces only E .

When the disc is spinning, there is an associated dipole moment. It produces E as well as B .

A constant current in a ring also produces E and B .

A time changing current produces a time changing B . A time changing B induces an electric field.

8. Refer to question 8 in worksheet 2 of the chapter "Electromagnetic Induction".

9. When current in the solenoid increases from 0 to I , it creates a magnetic field. The flux linked with the ring changes suddenly and a current is induced in it. The ring experiences a sudden repulsive force and jumps up.

The emf induced in the two rings will be same but induced current is higher for the one which has low resistance.

IIT JEE/JEE ADVANCED QUESTIONS

1. There is no emf because there is no change in flux.

2. The figure shows the cross section. The inner circle of radius a has a magnetic field directed perpendicular to the plane of the figure. The value of this field is

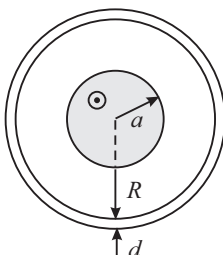
$$B = \mu_0 n i = \mu_0 n i_0 \sin \omega t$$

$$\text{Flux through circle of radius } R \text{ is } \phi = B \cdot \pi a^2 = \mu_0 \pi n i_0 a^2 \sin \omega t$$

Emf induced in the cylindrical shell will be

$$\varepsilon = -\frac{d\phi}{dt} = -\mu_0 \pi n i_0 a^2 \omega \cos(\omega t)$$

$$\text{Resistance of the shell is } R = \frac{\rho 2\pi R}{Ld}$$



If $\rho_A < \rho_B$, induced current in A is higher. It experiences bigger impulsive force. If $m_A = m_B$ or $m_A < m_B$, the ring will certainly go up to larger height.

10. Compare the tube to an ideal solenoid.

$$B = \mu_0 n i = \mu_0 \frac{I}{L}$$

(ni) in a solenoid is nothing but the total current along the surface of the cylinder on its unit length

Flux through the wire loop, $\phi = \pi r^2 \cdot B$

$$\text{or, } \phi = \frac{\mu_0 I \pi r^2}{L}$$

Induced emf in the wire loop is

$$\varepsilon = \frac{d\phi}{dt} = \frac{\mu_0 \pi r^2}{L} \frac{dI}{dt}$$

$$\text{Induced current, } i = \frac{\varepsilon}{R} = \frac{\mu_0 \pi r^2}{LR} \frac{dI}{dt}$$

$$\begin{aligned} \text{Magnetic moment} &= i \cdot \pi r^2 = \frac{\mu_0 (\pi r^2)^2}{LR} \frac{dI}{dt} \\ &= \frac{\mu_0 (\pi r^2)^2}{LR} I_0 \times 300 \times \cos(300t) \end{aligned}$$

$$\therefore N = \frac{300 \times \pi^2 \times 10^{-4}}{0.005 \times 10} = \frac{\pi^2 \times 3 \times 10^{-2}}{5 \times 10^{-3} \times 10} \simeq 6 \quad [\pi^2 \simeq 10]$$

11. Consider the circular loop to be primary. Assume a current i in it. The magnetic field on the axis at a distance $z = \sqrt{3} R$ is

$$B = \frac{\mu_0 i R^2}{2(R^2 + z^2)^{3/2}} = \frac{\mu_0 i R^2}{2(R^2 + 3R^2)^{3/2}} = \frac{\mu_0 i}{16R}$$

The field is directed along the axis. We can assume the flux to be uniform over the entire area of the square loop. The area vector of the square frame makes 45° with the field direction.

$$\therefore \phi = NBA \cos 45^\circ = 2 \cdot \frac{\mu_0 i}{16R} a^2 \cdot \frac{1}{\sqrt{2}}$$

$$\Rightarrow \phi = \frac{\mu_0 i a^2}{8\sqrt{2}R}$$

$$\therefore M = \frac{\mu_0 a^2}{2^{7/2} \cdot R}$$

$$\therefore p = 7$$

$$12. E \cdot 2\pi R = \pi R^2 \frac{dB}{dt} \quad \Rightarrow E = \frac{RB}{2} \quad \left[\because \frac{dB}{dt} = \frac{B}{1} \right]$$

$$13. \text{ Force due to induced electric field, } F = \frac{QRB}{2}$$

$$\text{Torque due to this force, } \tau = \frac{QR^2 B}{2}$$

This torque lasts for 1 s. The angular impulse of this

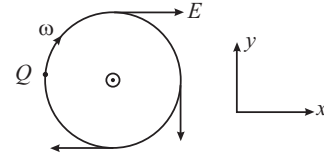
$$\text{torque is } = \tau \Delta t = \left(\frac{QR^2 B}{2} \right) \quad (1)$$

$\therefore$ Change in angular momentum is

$$\Delta L = \frac{QR^2 B}{2}$$

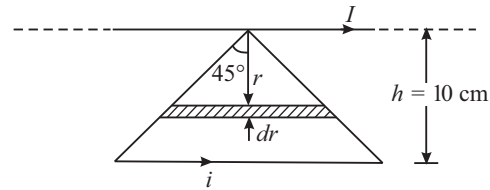
$$\Delta \mu = \gamma \Delta L = \frac{\gamma QR^2 B}{2}$$

Direction: when B increases in z direction, it will induce an electric field E as directed. If the charge was rotating clockwise, it will speed up. Change in angular momentum will be in negative z direction. The magnetic dipole moment will also increase (pointing in negative z direction).



If charge is rotating anticlockwise, the induced electric field E will slow it. Now angular momentum as well as magnetic dipole moment will decrease.

14. Refer to example 50 in the chapter “Electromagnetic Induction.”



Consider a current I in the long wire. Field at a distance r is

$$B = \frac{\mu_0 I}{2\pi r}$$

Flux in a strip of width dr is

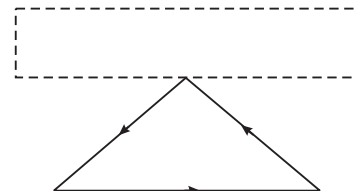
$$d\phi = B \cdot (2r \tan 45^\circ) dr$$

$$= \frac{\mu_0 I}{\pi} \int_0^L dr = \frac{\mu_0 h}{\pi} I$$

$$\therefore \text{Mutual inductance, } M = \frac{\mu_0 h}{\pi}$$

When a current is given to the triangle, induced emf in the long wire is

$$\varepsilon = M \frac{di}{dt} = \frac{\mu_0 h}{\pi} \times 10 = \frac{\mu_0}{\pi} \quad [h = 0.1\text{m}]$$

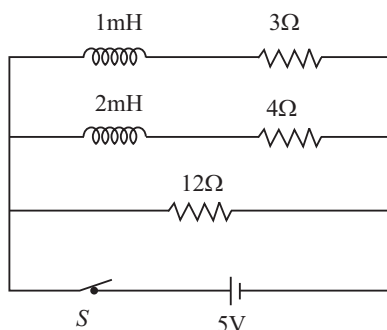


Think of the long wire as a part of an infinite rectangular loop. When the current in triangle is increased, it causes the flux through the rectangle to increase. A current is induced which produces repulsion. This has a tendency to increase the distance between the loops and decrease the flux.

15. As the loop enters, a counter clockwise current gets induced in it. The magnetic force on the front arm slows it down. This reduced the emf and hence the current. Once the loop is completely inside the field region, there is no emf. There is no current. There is no force on the loop and velocity becomes constant. When the loop begins to exit, again an emf is induced. This time direction of induced current is clockwise. The emf decreases in magnitude. The loop will again retard.

16. At $t = 0^+$, there is current only in 12Ω resistance.

$$I_{\min} = \frac{5}{12} \text{ A}$$



At $t \rightarrow \infty$, the inductance has no role. The equivalent resistance is

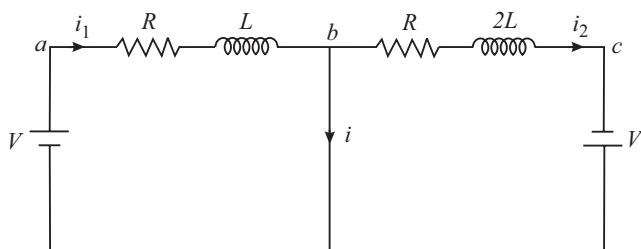
$$\frac{1}{R} = \frac{1}{3} + \frac{1}{4} + \frac{1}{12} = \frac{2}{3}$$

$$\therefore R = \frac{3}{2} \Omega$$

$$I_{\max} = \frac{5}{3/2} = \frac{10}{3} \text{ A}$$

$$\therefore \frac{I_{\max}}{I_{\min}} = 8$$

17. The L-R combination between a and b is continuously subjected to a potential difference of V .



$$\therefore i_1 = \frac{V}{R} \left[1 - e^{-\frac{Rt}{L}} \right]$$

Similarly, the L-R combination between b and c always sees an emf of $2V$.

$$\therefore i_2 = \frac{V}{R} \left[1 - e^{-\frac{Rt}{2L}} \right]$$

$$\text{Now, } i = i_1 - i_2 = \frac{V}{R} \left[e^{-\frac{Rt}{2L}} - e^{-\frac{Rt}{L}} \right] \quad \dots(i)$$

For i to be maximum $\frac{di}{dt} = 0$

$$\text{or, } -\frac{R}{2L} e^{-\frac{Rt}{2L}} + \frac{R}{L} e^{-\frac{Rt}{L}} = 0$$

$$\text{or, } e^{-\frac{Rt}{2L}} = \frac{1}{2}$$

$$\text{or, } \frac{Rt}{2L} = \ln 2 \Rightarrow t = \frac{2L}{R} (\ln 2)$$

$$\text{Put } e^{-\frac{Rt}{2L}} = \frac{1}{2} \text{ in (i)}$$

$$i_{\max} = \frac{V}{R} \left[\frac{1}{2} - \frac{1}{4} \right] = \frac{V}{4R}$$

ALTERNATING CURRENT CIRCUITS

AIEEE/JEE Main Questions

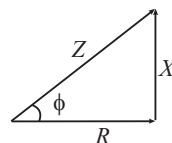
1. The current flow and power delivered will be maximum if the LCR circuit is in resonance

$$\frac{1}{\sqrt{LC}} = 2\pi f$$

$$\text{or, } C = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4 \times 10^2 \times 50^2 \times 10} \\ = 10^{-6} \text{ F} = 1\mu\text{F}$$

$$[\pi^2 \simeq 10]$$

$$2. \cos \phi = \frac{R}{Z} = \frac{12}{15} = 0.8$$



3. When the circuit has R and L

$$\phi < 90^\circ$$

$$4. IR = V_R$$

$$\Rightarrow I = \frac{100}{1000} = 0.1 \text{ A}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{200 \times 2 \times 10^{-6}} = 250\Omega$$

Circuit is in resonance

$$\therefore X_L = X_C = 250\Omega$$

6. $\phi = \frac{\pi}{2}$; $\cos \phi = 0$

$\therefore P = 0$

7. With only L and R , $\tan \phi = \frac{X_L}{R}$

or, $\frac{X_L}{R} = \frac{1}{\sqrt{3}}$

with only C and R ; $\tan \phi = \frac{X_C}{R}$

or, $\frac{X_L}{R} = \frac{1}{\sqrt{3}}$

With L , C and R the circuit will be in resonance as $X_C = X_L$.

$\therefore P = \frac{V^2}{R} = \frac{(220)^2}{200} = 242\text{W}$

8. $V_L^2 + V_R^2 = (220)^2 \Rightarrow V_L^2 = (220)^2 - (80)^2$

or, $V_L = 204.9\text{ V}$

$IX_L = 204.9\text{V} \Rightarrow X_L = \frac{204.9}{10} = 20.49\Omega$

or, $2\pi f \cdot L = 20.49 \Rightarrow L = \frac{20.49}{2 \times 3.14 \times 50} = 0.065\text{H}$

9. $Q = \frac{\omega_0}{\Delta\omega} = \frac{\omega_0 L}{R}$

where ω_0 is resonance frequency.

(t) $R = 1000\Omega$, $X_C = 1062\Omega$

$X_C > R$

$\therefore V_2 > V_1$

3. In case (A) capacitance is C and $X_C = \frac{1}{\omega C} = X_0$ (say)

In case (B) capacitance is $4C$ and $X_C = \frac{1}{4\omega C} = \frac{X_0}{4}$

Impedance is lower in case (B). Current will be higher in this case.

$\therefore I_R^A < I_R^B$

$V_R^A = I_R^A \cdot R$ and $V_R^B = I_R^B \cdot R$

$\therefore V_R^A < V_R^B$

Since voltage across R is higher in case B , the voltage across C must be lower.

$V_C^A > V_C^B$

4. $Z = \sqrt{R^2 + X_C^2}$ [Z = 1.25R]

or, $R^2 + X_C^2 = 1.25R^2 \Rightarrow X_C = \frac{R}{2}$

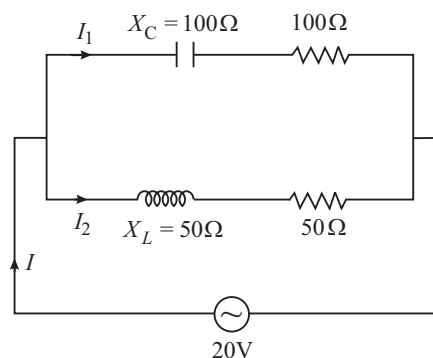
or, $\frac{1}{\omega C} = \frac{R}{2}$

Time constant $\tau = RC = \frac{2}{\omega} = \frac{2}{500} = 4\text{ms}$

5. $X_C = \frac{1}{\omega C} = \frac{1}{100 \times 100 \times 10^{-6}} = 100\Omega$

$X_L = \omega L = 100 \times 0.5 = 50\Omega$

Impedance of path having capacitor is



$Z_1 = \sqrt{100^2 + 100^2} = 100\sqrt{2}\Omega$

$I_1 = \frac{20V}{100\sqrt{2}\Omega} = \frac{1}{5\sqrt{2}}\text{A}$

Current I_1 leads in phase to the source voltage by an angle

$\phi_1 = \tan^{-1} \left(\frac{X_C}{R} \right) = \tan^{-1}(1) = \frac{\pi}{4}$

IIT JEE/JEE ADVANCED QUESTIONS

1. $X_C = \frac{1}{\omega C}$

When ω is increased, X_C decreases. The impedance of the circuit decreased and the current increases. The bulb glows brighter.

2. (p) There is no current in steady state as the capacitor will not allow a current through it.

$V_1 = 0$; $V_2 = V$

(q) $I = \frac{V}{R}$ and $V_1 = 0$, $V_2 = V$

(r) $X_L = \omega L = 2\pi fL = 2 \times 3.14 \times 50 \times 6 \times 10^{-3} = 1.88\Omega$

$V_1 = IX_L$; $V_2 = IR$

$\therefore R > X_L \therefore V_2 > V_1$

(s) $X_L = 1.88\Omega$

$X_C = \frac{1}{2\pi fC} = \frac{1}{2 \times 3.14 \times 50 \times 3 \times 10^{-6}} = 1062\Omega$

$V_2 = IX_C$; $V_1 = IX_L$

$\therefore X_C > X_L \therefore V_2 > V_1$

Similarly, current in path having inductor is

$$I_2 = \frac{20V}{Z_2} = \frac{20}{\sqrt{50^2 + 50^2}} = \frac{2}{5\sqrt{2}} \text{ A}$$

Current I_2 lags in phase to the source voltage by an angle

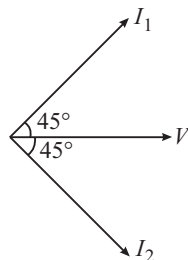
$$\phi_2 = \tan^{-1} \left(\frac{X_L}{R} \right) = \tan^{-1}(1) = \frac{\pi}{4}$$

Current I is sum of I_1 and I_2 which are at a phase difference of $\frac{\pi}{2}$.

$$\therefore I = \sqrt{I_1^2 + I_2^2} = \frac{1}{\sqrt{10}} \text{ A} \simeq 0.3 \text{ A}$$

Voltage across 100Ω resistance is

$$V_R = RI_1 = 100 \times \frac{1}{5\sqrt{2}} = 10\sqrt{2} \text{ V}$$



6. At the power plant, a step-up transformer has been used. Therefore, the voltage (V_0) used for transmission is given by

$$\frac{V_0}{4000} = \frac{10}{1} \Rightarrow V_0 = 40,000 \text{ V}$$

This 40,000 V is step down to 200V at the consumer end.

$$\frac{40000}{200} = \frac{N_p}{N_s}$$

$$\Rightarrow \frac{N_p}{N_s} = 200$$

7. $i = \frac{P}{V} = \frac{600 \times 10^3}{4000} = 150 \text{ A}$

When the power is transmitted at 4000 V, the transmission cables carry a current of 150A.

Resistance of cable, $R = 0.4 \times 20 = 8\Omega$

$$\therefore \text{Power loss} = i^2 R = (150)^2 \times 8 = 18000 \text{ W} = 180 \text{ kW}$$

$$\text{Loss percentage} = \frac{180}{600} \times 100 = 30\%$$

8. Total charge flow to the capacitor in time $t = \frac{7\pi}{6\omega}$ is

$$\begin{aligned} Q &= \int_0^t I dt = I_0 \int_0^t \cos(\omega t) dt \\ &= \frac{I_0}{\omega} [\sin \omega t]_0^t = \frac{1}{500} \left[\sin \left(\frac{7\pi}{6} \right) - \sin 0 \right] \\ &= \frac{1}{500} \left[-\frac{1}{2} - 0 \right] = -\frac{1}{1000} \text{ C} \end{aligned}$$

It is easy to see that charge Q will be maximum when

$$\omega t = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{2\omega}$$

$\therefore$ At $t = \frac{\pi}{2\omega}$ (which is before the end of the charging interval at $t = \frac{7\pi}{6\omega}$) the charge on the capacitor is maximum.

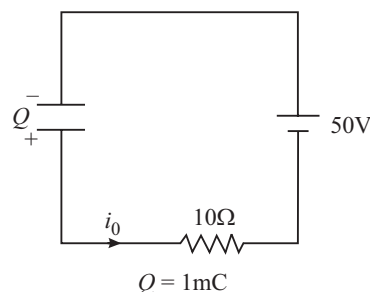
$$Q_{\max} = \frac{I_0}{\omega} \left[\sin \frac{\pi}{2} - \sin 0 \right] = \frac{1}{500} = 2 \times 10^{-3} \text{ C}$$

However, at the instant the switch is shifted to D , the charge on the capacitor is $Q = -\frac{1}{1000} \text{ C}$. The negative sign indicates that the upper plate is negative.

$$\text{At } t = \frac{7\pi}{6\omega}$$

$$I = I_0 \cos \left(\frac{7\pi}{6} \right) \rightarrow \text{negative}$$

$\therefore$ Current is anticlockwise just before shifting the switch. Just after the switch is moved to D , the circuit is as shown voltage across the capacitor is



$$V_C = \frac{Q}{C} = \frac{1\text{mC}}{20\mu\text{F}} = 50\text{V}$$

$$\therefore i_0 \cdot R = 50 + 50$$

$$\text{or, } i_0 = \frac{100}{10} = 10 \text{ A}$$

ELECTROMAGNETIC WAVES

AIEEE/JEE Main Questions

$$1. E_0 = cB_0 = 3 \times 10^8 \times 20 \times 10^{-9} = 6 \text{ V m}^{-1}$$

3. Infrared waves, known as heat waves, can penetrate deep into your skin and warms the muscles. It is used to relieve muscular pains and stiff joints.

Radio waves, due to their long wavelengths are suitable for broadcasting.

X-rays can penetrate our skin and flesh but get reflected from our bones. This helps in taking pictures of bones.

Harmful UV rays from the Sun are absorbed by the ozone layer in the atmosphere.

4. In air $E_1 = E_{01} \cos \left[2\pi v \left(\frac{z}{c} - t \right) \right]$

Angular wave number $k_1 = \frac{2\pi v}{c}$

Angular frequency $\omega_1 = 2\pi v$

Speed of wave in air, $v_1 = \frac{\omega_1}{k_1} = c$

In medium $E_2 = E_{02} \cos [k (2z - ct)]$

$k_2 = 2k, \omega_2 = kc$

$\therefore$ Speed in medium, $v_2 = \frac{\omega_2}{k_2} = \frac{c}{2}$

But speed of an electromagnetic wave in a medium is given by

$$v = \frac{1}{\sqrt{\epsilon\mu}}$$

Medium is non-magnetic, means $\mu_1 = \mu_2 = \mu_0$

$$\therefore v \propto \frac{1}{\sqrt{\epsilon}} = \frac{1}{\sqrt{\epsilon_r \epsilon_0}}$$

$\therefore$ Speed getting halved means that ϵ_r has become four times.